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Patryk's Notes

Patryk Kozlowski

October 10, 2026

Chapter 1

G_0W_0 Derivations: 11/9

1.1 Deriving Hedin's equations

1.1.1 Time-Domain Definition of the Green's Function

Start by considering the equation of motion for the field operators

$$i\frac{\partial}{\partial t}\hat{\psi}(\mathbf{x},t) = [\hat{\psi}(\mathbf{x},t), \hat{H}_{elec}] = [\hat{\psi}(\mathbf{x},t), \hat{H}_0 + \hat{H}_{int}] = [\hat{\psi}(\mathbf{x},t), \hat{H}_0] + [\hat{\psi}(\mathbf{x},t), \hat{H}_{int}] \tag{1.1}$$

For the non-interacting part, we have

$$[\hat{\psi}(\mathbf{x},t), \hat{H}_0] = [\hat{\psi}(\mathbf{x},t), \int d\mathbf{x}'\hat{\psi}^\dagger(\mathbf{x}',t)\hat{h}^0(\mathbf{x}')\hat{\psi}(\mathbf{x}',t)] \tag{1.2}$$

$$= \int d\mathbf{x}'[\hat{\psi}(\mathbf{x},t), \hat{\psi}^\dagger(\mathbf{x}',t)\hat{h}^0(\mathbf{x}')\hat{\psi}(\mathbf{x}',t)] \tag{1.3}$$

$$= \int d\mathbf{x}' \left(\underbrace{[\hat{\psi}(\mathbf{x},t), \hat{\psi}^\dagger(\mathbf{x}',t)]}_{\delta(\mathbf{x}-\mathbf{x}')} \hat{h}^0(\mathbf{x}')\hat{\psi}(\mathbf{x}',t) + \hat{\psi}^\dagger(\mathbf{x}',t)\hat{h}^0(\mathbf{x}') \underbrace{[\hat{\psi}(\mathbf{x},t), \hat{\psi}(\mathbf{x}',t)]}_{0} \right) \tag{1.4}$$

$$= h^0(\mathbf{x})\hat{\psi}(\mathbf{x},t). \tag{1.5}$$

For the interacting part, we have

$$[\hat{\psi}(\mathbf{x},t), \hat{H}_{int}] = \frac{1}{2}\int d\mathbf{x}'d\mathbf{x}''v(\mathbf{x},\mathbf{x}')[\hat{\psi}(\mathbf{x},t), \hat{\psi}^\dagger(\mathbf{x}',t)\hat{\psi}^\dagger(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}',t)] \tag{1.6}$$

$$= \frac{1}{2}\int d\mathbf{x}'d\mathbf{x}''v(\mathbf{x},\mathbf{x}') \left(\delta(\mathbf{x}-\mathbf{x}')\hat{\psi}^\dagger(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}',t) + \hat{\psi}^\dagger(\mathbf{x}',t)\delta(\mathbf{x}-\mathbf{x}'')\hat{\psi}(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}',t) \right) \tag{1.7}$$

$$= \int d\mathbf{x}'\hat{\psi}^\dagger(\mathbf{x}',t)v(\mathbf{x},\mathbf{x}')\hat{\psi}(\mathbf{x}',t)\hat{\psi}(\mathbf{x},t) \tag{1.8}$$

so overall we have

$$i\frac{\partial}{\partial t}\hat{\psi}(\mathbf{x},t) = \left(\hat{h}^0(\mathbf{x}) + \int d\mathbf{x}'v(\mathbf{x},\mathbf{x}')\hat{\psi}^\dagger(\mathbf{x}',t)\hat{\psi}(\mathbf{x}',t) \right) \hat{\psi}(\mathbf{x},t) \tag{1.9}$$

Now we can consider the equation of motion for the Green's function, defined as $G(\mathbf{x}t, \mathbf{x}'t') = -i \langle N| \mathcal{T}\left(\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right)|N\rangle$, where $\mathcal{T}$ is the time-ordering operator.

$$\frac{\partial}{\partial t}G(\mathbf{x}t,\mathbf{x}'t') = -i\langle N|\frac{\partial}{\partial t}\mathcal{T}\left(\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right)|N\rangle \tag{1.10}$$

$$\tag{1.11}$$

Now,

$$\frac{\partial}{\partial t}\mathcal{T}\left(\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right) = \frac{\partial}{\partial t}\left(\theta(t-t')\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t') - \theta(t'-t)\hat{\psi}^\dagger(\mathbf{x}',t')\hat{\psi}(\mathbf{x},t)\right) \tag{1.12}$$

$$= \underbrace{\frac{\partial\theta(t-t')}{\partial t}}_{\delta(t-t')}\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t') + \theta(t-t')\frac{\partial}{\partial t}\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t') - \underbrace{\frac{\partial\theta(t'-t)}{\partial t}}_{-\delta(t'-t)}\hat{\psi}^\dagger(\mathbf{x}',t')\hat{\psi}(\mathbf{x},t) - \theta(t'-t)\frac{\partial}{\partial t}\hat{\psi}^\dagger(\mathbf{x}',t \tag{1.13}$$

$$= \underbrace{\delta(t-t')\left\{\hat{\psi}(\mathbf{x},t),\hat{\psi}^\dagger(\mathbf{x}',t')\right\}}_{\delta(\mathbf{x}-\mathbf{x}')\delta(t-t')} + \mathcal{T}\left(\frac{\partial}{\partial t}\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right) \tag{1.14}$$

$$\tag{1.15}$$

So now consider plugging in the equation of motion for $\hat{\psi}(\mathbf{x},t)$ into the above expression

$$\mathcal{T}\left(\frac{\partial}{\partial t}\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right) = \mathcal{T}\left(-i\left(\hat{h}^0(\mathbf{x}) + \int d\mathbf{x}''v(\mathbf{x},\mathbf{x}'')\hat{\psi}^\dagger(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}'',t)\right)\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right) \tag{1.16}$$

$$= -i\hat{h}^0(\mathbf{x})\mathcal{T}\left(\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}',t')\right) - i\int d\mathbf{x}''v(\mathbf{x},\mathbf{x}'')\mathcal{T}\left(\hat{\psi}(\mathbf{x},t)\hat{\psi}^\dagger(\mathbf{x}'',t)\hat{\psi}(\mathbf{x}'',t)\hat{\psi}^\dagger(\mathbf{x}',t')\right) \tag{1.17}$$

So we have

$$\left[i\frac{\partial}{\partial t} - \hat{h}^0(\mathbf{x})\right]G(\mathbf{x}t,\mathbf{x}'t') = \delta(\mathbf{x}-\mathbf{x}')\delta(t-t') - i\int d\mathbf{x}''v(\mathbf{x},\mathbf{x}'')\underbrace{G_2(\mathbf{x}t,\mathbf{x}''t,\mathbf{x}'t^+,\mathbf{x}'t')}_{\langle N|\mathcal{T}\left(\hat{\psi}(\mathbf{x},t)\hat{\psi}(\mathbf{x}'',t)\hat{\psi}^\dagger(\mathbf{x}'',t')\hat{\psi}^\dagger(\mathbf{x}',t')\right)|N\rangle} \tag{1.18}$$

and we notice that in order to compute the single-particle Green's function, we need to know the two-particle Green's function, which needs the three-particle Green's function, and so on. So to simplify we introduce a nonlocal, time-dependent self-energy $\Sigma(\mathbf{x}t,\mathbf{x}'t')$ that satisfies

$$-i\int d\mathbf{x}''v(\mathbf{x},\mathbf{x}'')G_2(\mathbf{x}t,\mathbf{x}''t,\mathbf{x}'t^+,\mathbf{x}'t') \equiv \int dt''\int d\mathbf{x}''\overline{\Sigma}(\mathbf{x}t,\mathbf{x}''t'')G(\mathbf{x}''t'',\mathbf{x}'t') \tag{1.19}$$

and further define $\Sigma = \overline{\Sigma} - v_H$ with

$$v_H(\mathbf{x},t) = \int d\mathbf{x}'v(\mathbf{x},\mathbf{x}')\underbrace{\langle N|\hat{\psi}^\dagger(\mathbf{x}')\hat{\psi}(\mathbf{x}')|N\rangle}_{-\frac{1}{i}G(\mathbf{x}'t,\mathbf{x}'t)} = i\int d\mathbf{x}'v(\mathbf{x},\mathbf{x}')G(\mathbf{x}'t,\mathbf{x}'t) \tag{1.20}$$

and we can rewrite the equation of motion as

$$\left[i\frac{\partial}{\partial t} - \hat{h}^0(\mathbf{x}) - v_H(\mathbf{x},t)\right] G(\mathbf{x}t,\mathbf{x}'t') = \delta(\mathbf{x}-\mathbf{x}')\delta(t-t') + \int dt'' \int d\mathbf{x}''\Sigma(\mathbf{x}t,\mathbf{x}''t'')G(\mathbf{x}''t'',\mathbf{x}'t') \tag{1.21}$$

Now consider defining the G_0 of the non-interacting system

$$\left[i\frac{\partial}{\partial t} - \hat{h}^0(\mathbf{x}) - v_H(\mathbf{x},t)\right] G_0(\mathbf{x}t,\mathbf{x}'t') = \delta(\mathbf{x}-\mathbf{x}')\delta(t-t') \tag{1.22}$$

So we can write equations 1.21 and 1.22 symbolically as

$$\hat{O}G = \delta + \Sigma G \quad \text{and} \quad \hat{O}G_0 = \delta \tag{1.23}$$

$$\implies G_0 = \hat{O}^{-1} \implies G = G_0 + G_0\Sigma G \tag{1.24}$$

$$\implies G(1,2) = G_0(1,2) + \int d3d4 G_0(1,3)\Sigma(3,4)G(4,2) \tag{1.25}$$

where we use the space-time notation $1 = (\mathbf{x}_1, t_1)$ etc.

1.1.2 Hedin's Equations

Schwinger chose to introduce a potential φ that we will later set to zero, in order to rewrite the two-particle Green's function as

$$G_2(1,3,2,3^+) = G(1,2)\,G(3,3^+) - \frac{\delta G(1,2)}{\delta\varphi(3)}, \tag{1.26}$$

So

$$\bar{\Sigma}(1,2) = -i\int d(3)\; v(1,3)\,G_2(1,3,2,3^+) \tag{1.27}$$

$$= -i\int d(3)\, v(1,3)\left[G(1,2)\,G(3,3^+) - \frac{\delta G(1,2)}{\delta\varphi(3)}\right] \tag{1.28}$$

$$= -i\,G(1,2)\underbrace{\int d(3)\, v(1,3)\,G(3,3^+)}_{-iv_H(1)} + i\int d(3)\, v(1,3)\,\frac{\delta G(1,2)}{\delta\varphi(3)}. \tag{1.29}$$

Now because $\delta G = -G\,(\delta G^{-1})\,G$ we can write the identity

$$\frac{\delta G(1,2)}{\delta\varphi(3)} = -\int d(4)d(5)\; G(1,4)\,\frac{\delta G^{-1}(4,5)}{\delta\varphi(3)}\,G(5,2). \tag{1.30}$$

So the second term in Eq. (1.29) gives

$$i\int d(3)\; v(1,3)\,\frac{\delta G(1,2)}{\delta\varphi(3)} = -i\int d(3)\, v(1,3)\int d(4)d(5)\; G(1,4)\,\frac{\delta G^{-1}(4,5)}{\delta\varphi(3)}\,G(5,2)$$

$$= -i\int d(3,4,5)\; v(1,3)\,G(1,4)\,\frac{\delta G^{-1}(4,5)}{\delta\varphi(3)}\,G(5,2). \tag{1.31}$$

Now we can get rid of a $G(1,2)$ dependence by multiplying with G^{-1}, yielding

$$\overline{\Sigma}(1,2) = -\delta(1,2)\, v_H(1) - i \int d(3,4)\; v(1,3)\, G(1,4)\, \frac{\delta G^{-1}(4,2)}{\delta \varphi(3)}. \tag{1.32}$$

Introduce $V(1) = \varphi(1) + v_H(1)$ as the total potential that electrons experience. Consider

$$\frac{\delta G^{-1}(1,2)}{\delta \varphi(3)} \equiv \underbrace{\frac{\delta G^{-1}(1,2)}{\delta V(5)}}_{-\Gamma(1,2,5)} \underbrace{\frac{\delta V(5)}{\delta \varphi(3)}}_{\varepsilon^{-1}(5,3)}. \tag{1.33}$$

So

$$\overline{\Sigma}(1,2) = -\delta(1,2)\, v_H(1) + i \int d(5) \underbrace{\int d(3) v(1,3)\, \varepsilon^{-1}(3,5)}_{W(1,5)} \int d(4) G(1,4)\, \Gamma(4,5,2). \tag{1.34}$$

and if we further make the GW approximation where $\Gamma(4,5,2) \approx \delta(4,5)\delta(2,5)$ we get

$$\overline{\Sigma}(1,2) = -\delta(1,2)\, v_H(1) + iW(1,2)G(1,2) \tag{1.35}$$

and if we just care about the exchange-correlation part, we can define

$$\Sigma_{xc}(1,2) = \overline{\Sigma}(1,2) + \delta(1,2)\, v_H(1) = iW(1,2)G(1,2) \implies \Sigma_{xc}(\tau) = iW(\tau)G(\tau) \tag{1.36}$$

where $\tau = t_1 - t_2$. Define $G(\tau) = \int \frac{d\omega'}{2\pi} e^{-i\omega'\tau} G(\omega')$ and $W(\tau) = \int \frac{d\omega''}{2\pi} e^{-i\omega''\tau} W(\omega'')$ to get

$$\Sigma_{xc}(\tau) = i \int \frac{d\omega'}{2\pi} \int \frac{d\omega''}{2\pi} e^{-i(\omega'+\omega'')\tau} G(\omega') W(\omega'') \tag{1.37}$$

Taking the inverse Fourier transform of $\Sigma_{xc}(\tau)$ we get

$$\Sigma_{xc}(\omega) = \int \frac{d\tau}{2\pi} e^{i\omega\tau} \Sigma_{xc}(\tau) = i \int \frac{d\omega'}{2\pi} \int \frac{d\omega''}{2\pi} G(\omega') W(\omega'') \underbrace{\int d\tau e^{i(\omega-\omega'-\omega'')\tau}}_{2\pi\delta(\omega-\omega'-\omega'')} = i \int \frac{d\omega'}{2\pi} G(\omega') W(\omega-\omega') \tag{1.38}$$

Now in G_0W_0 one applies the Cauchy residue theorem to solve this convolution integral, yielding the known form.

1.2 Final expressions

1.2.1 Fully analytic

I follow the notation of Tianyu's paper throughout this section [5]. We want to solve for the self-energy whose form along the real axis is:

$$\Sigma(\mathbf{r}, \mathbf{r}', \omega) = \frac{i}{2\pi} \int_{-\infty}^{\infty} e^{i\omega'\eta} d\omega' G_0(\mathbf{r}, \mathbf{r}', \omega + \omega') W_0(\mathbf{r}, \mathbf{r}', \omega') \tag{1.39}$$

In the molecular brutal basis, the self energy is given as:

$$\Sigma_{nn'}(\mathbf{k},\omega) = \iint d\mathbf{r}d\mathbf{r}'\psi^*_{n\mathbf{k}}(\mathbf{r})\Sigma(\mathbf{r},\mathbf{r}',\omega)\psi_{n'\mathbf{k}}(\mathbf{r}') \tag{1.40}$$

Also, recall that the Lehmann representation of the noninteracting Green's function is:

$$G_0\left(\mathbf{r},\mathbf{r}',\omega\right) = \sum_{o\mathbf{q}} \frac{\psi_{o\mathbf{q}}(\mathbf{r})\psi^*_{o\mathbf{q}}\left(\mathbf{r}'\right)}{\omega - \epsilon_{o\mathbf{q}} + i\eta\,\mathrm{sgn}\left(\epsilon_{o\mathbf{q}} - \mu\right)} \tag{1.41}$$

Now plugging both of these back into the original expression, we find:

$$\begin{aligned}
\Sigma_{nn'}(\mathbf{k},\omega) &= \frac{i}{2\pi}\sum_{o\mathbf{q}}\int_{-\infty}^{\infty} d\omega' \frac{e^{i\omega'\eta}}{\omega+\omega'-\epsilon_{o\mathbf{q}}+i\eta\,\mathrm{sgn}\left(\epsilon_{o\mathbf{q}}-\mu\right)} \\
&\quad\times \iint d\mathbf{r}d\mathbf{r}'\psi^*_{n\mathbf{k}}(\mathbf{r})\psi_{o\mathbf{q}}(\mathbf{r})W_0(\mathbf{r},\mathbf{r}',\omega')\psi^*_{o\mathbf{q}}\left(\mathbf{r}'\right)\psi_{n'\mathbf{k}}(\mathbf{r}') \qquad (1.42)\\
&= \frac{i}{2\pi}\sum_{o\mathbf{q}}\int_{-\infty}^{\infty} d\omega' \frac{e^{i\omega'\eta}}{\omega+\omega'-\epsilon_{o\mathbf{k}-\mathbf{q}}+i\eta\,\mathrm{sgn}\left(\epsilon_{o\mathbf{k}-\mathbf{q}}-\mu\right)}(n_{\mathbf{k}}o_{\mathbf{k}-\mathbf{q}}|W_0|o_{\mathbf{k}-\mathbf{q}}n'_{\mathbf{k}}) \qquad (1.43)
\end{aligned}$$

Where we have used the fact that the momentum index $\mathbf{q}$ is the same as $\mathbf{k}-\mathbf{q}$, given that we are looping over both $\mathbf{k}$ and $\mathbf{q}$ anyways.

So the Green's function will bring poles at $\omega' = \epsilon_{0\mathbf{k}-\mathbf{q}} - \omega + i\eta\,\mathrm{sgn}(\mu - \epsilon_{0\mathbf{k}-\mathbf{q}})$. Now, we know that the screened Coulomb interaction has the expansion in terms of the bare Coulomb potential v and the density response function χ_0 as $W_0 = v + v\chi_0 v + v\chi_0 v\chi_0 v + \cdots = v\left(1+\chi_0 v+\chi_0 v\chi_0 v+\cdots\right) = v\left(1-\chi_0 v\right)^{-1}$, where we recognize the dielectric function as $\epsilon_0 = 1-\chi_0 v$ so we can express the screened Coulomb interaction as

$$W_0(\mathbf{r},\mathbf{r}',\omega) = \frac{v(\mathbf{r},\mathbf{r}')}{1-\left(\chi_0 v\right)\left(\mathbf{r},\mathbf{r}',\omega\right)} \tag{1.44}$$

recalling that the bare Coulomb interaction should be independent of frequency. A discussion of how to compute the screened Coulomb interaction can be found in this old work [3]. To simplify notation let us define a polarizability $\mathbf{\Pi}\left(\mathbf{r},\mathbf{r}',\omega\right) = \left(\chi_0 v\right)\left(\mathbf{r},\mathbf{r}',\omega\right)$, so that we can rewrite the screened Coulomb interaction as:

$$(n_{\mathbf{k}}o_{\mathbf{k}-\mathbf{q}}|W_0|o_{\mathbf{k}-\mathbf{q}}n'_{\mathbf{k}}) = \iint d\mathbf{r}d\mathbf{r}'\psi^*_{n\mathbf{k}}(\mathbf{r})\psi_{o\mathbf{q}}(\mathbf{r})W_0(\omega)\psi^*_{o\mathbf{q}}\left(\mathbf{r}'\right)\psi_{n'\mathbf{k}}(\mathbf{r}') \tag{1.45}$$

At this point, we recognize the decomposition of the ERIs with the Cholesky vectors as:

$$(p_{\mathbf{k}_p}q_{\mathbf{k}_q}|\frac{1}{|\mathbf{r}-\mathbf{r}'|}|r_{\mathbf{k}_r}s_{\mathbf{k}_s}) = \sum_{PQ} v^{p\mathbf{k_p}q\mathbf{k_q}}_{P\mathbf{q}} v^{r\mathbf{k_r}s\mathbf{k_s}}_{Q(-\mathbf{q})} \tag{1.46}$$

so each Cholesky brings a factor of $\mathbf{J}^{\frac{1}{2}}$. Each Cholesky is defined as:

$$v^{p\mathbf{k_p}q\mathbf{k_q}}_{P\mathbf{q}} = \sum_R \mathbf{J}^{-\frac{1}{2}}_{RP}(\mathbf{q})\left(R\mathbf{q}\mid p\mathbf{k}_p q\mathbf{k}_q\right) \tag{1.47}$$

where

$$\begin{aligned}\mathbf{J}_{PQ}(\mathbf{k}) &= \iint d\mathbf{r}d\mathbf{r}'\phi_{P(-\mathbf{k})}(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}\phi_{Q\mathbf{k}}(\mathbf{r}') \\ (Q\mathbf{k}_{rs} \mid r\mathbf{k}_r s\mathbf{k}_s) &= \iint d\mathbf{r}d\mathbf{r}'\phi_{Q\mathbf{k}_{rs}}(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}\phi^*_{r\mathbf{k}_r}(\mathbf{r}')\,\phi_{s\mathbf{k}_s}(\mathbf{r}')\end{aligned} \tag{1.48}$$

So the simplest thing now will be to derive an expression for the columb interaction in terms of an auxiliary basis:

$$W_{0,PQ}(\omega) = \left[\mathbf{J}(\mathbf{I}-\mathbf{\Pi}(\mathbf{q},\omega))^{-1}\right]_{PQ} \tag{1.49}$$

and then we need to contract with the Choleskies to get the matrix element:

$$(n_{\mathbf{k}}o_{\mathbf{k}-\mathbf{q}}|W_0|o_{\mathbf{k}-\mathbf{q}}n'_{\mathbf{k}}) = \sum_{PQ} v_P^{nm}\left[\mathbf{I}-\mathbf{\Pi}(\mathbf{q},\omega)\right]^{-1}_{PQ} v_Q^{mn'} \tag{1.50}$$

So in our quest to find poles of W_0, we are really just looking for poles of the χ_0. χ_0 is given by:

$$\chi_0(\mathbf{r},\mathbf{r}',\omega) = \sum_{r\mathbf{k}s\mathbf{k}'}(f_{r\mathbf{k}}-f_{s\mathbf{k}'})\frac{\psi_{r\mathbf{k}}(\mathbf{r})\psi^*_{r\mathbf{k}}(\mathbf{r}')\,\psi_{s\mathbf{k}'}(\mathbf{r}')\,\psi^*_{s\mathbf{k}'}(\mathbf{r})}{\omega-(\epsilon_{r\mathbf{k}}-\epsilon_{s\mathbf{k}'})+i\eta\,\mathrm{sgn}(\epsilon_{r\mathbf{k}}-\epsilon_{s\mathbf{k}'}-\mu)} \tag{1.51}$$

where the occupations of the KS states $r\mathbf{k}(s\mathbf{k}')$ with energies $\epsilon_{r\mathbf{k}}(\epsilon_{s\mathbf{k}'})$ are given by the Fermi-Dirac distribution $f_{r\mathbf{k}}(f_{s\mathbf{k}'})$, which is just a step function at zero temperature. Notice that the occupation factor will always be 0 unless rs form an occupied-virtual pair. So we can separate the density response into two terms, one where δ_{ri} and δ_{sa} and the other with δ_{ra} and δ_{si}, where i and a are occupied and virtual indices, respectively. This allows us to now combine with the bare Coulomb potential in order to form the polarizability $\Pi \equiv \chi_0 v$ as:

$$\Pi(\mathbf{r},\mathbf{r}',\omega) = \sum_{i\mathbf{k}a\mathbf{k}'}\frac{\psi_{i\mathbf{k}}(\mathbf{r})\psi^*_{i\mathbf{k}}(\mathbf{r}')\frac{1}{|\mathbf{r}-\mathbf{r}'|}\psi_{a\mathbf{k}'}(\mathbf{r}')\,\psi^*_{a\mathbf{k}'}(\mathbf{r})}{\omega+(\Omega_{i\mathbf{k}a\mathbf{k}'})-i\eta} - \sum_{a\mathbf{k}i\mathbf{k}'}\frac{\psi_{a\mathbf{k}}(\mathbf{r})\psi^*_{a\mathbf{k}}(\mathbf{r}')\frac{1}{|\mathbf{r}-\mathbf{r}'|}\psi_{i\mathbf{k}'}(\mathbf{r}')\,\psi^*_{i\mathbf{k}'}(\mathbf{r})}{\omega-(\Omega_{i\mathbf{k}a\mathbf{k}'})+i\eta}, \tag{1.52}$$

where we define the KS eigenvalue differences as $\Omega_{i\mathbf{k}a\mathbf{k}'} = \epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}'}$, which will eventually become the excitation energies from RPA. So sandwiching this operator in between the molecular or brutal bases gives:

$$\langle n\mathbf{k}|\,\Pi(\omega)\,|n'\mathbf{k}\rangle = \sum_{iajb\mathbf{k}\mathbf{k}'}\frac{(ia\mid jb)}{(\omega+\Omega^\mu_{\mathbf{k}})-i\eta} - \sum_{aibj\mathbf{k}\mathbf{k}'}\frac{(ai\mid bj)}{(\omega-\Omega^\mu_{\mathbf{k}})+i\eta} \tag{1.53}$$

So we see that we can get the poles of the screened Coulomb interaction by the poles of the polarizability, which are $\omega = \Omega^\mu_{\mathbf{k}} - i\eta$ and $\omega = \Omega^\mu_{\mathbf{k}} + i\eta$, suggesting that they are in the upper complex plane for excitations and vice versa for deexcitations. See the figure 1.1 for a picture. For a more comprehensive picture, this should be juxtaposed with the figure from Tianyu's paper for CD. In the literature, they talk about approximating the dielectric function by a multiple one or a single pole approximation, so which one would I want to implement? This suggests that the notation in the G_0W_0 literature is confusing because they always say that to solve for the χ_0 in the RPA, but if we are actually dealing with χ_0, which is the Kohn-Sham density response function, then we don't use the RPA, where the density response function is solved for using a Dyson-like equation [4]:

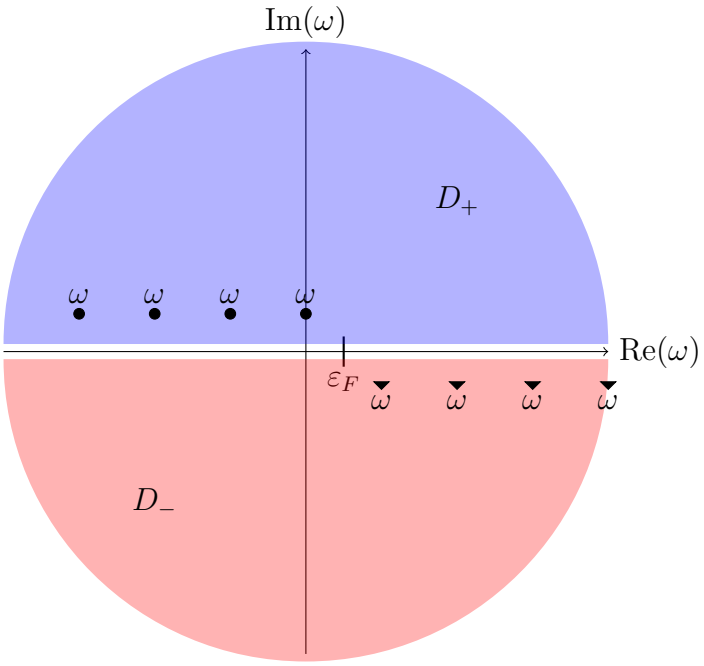


Figure 1.1: Contour for the complex frequency integral. The poles are denoted by the various ω. The Fermi energy is denoted by ε_F. The integration contour D_+ is the semicircle in the upper complex plane, while D_- is the semicircle in the lower complex plane.

$$\chi^{\lambda}\left(\mathbf{r},\mathbf{r}',i\omega\right)=\chi^{0}\left(\mathbf{r},\mathbf{r}',i\omega\right)+\int d\mathbf{r}_1 d\mathbf{r}_2 \chi^{0}\left(\mathbf{r},\mathbf{r}_1,i\omega\right)\left[\frac{\lambda}{|\mathbf{r}_1-\mathbf{r}_2|}+f_{\mathrm{xc}}^{\lambda}\left(\mathbf{r}_1,\mathbf{r}_2,i\omega\right)\right]\chi^{\lambda}\left(\mathbf{r}_2,\mathbf{r}',\omega\right) \tag{1.54}$$

where the parameter λ controls the amount of interaction in the system, ranging from $\lambda=0$ for the KS reference system to $\lambda=1$ for the fully interacting system. The $f_{\mathrm{xc}}^{\lambda}$ is the exchange-correlation kernel, which is set to zero for the RPA. But we will proceed with an RPA calculation anyways in order to solve for the excitation energies and their corresponding eigenvectors. So it makes sense that the numerator of the expression for the screened Coulomb interaction should be given a construction of the ERIs with the excitation factors in a transition density defined as:

$$w_{pq}^{\mu}=\sum_{ia}(pq|ia)\left(X_{ia}^{\mu}+Y_{ai}^{\mu}\right) \tag{1.55}$$

where we have defined the excitation and de-excitation vectors at the excitation index μ as X_{ia}^{μ} and Y_{ai}^{μ}, respectively. I am not sure how to connect this with the known expression $v\epsilon^{-1}$; I see the similarities given that we are contracting an ERI with what we get from the RPA calculation that is connected to the polarizability, but can't connect exactly. We want to figure out how this matches with my previous $O(N^6)$ expression, which was

$$\Sigma_{pp}^{\mathrm{corr}}(\omega)=\sum_{\mu}^{\mathrm{RPA}}\left(\sum_{i}^{\mathrm{occupied}}\frac{w_{pi}^{\mu}w_{ip}^{\mu}}{\omega-(\epsilon_i-\Omega_{\mu})}+\sum_{a}^{\mathrm{virtual}}\frac{w_{pa}^{\mu}w_{ap}^{\mu}}{\omega-(\epsilon_a+\Omega_{\mu})}\right) \tag{1.56}$$

for the molecular case. Today I want us to dissect how this equation came about, so that I can understand for my k-point version.

1.2.2 Analytic continuation

We can describe the final equations to implement. The correlation self energy can be computed on the imaginary frequency axis as:

$$\Sigma^c_{nn'}(\mathbf{k}, i\omega) = -\frac{1}{\pi N_{\mathbf{k}}} \sum_{m\mathbf{q}} \int_0^\infty d\omega' \frac{i\omega + \epsilon_F - \epsilon_{m\mathbf{k}-\mathbf{q}}}{\left(i\omega + \epsilon_F - \epsilon_{m\mathbf{k}-\mathbf{q}}\right)^2 + \omega'^2} \sum_{PQ} v_P^{nm} \left[\left[\mathbf{I} - \mathbf{\Pi}\left(\mathbf{q}, i\omega'\right) \right]^{-1}_{PQ} - \delta_{PQ} \right] v_Q^{mn'} \tag{1.57}$$

with the polarizability given by

$$\mathbf{\Pi}_{PQ}\left(\mathbf{q}, i\omega'\right) = \frac{2}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} v_P^{ia} \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}-\mathbf{q}}}{\omega'^2 + \left(\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}-\mathbf{q}}\right)^2} v_Q^{ai}. \tag{1.58}$$

The integration in Eq. 27 must be performed on a numerical grid. Following Ref., [43] we employ a 9

modified Gauss-Legendre grid that transforms a standard Gauss-Legendre grid in the range $[\,-1, 1\,]$ to $[0, \infty]$:

$$\tilde{x}_i = \frac{1 + x_i}{2\left(1 - x_i\right)}, \quad \tilde{w}_i = \frac{w_i}{\left(1 - x_i\right)^2}.$$

$\{x_i, w_i\}$ are the original Gauss-Legendre abscissas and weights of the grid points in the range $[-1, 1]$, and $\{\tilde{x}_i, \tilde{w}_i\}$ are the transformed abscissas and weights of the grid points used for integrating Eq. 27. In this work, $N_G = 100$ grid points were used for all reported results unless specified.

The real-frequency self-energy is then obtained through an analytic continuation scheme by fitting to an analytic function defined on the full complex plane. We fit the self-energy matrix elements to N-point Padé approximants using Thiele's reciprocal difference method (see Appendix of Ref. [62] for the detailed algorithm):

$$\Sigma_{nr'}(\mathbf{k}, \mathbf{z}) = \frac{a_0 + a_1 \cdot \mathbf{z} + \ldots + a_{(N-1)/2} \cdot z^{(N-1)/2}}{1 + b_1 \cdot \mathbf{z} + \ldots + b_{N/2} \cdot z^{N/2}}$$

where z is any complex frequency, $\{a_i, b_i\}$ are Padé coefficients that need to be fitted. In this work, we use $N = 18$ Padé approximants, which requires $18 \mathbf{\Sigma}_{mn'}(\mathbf{k}, \mathbf{z} = i\omega)$ data points as input to the fit. We choose these 18 imaginary frequency points to be the positions of the modified GaussLegendre grid points. Once the Padé polynomials are fitted, the real-frequency self-energy is easily calculated:

$$\mathbf{\Sigma}_{nn'}(\mathbf{k}, \omega) = \frac{a_0 + a_1 \cdot \omega + \ldots + a_{(N-1)/2} \cdot \omega^{(N-1)/2}}{1 + b_1 \cdot \omega + \ldots + b_{N/2} \cdot \omega^{N/2}}$$

We start with the original form for the self-energy along the real axis:

$$\Sigma\left(\mathbf{r}, \mathbf{r}', \omega\right) = \frac{i}{2\pi} \int_{-\infty}^{\infty} d\omega' e^{i\omega'\eta} G_0\left(\mathbf{r}, \mathbf{r}', \omega + \omega'\right) W_0\left(\mathbf{r}, \mathbf{r}', \omega'\right) \tag{1.59}$$

But to avoid the poles, we need to evaluate along the imaginary axis, so the problem becomes:

$$\Sigma\left(\mathbf{r},\mathbf{r}',i\omega\right)=-\frac{1}{2\pi}\int_{-\infty}^{\infty}d\omega' G_0\left(\mathbf{r},\mathbf{r}',i\omega+i\omega'\right)W_0\left(\mathbf{r},\mathbf{r}',i\omega'\right)$$

We are interested in evaluating the matrix elements of this operator in the molecular orbital basis. Note that both molecular orbitals must have the same crystal momentum in order for it to be conserved in this process. We also apply the identity operator:

$$\left\langle n\mathbf{k}\right|\Sigma(i\omega)\left|n'\mathbf{k}\right\rangle=-\frac{1}{2\pi}\sum_{m\mathbf{k}'}\int_{-\infty}^{\infty}d\omega'\left\langle n\mathbf{k}\right|G_0(i\omega+i\omega')\left|m\mathbf{k}'\right\rangle\left\langle m\mathbf{k}'\right|W_0(i\omega')\left|n'\mathbf{k}\right\rangle \tag{1.60}$$

The noninteracting Green's function has the form:

$$G_0\left(\mathbf{r},\mathbf{r}',i\omega\right)=\sum_{m\mathbf{k}_m}\frac{\psi_{m\mathbf{k}_m}(\mathbf{r})\psi^*_{m\mathbf{k}_m}\left(\mathbf{r}'\right)}{i\omega+\epsilon_F-\epsilon_{m\mathbf{k}_m}}\implies G_0\left(\mathbf{k}-\mathbf{q},i\omega+i\omega'\right)=\sum_{m\mathbf{k}-\mathbf{q}}\frac{\psi_{m\mathbf{k}-\mathbf{q}}\psi^*_{m\mathbf{k}-\mathbf{q}}}{i\left(\omega+\omega'\right)+\epsilon_F-\epsilon_{m\mathbf{k}-\mathbf{q}}}$$

so that the above equation simplifies to:

$$\mathbf{\Sigma}_{nn'}(\mathbf{k},i\omega)=-\frac{1}{2\pi N_{\mathbf{k}}}\sum_{m\mathbf{q}}\int_{-\infty}^{\infty}d\omega'\frac{(n\mathbf{k},m\mathbf{k}-\mathbf{q}\mid W_0(\mathbf{q},i\omega)\mid m\mathbf{k}-\mathbf{q},n'\mathbf{k})}{i\left(\omega+\omega'\right)+\epsilon_F-\epsilon_{m\mathbf{k}-\mathbf{q}}} \tag{1.61}$$

Screened Coulomb Interaction

$$(n\mathbf{k},m\mathbf{k}-\mathbf{q}\left|W_0(\mathbf{q},i\omega)\right|m\mathbf{k}-\mathbf{q},n'\mathbf{k})=\int\int d\mathbf{r}_1d\mathbf{r}_2\psi^*_{n\mathbf{k}}(\mathbf{r}_1)\psi_{m\mathbf{k}-\mathbf{q}}(\mathbf{r}_1)W_0(\mathbf{q},\mathbf{r}_1,\mathbf{r}_2,i\omega)\psi^*_{m\mathbf{k}-\mathbf{q}}(\mathbf{r}_2)\psi_{n'\mathbf{k}}(\mathbf{r}_2)$$

We expand the orbital pair product $\psi^*_{n\mathbf{k}}(\mathbf{r})\psi_{m\mathbf{k}-\mathbf{q}}(\mathbf{r})$ in the auxiliary basis

$$\psi^*_{n\mathbf{k}}(\mathbf{r})\psi_{m\mathbf{k}-\mathbf{q}}(\mathbf{r})=\sum_P b^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}_{P\mathbf{q}}\phi_{P\mathbf{q}}(\mathbf{r})$$

and

$$\psi^*_{m\mathbf{k}-\mathbf{q}}(\mathbf{r})\psi_{n'\mathbf{k}}(\mathbf{r})=\sum_Q b^{m\mathbf{k}-\mathbf{q},n'\mathbf{k}}_{Q(-\mathbf{q})}\phi_{Q(-\mathbf{q})}(\mathbf{r}) \tag{1.62}$$

where we have recognized the fact that in the former there is a momentum transfer of $\mathbf{q}$, and in the latter, there is a momentum transfer of $-\mathbf{q}$. Substituting in gives

$$(n\mathbf{k},m\mathbf{k}-\mathbf{q}\left|W_0(\mathbf{q},i\omega)\right|m\mathbf{k}-\mathbf{q},n'\mathbf{k}) \tag{1.63}$$

$$=\sum_{PQ}b^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}_{P\mathbf{q}}\left[\iint d\mathbf{r}_1d\mathbf{r}_2\phi_{P\mathbf{q}}(\mathbf{r}_1)W_0(\mathbf{q},\mathbf{r}_1,\mathbf{r}_2,i\omega)\phi_{Q(-\mathbf{q})}(\mathbf{r}_2)\right]b^{m\mathbf{k}-\mathbf{q},n'\mathbf{k}}_{Q(-\mathbf{q})} \tag{1.64}$$

with

$$b^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}_{P\mathbf{q}}=\sum_R(n\mathbf{k},m\mathbf{k}-\mathbf{q}\mid R\mathbf{q})\cdot\mathbf{J}^{-1}_{RP}(\mathbf{q}) \tag{1.65}$$

Now is a good place to recall their definition of the density fitting where the ERIs are represented as:

$$(p\mathbf{k}_p q\mathbf{k}_q \mid r\mathbf{k}_r s\mathbf{k}_s) = \sum_{PQ} (p\mathbf{k}_p q\mathbf{k}_q \mid P\mathbf{k}_{pq})\, \mathbf{J}^{-1}_{PQ}\, (Q\mathbf{k}_{rs} \mid r\mathbf{k}_r s\mathbf{r}_s)\,,$$

with

$$\mathbf{J}_{PQ}(\mathbf{k}) = \iint d\mathbf{r}d\mathbf{r}'\phi_{P(-\mathbf{k})}(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}\phi_{Q\mathbf{k}}\,(\mathbf{r}')\,,$$
$$(Q\mathbf{k}_{rs} \mid r\mathbf{k}_r s\mathbf{k}_s) = \iint d\mathbf{r}d\mathbf{r}'\phi_{Q\mathbf{k}_{rs}}(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}\phi^*_{r\mathbf{k}_r}\,(\mathbf{r}')\,\phi_{s\mathbf{k}_s}\,(\mathbf{r}')\,.$$

Note that these b are then not yet our Cholesky vectors, since each one contains $\frac{|\mathbf{r}-\mathbf{r}'|}{|\mathbf{r}-\mathbf{r}'|}$ scaling, i.e., there should be a factor of $\mathbf{J}^{-\frac{1}{2}}$ instead of $\mathbf{J}^{-1}$ in 1.65 if we are to apply the Cholesky vectors. At this point, we use the expansion of the screened Coulomb interaction:

$$W_0 = v + v\chi_0 v + v\chi_0 v\chi_0 v + \cdots \tag{1.66}$$
$$= v(1 + \chi_0 v + \chi_0 v\chi_0 v + \cdots) \tag{1.67}$$
$$= v^{1/2}\,(1-\chi_0)^{-1}\,v^{1/2} \tag{1.68}$$

simplifying to

$$(n\mathbf{k}, m\mathbf{k}-\mathbf{q}\,|W_0(\mathbf{q},i\omega)|\,m\mathbf{k}-\mathbf{q}, n'\mathbf{k}) = \sum_{PQ} b^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}_{P\mathbf{q}} \left[\mathbf{J}^{\frac{\mathbf{1}}{\mathbf{2}}}\,(\mathbf{I}-\mathbf{\Pi}(\mathbf{q},i\omega))\,\mathbf{J}^{\frac{\mathbf{1}}{\mathbf{2}}}\right]^{-1}_{PQ} b^{m\mathbf{k}-\mathbf{q},n'\mathbf{k}}_{Q(-\mathbf{q})} \tag{1.69}$$
$$= \sum_{PQ} v^{nm}_P\,[\mathbf{I}-\mathbf{\Pi}\,(\mathbf{q},i\omega')]^{-1}_{PQ}\,v^{mn'}_Q \tag{1.70}$$

where $\mathbf{J}_{PQ}$ is the Coulomb interaction projected onto the auxiliary basis, and we have defined

$$v^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}_{P\mathbf{q}} = \sum_{pq} C_{pn}(\mathbf{k})C_{qm}(\mathbf{k}-\mathbf{q})v^{p\mathbf{k},q\mathbf{k}-\mathbf{q}}_{P\mathbf{q}} \tag{1.71}$$

with

$$v^{p\mathbf{k},q\mathbf{k}-\mathbf{q}}_{P\mathbf{q}} = \sum_R (p\mathbf{k}, q\mathbf{k}-\mathbf{q} \mid R\mathbf{q})\mathbf{J}^{-1/2}_{RP}(\mathbf{q}) \tag{1.72}$$

If we first rename $\mathbf{k}' = \mathbf{k}-\mathbf{q} \implies \mathbf{k} = \mathbf{k}'+\mathbf{q}$, and then we are free to redefine $\mathbf{q}\to-\mathbf{q}$, so that 1.71 becomes

$$v^{n\mathbf{k}-\mathbf{q},m\mathbf{k}}_{P-\mathbf{q}} = \sum_{pq} C_{pn}(\mathbf{k}-\mathbf{q})C_{qm}(\mathbf{k})v^{p\mathbf{k}-\mathbf{q},q\mathbf{k}}_{P-\mathbf{q}} \tag{1.73}$$

but we know that the bare Coulomb potential projected onto the auxiliary basis is given by

$$v^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}_{P\mathbf{q}} = \sum_{pq} C_{pn}(\mathbf{k})C_{qm}(\mathbf{k}-\mathbf{q})v^{p\mathbf{k},q\mathbf{k}-\mathbf{q}}_{P\mathbf{q}} \tag{1.74}$$

To ease notation, some momentum labels are suppressed in the above and following equations (e.g., we will use b_P^{nm} to denote $b_{P\mathbf{q}}^{n\mathbf{k},m\mathbf{k}-\mathbf{q}}$). Using Eqs. 19-21, the matrix elements of W_0 are computed as

$$\begin{aligned}
&(n\mathbf{k}, m\mathbf{k}-\mathbf{q}\,|W_0|\,m\mathbf{k}-\mathbf{q}, n'\mathbf{k}) \\
&= \sum_{PQ} b_P^{nm} \left[\iint d\mathbf{r} d\mathbf{r}' \phi_{P\mathbf{q}}(\mathbf{r}) W_0\left(\mathbf{r}, \mathbf{r}', i\omega'\right) \phi_{Q(-\mathbf{q})}\left(\mathbf{r}'\right) \right] b_Q^{mn'} \\
&= \sum_{PQ} b_P^{nm} \left[\mathbf{J}_{PQ}(\mathbf{q}) + \left(\mathbf{J}^{1/2}\mathbf{\Pi}\mathbf{J}^{1/2}\right)_{PQ}(\mathbf{q}) + \ldots \right] b_Q^{mn'} \\
&= \sum_{PQ} v_P^{nm} \left[\mathbf{I} - \mathbf{\Pi}\left(\mathbf{q}, i\omega'\right)\right]^{-1}_{PQ} v_Q^{mn'}
\end{aligned}$$

The 3-center 2-electron integral v_P^{nm} between auxiliary basis function P and molecular orbital pairs nm is obtained from an AO to MO transformation of the GDF AO integrals defined in Eq. 15:

$$v_P^{nm} = \sum_p \sum_q C_{pn}(\mathbf{k}) C_{qm}(\mathbf{k}-\mathbf{q}) v_{P\mathbf{q}}^{p\mathbf{k},q\mathbf{k}-\mathbf{q}}$$

where $C(\mathbf{k})$ refers to the MO coefficients in the AO basis. $\Pi\left(\mathbf{q}, i\omega'\right)$ in Eq. 22 is an auxiliary density response function:

$$\mathbf{\Pi}_{PQ}\left(\mathbf{q}, i\omega'\right) = \frac{2}{N_{\mathbf{k}}} \sum_{\mathbf{k}} \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} v_P^{ia} \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}-\mathbf{q}}}{\omega'^2 + \left(\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}-\mathbf{q}}\right)^2} v_Q^{ai}$$

Final expression

We can arrive at

$$\Sigma^c_{nn'}(\mathbf{k}, i\omega) = -\frac{1}{\pi N_{\mathbf{k}}} \sum_{m\mathbf{q}} \int_0^\infty d\omega' \frac{i\omega + \epsilon_F - \epsilon_{m\mathbf{k}-\mathbf{q}}}{\left(i\omega + \epsilon_F - \epsilon_{m\mathbf{k}-\mathbf{q}}\right)^2 + \omega'^2} \tag{1.75}$$

$$\times \sum_{PQ} v_P^{nm} \left[\left[\mathbf{I} - \mathbf{\Pi}\left(\mathbf{q}, i\omega'\right)\right]^{-1}_{PQ} - \delta_{PQ} \right] v_Q^{mn'} \tag{1.76}$$

1.2.3 Finite size corrections

The head correction to the correlation self-energy, which is just diagonal, is given as

$$\mathbf{\Sigma}^{\text{head}}_{nn}(\mathbf{k}, i\omega) = -\frac{1}{\pi^2} \left(\frac{6\pi^2}{\Omega_{\text{cell}} N_{\mathbf{k}}}\right)^{1/3} \int_{-\infty}^{\infty} d\omega' \frac{\epsilon^{-1}_{00}(\mathbf{q}=\mathbf{0}, i\omega) - 1}{i\left(\omega + \omega'\right) + \epsilon_F - \epsilon_{n\mathbf{k}}}. \tag{1.77}$$

Then the wings contribution to the correlation self-energy is given as

$$\mathbf{\Sigma}^{\text{wings}}_{nn'}\left(\mathbf{k}, i\omega\right) = -\frac{1}{\pi} \sqrt{\frac{\Omega_{\text{cell}}}{4\pi^3}} \left(\frac{6\pi^2}{\Omega_{\text{cell}} N_{\mathbf{k}}}\right)^{2/3} \sum_P \int_{-\infty}^{\infty} d\omega' \frac{\text{Re}\left[v_P^{nn'} \epsilon^{-1}_{P0}(\mathbf{q}=\mathbf{0}, i\omega)\right]}{i\left(\omega + \omega'\right) + \epsilon_F - \epsilon_{n'\mathbf{k}}}. \tag{1.78}$$

For the exchange self-energy, we only apply the following corr ection to the diagonal elements of the occupied orbitals:

$$\mathbf{\Sigma}^x_{nn}(\mathbf{k}, i\omega) = -\frac{2}{\pi}\left(\frac{6\pi^2}{\Omega_{\text{cell}} N_{\mathbf{k}}}\right)^{1/3} \tag{1.79}$$

where

$$\epsilon^{-1}_{\mathbf{00}} = 1/\left(h_{\mathbf{00}} - \sum_{PQ} w^{\dagger}_{P\mathbf{0}} B^{-1}_{PQ} w_{Q\mathbf{0}}\right) \tag{1.80}$$

$$\epsilon^{-1}_{P\mathbf{0}} = -\sum_{Q} B^{-1}_{PQ} w_{Q\mathbf{0}} \cdot \epsilon^{-1}_{\mathbf{00}} \tag{1.81}$$

with

$$h_{00}(\mathbf{q}=\mathbf{0}, i\omega) = 1 - \lim_{\mathbf{q}\to\mathbf{0}} \frac{4\pi}{|\mathbf{q}|^2}\chi_{00}(\mathbf{q}\to\mathbf{0}, i\omega), \tag{1.82}$$

$$w_{P0}(\mathbf{q}=\mathbf{0}, i\omega) = \lim_{\mathbf{q}\to\mathbf{0}} \frac{\sqrt{4\pi}}{|\mathbf{q}|}\chi_{P0}(\mathbf{q}\to\mathbf{0}, i\omega). \tag{1.83}$$

and $B_{PQ} = [\mathbf{I} - \mathbf{\Pi}(\mathbf{q}=\mathbf{0}, i\omega)]_{PQ}$ is the auxiliary density response function in the long-wavelength limit. $\chi_{00}(\mathbf{q}\to\mathbf{0}, i\omega)$ and $\chi_{P\mathbf{0}}(\mathbf{q}\to\mathbf{0}, i\omega)$ are the head and wings of the polarizability in the long-wavelength limit given by

$$\chi_{00}(\mathbf{q}\to\mathbf{0}, i\omega) = \frac{2}{N_{\mathbf{k}}\Omega_{\text{cell}}} \sum_{\mathbf{k}} \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} \rho^{i\mathbf{k}}_{a\mathbf{k}-\mathbf{q}}(\mathbf{0}) \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}}}{\omega^2 + (\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}})^2} \rho^{i\mathbf{k}*}_{a\mathbf{k}-\mathbf{q}}(\mathbf{0}), \tag{1.84}$$

$$\chi_{P\mathbf{0}}(\mathbf{q}\to\mathbf{0}, i\omega) = \frac{2}{N_{\mathbf{k}}\Omega^{1/2}_{\text{cell}}} \sum_{\mathbf{k}} \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} v^{ia}_{P} \frac{\epsilon_{i\ mathbfk} - \epsilon_{a\ mathbfk}}{\omega^2 + (\epsilon_{i\ mathbfk} - \epsilon_{a\ mathbfk})^2} \rho^{i\ mathbfk*}_{a\ mathbfk-\ mathbfq}(\mathbf{0}). \tag{1.85}$$

and

$$\rho^{i\mathbf{k}}_{a\mathbf{k}-\mathbf{q}}(\mathbf{0})\big|_{\mathbf{q}\to\mathbf{0}} = \frac{-i\mathbf{q}\cdot(\psi_{i\mathbf{k}}|\nabla|\psi_{a\mathbf{k}})}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}}. \tag{1.86}$$

We evaluate the limit using $\mathbf{q} = (0.001, 0, 0)$.

1.3 UHF formalism

The first thing to do is to solve the Casida equation for the polarizability in the direct formulation of the RPA:

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^* & \mathbf{A}^* \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} = \begin{pmatrix} \Omega & 0 \\ 0 & -\Omega \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \tag{1.87}$$

with $\mathbf{A}$ and $\mathbf{B}$ given by

$$\begin{aligned} \mathbf{A}^{\sigma\sigma'}_{ia,jb} &= \delta_{ij}\delta_{ab}\delta_{\sigma\sigma'}(\varepsilon_a - \varepsilon_i) + (i_\sigma a_\sigma | b_{\sigma'} j_{\sigma'}) \\ \mathbf{B}^{\sigma\sigma'}_{ia,jb} &= (i_\sigma a_\sigma | j_{\sigma'} b_{\sigma'}) \end{aligned} \tag{1.88}$$

Therefore, with the different spins we form a super matrix:

$$\begin{pmatrix} \begin{pmatrix} \mathbf{A}_{\alpha\alpha} & \mathbf{A}_{\alpha\beta} \\ \mathbf{A}_{\beta\alpha} & \mathbf{A}_{\beta\beta} \end{pmatrix} & \begin{pmatrix} \mathbf{B}_{\alpha\alpha} & \mathbf{B}_{\alpha\beta} \\ \mathbf{B}_{\beta\alpha} & \mathbf{B}_{\beta\beta} \end{pmatrix} \\ \begin{pmatrix} \mathbf{B}^*_{\alpha\alpha} & \mathbf{B}^*_{\alpha\beta} \\ \mathbf{B}^*_{\beta\alpha} & \mathbf{B}^*_{\beta\beta} \end{pmatrix} & \begin{pmatrix} \mathbf{A}^*_{\alpha\alpha} & \mathbf{A}^*_{\alpha\beta} \\ \mathbf{A}^*_{\beta\alpha} & \mathbf{A}^*_{\beta\beta} \end{pmatrix} \end{pmatrix} \begin{pmatrix} \mathbf{X}_{\alpha\alpha} & \mathbf{X}_{\alpha\beta} \\ \mathbf{X}_{\beta\alpha} & \mathbf{X}_{\beta\beta} \\ \mathbf{Y}_{\alpha\alpha} & \mathbf{Y}_{\alpha\beta} \\ \mathbf{Y}_{\beta\alpha} & \mathbf{Y}_{\beta\beta} \end{pmatrix} = \begin{pmatrix} \Omega & 0 & 0 & 0 \\ 0 & \Omega & 0 & 0 \\ 0 & 0 & -\Omega & 0 \\ 0 & 0 & 0 & -\Omega \end{pmatrix} \begin{pmatrix} \mathbf{X}_{\alpha\alpha} & \mathbf{X}_{\alpha\beta} \\ \mathbf{X}_{\beta\alpha} & \mathbf{X}_{\beta\beta} \\ \mathbf{Y}_{\alpha\alpha} & \mathbf{Y}_{\alpha\beta} \\ \mathbf{Y}_{\beta\alpha} & \mathbf{Y}_{\beta\beta} \end{pmatrix} \tag{1.89}$$

Now, there will be $2OV$ unique excitation energies; we sort them into singlets or triplets as follows: for each excitation we compute the overlap between the α and β excitation. For TDA, this is just the νth column of $\begin{pmatrix} \mathbf{X}_{\alpha\alpha} & \mathbf{X}_{\alpha\beta} \end{pmatrix}$ with the νth row of $\begin{pmatrix} \mathbf{X}_{\beta\alpha} \\ \mathbf{X}_{\beta\beta} \end{pmatrix}$. If greater than 0, we have a singlet excitation, otherwise we have a triplet excitation. For later use in GW, we just want the neutral excitation energies, so we only care about the singlet excitations.

1.4 Deriving linear response: 11/22

1.4.1 The Fundamentals

The motivation for this is to be able to understand why the poles of the screened Coulomb interaction are the same as those of the fully interacting polarizability, which are given by the frequencies of the RPA, obtained by diagonalizing the Casida equation. And then we want to be able to connect why $W_0 = v + v\chi_0 v + \ldots = \frac{v}{1-\chi_0 v}$ where W_0 is the screened Coulomb interaction and χ_0 is the non-interacting polarizability with Lehmann representation.

$$\chi_0\left(\mathbf{r}, \mathbf{r}', \omega\right) = \sum_{ia} \frac{\psi_i(\mathbf{r})\psi_a^*(\mathbf{r}')\psi_i(\mathbf{r}')\psi_a^*(\mathbf{r})}{\omega - (\epsilon_a - \epsilon_i) + i\eta\,\mathrm{sgn}\left(\epsilon_a - \epsilon_i - \mu\right)} \tag{1.90}$$

to do so, one must understand the reformulation of based on the density matrix as for posed by Furche [1]. Alternatively, let us start from the known Dyson equation that relates the fully interacting Green's function to the non-interacting one. We know the integral form of the Dyson equation is

$$G(1,2) = G_0(1,2) + \int d3d4 G_0(1,3)\Sigma(3,4)G(4,2) \tag{1.91}$$

but we proceed symbolically to get

$$G = G_0 + G_0\Sigma G \tag{1.92}$$

$$\left(I - G_0\Sigma\right) G = G_0 \tag{1.93}$$

$$G = \left(I - G_0\Sigma\right)^{-1} G_0 \tag{1.94}$$

$$G = \left(G_0\left(G_0^{-1} - \Sigma\right)\right)^{-1} G_0 \tag{1.95}$$

$$G = \left(G_0^{-1} - \Sigma\right)^{-1} \tag{1.96}$$

Now, we also know the Dyson equation for the polarizability is

$$\begin{aligned} \chi\left(\omega, x_1, x_2\right) =& \chi_0\left(\omega, x_1, x_2\right) + \int dx dx' \chi_0\left(\omega, x_1, x\right) \\ & \times \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} + f_{\mathrm{xc}}\left(\omega, x, x'\right)\right) \chi\left(\omega, x', x_2\right) \end{aligned} \tag{1.97}$$

In the RPA, we neglect the exchange correlation kernel f_{xc}, so we have

$$\chi_{RPA} = \chi_0 + \chi_0 v \chi_{RPA} = \frac{\chi_0}{1 - v\chi_0} \tag{1.98}$$

where in the final step we used the symbolic manipulation that was used before. So $W_0 = \frac{v}{1-v\chi_0} = v\chi_{RPA}\chi_0^{-1} = v\left(\frac{\chi_0\chi_0^{-1}}{1-v\chi_0}\right) = \frac{v}{1-v\chi_0}$. Therefore, we see why the poles of W_0 are the same as those of χ_{RPA}, since they have a linear relationship. Now, we will proceed to derive the poles of χ_{RPA}. But first we must introduce the density matrix based linear response theory.

1.4.2 Introduction to TDKS

In time-dependent density functional theory (TDDFT), the **Time-Dependent Kohn-Sham (TDKS)** equations describe a system of N noninteracting fermions that reproduce the same time-dependent density $\rho(t, \mathbf{r})$ as the interacting system. The TDKS equations are given by:

$$i\frac{\partial}{\partial t}\varphi_j(t, \mathbf{r}) = H[\rho](t, \mathbf{r})\varphi_j(t, \mathbf{r}) \tag{1.99}$$

where $j = 1, \ldots, N$ indexes the Kohn-Sham orbitals $\varphi_j(t, \mathbf{r})$, and $H[\rho](t, \mathbf{r})$ is the effective one-particle Hamiltonian defined as:

$$H[\rho](t, \mathbf{r}) = \frac{\boldsymbol{\pi}^2(t, \mathbf{r})}{2} + v_{\text{eff}}[\rho](t, \mathbf{r}) \tag{1.100}$$

where $v_{\text{eff}}[\rho](t, \mathbf{r}) = v_{\text{ext}}(t, \mathbf{r}) + v_{\text{H}}[\rho](t, \mathbf{r}) + v_{\text{xc}}[\rho](t, \mathbf{r})$.

The operator $\boldsymbol{\pi}(t, \mathbf{r})$ is known as the **kinetic momentum operator**. In the presence of an electromagnetic field, the kinetic momentum operator is modified from the canonical momentum operator $\mathbf{p}$ to include the effects of the vector potential $\mathbf{A}_{\text{ext}}(t, \mathbf{r})$:

$$\boldsymbol{\pi}(t, \mathbf{r}) = \mathbf{p} + \frac{1}{c}\mathbf{A}_{\text{ext}}(t, \mathbf{r}) \tag{1.101}$$

Here, $\mathbf{p} = -i\hbar\nabla$ is the canonical momentum operator, and c is the speed of light. The vector potential $\mathbf{A}_{\text{ext}}(t, \mathbf{r})$ accounts for the influence of external perturbative electromagnetic fields on the system. *Why it does the influence of the vector potential not just all go into the v_{eff}?*

But we know that under a gauge transformation, the physical observables will be invariant, while the orbits will merely acquire a **phase factor**:

$$\varphi_j(t, \mathbf{r}) \to \varphi_j'(t, \mathbf{r}) = \varphi_j(t, \mathbf{r}) \exp\left(-\frac{i}{c}\psi(t, \mathbf{r})\right) \tag{1.102}$$

Therefore observables, like the density or current density will be unaffected by this gauge transformation.

1.4.3 Density Matrix Formulation in TDKS Theory

In Time-Dependent Density Functional Theory (TDDFT), the **Time-Dependent Kohn-Sham (TDKS)** equations 1.99 describe a system of N noninteracting fermions that reproduce the same time-dependent electron density $\rho(t, \mathbf{r})$ as the interacting system. An alternative formulation of TDKS theory utilizes the one-particle density matrix $\gamma(t, \mathbf{r}, \mathbf{r}')$, which offers advantage because it introduces a basis that one can exploit computationally. The one-particle density matrix is defined as:

$$\gamma(t, \mathbf{r}, \mathbf{r}') = \sum_{j=1}^{N} \varphi_j(t, \mathbf{r}) \varphi_j^*(t, \mathbf{r}') \tag{1.103}$$

and it is idempotent, meaning that

$$\gamma^2(t, \mathbf{r}, \mathbf{r}') = \gamma(t, \mathbf{r}, \mathbf{r}') \tag{1.104}$$

See section **??** for a proof.

Now, we would like to derive an evolution equation for the density matrix in analogy with the one we already have for the KS orbitals in equation 1.99. The result is

$$i\frac{\partial}{\partial t}\gamma(t) = [H[\rho](t), \gamma(t)] \tag{1.105}$$

See section **??** for proof. For the purposes of response theory, it is convenient to consider external scalar potentials,

$$v_{\text{ext}}(t, x) = v^{(0)}(x) + \sum_{\alpha} \lambda_\alpha \left(v^{(\alpha)}(\omega_\alpha, x)\, e^{i\omega_\alpha t} + v^{(\alpha)}(-\omega_\alpha, x)\, e^{-i\omega_\alpha t} \right) \tag{1.106}$$

and longitudinal vector potentials

$$\mathbf{A}_{\text{ext}}(t, x) = \sum_{\alpha} \lambda_\alpha \left(\mathbf{A}^{(\alpha)}(\omega_\alpha, x)\, e^{i\omega_\alpha t} + \mathbf{A}^{(\alpha)}(-\omega_\alpha, x)\, e^{-i\omega_\alpha t} \right) \tag{1.107}$$

Note that we are not considering transverse vector potentials, as we would get if we had a magnetic field. The cemetery of the Fourier component is fate in section **??**. Next, we need to determine the derivatives of the observables with respect to a perturbation. We have Any time-dependent expectation value $f_\lambda(t)$ is a function of the coupling strength vector $\boldsymbol{\lambda}$. Its response to the external perturbation is defined by its derivatives with respect to $\boldsymbol{\lambda}$ at $\boldsymbol{\lambda} = 0$. For monochromatic perturbations, the derivatives exhibit a characteristic time dependence,

$$f_\lambda(t)|_{\boldsymbol{\lambda}=0} = f^{(0)}, \tag{1.108}$$

$$\frac{\partial}{\partial \lambda_\alpha} f_\lambda(t)\bigg|_{\boldsymbol{\lambda}=0} = f^{(\alpha)}(\omega_\alpha)\, e^{i\omega_\alpha t} + f^{(\alpha)}(-\omega_\alpha)\, e^{-i\omega_\alpha t}, \tag{1.109}$$

$$\tag{1.110}$$

Proof of the second expression is given in section **??**. The third expression follows by taking the derivative of the second expression.

$$\left.\frac{\partial^2}{\partial\lambda_\alpha\partial\lambda_\beta}f_\lambda(t)\right|_{\boldsymbol{\lambda}=0} = f^{(\alpha\beta)}\left(\omega_\alpha,\omega_\beta\right)e^{i\left(\omega_\alpha+\omega_\beta\right)t} + f^{(\alpha\beta)}\left(\omega_\alpha,-\omega_\beta\right)e^{i\left(\omega_\alpha-\omega_\beta\right)t} \tag{1.111}$$

$$+ f^{(\alpha\beta)}\left(-\omega_\alpha,\omega_\beta\right)e^{i\left(-\omega_\alpha+\omega_\beta\right)t} + f^{(\alpha\beta)}\left(-\omega_\alpha,-\omega_\beta\right)e^{-i\left(\omega_\alpha+\omega_\beta\right)t} \tag{1.112}$$

These expressions define the frequency dependent response of f_λ up to second order. The key step is to realize that when evaluating the expectation value of some observable O, we must consider the coupling to the density matrix, i.e. $f_\lambda(t) = \mathrm{tr}\left(O(t)\gamma_\lambda(t)\right)$.

The route to frequency-dependent response properties is then obvious: (1) Calculate the frequencydependent KS density matrix response by differentiation of Eqs. (8) and (10); (2) Take the trace with O. The interacting response can be calculated from the noninteracting KS system because the TDKS density matrix yields the interacting density and current density as it follows from Eq. (9). **Better understanding needed.**

For the idempotency constraint, expansion up to second order yields, in shorthand notation,

$$\gamma^{(0)} = \gamma^{(0)}\gamma^{(0)}, \tag{16}$$

$$\gamma^{(\alpha)} = \gamma^{(0)}\gamma^{(\alpha)} + \gamma^{(\alpha)}\gamma^{(0)}, \tag{17}$$

$$\gamma^{(\alpha\beta)} = \gamma^{(0)}\gamma^{(\alpha\beta)} + \gamma^{(\alpha)}\gamma^{(\beta)} + \gamma^{(\beta)}\gamma^{(\alpha)} + \gamma^{(\alpha\beta)}\gamma^{(0)}. \tag{18}$$

The equations of motion up to second order read

$$0 = \left[H^{(0)},\gamma^{(0)}\right] \tag{1.113}$$

$$\omega_\alpha\gamma^{(\alpha)} = \left[H^{(0)},\gamma^{(\alpha)}\right] + \left[H^{(\alpha)},\gamma^{(0)}\right] \tag{1.114}$$

$$\left(\omega_\alpha+\omega_\beta\right)\gamma^{(\alpha\beta)} = \left[H^{(0)},\gamma^{(\alpha\beta)}\right] + \left[H^{(\alpha)},\gamma^{(\beta)}\right] + \left[H^{(\beta)},\gamma^{(\alpha)}\right] + \left[H^{(\alpha\beta)},\gamma^{(0)}\right] \tag{1.115}$$

Proof of this series is given in section **??**.

1.5 GW Density Matrix

The expression for the density matrix γ is

$$\gamma\left(\mathbf{r},\mathbf{r}'\right) = -\frac{i}{2\pi}\int_{-\infty}^{+\infty} d\omega e^{i\eta\omega}G\left(\mathbf{r},\mathbf{r}',\omega\right) \tag{1.116}$$

Next, we want to consider the expression for the linearized Dyson equation, which is given by

$$G = G_0 + G_0\left(\Sigma_{xc} - V_{xc}\right)G_0 \tag{1.117}$$

So by inserting this LDE into the expression for the density matrix, we can identify a few different terms. First, consider the term just corresponding to G_0:

$$\gamma^{\mathrm{gKS}}_{\mathbf{k}ij} = 2\delta_{ij}\theta\left(\mu - \epsilon_{\mathbf{k}i}\right) \tag{1.118}$$

We can take a similar approach to treat the static portion of 1.117, i.e., the Hartree-Fock term, which is given by

$$\Delta\gamma^{\mathrm{HF}}_{\mathbf{k}ij} = 2\theta\left(\mu - \epsilon_{\mathbf{k}i}\right)\theta\left(\epsilon_{\mathbf{k}j} - \mu\right)\frac{\langle \mathbf{k}i|\Sigma_x - V_{xc}|\mathbf{k}j\rangle}{\epsilon_{\mathbf{k}i} - \epsilon_{\mathbf{k}j}} \tag{1.119}$$

Note that for a HF noninteracting Green's function, the static term is given by the Hartree-Fock self-energy, but this cancels out exactly with the exchange correlation potential V_{xc}, so this term is zero. Finally we have the most complicated term involving the insertion of $G_0\Sigma_c G_0$. The correlation self energy has a frequency dependence, so what they do is Finally, we get

$$\gamma^{GW} = \gamma^{\mathrm{gKS}} + \Delta\gamma^{\mathrm{HF}} + \Delta\gamma^{GW} \tag{1.120}$$

1.6 Summary of GW-RPA implementations

Note that mainly we will be talking about G_0W_0 implementations, but it will be commented on as to the whether self-consistency is possible. We will also discuss scaling and whether it can be applied to extended systems.

1.6.1 Complex integration approaches

Treatment of extended systems is possible for the below. Self-consistency is also possible in all cases, as there is no restriction for what the reference state can be. For a discussion of all of the below approaches, see [2]. They all share the desire of performing the frequency integration in the expression for the self-energy, which is given by

$$\Sigma_\sigma\left(\mathbf{r}, \mathbf{r}', \omega\right) = \frac{i}{2\pi}\int d\omega' e^{i\omega'\eta} G_0^\sigma\left(\mathbf{r}, \mathbf{r}', \omega + \omega'\right) W_0\left(\mathbf{r}, \mathbf{r}', \omega'\right) \tag{1.121}$$

Fully analytic

Here we will perform the frequency integration over the real axis. If we choose the upper or lower contour, we encounter many poles of G and W, so we must apply Cauchy's residue theorem repeatedly. Therefore this method must explicitly calculate the reducible polarizability for all frequencies, or in other words, if the RPA approximation is employed, diagonalize the RPA matrix, which scales as $O(N^6)$. Because of this prohibitive cost, the method is never used.

Analytic continuation (AC)

The idea is to not perform the frequency integration over the real axis in order to avoid the poles of G and W, but instead to perform this integration completely on the imaginary axis. This has positive effects on the scaling, but at the expense of a poor description of core states. For example, to accurately evaluate the QP equation for a core state, we need the self-energy at frequencies close to the core solution, where there is a fine pole structure, which AC cannot capture. By a similar reasoning, satellite features are also not captured.

Scaling analysis for extended systems

It is true that if only G_0W_0 QP energies are required, one only needs to compute the diagonal self-energy matrix elements. The most expensive step is then computing the auxiliary density response function Π, whose cost scales as $O\left(N_{\boldsymbol{k}}^2 N_O N_V N_{\text{aux}}^2\right)$. If the full G_0W_0 Green's function and off-diagonal self-energy matrix at all k-points are sought, computation of Σ^c becomes the most time-consuming step, with a cost scaling of $O\left(N_{\boldsymbol{k}}^2 N_{AO}^2 N_{\text{aux}}^2\right)$. For more detail, see [5].

Note that one must apply a finite size correction to the dielectric function to get a reasonable convergence towards the TDL, known as the head and wings correction. This is a result of the fact that RI is used for this method, where the 3-center integrals have a divergence at $\mathbf{q} = 0$.

Contour deformation (CD)

The real frequency integration is decomposed into an integral over the deformed contour minus an integral over the imaginary axis. By doing so, we avoid all poles of W, but poles of G below the Fermi level remain. This has a negative effect on the scaling if core states are sought, because we will have to evaluate the self-energy at frequencies well below the Fermi level, where we may encounter many of the occupied poles of G, which are positioned above the real axis. However the description of core states is known to be accurate, and the GW satellite structure is well-preserved.

Scaling analysis for extended systems

As is also the case for AC, if only G_0W_0 QP energies are required, one only needs to compute the diagonal self-energy matrix elements. In the decomposition of the real frequency integral in CD, the imaginary integration has a similar computational cost to the G_0W_0-AC scheme. But in the contour integral, we encounter many poles if we want core states, which are far away from the Fermi level, and so we must compute many residues. So for deep core excitations, the scaling becomes $O\left(N_{\boldsymbol{k}}^3 N_{\text{O}}^2 N_{\text{V}} N_{\text{aux}}^2\right)$. For more detail, see [5].

In analogy to AC, we must apply the finite size correction to the dielectric function for reasonable convergence towards the TDL.

1.6.2 Contour deformation (CD)

Spectral function

We know

$$A(\mathbf{k}, \omega) = \frac{1}{\pi} |\operatorname{Im} G^T(\mathbf{k}, \tilde{\omega})| \tag{1.122}$$

$$= \frac{1}{\pi} \frac{|\operatorname{Im} \Sigma^T(\mathbf{k}, \tilde{\omega})|}{\left(\omega - \epsilon_{\mathbf{k}} - \operatorname{Re} \Sigma^T(\mathbf{k}, \tilde{\omega})\right)^2 + \left(\operatorname{Im} \Sigma^T(\mathbf{k}, \tilde{\omega})\right)^2} \tag{1.123}$$

where $\tilde{\omega} = \omega - i\eta \,\mathrm{sgn}\,(\epsilon_F - \omega)$, but also in the retarded case we have

$$A(\mathbf{k}, \bar{\omega}) = \frac{1}{\pi} |\operatorname{Im} G^R(\mathbf{k}, \bar{\omega})| \tag{1.124}$$

$$= \frac{1}{\pi} \frac{|\operatorname{Im} \Sigma^R(\mathbf{k}, \bar{\omega})|}{\left(\bar{\omega} - \epsilon_{\mathbf{k}} - \operatorname{Re} \Sigma^R(\mathbf{k}, \bar{\omega})\right)^2 + \left(\operatorname{Im} \Sigma^R(\mathbf{k}, \bar{\omega})\right)^2} \tag{1.125}$$

where $\bar{\omega} = \omega + i\eta$. These two schemes have been shown to be equivalent in the TDL, but in finite systems, it is correct to choose the retarded path.

Retarded self-energy

We begin from the contour-deformation expression for the GW correlation self-energy,

$$\begin{aligned} \Sigma^c(\mathbf{r}, \mathbf{r}', \bar{\omega}) = \frac{i}{2\pi} \oint d\omega' \, G(\mathbf{r}, \mathbf{r}', \bar{\omega} + \omega') \, W^c(\mathbf{r}, \mathbf{r}', \omega') \\ - \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' \, G(\mathbf{r}, \mathbf{r}', \bar{\omega} + i\omega') \, W^c(\mathbf{r}, \mathbf{r}', i\omega') \end{aligned} \tag{1.126}$$

$$= \Sigma^C(\mathbf{r}, \mathbf{r}', \bar{\omega}) + \Sigma^I(\mathbf{r}, \mathbf{r}', \bar{\omega}) \tag{1.127}$$

Imaginary part

Numerical issues may arise from the latter piece, where we are evaluating the integral on the imaginary axis,so we first consider that. Projection onto the MO basis gives

$$\Sigma^I(\mathbf{k}_m, \bar{\omega}) = -\frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega' \int d\mathbf{r} d\mathbf{r}' \, \phi^*_{\mathbf{k}_m}(\mathbf{r}) G(\mathbf{r}, \mathbf{r}', \bar{\omega} + i\omega') \, W^c(\mathbf{r}, \mathbf{r}', i\omega') \, \phi_{\mathbf{k}_m}(\mathbf{r}') \tag{1.128}$$

The reference Green's function is taken to be diagonal in the MO basis, so we have

$$G(\mathbf{r}, \mathbf{r}', \bar{\omega} + i\omega') = \sum_{\boldsymbol{k}_n} \phi_{\mathbf{k}_n}(\mathbf{r}) G(\mathbf{k}_n, \bar{\omega} + i\omega') \, \phi^*_{\mathbf{k}_n}(\mathbf{r}') \tag{1.129}$$

and substituting this into the expression for Σ^I gives[1]

$$\Sigma^I(\mathbf{k}_m, \bar{\omega}) = -\frac{1}{2\pi} \sum_{\mathbf{k}_n} \int_{-\infty}^{\infty} d\omega' \, G(\mathbf{k}_n, \bar{\omega} + i\omega') \int d\mathbf{r} d\mathbf{r}' \, \phi^*_{\mathbf{k}_m}(\mathbf{r}) \phi_{\mathbf{k}_n}(\mathbf{r}) W^c(\mathbf{r}, \mathbf{r}', i\omega') \, \phi^*_{\mathbf{k}_n}(\mathbf{r}') \phi_{\mathbf{k}_m}(\mathbf{r}') \tag{1.130}$$

$$= -\frac{1}{2\pi} \sum_{\mathbf{k}_n} \int_{-\infty}^{\infty} d\omega' \, G(\mathbf{k}_n, \bar{\omega} + i\omega') \, W^c(\mathbf{k}_m, \mathbf{k}_n; i\omega') \,. \tag{1.131}$$

Now, we know that the retarded, reference Green's function is given by

$$G(\mathbf{k}_n, \bar{\omega} + i\omega') = \frac{1}{a_n + i\omega'}, \tag{1.132}$$

[1] Is this related to your derivation of BWs2 with screening earlier this year? Would that theory be equivalent if we used BWs2 as our reference Green's function? How would we evaluate W^c? Would we still do dRPA, or more like SOSEX?

where, at zero temperature, $a_n = \omega - \epsilon_{\mathbf{k}_n} + i\eta$. And elements of the screened interaction are given by

$$W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right) = \sum_{PQ} v_P^{\mathbf{k}_m \mathbf{k}_n} \left[\left[\mathbf{I} - \mathbf{\Pi}(i\omega')\right]_{PQ}^{-1} - \delta_{PQ}\right] v_Q^{\mathbf{k}_n \mathbf{k}_m} \tag{1.133}$$

where, in the restricted formulation, the polarizability on the imaginary axis is

$$\mathbf{\Pi}_{PQ}\left(i\omega'\right) = 2 \sum_{\mathbf{k}_i}^{\text{occ}} \sum_{\mathbf{k}_a}^{\text{vir}} v_P^{\mathbf{k}_i \mathbf{k}_a} \frac{\epsilon_{\mathbf{k}_i} - \epsilon_{\mathbf{k}_a}}{\omega'^2 + \left(\epsilon_{\mathbf{k}_i} - \epsilon_{\mathbf{k}_a}\right)^2} v_Q^{\mathbf{k}_a \mathbf{k}_i}. \tag{1.134}$$

Since $\mathbf{\Pi}(i\omega')$ depends only on ω'^2, $\mathbf{\Pi}\left(i\omega'\right) = \mathbf{\Pi}\left(-i\omega'\right) \implies W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right) = W^c\left(\mathbf{k}_m, \mathbf{k}_n; -i\omega'\right)$. Using this evenness of $W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right)$, we can fold the integral as:

$$\Sigma^I\left(\mathbf{k}_m, \bar{\omega}\right) = -\frac{1}{2\pi} \sum_{\mathbf{k}_n} \int_{-\infty}^{\infty} d\omega' \, \frac{W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right)}{a_n + i\omega'} \tag{1.135}$$

$$= -\frac{1}{2\pi} \sum_{\mathbf{k}_n} \int_0^{\infty} d\omega' \, W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right) \left[\frac{1}{a_n + i\omega'} + \frac{1}{a_n - i\omega'}\right] \tag{1.136}$$

$$= -\frac{1}{2\pi} \sum_{\mathbf{k}_n} \int_0^{\infty} d\omega' \, \frac{2a_n}{a_n^2 + \omega'^2} W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right) \tag{1.137}$$

$$= -\frac{1}{\pi} \sum_{\mathbf{k}_n} \int_0^{\infty} d\omega' \, \frac{a_n}{a_n^2 + \omega'^2} W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega'\right) \tag{1.138}$$

$$\approx -\frac{1}{\pi} \sum_{\mathbf{k}_n} \sum_{j=1}^{N_\omega} w_j \frac{a_n}{a_n^2 + \omega_j^2} W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega_j\right). \tag{1.139}$$

When $\omega \approx \epsilon_{\mathbf{k}_n}$, we have $a_n \approx i\eta$. But then the denominator becomes

$$a_n^2 + \omega_j^2 \approx -\eta^2 + \omega_j^2 \tag{1.140}$$

So we encounter a singularity if $\omega_j \approx \eta$. The scatter plot shows this numerically.

Regularization

To remedy this, one can regularize the denominator by performing a subtraction. By writing

$$W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega_j\right) = \left[W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\omega_j\right) - W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\eta\right)\right] + W^c\left(\mathbf{k}_m, \mathbf{k}_n; i\eta\right) \tag{1.141}$$

then starting from (1.138), we have[2]

$$\Sigma^I(\mathbf{k}_m,\bar{\omega}) = -\frac{1}{\pi}\sum_{\mathbf{k}_n}\int_0^\infty d\omega' \frac{a_n}{a_n^2+\omega'^2}\left[W^c(\mathbf{k}_m,\mathbf{k}_n;i\omega') - W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta)\right] \tag{1.142}$$

$$-\frac{1}{\pi}\sum_{\mathbf{k}_n} W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta)\int_0^\infty d\omega' \frac{a_n}{a_n^2+\omega'^2} \tag{1.143}$$

$$= -\frac{1}{\pi}\sum_{\mathbf{k}_n}\int_0^\infty d\omega' \frac{a_n}{a_n^2+\omega'^2}\left[W^c(\mathbf{k}_m,\mathbf{k}_n;i\omega') - W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta)\right] \tag{1.144}$$

$$-\frac{1}{2}\sum_{\mathbf{k}_n} W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta)\frac{a_n}{\sqrt{a_n^2}} \tag{1.145}$$

$$\approx -\frac{1}{\pi}\sum_{\mathbf{k}_n}\sum_{j=1}^{N_\omega} w_j \frac{a_n}{a_n^2+\omega_j^2}\left[W^c(\mathbf{k}_m,\mathbf{k}_n;i\omega_j) - W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta)\right] \tag{1.146}$$

$$-\frac{1}{2}\sum_{\mathbf{k}_n} W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta)\frac{a_n}{\sqrt{a_n^2}} \tag{1.147}$$

Now the integral is smoother because $W^c(\mathbf{k}_m,\mathbf{k}_n;i\omega_j) - W^c(\mathbf{k}_m,\mathbf{k}_n;i\eta) \to 0$ as $\omega_j \to \eta$.

Contour Integral

This part is exact.

Derivative of the contour-deformation self-energy with respect to frequency

First, we can determine the derivative with respect to the frequency.

$$\frac{d\Sigma_m^{\mathrm{CD}}(\omega)}{d\omega} = \frac{d\Sigma_m^C(\omega)}{d\omega} + \frac{d\Sigma_m^I(\omega)}{d\omega}. \tag{1.148}$$

Contour-residue contribution. The residue contribution to the correlation self-energy is

$$\Sigma_m^C(\omega) = \sum_n f_n(\omega)\sum_{PQ} v_P^{mn}\left\{\left[\mathbf{I}-\mathbf{\Pi}(z_n)\right]^{-1}-\mathbf{I}\right\}_{PQ} v_Q^{nm}, \tag{1.149}$$

where

$$z_n = \epsilon_n - \omega, \tag{1.150}$$

and the residue weight is

$$f_n(\omega) = \begin{cases} +1, & \omega > \epsilon_F, \quad \epsilon_F < \epsilon_n < \omega, \\ -1, & \omega < \epsilon_F, \quad \omega < \epsilon_n < \epsilon_F, \\ 0, & \text{otherwise.} \end{cases} \tag{1.151}$$

[2]We use the identity $\int_0^\infty d\omega' \frac{a}{a^2+\omega'^2} = \frac{\pi}{2}\frac{a}{\sqrt{a^2}}$

Away from threshold points $\omega = \epsilon_n$, the residue weight is piecewise constant, so

$$\frac{df_n}{d\omega} = 0. \tag{1.152}$$

Define

$$\mathbf{M}(z) = \mathbf{I} - \mathbf{\Pi}(z), \tag{1.153}$$

so that

$$\mathbf{W}_c(z) = \mathbf{M}(z)^{-1} - \mathbf{I}. \tag{1.154}$$

Then

$$\Sigma_m^C(\omega) = \sum_n f_n \sum_{PQ} v_P^{mn} \left[\mathbf{W}_c(z_n)\right]_{PQ} v_Q^{nm}. \tag{1.155}$$

The derivative of an inverse matrix is

$$\frac{d\mathbf{M}^{-1}}{dz} = -\mathbf{M}^{-1} \frac{d\mathbf{M}}{dz} \mathbf{M}^{-1}. \tag{1.156}$$

Since

$$\mathbf{M}(z) = \mathbf{I} - \mathbf{\Pi}(z), \tag{1.157}$$

we have

$$\frac{d\mathbf{M}}{dz} = -\frac{d\mathbf{\Pi}}{dz}. \tag{1.158}$$

Therefore

$$\frac{d\mathbf{M}^{-1}}{dz} = \mathbf{M}^{-1} \frac{d\mathbf{\Pi}}{dz} \mathbf{M}^{-1}. \tag{1.159}$$

Because

$$z_n = \epsilon_n - \omega, \tag{1.160}$$

we also have

$$\frac{dz_n}{d\omega} = -1. \tag{1.161}$$

Thus

$$\frac{d\mathbf{W}_c(z_n)}{d\omega} = \frac{d}{d\omega}\left[\mathbf{M}(z_n)^{-1} - \mathbf{I}\right] \tag{1.162}$$

$$= \frac{d\mathbf{M}(z_n)^{-1}}{dz_n} \frac{dz_n}{d\omega} \tag{1.163}$$

$$= -\mathbf{M}(z_n)^{-1} \mathbf{\Pi}'(z_n) \mathbf{M}(z_n)^{-1}, \tag{1.164}$$

where

$$\mathbf{\Pi}'(z) = \frac{d\mathbf{\Pi}(z)}{dz}. \tag{1.165}$$

The contour-residue derivative is therefore

$$\boxed{\frac{d\Sigma_m^C(\omega)}{d\omega} = -\sum_n f_n \sum_{PQ} v_P^{mn} \left[\mathbf{M}(z_n)^{-1} \mathbf{\Pi}'(z_n) \mathbf{M}(z_n)^{-1}\right]_{PQ} v_Q^{nm}}. \tag{1.166}$$

Derivative of the polarizability. For a general complex frequency z, the independent-particle polarizability is

$$\Pi_{PQ}(z) = \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} v_P^{ia} \left[\frac{1}{z - \Delta_{ia}} - \frac{1}{z + \Delta_{ia}} \right] v_Q^{ai}, \tag{1.167}$$

where

$$\Delta_{ia} = \epsilon_a - \epsilon_i. \tag{1.168}$$

Taking the derivative with respect to z gives

$$\Pi'_{PQ}(z) = \frac{d\Pi_{PQ}(z)}{dz} \tag{1.169}$$

$$= \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} v_P^{ia} \left[-\frac{1}{(z - \Delta_{ia})^2} + \frac{1}{(z + \Delta_{ia})^2} \right] v_Q^{ai}. \tag{1.170}$$

Equivalently,

$$\boxed{\Pi'_{PQ}(z) = \sum_{i}^{\text{occ}} \sum_{a}^{\text{vir}} v_P^{ia} \left[\frac{1}{(z + \Delta_{ia})^2} - \frac{1}{(z - \Delta_{ia})^2} \right] v_Q^{ai}}. \tag{1.171}$$

Imaginary-axis contribution. The imaginary-axis contribution can be written as

$$\Sigma_m^I(\omega) = -\frac{1}{\pi} \sum_n \int_0^\infty d\Omega \, K_n(\omega, \Omega) \sum_{PQ} v_P^{mn} \left[\mathbf{W}_c(i\Omega) \right]_{PQ} v_Q^{nm}, \tag{1.172}$$

where

$$\mathbf{W}_c(i\Omega) = \left[\mathbf{I} - \mathbf{\Pi}(i\Omega) \right]^{-1} - \mathbf{I}. \tag{1.173}$$

The scalar Green's-function kernel is

$$K_n(\omega, \Omega) = \frac{x_n(\omega)}{x_n(\omega)^2 + \Omega^2}. \tag{1.174}$$

If ω is measured relative to the Fermi energy, then

$$x_n(\omega) = \omega + \epsilon_F - \epsilon_n. \tag{1.175}$$

If instead ω is an absolute frequency, then

$$x_n(\omega) = \omega - \epsilon_n. \tag{1.176}$$

Since $\mathbf{W}_c(i\Omega)$ depends only on the integration frequency Ω, not on the external frequency ω, the derivative of Σ_m^I comes only from the derivative of K_n. We have

$$\frac{dK_n}{d\omega} = \frac{d}{d\omega} \left[\frac{x_n}{x_n^2 + \Omega^2} \right] \tag{1.177}$$

$$= \frac{dx_n}{d\omega} \frac{\Omega^2 - x_n^2}{\left(x_n^2 + \Omega^2 \right)^2}. \tag{1.178}$$

For the usual choices above,

$$\frac{dx_n}{d\omega} = 1, \tag{1.179}$$

so

$$\boxed{\frac{dK_n}{d\omega} = \frac{\Omega^2 - x_n(\omega)^2}{\left[x_n(\omega)^2 + \Omega^2\right]^2}}. \tag{1.180}$$

Therefore,

$$\boxed{\frac{d\Sigma_m^I(\omega)}{d\omega} = -\frac{1}{\pi}\sum_n \int_0^\infty d\Omega\, \frac{\Omega^2 - x_n(\omega)^2}{\left[x_n(\omega)^2 + \Omega^2\right]^2} \sum_{PQ} v_P^{mn} \left[\mathbf{W}_c(i\Omega)\right]_{PQ} v_Q^{nm}}. \tag{1.181}$$

Total derivative. Combining the contour-residue and imaginary-axis pieces gives

$$\boxed{\frac{d\Sigma_m^{\mathrm{CD}}(\omega)}{d\omega} = -\sum_n f_n \sum_{PQ} v_P^{mn} \left[\mathbf{M}(z_n)^{-1}\mathbf{\Pi}'(z_n)\mathbf{M}(z_n)^{-1}\right]_{PQ} v_Q^{nm}}$$

$$\boxed{-\frac{1}{\pi}\sum_n \int_0^\infty d\Omega\, \frac{\Omega^2 - x_n(\omega)^2}{\left[x_n(\omega)^2 + \Omega^2\right]^2} \sum_{PQ} v_P^{mn} \left[\mathbf{W}_c(i\Omega)\right]_{PQ} v_Q^{nm}}. \tag{1.182}$$

Here

$$\mathbf{M}(z) = \mathbf{I} - \mathbf{\Pi}(z), \tag{1.183}$$

$$\mathbf{W}_c(i\Omega) = \left[\mathbf{I} - \mathbf{\Pi}(i\Omega)\right]^{-1} - \mathbf{I}, \tag{1.184}$$

and

$$z_n = \epsilon_n - \omega. \tag{1.185}$$

Derivative of the contour-deformation self-energy with respect to single-particle energies

We now consider derivatives of the contour-deformation self-energy with respect to a single-particle energy ϵ_p. To distinguish the different sources of energy dependence, we write the energies entering the Green's function as ϵ^G, and the energies entering the screened interaction as ϵ^W. The external frequency ω is held fixed in this derivative.

The contour-deformation self-energy is written as

$$\Sigma_m^{\mathrm{CD}}(\omega) = \Sigma_m^C(\omega) + \Sigma_m^I(\omega), \tag{1.186}$$

where

$$\Sigma_m^C(\omega) = \sum_n f_n \sum_{PQ} v_P^{mn} \left[\mathbf{W}_c(z_n)\right]_{PQ} v_Q^{nm}, \tag{1.187}$$

with

$$z_n = \epsilon_n^G - \omega, \tag{1.188}$$

and

$$\mathbf{W}_c(z) = \left[\mathbf{I} - \mathbf{\Pi}(z)\right]^{-1} - \mathbf{I}. \tag{1.189}$$

The imaginary-axis contribution is

$$\Sigma_m^I(\omega) = -\frac{1}{\pi}\sum_n \int_0^\infty d\Omega\, K_n(\omega,\Omega) \sum_{PQ} v_P^{mn}\, [\mathbf{W}_c(i\Omega)]_{PQ}\, v_Q^{nm}. \tag{1.190}$$

The scalar Green's-function kernel is

$$K_n(\omega,\Omega) = \frac{x_n}{x_n^2+\Omega^2}. \tag{1.191}$$

If ω is measured relative to the Fermi energy, then

$$x_n = \omega + \epsilon_F - \epsilon_n^G. \tag{1.192}$$

If ω is an absolute frequency, then

$$x_n = \omega - \epsilon_n^G. \tag{1.193}$$

We define

$$\mathbf{M}(z) = \mathbf{I} - \mathbf{\Pi}(z), \tag{1.194}$$

so that

$$\mathbf{W}_c(z) = \mathbf{M}(z)^{-1} - \mathbf{I}. \tag{1.195}$$

The derivative of the screened interaction with respect to a parameter α that enters the polarizability is

$$\frac{\partial \mathbf{W}_c(z)}{\partial \alpha} = \mathbf{M}(z)^{-1}\frac{\partial \mathbf{\Pi}(z)}{\partial \alpha}\mathbf{M}(z)^{-1}. \tag{1.196}$$

If the parameter enters through the frequency argument z, then

$$\frac{\partial \mathbf{W}_c(z)}{\partial z} = \mathbf{M}(z)^{-1}\mathbf{\Pi}'(z)\mathbf{M}(z)^{-1}, \tag{1.197}$$

where

$$\mathbf{\Pi}'(z) = \frac{d\mathbf{\Pi}(z)}{dz}. \tag{1.198}$$

The independent-particle polarizability is

$$\Pi_{PQ}(z) = \sum_i^{\text{occ}}\sum_a^{\text{vir}} v_P^{ia}\left[\frac{1}{z-\Delta_{ia}} - \frac{1}{z+\Delta_{ia}}\right] v_Q^{ai}, \tag{1.199}$$

where

$$\Delta_{ia} = \epsilon_a^W - \epsilon_i^W. \tag{1.200}$$

Its derivative with respect to its frequency argument is

$$\Pi'_{PQ}(z) = \sum_i^{\text{occ}}\sum_a^{\text{vir}} v_P^{ia}\left[\frac{1}{(z+\Delta_{ia})^2} - \frac{1}{(z-\Delta_{ia})^2}\right] v_Q^{ai}. \tag{1.201}$$

We neglect derivatives of the residue weights f_n, assuming that the frequency is away from the threshold points where $\omega = \epsilon_n$ or $\epsilon_n = \epsilon_F$. Thus f_n is treated as piecewise constant.

Case 1: updating energies entering G, with W fixed. In this case, the differentiated energy is ϵ_p^G. The screened interaction W_c is not changed through its internal polarizability, because the energies entering W are held fixed. However, the contour argument

$$z_n = \epsilon_n^G - \omega \tag{1.202}$$

does depend on ϵ_p^G. Therefore

$$\frac{\partial z_n}{\partial \epsilon_p^G} = \delta_{np}. \tag{1.203}$$

The contour derivative is

$$\frac{\partial \Sigma_m^C(\omega)}{\partial \epsilon_p^G} = \sum_n f_n \sum_{PQ} v_P^{mn} \frac{\partial}{\partial \epsilon_p^G} \left[\mathbf{W}_c(z_n)\right]_{PQ} v_Q^{nm} \tag{1.204}$$

$$= \sum_n f_n \delta_{np} \sum_{PQ} v_P^{mn} \left[\frac{\partial \mathbf{W}_c(z_n)}{\partial z_n}\right]_{PQ} v_Q^{nm}. \tag{1.205}$$

Hence

$$\boxed{\frac{\partial \Sigma_m^C(\omega)}{\partial \epsilon_p^G} = f_p \sum_{PQ} v_P^{mp} \left[\mathbf{M}(z_p)^{-1} \mathbf{\Pi}'(z_p) \mathbf{M}(z_p)^{-1}\right]_{PQ} v_Q^{pm}}. \tag{1.206}$$

For the imaginary-axis contribution, $W_c(i\Omega)$ is fixed, and only the kernel K_n depends on ϵ_p^G. Since

$$K_n(\omega, \Omega) = \frac{x_n}{x_n^2 + \Omega^2}, \tag{1.207}$$

we have

$$\frac{\partial K_n}{\partial x_n} = \frac{\Omega^2 - x_n^2}{\left(x_n^2 + \Omega^2\right)^2}. \tag{1.208}$$

For the relative-frequency convention,

$$x_n = \omega + \epsilon_F - \epsilon_n^G, \tag{1.209}$$

and, holding ϵ_F fixed,

$$\frac{\partial x_n}{\partial \epsilon_p^G} = -\delta_{np}. \tag{1.210}$$

Therefore

$$\frac{\partial K_n}{\partial \epsilon_p^G} = -\delta_{np} \frac{\Omega^2 - x_n^2}{\left(x_n^2 + \Omega^2\right)^2}. \tag{1.211}$$

The imaginary-axis derivative is then

$$\frac{\partial \Sigma_m^I(\omega)}{\partial \epsilon_p^G} = -\frac{1}{\pi} \sum_n \int_0^\infty d\Omega \, \frac{\partial K_n}{\partial \epsilon_p^G} \sum_{PQ} v_P^{mn} \left[\mathbf{W}_c(i\Omega)\right]_{PQ} v_Q^{nm} \tag{1.212}$$

$$= \boxed{\frac{1}{\pi} \int_0^\infty d\Omega \, \frac{\Omega^2 - x_p^2}{\left(x_p^2 + \Omega^2\right)^2} \sum_{PQ} v_P^{mp} \left[\mathbf{W}_c(i\Omega)\right]_{PQ} v_Q^{pm}}. \tag{1.213}$$

Thus, for G-only energy updates,

$$\boxed{\frac{\partial \Sigma_m^{\mathrm{CD}}(\omega)}{\partial \epsilon_p^G} = \frac{\partial \Sigma_m^C(\omega)}{\partial \epsilon_p^G} + \frac{\partial \Sigma_m^I(\omega)}{\partial \epsilon_p^G}}. \tag{1.214}$$

Case 2: updating energies entering W, with G fixed. In this case, the differentiated energy is ϵ_p^W. The Green's-function quantities z_n, x_n, K_n, and f_n are fixed. The only dependence on ϵ_p^W enters through the polarizability $\mathbf{\Pi}$, and therefore through W_c.

The excitation energy is

$$\Delta_{ia} = \epsilon_a^W - \epsilon_i^W, \tag{1.215}$$

so

$$\frac{\partial \Delta_{ia}}{\partial \epsilon_p^W} = \delta_{ap} - \delta_{ip}. \tag{1.216}$$

Define

$$A_{ia}(z) = \frac{1}{z - \Delta_{ia}} - \frac{1}{z + \Delta_{ia}}. \tag{1.217}$$

Then

$$\frac{\partial A_{ia}(z)}{\partial \Delta_{ia}} = \frac{\partial}{\partial \Delta_{ia}} \left[\frac{1}{z - \Delta_{ia}} - \frac{1}{z + \Delta_{ia}} \right] \tag{1.218}$$

$$= \frac{1}{(z - \Delta_{ia})^2} + \frac{1}{(z + \Delta_{ia})^2}. \tag{1.219}$$

Therefore

$$\boxed{\frac{\partial \Pi_{PQ}(z)}{\partial \epsilon_p^W} = \sum_i^{\mathrm{occ}} \sum_a^{\mathrm{vir}} v_P^{ia} \left(\delta_{ap} - \delta_{ip}\right) \left[\frac{1}{(z - \Delta_{ia})^2} + \frac{1}{(z + \Delta_{ia})^2} \right] v_Q^{ai}}. \tag{1.220}$$

Using

$$\frac{\partial \mathbf{W}_c(z)}{\partial \epsilon_p^W} = \mathbf{M}(z)^{-1} \frac{\partial \mathbf{\Pi}(z)}{\partial \epsilon_p^W} \mathbf{M}(z)^{-1}, \tag{1.221}$$

the contour derivative is

$$\boxed{\frac{\partial \Sigma_m^C(\omega)}{\partial \epsilon_p^W} = \sum_n f_n \sum_{PQ} v_P^{mn} \left[\mathbf{M}(z_n)^{-1} \frac{\partial \mathbf{\Pi}(z_n)}{\partial \epsilon_p^W} \mathbf{M}(z_n)^{-1} \right]_{PQ} v_Q^{nm}}. \tag{1.222}$$

The imaginary-axis derivative is

$$\boxed{\frac{\partial \Sigma_m^I(\omega)}{\partial \epsilon_p^W} = -\frac{1}{\pi} \sum_n \int_0^\infty d\Omega \, K_n(\omega, \Omega) \sum_{PQ} v_P^{mn} \left[\mathbf{M}(i\Omega)^{-1} \frac{\partial \mathbf{\Pi}(i\Omega)}{\partial \epsilon_p^W} \mathbf{M}(i\Omega)^{-1} \right]_{PQ} v_Q^{nm}}. \tag{1.223}$$

Thus, for W-only energy updates,

$$\boxed{\frac{\partial \Sigma_m^{\mathrm{CD}}(\omega)}{\partial \epsilon_p^W} = \frac{\partial \Sigma_m^C(\omega)}{\partial \epsilon_p^W} + \frac{\partial \Sigma_m^I(\omega)}{\partial \epsilon_p^W}}. \tag{1.224}$$

Case 3: updating energies entering both G and W. In this case, the same energy ϵ_p enters both the Green's function and the screened interaction. Therefore the derivative is the sum of the G-side derivative and the W-side derivative:

$$\boxed{\frac{\partial \Sigma_m^{\mathrm{CD}}(\omega)}{\partial \epsilon_p} = \left.\frac{\partial \Sigma_m^{\mathrm{CD}}(\omega)}{\partial \epsilon_p^G}\right|_W + \left.\frac{\partial \Sigma_m^{\mathrm{CD}}(\omega)}{\partial \epsilon_p^W}\right|_G}. \tag{1.225}$$

The contour derivative is

$$\begin{aligned}\frac{\partial \Sigma_m^{C}(\omega)}{\partial \epsilon_p} &= \left.\frac{\partial \Sigma_m^{C}(\omega)}{\partial \epsilon_p^G}\right|_W + \left.\frac{\partial \Sigma_m^{C}(\omega)}{\partial \epsilon_p^W}\right|_G &(1.226)\\ &= \boxed{f_p \sum_{PQ} v_P^{mp} \left[\mathbf{M}(z_p)^{-1}\boldsymbol{\Pi}'(z_p)\mathbf{M}(z_p)^{-1}\right]_{PQ} v_Q^{pm}} \\ &\quad \boxed{+\sum_n f_n \sum_{PQ} v_P^{mn} \left[\mathbf{M}(z_n)^{-1}\frac{\partial \boldsymbol{\Pi}(z_n)}{\partial \epsilon_p}\mathbf{M}(z_n)^{-1}\right]_{PQ} v_Q^{nm}}. &(1.227)\end{aligned}$$

The imaginary-axis derivative is

$$\begin{aligned}\frac{\partial \Sigma_m^{I}(\omega)}{\partial \epsilon_p} &= \left.\frac{\partial \Sigma_m^{I}(\omega)}{\partial \epsilon_p^G}\right|_W + \left.\frac{\partial \Sigma_m^{I}(\omega)}{\partial \epsilon_p^W}\right|_G &(1.228)\\ &= \boxed{\frac{1}{\pi}\int_0^\infty d\Omega\, \frac{\Omega^2 - x_p^2}{\left(x_p^2+\Omega^2\right)^2} \sum_{PQ} v_P^{mp} \left[\mathbf{W}_c(i\Omega)\right]_{PQ} v_Q^{pm}} \\ &\quad \boxed{-\frac{1}{\pi}\sum_n \int_0^\infty d\Omega\, K_n(\omega,\Omega) \sum_{PQ} v_P^{mn} \left[\mathbf{M}(i\Omega)^{-1}\frac{\partial \boldsymbol{\Pi}(i\Omega)}{\partial \epsilon_p}\mathbf{M}(i\Omega)^{-1}\right]_{PQ} v_Q^{nm}}. &(1.229)\end{aligned}$$

Here

$$\frac{\partial \Pi_{PQ}(z)}{\partial \epsilon_p} = \sum_i^{\mathrm{occ}} \sum_a^{\mathrm{vir}} v_P^{ia} \left(\delta_{ap} - \delta_{ip}\right) \left[\frac{1}{(z-\Delta_{ia})^2} + \frac{1}{(z+\Delta_{ia})^2}\right] v_Q^{ai}. \tag{1.230}$$

Therefore the full derivative for simultaneous G- and W-energy updates is

$$\boxed{\frac{\partial \Sigma_m^{\mathrm{CD}}(\omega)}{\partial \epsilon_p} = \frac{\partial \Sigma_m^{C}(\omega)}{\partial \epsilon_p} + \frac{\partial \Sigma_m^{I}(\omega)}{\partial \epsilon_p}}. \tag{1.231}$$

Remarks on nonsmooth terms. The expressions above neglect derivatives of the residue weights f_n. This is appropriate away from threshold points. Formally, the derivative of the piecewise residue window produces delta-function contributions when $\epsilon_n = \omega$ or $\epsilon_n = \epsilon_F$. In numerical self-consistency iterations, these terms are usually omitted and the residue window is updated discretely between evaluations.

If the Fermi energy is updated together with the single-particle energies, then the kernel derivative must be modified. For the relative-frequency convention,

$$x_n = \omega + \epsilon_F - \epsilon_n, \tag{1.232}$$

and therefore

$$\frac{\partial x_n}{\partial \epsilon_p} = \frac{\partial \epsilon_F}{\partial \epsilon_p} - \delta_{np}. \tag{1.233}$$

The formulas above assume

$$\frac{\partial \epsilon_F}{\partial \epsilon_p} = 0 \tag{1.234}$$

during the derivative evaluation.

Check against Python

I have added a number of features to my Python implementation, and now I want these to be reflected in my C++ code. The best way to do this will be to run my python code for 14e/19o, and then check that the C++ code produces the same results up to numerical precision.

1. Check that the contour deformation quasi-particle energy is equivalent to the python one using $nw_{cd} = 400$.

2. Implement a BFGS optimization procedure to do self-consistency for GW0, G0W, and GW. This will involve implementing the analytic gradients (cite python functions). I was thinking of doing the self-consistency scheme with BFGS as I did with SRG-MP2. I already have an efficient C++ function for computing the contour deformation self-energy, so maybe this function can be modified a bit to also get the analytic gradients needed to drive the BFGS procedure. It is imperative that with these modifications we don't introduce any significant slowdowns in the self-energy evaluation, i.e. we still shouldn't store the full Choleskies and we should continue to GEMM things efficiently. Although the logic is quite different, model this after what I did with the $srg_mp2_with_gradient$ fn in qcpbc/libpbc/libpbc/ueg/$fock_pw_utils.C$, Where I computed the self-energy and analytic gradientin one fell swoop to be used in tandem in a BFGS optimization procedure as in qcpbc/libpbc/libpbc/ueg/bfgs4selfcons.C.

3. Compute the spectral functions for all of the GW variants like in

```
    gw_spec_cd = gw_spec_fn_fast(omegas, moes, mochol, nocc, 0, p_idx
, eta_sigma=eta, eta_plot=eta, cd=True, e_eff=e_eff, nw=nw_cd)
    print(f"\nComputed CD spectral function for {method_name}+GW; now
    computing other CD flavors...")
    gw0_spec_cd  = gw_spec_fn_fast(omegas, qpes_gw0_nt, mochol, nocc,
    0, p_idx, eta_sigma=eta, eta_plot=eta, cd=True, e_eff=e_eff,
moes_W=moes, nw=nw_cd)
    print(f"\nComputed CD spectral function for {method_name}+GW0;
now computing G0W CD flavor...")
    g0w_spec_cd  = gw_spec_fn_fast(omegas, moes,        mochol, nocc,
    0, p_idx, eta_sigma=eta, eta_plot=eta, cd=True, e_eff=e_eff,
moes_W=qpes_g0w_nt, nw=nw_cd)
    print(f"\nComputed CD spectral function for {method_name}+G0W;
now computing scGW CD flavor...")
    scgw_spec_cd = gw_spec_fn_fast(omegas, qpes_gw_nt,  mochol, nocc,
    0, p_idx, eta_sigma=eta, eta_plot=eta, cd=True, e_eff=e_eff,
moes_W=qpes_gw_nt, nw=nw_cd)
```

```
    print(f"\nComputed CD spectral functions for {method_name}
    reference; now computing cumulant spectral functions...")
```

of $qcpbc/libpbc/gw_tests/gw-ac/tests/compare_methods.py L363$.

4. Compute the Cumulant Spectral Functions as

```
      # get g0w0+c, gw0+c, g0w+c, and gw+c spectral fns
    ef_g0 = 0.5 * (moes[nocc - 1] + moes[nocc])
    omegaspeqp_g0w0 = ce_omegas + qp_cd
    betaspep_g0w0 = (1.0 / np.pi) * np.abs(sigma_cd(omegaspeqp_g0w0,
    ef_g0, moes, moes, mochol, nocc, p_idx, eta, nw=nw_cd, ueg=ueg).
    imag)
    ct, spec_fn_g0w0pc = compute_cumulant_spectral_function_numeric(
    betaspep_g0w0, ce_omegas, times, e_eff, eta,cumulant=True,
    time_ordered=False)
    print(f"  Done G0W0+C; now G0W+C...")

    omegaspeqp_g0w = ce_omegas + qpes_g0w_nt[p_idx]
    betaspep_g0w = (1.0 / np.pi) * np.abs(sigma_cd(omegaspeqp_g0w,
    ef_g0, moes, qpes_g0w_nt, mochol, nocc, p_idx, eta, nw=nw_cd, ueg=
    ueg).imag)
    ct, spec_fn_g0wpc = compute_cumulant_spectral_function_numeric(
    betaspep_g0w, ce_omegas, times, e_eff, eta,cumulant=True,
    time_ordered=False)
    print(f"  Done G0W+C; now GW0+C...")

    ef_gw0 = 0.5 * (qpes_gw0_nt[nocc - 1] + qpes_gw0_nt[nocc])
    omegaspeqp_gw0 = ce_omegas + qpes_gw0_nt[p_idx]
    betaspep_gw0 = (1.0 / np.pi) * np.abs(sigma_cd(omegaspeqp_gw0,
    ef_gw0, qpes_gw0_nt, moes, mochol, nocc, p_idx, eta, nw=nw_cd, ueg
    =ueg).imag)
    ct, spec_fn_gw0pc = compute_cumulant_spectral_function_numeric(
    betaspep_gw0, ce_omegas, times, e_eff, eta,cumulant=True,
    time_ordered=False)
    print(f"  Done GW0+C; now scGW+C...")

    ef_gw = 0.5 * (qpes_gw_nt[nocc - 1] + qpes_gw_nt[nocc])
    omegaspeqp_gw = ce_omegas + qpes_gw_nt[p_idx]
    betaspep_gw = (1.0 / np.pi) * np.abs(sigma_cd(omegaspeqp_gw,
    ef_gw, qpes_gw_nt, qpes_gw_nt, mochol, nocc, p_idx, eta, nw=nw_cd,
     ueg=ueg).imag)
    ct, spec_fn_gwpc = compute_cumulant_spectral_function_numeric(
    betaspep_gw, ce_omegas, times, e_eff, eta,cumulant=True,
    time_ordered=False)
    # print(f"  Done scGW+C; now plotting...")
```

of $qcpbc/libpbc/gw_tests/gw-ac/tests/compare_methods.py L495$. **Note: We also want to experiment with what we get when we change the omegaspep used to build betaspep. Because right now we just have omegaspeqp = ceomegas + qpe, But I also want to try omegaspeqp = ceomegas + mfsp.**

1.6.3 Plasmon pole models (PPMs)

For the homogeneous electron gas, we are trying to solve for

$$\Sigma(\boldsymbol{q},\omega) = \frac{i}{2\pi}\int d\omega' e^{i\eta\omega'} G_0(\boldsymbol{q},\omega+\omega') W_0(\boldsymbol{q},\omega'). \tag{1.235}$$

First, we can figure out how to solve for $W_0(\boldsymbol{q},\omega)$.

$$W_0(\mathbf{q},\omega) = \varepsilon^{-1}(\mathbf{q},\omega)\times v(\mathbf{q}) \tag{1.236}$$

The purpose of the PPM is to avoid solving for the full frequency dependence of the inverse dielectric function. Instead, it postulates that

$$\operatorname{Re}\varepsilon^{-1}(\mathbf{q},\omega) \approx 1 + \frac{\Omega^2(\mathbf{q})}{\omega^2 - \tilde{\omega}^2(\mathbf{q})} \tag{1.237}$$

and

$$\operatorname{Im}\varepsilon^{-1}(\mathbf{q},\omega) = A(\mathbf{q})\left\{\delta\left[\omega - \widetilde{\omega}(\mathbf{q})\right] - \delta\left[\omega + \widetilde{\omega}(\mathbf{q})\right]\right\}. \tag{1.238}$$

For the uniform electron gas we will always have

$$\Omega^2(\mathbf{q}) = \omega_p^2 \frac{\mathbf{G}\cdot\mathbf{G}'}{|\mathbf{G}|^2}\frac{\rho(\mathbf{G}-\mathbf{G}')}{\rho(\mathbf{0})} \tag{1.239}$$

$$= \begin{cases} 4\pi\dfrac{N}{V}, & \text{TDL,} \\ \dfrac{v(\mathbf{q})}{\Omega}\displaystyle\sum_{\substack{i\in\text{occ}\\ a\in\text{vir}\\ \mathbf{k}_a=\mathbf{k}_i+\mathbf{q}}} (\epsilon_a - \epsilon_i), & \text{finite cell.} \end{cases} \tag{1.240}$$

Now we can proceed to define the quantities In specific PPMs.

Hybertsen and Louie parametrization from 1986

Their model obeys the Kramers-Kronig relation, and the f-sum rule. The former states

$$\operatorname{Re}\varepsilon^{-1}(\mathbf{q},\omega) = 1 + \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty}\frac{\operatorname{Im}\varepsilon^{-1}(\mathbf{q},\omega')}{\omega'-\omega} \tag{1.241}$$

where $\mathcal{P}$ denotes the Cauchy principal value. The f-sum rule states

$$\int_{-\infty}^{\infty}\omega\operatorname{Im}\varepsilon^{-1}(\mathbf{q},\omega)d\omega = -\pi\Omega^2(\mathbf{q}). \tag{1.242}$$

This implies that

$$\tilde{\omega}^2(\mathbf{q}) = \frac{\Omega^2(\mathbf{q})}{1-\varepsilon^{-1}(\mathbf{q},\omega=0)} \tag{1.243}$$

$$\tag{1.244}$$

where the RPA form for the dielectric function used is

$$\varepsilon(\mathbf{q},\omega=0) = 1 - v(\mathbf{q})P(\mathbf{q},\omega=0) \tag{1.245}$$
$$\approx 1 - v(\mathbf{q})P_0(\mathbf{q},\omega=0). \tag{1.246}$$

The static, non-interacting polarizability is given by

$$P_0(\mathbf{q},\omega=0) = \sum_{\mathbf{k},\mathbf{k}'} \langle \mathbf{k}|e^{-i(\mathbf{q})\cdot\mathbf{r}}|\mathbf{k}'\rangle\langle\mathbf{k}'|e^{i(\mathbf{q})\cdot\mathbf{r}'}|\mathbf{k}\rangle \frac{f_{\mathbf{k}+\mathbf{q}} - f_{\mathbf{k}}}{\epsilon_{\mathbf{k}+\mathbf{q}} - \epsilon_{\mathbf{k}}} \tag{1.247}$$
$$\overset{\text{UEG}}{=} \frac{1}{V}\sum_{\mathrm{k}} \frac{f_{\mathrm{k}+\mathbf{q}} - f_{\mathrm{k}}}{\epsilon_{\mathrm{k}+\mathbf{q}} - \epsilon_{\mathrm{k}}}, \tag{1.248}$$

where we have a sum over the momentum transfers $\mathbf{k}$, and f is the Fermi-Dirac distribution, which is 1 for occupied states and 0 for unoccupied states at zero temperature. The single particle energies can be due to any mean field calculation. Finally,

$$A(\mathbf{q}) = -\frac{\pi}{2}\frac{\Omega^2(\mathbf{q})}{\tilde{\omega}(\mathbf{q})}. \tag{1.249}$$

Godby and Needs parametrization from 1995

In the GN approach, the Chosen Constraints are such that $\varepsilon_{\boldsymbol{GG'}}(\boldsymbol{q},\omega)$ is correctly reproduced at two different frequencies: the static limit ($\omega = 0$) and an additional imaginary point located at the Sommerfeld plasma frequency $i\omega_p$, where $\omega_p = \sqrt{4\pi\rho}$ with ρ the number of electrons per volume, As previously derived above . After some algebra, the following set of equations defining the plasmon-pole coefficients can be derived:

$$\begin{cases} A_{\boldsymbol{GG'}}(\boldsymbol{q}) = \varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{q},\omega=0) - \delta_{\boldsymbol{GG'}} \\ \tilde{\omega}^2_{\boldsymbol{GG'}} = \omega_p^2\left[\frac{A_{\boldsymbol{GG'}}(\boldsymbol{q})}{\varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{q},\omega=0)-\varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{q},i\omega_p)} - 1\right] \\ \Omega^2_{\boldsymbol{GG'}}(\boldsymbol{q}) = -A_{\boldsymbol{GG'}}(\boldsymbol{q})\tilde{\omega}^2_{\boldsymbol{GG'}}(\boldsymbol{q}) \end{cases}$$

Finite size corrections to the inverse dielectric function

The dielectric function on the real axis is defined as

$$\epsilon_{\mathrm{GG'}}(\mathbf{q},\omega) = \delta_{\mathrm{GG'}} - \frac{\sqrt{4\pi}}{|\mathbf{q}+\mathbf{G}|}\,\chi_{\mathrm{GG'}}(\mathbf{q},\omega)\,\frac{\sqrt{4\pi}}{|\mathbf{q}+\mathbf{G'}|}, \tag{41R}$$

where $\chi_{\mathrm{GG'}}(\mathbf{q},\omega)$ is Retarded polarizability, Kevin as

$$\chi_{\mathrm{GG'}}(\mathbf{q},\omega) = \frac{2}{N_{\mathrm{k}}\,\Omega_{\mathrm{cell}}}\sum_{\mathbf{k}}\sum_{i}^{\mathrm{occ}}\sum_{a}^{\mathrm{vir}} \rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(\mathbf{G}) \times \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a,\,\mathbf{k}-\mathbf{q}}}{(\omega+i\eta)^2 - \left(\epsilon_{i\mathbf{k}} - \epsilon_{a,\,\mathbf{k}-\mathbf{q}}\right)^2}\,\rho^{i\mathbf{k}*}_{a,\,\mathbf{k}-\mathbf{q}}\big(\mathbf{G'}\big). \tag{42R'}$$

The head and wings of the dielectric function in the $\mathbf{q} = \mathbf{0}$ limit are computed as

$$h_{00}(\mathbf{q} = 0, \omega) = 1 - \lim_{\mathbf{q}\to 0} \frac{4\pi}{|\mathbf{q}|^2} \chi_{00}(\mathbf{q} \to 0, \omega), \tag{47}$$

$$w_{P0}(\mathbf{q} = 0, \omega) = \lim_{\mathbf{q}\to 0} \frac{\sqrt{4\pi}}{|\mathbf{q}|} \chi_{P0}(\mathbf{q} \to 0, \omega). \tag{48}$$

where the polarizability is

$$\chi_{00}(\mathbf{q} \to 0, \omega) = \frac{2}{N_\mathrm{k}\,\Omega_\mathrm{cell}} \sum_{\mathbf{k}} \sum_{i}^{\mathrm{occ}} \sum_{a}^{\mathrm{vir}} \rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(0) \tag{49}$$

$$\times \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}}}{\left(\omega + i\eta\right)^2 + \left(\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}}\right)^2} \rho^{i\mathbf{k}*}_{a,\,\mathbf{k}-\mathbf{q}}(0), \tag{1.250}$$

$$\chi_{P0}(\mathbf{q} \to 0, \omega) = \frac{2}{N_\mathrm{k}\,\Omega_\mathrm{cell}^{1/2}} \sum_{\mathbf{k}} \sum_{i}^{\mathrm{occ}} \sum_{a}^{\mathrm{vir}} v_P^{ia} \tag{50}$$

$$\times \frac{\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}}}{\left(\omega + i\eta\right)^2 + \left(\epsilon_{i\mathbf{k}} - \epsilon_{a\mathbf{k}}\right)^2} \rho^{i\mathbf{k}*}_{a,\,\mathbf{k}-\mathbf{q}}(0). \tag{1.251}$$

We can evaluate the limit using $\mathbf{q} = (0.001, 0, 0)$, neglecting anisotropy for now. The pair density matrix in the long-wavelength limit is computed using $\mathbf{k} \cdot \mathbf{p}$ perturbation theory:

$$\rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(0)\big|_{\mathbf{q}\to 0} = \frac{-i\,\mathbf{q} \cdot \left(\psi_{i\mathbf{k}}|\nabla|\psi_{a\mathbf{k}}\right)}{\epsilon_{a\mathbf{k}} - \epsilon_{i\mathbf{k}}}. \tag{51}$$

Note that the head and wing corrections to the inverse dielectric function are not simple inverses of h_{00} and w_{P0}. Instead, they are matrix elements of the inverse of the full dielectric matrix

$$\epsilon_\mathrm{full}(\mathbf{q} = 0, i\omega) = \begin{bmatrix} h_{00} & w_{P0}^\dagger \\ w_{P0} & B_{PQ} \end{bmatrix}, \tag{52}$$

where $B_{PQ} = [\mathbf{I} - \mathbf{\Pi}]_{PQ}$ is the body of the dielectric function In the basis of choice.

Head corrections for the UEG

In the case of the UEG in the long-wavelength limit, the $\mathbf{k} \cdot \mathbf{p}$ perturbation theory can be used to compute the head corrections to the polarizability, but extra considerations have to

be taken. The idea is to do:

$$\rho^{n\mathrm{k}}_{m\mathrm{k-q}}(\mathrm{G}) = \left(n\mathrm{k}\left|e^{i(\mathrm{q+G})\mathrm{r}}\right|m\mathrm{k}-\mathrm{q}\right) \tag{1.252}$$

$$\implies \rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(0)\Big|_{\mathbf{q}\to 0} = \frac{-i\,\mathbf{q}\cdot\left(\psi_{i\mathbf{k}}|\nabla|\psi_{a\mathbf{k}}\right)}{\epsilon_{a\mathbf{k}}-\epsilon_{i\mathbf{k}}} \tag{1.253}$$

$$\implies \rho^{\mathbf{k_F}}_{\mathbf{k_F}-\mathbf{q}}(0)\Big|_{\mathbf{q}\to 0} \approx \frac{1}{\Omega_{\mathrm{cell}}}\frac{-i\,\mathbf{q}\cdot\left(e^{-i\mathbf{k_F}\cdot\mathbf{r}}|\nabla_{\boldsymbol{r}}|e^{i(\mathbf{k_F}-\boldsymbol{q})\cdot\mathbf{r}}\right)}{\epsilon_{\mathbf{k_F}-\mathbf{q}}-\epsilon_{\mathbf{k_F}}} \tag{1.254}$$

$$\approx \frac{2}{\Omega_{\mathrm{cell}}}\frac{-i\,\mathbf{q}\cdot i\,(\mathbf{k_F}-\mathbf{q})\left(e^{-i\mathbf{k_F}\cdot\mathbf{r}}|e^{i\mathbf{k_F}\cdot\mathbf{r}}\right)}{|\mathbf{k_F}-\mathbf{q}|^2-|\mathbf{k_F}|^2} \tag{1.255}$$

$$\approx \frac{1}{\Omega_{\mathrm{cell}}}\frac{\mathbf{q}\cdot(\mathbf{k_F}-\mathbf{q})}{\mathbf{k_F}\cdot\mathbf{q}+\mathbf{q}\cdot\mathbf{q}} \tag{1.256}$$

$$\approx -\frac{1}{\Omega_{\mathrm{cell}}}. \tag{1.257}$$

so then

$$\chi_{00}(\mathbf{q}\to 0,\omega) = \frac{2}{N_{\mathrm{k}}\,\Omega_{\mathrm{cell}}}\sum_{\mathbf{k}}\sum_{i}^{\mathrm{occ}}\sum_{a}^{\mathrm{vir}}\rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(0)\frac{\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}}{\left(\omega+i\eta\right)^2+\left(\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}\right)^2}\rho^{i\mathbf{k}*}_{a,\,\mathbf{k}-\mathbf{q}}(0) \tag{1.258}$$

$$\approx -\frac{4}{\Omega_{\mathrm{cell}}}\frac{1}{\Omega^2_{\mathrm{cell}}}\frac{\boldsymbol{k_F}\cdot\mathbf{q}+\mathbf{q}\cdot\mathbf{q}}{\left(\omega+i\eta\right)^2+\frac{1}{4}\left(\boldsymbol{k_F}\cdot\mathbf{q}+\mathbf{q}\cdot\mathbf{q}\right)^2} \tag{1.259}$$

$$\tag{1.260}$$

Because Tianyu's code seems to be working so well for this, we can just Reuse what I did previously to compute B_{PQ}. By inverting This matrix , one obtains $\epsilon^{-1}_{00}(\mathbf{q}=\mathbf{0},\omega)$ and $\epsilon^{-1}_{P0}(\mathbf{q}=\mathbf{0},\omega)$ as

$$\epsilon^{-1}_{00} = \frac{1}{h_{00}-\sum_{PQ} w^{\dagger}_{P0}\,B^{-1}_{PQ}\,w_{Q0}}, \tag{53}$$

$$\epsilon^{-1}_{P0} = -\,\epsilon^{-1}_{00}\sum_{Q} B^{-1}_{PQ} w_{Q0}. \tag{54}$$

The conclusion is that we can reuse the code we used previously to compute the finite source corrections to the inverse dielectric function.

Getting the head of the polarizability

The Adler-Wiser formula for the polarisability is

$$P_{\mathbf{GG'}}(\mathbf{k},\mathrm{i}\omega) = -\,\frac{4}{(2\pi)^3}\sum_{v,c}\int\,\mathrm{d}^3q\frac{\epsilon_{c\mathbf{q}+\mathbf{k}}-\epsilon_{v\mathbf{q}}}{\left(\epsilon_{c\mathbf{q}+\mathbf{k}}-\epsilon_{v\mathbf{q}}\right)^2+\omega^2} \tag{1.261}$$

$$\times\left\langle\varphi_{v\mathbf{q}}\right|e^{-\mathrm{i}(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}\left|\varphi_{c\mathbf{q}+\mathbf{k}}\right\rangle\left\langle\varphi_{c\mathbf{q}+\mathbf{k}}\right|e^{\mathrm{i}(\mathbf{k}+\mathbf{G'})\cdot\mathbf{r}}\left|\varphi_{v\mathbf{q}}\right\rangle, \tag{1.262}$$

where the sum over v and c runs over occupied and unoccupied states, respectively. For $\mathrm{G}' = 0$ and $\mathrm{k} \to 0$ this leads to a wings correction of

$$P_{\mathbf{G0}}(\mathbf{k}, \mathrm{i}\omega) \to -\frac{4\mathrm{i}}{(2\pi)^3} \sum_{v,c} \int \mathrm{d}^3 q \frac{\epsilon_{c\mathbf{q}} - \epsilon_{v\mathbf{q}}}{\left(\epsilon_{c\mathbf{q}} - \epsilon_{v\mathbf{q}}\right)^2 + \omega^2} \tag{1.263}$$

$$\times \langle \varphi_{v\mathbf{q}} | \, e^{-\mathrm{i}\mathbf{G}\cdot\mathbf{r}} \, | \varphi_{c\mathbf{q}} \rangle \left(\mathbf{k} \cdot \langle \varphi_{c\mathbf{q}} | \, \mathbf{r} \, | \varphi_{v\mathbf{q}} \rangle \right) \tag{1.264}$$

while for $\mathbf{G} = \mathbf{G}' = \mathbf{0}$ and $\mathbf{k} \to \mathbf{0}$, we get the head correction as

$$P_{\mathbf{00}}(\mathrm{k}, \mathrm{i}\omega) \to -\frac{4}{(2\pi)^3} \sum_{v,c} \int \mathrm{d}^3 q \frac{\epsilon_{c\mathbf{q}} - \epsilon_{v\mathbf{q}}}{\left(\epsilon_{c\mathbf{q}} - \epsilon_{v\mathbf{q}}\right)^2 + \omega^2} \tag{1.265}$$

$$\times \left(\mathrm{k} \cdot \langle \varphi_{v\mathbf{q}} | \, \mathrm{r} \, | \varphi_{c\mathbf{q}} \rangle \right) \left(\mathrm{k} \cdot \langle \varphi_{c\mathbf{q}} | \, \mathrm{r} \, | \varphi_{v\mathbf{q}} \rangle \right) \tag{1.266}$$

The matrix elements of $\mathbf{r}$ can be calculated via the commutator of $\mathbf{r}$ with the Kohn-Sham Hamiltonian h^{KS} as

$$\langle \varphi_{cq} | \, \mathbf{r} \, | \varphi_{vq} \rangle = \frac{\langle \varphi_{cq} | \left[h^{\mathrm{KS}}, \mathbf{r} \right] | \varphi_{vq} \rangle}{\epsilon_{cq} - \epsilon_{vq}} \tag{1.267}$$

$$\overset{\mathrm{UEG}}{=} -\frac{\langle \varphi_{cq} | \, \nabla \, | \varphi_{vq} \rangle}{\epsilon_{cq} - \epsilon_{vq}} \tag{1.268}$$

$$= 0. \tag{1.269}$$

By a similar reason, the wings correction will vanish as well for the UEG.

Finite size corrections to the correlation self-energy

In a plane-wave expansion, the correlation self-energy can be written as

$$\mathbf{\Sigma}^c_{nn'}(\mathbf{k}, i\omega) = -\frac{1}{2\pi N_{\mathbf{k}} \Omega_{\mathrm{cell}}} \sum_{mq} \sum_{\mathbf{GC}'} \int_{-\infty}^{\infty} d\omega' W_{\mathbf{GG}'}(\mathbf{q}, i\omega') \tag{1.270}$$

$$\times \frac{\rho^{n\mathbf{k}}_{m\mathrm{k}-\mathbf{q}}(\mathbf{G}) \rho^{n'\mathbf{k}*}_{m\mathrm{k}-\mathbf{q}}(\mathbf{G}')}{i\left(\omega + \omega'\right) + \epsilon_F - \epsilon_{m\mathrm{k}-\mathrm{q}}}, \tag{1.271}$$

where Ω_{cell} is the unit cell volume. The pair density nm is now expanded in plane waves with the matrix elements

$$\rho^{nk}_{mk-q}(\mathbf{G}) = \left(n\mathbf{k} \left| e^{i(\mathbf{q}+\mathbf{G})\mathbf{r}} \right| m\mathbf{k} - \mathbf{q} \right). \tag{1.272}$$

Tianyu then *derives* the head correction to the correlation self-energy to be purely diagonal, given as

$$\mathbf{\Sigma}^{\mathrm{head}}_{nn}(\mathbf{k}, i\omega) = -\frac{1}{\pi^2} \left(\frac{6\pi^2}{\Omega_{\mathrm{cell}} N_{\mathbf{k}}} \right)^{1/3} \int_{-\infty}^{\infty} d\omega' \frac{\epsilon^{-1}_{00}(\mathbf{q} = \mathbf{0}, i\omega) - 1}{i\left(\omega + \omega'\right) + \epsilon_F - \epsilon_{n\mathbf{k}}}. \tag{1.273}$$

Then the wings contribution to the correlation self-energy is given as

$$\mathbf{\Sigma}^{\mathrm{wings}}_{nn'}(\mathbf{k}, i\omega) = -\frac{1}{\pi} \sqrt{\frac{\Omega_{\mathrm{cell}}}{4\pi^3}} \left(\frac{6\pi^2}{\Omega_{\mathrm{cell}} N_{\mathbf{k}}} \right)^{2/3} \sum_P \int_{-\infty}^{\infty} d\omega' \frac{\mathrm{Re}\left[v^{nn'}_P \epsilon^{-1}_{P0}(\mathbf{q} = \mathbf{0}, i\omega) \right]}{i\left(\omega + \omega'\right) + \epsilon_F - \epsilon_{n'\mathbf{k}}}. \tag{1.274}$$

In order to obtain the head contribution to the inverse dielectric matrix that we need, we must first invert the full dielectric matrix, which is given by

$$\epsilon_{\text{full}}(\mathbf{q}=0,i\omega)=\begin{bmatrix} h_{00} & w_{P0}^{\dagger} \\ w_{P0} & B_{PQ} \end{bmatrix}, \tag{1.275}$$

where $B_{PQ}=[\mathbf{I}-\mathbf{\Pi}]_{PQ}$ is the body of the dielectric function (this can also be obtained from the Casida diagonalization, as we have done previously). The head and wings of the dielectric function in the $\mathbf{q}=\mathbf{0}$ limit are computed as

$$h_{00}(\mathbf{q}=0,\omega)=1-\lim_{\mathbf{q}\to 0}\frac{4\pi}{|\mathbf{q}|^2}\,\chi_{00}(\mathbf{q}\to 0,\omega), \tag{47}$$

$$w_{P0}(\mathbf{q}=0,\omega)=\lim_{\mathbf{q}\to 0}\frac{\sqrt{4\pi}}{|\mathbf{q}|}\,\chi_{P0}(\mathbf{q}\to 0,\omega). \tag{48}$$

where the polarizability is

$$\chi_{00}(\mathbf{q}\to 0,\omega)=\frac{2}{N_{\text{k}}\,\Omega_{\text{cell}}}\sum_{\mathbf{k}}\sum_{i}^{\text{occ}}\sum_{a}^{\text{vir}}\rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(0) \tag{49}$$

$$\times\frac{\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}}{(\omega+i\eta)^2+\left(\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}\right)^2}\rho^{i\mathbf{k}*}_{a,\,\mathbf{k}-\mathbf{q}}(0), \tag{1.276}$$

$$\chi_{P0}(\mathbf{q}\to 0,\omega)=\frac{2}{N_{\text{k}}\,\Omega_{\text{cell}}^{1/2}}\sum_{\mathbf{k}}\sum_{i}^{\text{occ}}\sum_{a}^{\text{vir}}v_P^{ia} \tag{50}$$

$$\times\frac{\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}}{(\omega+i\eta)^2+\left(\epsilon_{i\mathbf{k}}-\epsilon_{a\mathbf{k}}\right)^2}\rho^{i\mathbf{k}*}_{a,\,\mathbf{k}-\mathbf{q}}(0). \tag{1.277}$$

He chooses to evaluate the limit using $\mathbf{q}=(0.001,0,0)$, *which neglects anisotropy.* In the long-wavelength limit, he computes the pair density matrix using $\mathbf{k}\cdot\mathbf{p}$ perturbation theory:

$$\rho^{i\mathbf{k}}_{a,\,\mathbf{k}-\mathbf{q}}(0)\big|_{\mathbf{q}\to 0}=\frac{-i\,\mathbf{q}\cdot\left(\psi_{i\mathbf{k}}|\nabla|\psi_{a\mathbf{k}}\right)}{\epsilon_{a\mathbf{k}}-\epsilon_{i\mathbf{k}}} \tag{1.278}$$

$$\overset{\text{UEG}}{=}\frac{1}{\Omega_{\text{cell}}}\frac{-i\,\mathbf{q}\cdot\left(e^{-i\mathbf{G}_i\cdot\mathbf{r}}|\nabla_{\boldsymbol{r}}|e^{i(\mathbf{G}_a)\cdot\mathbf{r}}\right)}{\epsilon_{\mathbf{G}_a}-\epsilon_{\mathbf{G}_i}} \tag{1.279}$$

$$\overset{\text{UEG}}{=}\frac{2}{\Omega_{\text{cell}}}\frac{-i\,\mathbf{q}\cdot i\,(\mathbf{G}_a)\,\delta_{\mathbf{G}_a,\mathbf{G}_i}}{|\mathbf{G}_a|^2-|\mathbf{G}_i|^2} \tag{1.280}$$

$$=0. \tag{1.281}$$

So, following this logic tells that the head and wing corrections should vanish for the homogeneous electron gas.

Final form for the self-energy

We have seen that, given some choice of basis, the retarded GW self-energy can be written as

$$\Sigma^{\text{corr}}_{pp}(\omega)=\sum_{\mu}^{\text{RPA}}\left(\sum_{i}^{\text{occupied}}\frac{M^{\mu}_{pi}M^{\mu}_{ip}}{\omega-(\epsilon_i-\Omega_\mu)+i\eta}+\sum_{a}^{\text{virtual}}\frac{M^{\mu}_{pa}M^{\mu}_{ap}}{\omega-(\epsilon_a+\Omega_\mu)+i\eta}\right) \tag{1.282}$$

We can rewrite this In the Plain Wake Bases as

$$\Sigma^c(\mathbf{k},\omega) = \sum_{\mathbf{q}} \sum_{\mu}^{\text{RPA}} \left(f_{\mathbf{k}-\mathbf{q}} \frac{\left|M^{\mu}_{\mathbf{k},\mathbf{k}-\mathbf{q}}(\mathbf{q})\right|^2}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} + \Omega_{\mu}(\mathbf{q}) + i\eta} + [1 - f_{\mathbf{k}-\mathbf{q}}] \frac{\left|M^{\mu}_{\mathbf{k},\mathbf{k}-\mathbf{q}}(\mathbf{q})\right|^2}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \Omega_{\mu}(\mathbf{q}) + i\eta} \right), \tag{1.283}$$

PPMs choose To approximate this expression as

$$\Sigma^c(\mathbf{k},\omega) = \sum_{\mathbf{q}} v(q)A(q) \left[\frac{f_{\mathbf{k}-\mathbf{q}}}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} + \tilde{\omega}(q) + i\eta} + \frac{1 - f_{\mathbf{k}-\mathbf{q}}}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \tilde{\omega}(q) + i\eta} \right]. \tag{1.284}$$

This expression can be derived from PPMs postulation that The inverse dielectric function can be determined as

$$\operatorname{Re} \varepsilon^{-1}(\mathbf{q},\omega) \approx 1 + \frac{\Omega^2(\mathbf{q})}{\omega^2 - \tilde{\omega}^2(\mathbf{q})} \tag{1.285}$$

and

$$\operatorname{Im} \varepsilon^{-1}(\mathbf{q},\omega) = A(\mathbf{q}) \left\{ \delta\left[\omega - \widetilde{\omega}(\mathbf{q})\right] - \delta\left[\omega + \widetilde{\omega}(\mathbf{q})\right] \right\}. \tag{1.286}$$

These quantities can be defined up to some parameterization. Two popular ones have been introduced by Hybertsen and Louie in 1986, and Godby and Needs in 1995. HL postulate

$$\begin{cases} \Omega^2_{GG'}(q) = \omega_p^2 \frac{(q+G)\cdot(q+G')}{|q+G|^2} \frac{n(G-G')}{n(0)} \overset{UEG}{=} \omega_p^2 = 4\pi n \\ \tilde{\omega}^2_{GG'}(q) = \frac{\Omega^2_{GG'}(q)}{\delta_{GG'} - \varepsilon^{-1}_{GG'}(q,\omega=0)} \overset{UEG}{=} \frac{4\pi n}{1-\varepsilon^{-1}(q,\omega=0)} \\ A_{GG'}(q) = -\frac{\pi}{2} \frac{\Omega_{GG'}(q)}{\tilde{\omega}_{GG'}(q)}. \end{cases} \tag{1.287}$$

while GN postulate

$$\begin{cases} A_{\boldsymbol{GG'}}(\boldsymbol{q}) = \varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{q},\omega=0) - \delta_{\boldsymbol{GG'}} \\ \tilde{\omega}^2_{\boldsymbol{GG'}} = \omega_p^2 \left[\frac{A_{\boldsymbol{GG'}}(\boldsymbol{q})}{\varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{q},\omega=0) - \varepsilon^{-1}_{\boldsymbol{GG'}}(\boldsymbol{q},i\omega_p)} - 1 \right] \\ \Omega^2_{\boldsymbol{GG'}}(\boldsymbol{q}) = -A_{\boldsymbol{GG'}}(\boldsymbol{q}) \tilde{\omega}^2_{\boldsymbol{GG'}}(\boldsymbol{q}) \end{cases} \tag{1.288}$$

Note that in the uniform electron gas, we actually have G = G'. Clearly, these parametrizations depend on a description of the inverse dielectric function at $\mathbf{q}$. It has been shown that when using a Gaussian basis The dielectric function is not just a scalar quantity, but actually a matrix. In particular, we have

$$\boldsymbol{\epsilon}(\mathbf{q},\omega) = \begin{pmatrix} h_{00} & w^{\dagger}_{P0} \\ w_{P0} & B_{PQ} \end{pmatrix} \tag{1.289}$$

where $B_{PQ} = [\mathrm{I} - \Pi]_{PQ}$ is the body of the dielectric function with head and wing correction h_{00} and w_{P0}. For homogeneous systems, we can take the wing corrections to be zero, and thus

$$\boldsymbol{\epsilon}^{-1}(\mathbf{q},\omega) = \begin{pmatrix} 1/h_{00} & 0 \\ 0 & 1/B_{PQ} \end{pmatrix} \tag{1.290}$$

The dimension of the above matrix is $N_{\text{aux}} + 1$, where N_{aux} is the number of auxiliary basis functions, when $\mathbf{G} = 0$ is excluded. It happens to be the case that in the finite plane wave basis, the uniform electron gas Admits N_{aux} or $N_{\text{aux}} + 1$ Momentum transfer vectors, depending on whether the Divergent component is included or not. Recall that diagonalizing the Casada equation gave us one dominant plus one excitation at each momentum transfer vector.

Debarking steps

1. $\chi_{\text{RPA}}(\mathbf{q},\omega)$ should be identical regardless of whether we build it As a sum over poles via the Casida equation or via $\chi_{\text{RPA}} = \dfrac{\chi_0}{1 - v\chi_0}$, where χ_0 is the non-interacting polarizability. This has been done analytically and now numerically .
2. The PPM forms for $\boldsymbol{\epsilon}^{-1}(\mathbf{q},\omega)$ should be a good match with $1 - v\chi_{\text{RPA}}(\mathbf{q},\omega)$.
3. Garnet's spectral functions should be reproduced. Should we be able to do this with and without finite size corrections?

Therefore, at a given momentum transfer vector, we should be able to directly compare the PPM form for the inverse dielectric function with the one obtained from diagonalizing the Casida equation. Also, we can do a comparison of the self energies. Carrying on with this derivation leads to

$$\Sigma^c(\mathbf{k},\omega) = \sum_{\mathbf{q}} v(q)A(q)\left[\frac{1 - f_{\mathbf{k}-\mathbf{q}}}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \tilde{\omega}(q) + i\eta} + \frac{f_{\mathbf{k}-\mathbf{q}}}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} + \tilde{\omega}(q) + i\eta}\right]. \tag{1.291}$$

To get to the exact form for the GW soft energy of this form, consider that we can write a Casida-equation for each momentum transfer $\mathbf{q}$, which is given by

$$\begin{pmatrix} \boldsymbol{A}(\mathbf{q}) & \boldsymbol{B}(\mathbf{q}) \\ -\boldsymbol{B}(\mathbf{q}) & -\boldsymbol{A}(\mathbf{q}) \end{pmatrix} \begin{pmatrix} \boldsymbol{X}(\mathbf{q}) \\ \boldsymbol{Y}(\mathbf{q}) \end{pmatrix} = \boldsymbol{\Omega}(\mathbf{q}) \begin{pmatrix} \boldsymbol{X}(\mathbf{q}) \\ \boldsymbol{Y}(\mathbf{q}) \end{pmatrix} \tag{1.292}$$

where The matrix elements are defined as $A_{ia,jb}(\mathbf{q}) = \delta_{ij}\delta_{ab}(\varepsilon_a - \varepsilon_i) + 2(ia|bj)$ and $B_{ia,jb}(\mathbf{q}) = 2(ia|jb)$ With the requirement that there be a momentum transfer of $\mathbf{q}$ to get from the occupied plane with State i to the unoccupied plane with state a. At the end of this step, we are left with RPA Excitation Energies $\Omega_\nu(\mathbf{q})$ and RPA eigenvectors $X^\nu_{ia}(\mathbf{q})$ and $Y^\nu_{ia}(\mathbf{q})$ For the complete set $\{\mathbf{q},\nu\}$. At this point, we can arrive at the usual GW form, which is given by

$$\Sigma^c(\mathbf{k},\omega) = \sum_{\mathbf{q}}\sum_{\mu}^{\text{RPA}}\left(f_{\mathbf{k}-\mathbf{q}}\frac{\left|M^\mu_{\mathbf{k},\mathbf{k}-\mathbf{q}}(\mathbf{q})\right|^2}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} + \Omega_\mu(\mathbf{q}) - i\eta} + [1 - f_{\mathbf{k}-\mathbf{q}}]\frac{\left|M^\mu_{\mathbf{k},\mathbf{k}-\mathbf{q}}(\mathbf{q})\right|^2}{\omega - \varepsilon_{\mathbf{k}-\mathbf{q}} - \Omega_\mu(\mathbf{q}) + i\eta}\right), \tag{1.293}$$

Where the transition densities are the $M^\mu_{\mathbf{k},\mathbf{k}-\mathbf{q}}(\mathbf{q})$. There is an exact correspondence between the wave vector $\mathbf{k}$ and an MO index.

1.6.4 Showing that the dimension of the dielectric matrix is the number of momentum transfer vectors

The two-body contribution to the ab initio Hamiltonian, which is due to the electron-electron interactions, can be written in the standard Ewald summation as

$$\hat{H}^{(2)} = \frac{1}{2\Omega}\sum_{\mathbf{Q}\neq 0}\frac{4\pi}{Q^2}\hat{\rho}^\dagger(\mathbf{Q})\hat{\rho}(\mathbf{Q}) \tag{1.294}$$

where $\hat{\rho}(\mathbf{Q})$ is the one-body density operator given by

$$\hat{\rho}(\mathbf{Q}) = \sum_{G,\lambda} c^{\dagger}_{\mathrm{G+Q},\lambda} c_{\mathrm{G},\lambda}, \theta\left(E_{\mathrm{cut}} - |\mathbf{G}+\mathbf{Q}|^2/2\right), \lambda \in \alpha, \beta. \tag{1.295}$$

This is why we determine the momentum transfer vectors $\mathbf{Q}$ in a finite basis under the requirement that $(\boldsymbol{G}+\mathbf{Q})$ lies within the planewave basis (with an energy less than E_{cut}), where $\boldsymbol{G}$ is already a member of the planewave basis. From the form of equation 1.294, we can see that a full Coloumb kernel gets shared between two density operators. This is the same idea of the resolution of the identity approach, where the full Coulomb kernel is decomposed into two objects as $v_{pqrs} = \sum_T M^T_{pq} M^T_{rs}$. Therefore, we identify that the number of momentum transfer vectors $\mathbf{Q}$ is the same as the number of auxiliary basis functions in the finite planewave basis. Thus, in the following, we will the momentum transfer vectors $\boldsymbol{Q}$ and the auxiliary basis functions P, T interchangeably.

What this means for the dimension of the dielectric matrix

It is illustrative to consider a concrete example. On the imaginary axis, the GW self-energy is computed at the GPT, in the absence of finite size corrections, as

$$\Sigma^{\mathrm{c}}_n(i\omega) = -\frac{1}{2\pi}\sum_m \int_{-\infty}^{\infty} d\omega' \frac{1}{i\left(\omega-\omega'\right)+\varepsilon_{\mathrm{F}}-\varepsilon_m} \sum_{PT} A^{nm}_P \left[\left[1-\Pi\left(i\omega'\right)\right]^{-1}_{PT} - \delta_{PT}\right] A^{mn}_T, \tag{1.296}$$

so we can identify the dielectric matrix as $\tilde{\boldsymbol{B}}(i\omega) \equiv \boldsymbol{\epsilon}(i\omega) = \mathbf{1} - \boldsymbol{\Pi}(i\omega)$ where $\Pi_{PT}(i\omega) = 2\sum_{ia} B^{ia}_P \frac{\varepsilon_i-\varepsilon_a}{\omega^2+(\varepsilon_i-\varepsilon_a)^2} B^{ia}_T$ with $B^{ia}_P = \sum_{TR}(iaT) S^{-1}_{TR} L_{RP}$, where $\mathbf{L}$ is the Cholesky decomposition of the Coulomb interaction matrix in the RI basis. Now, the finite size corrected dielectric function can be written as

$$\boldsymbol{\epsilon}(\omega) = \begin{bmatrix} h_{00}(\omega) & w^{\dagger}_{P0}(\omega) \\ w_{P0}(\omega) & \tilde{B}_{PT}(\omega) \end{bmatrix} \tag{1.297}$$

$$\overset{\mathrm{UEG}}{=} \begin{bmatrix} h_{00}(\omega) & 0 \\ 0 & \tilde{B}_{PP}(\omega) \end{bmatrix} \tag{1.298}$$

$$\implies \boldsymbol{\epsilon}^{-1}(\omega) \overset{\mathrm{UEG}}{=} \begin{bmatrix} 1/h_{00}(\omega) & 0 \\ 0 & 1/\tilde{B}_{PP}(\omega) \end{bmatrix} \tag{1.299}$$

with head, wing, and body elements $h_{00}(\omega)$, $w_{P0}(\omega)$, and $\tilde{B}_{PT}(\omega)$.

> I have verified numerically that for the UEG the wing corrections are zero. Furthermore, I have verified numerically that for the UEG the body element is diagonal, and so it is appropriate to take the direct inverse.

In the absence of finite size corrections, this reduces to

$$\boldsymbol{\epsilon}^{-1}(\omega) = \left[1/\tilde{B}_{PP}(\omega)\right] \tag{1.300}$$

I have verified numerically that $\epsilon^{-1}(P,\omega)$ from eqn. (1.300) is equivalent to

$$\epsilon^{-1}(\boldsymbol{Q},\omega) = 1 + v(\boldsymbol{Q})\chi_{RPA}(\boldsymbol{Q},\omega), \tag{1.301}$$

where the RPA polarizability is $\chi_{RPA}(\boldsymbol{Q},\omega) = \dfrac{\chi_0(\omega)}{1 - v(\boldsymbol{Q})\chi_0(\omega)}$, with $\chi_0(\omega)$ as the non-interacting polarizability.

Therefore, I have verified that in the UEG the momentum transfer vectors $\boldsymbol{Q}$ are in one-to-one correspondence with the auxiliary basis functions P.

1.6.5 Evaluating a dynamical COHSEX self-energy for GW with PPMs

We know that $\mathrm{Re}\Sigma = \Sigma_{\mathrm{SEX}} + \Sigma_{\mathrm{COH}}$. Assuming that $\boldsymbol{G}$ and $\boldsymbol{G_n}$ represent valid plane wave indices with the indices $\boldsymbol{Q_1}, \boldsymbol{Q_2}$ running over all possible momentum transfers, we can write

$$\Sigma^{\mathrm{SEX}}_{GG}(\omega) = -\frac{1}{\Omega}\sum_{G_n}^{\mathrm{occ}}\sum_{Q_1Q_2}\psi_{G_n}(G-Q_1)\psi^*_{G_n}(G-Q_2)\left(1 + \frac{\Omega^2_{Q_1Q_2}}{(\omega-\varepsilon_{G_n})^2 - \tilde{\omega}^2_{Q_1Q_2}}\right)v\left(Q_2\right) \tag{1.302}$$

$$= -\frac{1}{\Omega}\sum_{G_n}^{\mathrm{occ}}\sum_{Q_1Q_2}\delta_{G_n,G-Q_1}\delta_{G_n,G-Q_2}\left(1 + \frac{\Omega^2_{Q_1Q_2}}{(\omega-\varepsilon_{G_n})^2 - \tilde{\omega}^2_{Q_1Q_2}}\right)v\left(Q_2\right) \tag{1.303}$$

$$= -\frac{1}{\Omega}\sum_{G_n}^{\mathrm{occ}}\left(1 + \frac{\Omega^2_{G-G_n,G-G_n}}{(\omega-\varepsilon_{G_n})^2 - \tilde{\omega}^2_{G-G_n,G-G_n}}\right)v\left(G-G_n\right) \tag{1.304}$$

since the delta functions require that $G - Q_1 = G_n = G - Q_2 \implies Q_1 = Q_2 = G - G_n$. Next,

$$\Sigma^{\mathrm{COH}}_{GG}(\omega) = \frac{1}{\Omega}\sum_{G_n}\sum_{Q_1Q_2}\psi_{G_n}(G-Q_1)\psi^*_{G_n}(G-Q_2)\frac{1}{2}\frac{\Omega^2_{Q_1Q_2}}{\tilde{\omega}_{Q_1Q_2}\left[\omega - \varepsilon_{G_n} - \tilde{\omega}_{Q_1Q_2}\right]}v\left(Q_2\right) \tag{1.305}$$

$$= \frac{1}{2\Omega}\sum_{G_n}\frac{\Omega^2_{G-G_n,G-G_n}}{\tilde{\omega}_{G-G_n,G-G_n}\left[\omega - \varepsilon_{G_n} - \tilde{\omega}_{G-G_n,G-G_n}\right]}v\left(G-G_n\right) \tag{1.306}$$

So the procedure will be to generate a set of $\Omega_{Q_1Q_1}$ and $\tilde{\omega}_{Q_1Q_1}$ for each momentum transfer vector Q_1 via the PPM and then to insert these into the above expressions for the self-energy.

1.6.6 Spectral functions with PPMs

All of my cumulant spectral functions are solving for the retarded Green's function as

$$G(\mathbf{k},t) = -i\theta(t)e^{-i\varepsilon^{\mathrm{eff}}_{\boldsymbol{k}}t}e^{C(\mathbf{k},t)}, \tag{1.307}$$

where $\varepsilon^{\mathrm{eff}}_{\boldsymbol{k}} = \varepsilon^{\mathrm{LDA}}_{\boldsymbol{k}} + \Sigma^x(\mathbf{k}) - V_{xc}(\mathbf{k})$. $\Sigma^x(\mathbf{k})$ is the HF exchange potential and $V_{xc}(\mathbf{k})$ is the LDA exchange-correlation potential, both evaluated in the LDA MO basis. Note that the

maximum off-diagonal component of these potentials is 1e-7 Ha, so it is appropriate to just consider the diagonal component for each MO. The cumulant function $C(\mathbf{k}, t)$ is given by

$$C(\mathbf{k}, t) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\omega \frac{|\operatorname{Im} \Sigma_R^c(\mathbf{k}, \omega + \varepsilon_{\boldsymbol{k}}^{\mathrm{GW}})|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1 \right), \tag{1.308}$$

were the manual broadening of η has been applied in the computation of $\operatorname{Im} \Sigma_R^c(\mathbf{k}, \omega)$. When I say iterative in my plot, I mean that I am solving for $\varepsilon_{\boldsymbol{k}}^{\mathrm{GW}}$ self-consistently via the equation

$$\varepsilon_{\boldsymbol{k}}^{\mathrm{GW}} = \varepsilon_{\boldsymbol{k}}^{\mathrm{eff}} + \operatorname{Re} \Sigma^c(\mathbf{k}, \varepsilon_{\boldsymbol{k}}^{\mathrm{GW}}). \tag{1.309}$$

When I say linearized, this means that I am doing

$$\varepsilon_{\boldsymbol{k}}^{\mathrm{GW}} = \varepsilon_{\boldsymbol{k}}^{\mathrm{LDA}} + Z \times \left[\Sigma^x(\mathbf{k}) - V_{xc}(\mathbf{k}) + \operatorname{Re} \Sigma^c(\mathbf{k}, \varepsilon_{\boldsymbol{k}}^{\mathrm{LDA}}) \right], \tag{1.310}$$

where

$$Z = \left[1 - \left. \frac{\partial \operatorname{Re} \Sigma_R^c(\mathbf{k}, \omega)}{\partial \omega} \right|_{\omega = \varepsilon_{\boldsymbol{k}}^{\mathrm{LDA}}} \right]^{-1}. \tag{1.311}$$

Also, I have verified that the HL-PPM result frequency integration scheme gives the same answer as the exact and AC schemes, for both LDA+GW and LDA+GW+C. This is possible for this system because the inverse dielectric can be well approximated by a single pole model, as exemplified by the match between the HL-PPM and exact (due to solving the Casida equation) answers for some low-energy momentum transfer vector.

1.7 Attempting to reproduce the cumulant spectral function from Garnet's paper

I am trying to reproduce The reference LDA+GW+C curve from Figure 1A in McClain's paper. I have manually extracted the data, and its features are

```
For LDA+GW+C (ref):  QP: omega = -5.612 eV, A = 0.5149 eV^{-1}
For LDA+GW+C (ref):  sat 1: omega = -7.849 eV, A = 0.3140 eV^{-1}
For LDA+GW+C (ref):  sat 2: omega = -10.184 eV, A = 0.1303 eV^{-1}
For LDA+GW+C (ref):  sat 3: omega = -12.421 eV, A = 0.0345 eV^{-1}
```

I have attempted to implement LDA + GW + C in the retarded cumulant formulation, but I get

```
Peaks for LDA+GW+C retarded [delta_omega=0.001, omega_int=1, tol=3e-06, skip=0]:
Peak 0: omega = -5.637 eV, A = 0.2119 eV^{-1}, LHWHM = 0.6731, RHWHM = 0.4438, FWHM = 1.
Peak 1: omega = -7.870 eV, A = 0.2069 eV^{-1}, LHWHM = 0.9651, RHWHM = 0.8672, FWHM = 1.
Peak 2: omega = -10.035 eV, A = 0.1166 eV^{-1}, LHWHM = 1.1919, RHWHM = 5.0931, FWHM = 6
```

with

```
def analyze_spectral_peaks(omegas, spectral, min_height_frac=0.1, min_distance_ev=0.
"""Per-peak geometry: center, height, left/right HWHM, and asymmetry index.

Asymmetry index = (RHWHM - LHWHM) / (RHWHM + LHWHM).
  > 0 : peak is wider on the high-frequency side (right-skewed)
  < 0 : peak is wider on the low-frequency side (left-skewed)
  = 0 : symmetric

Returns a list of dicts, one per peak, sorted high → low frequency.
Each dict has keys:
  omega_ev, height_ev, lhwhm_ev, rhwhm_ev, asym, fwhm_ev
"""
from scipy.signal import find_peaks

omegas = np.asarray(omegas, dtype=float)
spectral = np.asarray(spectral, dtype=float)
spec_ev = spectral / H2EV
d_omega_ev = np.abs(omegas[1] - omegas[0]) * H2EV
min_distance_pts = max(1, int(min_distance_ev / d_omega_ev))

global_max = spec_ev.max()
if global_max <= 0:
    return []

indices, _ = find_peaks(
    spec_ev,
    height=min_height_frac * global_max,
    distance=min_distance_pts,
)
order = np.argsort(omegas[indices])[::-1]
indices = indices[order]

results = []
for idx in indices:
    height = spec_ev[idx]
    half = height / 2.0

    # left HWHM: walk left until we drop below half-max
    li = idx
    while li > 0 and spec_ev[li] > half:
        li -= 1
    if spec_ev[li + 1] - spec_ev[li] != 0:
        frac = (half - spec_ev[li]) / (spec_ev[li + 1] - spec_ev[li])
        left_omega = omegas[li] + frac * (omegas[li + 1] - omegas[li])
    else:
```

```
            left_omega = omegas[li]

        # right HWHM: walk right
        ri = idx
        while ri < len(spec_ev) - 1 and spec_ev[ri] > half:
            ri += 1
        if spec_ev[ri] - spec_ev[ri - 1] != 0:
            frac = (half - spec_ev[ri - 1]) / (spec_ev[ri] - spec_ev[ri - 1])
            right_omega = omegas[ri - 1] + frac * (omegas[ri] - omegas[ri - 1])
        else:
            right_omega = omegas[ri]

        lhwhm = (omegas[idx] - left_omega) * H2EV
        rhwhm = (right_omega - omegas[idx]) * H2EV
        fwhm = lhwhm + rhwhm
        asym = (rhwhm - lhwhm) / fwhm if fwhm > 0 else 0.0

        results.append(dict(
            omega_ev=omegas[idx] * H2EV,
            height_ev=height,
            lhwhm_ev=lhwhm,
            rhwhm_ev=rhwhm,
            fwhm_ev=fwhm,
            asym=asym,
        ))
    return results
```

So, although the peak positions may seem reasonable, the peak heights are way off. For a few days now, I have been trying to hack at this retarded cumulant formulation, trying to change some parameters, but no success so far. A closer read of McClain's paper suggests that they might be using the time-ordered cumulant formulation to compute the photo emission spectrum for this core state. In particular, they write that they obtain the Green's function via

$$G_k(t) = G_{0,k}(t) \exp\left[-i\Sigma_k^{\mathrm{x}} t + C_k(t)\right] \tag{1.312}$$

where G_0 is the Green's function from mean-field theory and $\Sigma_{\boldsymbol{k}}^{\mathrm{X}}$ is the bare exchange self-energy. Because we Are trying to obtain the spectral function for the deep core state at $k = [0, 0, 0]$ inf fig. 1a we only want to obtain the Green's function for occupied states, my interpretation is that they mean

$$\left\{ G_{0,k}^{\mathrm{TO}}(t) = i\Theta(-t) e^{-i\varepsilon_k^0 t} \right. \tag{1.313}$$

where $\varepsilon_k^0 = \varepsilon_k^{\mathrm{LDA}} - V_{xc}(\mathbf{k}) = \frac{k^2}{2}$. Note that in my code I set the pole to be $\varepsilon_k^{\mathrm{eff}} = \varepsilon_k^{\mathrm{LDA}} + \Sigma_k^x - V_{xc}(\mathbf{k})$ as

```
Vx_mo = C_lda.T @ lda_vx_ao_corr @ C_lda
```

```
Vc_mo = C_lda.T @ lda_vc_ao_corr @ C_lda

dm_lda_alpha = C_lda[:, :NOCC] @ C_lda[:, :NOCC].T
K_chol = d['cholK'].transpose(1,2,0)
K_hf_ao = -np.einsum('prX,qsX,rs->pq', K_chol, K_chol, dm_lda_alpha, optimize=True)
K_hf_mo = C_lda.T @ K_hf_ao @ C_lda
# check if K_hf_mo and Vx_mo are Hermitian

lda_sps_eff = lda_sps + np.diag(K_hf_mo - Vx_mo - Vc_mo)
```

Note that I Match the GW spectral functions Of the reference via

$$A(\mathbf{k},\omega) = \frac{1}{\pi}|\operatorname{Im} G^R(\mathbf{k},\omega)| = \frac{1}{\pi}|\frac{\operatorname{Im}\Sigma_R^c(\mathbf{k},\omega)+\eta_P}{\left[\omega-(\varepsilon_{\mathbf{k}}^0+\Sigma_k^{\mathrm{x}}+\operatorname{Re}\Sigma_R^c(\mathbf{k},\omega))\right]^2+\left[\operatorname{Im}\Sigma_R^c(\mathbf{k},\omega)+\eta_P\right]^2}|, \tag{1.314}$$

where I used $\eta_s = \eta_P = 0.2$ eV in the computation of $\operatorname{Im}\Sigma_R^c(\mathbf{k},\omega)[\eta_s]$. This has suggested that my implementation of the retarded correlation self-energy may be done right. Next, they write that

$$C_k(t) = \pi^{-1}\int d\omega \left|\operatorname{Im}\Sigma_k\left(\omega+E_k^{GW}\right)\right| \frac{e^{-i\omega t}+i\omega t-1}{\omega^2} \tag{1.315}$$

Up until now, I have believed that they are working in the retarded formalism, but perhaps not. In the time-ordered formalism, the Green's function for a hole state is

$$G_k^h(t) = i\Theta(-t)e^{-i\varepsilon_k^0 t}\exp\left[-i\Sigma_k^{\mathrm{x}}t + C_k^h(t)\right] \tag{1.316}$$

with

$$C_k^h(t) = \frac{1}{\pi}\int_{-\infty}^{0} d\omega \frac{\left|\operatorname{Im}\Sigma_k^R\left(E_k^{GW}+\omega\right)\right|}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right). \tag{1.317}$$

Then, we can use the numerical FFT to get $G_k^h(\omega)$ and then $A_k^h(\omega) = \frac{1}{\pi}|\operatorname{Im} G_k^h(\omega)|$. In that case, the code I have used is

```
    cumulant_results = []  # list of (label, c_omegas, spec_fn_gwc)
start=-15/H2EV; stop=15/H2EV; num=10000
ce_omegas = np.linspace(start, stop, num)
ce_omegas_to = np.linspace(start, 0, num, endpoint=False)  # only negative frequenci
# print the 10 ce_omegas with smallest absolute value
print("10 cumulant omegas with smallest absolute value in ce_omegas:", ce_omegas[np.
print("10 cumulant omegas with smallest absolute value in ce_omegas_to:", ce_omegas_
omegaspep_iter = ce_omegas + qp_exact
betaspep_iter = 1/np.pi * np.abs(dyn_cohsex_batch(
    omegaspep_iter, p_idx, per_q_data_hl, moes, nocc, eta, ueg).imag)
betaspep_iter_to = 1/np.pi * np.abs(dyn_cohsex_batch(
    ce_omegas_to, p_idx, per_q_data_hl, moes, nocc, eta, ueg, time_ordered=False).im
omegaspep_lin = ce_omegas + qp_hl_lin
```

```
omegaspep_lin_to = ce_omegas_to + qp_hl_lin
betaspep_lin = 1/np.pi * np.abs(dyn_cohsex_batch(
    omegaspep_lin, p_idx, per_q_data_hl, moes, nocc, eta, ueg).imag)
betaspep_lin_to = 1/np.pi * np.abs(dyn_cohsex_batch(
    omegaspep_lin_to, p_idx, per_q_data_hl, moes, nocc, eta, ueg).imag)
# plot betaspep_iter and betaspep_lin to check they look reasonable
plt.figure(figsize=(8,4))
plt.plot(ce_omegas * H2EV, betaspep_lin, label='betaspep_lin', alpha=0.7)
plt.plot(ce_omegas_to * H2EV, betaspep_lin_to, label='betaspep_lin_to', linestyle='-
plt.legend()
plt.xlim(-20,20)
plt.xlabel('Frequency (eV)')

plt.savefig(os.path.join(GARNET, f"beta_sweep_pidx{p_idx}.png"))
for cp in cumulant_sweep:
    tol_c = cp['tol']
    if 'dt' in cp and 'N_t' in cp:
        dt_c, N_c = cp['dt'], cp['N_t']
        lbl_params = f"dt={dt_c:.4f}, N_t={N_c}"
    else:
        dw, wi = cp['Delta_omega'], cp['omega_interest']
        dt_c = np.pi / wi
        N_c = int(np.floor(2 * np.pi / dw / dt_c))

    c_times = np.arange(N_c) * dt_c
    c_neg_times = -c_times[::-1]
    t_fft_in = np.concatenate([c_times, -c_times[1:][::-1]])
    plot_omegas = np.fft.fftshift(np.fft.fftfreq(2 * N_c - 1, d=dt_c) * 2.0 * np.pi)

    n_skipped = int(np.sum(np.abs(ce_omegas) < tol_c))
    print(f"[cumulant] {lbl_params}, tol={tol_c} → N={N_c}, dt={dt_c:.6f}, max_t={c_

    # gt, spec_fn_gwc_c = compute_cumulant_spectral_function_numeric(
    #     betaspep_iter, ce_omegas, c_times, e_eff, eta, green=True, tol=tol_c)
    gt_lin, spec_fn_gwc_lin = compute_cumulant_spectral_function_numeric(
        betaspep_lin, ce_omegas, c_times, e_eff, eta, green=True, tol=tol_c)
    # plt plot gt and gt_lin to check they look reasonable
    plt.figure(figsize=(8,4))
    # plt.plot(c_times, gt, label='gt_iter', alpha=0.7)
    plt.plot(t_fft_in, gt_lin, label='gt_lin', linestyle='--', alpha=0.7)
    plt.legend()
    plt.xlabel('Time (Hartree^-1)')
    plt.ylabel('Green\'s function')
    plt.title(f"G(t) for p_idx={p_idx}")
    plt.savefig(os.path.join(GARNET, f"gt_sweep_pidx{p_idx}_tol{tol_c:.0e}.png"))
```

```
        # gt_to, spec_fn_gwc_to = compute_cumulant_spectral_function_numeric(
        #     betaspep_iter_to, ce_omegas_to, c_neg_times, e_eff, eta, green=True, tol=t
        pole=e_eff
        gt_lin_to, spec_fn_gwc_lin_to = compute_cumulant_spectral_function_numeric(
            betaspep_lin_to, ce_omegas_to, c_neg_times, pole, eta, green=True, tol=tol_c
```

using

```
    def dyn_cohsex_batch(omegas, p_idx, per_q_data, moes, nocc, eta, ueg, time_ordered=F
    """Vectorised dyn_cohsex over an array of frequencies; numba-accelerated inner loop.
    omegas = np.asarray(omegas, dtype=np.float64)
    vqvecs = ueg.vqvecJ
    vol    = ueg.vol

    eps_list, Omega_sq_list, wt_list, v_list, is_occ_list = [], [], [], [], []
    for idx_gn, gn in enumerate(ueg.basis):
        idx_gmgn = np.where((ueg.qvecs == ueg.basis[p_idx] - gn).all(axis=1))[0][0]
        if idx_gmgn == 0:
            continue
        wt = per_q_data[1][idx_gmgn]
        if wt == 0.0:
            continue
        eps_list.append(float(moes[idx_gn]))
        Omega_sq_list.append(float(per_q_data[0][idx_gmgn]))
        wt_list.append(float(wt))
        v_list.append(float(vqvecs[idx_gmgn]) / vol)
        is_occ_list.append(idx_gn < nocc)

    eps_arr     = np.array(eps_list,     dtype=np.float64)
    Omega_sq_arr = np.array(Omega_sq_list, dtype=np.float64)
    wt_arr      = np.array(wt_list,      dtype=np.float64)
    v_arr       = np.array(v_list,       dtype=np.float64)
    is_occ_arr  = np.array(is_occ_list,  dtype=np.bool_)

    if time_ordered:
        return _dyn_cohsex_batch_kernel_to(omegas, eta, eps_arr, Omega_sq_arr, wt_arr, v
    return _dyn_cohsex_batch_kernel(omegas, eta, eps_arr, Omega_sq_arr, wt_arr, v_arr, i
```

and

```
    def compute_cumulant_spectral_function_numeric(betaspep, freq_grid, times, pole, eta
    # Make contiguous for Numba speed
    betaspep = np.array(betaspep, dtype=np.float64)
    freq_grid = np.array(freq_grid, dtype=np.float64)
    times = np.array(times, dtype=np.float64)
```

```
    dt = times[1] - times[0]
    hartree_to_ev = 27.2114
    assert np.allclose(eta, 2e-1/hartree_to_ev), f"Expected eta to be {2e-1/hartree_to_e
    c_t = _compute_c_t_numba(betaspep, freq_grid, times, tol)
    if not time_ordered:
        assert np.all(times >= 0), "For retarded Green's function, all times should be >
        g_t = -1j * np.exp(-times * (1j * pole)) * np.exp(c_t) * np.exp(-eta * times)
        print(f"g_t[0] = {g_t[0]}, g_t[-1] = {g_t[-1]}")
        g_fft_in = np.concatenate(
            (g_t, -1.0 * np.flip(np.conjugate(g_t[1:]), axis=-1)),
            axis=-1
        )
        print(f"g_fft_in[0] = {g_fft_in[0]}, g_fft_in[-1] = {g_fft_in[-1]}")
        g_w = np.fft.ifft(g_fft_in, n=len(g_fft_in), norm='forward') * dt / 2.0
        g_w = np.fft.fftshift(g_w)
        spec_fn = -(1.0 / np.pi) * np.imag(g_w)
    else:
        # $G_{i i}^{\mathrm{T}}(t)=\mathrm{i} \Theta(-t) e^{-\mathrm{i} \epsilon_i t} e^
        assert np.all(times <= 0), "For time-ordered Green's function, all times should
        N_t = len(times)
        if exchange is not None:
            g_t = 1j * np.exp(-times * (1j * pole)) * np.exp(c_t + times * 1j * exchange
        elif mf:
            g_t = 1j * np.exp(-times * (1j * pole)) * np.exp(-eta * np.abs(times))
        else:
            g_t = 1j * np.exp(-times * (1j * pole)) * np.exp(c_t)# * np.exp(-eta * np.ab
        # Print the first and last elements of g_t for debugging
        print(f"g_t[0] = {g_t[0]}, g_t[-1] = {g_t[-1]}")
        g_fft_in = np.concatenate(
            (-1.0 * np.flip(np.conjugate(g_t[:-1]), axis=-1), g_t),
            axis=-1
        )
        print(f"g_fft_in[0] = {g_fft_in[0]}, g_fft_in[-1] = {g_fft_in[-1]}")
        g_w = np.fft.fftshift(np.fft.ifft(g_fft_in, norm='forward')) * dt
        spec_fn = (1.0 / np.pi) * np.abs(np.imag(g_w))

    if cumulant:
        return c_t,spec_fn
    elif green:
        return g_fft_in, spec_fn
    else:
        return spec_fn
```

with

```
    @njit(parallel=True, fastmath=True)
```

```
def _compute_c_t_numba(betaspep, freq_grid, times, tol, time_ordered=False):
    nfreq = freq_grid.shape[0]
    nt = times.shape[0]
    c_t = np.empty(nt, dtype=np.complex128)

    dw = freq_grid[1] - freq_grid[0]

    for it in range(nt):
        t = times[it]
        re = 0.0
        im = 0.0

        for i in prange(nfreq):
            w = freq_grid[i]
            b = betaspep[i]

            if abs(w * t) < tol:
            # if abs(w) < tol:
                landau_re = -0.5 * t * t + (w * w * t**4) / 24.0
                landau_im = (w * t**3) / 6.0
            else:
                x = w * t
                inv_w2 = 1.0 / (w * w)

                # exp(-i x) + i x - 1
                # = cos(x) - 1 + i(x - sin(x))
                landau_re = (np.cos(x) - 1.0) * inv_w2
                landau_im = (x - np.sin(x)) * inv_w2

            re += dw * b * landau_re
            im += dw * b * landau_im

        c_t[it] = re + 1j * im

    return c_t
```

As I said before, note that I am pretty confident in my implementation of the GW self-energy using the Plasma-Pole approximation because I have checked that it is equivalent to the self-energy for my system. And the match I get with the reference for GW in fig1a is

```
Peaks for LDA+GW (ref):
QP: omega = -5.601 eV, A = 0.7323 eV^-1, LHWHM=0.1873, RHWHM=0.1803, FWHM=0.3676, asym=-
sat 1: omega = -10.779 eV, A = 0.9076 eV^-1, LHWHM=0.1950, RHWHM=0.2102, FWHM=0.4052, as
```

vs

```
    Peaks for LDA+GW (PPM-HL, eta_sigma=eta_plot=eta):
```

```
Peak 0: omega=-5.585 eV, A=0.6937 eV^-1, LHWHM=0.2367, RHWHM=0.1782, FWHM=0.4149, asym=-
Peak 1: omega=-10.762 eV, A=0.8651 eV^-1, LHWHM=0.2134, RHWHM=0.1914, FWHM=0.4048, asym=
```

I want you to tinker with my cumulant formulation until you get my LDA+GW+C to match the reference as closely as the match for LDA+GW. I guess first you want to verify that I am indeed telling the truth when I say that the attached paper is used in the time-ordered ce formulation. And then I want you to design and then run a robust test tweet in order to identify a LDA+GW+C formulation that will reproduce the reference with A similar accuracy to what we got for LDA+GW . I understand that the retarded cumulant formulation might be better, but I just want to reproduce the reference curve, so if we determine that it is a time-ordered formulation, I want us to experiment with that. Note that I am working in a system which uses a finite plane-wave basis. I only say that because it seems that you have mostly been trained on information which just considers the UEG in its thermodynamic limit. However, for this cumulant spectral function formulation, I don't really see how the difference between the finite basis and the TDL could play a major role. If I have been able to reproduce the LDA+GW curve.

1.7.1 Supermatrix approaches

These all make use of Löwdin's partitioning technique to form an upfolded supermatrix, whose eigenpairs are the QP energies and Dyson orbitals. It is given by

$$\boldsymbol{H}^{GW}_{\text{Upfolded}} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W}^{<} & \boldsymbol{W}^{>} \\ \boldsymbol{W}^{<,\dagger} & \boldsymbol{d}^{<} & 0 \\ \boldsymbol{W}^{>,\dagger} & 0 & \boldsymbol{d}^{>} \end{pmatrix} \tag{1.318}$$

where $\boldsymbol{F}$ is the HF Fock matrix, $\boldsymbol{W}^{<}_{pk\nu} = \sum_{ia}(pk|ia)\,(X^{\nu}_{ia} + Y^{\nu}_{ia})$, $\boldsymbol{W}^{>}_{pc\nu} = \sum_{ia}(pc|ia)\,(X^{\nu}_{ia} + Y^{\nu}_{ia})$, $\boldsymbol{d}^{<}_{k\nu,l\nu'} = (\epsilon_k - \Omega_\nu)\,\delta_{k,l}\delta_{\nu,\nu'}$, and $\boldsymbol{d}^{>}_{c\nu,d\nu'} = (\epsilon_c + \Omega_\nu)\,\delta_{c,d}\delta_{\nu,\nu'}$. Recall that the X^{ν}_{ia} and Y^{ν}_{ia} are the RPA eigenvectors and the Ω_ν are the RPA excitation energies. Just constructing this supermatrix requires diagonalizing the RPA matrix, which scales as $O(N^6)$. Therefore, the below methods use Krylov subspace procedures to reduce this scaling. Because the QPs are interior eigenpairs of the supermatrix, the strategy is to identify the QPs by a root-following procedure, where we look to maximize overlap with the mean field eigenvectors.

Moment conserving

Moments of the self-energy are computed, which are then used to power a Lanczos procedure. To learn more, see [**?**]. Crucially, the Krylov subspace is not reorthogonalized throughout the iteration. If the lack of reorthogonalization did not interfere with the convergence of the Ritz values to the exact solution to within chemical accuracy, this would not be a problem, but as can be seen, it is a problem for some small molecules, invalidating the approach.

> The Casida transformation is used in this, so extension to extended systems is not possible. Self-consistency is possible, as shown in [**?**]. Scaling is $O(N^4)$ with RI. No diagonal approximation to the self-energy must be made.

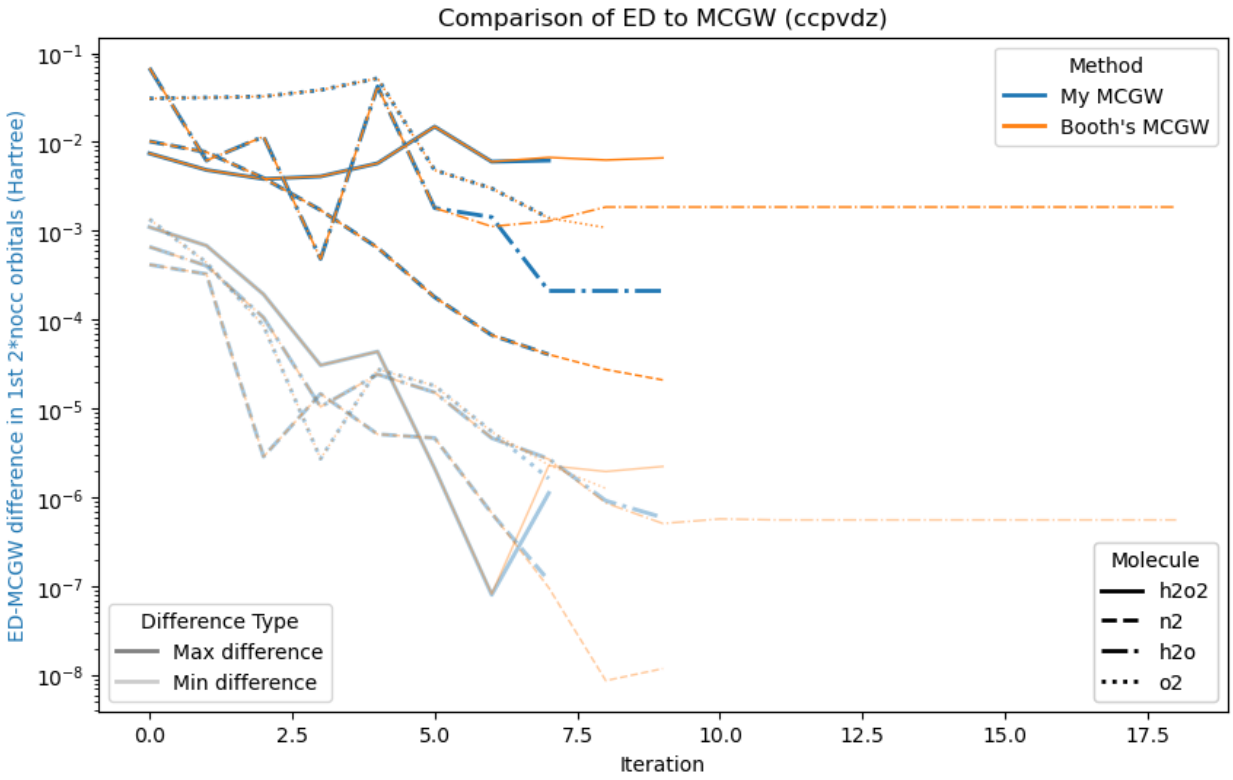


Figure 1.2: From the first $2N_{occ}$ orbitals, representing the space of interest, the maximum and minimum differences between the Ritz values and the exact eigenvalues are plotted as a function of the number of Lanczos iterations used. A cc-pvdz basis is used. An exact match between the blue (my implementation) and the orange curves (George Booth's open source code, momentGW) is shown for all early Lanczos iterations. For later iterations, there start to become no new directions for Lanczos to explore, and so the match no longer is exact, reflecting the different numerical handlings of this limiting case.

Auxiliary boson (AB)

The known connection of CC to G_0W_0 discussed in [**?**] is exploited in [**?**] to form a supermatrix. Then, a series of matrix-vector products is proposed, which can be used in a Davidson procedure.

> The study is done for finite systems so the Casida transformation is used, but it can not be used too, so extension to extended systems is possible. Self-consistency is not possible; in CC, the reference determinant must always be HF, so there can no self-consistency in the exactly related AB method. Scaling is $O(N^4)$ with RI. No diagonal approximation to the self-energy must be made. In principle, the method is exact, but poor convergence with respect to the basis size is observed, likely due to the AB expansion ansatz.

Tim's method

As discussed in [**?**], this method is exact for the case of TDA screening. The supplementary material proposes an extension to the full RPA screening, by postulating a non-symmetric supermatrix, which can be iteratively diagonalized by a non-symmetric Davidson procedure. I was not able to find a similarity transformation that relates their supermatrix to the exact solution, so the validity of this method is doubtful.

> It can be applied to extended systems. Self-consistency is possible. No diagonal approximation to the self-energy must be made. Focusing the discussion here just on finite systems, the scaling could be brought down to $O(N^4)$ with RI, but I already observe the numerical issues with my $O(N^6)$ implementation due to the introduction of a large η parameter, so I did not pursue the scaling reduction. A few words about this potential scaling reduction; the construction of the matrix-vector products requires building a step function, which requires diagonalizing the RPA matrix, conventionally at a $O(N^6)$ cost, but an approximation to the step function with Chebyshev polynomials could be made, reducing the scaling.

Numerical Lanczos

The method described in [**?**] does not use a supermatrix per se, but it fits better conceptually in this section. This method reformulates the correlation part of the GW self-energy as a resolvent of a Hermitian matrix to which a Lanczos procedure can be applied.

> It cannot be applied to extended systems, since the method relies on the Casida transformation. Self-consistency is possible. The diagonal approximation to the self-energy must be made. The scaling is $O(N^4)$ with RI.

Chapter 2

Dyson to GQME: 12/13

The Generalized Quantum Master Equation (GQME) is given by

$$\dot{\mathcal{C}}(t) = \mathcal{C}(t)\Omega_1 - \int_0^t d\tau\, \mathcal{C}(t-\tau)\mathcal{K}_1(\tau) + D(t) \tag{2.1}$$

where the correlation function is defined as

$$\mathcal{C}(t) = (\hat{\mu} \mid \hat{\mu}(t)), \tag{2.2}$$

the higher-order moments are

$$\Omega_n \equiv \left((i\mathcal{L})^n \hat{\mu}, \hat{\mu}\right) / (\hat{\mu}, \hat{\mu}) \tag{2.3}$$

with the auxiliary kernels

$$K_n(t) \equiv \left((i\mathcal{L})^n \hat{f}(t), \hat{\mu}\right) / (\hat{\mu}, \hat{\mu}) \tag{2.4}$$

$\hat{f}(t)$ in the above equation is referred to as the random force operator

$$\hat{f}(t) \equiv e^{it\mathcal{L}\mathcal{L}} \mathcal{Q} i\mathcal{L}\hat{\mu} \tag{2.5}$$

with $\mathcal{Q} = \mathcal{I} - \mathcal{P}$ being the complementary projection operator. But this becomes complicated, so Wenjie found that we can express $\mathcal{K}_1(t)$ without time evolution using $\hat{f}(0) = \mathcal{Q} i\mathcal{L}\hat{\mu}$ and, we get

$$K_n(0) = \Omega_{n+1} - \Omega_n \Omega_1 \tag{2.6}$$

Therefore, we only need to consider $\dot{K}_1(t)$, which can be obtained directly:

$$\dot{K}_1(t) = \frac{(i\mathcal{L}\dot{\hat{f}}(t), \hat{\mu})}{(\hat{\mu}, \hat{\mu})} = K_2(t) - \Omega_1 K_1(t) \tag{2.7}$$

Similarly, we can show that the auxiliary kernels are coupled through

$$\dot{K}_n(t) = K_{n+1}(t) - \Omega_n K_1(t). \tag{2.8}$$

We expect that the higher order auxiliary kernels will decay quickly, so we can truncate the series at some finite n. The moments of the memory kernel are

$$\Omega_n = \frac{((i\mathcal{L})^n \hat{\mu}, \hat{\mu})}{(\hat{\mu}, \hat{\mu})}, \tag{2.9}$$

with $\mathcal{L}$ being the Liouville superoperator with $\mathcal{L}\hat{\mu} = \left[\hat{H}, \hat{\mu}\right]$. The construction of the numerator in equation 2.9 can be thought of as the generation of a Krylov subspace up to level n, i.e. we need to build up $\mathcal{K}_n(\mathcal{L}, \hat{\mu}) = \text{span}\{\hat{\mu}, (i\mathcal{L})\hat{\mu}, (i\mathcal{L})^2\hat{\mu}, \ldots, (i\mathcal{L})^{n-1}\hat{\mu}\}$, where $\hat{\mu} = \hat{c}$ or $\hat{c}^\dagger$. In the case if we choose $\hat{\mu} = \hat{c}$, we get the lesser Green's function

$$\mathcal{C}(t) = (\hat{c}, \hat{c}(t)) \equiv \langle \hat{c}^\dagger(0)\hat{c}(t)\rangle = \frac{G^<(t)}{i} \tag{2.10}$$

whereas if we chose $\hat{\mu} = \hat{c}^\dagger$, we get the greater Green's function

$$\mathcal{C}(t) = (\hat{c}^\dagger, \hat{c}^\dagger(t)) \equiv \langle \hat{c}(0)\hat{c}^\dagger(t)\rangle = -\frac{G^>(t)}{i} \tag{2.11}$$

Then we can construct the retarded Green's function as

$$G_R(t) = \Theta(t)\left(G^<(t) - G^>(t)\right) \tag{2.12}$$

Using Krylov subspace methods, one never has to construct the Liouvillian matrix, but instead can directly compute the extremal eigenvalues and eigenvectors of $\mathcal{L}$ by considering the action of $\mathcal{L}$ on the Krylov subspace.

2.1 Explicit Construction of the Liouville Superoperator

Consider that we are working with the upfolded Hamiltonian

$$\mathbf{H} = \begin{pmatrix} \mathbf{f} & \mathbf{W} \\ \mathbf{W}^\dagger & \mathbf{d} \end{pmatrix} \tag{2.13}$$

where we again have a physical space $\mathbf{f}$ and a bath space $\mathbf{d}$, whose coupling is given by $\mathbf{W}$. Tell me what would happened if we considered the action of this on the composite operator vector defined lower? Lets consider making a Krylov subspace, corresponding to repeated applications of the Liouville superoperator to the initial operator $\hat{\boldsymbol{\mu}}$. Now, The idea is to define a composite operator vector

$$\hat{\boldsymbol{\mu}} \equiv \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = \begin{pmatrix} \hat{c} \\ \hat{c}^\dagger \end{pmatrix}, \tag{2.14}$$

where $\hat{c}$ is the annihilation operator and $\hat{c}^\dagger$ is the creation operator. Notice that the equation of motion for the Green's function is

$$(i\partial_t - \hat{H}_0)G(t,t') = \delta(t-t') + \int_{-\infty}^{\infty} d\tau\, \Sigma(t,\tau)\, G(\tau,t') \tag{2.15}$$

$$\rightarrow G(\dot{t},t') = -i\hat{H}_0 G(t,t') - i\delta(t-t') + \int_{-\infty}^{\infty} d\tau'\, \Sigma(t,\tau')\, G(\tau',t') \tag{2.16}$$

So our task becomes to figure out how

$$-i\hat{H}_0 G(t,t') - i\delta(t-t') + \int_{-\infty}^{\infty} d\tau'\, \Sigma(t,\tau')\, G(\tau',t') = \mathcal{C}(t)\Omega_1 - \int_0^t d\tau\, \mathcal{C}(t-\tau)\mathcal{K}(\tau) + D(t) \quad (2.17)$$

I feel like it should be the case that $-i\hat{H}_0 G(t,t') = \mathcal{C}(t)\Omega_1$. Do you think that this should be the case or no? Because I think we can agree that the equation that comes first should be the same as the equation of motion for the greens function. And then try to apply a decomposed Hamiltonia like $H = H_0 + V$ to the first equation, so that we can see what happens.

To simple by things as much as possible initially consider that we only use the noninteracting Hamiltonian $\hat{H}_0 = \epsilon \hat{c}^\dagger \hat{c}$ in the action of the Liouvillian.

$$\Omega_1 = \frac{((i\mathcal{L})\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}})}{(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}})} = \frac{\Big((i[\hat{H}_0, \hat{\boldsymbol{\mu}}], \hat{\boldsymbol{\mu}}\Big)}{(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}})} \quad (2.18)$$

If we just consider the numerator, we see that

$$\Big((i[\hat{H}_0, \hat{\boldsymbol{\mu}}], \hat{\boldsymbol{\mu}}\Big) = \Big(i[\hat{H}_0, \hat{c}], \hat{c}\Big) + \Big(i[\hat{H}_0, \hat{c}^\dagger], \hat{c}^\dagger\Big) \quad (2.19)$$

Considering just the first term

$$\Big(i[\hat{H}_0, \hat{c}], \hat{c}\Big) = -i\epsilon([\hat{c}^\dagger \hat{c}, \hat{c}], \hat{c}) = -i\epsilon(\hat{c}, \hat{c}) = -i\epsilon\,(1 - f(\epsilon)) \quad (2.20)$$

and the second term

$$\Big(i[\hat{H}_0, \hat{c}^\dagger], \hat{c}^\dagger\Big) = -i\epsilon([\hat{c}^\dagger \hat{c}, \hat{c}^\dagger], \hat{c}^\dagger) = -i\epsilon(\hat{c}^\dagger, \hat{c}^\dagger) = -i\epsilon f(\epsilon) \quad (2.21)$$

which can be summarized as

$$\Big(i[\hat{H}_0, \hat{\boldsymbol{\mu}}], \hat{\boldsymbol{\mu}}\Big) = -i\epsilon \implies \Omega_1 = -i\epsilon \quad (2.22)$$

Now, the equation of motion for the interacting Green's function is given by

$$\left(i\frac{\partial}{\partial t} - h_0\right) G(t,t') = \delta(t-t') + \int dt'' \Sigma(t,t'') G(t'',t') \quad (2.23)$$

$$\frac{\partial}{\partial t} G(t,t') = \underbrace{-ih_0 G(t,t')}_{\Omega_1 C(t)} - i\delta(t-t') + \int dt'' \Sigma(t,t'') G(t'',t') \quad (2.24)$$

Now come if we consider the higher-order moments

$$\Omega_n \equiv \frac{((i\mathcal{L})^n \hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}})}{(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}})} = \frac{(i)^n \left(\mathcal{L}^n \hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}}\right)}{(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\mu}})} = (i)^n \Big([\hat{H}, [\hat{H}, [\hat{H}, \cdots, \hat{\boldsymbol{\mu}}]]] \cdots\Big) \quad (2.25)$$

where it is implied that we are applying the commutator n times. We want to answer the form for the $\hat{\mathbf{H}}^{G_0 W_0}$ Hamiltonian, which has the super matrix form of

$$\begin{bmatrix} \mathbf{f} + \boldsymbol{\Sigma}_\infty & \mathbf{W} \\ \mathbf{W}^\dagger & \mathbf{d} \end{bmatrix} \quad (2.26)$$

and the memory kernel

$$\mathcal{K}(t) = \left(\mathbf{A} \left| \mathcal{L}\mathcal{Q} e^{i\mathcal{Q}\mathcal{L}t} \mathcal{Q}\mathcal{L} \right| \mathbf{A}\right), \quad (2.27)$$

where $\mathcal{Q} = \mathcal{I} - \mathcal{P}$ is the complementary projection operator.

Chapter 3

Moment-Conserving GW

3.1 Löwdin Downfolding: 2/2/2025

We know that the definition of a Green's function associated with some Hamiltonian $\boldsymbol{H}$ is given by:

$$(\omega - \boldsymbol{H})\,\boldsymbol{G} = \boldsymbol{I} \tag{3.1}$$

where we can consider both cases to be fully interacting. The downfolding tells us to separate into a system $\mathcal{S}$ and auxiliary space $\mathcal{L}$, so we have:

$$\left(\omega - \begin{pmatrix} \boldsymbol{H}_{\mathcal{SS}} & \boldsymbol{H}_{\mathcal{SL}} \\ \boldsymbol{H}_{\mathcal{LS}} & \boldsymbol{H}_{\mathcal{LL}} \end{pmatrix}\right) \begin{pmatrix} \boldsymbol{G}_{\mathcal{SS}} & \boldsymbol{G}_{\mathcal{SL}} \\ \boldsymbol{G}_{\mathcal{LS}} & \boldsymbol{G}_{\mathcal{LL}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{I}_{\mathcal{SS}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}_{\mathcal{LL}} \end{pmatrix}.$$

or:

$$\begin{pmatrix} \omega - \boldsymbol{H}_{\mathcal{SS}} & -\boldsymbol{H}_{\mathcal{SL}} \\ -\boldsymbol{H}_{\mathcal{LS}} & \omega - \boldsymbol{H}_{\mathcal{LL}} \end{pmatrix} \begin{pmatrix} \boldsymbol{G}_{\mathcal{SS}} & \boldsymbol{G}_{\mathcal{SL}} \\ \boldsymbol{G}_{\mathcal{LS}} & \boldsymbol{G}_{\mathcal{LL}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{I}_{\mathcal{SS}} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}_{\mathcal{LL}} \end{pmatrix}.$$

Multiplying out the matrices, the $\mathcal{S}$ block gives:

$$(\omega - \boldsymbol{H}_{\mathcal{SS}})\,\boldsymbol{G}_{\mathcal{SS}} - \boldsymbol{H}_{\mathcal{SL}}\,\boldsymbol{G}_{\mathcal{LS}} = \boldsymbol{I}_{\mathcal{SS}}\,. \tag{3.2}$$

Similarly, the $\mathcal{L}$ block gives:

$$-\boldsymbol{H}_{\mathcal{LS}}\,\boldsymbol{G}_{\mathcal{SS}} + (\omega - \boldsymbol{H}_{\mathcal{LL}})\,\boldsymbol{G}_{\mathcal{LS}} = \boldsymbol{0}\,. \tag{3.3}$$

Assuming $(\omega - \boldsymbol{H}_{\mathcal{LL}})$ is invertible, from the second equation we obtain:

$$\boldsymbol{G}_{\mathcal{LS}} = (\omega - \boldsymbol{H}_{\mathcal{LL}})^{-1}\,\boldsymbol{H}_{\mathcal{LS}}\,\boldsymbol{G}_{\mathcal{SS}}\,.$$

Substituting this into the first equation:

$$(\omega - \boldsymbol{H}_{\mathcal{SS}})\,\boldsymbol{G}_{\mathcal{SS}} - \boldsymbol{H}_{\mathcal{SL}}\,(\omega - \boldsymbol{H}_{\mathcal{LL}})^{-1}\,\boldsymbol{H}_{\mathcal{LS}}\,\boldsymbol{G}_{\mathcal{SS}} = \boldsymbol{I}_{\mathcal{SS}}\,.$$

Factorizing $\boldsymbol{G}_{\mathcal{SS}}$:

$$\left[(\omega - \boldsymbol{H}_{\mathcal{SS}}) - \boldsymbol{H}_{\mathcal{SL}}\,(\omega - \boldsymbol{H}_{\mathcal{LL}})^{-1}\,\boldsymbol{H}_{\mathcal{LS}}\right]\,\boldsymbol{G}_{\mathcal{SS}} = \boldsymbol{I}_{\mathcal{SS}}\,.$$

Thus:

$$\boldsymbol{G}_{\mathcal{SS}} = \left[(\omega - \boldsymbol{H}_{\mathcal{SS}}) - \boldsymbol{H}_{\mathcal{SL}}\,(\omega - \boldsymbol{H}_{\mathcal{LL}})^{-1}\,\boldsymbol{H}_{\mathcal{LS}}\right]^{-1}\,.$$

Now, notice that:

$$[\boldsymbol{G}^0_{\mathcal{SS}}(\omega)]^{-1} \equiv \omega - \boldsymbol{H}_{\mathcal{SS}} = \omega - (\boldsymbol{F} + \boldsymbol{\Sigma}(\infty)),\implies \boldsymbol{H}_{\mathcal{SS}} = \boldsymbol{F} + \boldsymbol{\Sigma}(\infty).$$

where $\boldsymbol{\Sigma}(\infty)$ is the static self-energy (0 for a HF mean-field reference), and $\boldsymbol{F}$ is the Fock matrix. Identifying the coupling matrices:

$$\boldsymbol{H}_{\mathcal{SL}} = \boldsymbol{W},\quad \boldsymbol{H}_{\mathcal{LS}} = \boldsymbol{W}^\dagger\,.$$

and

$$\boldsymbol{H}_{\mathcal{LL}} = \boldsymbol{d},$$

Now let us make an ansatz for the upfolded Hamiltonian:

$$\boldsymbol{H}_{\text{Upfolded}} = \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix}.$$

Then, the resolvent is given by:

$$(\omega - \boldsymbol{H}_{\text{Upfolded}}) = \begin{pmatrix} \omega - \boldsymbol{F} - \boldsymbol{\Sigma}(\infty) & -\boldsymbol{W} \\ -\boldsymbol{W}^\dagger & \omega - \boldsymbol{d} \end{pmatrix} \tag{3.4}$$

Because we are interested in $\boldsymbol{G}_{\mathcal{SS}}(\omega)$, we care about $(\omega - \boldsymbol{H}_{\text{Upfolded}})^{-1}$ in the upper left block, which is the Schur complement of $(\omega - \boldsymbol{H}_{\text{Upfolded}})$ with respect to $\omega - \boldsymbol{d}$, defined as:

$$\boldsymbol{G}_{\mathcal{SS}}(\omega) = \left(\frac{(\omega - \boldsymbol{H}_{\text{Upfolded}})}{\omega - \boldsymbol{d}}\right)^{-1}_{\mathcal{SS}} = \left(\omega - (\boldsymbol{F} + \boldsymbol{\Sigma}(\infty)) - \boldsymbol{W}\,[\omega - \boldsymbol{d}]^{-1}\,\boldsymbol{W}^\dagger\right)^{-1} \tag{3.5}$$

so the ansatz is correct.

3.2 Cumulant Idea

The definition of the cumulant ansatz for the Green's function is given by:

$$\boldsymbol{G}_{\mathcal{SS}}(t) = \boldsymbol{G}^0_{\mathcal{SS}}(t) e^{\boldsymbol{C}(t)} \tag{3.6}$$

where $\boldsymbol{C}(t)$ is the cumulant and $\boldsymbol{G}^0_{\mathcal{SS}}(t)$ is the HF Green's function. By relating the Dyson equation to the Taylor series expansion of the exponential, we can write:

$$\boldsymbol{G}^0_{\mathcal{SS}}(t)\boldsymbol{C}(t) = \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{\mathcal{SS}}(t - t_1)\boldsymbol{\Sigma}^c(t_1 - t_2)\boldsymbol{G}^0_{\mathcal{SS}}(t_2) \tag{3.7}$$

Projecting to the spin-orbital basis and inserting the resolution of the identity, we get:

$$\sum_r \langle p| \boldsymbol{G}^0_{\mathcal{SS}}(t) |r\rangle \langle r| \boldsymbol{C}(t) |q\rangle = \sum_{rs} \iint \mathrm{d}t_1 \mathrm{d}t_2 \langle p| \boldsymbol{G}^0_{\mathcal{SS}}(t-t_1) |r\rangle \langle r| \boldsymbol{\Sigma}^c(t_1-t_2) |s\rangle \langle s| \boldsymbol{G}^0_{\mathcal{SS}}(t_2) |q\rangle \tag{3.8}$$

$$\sum_r \boldsymbol{G}^0_{pr}(t)\boldsymbol{C}_{rq}(t) = \sum_{rs} \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{ps}(t-t_1)\boldsymbol{\Sigma}^c_{sr}(t_1-t_2)\boldsymbol{G}^0_{rq}(t_2) \tag{3.9}$$

$$\boldsymbol{G}^0_{pp}(t)\boldsymbol{C}_{pq}(t) = \underbrace{\iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{pp}(t-t_1)\boldsymbol{\Sigma}^c_{pq}(t_1-t_2)\boldsymbol{G}^0_{qq}(t_2)}_{*} \tag{3.10}$$

where $\boldsymbol{G}^0_{\mathcal{SS}}(t) \equiv \boldsymbol{G}^0(t)$ and in the last step we used the fact that the HF Green's function is diagonal in the spin-orbital basis, specifically $\boldsymbol{G}^0_{pp}(t) = -i\Theta(t)e^{-i\epsilon_p t}$, where ϵ_p is the HF energy of the p-th spin-orbital. The formula for the inverse Fourier transform is given by:

$$f(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} f(\omega) \tag{3.11}$$

which implies that

$$\boldsymbol{G}^0_{pp}(t-t_1) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega) \tag{3.12}$$

$$\boldsymbol{\Sigma}^c_{pq}(t_1-t_2) = \int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega') \tag{3.13}$$

$$\boldsymbol{G}^0_{qq}(t_2) = \int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'') \tag{3.14}$$

and plugging into the double time integral *, we get:

$$* = \iint \mathrm{d}t_1 \mathrm{d}t_2 \left[\int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega)\right] \left[\int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega')\right] \left[\int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'')\right] \tag{3.15}$$

$$= \underbrace{\int \mathrm{d}t_1 e^{-i(\omega'-\omega)t_1} \int \mathrm{d}t_2 e^{-i(\omega''-\omega')t_2}}_{4\pi^2\delta(\omega'-\omega)\delta(\omega''-\omega')} \iiint \mathrm{d}\omega \mathrm{d}\omega' \mathrm{d}\omega'' \frac{e^{-i\omega t}}{8\pi^3} \boldsymbol{G}^0_{pp}(\omega)\boldsymbol{\Sigma}^c_{pq}(\omega')\boldsymbol{G}^0_{qq}(\omega'') \tag{3.16}$$

$$= \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} \boldsymbol{G}^0_{pp}(\omega)\boldsymbol{\Sigma}^c_{pq}(\omega)\boldsymbol{G}^0_{qq}(\omega) \tag{3.17}$$

But now note that from the left hand side of eqn. 6.1376, we can divide out the HF Green's function to get:

$$\boldsymbol{C}_{pq}(t) = i \int \frac{\mathrm{d}\omega}{2\pi} e^{-i(\omega-\epsilon_p)t} \boldsymbol{G}^0_{pp}(\omega)\boldsymbol{\Sigma}^c_{pq}(\omega)\boldsymbol{G}^0_{qq}(\omega) \tag{3.18}$$

Now, we insert the upfolded form for the self-energy, which is frequency independent as $\boldsymbol{\Sigma}^c_{pq}(\omega) \equiv \boldsymbol{\Sigma}^c_{pq} = \begin{pmatrix} \boldsymbol{\Sigma}(\infty) & \boldsymbol{W}^< & \boldsymbol{W}^> \\ \boldsymbol{W}^{\dagger <} & \boldsymbol{d}^< & \mathbf{0} \\ \boldsymbol{W}^{\dagger >} & \mathbf{0} & \boldsymbol{d}^> \end{pmatrix}_{pq}$. *Customarily this is the point where the diagonal*

approximation for the self-energy is introduced instead. We also know that $\boldsymbol{G}^0_{pp}(\omega) = \frac{\boldsymbol{I}}{\omega - \epsilon_p}$. We can then write:

$$\boldsymbol{C}_{pq}(t) = i\boldsymbol{\Sigma}^c_{pq} \int \frac{\mathrm{d}\omega}{2\pi} \frac{e^{-i(\omega-\epsilon_p)t}}{(\omega-\epsilon_p)(\omega-\epsilon_q)} \tag{3.19}$$

$$= i\frac{\boldsymbol{\Sigma}^c_{pq}}{\epsilon_q - \epsilon_p} \left[\underbrace{\int \frac{\mathrm{d}\omega}{2\pi} \frac{e^{-i(\omega-\epsilon_p)t}}{\omega-\epsilon_p}}_{-i\Theta(t)} - \underbrace{\int \frac{\mathrm{d}\omega}{2\pi} \frac{e^{-i(\omega-\epsilon_q)t}}{\omega-\epsilon_q}}_{e^{-i(\epsilon_q-\epsilon_p)t}(-i\Theta(t))} \right] \tag{3.20}$$

$$= \frac{\boldsymbol{\Sigma}^c_{pq}}{\epsilon_q - \epsilon_p} \Theta(t) \left[1 - e^{-i(\epsilon_q-\epsilon_p)t}\right] \tag{3.21}$$

$$\tag{3.22}$$

So now we insert this expression into our original ansatz for the cumulant, and we get:

$$G(t) = G^0(t) e^{\frac{\boldsymbol{\Sigma}^c_{pq}}{\epsilon_q-\epsilon_p}\Theta(t)\left[1-e^{-i(\epsilon_q-\epsilon_p)t}\right]} \tag{3.23}$$

3.3 Lanczos Iteration

The block tridiagonal form can be expressed as:

$$\tilde{\mathbf{H}}_{\text{tri}} = \tilde{\mathbf{q}}^{(j),\dagger} \begin{bmatrix} \mathbf{f}+\boldsymbol{\Sigma}_\infty & \mathbf{W} \\ \mathbf{W}^\dagger & \mathbf{d} \end{bmatrix} \tilde{\mathbf{q}}^{(j)} \tag{3.24}$$

$$= \begin{bmatrix} \mathbf{f}+\boldsymbol{\Sigma}_\infty & \mathbf{L} & & & & \mathbf{0} \\ \mathbf{L}^\dagger & \mathbf{H}_1 & \mathbf{C}_1 & & & \\ & \mathbf{C}_1^\dagger & \mathbf{H}_2 & \mathbf{C}_2 & & \\ & & \mathbf{C}_2^\dagger & \mathbf{H}_3 & \ddots & \\ & & & \ddots & \ddots & \mathbf{C}_{j-1} \\ \mathbf{0} & & & & \mathbf{C}_{j-1}^\dagger & \mathbf{H}_j \end{bmatrix} \tag{3.25}$$

where we define $\tilde{\mathbf{q}}^{(j)}$ as

$$\tilde{\mathbf{q}}^{(j)} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{q}^{(j)} \end{bmatrix} \tag{3.26}$$

This formulation becomes exact when the level j equals N, the dimension of the original hamiltonian; this corresponds to considering up to the highest moment of the self-energy, i.e. n france from $1, \ldots, N$. in practice, however, we always truncate the Krylov subspace to some $j < N$. Note that the tridiagonal form never actually forces us to compute $\mathbf{W}$ or $\mathbf{d}$, as desired.

3.3.1 Creation of Krylov Subspace

Formally, the Krylov subspace of level j is given as $\mathbf{q}^{(j)} = [\mathbf{q}_1, \mathbf{q}_2, \cdots, \mathbf{q}_j]$, where the projection of the full Hamiltonian onto this subspace gives the tridiagonal form of equation 3.25. To start building up the subspace, we need to determine $\mathbf{q}_1$ via QR decomposition of the exact GW couplings as $\mathbf{W}^\dagger = \mathbf{q}_1\mathbf{L}^\dagger \rightarrow \mathbf{q}_1 = \mathbf{W}^\dagger\mathbf{L}^{\dagger,-1}$. $\mathbf{L}$ is defined in terms of just the 0th order self-energy moment as $\mathbf{L}^\dagger = \left(\mathbf{\Sigma}^{(0)}\right)^{\frac{1}{2}}$. We build up the subsequent q_i vectors through a three-term recurrence

$$\mathbf{q}_{i+1}\mathbf{C}_i^\dagger = \left[\mathbf{d}\mathbf{q}_i - \mathbf{q}_i\mathbf{H}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}\right], \tag{3.27}$$

where the on-diagonal blocks are defined as

$$\mathbf{H}_i = \mathbf{q}_i^\dagger \mathbf{d}\mathbf{q}_i \tag{3.28}$$

Notice that to form the initial vector $\mathbf{q}_1$ we would need $\mathbf{W}^\dagger$ and to continue building the subspace, we would need $\mathbf{d}$, so to avoid this, we introduce the self-energy moments.

3.3.2 A sketch of the implicit Lanczos method

Due to Garnet's paper on the quasi-boson G_0W_0 method, we know that we have a form for an upfolded G_0W_0 Hamiltonian as

$$\boldsymbol{H}^{G_0W_0}_{\text{Upfolded}} = \begin{pmatrix} \boldsymbol{F} + \mathbf{\Sigma}(\infty) & \boldsymbol{W}^< & \boldsymbol{W}^> \\ \boldsymbol{W}^{<,\dagger} & \boldsymbol{d}^< & 0 \\ \boldsymbol{W}^{>,\dagger} & 0 & \boldsymbol{d}^> \end{pmatrix} \tag{3.29}$$

with quantities defined separately for lesser and greater parts. But we will just focus on the lesser part for now, where matrix elements of the screened interaction $\boldsymbol{W}^<$ are given by

$$W^<_{pkv} = \sum_{ia}(pk|ia)\left(X^v_{ia} + Y^v_{ia}\right) \tag{3.30}$$

and

$$d^<_{kv,lv'} = \left(\epsilon_k - \Omega_v\right)\delta_{k,l}\delta_{v,v'} \tag{3.31}$$

Note that we get the factor of $\sqrt{2}$ accompanying all ERIs in RHF because we are considering the expectation value of the form

$$\frac{1}{2}\sum_{pqrs}\langle pq||rs\rangle \left\langle\Psi_0\right| \left(\hat{T}_i^{a,\alpha} + \hat{T}_i^{a,\beta}\right)^\dagger \left(a_p^\dagger a_q^\dagger a_s a_r \left(\hat{T}_i^{a,\alpha} + \hat{T}_i^{a,\beta}\right)\right) \left|\Psi_0\right\rangle \tag{3.32}$$

where we have the excitation operator for a given spin channel as $\hat{T}_i^{a,\sigma} = a_a^{\dagger,\sigma}a_i^\sigma$ and a singlet state for the RHF ground state carrying a factor of $\frac{1}{\sqrt{2}}$. If we apply Wick's theorem to this string, we get a contribution from both the α and β channels, so $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$.
Through the introduction of a Krylov subspace as

$$\tilde{\boldsymbol{H}}^{G_0W_0}_{\text{Upfolded}} = \tilde{\boldsymbol{Q}}^{(n,\dagger)}\boldsymbol{H}^{G_0W_0}_{\text{Upfolded}}\tilde{\boldsymbol{Q}}^{(n)} \tag{3.33}$$

where $\boldsymbol{Q}^{(n)} \equiv \begin{pmatrix} \boldsymbol{q}_1 & \boldsymbol{q}_2 & \cdots & \boldsymbol{q}_n \end{pmatrix}$ is the block Krylov subspace spanned by the Lanczos vectors, but we want to preserve the physical space of $\boldsymbol{F} + \boldsymbol{\Sigma}(\infty)$ so we are really interested in the projection matrix

$$\tilde{\boldsymbol{Q}}^{(n)} = \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{Q}^{(n)} \end{pmatrix} \tag{3.34}$$

To get a gist of what the block Lanczos will do, let's just consider the case where we have two Lanczos vectors, so that we have a projection matrix of the form

$$\tilde{\boldsymbol{Q}}^{(2)} = \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{Q}^{(2)} \end{pmatrix} \tag{3.35}$$

So

$$\tilde{\boldsymbol{H}}^{\text{Lanczos Iter 2}}_{\text{upfolded}} = \tilde{\boldsymbol{Q}}^{(2,\dagger)} \boldsymbol{H}^{G_0W_0}_{\text{Upfolded}} \tilde{\boldsymbol{Q}}^{(2)} \tag{3.36}$$

$$= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{Q}^{(2)\dagger} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix} \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{Q}^{(2)} \end{pmatrix} \tag{3.37}$$

$$= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1^\dagger \quad \boldsymbol{q}_2^\dagger] \end{pmatrix} \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix} \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1 \quad \boldsymbol{q}_2] \end{pmatrix} \tag{3.38}$$

$$= \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & \boldsymbol{W} \\ [\boldsymbol{q}_1^\dagger \quad \boldsymbol{q}_2^\dagger]\boldsymbol{W}^\dagger & [\boldsymbol{q}_1^\dagger \quad \boldsymbol{q}_2^\dagger]\boldsymbol{d} \end{pmatrix} \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1 \quad \boldsymbol{q}_2] \end{pmatrix} \tag{3.39}$$

$$= \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & [\underbrace{\boldsymbol{W}\boldsymbol{q}_1}_{\boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{Q}}^\dagger\boldsymbol{\mathcal{Q}}=\boldsymbol{\mathcal{R}}} \quad \underbrace{\boldsymbol{W}\boldsymbol{q}_2}_{\boldsymbol{0}}] \\ [\underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{\mathcal{Q}}^\dagger\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{R}}^\dagger=\boldsymbol{\mathcal{R}}^\dagger} \quad \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{0}}] & \begin{pmatrix} \underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{d}\boldsymbol{q}_1}_{\boldsymbol{M}_1} & \underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{d}\boldsymbol{q}_2}_{\boldsymbol{C}_1} \\ \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{d}\boldsymbol{q}_1}_{\boldsymbol{C}_1^\dagger} & \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{d}\boldsymbol{q}_2}_{\boldsymbol{M}_2} \end{pmatrix} \end{pmatrix} \tag{3.40}$$

where in the last line we used the fact that we are free to choose our $\boldsymbol{q}_1 \equiv \boldsymbol{\mathcal{Q}}$ as the same orthogonal matrix employed in constructing the QR decomposition of $\boldsymbol{W}^\dagger$ as

$$\boldsymbol{W}^\dagger = \boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{R}}^\dagger \implies \boldsymbol{q}_1 = \boldsymbol{W}^\dagger \boldsymbol{\mathcal{R}}^{-1,\dagger} \tag{3.41}$$

3.3.3 Physical-auxiliary coupling W

Now consider $\boldsymbol{W}\boldsymbol{W}^\dagger = \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{Q}}^\dagger\boldsymbol{\mathcal{R}}^\dagger = \boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{R}}^\dagger$. So if we can do a Cholesky decomposition of $\boldsymbol{W}\boldsymbol{W}^\dagger$ we would get access to $\boldsymbol{\mathcal{R}}$. But consider that

$$\Sigma^{(n,<)}_{pq} = \sum_{ia,jb,k} \sum_{\mu} (pk|ia) \left(X^\mu_{ia} + Y^\mu_{ia}\right) \left(X^\mu_{jb} + Y^\mu_{jb}\right) (qk|jb)[(\epsilon_k - \Omega_\mu)]^n \tag{3.42}$$

so it becomes clear that $\boldsymbol{W}\boldsymbol{W}^\dagger = \boldsymbol{\Sigma}^0 \implies \boldsymbol{\mathcal{R}} = \left(\boldsymbol{W}\boldsymbol{W}^\dagger\right)^{1/2} = \left(\boldsymbol{\Sigma}^0\right)^{1/2}$, which we hope to be able to accomplish with the Cholesky decomposition.

3.3.4 Auxiliary-auxiliary space d: the working equations

From above, notice that

$$\boldsymbol{M}_i = \boldsymbol{q}_i^\dagger \boldsymbol{d}\boldsymbol{q}_i \quad \text{and} \quad \boldsymbol{C}_i = \boldsymbol{q}_i^\dagger \boldsymbol{d}\boldsymbol{q}_{i+1} \tag{3.43}$$

To get these, define

$$\boldsymbol{S}_{i,j}^{(n)} = \boldsymbol{q}_i^\dagger \boldsymbol{d}^n \boldsymbol{q}_j \tag{3.44}$$

So it must be that $\boldsymbol{S}_{0,j}^{(n)} = \boldsymbol{S}_{i,0}^{(n)} = 0$ for all i, j, $\boldsymbol{S}_{i,j}^{(0)} = \delta_{ij}\boldsymbol{I}$, and we can demand Hermiticity, so that $\boldsymbol{S}_{i,j}^{(n)} = \boldsymbol{S}_{j,i}^{(n)\dagger}$. We are able to initialize $\boldsymbol{S}$ using

$$\boldsymbol{S}_{1,1}^{(n)} = \boldsymbol{q}_1^\dagger \boldsymbol{d}^n \boldsymbol{q}_1 = \boldsymbol{R}^{-1}\boldsymbol{W}\boldsymbol{d}^n\boldsymbol{W}^\dagger\boldsymbol{R}^{-1,\dagger} = \boldsymbol{R}^{-1}\boldsymbol{\Sigma}^{(n)}\boldsymbol{R}^{-1,\dagger} \tag{3.45}$$

So looking at eqn. 3.117, we see that the on-diagonal elements are $\boldsymbol{M}_i = \boldsymbol{S}_{i,i}^{(1)}$ while the off-diagonal elements are $\boldsymbol{C}_i = \boldsymbol{S}_{i,i+1}^{(1)}$. The familiar three-term Lanczos recurrence is

$$\mathbf{q}_{i+1}\mathbf{C}_i^\dagger = [\mathbf{dq}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}] \implies \mathbf{q}_{i+1} = [\mathbf{dq}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}]\,\boldsymbol{C}_i^{\dagger,-1} \tag{3.46}$$

where the participants are block vectors and we have assumed that $\boldsymbol{C}_i$ is invertible. Let us start by considering the form of

$$\boldsymbol{S}_{i+1,i}^n \equiv \boldsymbol{q}_{i+1}^\dagger \boldsymbol{d}^n \boldsymbol{q}_i = \left[[\mathbf{dq}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}]\,\boldsymbol{C}_i^{\dagger,-1}\right]^\dagger \boldsymbol{d}^n\boldsymbol{q}_i \tag{3.47}$$

$$= \boldsymbol{C}_i^{-1}\left[\underbrace{\boldsymbol{q}_i^\dagger \boldsymbol{d}^{n+1}\boldsymbol{q}_i}_{\boldsymbol{S}_{i,i}^{n+1}} - \boldsymbol{M}_i\underbrace{\boldsymbol{q}_i^\dagger \boldsymbol{d}^n\boldsymbol{q}_i}_{\boldsymbol{S}_{i,i}^{n}} - \boldsymbol{C}_{i-1}^\dagger\underbrace{\boldsymbol{q}_{i-1}^\dagger \boldsymbol{d}^n\boldsymbol{q}_i}_{\boldsymbol{S}_{i-1,i}^{n}}\right] \tag{3.48}$$

$$= \boxed{\boldsymbol{C}_i^{-1}\left[\boldsymbol{S}_{i,i}^{n+1} - \boldsymbol{M}_i\boldsymbol{S}_{i,i}^n - \boldsymbol{C}_{i-1}^\dagger\boldsymbol{S}_{i-1,i}^n\right]} \tag{3.49}$$

Similarly

$$\boldsymbol{S}_{i+1,i+1}^n \equiv \boldsymbol{q}_{i+1}^\dagger \boldsymbol{d}^n \boldsymbol{q}_{i+1} = \left[[\mathbf{dq}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}]\,\boldsymbol{C}_i^{\dagger,-1}\right]^\dagger \boldsymbol{d}^n\,[\mathbf{dq}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}]\,\boldsymbol{C}_i^{\dagger,-1} \tag{3.50}$$

$$\tag{3.51}$$

$$
\begin{aligned}
\boldsymbol{S}^{n}_{i+1,i+1} = \boldsymbol{C}_i^{-1} \Big[& \boldsymbol{S}^{n+2}_{i,i} + \boldsymbol{M}_i \boldsymbol{S}^{n}_{i,i} \boldsymbol{M}_i + \boldsymbol{C}^{\dagger}_{i-1} \boldsymbol{S}^{n}_{i-1,i-1} \boldsymbol{C}_{i-1} \\
& - \underbrace{\boldsymbol{q}_i^{\dagger} \boldsymbol{d}^{n+1} \boldsymbol{q}_i}_{\boldsymbol{S}^{n+1}_{i,i}} \boldsymbol{M}_i - \boldsymbol{M}_i \underbrace{\boldsymbol{q}_i^{\dagger} \boldsymbol{d}^{n+1} \boldsymbol{q}_i}_{\boldsymbol{S}^{n+1}_{i,i}} \\
& - \underbrace{\boldsymbol{q}_i^{\dagger} \boldsymbol{d}^{n+1} \boldsymbol{q}_{i-1}}_{\boldsymbol{S}^{n+1}_{i,i-1}} \boldsymbol{C}_{i-1} - \boldsymbol{C}^{\dagger}_{i-1} \underbrace{\boldsymbol{q}_{i-1}^{\dagger} \boldsymbol{d}^{n+1} \boldsymbol{q}_i}_{\boldsymbol{S}^{n+1}_{i-1,i}} \\
& + \boldsymbol{M}_i \underbrace{\boldsymbol{q}_i^{\dagger} \boldsymbol{d}^{n} \boldsymbol{q}_{i-1}}_{\boldsymbol{S}^{n}_{i,i-1}} \boldsymbol{C}_{i-1} + \boldsymbol{C}^{\dagger}_{i-1} \underbrace{\boldsymbol{q}_{i-1}^{\dagger} \boldsymbol{d}^{n} \boldsymbol{q}_i}_{\boldsymbol{S}^{n}_{i-1,i}} \boldsymbol{M}_i \Big] \boldsymbol{C}_i^{\dagger,-1} \qquad (3.52) \\
= \boldsymbol{C}_i^{-1} \Big[& \boldsymbol{S}^{n+2}_{i,i} + \boldsymbol{M}_i \boldsymbol{S}^{n}_{i,i} \boldsymbol{M}_i + \boldsymbol{C}^{\dagger}_{i-1} \boldsymbol{S}^{n}_{i-1,i-1} \boldsymbol{C}_{i-1} \\
& - P(\boldsymbol{S}^{n+1}_{i,i} \boldsymbol{M}_i) - P(\boldsymbol{S}^{n+1}_{i,i-1} \boldsymbol{C}_{i-1}) + P(\boldsymbol{M}_i \boldsymbol{S}^{n}_{i,i-1} \boldsymbol{C}_{i-1}) \Big] \boldsymbol{C}_i^{\dagger,-1} \qquad (3.53)
\end{aligned}
$$

where we introduced the permutation operator as $P(\boldsymbol{A}) = \boldsymbol{A} + \boldsymbol{A}^{\dagger}$. Setting $n = 0$ we get

$$
\boldsymbol{I} = \boldsymbol{C}_i^{-1} \Big[\boldsymbol{S}^{2}_{i,i} + \boldsymbol{M}_i^2 + \boldsymbol{C}^{\dagger}_{i-1} \boldsymbol{C}_{i-1} - P\left(\boldsymbol{M}_i^2\right) - P\left(\boldsymbol{S}^{1}_{i,i-1} \boldsymbol{C}_{i-1}\right) + 0 \Big] \boldsymbol{C}_i^{\dagger,-1} \qquad (3.54)
$$

$$
\implies \boldsymbol{C}_i^2 = \boldsymbol{C}_i \boldsymbol{C}_i^{\dagger} = \boxed{\boldsymbol{S}^{2}_{i,i} + \boldsymbol{M}_i^2 + \boldsymbol{C}^{\dagger}_{i-1} \boldsymbol{C}_{i-1} - P\left(\boldsymbol{S}^{1}_{i,i} \boldsymbol{M}_i\right) - P\left(\boldsymbol{S}^{1}_{i,i-1} \boldsymbol{C}_{i-1}\right)} \qquad (3.55)
$$

$$
\implies \boldsymbol{C}_i \boldsymbol{C}_i^{\dagger} \equiv \boldsymbol{q}_i^{\dagger} \boldsymbol{d} \boldsymbol{q}_{i+1} \boldsymbol{q}_{i+1}^{\dagger} \boldsymbol{d} \boldsymbol{q}_i = \boldsymbol{q}_i^{\dagger} \boldsymbol{d}^2 \boldsymbol{q}_i \qquad (3.56)
$$

The hope is that we can use the Cholesky QR algorithm to take the effective matrix square root to get the $\boldsymbol{C}_i$.

3.4 Efficient generation of the self-energy moments

Since the self-energy is defined as a convolution between the interacting Green's function and the screened Coulomb potential in GW we can use the known formula for the moment distribution of a convolution of two quantities to get

$$
\Sigma^{(n,<)}_{pq} = \sum_{ia,jb,k} \sum_{t=0}^{n} \binom{n}{t} (-1)^t \epsilon_k^{n-t} (pk \mid ia) \eta^{(t)}_{ia,jb} (qk \mid jb), \qquad (3.57)
$$

where we have defined the density-density response moment as

$$
\eta^{(n)}_{ia,jb} = \sum_{v} \left(X^{v}_{ia} + Y^{v}_{ia}\right) \Omega_v^n \left(X^{v}_{jb} + Y^{v}_{jb}\right) \qquad (3.58)
$$

3.4.1 An alternative formulation of the RPA polarizability with the density response moments

The Casida equation is given by

$$
\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ -\boldsymbol{B} & -\boldsymbol{A} \end{pmatrix} \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix} \begin{pmatrix} \boldsymbol{\Omega} & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{\Omega} \end{pmatrix} = \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix} \qquad (3.59)
$$

where the excitation $\boldsymbol{X}$ and de-excitation $\boldsymbol{Y}$ eigenvectors form the biorthogonal set as

$$(\boldsymbol{X}+\boldsymbol{Y})(\boldsymbol{X}-\boldsymbol{Y})^T=(\boldsymbol{X}+\boldsymbol{Y})^T(\boldsymbol{X}-\boldsymbol{Y})=\boldsymbol{I} \tag{3.60}$$

The $\boldsymbol{A}$ and $\boldsymbol{B}$ matrices are defined as

$$A_{ia,jb}=(\epsilon_a-\epsilon_i)\,\delta_{ij}\delta_{ab}+\mathcal{K}_{ia,bj} \tag{3.61}$$
$$B_{ia,jb}=\mathcal{K}_{ia,jb} \tag{3.62}$$

and I take the direct approximation, so $\mathcal{K}_{ia,jb}=(ia\mid jb)=\mathcal{K}_{ia,bj}$. As I showed earlier this year, these neutral excitation energies $\boldsymbol{\Omega}$ define the poles of the polarizability $\chi_{\mathrm{RPA}}(\omega)$ as

$$\chi_{\mathrm{RPA}}(\omega)=\begin{bmatrix}\boldsymbol{X}&\boldsymbol{Y}\\ \boldsymbol{Y}&\boldsymbol{X}\end{bmatrix}\begin{bmatrix}\omega\boldsymbol{I}-\boldsymbol{\Omega}&\boldsymbol{0}\\ \boldsymbol{0}&-\omega\boldsymbol{I}-\boldsymbol{\Omega}\end{bmatrix}^{-1}\begin{bmatrix}\boldsymbol{X}&\boldsymbol{Y}\\ \boldsymbol{Y}&\boldsymbol{X}\end{bmatrix}^{T}. \tag{3.63}$$

where $\chi_{\mathrm{RPA}}(\omega)=(\chi_0(\omega)^{-1}-\mathcal{K})^{-1}$, where $\chi_0(\omega)$ is the irreducible polarizability of the reference state. Because of the bosonic-like symmetry in equation 3.63, we can write a more compact form of the RPA polarizability as

$$\eta(\omega)=(\mathbf{X}+\mathbf{Y})(\omega\mathbf{I}-\boldsymbol{\Omega})^{-1}(\mathbf{X}+\mathbf{Y})^T. \tag{3.64}$$

However, here we are interested in the spectral moments of the compactified RPA polarizability in equation 3.64 over all RPA excitation energies, which is given as

$$\eta^{(n)}_{ia,jb}=-\frac{1}{\pi}\int_0^\infty \mathrm{Im}\left[\eta_{ia,jb}(\omega)\right]\omega^n d\omega=-\frac{1}{\pi}\left(\boldsymbol{X}+\boldsymbol{Y}\right)\left[\int_0^\infty \mathrm{Im}\left[(\omega\mathbf{I}-\boldsymbol{\Omega})^{-1}\right]\omega^n d\omega\right]\left(\boldsymbol{X}+\boldsymbol{Y}\right)^T. \tag{3.65}$$

where in the final equality we have just factored out the frequency independent piece. Now, we can use the identity from complex analysis that $\frac{1}{-\boldsymbol{\Omega}+\boldsymbol{I}(\omega+i\eta)}=P\left(\frac{1}{\omega\boldsymbol{I}-\boldsymbol{\Omega}}\right)-i\pi\delta(\omega-\boldsymbol{\Omega})\implies \mathrm{Im}\left[\frac{1}{-\boldsymbol{\Omega}+\boldsymbol{I}(\omega+i\eta)}\right]=-\pi\delta(\omega-\boldsymbol{\Omega})$, where P is the Cauchy principal value, so

$$\int_0^\infty \mathrm{Im}\left[(\omega\mathbf{I}-\boldsymbol{\Omega})^{-1}\right]\omega^n d\omega=-\pi\int_0^\infty \delta(\omega-\boldsymbol{\Omega})\omega^n d\omega=-\pi\boldsymbol{\Omega}^n. \tag{3.66}$$

and plugging back and, we get $\boldsymbol{\eta}^{(n)}=\left(\boldsymbol{X}+\boldsymbol{Y}\right)\left(\boldsymbol{\Omega}^n\right)\left(\boldsymbol{X}+\boldsymbol{Y}\right)^T$.

3.4.2 Getting η with lower scaling

Now we will make use of the symmetric formulation of the RPA problem by Fillip Furche. The RPA eigenvalue problem is given by

$$\begin{pmatrix}\boldsymbol{A}&\boldsymbol{B}\\ -\boldsymbol{B}&-\boldsymbol{A}\end{pmatrix}\begin{pmatrix}\boldsymbol{X}\\ \boldsymbol{Y}\end{pmatrix}=\boldsymbol{\Omega}\begin{pmatrix}\boldsymbol{X}\\ \boldsymbol{Y}\end{pmatrix} \tag{3.67}$$

so we can get the coupled equations

$$\boldsymbol{A}\boldsymbol{X}+\boldsymbol{B}\boldsymbol{Y}=\boldsymbol{\Omega}\boldsymbol{X} \tag{3.68}$$
$$-\boldsymbol{B}\boldsymbol{X}-\boldsymbol{A}\boldsymbol{Y}=\boldsymbol{\Omega}\boldsymbol{Y} \tag{3.69}$$

Adding and subtracting these two equations, we get

$$(\boldsymbol{A}-\boldsymbol{B})(\boldsymbol{X}-\boldsymbol{Y}) = (\boldsymbol{X}+\boldsymbol{Y})\,\boldsymbol{\Omega} \implies (\boldsymbol{A}-\boldsymbol{B}) = (\boldsymbol{X}+\boldsymbol{Y})\,\boldsymbol{\Omega}\,(\boldsymbol{X}-\boldsymbol{Y})^{\dagger} \implies \boxed{\boldsymbol{\eta}^{(1)} = \boldsymbol{A}-\boldsymbol{B}} \tag{3.70}$$

$$(\boldsymbol{A}+\boldsymbol{B})(\boldsymbol{X}+\boldsymbol{Y}) = (\boldsymbol{X}-\boldsymbol{Y})\,\boldsymbol{\Omega} \implies (\boldsymbol{A}+\boldsymbol{B}) = (\boldsymbol{X}-\boldsymbol{Y})\,\boldsymbol{\Omega}\,(\boldsymbol{X}+\boldsymbol{Y})^{\dagger} \tag{3.71}$$

But then also notice that since $\boldsymbol{\eta}^{(0)} = (\boldsymbol{X}+\boldsymbol{Y})\,(\boldsymbol{X}+\boldsymbol{Y})^{\dagger}$, we can write $\boldsymbol{\eta}^{(1)} = \boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)}$. Continuing on, we arrive at Furche's symmetric formulation of the RPA problem as

$$(\boldsymbol{A}-\boldsymbol{B})(\boldsymbol{A}+\boldsymbol{B})(\boldsymbol{X}+\boldsymbol{Y}) = (\boldsymbol{X}+\boldsymbol{Y})\,\boldsymbol{\Omega}^2 \tag{3.72}$$

By right multiplying by $(\boldsymbol{X}+\boldsymbol{Y})^{\dagger}$, this leads to

$$(\boldsymbol{A}-\boldsymbol{B})(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)} = \boldsymbol{\eta}^{(2)} \tag{3.73}$$

This is suggestive of a recursive relation with the form

$$\boldsymbol{\eta}^{(m)} = (\boldsymbol{A}-\boldsymbol{B})(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(m-2)} \tag{3.74}$$

$$\tag{3.75}$$

But we have already found that $\boldsymbol{\eta}^{(1)} = \boldsymbol{A}-\boldsymbol{B} \implies \boldsymbol{\eta}^{(2)} = \boldsymbol{\eta}^{(1)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)}$. Now plug in $\boldsymbol{\eta}^{(1)} = \boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)}$ to get

$$\boldsymbol{\eta}^{(2)} = \boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)} = \left[\boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\right]^2\boldsymbol{\eta}^{(0)} \tag{3.76}$$

Repeating gives

$$\boxed{\boldsymbol{\eta}^{(m)} = \left[\boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\right]^m\boldsymbol{\eta}^{(0)}} \tag{3.77}$$

and then to initialize, consider

$$\boldsymbol{A}-\boldsymbol{B} = \boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)} \tag{3.78}$$

$$(\boldsymbol{A}-\boldsymbol{B})(\boldsymbol{A}+\boldsymbol{B}) = \boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\,\boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B}) = \left[\boldsymbol{\eta}^{(0)}\,(\boldsymbol{A}+\boldsymbol{B})\right]^2 \tag{3.79}$$

$$\implies \boldsymbol{\eta}^{(0)} = \left[(\boldsymbol{A}-\boldsymbol{B})(\boldsymbol{A}+\boldsymbol{B})\right]^{1/2}(\boldsymbol{A}+\boldsymbol{B})^{-1} \tag{3.80}$$

We do assume a lot of things in going to equation 3.80, but I have

```
assert np.all(np.linalg.eigvals(ApB) > 0)
```

3.4.3 Bringing it back to outline an efficient procedure for Computing Self-Energy Moments

The moments of the lesser and greater parts of the self-energy have the form

$$\begin{aligned}
\Sigma^{(n,<)}_{pq} &= \sum_{ia,jb,k}\sum_{t=0}^{n}\binom{n}{t}(-1)^t\epsilon_k^{n-t}(pk\mid ia)\eta^{(t)}_{ia,jb}(qk\mid jb),\\
\Sigma^{(n,>)}_{pq} &= \sum_{ia,jb,c}\sum_{t=0}^{n}\binom{n}{t}\epsilon_c^{n-t}(pc\mid ia)\eta^{(t)}_{ia,jb}(qc\mid jb),
\end{aligned} \tag{3.81}$$

respectively. The energies ϵ and integrals $(pq \mid rs)$ are known to us from the prior mean-field calculation, but the density-density moments $\eta^{(t)}$ are not, so we need to compute them. We find them to be defined as

$$\boldsymbol{\eta}^{(m)} = \left[\boldsymbol{\eta}^{(0)}(\mathbf{A}+\mathbf{B})\right]^m \boldsymbol{\eta}^{(0)}, \tag{3.82}$$

with the initial

$$\boldsymbol{\eta}^{(0)} = [(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B})]^{\frac{1}{2}}(\mathbf{A}+\mathbf{B})^{-1} \tag{3.83}$$

where we have $\mathbf{A}$ and $\mathbf{B}$ with components $A_{ia,jb} = \delta_{ij}\delta_{ab}\left(\epsilon_a - \epsilon_i\right) + (ia \mid jb)$ and $B_{ia,jb} = (ia \mid jb)$. Note that the scaling of the above would be improved by introducing low-rank approximations to the electron repulsion integrals (ERIs) via the Cholesky decomposition or tensor hyper-contraction, which express these 4-index quantities as a sum of two 3- or five 2-index tensors, respectively.

3.5 Avoiding auxiliary space

To avoid using the large auxiliary quantities, we introduce an operator which will project onto the block Lanczos space as $\mathbf{S}^{(n)}_{i,j} = \mathbf{q}_i^\dagger \mathbf{d}^n \mathbf{q}_j$. But due to the orthogonality of the Krylov subspace, we have $q_i^\dagger q_j = 0$ for $i \neq j$, so $\mathbf{S}^{(n)}_{i,j} = 0$ for $|i-j| > 1$. Furthermore, due to Hermiticity, $\mathbf{S}^{(n)}_{i,j} = \mathbf{S}^{(n)\dagger}_{j,i}$. We can start with the initial definition $\mathbf{S}^{(n)}_{1,1} = \mathbf{q}_1^\dagger \mathbf{d}^n \mathbf{q}_1 = \mathbf{L}^{-1}\boldsymbol{\Sigma}^{(n)}\mathbf{L}^{-1,\dagger}$.

Then we know that the recurrence relation is given by

$$\mathbf{q}_{i+1}\mathbf{C}_i^\dagger = [\mathbf{d}\mathbf{q}_i - \mathbf{q}_i\mathbf{H}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}] \tag{3.84}$$

I will not repeat the algebraic manipulations, but Backhouse shows in his thesis that the following relations follow

$$\begin{aligned}
\mathbf{S}^{(n)}_{i+1,i} =& \mathbf{q}_{i+1}^\dagger \mathbf{d}^n \mathbf{q}_i = \mathbf{C}_i^{-1}\left[\mathbf{S}^{(n+1)}_{i,i} - \mathbf{H}_i\mathbf{S}^{(n)}_{i,i} - \mathbf{C}^\dagger_{i-1}\mathbf{S}^{(n)}_{i-1,i}\right] \\
\mathbf{S}^{(n)}_{i+1,i+1} =& \mathbf{q}_{i+1}^\dagger \mathbf{d}^n \mathbf{q}_{i+1} = \mathbf{C}_i^{-1}\Big[\mathbf{S}^{(n+2)}_{i,i} + \mathbf{H}_i\mathbf{S}^{(n)}_{i,i}\mathbf{H}_i + \mathbf{C}^\dagger_{i-1}\mathbf{S}^{(n)}_{i-1,i-1}\mathbf{C}_{i-1} \\
& - P\left(\mathbf{S}^{(n+1)}_{i,i}\mathbf{H}_i\right) + P\left(\mathbf{H}_i\mathbf{S}^{(n)}_{i,i-1}\mathbf{C}_{i-1}\right) - P\left(\mathbf{S}^{(n+1)}_{i,i-1}\mathbf{C}_{i-1}\right)\Big]\mathbf{C}_i^{-1,\dagger},
\end{aligned} \tag{3.85}$$

and then solving for $\mathbf{C}_i^2$

$$\mathbf{C}_i^2 = \left[\mathbf{S}^{(2)}_{i,i} + \mathbf{H}_i^2 + \mathbf{C}^\dagger_{i-1}\mathbf{C}_{i-1} - P\left(\mathbf{S}^{(1)}_{i,i}\mathbf{H}_i\right) - P\left(\mathbf{S}^{(1)}_{i,i-1}\mathbf{C}_{i-1}\right)\right] \tag{3.86}$$

and we can also find the on-diagonal $\mathbf{H}$ matrices using

$$\mathbf{H}_i = \mathbf{q}_i^\dagger \mathbf{d}\mathbf{q}_i = \mathbf{S}^{(1)}_{i,i}. \tag{3.87}$$

Now, we know how to compute every quantity of equation 3.25.

3.6 Spectral Function

When we diagonalize the upfolded Hamiltonian, the eigenpairs are composed of charged excitation energies E_k (analogous to QP energies from the QP equation, but without a diagonal approximation to the self-energy) and eigenvectors $\boldsymbol{u}_k$, which can be transformed into the Dyson orbitals $\boldsymbol{\psi}_k$ via

$$\boldsymbol{\psi}_k = \boldsymbol{L}\boldsymbol{P}\boldsymbol{u}_k \tag{3.88}$$

where $\boldsymbol{P}$ is the projection operator into the physical space

```
eigvec_phys = extracted_eigvecs[j][:mf.nbsf]
```

and then $\boldsymbol{L}$, depending on whether we have an occupied or virtual state, is

```
        moments_less = [form_moment(i, "lesser", mf) for i in range(2*
    n_initial+2)]
        moments_great = [form_moment(i, "greater", mf) for i in range(2*
    n_initial+2)]
        L_less = np.linalg.cholesky(moments_less[0], upper=False)
        L_great = np.linalg.cholesky(moments_great[0], upper=False)
        L = (L_less, L_great)
```

Then, we can construct the spectral function via

$$A(\omega) = \sum_k ||\boldsymbol{\psi}_k||^2 \underbrace{\frac{1}{\pi}\frac{\eta}{\left(\omega - E_k\right)^2 + \eta^2}}_{\text{Lorentzian}} \tag{3.89}$$

where η is a broadening parameter. In general, the spectral function is known to have the form

$$A(\omega) = -\frac{1}{\pi}\text{Im}G\left(\omega\right) \tag{3.90}$$

and the Dyson equation is

$$G(\omega) = G_0(\omega) + G_0(\omega)\Sigma(\omega)G(\omega) = \frac{1}{\underbrace{\omega - \epsilon_0}_{G_0(\omega)^{-1}} - \Sigma(\omega)} \tag{3.91}$$

but then we know that $\Sigma(\omega) = \Sigma_R(\omega) + i\Sigma_I(\omega)$, so we can write

$$G(\omega) = \frac{1}{\omega - \epsilon_0 - \Sigma_R(\omega) - i\Sigma_I(\omega)} \tag{3.92}$$

Defining $x(\omega) = \omega - \epsilon_0 - \Sigma_R(\omega)$ and $y(\omega) = \Sigma_I(\omega)$, we can rewrite the Dyson equation as

$$G(\omega) = \frac{1}{x(\omega) - iy(\omega)} \times \frac{x(\omega) + iy(\omega)}{x(\omega) + iy(\omega)} = \frac{x(\omega) + iy(\omega)}{\left(x(\omega)\right)^2 + \left(y(\omega)\right)^2} \implies \text{Im}G(\omega) = \frac{y(\omega)}{\left(x(\omega)\right)^2 + \left(y(\omega)\right)^2} \tag{3.93}$$

We have the following fully analytic expression for the self-energy

$$\Sigma_{pp}^{\text{corr}}(\omega) = \sum_{\mu}^{\text{RPA}} \left(\sum_{i}^{\text{occupied}} \frac{w_{pi}^{\mu} w_{ip}^{\mu}}{\omega - (\epsilon_i - \Omega_\mu) + i\eta} + \sum_{a}^{\text{virtual}} \frac{w_{pa}^{\mu} w_{ap}^{\mu}}{\omega - (\epsilon_a + \Omega_\mu) - i\eta} \right) \tag{3.94}$$

with

$$w_{pq}^{\mu} = \sum_{jb} (pq|jb) \left(X_{jb}^{\mu} + Y_{jb}^{\mu} \right) \tag{3.95}$$

So

$$\Sigma_{pp}^{R}(\omega) = \sum_{\mu}^{\text{RPA}} \left(\sum_{i}^{\text{occupied}} \frac{w_{pi}^{\mu} w_{ip}^{\mu}}{\omega - (\epsilon_i - \Omega_{\mu})} + \sum_{a}^{\text{virtual}} \frac{w_{pa}^{\mu} w_{ap}^{\mu}}{\omega - (\epsilon_a + \Omega_{\mu})} \right) \tag{3.96}$$

and because $\frac{1}{\omega \pm i\eta} = \mathcal{P}\frac{1}{\omega} \mp i\pi\delta(\omega)$, we can write

$$\Sigma_{pp}^{I}(\omega) = \pi \sum_{\mu}^{\text{RPA}} \left(\sum_{a}^{\text{virtual}} w_{pa}^{\mu} w_{ap}^{\mu} \delta(\omega - (\epsilon_a + \Omega_{\mu}) - \sum_{i}^{\text{occupied}} w_{pi}^{\mu} w_{ip}^{\mu} \delta(\omega - (\epsilon_i - \Omega_{\mu})) \right) \tag{3.97}$$

This brings us back to a Lorentzian

$$A(\omega) = -\frac{1}{\pi} \sum_{p} \frac{\eta(\omega)}{\left(\omega - \epsilon_0 - \Sigma_{pp}^{R}(\omega)\right)^2 + \eta(\omega)^2} \tag{3.98}$$

with $\eta(\omega) \equiv \Sigma_{pp}^{I}(\omega)$. We need to compute

$$\eta(\omega) = \left(\sum_{\mu}^{\text{RPA}} \left(\sum_{a}^{\text{virtual}} w_{pa}^{\mu} w_{ap}^{\mu} \left(\frac{\gamma}{\left(\omega - (\epsilon_a + \Omega_{\mu})\right)^2 + \gamma^2} \right) - \sum_{i}^{\text{occupied}} w_{pi}^{\mu} w_{ip}^{\mu} \left(\frac{\gamma}{\left(\omega - (\epsilon_i - \Omega_{\mu})\right)^2 + \gamma^2} \right) \right) \right)_p \tag{3.99}$$

at each $\omega = E_{QP}$ that we found in the QP equation. Meanwhile, we know that the QP renormalization factor is given by

$$Z_p = \left(1 - \frac{\partial \Sigma_{pp}^{R}(\omega)}{\partial \omega} \bigg|_{\omega = E_{QP}} \right)^{-1} \tag{3.100}$$

So I need to prove the equivalence between this form and the one I have above using $\eta(\omega)$.

3.7 6/2 Conclusions

3.7.1 Hypothesis

If one maintains orthogonality in the Krylov subspace, the Ritz eigenpairs will converge to exact diagonalization within numerical precision. However, it is a fact that if the residual of the recurrence relation 3.113 loses full rank, the off-diagonal elements get a high condition number and are no longer invertible. So we have seen that MC-GW can replicate the block Lanczos up until this condition is met, but no longer afterwards. So even though MC-GW enforces orthogonality of the Krylov subspace even after the residual does lose its full rank, the off-diagonal elements are then no longer invertible, which explains the numerical issues that I observe after this point.

3.7.2 Systems of Study

Given my physical dimension n_{nocc}, the auxiliary dimension is of size $n_{\text{channel}} \times n_{\text{virt}} \times n_{\text{occ}}$, where n_{channel} is either n_{occ} or n_{virt}, depending on whether we are dealing with the lesser or greater channel; it becomes clear that the block size, which is determined by the physical dimension, will be $O(n_{\text{occ}})$, whereas the size of the auxiliary dimension will be $O(n_{\text{channel}} \times n_{\text{virt}} \times n_{\text{occ}})$. I have been doing my tests up until now with a minimal basis, where at least one of these ns is small, so I wonder if when I move to a larger basis there will be a noticeable difference with the large discrepancy between n and n^3. Perhaps we will be able to exactly converge the low-lying orbital MO QPEs, but not the higher ones, but this is typically all that is desired of the method.

3.8 6/10 plots

As you probably have seen, in the theory of MC-GW, the equations come in two forms; the first gives the explicit definition of the quantity in terms of a projection on the Krylov subspace and the second gives the recurrence relation that is used in MC-GW.

$$\boldsymbol{S}^n_{i+1,i} \equiv \boldsymbol{q}^\dagger_{i+1} \boldsymbol{d}^n \boldsymbol{q}_i = \boldsymbol{C}_i^{-1} \left[\boldsymbol{S}^{n+1}_{i,i} - \boldsymbol{M}_i \boldsymbol{S}^n_{i,i} - \boldsymbol{C}^\dagger_{i-1} \boldsymbol{S}^n_{i-1,i} \right] \tag{3.101}$$

or

$$\begin{aligned} \boldsymbol{S}^n_{i+1,i+1} \equiv \boldsymbol{q}^\dagger_{i+1} \boldsymbol{d}^n \boldsymbol{q}_{i+1} = \boldsymbol{C}_i^{-1} \Big[& \boldsymbol{S}^{n+2}_{i,i} + \boldsymbol{M}_i \boldsymbol{S}^n_{i,i} \boldsymbol{M}_i + \boldsymbol{C}^\dagger_{i-1} \boldsymbol{S}^n_{i-1,i-1} \boldsymbol{C}_{i-1} \\ & - P(\boldsymbol{S}^{n+1}_{i,i} \boldsymbol{M}_i) - P(\boldsymbol{S}^{n+1}_{i,i-1} \boldsymbol{C}_{i-1}) + P(\boldsymbol{M}_i \boldsymbol{S}^n_{i,i-1} \boldsymbol{C}_{i-1}) \Big] \boldsymbol{C}_i^{\dagger,-1} \end{aligned} \tag{3.102}$$

or

$$\boldsymbol{C}_i^2 = \boldsymbol{C}_i \boldsymbol{C}_i^\dagger = \boldsymbol{S}^2_{i,i} + \boldsymbol{M}_i^2 + \boldsymbol{C}^\dagger_{i-1} \boldsymbol{C}_{i-1} - P\left(\boldsymbol{S}^1_{i,i} \boldsymbol{M}_i\right) - P\left(\boldsymbol{S}^1_{i,i-1} \boldsymbol{C}_{i-1}\right) \tag{3.103}$$

$$\tag{3.104}$$

with

$$\boldsymbol{C}_i = \boldsymbol{q}_i^\dagger \boldsymbol{d} \boldsymbol{q}_{i+1} \tag{3.105}$$

The curve that I label as MCGW dense corresponds to the middle equality, MCGW sparse corresponds to the right equality, and the black curve is the standard block Lanczos with matrix vector products. Then, I also display the solution that Booth code gives. There is a reason why I am not able to get the MCGW sparse curve to match the dense curve. Recall that to solve 3.107, we want to perform a Cholesky decomposition of the matrix $\boldsymbol{C}_i^2$ to get $\boldsymbol{C}_i$. So the matrix should be symmetric and also positive semidefinite. I always begin by trying to use the modified Cholesky algorithm to perform this matrix square root. However, even in the exact dense theory, the matrix $\boldsymbol{C}_i^2$ is not always even positive semidefinite (it has negative eigenvalues), so I cannot use the modified Cholesky algorithm to solve in that case. In addition, due to the rise in numerical imprecision, the matrix also becomes non-symmetric, so in these cases I just symmetrize it, get the eigendecomposition, and do the matrix square root that way. But in exact arithmetic, this is a Hermitian theory, so I shouldn't have to symmetrize like this and in doing so I introduce some error. As can be seen, this gives a fairly good match with what Booth codes predict; we are probably doing similar things.

3.9 Canonical orthogonalization

The numerical issue with the theory ofMCGW seems to lie with then obtaining the appropriate off-diagonals from

$$\boldsymbol{C}_i^2 = \boldsymbol{C}_i\boldsymbol{C}_i^\dagger = \boldsymbol{S}_{i,i}^2 + \boldsymbol{M}_i^2 + \boldsymbol{C}_{i-1}^\dagger\boldsymbol{C}_{i-1} - P\left(\boldsymbol{S}_{i,i}^1\boldsymbol{M}_i\right) - P\left(\boldsymbol{S}_{i,i-1}^1\boldsymbol{C}_{i-1}\right) \tag{3.106}$$

$$\tag{3.107}$$

Let us now try to achieve the same result using canonical orthogonalization. Once we build $\boldsymbol{C}_i^2$ we can take the eigendecomposition of the matrix to get

$$\boldsymbol{C}_i^2 = \boldsymbol{U}_i\boldsymbol{\Lambda}_i\boldsymbol{U}_i^\dagger \tag{3.108}$$

3.10 Krylov subspace orthogonality contradiction

3.10.1 Theoretical background

There seems to be a contradiction regarding the orthogonality of the Krylov vectors in MCGW. We know that I am able to see convergence to ED to within machine precision when I explicitly build up a Krylov subspace, taking care to ensure orthogonality between Krylov vectors, by both doing Gram-Schmidt on each new vector with all previous ones and then throwing away the small singular values if there is a lack of new directions. This is the exact block Lanczos with matrix vector products curve that I sent Tuesday. Now, MCGW claims to enforce orthogonality in the Krylov subspace by doing $\boldsymbol{S}_{i,j}^{(0)} = \delta_{ij}\boldsymbol{I}$ with

$$\boldsymbol{S}_{i,j}^{(n)} = \boldsymbol{q}_i^\dagger \boldsymbol{d}^n \boldsymbol{q}_j \tag{3.109}$$

This is what rationalizes the simplification in the rewriting of the coupling block $\boldsymbol{W}$ as $\boldsymbol{R}$ padded with 0s. For a brief refresher, see the flow in the equations below in the example of a Krylov subspace which is containing only two block vectors.

$$\tilde{\boldsymbol{H}}_{\text{upfolded}}^{\text{Lanczos Iter 2}} = \tilde{\boldsymbol{Q}}^{(2,\dagger)}\boldsymbol{H}_{\text{Upfolded}}^{G_0W_0}\tilde{\boldsymbol{Q}}^{(2)} \tag{3.110}$$

$$= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1^\dagger \quad \boldsymbol{q}_2^\dagger] \end{pmatrix} \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix} \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1 \quad \boldsymbol{q}_2] \end{pmatrix} \tag{3.111}$$

$$= \begin{pmatrix} \boldsymbol{F} + \boldsymbol{\Sigma}(\infty) & [\underbrace{\boldsymbol{W}\boldsymbol{q}_1}_{\boldsymbol{\mathcal{R}}\boldsymbol{\mathcal{Q}}^\dagger\boldsymbol{\mathcal{Q}}=\boldsymbol{\mathcal{R}}} \quad \underbrace{\boldsymbol{W}\boldsymbol{q}_2}_{\boldsymbol{0}}] \\ [\underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{\mathcal{Q}}^\dagger\boldsymbol{\mathcal{Q}}\boldsymbol{\mathcal{R}}^\dagger=\boldsymbol{\mathcal{R}}^\dagger} \quad \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{0}}] & \begin{pmatrix} \underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{d}\boldsymbol{q}_1}_{\boldsymbol{M}_1} & \underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{d}\boldsymbol{q}_2}_{\boldsymbol{C}_1} \\ \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{d}\boldsymbol{q}_1}_{\boldsymbol{C}_1^\dagger} & \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{d}\boldsymbol{q}_2}_{\boldsymbol{M}_2} \end{pmatrix} \end{pmatrix} \tag{3.112}$$

But now recall that MCGW relies on the three-term Lanczos recurrence of

$$\mathbf{q}_{i+1}\mathbf{C}_i^\dagger = [\mathbf{d}\mathbf{q}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}] \implies \mathbf{q}_{i+1} = [\mathbf{d}\mathbf{q}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}]\,\boldsymbol{C}_i^{\dagger,-1} \tag{3.113}$$

which is used in the definition of the $\boldsymbol{S}$es. Crucially, orthogonality is never enforced in these relations, so after sufficient iterations we get $\boldsymbol{q}_i^\dagger \boldsymbol{q}_j \neq 0$ for $i \neq j$. This is in contradiction with our initial assumption that the coupling block $\boldsymbol{W}$ can be written as $\boldsymbol{R}$ padded with zeros. So it is perfectly natural to not see MCGW converge to ED to within machine precision as is seen in the MCGW sparse curve of my plots. Recall that the MCGW dense curve was obtained by explicitly plugging in the Krylov vector from the exact block Lanczos with matrix vector products into 3.109; this was done with genuinely orthogonalized Krylov vectors and hence we saw an exact convergence to ED similar to that of the matrix vector product implementation.

3.10.2 Comment about the effect of minimal basis

It is true that the fact that I was experiencing a lack of new directions was an artifact of me using a minimal basis in my calculations. This put the size of the physical space, which scales as $O(O+V)$, on par with the auxiliary space, which scales as $O(O^2V)$ or $O(V^2O)$, depending on whether we are dealing with the lesser or greater channel, respectively. The dimension of the auxiliary space will grow faster than that of the physical space as we move to larger system sizes, so it is expected that this effect would be negligible by using a larger basis in my calculation. However, this does not solve the issue of not being able to maintain orthogonality by doing Gram-Schmidt of a new vector on all previous vectors. But, I did a numerical tests of the effect of a lack of reorthogonalization on the Ritz values a few months ago, and the effect was negligible.

3.10.3 Next steps

Should I run this with double zeta basis in C++? At the end of the day, I have just speculated, and I haven't proven anything, so it would be nice to see if what I hypothesis plays out in practice.

3.11 A hierarchy of approximations to block Lanczos with reorthogonalization

To simplify the discussion, I will just consider a single channel here.

3.11.1 Exact solution

The QP energies are given by diagonalizing the upfolded Hamiltonian

$$\boldsymbol{H}_{\text{upfolded}} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix} \tag{3.114}$$

We can accomplish the same thing by considering the Ritz values as obtained by projecting the upfolded Hamiltonian, which is called exact $\boldsymbol{d}$, onto an orthonormal Krylov subspace as

$$\tilde{\boldsymbol{H}}^{\text{Lanczos Iter 3}}_{\text{upfolded}} = \tilde{\boldsymbol{Q}}^{(3,\dagger)} \boldsymbol{H}_{\text{upfolded}} \tilde{\boldsymbol{Q}}^{(3)} \tag{3.115}$$

$$= \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1^\dagger \quad \boldsymbol{q}_2^\dagger \quad \boldsymbol{q}_3^\dagger] \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix} \begin{pmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & [\boldsymbol{q}_1 \quad \boldsymbol{q}_2 \quad \boldsymbol{q}_3] \end{pmatrix} \tag{3.116}$$

$$= \begin{pmatrix} \boldsymbol{F} & [\underbrace{\boldsymbol{W}\boldsymbol{q}_1}_{\boldsymbol{\mathcal{R}}\boldsymbol{q}_1^\dagger\boldsymbol{q}_1=\boldsymbol{\mathcal{R}}} \quad \underbrace{\boldsymbol{W}\boldsymbol{q}_2}_{\boldsymbol{\mathcal{R}}\boldsymbol{q}_1^\dagger\boldsymbol{q}_2=\boldsymbol{0}} \quad \underbrace{\boldsymbol{W}\boldsymbol{q}_3}_{\boldsymbol{\mathcal{R}}\boldsymbol{q}_1^\dagger\boldsymbol{q}_3=\boldsymbol{0}}] \\ [\underbrace{\boldsymbol{q}_1^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{q}_1^\dagger\boldsymbol{q}_1\boldsymbol{\mathcal{R}}^\dagger=\boldsymbol{\mathcal{R}}^\dagger} \quad \underbrace{\boldsymbol{q}_2^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{q}_1^\dagger\boldsymbol{q}_2\boldsymbol{\mathcal{R}}^\dagger=\boldsymbol{0}^\dagger} \quad \underbrace{\boldsymbol{q}_3^\dagger\boldsymbol{W}^\dagger}_{\boldsymbol{q}_1^\dagger\boldsymbol{q}_3\boldsymbol{\mathcal{R}}^\dagger=\boldsymbol{0}^\dagger}] & \begin{pmatrix} \underbrace{\boldsymbol{q}_1^\dagger d\boldsymbol{q}_1}_{\boldsymbol{M}_1} & \underbrace{\boldsymbol{q}_1^\dagger d\boldsymbol{q}_2}_{\boldsymbol{C}_1} & \underbrace{\boldsymbol{q}_1^\dagger d\boldsymbol{q}_3}_{\boldsymbol{0}} \\ \underbrace{\boldsymbol{q}_2^\dagger d\boldsymbol{q}_1}_{\boldsymbol{C}_1^\dagger} & \underbrace{\boldsymbol{q}_2^\dagger d\boldsymbol{q}_2}_{\boldsymbol{M}_2} & \underbrace{\boldsymbol{q}_2^\dagger d\boldsymbol{q}_3}_{\boldsymbol{C}_2} \\ \underbrace{\boldsymbol{q}_3^\dagger d\boldsymbol{q}_1}_{\boldsymbol{0}} & \underbrace{\boldsymbol{q}_3^\dagger d\boldsymbol{q}_2}_{\boldsymbol{C}_2^\dagger} & \underbrace{\boldsymbol{q}_3^\dagger d\boldsymbol{q}_3}_{\boldsymbol{M}_3} \end{pmatrix} \end{pmatrix} \tag{3.117}$$

This is what the matrix-vector product and exact $\boldsymbol{d}$ both do when we apply reorthogonalization.

3.11.2 Matrix-vector product without reorthogonalization

This is one step down from the exact answer. The only reason that we are able to set $\boldsymbol{q}_i d\boldsymbol{q}_j = \boldsymbol{0}$ for $i > j+1$ in 3.117 is because we are assuming an orthogonal basis where

$$d\boldsymbol{q}_j = \boldsymbol{C}_{j-1}\boldsymbol{q}_{j-1} + \boldsymbol{M}_j\boldsymbol{q}_j + \boldsymbol{C}_{j+1}\boldsymbol{q}_{j+1} \implies \boldsymbol{q}_{j+2}^\dagger d\boldsymbol{q}_j = \boldsymbol{q}_{j+2}^\dagger \left(\boldsymbol{C}_{j-1}^\dagger \boldsymbol{q}_{j-1} + \boldsymbol{M}_j^\dagger \boldsymbol{q}_j + \boldsymbol{C}_{j+1}^\dagger \boldsymbol{q}_{j+1}\right) = \boldsymbol{0} \tag{3.118}$$

We will call this assumption 1. But in general, with a non-orthogonal basis, we have

$$d\boldsymbol{q}_j = \sum_{\ell=1}^{j+1} \boldsymbol{H}_{j,\ell}\boldsymbol{q}_\ell. \tag{3.119}$$

This becomes the case after a few Lanczos iterations, which I have shown before. However, the fact that we have written the coupling block as $\boldsymbol{R}$ padded with zeros (I will refer to this as assumption 2) is consistent with the way that we have constructed the auxiliary block, so even though we have violated assumption 1, assumption 2 still holds, and we get results one level down from the exact solution.

3.11.3 Exact d without reorthogonalization

Because this is just projecting $\boldsymbol{d}$ onto an increasingly non-orthogonal Krylov subspace, we are no longer making assumption 1. To make it more clear what I mean here, consider that we are no longer assuming a tridiagonal form for the auxiliary block in 3.117, as we are allowing for nonzero $\boldsymbol{q}_i d\boldsymbol{q}_j$ with $i > j+1$. But now we are violating assumption 2, which

is why this gives different results than 3.11.2. **Do we want to understand under what conditions approximation 1 seems to do better than 2, and if so, then how? It doesn't seem to matter in the context of MC-GW, but maybe you have a broader vision.**

3.11.4 MCGW with recurrence relation

In our derivation of the extended recurrence relation, there were 3 instances where we assumed that we could substitute in our expression for $\boldsymbol{q}_{j+1}$ from 3.118 as

$$\mathbf{q}_{i+1} = \left[\mathbf{d}\mathbf{q}_i - \mathbf{q}_i\mathbf{M}_i - \mathbf{q}_{i-1}\mathbf{C}_{i-1}\right] \boldsymbol{C}_i^{\dagger,-1} \tag{3.120}$$

2 were when defined a $\boldsymbol{S}^n_{i+1,i+1}$ and 1 was when we defined $\boldsymbol{S}^n_{i+1,i}$. This uses more approximations than 3.11.3, so it makes sense that we get worse results.

3.12 Next steps

I have a C++ code that I can run now for the dzvp basis, but it is still to slow, which was why I was only able to show you a limited amount of results last time. I will talk to people in the group about how to make it faster. I guess you want me to generate plots like this one for the larger system sizes.

Chapter 4

Berkelbach's GW

4.1 Formulation for the TDA

4.1.1 Definitions

For simplicity, we will just work with a single channel, the lesser one. In the dTDA case, Booth's formulation for the upfolded Hamiltonian is

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{d} \end{pmatrix} \tag{4.1}$$

where we have the definitions

$$W_{pkv} = \sum_{ia} (pk|ia) X^v_{ia} \quad \text{and} \quad d_{kv,lv'} = (\epsilon_k - \Omega_v)\, \delta_{k,l} \delta_{v,v'} \tag{4.2}$$

Now, Tim's version of the Hamiltonian is given by

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{V}^{\text{2h1p}} \\ (\boldsymbol{V}^{\text{2 h1p}})^\dagger & \boldsymbol{C}^{\text{2hlp}} \end{pmatrix} \tag{4.3}$$

where the definitions of the matrix elements are

$$V^{\text{2 h1p}}_{p,k[lc]} = \langle pc|kl\rangle \equiv (pk|lc) \tag{4.4}$$

$$C^{\text{2 h1p}}_{i[ja],k[lc]} = \left[(\epsilon_i + \epsilon_j - \epsilon_a)\, \delta_{jl}\delta_{ac} - \langle jc|al\rangle\right] \delta_{ik} \tag{4.5}$$

and in particular, we have a definition

$$\boldsymbol{C}^{\text{2hlp}} = \epsilon^{\text{1 h}} \oplus (-\boldsymbol{A}) = \epsilon^{\text{1 h}} \otimes \boldsymbol{1} + \boldsymbol{1} \otimes (-\boldsymbol{A}) \tag{4.6}$$

Let us show how 6.21 comes from 4.6. If $A_{[ja],[lc]} = (\epsilon_j - \epsilon_a)\, \delta_{jl}\delta_{ac} + \langle jc|al\rangle$, then we can write

$$C^{\text{2 h1p}}_{i[ja],k[lc]} = \left[\epsilon_i \delta_{ik}\delta_{jl}\delta_{ac} + (-A)_{[ja],[lc]}\right] \tag{4.7}$$

$$= \delta_{ik}\left[\epsilon_i\, \delta_{jl}\delta_{ac} + (-A)_{[ja],[lc]}\right] \tag{4.8}$$

$$= \delta_{ik}\left[(\epsilon_i + \epsilon_j - \epsilon_a)\, \delta_{jl}\delta_{ac} - \langle jc|al\rangle\right] \tag{4.9}$$

4.1.2 Similarity transformation

Let us define a unitary rotation $\boldsymbol{U} = \mathbf{1}_P \oplus_{\text{diag}} (\mathbf{1}_O \otimes \boldsymbol{X}_{OV})$. Application of this unitary to the Hamiltonian will not change the spectrum and actually transforms the problem into

$$\boldsymbol{H}^{2h1p,\prime} = \boldsymbol{U}^\dagger \boldsymbol{H}^{2h1p} \boldsymbol{U} = \begin{pmatrix} \mathbf{1}_P & \mathbf{0} \\ \mathbf{0} & (\mathbf{1}_O \otimes \boldsymbol{X}_{OV}) \end{pmatrix}^\dagger \begin{pmatrix} \boldsymbol{F}_P & \boldsymbol{V}^{2\text{h1p}}_{P,O^2V} \\ \left(\boldsymbol{V}^{2\text{ h1p}}\right)^\dagger_{O^2V,P} & \boldsymbol{C}^{2\text{hlp}}_{O^2V,O^2V} \end{pmatrix} \begin{pmatrix} \mathbf{1}_P & \mathbf{0} \\ \mathbf{0} & (\mathbf{1}_O \otimes \boldsymbol{X}_{OV}) \end{pmatrix} \tag{4.10}$$

$$= \begin{pmatrix} \boldsymbol{F}_P & \boldsymbol{V}^{2\text{h1p}}_{P,O^2V} (\mathbf{1} \otimes \boldsymbol{X})_{O^2V,O^2V} \\ \left(\boldsymbol{V}^{2\text{h1p}}_{P,O^2V} (\mathbf{1} \otimes \boldsymbol{X})_{O^2V,O^2V}\right)^\dagger & \left((\mathbf{1} \otimes \boldsymbol{X})_{O^2V,O^2V}\right)^\dagger \boldsymbol{C}^{2\text{hlp}}_{O^2V,O^2V} \left((\mathbf{1} \otimes \boldsymbol{X})_{O^2V,O^2V}\right) \end{pmatrix}. \tag{4.11}$$

Now let us evaluate $(\mathbf{1} \otimes \boldsymbol{X})^\dagger \boldsymbol{C}^{2\text{hlp}} (\mathbf{1} \otimes \boldsymbol{X})$ in the TDA case. We have

$$(\mathbf{1} \otimes \boldsymbol{X})^\dagger \boldsymbol{C}^{2\text{hlp}} (\mathbf{1} \otimes \boldsymbol{X}) = (\mathbf{1} \otimes \boldsymbol{X})^\dagger \left[\epsilon^{1\text{ h}} \otimes \mathbf{1} + \mathbf{1} \otimes (-\boldsymbol{A})\right] (\mathbf{1} \otimes \boldsymbol{X}) \tag{4.12}$$

$$= \left(\mathbf{1} \otimes \boldsymbol{X}^\dagger\right) \left[\boldsymbol{\epsilon}^{1\text{ h}} \otimes \mathbf{1}\right] (\mathbf{1} \otimes \boldsymbol{X}) + \left(\mathbf{1} \otimes \boldsymbol{X}^\dagger\right) [\mathbf{1} \otimes (-\boldsymbol{A})] (\mathbf{1} \otimes \boldsymbol{X}) \tag{4.13}$$

$$= \mathbf{1}\boldsymbol{\epsilon}^{1\text{ h}}\mathbf{1} \otimes \boldsymbol{X}^\dagger \mathbf{1} \boldsymbol{X} + \mathbf{1}\mathbf{1} \otimes (-\boldsymbol{X}^\dagger \boldsymbol{A} \boldsymbol{X}) \tag{4.14}$$

$$= \boldsymbol{\epsilon}^{1\text{ h}} \otimes \mathbf{1} + \mathbf{1} \otimes (-\boldsymbol{X}^\dagger \boldsymbol{A} \boldsymbol{X}) \tag{4.15}$$

$$= \boldsymbol{\epsilon}^{1\text{ h}} \otimes \mathbf{1} + \mathbf{1} \otimes (-\boldsymbol{\Omega}) \tag{4.16}$$

$$= \boldsymbol{\epsilon}^{1\text{ h}} \oplus (-\boldsymbol{\Omega}) \tag{4.17}$$

where we have used the fact that $\boldsymbol{X}^\dagger \boldsymbol{X} = \mathbf{1}$, since $\boldsymbol{X}$ is unitary. Similarly, we can evaluate the other term

$$\boldsymbol{V}^{2\text{h1p}}_{P,O^2V} (\mathbf{1} \otimes \boldsymbol{X})_{O^2V,O^2V} = \sum_{k'lc} (pk|lc) X^v_{lc} \delta_{kk'} \tag{4.18}$$

$$= \sum_{lc} (pk|lc) X^v_{lc} \equiv \boldsymbol{W}^< \tag{4.19}$$

$$\tag{4.20}$$

So because Booth's and Tim's forms for TDA are related by a similarity transformation, we know that they have the same spectrum.

4.1.3 Deriving the matrix vector products

Now, we can define a vector $\boldsymbol{R} = (r_i,\ r_a,\ r_{i[jb]},\ r_{[jb]a})$. Application of the Hamiltonian to this vector gives us the matrix-vector product $\boldsymbol{HR} = \boldsymbol{\sigma}$, where $\boldsymbol{\sigma} = (\sigma_i,\ \sigma_a,\ \sigma_{i[jb]},\ \sigma_{[jb]a})$. Consider

$$\boldsymbol{HR} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{V}^{2\text{h1p}} & \boldsymbol{V}^{2\text{p1 h}} \\ \left(\boldsymbol{V}^{2\text{ h1p}}\right)^\dagger & \boldsymbol{C}^{2\text{hlp}} & \mathbf{0} \\ \left(\boldsymbol{V}^{2\text{plh}}\right)^\dagger & \mathbf{0} & \boldsymbol{C}^{2\text{plh}} \end{pmatrix} \begin{pmatrix} r_i \\ r_a \\ r_{i[jb]} \\ r_{[jb]a} \end{pmatrix} = \begin{pmatrix} \sigma_i \\ \sigma_a \\ \sigma_{i[jb]} \\ \sigma_{[jb]a} \end{pmatrix} \tag{4.21}$$

$$\tag{4.22}$$

Let us enumerate now what we actually will get:

$$\sigma_i = \sum_j f_{ij}\, r_j + \sum_b f_{ib}\, r_b + \sum_{klc} \big\langle i\, c | k\, l \big\rangle\, r_{k[lc]} + \sum_{kcd} \big\langle i\, k | d\, c \big\rangle\, r_{[kc]d}, \tag{4.23}$$

$$\sigma_a = \sum_j f_{aj}\, r_j + \sum_b f_{ab}\, r_b + \sum_{klc} \big\langle a\, c | k\, l \big\rangle\, r_{k[lc]} + \sum_{kcd} \big\langle a\, k | d\, c \big\rangle\, r_{[kc]d}, \tag{4.24}$$

$$\sigma_{i[ja]} \sum_k \big\langle k\, a | i\, j \big\rangle\, r_k + \sum_b \big\langle b\, a | i\, j \big\rangle\, r_b + (\epsilon_i + \epsilon_j - \epsilon_a)\, r_{i[ja]} - \sum_{lc} \big\langle j\, c | a\, l \big\rangle\, r_{i[lc]} \tag{4.25}$$

$$\sigma_{[ia]b} = \sum_j \big\langle j\, i | b\, a \big\rangle\, r_j + \sum_c \big\langle c\, i | b\, a \big\rangle\, r_c + (\epsilon_a + \epsilon_b - \epsilon_i)\, r_{[ia]b} + \sum_{kc} \big\langle a\, k | i\, c \big\rangle\, r_{[kc]b}. \tag{4.26}$$

4.2 Formulation for the dRPA

Exact form

The exact form is

$$\boldsymbol{H}^{G_0W_0}_{\text{Upfolded}} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W}^{<} & \boldsymbol{W}^{>} \\ \boldsymbol{W}^{<,\dagger} & \boldsymbol{d}^{<} & 0 \\ \boldsymbol{W}^{>,\dagger} & 0 & \boldsymbol{d}^{>} \end{pmatrix} \tag{4.27}$$

where $\boldsymbol{F}$ is the HF Fock matrix

$$\boldsymbol{W}^{<}_{pk\nu} = \sum_{ia} (pk|ia)\, (X^{\nu}_{ia} + Y^{\nu}_{ia}) \quad \text{and} \quad \boldsymbol{W}^{>}_{pc\nu} = \sum_{ia} (pc|ia)\, (X^{\nu}_{ia} + Y^{\nu}_{ia}) \tag{4.28}$$

and

$$\boldsymbol{d}^{<}_{k\nu,l\nu'} = (\epsilon_k - \Omega_\nu)\, \delta_{k,l}\delta_{\nu,\nu'} \quad \text{and} \quad \boldsymbol{d}^{>}_{c\nu,d\nu'} = (\epsilon_c + \Omega_\nu)\, \delta_{c,d}\delta_{\nu,\nu'}. \tag{4.29}$$

Downfolding exercise

Let us confirm that this downfolds to the correct result. First, we can define a excitation vector $\boldsymbol{R}$ as

$$\boldsymbol{R} = \begin{pmatrix} \boldsymbol{R}^{\mathrm{1h+1p}} \\ \boldsymbol{R}^{\mathrm{2h1p}} \\ \boldsymbol{R}^{\mathrm{2p1h}} \end{pmatrix} \tag{4.30}$$

and we can write

$$\begin{pmatrix} \boldsymbol{F} & \boldsymbol{W}^{<} & \boldsymbol{W}^{>} \\ \boldsymbol{W}^{<,\dagger} & \boldsymbol{d}^{<} & 0 \\ \boldsymbol{W}^{>,\dagger} & 0 & \boldsymbol{d}^{>} \end{pmatrix} \begin{pmatrix} \boldsymbol{R}^{\mathrm{1h+1p}} \\ \boldsymbol{R}^{\mathrm{2h1p}} \\ \boldsymbol{R}^{\mathrm{2p1h}} \end{pmatrix} = E \begin{pmatrix} \boldsymbol{R}^{\mathrm{1h+1p}} \\ \boldsymbol{R}^{\mathrm{2h1p}} \\ \boldsymbol{R}^{\mathrm{2p1h}} \end{pmatrix} \tag{4.31}$$

$$\begin{pmatrix} \boldsymbol{F}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{W}^{<}\boldsymbol{R}^{\mathrm{2h1p}} + \boldsymbol{W}^{>}\boldsymbol{R}^{\mathrm{2p1h}} \\ \boldsymbol{W}^{<,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{d}^{<}\boldsymbol{R}^{\mathrm{2h1p}} \\ \boldsymbol{W}^{>,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{d}^{>}\boldsymbol{R}^{\mathrm{2p1h}} \end{pmatrix} = E \begin{pmatrix} \boldsymbol{R}^{\mathrm{1h+1p}} \\ \boldsymbol{R}^{\mathrm{2h1p}} \\ \boldsymbol{R}^{\mathrm{2p1h}} \end{pmatrix} \tag{4.32}$$

This implies three coupled equations

$$\boldsymbol{F}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{W}^{<}\boldsymbol{R}^{\mathrm{2h1p}} + \boldsymbol{W}^{>}\boldsymbol{R}^{\mathrm{2p1h}} = E\boldsymbol{R}^{\mathrm{1h+1p}} \tag{4.33}$$

$$\boldsymbol{W}^{<,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{d}^{<}\boldsymbol{R}^{\mathrm{2h1p}} = E\boldsymbol{R}^{\mathrm{2h1p}} \tag{4.34}$$

$$\boldsymbol{W}^{>,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{d}^{>}\boldsymbol{R}^{\mathrm{2p1h}} = E\boldsymbol{R}^{\mathrm{2p1h}} \tag{4.35}$$

Solving the latter two equations gives

$$\boldsymbol{R}^{\mathrm{2h1p}} = (E\mathbf{1} - \boldsymbol{d}^{<})^{-1}\,\boldsymbol{W}^{<,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} \tag{4.36}$$

$$\boldsymbol{R}^{\mathrm{2p1h}} = (E\mathbf{1} - \boldsymbol{d}^{>})^{-1}\,\boldsymbol{W}^{>,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} \tag{4.37}$$

Substituting these into the first equation gives us

$$\boldsymbol{F}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{W}^{<}\,(E\mathbf{1} - \boldsymbol{d}^{<})^{-1}\,\boldsymbol{W}^{<,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} + \boldsymbol{W}^{>}\,(E\mathbf{1} - \boldsymbol{d}^{>})^{-1}\,\boldsymbol{W}^{>,\dagger}\boldsymbol{R}^{\mathrm{1h+1p}} = E\boldsymbol{R}^{\mathrm{1h+1p}} \tag{4.38}$$

from which we can get the eigenvalue equation

$$\left(\boldsymbol{F} + \boldsymbol{W}^{<}\,(E\mathbf{1} - \boldsymbol{d}^{<})^{-1}\,\boldsymbol{W}^{<,\dagger} + \boldsymbol{W}^{>}\,(E\mathbf{1} - \boldsymbol{d}^{>})^{-1}\,\boldsymbol{W}^{>,\dagger}\right)\boldsymbol{R}^{\mathrm{1h+1p}} = E\boldsymbol{R}^{\mathrm{1h+1p}} \tag{4.39}$$

so we can identify that the correlation self energy in this approach is

$$\boldsymbol{\Sigma}_c = \boldsymbol{W}^{<}\,(\omega\mathbf{1} - \boldsymbol{d}^{<})^{-1}\,\boldsymbol{W}^{<,\dagger} + \boldsymbol{W}^{>}\,(\omega\mathbf{1} - \boldsymbol{d}^{>})^{-1}\,\boldsymbol{W}^{>,\dagger} \tag{4.40}$$

which is the same as the known frequency dependent form for the real part of the correlation self energy:

$$\Sigma^{\mathrm{corr}}_{pq}(\omega) = \sum_{\mu}^{\mathrm{RPA}} \left(\sum_{i}^{\mathrm{occupied}} \frac{w^{\mu}_{pi} w^{\mu}_{iq}}{\omega - (\epsilon_i - \Omega_\mu)} + \sum_{a}^{\mathrm{virtual}} \frac{w^{\mu}_{pa} w^{\mu}_{aq}}{\omega - (\epsilon_a + \Omega_\mu)} \right) \tag{4.41}$$

with $w^{\mu}_{pq} = \sum_{ia} (pq|ia)\,(X^{\mu}_{ia} + Y^{\mu}_{ai})$.

Showing equivalence between excitation energies of M and Mtilde

So we start with this generalized eigenvalue equation

$$\mathbf{M}\boldsymbol{Z} = \mathbf{N}\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ \end{pmatrix}$$

where

$$\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} \quad \mathbf{N} = \begin{pmatrix} \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\mathbf{1} \end{pmatrix} \quad \boldsymbol{Z} = \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix}$$

and $\boldsymbol{\Omega}_+$is a diagonal matrix of positive excitation energies. Left multiplying both sides by $\boldsymbol{N}$ and right multiplying by $\boldsymbol{Z}^{-1}$ gives us

$$\boldsymbol{N}\boldsymbol{M}\underbrace{\boldsymbol{Z}\boldsymbol{Z}^{-1}}_{\mathbf{1}} = \underbrace{\boldsymbol{N}\boldsymbol{N}}_{\mathbf{1}}\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ \end{pmatrix}\boldsymbol{Z}^{-1} \implies -\boldsymbol{N}\boldsymbol{M} = -\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ \end{pmatrix}\boldsymbol{Z}^{-1} \tag{4.42}$$

Now we can use the fact that the action of a scalar function f, such as the step function, on a diagonalizable matrix $\boldsymbol{X} \equiv \boldsymbol{Y}\boldsymbol{\Lambda}\boldsymbol{Y}^{-1}$ can be expressed as

$$f(\boldsymbol{X}) = \boldsymbol{Y}f(\boldsymbol{\Lambda})\boldsymbol{Y}^{-1}$$

so we can write

$$\Theta(-\boldsymbol{N}\boldsymbol{M}) = \boldsymbol{Z}\begin{pmatrix} \Theta(-\boldsymbol{\Omega}_+) & 0 \\ 0 & \Theta(\boldsymbol{\Omega}_+) \end{pmatrix}\boldsymbol{Z}^{-1} = \boldsymbol{Z}\begin{pmatrix} \mathbf{0} & 0 \\ 0 & \mathbf{1} \end{pmatrix}\boldsymbol{Z}^{-1} \tag{4.43}$$

and so it becomes clear that if we define $\tilde{\boldsymbol{M}} = \boldsymbol{M} + \eta\boldsymbol{N}\Theta(-\boldsymbol{N}\boldsymbol{M})$, we can write

$$\tilde{\boldsymbol{M}}\boldsymbol{Z} = \boldsymbol{M}\boldsymbol{Z} + \eta\boldsymbol{N}\Theta(-\boldsymbol{N}\boldsymbol{M})\boldsymbol{Z} = \mathbf{N}\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ \end{pmatrix} + \boldsymbol{N}\boldsymbol{Z}\begin{pmatrix} \mathbf{0} & 0 \\ 0 & \boldsymbol{\eta} \end{pmatrix}\underbrace{\boldsymbol{Z}^{-1}\boldsymbol{Z}}_{\mathbf{1}} = \boldsymbol{N}\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ + \boldsymbol{\eta} \end{pmatrix} \tag{4.44}$$

4.2.1 Considering the supermatrices they give

It is understood that the notations <,> and 2h1p, 2p1h can be used interchangeably, respectively. Tim's supermatrix is given by

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2plh}} & \boldsymbol{V}^{\text{2plh}} \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger & & & & \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger & & \boldsymbol{C}^{\text{2hlp}} & & \mathbf{0} \\ \left(\boldsymbol{V}^{\text{2plh}}\right)^\dagger & & & & \\ \left(\boldsymbol{V}^{\text{2plh}}\right)^\dagger & & \mathbf{0} & & \boldsymbol{C}^{\text{2plh}} \end{pmatrix} \tag{4.45}$$

We are told that $\mathbf{C}^{\text{2hlp}} = \varepsilon^{1\ \text{h}} \oplus (-\tilde{\mathbf{M}})$ and $\mathbf{C}^{\text{2plh}} = \varepsilon^{\text{1p}} \oplus \tilde{\mathbf{M}}$ and further, that the super-metric is

$$\boldsymbol{\mathcal{N}} = \begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} \oplus \boldsymbol{N} & 0 \\ 0 & 0 & \mathbf{1} \oplus \boldsymbol{N} \end{pmatrix} \tag{4.46}$$

Let's just focus on the 2h1p sector for now, so the supermatrices are

$$\boldsymbol{H}^{\text{2hlp}} = \begin{pmatrix} \boldsymbol{F}_{P,P} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \boldsymbol{C}^{\text{2hlp}}_{2O^2V,2O^2V} \end{pmatrix}, \quad \boldsymbol{\mathcal{N}}^{\text{2hlp}} = \begin{pmatrix} \mathbf{1}_{P,P} & 0 \\ 0 & \mathbf{1}_O \oplus \boldsymbol{N}_{2OV} \end{pmatrix}, \quad \boldsymbol{R}^{\text{2hlp}} = \begin{pmatrix} r_i \\ r_a \\ r_{i[jb]} \\ \bar{r}_{i[jb]} \end{pmatrix} \tag{4.47}$$

In the following, we might sometimes use the notation $P = O + V$, $A = O^2V$ and T as the column dimension of the excitation vector.

Multiplication of the Hamiltonian by the excitation vector

First, we can take

$$\boldsymbol{H}^{\text{2hlp}}\boldsymbol{R}^{\text{2hlp}} \tag{4.48}$$

$$= \begin{pmatrix} \boldsymbol{F}_{P,P} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \boldsymbol{C}^{\text{2hlp}}_{2O^2V,2O^2V} \end{pmatrix} \begin{pmatrix} r_i \\ r_a \\ r_{i[jb]} \\ \bar{r}_{i[jb]} \end{pmatrix} \tag{4.49}$$

$$= \begin{pmatrix} \begin{pmatrix} \boldsymbol{F}_{OO} & \boldsymbol{F}_{OV} \\ \boldsymbol{F}_{VO} & \boldsymbol{F}_{VV} \end{pmatrix}_{P,P} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \end{pmatrix}_{P,T} + \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}}_{O,i[jb]} & \boldsymbol{V}^{\text{2h1p}}_{O,i[jb]} \\ \boldsymbol{V}^{\text{2h1p}}_{V,i[jb]} & \boldsymbol{V}^{\text{2h1p}}_{V,i[jb]} \end{pmatrix}_{P,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \\ \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}_{O,i[jb]}\right)^\dagger & \left(\boldsymbol{V}^{\text{2h1p}}_{V,i[jb]}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}_{O,i[jb]}\right)^\dagger & \left(\boldsymbol{V}^{\text{2h1p}}_{V,i[jb]}\right)^\dagger \end{pmatrix}_{2A,P} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \end{pmatrix}_{P,T} + \left[\boldsymbol{\epsilon}^{1\text{ h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV} + \mathbf{1}_{O,O} \otimes -\tilde{\boldsymbol{M}}_{2OV,2OV}\right]_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \end{pmatrix} \tag{4.50}$$

Just evaluate

$$\left[\boldsymbol{\epsilon}^{1\text{ h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV} + \mathbf{1}_{O,O} \otimes -\tilde{\boldsymbol{M}}_{2OV,2OV}\right]_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \tag{4.51}$$

$$\tag{4.52}$$

First, use the definition of the tensor product and also notice that $\boldsymbol{\epsilon}^{1\text{ h}}_{O,O}$ is a matrix with the hole energies on the diagonal and zeros elsewhere

$$\boldsymbol{\epsilon}^{1\text{ h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV} = \begin{pmatrix} \epsilon^{1\text{ h}}_{11}\mathbf{1}_{2OV,2OV} & \epsilon^{1\text{ h}}_{12}\mathbf{1}_{2OV,2OV} & \dots & \epsilon^{1\text{ h}}_{1O}\mathbf{1}_{2OV,2OV} \\ \epsilon^{1\text{ h}}_{21}\mathbf{1}_{2OV,2OV} & \epsilon^{1\text{ h}}_{22}\mathbf{1}_{2OV,2OV} & \dots & \epsilon^{1\text{ h}}_{2O}\mathbf{1}_{2OV,2OV} \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon^{1\text{ h}}_{O1}\mathbf{1}_{2OV,2OV} & \epsilon^{1\text{ h}}_{O2}\mathbf{1}_{2OV,2OV} & \dots & \epsilon^{1\text{ h}}_{OO}\mathbf{1}_{2OV,2OV} \end{pmatrix}_{2A,2A} \tag{4.53}$$

$$= \begin{pmatrix} \epsilon^{1\text{ h}}_{11}\mathbf{1}_{2OV,2OV} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \epsilon^{1\text{ h}}_{22}\mathbf{1}_{2OV,2OV} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \epsilon^{1\text{ h}}_{OO}\mathbf{1}_{2OV,2OV} \end{pmatrix}_{2A,2A} \tag{4.54}$$

$$\implies \left(\boldsymbol{\epsilon}^{1\text{ h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV}\right)_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \tag{4.55}$$

$$= \begin{pmatrix} \epsilon^{1\text{ h}}_{11}\mathbf{1}_{2OV,2OV} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \epsilon^{1\text{ h}}_{22}\mathbf{1}_{2OV,2OV} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \epsilon^{1\text{ h}}_{OO}\mathbf{1}_{2OV,2OV} \end{pmatrix}_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \tag{4.56}$$

$$\tag{4.57}$$

For these, we can use the formula

$$(A \otimes B) = \begin{pmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ a_{21}B & a_{22}B & \dots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \dots & a_{mn}B \end{pmatrix} \implies (A \otimes B)\, K = \begin{bmatrix} \sum_{j=1}^{n} a_{1j} B K^{(j)} \\ \sum_{j=1}^{n} a_{2j} B K^{(j)} \\ \vdots \\ \sum_{j=1}^{n} a_{mj} B K^{(j)} \end{bmatrix} \tag{4.58}$$

where the $K^{(j)}$ are the rows of the matrix K. To simplify the notation let us define $t = O//2$. First, with

$$\boldsymbol{K} = \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[jb]} \end{pmatrix} = \begin{pmatrix} r_{1[jb]} \\ \vdots \\ r_{O[jb]} \\ \bar{r}_{1[jb]} \\ \vdots \\ \bar{r}_{O[jb]} \end{pmatrix} = \begin{pmatrix} K^{(1)} \\ \vdots \\ K^{(t)} \\ \vdots \\ K^{(O)} \end{pmatrix} \in \mathbb{R}^{2O^2V}, \tag{4.59}$$

$$K^{(m)} = \begin{pmatrix} r_{(2m-1)[jb]} \\ r_{(2m)[jb]} \end{pmatrix} \in \mathbb{R}^{2OV} \forall m \leq t, \quad K^{(m)} = \begin{pmatrix} \bar{r}_{(m-1-t)[jb]} \\ \bar{r}_{(m-t)[jb]} \end{pmatrix} \in \mathbb{R}^{2OV} \forall m > t+1 \tag{4.60}$$

Depending on whether O is even or odd, $K^{(t)}$ will be $\begin{pmatrix} r_{(O-1)[jb]} \\ r_{O[jb]} \end{pmatrix}$ or $\begin{pmatrix} r_{O[jb]} \\ \bar{r}_{1[jb]} \end{pmatrix}$, respectively. This version seems correct, but I am unsure about how to proceed. Next, I will consider a modification to lead to a familiar form, but I am not sure if it is correct.

Incorrect but suggestive form

$$\boldsymbol{K} = \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[jb]} \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} r_{1[jb]} \\ \bar{r}_{1[jb]} \\ r_{2[jb]} \\ \bar{r}_{2[jb]} \\ \vdots \\ r_{O[jb]} \\ \bar{r}_{O[jb]} \end{pmatrix} = \begin{pmatrix} K^{(1)} \\ K^{(2)} \\ \vdots \\ K^{(O)} \end{pmatrix} \in \mathbb{R}^{2O^2V}, \quad K^{(i)} = \begin{pmatrix} r_{i[jb]} \\ \bar{r}_{i[jb]} \end{pmatrix} \in \mathbb{R}^{2OV} \tag{4.61}$$

$$\implies \left(\boldsymbol{\epsilon}^{1\text{ h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV}\right)_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}b]} \end{pmatrix}_{2A,T} \stackrel{?}{=} \begin{pmatrix} \epsilon^{1\text{h}}_{11} K^{(1)} + \epsilon^{1\text{h}}_{12} K^{(2)} + \cdots + \epsilon^{1\text{h}}_{1O} K^{(O)} \\ \epsilon^{1\text{h}}_{21} K^{(1)} + \epsilon^{1\text{h}}_{22} K^{(2)} + \cdots + \epsilon^{1\text{h}}_{2O} K^{(O)} \\ \vdots \\ \epsilon^{1\text{h}}_{O1} K^{(1)} + \epsilon^{1\text{h}}_{O2} K^{(2)} + \cdots + \epsilon^{1\text{h}}_{OO} K^{(O)} \end{pmatrix} \tag{4.62}$$

$$= \begin{pmatrix} \epsilon^{1\text{h}}_{11} K^{(1)} + \cdots + 0 \\ 0 + \epsilon^{1\text{h}}_{22} K^{(2)} + \cdots + 0 \\ \vdots \\ 0 + \cdots + \epsilon^{1\text{h}}_{OO} K^{(O)} \end{pmatrix} = \begin{pmatrix} \epsilon^{1\text{h}}_{11} \begin{pmatrix} r_{1[jb]} \\ \bar{r}_{1[jb]} \end{pmatrix} \\ \epsilon^{1\text{h}}_{22} \begin{pmatrix} r_{2[jb]} \\ \bar{r}_{2[jb]} \end{pmatrix} \\ \vdots \\ \epsilon^{1\text{h}}_{OO} \begin{pmatrix} r_{O[jb]} \\ \bar{r}_{O[jb]} \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\epsilon}^{1\text{h}} \boldsymbol{r}_{i[jb]} \\ \boldsymbol{\epsilon}^{1\text{h}} \bar{\boldsymbol{r}}_{i[jb]} \end{pmatrix} \tag{4.63}$$

where we have used the fact that $\boldsymbol{\epsilon}^{1\text{h}}$ is a diagonal matrix with the hole energies. Next, it is useful to define (where the composite index $\mu \equiv OV$; it implies that the occupied-virtual pair is fixed)

$$\boldsymbol{K} = \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[jb]} \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} r_{i[\mu_1]} \\ \bar{r}_{i[\mu_2]} \\ r_{i[\mu_3]} \\ \bar{r}_{i[\mu_4]} \\ \vdots \\ r_{i[\mu_{2OV-1}]} \\ \bar{r}_{i[\mu_{2OV}]} \end{pmatrix} = \begin{pmatrix} K^{(1)} \\ K^{(2)} \\ \vdots \\ K^{(2OV)} \end{pmatrix} \in \mathbb{R}^{2O^2V}, \quad K^{(i)} = \left(r_{i[OV]} / \bar{r}_{i[OV]}\right) \in \mathbb{R}^{O} \tag{4.64}$$

$$\implies \left(\mathbf{1}_{O,O} \otimes \tilde{\boldsymbol{M}}_{2OV,2OV}\right)_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{jb}]} \end{pmatrix}_{2A,T} \stackrel{?}{=} \left(\tilde{\boldsymbol{M}}_{2OV,2OV} \otimes \mathbf{1}_{O,O}\right)_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{jb}]} \end{pmatrix}_{2A,T} \tag{4.65}$$

$$\stackrel{?}{=} \begin{pmatrix} \tilde{M}_{11}K^{(1)} + \cdots + \tilde{M}_{1,OV}K^{(OV)} + \tilde{M}_{1,OV+1}K^{(OV+1)} + \cdots + \tilde{M}_{1,2OV}K^{(2OV)} \\ \tilde{M}_{2,1}K^{(1)} + \cdots + \tilde{M}_{2,OV}K^{(OV)} + \tilde{M}_{2,OV+1}K^{(OV+1)} + \cdots + \tilde{M}_{2,2OV}K^{(2OV)} \\ \vdots \\ \tilde{M}_{OV,1}K^{(1)} + \cdots + \tilde{M}_{OV,OV}K^{(OV)} + \tilde{M}_{OV,OV+1}K^{(OV+1)} + \cdots + \tilde{M}_{OV,2OV}K^{(2OV)} \\ \tilde{M}_{OV+1,1}K^{(1)} + \cdots + \tilde{M}_{OV+1,OV}K^{(OV)} + \tilde{M}_{OV+1,OV+1}K^{(OV+1)} + \cdots + \tilde{M}_{OV+1,2OV}K^{(2OV)} \\ \vdots \\ \tilde{M}_{2OV,1}K^{(1)} + \cdots + \tilde{M}_{2OV,OV}K^{(OV)} + \tilde{M}_{2OV,OV+1}K^{(OV+1)} + \cdots + \tilde{M}_{2OV,2OV}K^{(2OV)} \end{pmatrix} \tag{4.66}$$

$$= \begin{pmatrix} \tilde{\boldsymbol{M}}_{xx}\boldsymbol{K}_{1\to OV} + \tilde{\boldsymbol{M}}_{xd}\boldsymbol{K}_{OV+1\to 2OV} \\ \tilde{\boldsymbol{M}}_{dx}\boldsymbol{K}_{1\to OV} + \tilde{\boldsymbol{M}}_{dd}\boldsymbol{K}_{OV+1\to 2OV} \end{pmatrix} \tag{4.67}$$

where $\tilde{\boldsymbol{M}}_{xx}$, $\tilde{\boldsymbol{M}}_{xd}$, $\tilde{\boldsymbol{M}}_{dx}$, and $\tilde{\boldsymbol{M}}_{dd}$ are the excitation-excitation, excitation-de-excitation, de-excitation-excitation, and de-excitation-de-excitation blocks of the matrix $\tilde{\boldsymbol{M}}$ respectively. $\boldsymbol{K}_{1\to OV}$ and $\boldsymbol{K}_{OV+1\to 2OV}$ are defined as

$$\boldsymbol{K}_{1\to OV} = \begin{pmatrix} K^{(1)} & K^{(2)} & \dots & K^{(OV)} \end{pmatrix} = \begin{pmatrix} r_{i[\mu_1]} & \bar{r}_{i[\mu_2]} & \dots & r_{i[\mu_{OV}]} \end{pmatrix} \tag{4.68}$$

$$\boldsymbol{K}_{OV+1\to 2OV} = \begin{pmatrix} K^{(OV+1)} & K^{(OV+2)} & \dots & K^{(2OV)} \end{pmatrix} = \begin{pmatrix} r_{i[\mu_{OV+1}]} & \bar{r}_{i[\mu_{OV+2}]} & \dots & r_{i[\mu_{2OV}]} \end{pmatrix} \tag{4.69}$$

$$\boldsymbol{H}^{\text{2hlp}}\boldsymbol{R}^{\text{2hlp}} \tag{4.70}$$

$$\stackrel{?}{=} \begin{pmatrix} \boldsymbol{F}_{OO}\boldsymbol{r}_i + \boldsymbol{F}_{OV}\boldsymbol{r}_a + \boldsymbol{V}^{\text{2h1p}}_{O,i[jb]}\boldsymbol{r}_{i[jb]} + \boldsymbol{V}^{\text{2h1p}}_{O,i[jb]}\bar{\boldsymbol{r}}_{i[\bar{jb}]} \\ \boldsymbol{F}_{VO}\boldsymbol{r}_i + \boldsymbol{F}_{VV}\boldsymbol{r}_a + \boldsymbol{V}^{\text{2h1p}}_{V,i[jb]}\boldsymbol{r}_{i[jb]} + \boldsymbol{V}^{\text{2h1p}}_{V,i[jb]}\bar{\boldsymbol{r}}_{i[\bar{jb}]} \\ \left(\boldsymbol{V}^{\text{2h1p}}_{O,i[jb]}\right)^\dagger \boldsymbol{r}_i + \left(\boldsymbol{V}^{\text{2h1p}}_{O,i[jb]}\right)^\dagger \boldsymbol{r}_a + \boldsymbol{\epsilon}^{1\text{ h}}_{O,O}\boldsymbol{r}_{i[jb]} - \tilde{\boldsymbol{M}}_{xx}\boldsymbol{K}_{1\to OV} - \tilde{\boldsymbol{M}}_{xd}\boldsymbol{K}_{OV+1\to 2OV} \\ \left(\boldsymbol{V}^{\text{2h1p}}_{V,i[jb]}\right)^\dagger \boldsymbol{r}_i + \left(\boldsymbol{V}^{\text{2h1p}}_{V,i[jb]}\right)^\dagger \boldsymbol{r}_a + \boldsymbol{\epsilon}^{1\text{ h}}_{O,O}\bar{\boldsymbol{r}}_{i[jb]} - \tilde{\boldsymbol{M}}_{dx}\boldsymbol{K}_{1\to OV} - \tilde{\boldsymbol{M}}_{dd}\boldsymbol{K}_{OV+1\to 2OV} \end{pmatrix}_{P+2A,T} \tag{4.71}$$

$$= \begin{pmatrix} \sum_j f_{ij}\, r_j + \sum_b f_{ib}\, r_b + \sum_{klc}\langle i\, c|k\, l\rangle\, r_{k[lc]} + \sum_{klc}\langle i\, c|k\, l\rangle\, \bar{r}_{k[lc]} \\ \sum_j f_{aj}\, r_j + \sum_b f_{ab}\, r_b + \sum_{klc}\langle a\, c|k\, l\rangle\, r_{k[lc]} + \sum_{klc}\langle a\, c|k\, l\rangle\, \bar{r}_{k[lc]} \\ \vdots \end{pmatrix}_{P+2A,T} \tag{4.72}$$

Multiplying the supermetric by the Hamiltonian

Alternatively, we can consider the LHS matrix multiplication

$$\boldsymbol{\mathcal{N}}^{\text{2hlp}}\boldsymbol{H}^{\text{2hlp}} = \begin{pmatrix} \mathbf{1}_{P,P} & 0 \\ 0 & \mathbf{1}_{O,O} \oplus_{\text{kron}} \boldsymbol{N}_{2OV,2OV} \end{pmatrix} \begin{pmatrix} \boldsymbol{F}_{P,P} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \boldsymbol{C}^{\text{2hlp}}_{2O^2V,2O^2V} \end{pmatrix} \tag{4.73}$$

$$= \begin{pmatrix} \boldsymbol{F}_{P,P} & \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2h1p}} \end{pmatrix}_{P,2O^2V} \\ (\mathbf{1} \oplus_{\text{kron}} \boldsymbol{N})_{2O^2V,2O^2V} \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} & (\mathbf{1} \oplus_{\text{kron}} \boldsymbol{N})_{2O^2V,2O^2V} \boldsymbol{C}^{\text{2hlp}}_{2O^2V,2O^2V} \end{pmatrix} \tag{4.74}$$

$$\tag{4.75}$$

First, we can consider the off diagonal term

$$(\mathbf{1} \oplus_{\text{kron}} \boldsymbol{N}_{2OV,2OV}) \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} = (\mathbf{1}_O \otimes \mathbf{1}_{2OV} + \mathbf{1}_O \otimes \boldsymbol{N}_{2OV}) \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} \tag{4.76}$$

$$= \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} + \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ -\left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} \tag{4.77}$$

$$= \begin{pmatrix} 2\left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ 0 \end{pmatrix}_{2O^2V,P} \tag{4.78}$$

Next, we can consider the auxiliary block

$$(\mathbf{1} \oplus_{\text{kron}} \boldsymbol{N})\, \boldsymbol{C}^{\text{2hlp}} = (\mathbf{1}_O \oplus_{\text{kron}} \boldsymbol{N}_{2OV}) \left(\boldsymbol{\epsilon}_O^{1\text{ h}} \oplus_{\text{kron}} (-\tilde{\boldsymbol{M}})_{2OV}\right) \tag{4.79}$$

$$= (\mathbf{1}_O \otimes \mathbf{1}_{2OV} + \mathbf{1}_O \otimes \boldsymbol{N}_{2OV}) \left(\boldsymbol{\epsilon}_O^{1\text{ h}} \otimes \mathbf{1}_{2OV} + \mathbf{1}_O \otimes (-\tilde{\boldsymbol{M}})_{2OV}\right) \tag{4.80}$$

$$= \boldsymbol{\epsilon}_O^{1\text{ h}} \otimes \mathbf{1}_{2OV} + \underbrace{\mathbf{1}_O \otimes -\tilde{\boldsymbol{M}}_{2OV} + \boldsymbol{\epsilon}_O^{1\text{ h}} \otimes \boldsymbol{N}_{2OV}}_{\text{cross terms}} + \mathbf{1}_O \otimes -\boldsymbol{N}\tilde{\boldsymbol{M}}_{2OV} \tag{4.81}$$

$$= \begin{pmatrix} \epsilon_1^{1\text{ h}} \mathbf{1}_{2OV} & 0 & \dots & 0 \\ 0 & \epsilon_2^{1\text{ h}} \mathbf{1}_{2OV} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \epsilon_O^{1\text{ h}} \mathbf{1}_{2OV} \end{pmatrix} - \begin{pmatrix} 1\tilde{\boldsymbol{M}}_{2OV} & 0 & \dots & 0 \\ 0 & 1\tilde{\boldsymbol{M}}_{2OV} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1\tilde{\boldsymbol{M}}_{2OV} \end{pmatrix} \tag{4.82}$$

$$+ \begin{pmatrix} \epsilon_1^{1\text{ h}} \boldsymbol{N}_{2OV} & 0 & \dots & 0 \\ 0 & \epsilon_2^{1\text{ h}} \boldsymbol{N}_{2OV} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \epsilon_O^{1\text{ h}} \boldsymbol{N}_{2OV} \end{pmatrix} - \begin{pmatrix} 1\left(\boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} & 0 & \dots & 0 \\ 0 & 1\left(\boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1\left(\boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} \end{pmatrix} \tag{4.83}$$

$$= \begin{pmatrix} \epsilon_1^{1\text{ h}} \left(\mathbf{1} + \boldsymbol{N}\right)_{2OV} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \epsilon_O^{1\text{ h}} \left(\mathbf{1} + \boldsymbol{N}\right)_{2OV} \end{pmatrix} - \begin{pmatrix} 1\left(\tilde{\boldsymbol{M}} + \boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1\left(\tilde{\boldsymbol{M}} + \boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} \end{pmatrix} \tag{4.84}$$

$$= \begin{pmatrix} \epsilon_1^{1\text{ h}} \left(\mathbf{1} + \boldsymbol{N}\right)_{2OV} - \left(\tilde{\boldsymbol{M}} + \boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \epsilon_O^{1\text{ h}} \left(\mathbf{1} + \boldsymbol{N}\right)_{2OV} - \left(\tilde{\boldsymbol{M}} + \boldsymbol{N}\tilde{\boldsymbol{M}}\right)_{2OV} \end{pmatrix} \tag{4.85}$$

$$= \begin{pmatrix} \epsilon_1^{1\text{ h}} \left(\mathbf{1} + \boldsymbol{N}\right)_{2OV} - \left[\left(\mathbf{1} + \boldsymbol{N}\right)\tilde{\boldsymbol{M}}\right]_{2OV} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \epsilon_O^{1\text{ h}} \left(\mathbf{1} + \boldsymbol{N}\right)_{2OV} - \left[\left(\mathbf{1} + \boldsymbol{N}\right)\tilde{\boldsymbol{M}}\right]_{2OV} \end{pmatrix} \tag{4.86}$$

I have plotted 3 different data points from each molecular calculation. The blue bar shows the average deviation of the Ritz values from the exact solution using the expressions for the matrix vector products given in the main paper. The orange bar shows the average deviation of the Ritz values from the exact solution using the expressions for the matrix vector products given in the supplemental material (Supp. MV), where we set B=0. The green bar is just showing the norm of the final residual (when I reach the maximum number of iterations) I get when I run the Davidson iteration using Supp. MV. I am running for up to 75 iterations without doing any restarting. I interpret this as there not being a bug in my implementation since I see the same behavior across different molecules, but perhaps the expressions for the matrix vector products that he proposes in the supplemental material are just not exact. Do you have a different opinion?

Reverse engineering

We have the problem $\boldsymbol{\sigma} = \boldsymbol{\mathcal{N}}\boldsymbol{H}\boldsymbol{R}$. I know that $\boldsymbol{\sigma}$ is given by

$$\sigma_i = \sum_j f_{ij}\, r_j + \sum_b f_{ib}\, r_b + \sum_{klc} \langle i\, c | k\, l \rangle\, r_{k[lc]} + \sum_{kcd} \langle i\, k | d\, c \rangle\, r_{[kc]d} + \sum_{klc} \langle i\, c | k\, l \rangle\, \bar{r}_{k[lc]} + \sum_{kcd} \langle i\, k | d\, c \rangle\, \bar{r}_{[kc]d} \tag{4.87}$$

$$\sigma_a = \sum_j f_{aj}\, r_j + \sum_b f_{ab}\, r_b + \sum_{klc} \langle a\, c | k\, l \rangle\, r_{k[lc]} + \sum_{kcd} \langle a\, k | d\, c \rangle\, r_{[kc]d} + \sum_{klc} \langle a\, c | k\, l \rangle\, \bar{r}_{k[lc]} + \sum_{kcd} \langle a\, k | d\, c \rangle\, \bar{r}_{[kc]d} \tag{4.88}$$

$$\sigma_{i[ja]} = \sum_k \langle k\, a | i\, j \rangle\, r_k + \sum_b \langle b\, a | i\, j \rangle\, r_b + \varepsilon_i r_{i[ja]} - \sum_{kb} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{xx}}_{jakb} r_{i[kb]} - \sum_{kb} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{xd}}_{jakb} \bar{r}_{i[kb]} \tag{4.89}$$

$$\bar{\sigma}_{i[ja]} = -\sum_k \langle k\, a | i\, j \rangle\, r_k - \sum_b \langle b\, a | i\, j \rangle\, r_b + \varepsilon_i \bar{r}_{i[ja]} - \sum_{kb} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{dx}}_{ja,kb} r_{i[kb]} - \sum_{kb} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{dd}}_{ja,kb} \bar{r}_{i[kb]} \tag{4.90}$$

$$\sigma_{[ia]b} = \sum_j \langle j\, i | b\, a \rangle\, r_j + \sum_c \langle c\, i | b\, a \rangle\, r_c + \varepsilon_b r_{[ia]b} + \sum_{jc} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{xx}}_{ia,jc} r_{[jc]b} + \sum_{jc} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{xd}}_{ia,jc} \bar{r}_{[jc]b} \tag{4.91}$$

$$\bar{\sigma}_{[ia]b} = -\sum_j \langle j\, i | b\, a \rangle\, r_j - \sum_c \langle c\, i | b\, a \rangle\, r_c + \varepsilon_b \bar{r}_{[ia]b} + \sum_{jc} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{dx}}_{ia,jc} r_{[jc]b} + \sum_{jc} \big[\mathbf{N}\tilde{\mathbf{M}}\big]^{\mathrm{dd}}_{ia,jc} \bar{r}_{[jc]b} \tag{4.92}$$

$\boldsymbol{R}^{2h1p} = \begin{pmatrix} r_i \\ r_a \\ r_{i[jb]} \\ \bar{r}_{i[\bar{jb}]} \end{pmatrix}$ is the excitation vector. We will assume the form $\boldsymbol{\mathcal{N}}^{2h1p} = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \boldsymbol{X} \end{pmatrix}$. We assume that $\boldsymbol{H}^{2h1p}$ is given by

$$\boldsymbol{H}^{2h1p} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{V}^{\mathrm{2h1p}} & \boldsymbol{V}^{\mathrm{2h1p}} \\ \big(\boldsymbol{V}^{\mathrm{2h1p}}\big)^\dagger & & \\ \big(\boldsymbol{V}^{\mathrm{2h1p}}\big)^\dagger & & \boldsymbol{C}^{\mathrm{2hlp}} \end{pmatrix} \tag{4.93}$$

In our notation, $P = O + V$ is the number of molecular orbitals, $A = O^2V$ and T is the column dimension of our excitation vector.

$$\boldsymbol{\mathcal{N}}^{2h1p}\boldsymbol{H}^{2h1p}\boldsymbol{R}^{2h1p} \tag{4.94}$$

$$= \begin{pmatrix} \mathbf{1}_{P,P} & 0 \\ 0 & \boldsymbol{X}_{2A,2A} \end{pmatrix}_{P+2A,P+2A} \begin{pmatrix} \boldsymbol{F}_{P,P} & \boldsymbol{V}^{2h1p}_{P,A} & \boldsymbol{V}^{2h1p}_{P,A} \\ \left(\boldsymbol{V}^{2h1p}\right)^{\dagger}_{A,P} & & \\ \left(\boldsymbol{V}^{2h1p}\right)^{\dagger}_{A,P} & & \boldsymbol{C}^{2hlp}_{2A,2A} \end{pmatrix}_{P+2A,P+2A} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \\ \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{P+2A,T} \tag{4.95}$$

$$= \begin{pmatrix} \mathbf{1}_{P,P} \begin{pmatrix} \boldsymbol{F}_{OO} & \boldsymbol{F}_{OV} \\ \boldsymbol{F}_{VO} & \boldsymbol{F}_{VV} \end{pmatrix}_{P,P} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \end{pmatrix}_{P,T} + \mathbf{1}_{P,P} \begin{pmatrix} \boldsymbol{V}^{2h1p}_{O,i[jb]} & \boldsymbol{V}^{2h1p}_{O,i[jb]} \\ \boldsymbol{V}^{2h1p}_{V,i[jb]} & \boldsymbol{V}^{2h1p}_{V,i[jb]} \end{pmatrix}_{P,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \\ \boldsymbol{X}_{2A,2A} \begin{pmatrix} \left(\boldsymbol{V}^{2h1p}_{O,i[jb]}\right)^{\dagger} & \left(\boldsymbol{V}^{2h1p}_{V,i[jb]}\right)^{\dagger} \\ \left(\boldsymbol{V}^{2h1p}_{O,i[jb]}\right)^{\dagger} & \left(\boldsymbol{V}^{2h1p}_{V,i[jb]}\right)^{\dagger} \end{pmatrix}_{2A,P} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \end{pmatrix}_{P,T} + \boldsymbol{X}_{2A,2A}\boldsymbol{C}^{2hlp}_{2A,2A} \begin{pmatrix} \boldsymbol{r}_{i[jb]} \\ \bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \end{pmatrix}_{2A,T} \end{pmatrix} \tag{4.96}$$

$$= \begin{pmatrix} \boldsymbol{F}_{OO}\boldsymbol{r}_i + \boldsymbol{F}_{OV}\boldsymbol{r}_a + \boldsymbol{V}^{2h1p}_{O,i[jb]}\boldsymbol{r}_{i[jb]} + \boldsymbol{V}^{2h1p}_{O,i[jb]}\bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \\ \boldsymbol{F}_{VO}\boldsymbol{r}_i + \boldsymbol{F}_{VV}\boldsymbol{r}_a + \boldsymbol{V}^{2h1p}_{V,i[jb]}\boldsymbol{r}_{i[jb]} + \boldsymbol{V}^{2h1p}_{V,i[jb]}\bar{\boldsymbol{r}}_{i[\bar{j}\bar{b}]} \\ ? \end{pmatrix}_{P+2A,T} \tag{4.97}$$

$$= \begin{pmatrix} \sum_j f_{ij}r_j + \sum_b f_{ib}r_b + \sum_{klc}\langle i\,c|k\,l\rangle r_{k[lc]} + \sum_{klc}\langle i\,c|k\,l\rangle \bar{r}_{k[lc]} \\ \sum_j f_{aj}r_j + \sum_b f_{ab}r_b + \sum_{klc}\langle a\,c|k\,l\rangle r_{k[lc]} + \sum_{klc}\langle a\,c|k\,l\rangle \bar{r}_{k[lc]} \\ ? \end{pmatrix} \tag{4.98}$$

$$\tag{4.99}$$

Now let's consider that $\boldsymbol{X}_{2A,2A} = \mathbf{1}_O \otimes \boldsymbol{N}_{2OV,2OV} = \mathbf{1}_O \otimes \begin{pmatrix} \mathbf{1}_{OV,OV} & 0 \\ 0 & -\mathbf{1}_{OV,OV} \end{pmatrix}$.Then, it becomes true that

$$(\mathbf{1}_O \otimes \boldsymbol{N}_{2OV,2OV}) \begin{pmatrix} \left(\boldsymbol{V}^{2h1p}_{O,i[jb]}\right)^{\dagger} & \left(\boldsymbol{V}^{2h1p}_{V,i[jb]}\right)^{\dagger} \\ \left(\boldsymbol{V}^{2h1p}_{O,i[jb]}\right)^{\dagger} & \left(\boldsymbol{V}^{2h1p}_{V,i[jb]}\right)^{\dagger} \end{pmatrix}_{2A,P} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \end{pmatrix}_{P,T} \tag{4.100}$$

$$= \begin{pmatrix} \left(\boldsymbol{V}^{2h1p}_{O,i[jb]}\right)^{\dagger} & \left(\boldsymbol{V}^{2h1p}_{V,i[jb]}\right)^{\dagger} \\ \left(-\boldsymbol{V}^{2h1p}_{O,i[jb]}\right)^{\dagger} & -\left(\boldsymbol{V}^{2h1p}_{V,i[jb]}\right)^{\dagger} \end{pmatrix}_{2A,P} \begin{pmatrix} \boldsymbol{r}_i \\ \boldsymbol{r}_a \end{pmatrix}_{P,T} \tag{4.101}$$

$$= \begin{pmatrix} \sum_k \langle k\,a|i\,j\rangle\, r_k + \sum_b \langle b\,a|i\,j\rangle\, r_b \\ -\sum_k \langle k\,a|i\,j\rangle\, r_k - \sum_b \langle b\,a|i\,j\rangle\, r_b \end{pmatrix}_{2A,T} \tag{4.102}$$

But it is still unclear how terms like 4.89, arising from the auxiliary space, come about.

4.2.2 My hypothesis for the supermatrix

By reverse engineering from the matrix vector products given in the supplemental material, I have found the form (we have defined $P = O + V$)

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{F}_{P,P} & \begin{pmatrix} \boldsymbol{V}^{\mathrm{2h1p}} & \boldsymbol{V}^{\mathrm{2h1p}} \end{pmatrix}_{P,2O^2V} & \begin{pmatrix} \boldsymbol{V}^{\mathrm{2p1h}} & \boldsymbol{V}^{\mathrm{2p1h}} \end{pmatrix}_{P,2OV^2} \\ \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^\dagger \\ -\left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} & \boldsymbol{C}^{\mathrm{2hlp}}_{2O^2V,2O^2V} & \boldsymbol{0} \\ \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2p1h}}\right)^\dagger \\ -\left(\boldsymbol{V}^{\mathrm{2p1h}}\right)^\dagger \end{pmatrix}_{2OV^2,P} & \boldsymbol{0} & \boldsymbol{C}^{\mathrm{2plh}}_{2OV^2,2OV^2} \end{pmatrix} \tag{4.103}$$

with

$$V^{2h1p}_{p,k[ia]} = \langle pa|ki\rangle \equiv (pk|ia) \tag{4.104}$$

$$V^{2p1h}_{p,[ia]c} = \langle pi|ac\rangle \equiv (pc|ia) \tag{4.105}$$

$$\tag{4.106}$$

and

$$C^{\mathrm{2hlp}}_{i[ka],j[lb]} = \begin{pmatrix} \varepsilon^{1h}_{ij}\delta_{kl}\delta_{ab} - \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xx}}_{ka,lb}\delta_{ij} & -\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xd}}_{ka,lb}\delta_{ij} \\ -\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dx}}_{ka,lb}\delta_{ij} & \varepsilon^{1h}_{ij}\delta_{kl}\delta_{ab} - \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dd}}_{ja,lb}\delta_{ij} \end{pmatrix} = \delta_{kl}\delta_{ab}\left(\boldsymbol{\varepsilon}^{1h} \oplus_{\mathrm{diag}} \boldsymbol{\varepsilon}^{1h}\right) - \delta_{ij}\boldsymbol{N}\tilde{\mathbf{M}} \tag{4.107}$$

$$C^{\mathrm{2plh}}_{[kc]a,[ld]b} = \begin{pmatrix} \varepsilon^{1p}_{ab}\delta_{kl}\delta_{cd} + \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xx}}_{kc,ld}\delta_{ab} & \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xd}}_{kc,ld}\delta_{ab} \\ \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dx}}_{kc,ld}\delta_{ab} & \varepsilon^{1p}_{ab}\delta_{kl}\delta_{cd} + \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dd}}_{kc,ld}\delta_{ab} \end{pmatrix} = \delta_{kl}\delta_{cd}\left(\boldsymbol{\varepsilon}^{1p} \oplus_{\mathrm{diag}} \boldsymbol{\varepsilon}^{1p}\right) + \delta_{ab}\boldsymbol{N}\tilde{\mathbf{M}} \tag{4.108}$$

with $\boldsymbol{\varepsilon}^{1h}$ and $\boldsymbol{\varepsilon}^{1p}$ diagonal matrices with the 1h and 1p energies on the diagonal, so actually $\varepsilon^{1h}_{ij} = \varepsilon^{1h}_{ii}\delta_{ij}$ and $\varepsilon^{1p}_{ab} = \varepsilon^{1p}_{aa}\delta_{ab}$, respectively and $\tilde{\boldsymbol{M}} = \boldsymbol{M} + \eta\boldsymbol{N}\Theta(-\boldsymbol{N}\boldsymbol{M})$ where

$$\boldsymbol{M} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{A} \end{pmatrix}_{2OV,2OV} \quad \boldsymbol{N} = \begin{pmatrix} \boldsymbol{1} & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{1} \end{pmatrix}_{2OV,2OV} \tag{4.109}$$

and η is some very large number.

Connecting to the exact form of GW-RPA

We know the eigendecomposition $\boldsymbol{N}\tilde{\mathbf{M}} = \boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ + \boldsymbol{\eta} \end{pmatrix}\boldsymbol{Z}^{-1}$. This means we can rewrite 4.107 and 4.108 as

$$C^{\mathrm{2hlp}}_{i[ka],j[lb]} = \delta_{kl}\delta_{ab}\left(\boldsymbol{\varepsilon}^{1h} \oplus_{\mathrm{diag}} \boldsymbol{\varepsilon}^{1h}\right) - \delta_{ij}\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ + \boldsymbol{\eta} \end{pmatrix}\boldsymbol{Z}^{-1} \tag{4.110}$$

$$C^{\mathrm{2plh}}_{[kc]a,[ld]b} = \delta_{kl}\delta_{cd}\left(\boldsymbol{\varepsilon}^{1p} \oplus_{\mathrm{diag}} \boldsymbol{\varepsilon}^{1p}\right) + \delta_{ab}\boldsymbol{Z}\begin{pmatrix} \boldsymbol{\Omega}_+ & 0 \\ 0 & -\boldsymbol{\Omega}_+ + \boldsymbol{\eta} \end{pmatrix}\boldsymbol{Z}^{-1} \tag{4.111}$$

Why would we want an Mtilde of this form?

Now, we know the following identity to be true

$$\boldsymbol{M}\boldsymbol{Z} = \boldsymbol{N}\boldsymbol{Z}\begin{pmatrix}\boldsymbol{\Omega}_+ & 0\\ 0 & -\boldsymbol{\Omega}_+\end{pmatrix} \tag{4.112}$$

with eigenvectors $\boldsymbol{Z} = \begin{pmatrix}\boldsymbol{X} & \boldsymbol{Y}\\ \boldsymbol{Y} & \boldsymbol{X}\end{pmatrix}_{2OV,2OV}$ and eigenvalues $\begin{pmatrix}\boldsymbol{\Omega}_+ & 0\\ 0 & -\boldsymbol{\Omega}_+\end{pmatrix}$. Now we can use the fact that the action of a scalar function f, such as the step function, on a diagonalizable matrix $\boldsymbol{X} \equiv \boldsymbol{Y}\boldsymbol{\Lambda}\boldsymbol{Y}^{-1}$ can be expressed as $f(\boldsymbol{X}) = \boldsymbol{Y}f(\boldsymbol{\Lambda})\boldsymbol{Y}^{-1}$ so we can write

$$\Theta(-\boldsymbol{N}\boldsymbol{M}) = \boldsymbol{Z}\begin{pmatrix}\Theta(-\boldsymbol{\Omega}_+) & 0\\ 0 & \Theta(\boldsymbol{\Omega}_+)\end{pmatrix}\boldsymbol{Z}^{-1} = \boldsymbol{Z}\begin{pmatrix}\boldsymbol{0} & 0\\ 0 & \boldsymbol{1}\end{pmatrix}\boldsymbol{Z}^{-1} \tag{4.113}$$

$$\implies \tilde{\boldsymbol{M}}\boldsymbol{Z} = \boldsymbol{N}\boldsymbol{Z}\begin{pmatrix}\boldsymbol{\Omega}_+ & 0\\ 0 & -\boldsymbol{\Omega}_+ + \boldsymbol{\eta}\end{pmatrix} \quad \text{and} \quad \tilde{\boldsymbol{M}} = \boldsymbol{Z}\boldsymbol{N}\begin{pmatrix}\boldsymbol{\Omega}_+ & 0\\ 0 & -\boldsymbol{\Omega}_+ + \boldsymbol{\eta}\end{pmatrix}\boldsymbol{Z}^{-1} \tag{4.114}$$

So now the purpose of the η becomes clear; it is removing the influence of the negative RPA roots, so that in the GW supermatrix we can easily disregard the portion of its spectrum which inherits from the negative RPA roots. After all, the expression for the GW correlation self-energy is

$$\Sigma^c_{pq}(\omega) = \sum_{\mu}^{RPA}\left[\underbrace{\sum_{i}^{occ}\frac{\rho^{\mu}_{pi}\rho^{\mu}_{qi}}{\omega - (\epsilon_i - \Omega_\mu) - i\eta}}_{2h1p} + \underbrace{\sum_{a}^{virt}\frac{\rho^{n}_{pa}\rho^{n}_{qa}}{\omega - (\epsilon_a + \Omega_\mu) + i\eta}}_{1h2p}\right] \tag{4.115}$$

with the transition densities $\rho^{\nu}_{rs} = \sum_{ia}(rs|ia)\,(X^{\nu}_{ia} + Y^{\nu}_{ia})$, where ν is the RPA index. The standard way to show that this maps to GW-RPA is via similarity transformations, like we did with GW-TDA in 4.1.2, but we can't do this her because $\boldsymbol{N}\tilde{\boldsymbol{M}}$ is not a Hermitian.

Avoid constructing Theta

Right now in order to construct $\Theta(-\boldsymbol{N}\boldsymbol{M})$, I have to diagonalize $\boldsymbol{N}\boldsymbol{M}$ first, which is $O(N^6)$ cost. To avoid this, we can approximate the step function by Chebyshev polynomials. But we don't want to form this Chebyshev polynomial at the beginning of the Davidson iteration, but rather just make an approximation, that will keep improving as we iterate. It is true, though, that we can efficiently estimate the bounds of the spectrum of $\boldsymbol{N}\boldsymbol{M}$ using a few iterations of Arnoldi, getting $\lambda_{\max}$ (because this is RPA, we know that $\lambda_{\min} = -\lambda_{\max}$). So we can rescale the matrix so that its spectrum lies within the interval $[-1, 1]$; this is accomplished by dividing by $\lambda_{\max}$. Then, at each iteration of Davidson, we can apply this polynomial to the current guess vectors. This will be $O(N^4)$ cost per iteration, which is not too bad. The hope is that as we iterate, the guess vectors will span more and more of the space spanned by the negative eigenvalues of $\boldsymbol{N}\boldsymbol{M}$, so that we are effectively applying the step function to the relevant subspace.

Attempting to transform to a Hermitian matrix

The supermatrix 4.103 is not pseudohermitian, or Hermitian under the super metric, so it really is just a similarity transform

$$\boldsymbol{\mathcal{N}} = \begin{pmatrix} \mathbf{1}_{P,P} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1}_{V,V} \otimes \boldsymbol{N}_{2OV,2OV} \end{pmatrix} \tag{4.116}$$

$$\implies \boldsymbol{H}^{\dagger} = \boldsymbol{\mathcal{N}} \boldsymbol{H} \boldsymbol{\mathcal{N}}^{-1} \tag{4.117}$$

$$= \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \otimes \boldsymbol{N} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \otimes \boldsymbol{N} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \begin{pmatrix} \boldsymbol{V}^{\mathrm{2h1p}} & \boldsymbol{V}^{\mathrm{2h1p}} \end{pmatrix} & \begin{pmatrix} \boldsymbol{V}^{\mathrm{2p1h}} & \boldsymbol{V}^{\mathrm{2p1h}} \end{pmatrix} \\ \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \\ -\left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \end{pmatrix} & \boldsymbol{C}^{\mathrm{2hlp}} & \mathbf{0} \\ \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2p1h}}\right)^{\dagger} \\ -\left(\boldsymbol{V}^{\mathrm{2p1h}}\right)^{\dagger} \end{pmatrix} & \mathbf{0} & \boldsymbol{C}^{\mathrm{2plh}} \end{pmatrix} \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \otimes \boldsymbol{N} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \otimes \boldsymbol{N} \end{pmatrix} \tag{4.118}$$

$$= \begin{pmatrix} \boldsymbol{F} & \begin{pmatrix} \boldsymbol{V}^{\mathrm{2h1p}} & \boldsymbol{V}^{\mathrm{2h1p}} \end{pmatrix} & \begin{pmatrix} \boldsymbol{V}^{\mathrm{2p1h}} & \boldsymbol{V}^{\mathrm{2p1h}} \end{pmatrix} \\ (\mathbf{1} \otimes \boldsymbol{N}) \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \\ -\left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \end{pmatrix} & (\mathbf{1} \otimes \boldsymbol{N})\, \boldsymbol{C}^{\mathrm{2hlp}} & \mathbf{0} \\ (\mathbf{1} \otimes \boldsymbol{N}) \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2p1h}}\right)^{\dagger} \\ -\left(\boldsymbol{V}^{\mathrm{2p1h}}\right)^{\dagger} \end{pmatrix} & \mathbf{0} & (\mathbf{1} \otimes \boldsymbol{N})\, \boldsymbol{C}^{\mathrm{2plh}} \end{pmatrix} \begin{pmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \otimes \boldsymbol{N} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \otimes \boldsymbol{N} \end{pmatrix} \tag{4.119}$$

$$\tag{4.120}$$

To simplify notation, from here on, we will just consider the 2h1p sector

$$\boldsymbol{\mathcal{N}}^{\mathrm{2h1p}} \boldsymbol{H}^{\mathrm{2h1p}} \left(\boldsymbol{\mathcal{N}}^{\mathrm{2h1p}}\right)^{-1} \tag{4.121}$$

$$= \begin{pmatrix} \boldsymbol{F}_P & \left[\begin{pmatrix} \boldsymbol{V}^{\mathrm{2h1p}} & \boldsymbol{V}^{\mathrm{2h1p}} \end{pmatrix} (\mathbf{1} \otimes \boldsymbol{N})\right]_{P,2O^2V} \\ \left[(\mathbf{1} \otimes \boldsymbol{N}) \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \\ -\left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \end{pmatrix}\right]_{2O^2V,P} & (\mathbf{1} \otimes \boldsymbol{N})_{2O^2V}\, \boldsymbol{C}^{\mathrm{2hlp}}_{2O^2V}\, (\mathbf{1} \otimes \boldsymbol{N})_{2O^2V} \end{pmatrix} \tag{4.122}$$

Consider just the multiplication

$$\begin{pmatrix} \boldsymbol{V}^{\mathrm{2h1p}} & \boldsymbol{V}^{\mathrm{2h1p}} \end{pmatrix} (\mathbf{1} \otimes \boldsymbol{N}) = \begin{pmatrix} \boldsymbol{V}^{\mathrm{2h1p}} & -\boldsymbol{V}^{\mathrm{2h1p}} \end{pmatrix} \tag{4.123}$$

$$\tag{4.124}$$

and

$$(\mathbf{1} \otimes \boldsymbol{N}) \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \\ -\left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \end{pmatrix} = \begin{pmatrix} \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \\ \left(\boldsymbol{V}^{\mathrm{2h1p}}\right)^{\dagger} \end{pmatrix} \tag{4.125}$$

But now we are left to evaluate

$$(\mathbf{1}\otimes\boldsymbol{N})_{2O^2V,2O^2V}\,\boldsymbol{C}^{\mathrm{2hlp}}_{2O^2V,2O^2V}\,(\mathbf{1}\otimes\boldsymbol{N})_{2O^2V,2O^2V} \tag{4.126}$$

$$=\sum_{i'[k'a']}\sum_{j'[l'b']}(\delta_{ii'}N_{k'a',ka})\left[\begin{pmatrix}\varepsilon^{1h}_{i'j'}\delta_{k'l'}\delta_{a'b'} & 0\\ 0 & \varepsilon^{1h}_{i'j'}\delta_{k'l'}\delta_{a'b'}\end{pmatrix}-\begin{pmatrix}\delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xx}}_{k'a',l'b'} & \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xd}}_{k'a',l'b'}\\ \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dx}}_{k'a',l'b'} & \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dd}}_{k'a',l'b'}\end{pmatrix}\right] \tag{4.127}$$

$$\times(\delta_{jj'}N_{lb,l'b'}) \tag{4.128}$$

$$=\sum_{i'[k'a']}\sum_{j'[l'b']}\begin{pmatrix}\delta_{ii'}\delta_{k'k}\delta_{a'a} & 0\\ 0 & -\delta_{ii'}\delta_{k'k}\delta_{a'a}\end{pmatrix}\left[\begin{pmatrix}\varepsilon^{1h}_{i'j'}\delta_{k'l'}\delta_{a'b'} & 0\\ 0 & \varepsilon^{1h}_{i'j'}\delta_{k'l'}\delta_{a'b'}\end{pmatrix}-\begin{pmatrix}\delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xx}}_{k'a',l'b'} & \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xd}}_{k'a',l'b}\\ \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dx}}_{k'a',l'b'} & \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dd}}_{k'a',l'b}\end{pmatrix}\right] \tag{4.129}$$

$$\times\begin{pmatrix}\delta_{jj'}\delta_{ll'}\delta_{aa'} & 0\\ 0 & -\delta_{jj'}\delta_{ll'}\delta_{aa'}\end{pmatrix} \tag{4.130}$$

$$=\left[\begin{pmatrix}\varepsilon^{1h}_{ij}\delta_{kl}\delta_{ab} & 0\\ 0 & \varepsilon^{1h}_{ij}\delta_{kl}\delta_{ab}\end{pmatrix}-\sum_{i'[k'a']}\sum_{j'[l'b']}\begin{pmatrix}\delta_{ii'}\delta_{k'k}\delta_{a'a} & 0\\ 0 & -\delta_{ii'}\delta_{k'k}\delta_{a'a}\end{pmatrix}\begin{pmatrix}\delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xx}}_{k'a',l'b'} & \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xd}}_{k'a',l'b'}\\ \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dx}}_{k'a',l'b'} & \delta_{i'j'}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dd}}_{k'a',l'b'}\end{pmatrix}\right. \tag{4.131}$$

$$\left.\times\begin{pmatrix}\delta_{jj'}\delta_{ll'}\delta_{aa'} & 0\\ 0 & -\delta_{jj'}\delta_{ll'}\delta_{aa'}\end{pmatrix}\right] \tag{4.132}$$

$$=\left[\begin{pmatrix}\varepsilon^{1h}_{ij}\delta_{kl}\delta_{ab} & 0\\ 0 & \varepsilon^{1h}_{ij}\delta_{kl}\delta_{ab}\end{pmatrix}-\delta_{ij}\begin{pmatrix}\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xx}}_{ka,lb} & -\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{xd}}_{ka,lb}\\ -\left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dx}}_{ka,lb} & \left[\mathbf{N}\tilde{\mathbf{M}}\right]^{\mathrm{dd}}_{ka,lb}\end{pmatrix}\right] \tag{4.133}$$

This is nothing new, but I have left it here in case it becomes useful.

Identification analysis

Some explanation of the terminology is in order. When I talk about a priori identification, I mean that I identify eigenvalues within the spectrum as QPEs by identifying the eigenpair whose eigenvector has the largest overlap with the MO. This is what I have and still am always doing to identify QPEs in the exact ED spectrum; mathematically, it seemed like the most straightforward approach. However, because there is not a unitary transformation to relate the Supp. supermatrix to the exact one, I suspect that this method may not work for the Supp. supermatrix. In a posteriori identification, I choose which eigenvalue of Supp. supermatrix is closest to the exact QPE, by computing all the differences after both diagonalizations have proceeded. Note that this is not a sustainable approach, but I have included it to highlight that my a priori identification method is not working for the Supp. supermatrix. Alternatively, I have seen in this GW supermatrix literature sometimes people adopt different methods for selecting QPEs from the spectrum, so maybe I can try out some other ones. Attached is a screenshot from Tim's paper. Maybe I can try one of the alternatives he proposes. So I see two things that I need to do:

1. Explore different methods for selecting QPEs from the spectrum of the Supp. supermatrix, like the ones proposed in Tim's paper.

2. Find a way to mathematically bound the difference of the Supp. supermatrix with the exact one. From Googling, I see that there are some different ways, for instance, from the idea for the pseudospectrum.

The first plot that I have attached has the exact same format as what I sent you earlier today, but with CO included, as you requested. The following plots look through all the roots for certain molecule/parameter pairings for CO and N2; these are the parameters that seem to be presenting me with the most identification trouble.

What to do next?

I find numerically that all the eigenvalues of the Supp. supermatrix 4.103 are real. This implies that there is some metric with respect to which the matrix is pseudo-Hermitian. I have not been able to find it yet. I have tried

$$\boldsymbol{\mathcal{N}} = \begin{pmatrix} \mathbf{1}_{P,P} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1}_{V,V} \otimes \boldsymbol{N}_{2OV,2OV} \end{pmatrix} \tag{4.134}$$

but this doesn't work (see 4.2.2). I probably want to inverstigate this further first, before I move on to a non-Hermitian Davidson, because if I can find the metric, I can use a Hermitian Davidson, which is simpler.

4.2.3 Alternate possibilities (assuming a typo was made)

Supermetric: Kronecker product, Auxiliary: Kronecker sum

Note that it makes the most sense for the auxiliary block to be a Kronecker sum, because this was the case for TDA as can be seen in 4.9. So here, I tried the case where the supermetric is a Kronecker product

$$\boldsymbol{\mathcal{N}}^{\text{2hlp}} = \begin{pmatrix} \mathbf{1}_{P,P} & 0 \\ 0 & \mathbf{1}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV} \end{pmatrix} \tag{4.135}$$

and the auxiliary block is a Kronecker sum. Now we can consider the matrix multiplication

$$\boldsymbol{\mathcal{N}}^{\text{2hlp}} \boldsymbol{H}^{\text{2hlp}} = \begin{pmatrix} \mathbf{1}_{P,P} & 0 \\ 0 & \mathbf{1}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV} \end{pmatrix} \begin{pmatrix} \boldsymbol{F}_{P,P} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} & \boldsymbol{V}^{\text{2h1p}}_{P,O^2V} \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \\ \left(\boldsymbol{V}^{\text{2h1p}}_{P,O^2V}\right)^\dagger & & \boldsymbol{C}^{\text{2hlp}}_{2O^2V,2O^2V} \end{pmatrix} \tag{4.136}$$

$$= \begin{pmatrix} \boldsymbol{F}_{P,P} & \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2h1p}} \end{pmatrix}_{P,2O^2V} \\ (\mathbf{1} \otimes \boldsymbol{N})_{2O^2V,2O^2V} \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} & (\mathbf{1} \otimes \boldsymbol{N})_{2O^2V,2O^2V} \boldsymbol{C}^{\text{2hlp}}_{2O^2V,2O^2V} \end{pmatrix} \tag{4.137}$$

$$\tag{4.138}$$

Evaluating the off-diagonal term, we have ($\boldsymbol{V}^{2\mathrm{h1p}}$ has the elements $V^{2\ \mathrm{h1p}}_{p,k[ia]} = \langle pa|ki\rangle \equiv (pk|ia)$ and $\boldsymbol{N}_{2OV,2OV} = \mathbf{1}_{OV,OV} \oplus_{\mathrm{direct}} -\mathbf{1}_{OV,OV}$):

$$(\mathbf{1}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV}) \begin{pmatrix} \left(\boldsymbol{V}^{2\mathrm{h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{2\mathrm{h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} = \begin{pmatrix} \left(\boldsymbol{V}^{2\mathrm{h1p}}\right)^\dagger \\ -\left(\boldsymbol{V}^{2\mathrm{h1p}}\right)^\dagger \end{pmatrix}_{2O^2V,P} \tag{4.139}$$

Evaluating the auxiliary term with $\boldsymbol{C}^{2\mathrm{hlp}}_{2O^2V,2O^2V} = \epsilon^{1\ \mathrm{h}} \oplus_{\mathrm{kron}} (-\tilde{\boldsymbol{M}}) = \boldsymbol{\epsilon}^{1\ \mathrm{h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV} + \mathbf{1}_{O,O} \otimes (-\tilde{\boldsymbol{M}})_{2OV,2OV}$ we have

$$\begin{aligned}
(\mathbf{1} \otimes \boldsymbol{N})\, \boldsymbol{C}^{2\mathrm{hlp}} &= (\mathbf{1}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV}) \left(\boldsymbol{\epsilon}^{1\ \mathrm{h}}_{O,O} \otimes \mathbf{1}_{2OV,2OV} + \mathbf{1}_{O,O} \otimes (-\tilde{\boldsymbol{M}})_{2OV,2OV}\right) && (4.140)\\
&= \boldsymbol{\epsilon}^{1\ \mathrm{h}}_{O,O} \otimes \boldsymbol{N}_{2OV,2OV} + \mathbf{1}_{O,O} \otimes -\boldsymbol{N}_{2OV,2OV}\tilde{\boldsymbol{M}}_{2OV,2OV} && (4.141)\\
&= \left[\left(\boldsymbol{\epsilon}^{1\ \mathrm{h}}_{O,O} \otimes \mathbf{1}_{OV,OV}\right) \oplus_{\mathrm{direct}} - \left(\boldsymbol{\epsilon}^{1\ \mathrm{h}}_{O,O} \otimes \mathbf{1}_{OV,OV}\right)\right] + \left(\mathbf{1}_{O,O} \otimes -\boldsymbol{N}_{2OV,2OV}\tilde{\boldsymbol{M}}_{2OV,2OV}\right) && (4.142)\\
&= \begin{pmatrix} \boldsymbol{\epsilon}^{1\ \mathrm{h}} \otimes \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\left(\boldsymbol{\epsilon}^{1\ \mathrm{h}} \otimes \mathbf{1}\right) \end{pmatrix}_{2O^2V} + \left(\mathbf{1}_O \otimes -\boldsymbol{Z}_{2OV} \begin{pmatrix} \boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & \eta - \boldsymbol{\Omega} \end{pmatrix}_{2OV} \boldsymbol{Z}^{-1}_{2OV}\right)_{2O^2V} && (4.143)\\
&= \begin{pmatrix} \boldsymbol{\epsilon}^{1\ \mathrm{h}} \otimes \mathbf{1} & \mathbf{0} \\ \mathbf{0} & -\left(\boldsymbol{\epsilon}^{1\ \mathrm{h}} \otimes \mathbf{1}\right) \end{pmatrix}_{2O^2V} - \left[(\mathbf{1} \otimes \boldsymbol{Z}) \begin{pmatrix} \mathbf{1} \otimes \boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} \otimes (\eta - \boldsymbol{\Omega}) \end{pmatrix} \left(\mathbf{1} \otimes \boldsymbol{Z}^{-1}\right)\right]_{2O^2V} && (4.144)\\
&= (\mathbf{1} \otimes \boldsymbol{Z}) \begin{pmatrix} \boldsymbol{\epsilon}^{1\ \mathrm{h}} \otimes \mathbf{1} + \mathbf{1} \otimes -\boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & -\left(\boldsymbol{\epsilon}^{1\ \mathrm{h}} \otimes \mathbf{1}\right) - (\mathbf{1} \otimes (\eta - \boldsymbol{\Omega})) \end{pmatrix} \left(\mathbf{1} \otimes \boldsymbol{Z}^{-1}\right) && (4.145)\\
& && (4.146)
\end{aligned}$$

where $\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix}$ is the eigenvector matrix and $\boldsymbol{\Omega}$ is the diagonal matrix of positive excitation energies.

Similarity transformation

Now let us define the rotation $\boldsymbol{\zeta} = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \otimes \boldsymbol{Z} \end{pmatrix}$. We can left multiply by $\boldsymbol{\zeta}^{-1}$ and right multiply by $\boldsymbol{\zeta}$ to get

$$\boldsymbol{\zeta}^{-1} \boldsymbol{\mathcal{N}}^{\text{2hlp}} \boldsymbol{H}^{\text{2hlp}} \boldsymbol{\zeta} = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \otimes \boldsymbol{Z}^{-1} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2h1p}} \end{pmatrix} \\ \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ -\left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix} & (\mathbf{1} \otimes \boldsymbol{N}) \boldsymbol{C}^{\text{2hlp}} \end{pmatrix} \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \otimes \boldsymbol{Z} \end{pmatrix} \tag{4.147}$$

$$= \begin{pmatrix} \boldsymbol{F} & \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2h1p}} \end{pmatrix} (\mathbf{1} \otimes \boldsymbol{Z}) \\ (\mathbf{1} \otimes \boldsymbol{Z}^{-1}) \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ -\left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix} & (\mathbf{1} \otimes \boldsymbol{Z}^{-1}\boldsymbol{Z}) \begin{pmatrix} \boldsymbol{\epsilon}^{1\text{ h}} \oplus_{\text{kron}} -\boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{\epsilon}^{1\text{ h}} \oplus_{\text{kron}} [\eta - \boldsymbol{\Omega}] \end{pmatrix} (\mathbf{1} \otimes \boldsymbol{Z}^{-1}\boldsymbol{Z}) \end{pmatrix} \tag{4.148}$$

$$= \begin{pmatrix} \boldsymbol{F} & \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}} & \boldsymbol{V}^{\text{2h1p}} \end{pmatrix} \left(\mathbf{1} \otimes \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix}\right) \\ \left(\mathbf{1} \otimes \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix}^{-1}\right) (\mathbf{1} \otimes \boldsymbol{N}) \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix} & \begin{pmatrix} \boldsymbol{\epsilon}^{1\text{ h}} \oplus_{\text{kron}} -\boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{\epsilon}^{1\text{ h}} \oplus_{\text{kron}} [\eta - \boldsymbol{\Omega}] \end{pmatrix} \end{pmatrix} \tag{4.149}$$

$$= \begin{pmatrix} \boldsymbol{F} & \begin{pmatrix} \boldsymbol{V}^{\text{2h1p}} (\boldsymbol{X} + \boldsymbol{Y}) & \boldsymbol{V}^{\text{2h1p}} (\boldsymbol{X} + \boldsymbol{Y}) \end{pmatrix} \\ \left(\mathbf{1} \otimes \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix}^{-1}\right) \begin{pmatrix} \left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \\ -\left(\boldsymbol{V}^{\text{2h1p}}\right)^\dagger \end{pmatrix} & \begin{pmatrix} \boldsymbol{\epsilon}^{1\text{ h}} \oplus_{\text{kron}} -\boldsymbol{\Omega} & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{\epsilon}^{1\text{ h}} \oplus_{\text{kron}} [\eta - \boldsymbol{\Omega}] \end{pmatrix} \end{pmatrix} \tag{4.150}$$

But because it is true that $\boldsymbol{Z}$ are the eigenvectors of a non-Hermitian eigenproblem, it is not true that $\boldsymbol{Z}^{-1} = \boldsymbol{Z}^\dagger$, so we cannot simplify the lower left block any further. Accordingly, doing ED of this form for the supermatrices gives the wrong spectrum, but I have included this derivation of the similarity transformation because it seems like the most promising approach I have considered so far.

Supermetric: Kronecker sum, Auxiliary: Kronecker product

Let us try the alternate case where $\boldsymbol{C}^{\text{2hlp}} = \left(\boldsymbol{\epsilon}^{1\text{ h}} \otimes -\tilde{\boldsymbol{M}}\right)$, which gives

$$(\mathbf{1} \oplus_{\text{kron}} \boldsymbol{N}) \boldsymbol{C}^{\text{2hlp}} = (\mathbf{1}_{O,O} \oplus_{\text{kron}} \boldsymbol{N}_{2OV,2OV}) \left(\boldsymbol{\epsilon}^{1\text{ h}}_{O,O} \otimes (-\tilde{\boldsymbol{M}})_{2OV,2OV}\right) \tag{4.151}$$

$$= (\mathbf{1}_O \otimes \mathbf{1}_{2OV} + \mathbf{1}_O \otimes \boldsymbol{N}_{2OV}) \left(\boldsymbol{\epsilon}^{1\text{ h}}_{,O} \otimes (-\tilde{\boldsymbol{M}})_{2OV,}\right) \tag{4.152}$$

$$= \boldsymbol{\epsilon}^{1\text{ h}} \otimes -\tilde{\boldsymbol{M}} + \boldsymbol{\epsilon}^{1\text{ h}} \otimes -\boldsymbol{N}\tilde{\boldsymbol{M}} \tag{4.153}$$

$$\tag{4.154}$$

This is not what we want and the form for the off-diagonal term is the same as in the previous subsubsection.

They say that $\boldsymbol{V}^{\mathrm{2h1p}}$ has the elements $V^{2\ \mathrm{h1p}}_{p,k[ia]} = \langle pa|ki\rangle \equiv (pk|ia)$ so we say that $\boldsymbol{v}^{\mathrm{2h1p}}_{O,O^2V}$ is the first O rows of $\boldsymbol{V}^{\mathrm{2h1p}}$, while $\boldsymbol{v}^{\mathrm{2h1p}}_{V,O^2V}$ is the latter V rows of $\boldsymbol{V}^{\mathrm{2h1p}}$.

Chapter 5

Auxiliary-boson GW

5.1 Constructing the Hamiltonian

The idea of this method is not to work in the MO basis, but rather to work in a basis of particle-hole excitations, which are approximated as bosons. So $\hat{a}_a^\dagger \hat{a}_i \approx \hat{b}_\nu^\dagger$ and $\hat{a}_i^\dagger \hat{a}_a \approx \hat{b}_\nu$, where in second quantization $\hat{a}$ are fermionic and $\hat{b}$ are bosonic operators, respectively. The drawback is that these bosonic operators no longer obey the Pauli exclusion principle; what was done above is known as the quasi-boson approximation. Then we define the electron-boson Hamiltonian as

$$\hat{H}^{\mathrm{eB}} = \hat{H}^{\mathrm{e}} + \hat{H}^{\mathrm{B}} + \hat{V}^{\mathrm{eB}} \tag{5.1}$$

where $\hat{H}^{\mathrm{e}}$ is the electronic Hamiltonian, $\hat{H}^{\mathrm{B}}$ is the bosonic Hamiltonian, and $\hat{V}^{\mathrm{eB}}$ is the electron-boson coupling term, given as

$$\hat{H}^{e} = \sum_{pq} f_{pq} \left\{ \hat{a}_p^\dagger \hat{a}_q \right\} \tag{5.2}$$

$$\hat{H}^{B} = \sum_{\nu\mu} A_{\nu\mu} \hat{b}_\nu^\dagger \hat{b}_\mu + \frac{1}{2} \sum_{\nu\mu} B_{\nu\mu} \left(\hat{b}_\nu^\dagger \hat{b}_\mu^\dagger + \hat{b}_\nu \hat{b}_\mu \right) \tag{5.3}$$

$$\hat{V}^{eB} = \sum_{pq,\nu} V_{pq\nu} \left\{ \hat{a}_p^\dagger \hat{a}_q \right\} \left(\hat{b}_\nu^\dagger + \hat{b}_\nu \right) \tag{5.4}$$

$A_{\nu\mu}$ and $B_{\nu\mu}$ denote the dRPA matrices, as

$$\begin{aligned} A_{\nu\mu} &= A_{ia,jb} = \delta_{ij}\delta_{ab} \left(\epsilon_a - \epsilon_i \right) + (ia|bj) \\ B_{v\mu} &= B_{ia,jb} = (ia|jb) \end{aligned} \tag{5.5}$$

and the electron-boson coupling term is defined as

$$V_{pq\nu} = V_{pq,ia} = (pq|ia) \tag{5.6}$$

As we will see shortly, this formalism has the ability to introduce the desired RPA screening. But the connection to Booth's ED is already clear; the physical space is represented by the electronic Hamiltonian, the auxiliary space by the bosonic Hamiltonian, and the coupling between them by the electron-boson coupling term. Right now the bosonic Hamiltonian

(specifically its second term) does not conserve the boson number. To remedy this, we perform a unitary transformation

$$\hat{U}^\dagger \hat{H}^{\mathrm{eB}} \hat{U} \to \tilde{H}^{\mathrm{eB}}. \tag{5.7}$$

5.1.1 Nature of the unitary transformation

First consider what the bosonic Hamiltonian looks like when expressed in the bosonic basis $\boldsymbol{b} = \left(\hat{b}_1, \hat{b}_2, \ldots\right)$ as

$$\hat{H}^{\mathrm{B}}\left(\hat{b}, \hat{b}^\dagger\right) = -\frac{1}{2}\operatorname{tr}\mathbf{A} + \frac{1}{2}\begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix}\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix}\begin{pmatrix} \mathbf{b} \\ \mathbf{b}^\dagger \end{pmatrix} \tag{5.8}$$

$$= -\frac{1}{2}\operatorname{tr}\mathbf{A} + \frac{1}{2}\begin{pmatrix} \boldsymbol{b}^\dagger & \boldsymbol{b} \end{pmatrix}\begin{pmatrix} \boldsymbol{A}\boldsymbol{b} + \boldsymbol{B}\boldsymbol{b}^\dagger \\ \boldsymbol{B}\boldsymbol{b} + \boldsymbol{A}\boldsymbol{b}^\dagger \end{pmatrix} \tag{5.9}$$

$$= -\frac{1}{2}\operatorname{tr}\mathbf{A} + \frac{1}{2}\left[\boldsymbol{b}^\dagger\boldsymbol{A}\boldsymbol{b} + \boldsymbol{b}^\dagger\boldsymbol{B}\boldsymbol{b}^\dagger + \boldsymbol{b}\boldsymbol{B}\boldsymbol{b} + \boldsymbol{b}\boldsymbol{A}\boldsymbol{b}^\dagger\right] \tag{5.10}$$

$$= \boldsymbol{b}^\dagger\boldsymbol{A}\boldsymbol{b} + \frac{1}{2}\left[\boldsymbol{b}^\dagger\boldsymbol{B}\boldsymbol{b}^\dagger + \boldsymbol{b}\boldsymbol{B}\boldsymbol{b}\right] + \mathbf{0} \tag{5.11}$$

$$= \sum_{\nu\mu} A_{\nu\mu}\hat{b}_\nu^\dagger\hat{b}_\mu + \frac{1}{2}\sum_{\nu\mu} B_{\nu\mu}\left(\hat{b}_\nu^\dagger\hat{b}_\mu^\dagger + \hat{b}_\nu\hat{b}_\mu\right) \tag{5.12}$$

Going from 5.10 to 5.11, we used the fact that the final term needs to be put into normal order so we can do $\boldsymbol{b}\boldsymbol{A}\boldsymbol{b}^\dagger = \sum_{\mu\nu} A_{\mu\nu} b_\mu b_\nu^\dagger = \sum_{\mu\nu} A_{\mu\nu}\left(b_\nu^\dagger b_\mu + \delta_{\mu\nu}\right) = \boldsymbol{b}^\dagger\boldsymbol{A}\boldsymbol{b} + \operatorname{Tr}(\boldsymbol{A})$. In the above we have showed equivalence to the previously defined form in 6.20. Within this representation of the bosonic Hamiltonian in the bosonic basis, in 5.8, we recognize the appearance of the RPA matrix. From 5.8, we can obtain a diagonalized form

$$\hat{H}^{\mathrm{B}}\left(\bar{b}, \bar{b}^\dagger\right) = -\frac{1}{2}\operatorname{tr}\mathbf{A} + \frac{1}{2}\left(\overline{\mathbf{b}}^\dagger \overline{\mathbf{b}}\right)\begin{pmatrix} \Omega\mathbf{1} & 0 \\ 0 & \Omega\mathbf{1} \end{pmatrix}\begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{b}}^\dagger \end{pmatrix} \tag{5.13}$$

through a redefinition of the bosonic operators as

$$\begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{b}}^\dagger \end{pmatrix} = \begin{pmatrix} \mathbf{X} & -\mathbf{Y} \\ -\mathbf{Y} & \mathbf{X} \end{pmatrix}^T \begin{pmatrix} \mathbf{b} \\ \mathbf{b}^\dagger \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \mathbf{b} \\ \mathbf{b}^\dagger \end{pmatrix} = \begin{pmatrix} \mathbf{X} & \mathbf{Y} \\ \mathbf{Y} & \mathbf{X} \end{pmatrix}\begin{pmatrix} \overline{\mathbf{b}} \\ \overline{\mathbf{b}}^\dagger \end{pmatrix}. \tag{5.14}$$

Effect on the bosonic Hamiltonian

Now by expanding, we see

$$\hat{H}^{\mathrm{B}}\left(\overline{\mathbf{b}},\overline{\mathbf{b}}^{\dagger}\right)=-\frac{1}{2}\operatorname{tr}\mathbf{A}+\frac{1}{2}\left(\overline{\mathbf{b}}^{\dagger}\overline{\mathbf{b}}\right)\begin{pmatrix}\Omega\mathbf{1} & 0\\ 0 & \Omega\mathbf{1}\end{pmatrix}\begin{pmatrix}\overline{\mathbf{b}}\\ \overline{\mathbf{b}}^{\dagger}\end{pmatrix} \tag{5.15}$$

$$=-\frac{1}{2}\operatorname{tr}\mathbf{A}+\frac{1}{2}\begin{pmatrix}\overline{\mathbf{b}}^{\dagger} & \overline{\mathbf{b}}\end{pmatrix}\begin{pmatrix}\boldsymbol{\Omega}\overline{\mathbf{b}}\\ \boldsymbol{\Omega}\overline{\mathbf{b}}^{\dagger}\end{pmatrix} \tag{5.16}$$

$$=-\frac{1}{2}\operatorname{tr}\mathbf{A}+\frac{1}{2}\left[\overline{\mathbf{b}}^{\dagger}\boldsymbol{\Omega}\overline{\mathbf{b}}+\underbrace{\overline{\mathbf{b}}\boldsymbol{\Omega}\overline{\mathbf{b}}^{\dagger}}_{\bar{b}^{\dagger}\boldsymbol{\Omega}\bar{b}+\mathrm{Tr}(\boldsymbol{\Omega})}\right] \tag{5.17}$$

$$=\overline{\mathbf{b}}^{\dagger}\boldsymbol{\Omega}\overline{\mathbf{b}}+\frac{1}{2}\operatorname{Tr}\left(\boldsymbol{\Omega}-\mathbf{A}\right) \tag{5.18}$$

$$=\sum_{\nu}\Omega_{\nu}\bar{b}_{\nu}^{\dagger}\bar{b}_{\nu}+E_{\mathrm{RPA}}^{c} \tag{5.19}$$

so we removed the non-boson conserving quality of the bosonic Hamiltonian (compare to 6.20).

Effect on the electron-boson coupling term

Originally, the electron-boson coupling term is given as

$$\hat{V}^{\mathrm{eB}}=\sum_{pq,\nu}V_{pq,\nu}\left\{\hat{a}_{p}^{\dagger}\hat{a}_{q}\right\}\left(\hat{b}_{\nu}^{\dagger}+\hat{b}_{\nu}\right) \tag{5.20}$$

Here, we will use that the redefinition of the bosonic operators such that $\hat{b_{\nu}}=\sum_{\mu}\left(\mathbf{X}_{\mu}^{\nu}\hat{\bar{b}}_{\nu}+\mathbf{Y}_{\mu}^{\nu}\hat{\bar{b}}_{\nu}^{\dagger}\right)$ and $\hat{b_{\nu}}^{\dagger}=\sum_{\mu}\left(\mathbf{X}_{\mu}^{\nu}\hat{\bar{b}}_{\nu}^{\dagger}+\mathbf{Y}_{\mu}^{\nu}\hat{\bar{b}}_{\nu}\right)$, which gives $\hat{b}_{\nu}+\hat{b}_{\nu}^{\dagger}=\sum_{\mu}\left(\mathbf{X}_{\mu}^{\nu}+\mathbf{Y}_{\mu}^{\nu}\right)\left(\hat{\bar{b}}_{\nu}+\hat{\bar{b}}_{\nu}^{\dagger}\right)$, so after we plug in

$$\hat{V}^{\mathrm{eB}}=\sum_{pq,\nu}V_{pq,\nu}\left\{\hat{a}_{p}^{\dagger}\hat{a}_{q}\right\}\left(\sum_{\mu}\left(\mathbf{X}_{\mu}^{\nu}+\mathbf{Y}_{\mu}^{\nu}\right)\right)\left(\hat{\bar{b}}_{\nu}+\hat{\bar{b}}_{\nu}^{\dagger}\right) \tag{5.21}$$

$$=\sum_{pq,\nu}W_{pq,\nu}\left\{\hat{a}_{p}^{\dagger}\hat{a}_{q}\right\}\left(\bar{b}_{\nu}+\bar{b}_{\nu}^{\dagger}\right) \tag{5.22}$$

where now we had identified the RPA screened Coulomb interaction $W_{pq,\nu}=V_{pq,\nu}\sum_{\mu}\left(\mathbf{X}_{\mu}^{\nu}+\mathbf{Y}_{\mu}^{\nu}\right)$.

5.2 Connection to Booth supermatrix

We then build the supermatrices $\mathbf{H}$ and $\mathbf{S}$ with matrix elements,

$$H_{IJ}=\langle 0_{\mathrm{F}}0_{\mathrm{B}}|\left[C_{I},\left[\tilde{H}^{\mathrm{eB}},C_{J}^{\dagger}\right]\right]|0_{\mathrm{F}}0_{\mathrm{B}}\rangle$$

$$S_{IJ}=\langle 0_{\mathrm{F}}0_{\mathrm{B}}|\left[C_{I},C_{J}^{\dagger}\right]|0_{\mathrm{F}}0_{\mathrm{B}}\rangle$$

where $\left\{C_I^\dagger\right\} = \left\{ \underbrace{a_i}_{1h}, \underbrace{a_a}_{1p}, \underbrace{a_i b_\nu^\dagger}_{2h1p}, \underbrace{a_a b_\nu}_{1p2p} \right\}$ and $|0\rangle_{\mathrm{F}}$ and $|0\rangle_{\mathrm{B}}$ are the Fermi and boson vacuums. Then constructing $-\boldsymbol{S}^{-1}\boldsymbol{H}$ yields Booth's ED, which is

$$\boldsymbol{H}^{G_0W_0} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W}^{<} & \boldsymbol{W}^{>} \\ \boldsymbol{W}^{\dagger<} & \boldsymbol{d}^{<} & \boldsymbol{0} \\ \boldsymbol{W}^{\dagger>} & \boldsymbol{0} & \boldsymbol{d}^{>} \end{pmatrix} \tag{5.23}$$

where $\boldsymbol{F}$ is the Fock matrix, $\boldsymbol{W}^{<}$ and $\boldsymbol{W}^{>}$ are the lesser and greater components of the RPA screened Coulomb interaction, defined as

$$W^{<}_{pk\nu} = \sum_{ia} (pk|ia)\left(X^{\nu}_{ia} + Y^{\nu}_{ia}\right) \quad \text{and} \quad W^{>}_{pc\nu} = \sum_{ia} (pc|ia)\left(X^{\nu}_{ia} + Y^{\nu}_{ia}\right) \tag{5.24}$$

and the auxiliary blocks $\boldsymbol{d}^{<}$ and $\boldsymbol{d}^{>}$ are defined as

$$d^{<}_{k\nu,l\nu'} = (\epsilon_k - \Omega_\nu)\,\delta_{k,l}\delta_{\nu,\nu'} \quad \text{and} \quad d^{>}_{c\nu,d\nu'} = (\epsilon_c + \Omega_\nu)\,\delta_{c,d}\delta_{\nu,\nu'} \tag{5.25}$$

5.2.1 Derivation of the supermatrices for the 2h1p sector

Overlap

Computing the matrix elements of the $\boldsymbol{S}$ gives:

$$\boldsymbol{S} = \begin{pmatrix} \delta_{ij} & 0 & 0 & 0 \\ 0 & -\delta_{ab} & 0 & 0 \\ 0 & 0 & \delta_{ij}\delta_{\nu\nu'} & 0 \\ 0 & 0 & 0 & -\delta_{ab}\delta_{\nu\nu'} \end{pmatrix} \tag{5.26}$$

$\boldsymbol{H}$ takes more care.

Physical

$$H_{ij} = \langle 0_\mathrm{F} 0_\mathrm{B}| \left[a_i^\dagger, \left[\hat{H}^\mathrm{eB}, a_j\right]\right] |0_\mathrm{F} 0_\mathrm{B}\rangle = \langle 0_\mathrm{F} 0_\mathrm{B}| \left[a_i^\dagger, \left[\hat{H}^\mathrm{e}, a_j\right]\right] |0_\mathrm{F} 0_\mathrm{B}\rangle \tag{5.27}$$

We can make this simplification because the electronic operators commute with all bosonic operators. Now

$$[\hat{H}^e,\ a_j] = \sum_{pq} f_{pq} \left[a_p^\dagger a_q,\ a_j\right] = \sum_{pq} f_{pq} \left(a_p^\dagger \left[\, a_q, a_j \,\right] + [a_p^\dagger, a_j] a_q\right) \tag{5.28}$$

$$= \sum_{pq} f_{pq} \left(a_p^\dagger a_q a_j - a_p^\dagger a_j a_q + a_p^\dagger a_j a_q - a_j a_p^\dagger a_q\right) \tag{5.29}$$

$$[a_i^\dagger,\ [\hat{H}^e,\ a_j]] = \sum_{pq} f_{pq} \left[a_i^\dagger, \left(a_p^\dagger a_q a_j - a_j a_p^\dagger a_q\right)\right] \tag{5.30}$$

$$= \sum_{pq} f_{pq} \left([a_i^\dagger, a_p^\dagger a_q a_j] - [a_i^\dagger, a_j a_p^\dagger a_q]\right) \tag{5.31}$$

$$= \sum_{pq} f_{pq} \left(a_i^\dagger a_p^\dagger a_q a_j - a_p^\dagger a_q a_j a_i^\dagger - a_i^\dagger a_j a_p^\dagger a_q + a_j a_p^\dagger a_q a_i^\dagger\right) \tag{5.32}$$

$$a_i^\dagger a_p^\dagger a_q a_j = \overbrace{a_i^\dagger \overbrace{a_p^\dagger a_q} a_j} + \overbrace{a_i^\dagger \overbrace{a_p^\dagger a_q a_j}} = -\delta_{iq}\delta_{jp} + \cancel{\delta_{ij}\delta_{pq}} \tag{5.33}$$

$$a_p^\dagger a_q a_j a_i^\dagger = 0 \tag{5.34}$$

$$a_i^\dagger a_j a_p^\dagger a_q = \overbrace{a_i^\dagger \overbrace{a_j \overbrace{a_p^\dagger a_q}}} + \overbrace{a_i^\dagger \overbrace{a_j a_p^\dagger a_q}} = 0 + \cancel{\delta_{ij}\delta_{pq}} \tag{5.35}$$

$$a_j a_p^\dagger a_q a_i^\dagger = 0 \tag{5.36}$$

$$= -\sum_{pq} f_{pq}\delta_{iq}\delta_{jp} = -f_{ji} \tag{5.37}$$

So $H_{ij} = -f_{ji}$ and then $\boxed{(-S^{-1}H)_{ij} = +\delta_{ij} f_{ji} = f_{ii} = \epsilon_i}$.

Auxiliary

$$H_{i\nu j\nu'} = \langle 0_\mathrm{F} 0_\mathrm{B}| \left[a_i^\dagger b_\nu, \left[\hat{H}^\mathrm{eB}, a_j b_{\nu'}^\dagger\right]\right] |0_\mathrm{F} 0_\mathrm{B}\rangle = \langle 0_\mathrm{F} 0_\mathrm{B}| \left[a_i^\dagger b_\nu, \left[\hat{H}^\mathrm{e} + \hat{H}^\mathrm{B} + \hat{V}^\mathrm{eB}, a_j b_{\nu'}^\dagger\right]\right] |0_\mathrm{F} 0_\mathrm{B}\rangle \tag{5.38}$$

$$\tag{5.39}$$

First

$$[\hat{H}^e,\ a_j b_\mu^\dagger] = \sum_{pq} f_{pq}\,[a_p^\dagger a_q, a_j b_\mu^\dagger] = \sum_{pq} f_{pq}[a_p^\dagger a_q a_j - a_j a_p^\dagger a_q] b_\mu^\dagger \tag{5.40}$$

$$[a_i^\dagger b_\nu,\ [\hat{H}^e,\ a_j b_\mu^\dagger]] = [a_i^\dagger b_\nu,\ \sum_{pq} f_{pq}[a_p^\dagger a_q a_j - a_j a_p^\dagger a_q] b_\mu^\dagger] = \sum_{pq} f_{pq}[a_i^\dagger, a_p^\dagger a_q a_j - a_j a_p^\dagger a_q][b_\nu, b_\mu^\dagger] \tag{5.41}$$

$$= \delta_{\nu\mu} \sum_{pq} f_{pq} \left(a_i^\dagger a_p^\dagger a_q a_j - a_p^\dagger a_q a_j a_i^\dagger - a_i^\dagger a_j a_p^\dagger a_q + a_j a_p^\dagger a_q a_i^\dagger \right) \tag{5.42}$$

$$a_i^\dagger a_p^\dagger a_q a_j = \overbrace{a_i^\dagger \overbrace{a_p^\dagger a_q} a_j} + \overbrace{a_i^\dagger \overbrace{a_p^\dagger a_q a_j}} = -\underbrace{\delta_{iq}\delta_{jp}}_{q,p\in O} + \cancel{\delta_{ij}\delta_{pq}} \tag{5.43}$$

$$a_p^\dagger a_q a_j a_i^\dagger = 0 \tag{5.44}$$

$$a_i^\dagger a_j a_p^\dagger a_q = \overbrace{a_i^\dagger \overbrace{a_j a_p^\dagger} a_q} + \overbrace{a_i^\dagger a_j} \overbrace{a_p^\dagger a_q} = 0 + \cancel{\delta_{ij}\delta_{pq}} \tag{5.45}$$

$$a_p^\dagger a_j a_q a_i^\dagger = 0 \tag{5.46}$$

$$\tag{5.47}$$

So $[a_i^\dagger b_\nu,\ [\hat{H}^e,\ a_j b_\mu^\dagger]] = -\delta_{\nu\mu} f_{ji}$. Next

$$[\hat{H}^B,\ a_j b_\mu^\dagger] = [\sum_\nu \Omega_\nu b_\nu^\dagger b_\nu + E^c_{\mathrm{RPA}}, a_j b_\mu^\dagger] = \sum_\nu \Omega_\nu a_j [b_\nu^\dagger b_\nu, b_\mu^\dagger] = \sum_\nu \Omega_\nu a_j b_\nu^\dagger [b_\nu, b_\mu^\dagger] = \Omega_\mu a_j b_\mu^\dagger \tag{5.48}$$

$$[a_i^\dagger b_\nu,\ [\hat{H}^B,\ a_j b_\mu^\dagger]] = \Omega_\mu [a_i^\dagger b_\nu, a_j b_\mu^\dagger] = \Omega_\mu [a_i^\dagger, a_j][b_\nu, b_\mu^\dagger] = \delta_{ij}\delta_{\nu\mu}\Omega_\mu \tag{5.49}$$

$$\tag{5.50}$$

So $[a_i^\dagger b_\nu,\ [\hat{H}^B,\ a_j b_\mu^\dagger]] = \delta_{ij}\delta_{\nu\mu}\Omega_\mu$. last

$$[a_i^\dagger b_\nu,\ [\hat{V}^{eB},\ a_j b_{\nu'}^\dagger]] = 0 \tag{5.51}$$

because we notice that in the coupling term, the number of bosons is not conserved. So $H_{i\nu j\nu'} = \delta_{\nu\nu'}\left(\delta_{ij}\Omega_{\nu'} - f_{ji}\right)$ and then $\boxed{(-S^{-1}H)_{i\nu j\nu'} = -\delta_{ij}\left(\delta_{\nu\nu'}\left(\delta_{ij}\Omega_\nu - f_{ji}\right)\right) = \delta_{ij}\delta_{\nu\nu'}\left(\epsilon_i - \Omega_\nu\right)}$.

Coupling

$$H_{i,p\nu} = \langle 0_{\mathrm{F}} 0_{\mathrm{B}}| \left[a_i^\dagger, \left[\hat{H}^{\mathrm{eB}}, a_p b_\nu^\dagger\right]\right] |0_{\mathrm{F}} 0_{\mathrm{B}}\rangle = \langle 0_{\mathrm{F}} 0_{\mathrm{B}}| \left[a_i^\dagger, \left[\hat{V}^{\mathrm{eB}}, a_p b_\nu^\dagger\right]\right] |0_{\mathrm{F}} 0_{\mathrm{B}}\rangle \tag{5.52}$$

Note that we have neglected the electronic and bosonic Hamiltonians, because using them

in this arrangement will not conserve the number of bosons. So we have

$$[V^{eB},\ a_p b_\nu^\dagger] = \sum_{rs,\nu'} W_{rs,\nu'} [a_r^\dagger a_s (b_{\nu'} + b_{\nu'}^\dagger), a_p b_\nu^\dagger] = \sum_{rs,\nu'} W_{rs,\nu'} [a_r^\dagger a_s, a_p][b_{\nu'}, b_\nu^\dagger] \tag{5.53}$$

$$= \sum_{rs,\nu} W_{rs,\nu} \left(a_r^\dagger a_s a_p - a_p a_r^\dagger a_s \right) \tag{5.54}$$

$$[a_i^\dagger,\ [\hat{V}^{eB},\ a_p b_\nu^\dagger]] = \sum_{rs} W_{rs,\nu} \left[a_i^\dagger \hat{a}_r^\dagger \hat{a}_s a_p - \hat{a}_r^\dagger \hat{a}_s a_p a_i^\dagger - \left(a_i^\dagger a_p a_r^\dagger a_s - a_p a_r^\dagger a_s a_i^\dagger \right) \right] \tag{5.55}$$

$$a_i^\dagger \hat{a}_r^\dagger \hat{a}_s a_p = \overbrace{a_i^\dagger \overbrace{a_r^\dagger a_s} a_p} + \overbrace{a_i^\dagger \overbrace{a_r^\dagger a_s a_p}} = -\underbrace{\delta_{is}\delta_{rp}}_{s \in O} + \cancel{\delta_{ip}\delta_{rs}} \tag{5.56}$$

$$\hat{a}_r^\dagger \hat{a}_s a_p a_i^\dagger = \overbrace{a_r^\dagger \overbrace{a_s a_p} a_i^\dagger} + \overbrace{a_r^\dagger a_s \overbrace{a_p a_i^\dagger}} = 0 \tag{5.57}$$

$$a_i^\dagger a_p a_r^\dagger a_s = \overbrace{a_i^\dagger a_p} \overbrace{a_r^\dagger a_s} + \overbrace{a_i^\dagger \overbrace{a_p a_r^\dagger} a_s} = \cancel{\delta_{ip}\delta_{rs}} + 0 \tag{5.58}$$

$$a_p a_r^\dagger a_s a_i^\dagger = 0 \tag{5.59}$$

$$\tag{5.60}$$

So we can write $H_{i,p\nu} = -W_{pi,\nu} = -W_{ip,\nu}$ because of the permutational symmetry of the ERIs. Then $\boxed{(-S^{-1}H)_{p,i\nu} = +W_{pi,\nu}}$.

5.3 Implementation for periodic systems

5.3.1 RI

We know the periodic integrals can be expressed as:

$$(i\mathbf{k}_1, a\mathbf{k}_2 \mid j\mathbf{k}_3, b\mathbf{k}_4) = \delta_{\mathbf{k}_1+\mathbf{k}_2,\mathbf{k}_3+\mathbf{k}_4+\mathbf{G}} (ia \mid jb)_{\mathbf{q}'} \tag{5.61}$$

where $\mathbf{q}' = \mathbf{k}_1 - \mathbf{k}_3$ and $\mathbf{G}$ is a reciprocal lattice vector. The RI coefficients are then given by

$$R_{jb}^L(\mathbf{q}) = \sum_Q \left[\mathbf{V}^{-1/2}(\mathbf{q}) \right]_{LQ} (Q\mathbf{q} \mid j\mathbf{k}_j, b\mathbf{k}_b) \tag{5.62}$$

It is understood that we must have $\boldsymbol{q} = \mathbf{k}_j - \mathbf{k}_b$ to ensure momentum conservation and $\mathbf{V}(\mathbf{q})$ is the Coulomb metric in the auxiliary basis for that $\mathbf{q}$:

$$V_{PQ}(\mathbf{q}) = (P\mathbf{q} \mid Q\mathbf{q}) = \sum_{\mathbf{R}} e^{-i\mathbf{q}\cdot\mathbf{R}} \int d\mathbf{r} d\mathbf{r}' \chi_P(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} \chi_Q(\mathbf{r}' - \mathbf{R}) \tag{5.63}$$

So we can recover the ERIs as

$$(ia \mid jb)_{\mathbf{q}} = \sum_L \left(R^L_{ia}(\mathbf{q})\right)^\dagger R^L_{jb}(\mathbf{q}) \tag{5.64}$$

$$= \sum_L \left(\sum_Q (i\mathbf{k}_i, a\mathbf{k}_a \mid Q\mathbf{q})[\mathbf{V}^{-1/2}(\mathbf{q})]_{QL} \right) \left(\sum_R [\mathbf{V}^{-1/2}(\mathbf{q})]_{LR} (R\mathbf{q} \mid j\mathbf{k}_j, b\mathbf{k}_b) \right) \tag{5.65}$$

$$= \sum_{QR} (i\mathbf{k}_i, a\mathbf{k}_a \mid Q\mathbf{q}) \left(\sum_L [\mathbf{V}^{-1/2}(\mathbf{q})]_{QL} [\mathbf{V}^{-1/2}(\mathbf{q})]_{LR} \right) (R\mathbf{q} \mid j\mathbf{k}_j, b\mathbf{k}_b) \tag{5.66}$$

$$= \sum_{QR} (i\mathbf{k}_i, a\mathbf{k}_a \mid Q\mathbf{q}) [\mathbf{V}^{-1}(\mathbf{q})]_{QR} (R\mathbf{q} \mid j\mathbf{k}_j, b\mathbf{k}_b) \tag{5.67}$$

$$= (i\mathbf{k}_i, a\mathbf{k}_a \mid j\mathbf{k}_j, b\mathbf{k}_b)_{\mathbf{q}}. \tag{5.68}$$

where in the last step we used the definition of 5.63 to make the appropriate cancellation. Then, we can define a basis of auxiliary bosons as

$$\hat{b}_\nu(\mathbf{q}) = \sum_Q^{N_{AB}} C^Q_\nu(\mathbf{q}) \hat{b}_Q(\mathbf{q}) \tag{5.69}$$

where we define the expansion coefficients as

$$C^Q_\nu(\mathbf{q}) = \sum_L R^L_\nu(\mathbf{q}) \left[\boldsymbol{S}^{-1/2}(\mathbf{q})\right]_{LM} P^Q_M \tag{5.70}$$

where $S_{LM}(\mathbf{q}) = \sum_\nu R^L_\nu(\mathbf{q}) R^M_\nu(\mathbf{q}) = \sum_Q P^Q_L(\mathbf{q}) E_Q(\mathbf{q}) P^Q_M(\mathbf{q})$ can be thought of as the overlap matrix in this AB basis where the final expression employs its eigendecomposition. Even though the axillary bases already is small, we can obtain a further truncated basis by only choosing the eigenvalues over a certain threshhold $E_Q(\mathbf{q}) > \epsilon_{AB}$. This is a similar idea to the one that was used to identify rank deficiency in Lanczos.

Scaling comments

Getting the RI coefficients into the MO basis scales as $O(N^3_{\text{orb}} N_{\text{aux}} N_{\mathbf{k}})$, the determination of the overlap matrix scales as $O(N_o N_v N^2_{\text{aux}} N^2_{\mathbf{k}})$, and then computing the AB basis will cost $O(N_{\text{AB}} N^2_{\text{aux}} N_{\mathbf{k}})$. In order to construct the transformed electron boson Hamiltonian in the AB basis we first need to solve the symmetrized dRPA eigenvalue problem

5.4 ECC on the electron-boson Hamiltonian

5.4.1 Extended Coupled Cluster

The extended CC energy bi-functional, with $\hat{T} = \sum_q t_q \hat{\tau}_q$ and $\hat{Z} = \sum_q z_q \hat{\tau}^\dagger_q$, is given by

$$E_{\text{ECC}} = \langle \Phi_0 | e^{\hat{Z}} e^{-\hat{T}} \hat{H} e^{\hat{T}} e^{-\hat{Z}} | \Phi_0 \rangle \tag{5.71}$$

$$= \langle \Phi_0 | e^{\hat{Z}} \bar{H} e^{-\hat{Z}} | \Phi_0 \rangle \tag{5.72}$$

where $\bar{H} = e^{-\hat{T}}\hat{H}e^{\hat{T}}$ is the similarity-transformed Hamiltonian. Differentiating the functional with respect to z_q gives the amplitude equations:

$$0 = \frac{\partial E_{\mathrm{ECC}}}{\partial z_q} \tag{5.73}$$

$$= \langle\Phi_0|\hat{\tau}_q^\dagger e^{\hat{Z}}\bar{H}e^{-\hat{Z}}|\Phi_0\rangle - \langle\Phi_0|e^{\hat{Z}}\bar{H}e^{-\hat{Z}}\hat{\tau}_q^\dagger|\Phi_0\rangle \tag{5.74}$$

$$= \langle\Phi_0|\hat{\tau}_q^\dagger e^{\hat{Z}}\bar{H}e^{-\hat{Z}}|\Phi_0\rangle \tag{5.75}$$

$$= \langle\Phi_q|e^{\hat{Z}}\bar{H}|\Phi_0\rangle \tag{5.76}$$

$$= \langle\Phi_q|e^{\hat{Z}}e^{-\hat{T}}\hat{H}e^{\hat{T}}|\Phi_0\rangle. \tag{5.77}$$

Differentiating with respect to t_q gives the adjoint amplitude equations:

$$0 = \frac{\partial E_{\mathrm{ECC}}}{\partial t_q} \tag{5.78}$$

$$= \langle\Phi_0|e^{\hat{Z}}\frac{\partial\bar{H}}{\partial t_q}e^{-\hat{Z}}|\Phi_0\rangle \tag{5.79}$$

$$= \langle\Phi_0|e^{\hat{Z}}\left(e^{-\hat{T}}[\hat{H},\hat{\tau}_q]e^{\hat{T}}\right)e^{-\hat{Z}}|\Phi_0\rangle \tag{5.80}$$

$$= \langle\Phi_0|e^{\hat{Z}}e^{-\hat{T}}[\hat{H},\hat{\tau}_q]e^{\hat{T}}|\Phi_0\rangle \tag{5.81}$$

$$= \langle\Phi_0|e^{\hat{Z}}[\bar{H},\hat{\tau}_q]|\Phi_0\rangle \tag{5.82}$$

This framework forces the t- and z-amplitudes to be solved simultaneously. Restricting to double excitations, that is, $\hat{T} = \hat{T}_2$, we can rewrite the ECC energy functional as

$$E_{\mathrm{ECC}} \approx \langle\Phi_0|e^{\hat{Z}}e^{-\hat{T}_2}\hat{H}e^{\hat{T}_2}e^{-\hat{Z}}|\Phi_0\rangle \tag{5.83}$$

$$= \langle\Phi_0|e^{\hat{Z}}e^{-\hat{T}_2}\hat{H}e^{\hat{T}_2}|\Phi_0\rangle \tag{5.84}$$

$$= \langle\Phi_0|e^{\hat{Z}}\left(\hat{H}e^{\hat{T}_2}\right)_{\mathrm{c}}|\Phi_0\rangle \tag{5.85}$$

$$= \langle\Phi_0|e^{\hat{Z}}\left(\hat{H} + \hat{H}\hat{T}_2 + \frac{1}{2}\hat{H}\hat{T}_2^2 + \frac{1}{6}\hat{H}\hat{T}_2^3 + \frac{1}{24}\hat{H}\hat{T}_2^4\right)_{\mathrm{c}}|\Phi_0\rangle \tag{5.86}$$

$$= \langle\Phi_0|\hat{H} + \hat{Z}\left(\hat{H} + \hat{H}\hat{T}_2 + \frac{1}{2}\hat{H}\hat{T}_2^2\right)_{\mathrm{c}} \tag{5.87}$$

$$+ \frac{1}{2}\hat{Z}^2\left(\frac{1}{2}\hat{H}\hat{T}_2^2 + \frac{1}{6}\hat{H}\hat{T}_2^3\right)_{\mathrm{c}} + \frac{1}{6}\hat{Z}^3\left(\frac{1}{24}\hat{H}\hat{T}_2^4\right)_{\mathrm{c}}|\Phi_0\rangle$$

Note that we can get to 5.85 by performing the BCH expansion of $e^{-\hat{T}_2}\hat{H}e^{\hat{T}_2}$ and then, due to the linked cluster theorem, only keeping the connected terms survive. Then, we can get to 5.86 because $\hat{H}$ contains at most two-body interactions, so the series naturally truncates at the fourth power of $\hat{T}_2$ in the energy expression. Finally, we can get to 5.87 by expanding $e^{\hat{Z}}$, but once again only keeping the terms that survive the vacuum expectation value.

Comparison with UCC

The ECC form, $e^{\hat{Z}}e^{-\hat{T}}\hat{H}e^{\hat{T}}e^{-\hat{Z}}$, can be recast as $e^{-\hat{t}}\hat{H}e^{\hat{t}}$ with $\hat{t} = \hat{T} - \hat{Z} + \frac{1}{2}\left[\hat{Z}, \hat{T}\right] + \cdots$, whereas UCC has $\hat{t} = \hat{T} - \hat{T}^\dagger$. In UCC, the anti-Hermiticity condition $\hat{t}^\dagger = -\hat{t}$ ensures a unitary transformation. In contrast, ECC does not enforce this constraint: $\hat{Z}$ is independent of $\hat{T}$, making $\hat{t}$ generally non-Hermitian, offering greater flexibility at the cost of Hermiticity.

5.4.2 ECC on the electron-boson Hamiltonian

$$\hat{H}_{\mathrm{eB}} = \hat{H}_{\mathrm{e}} + \hat{H}_{\mathrm{B}} + \hat{V}_{\mathrm{eB}} \tag{5.88}$$

where

$$\hat{H}_{\mathrm{e}} = \sum_{pq} f_{pq}\hat{a}_p^\dagger \hat{a}_q \tag{5.89}$$

$$\hat{H}_{\mathrm{B}} = \sum_{\mu\nu} A_{\mu\nu}\hat{b}_\mu^\dagger \hat{b}_\nu + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}\left(\hat{b}_\mu^\dagger \hat{b}_\nu^\dagger + \hat{b}_\mu \hat{b}_\nu\right) \tag{5.90}$$

$$\hat{V}_{\mathrm{eB}} = \sum_{pq\nu} V_{pq,\nu}\hat{a}_p^\dagger \hat{a}_q \left(\hat{b}_\nu^\dagger + \hat{b}_\nu\right) \tag{5.91}$$

We can perform the double similarity transformation on $\hat{H}_{\mathrm{eB}}$ as

$$\begin{aligned} \bar{\bar{H}}_{\mathrm{eB}} &= e^{\hat{Z}}e^{-\hat{T}}\hat{H}_{\mathrm{eB}}e^{\hat{T}}e^{-\hat{Z}} \\ &= \bar{\hat{H}}_{\mathrm{eB}} + [\hat{Z}, \bar{\hat{H}}_{\mathrm{eB}}] \\ &= \hat{H}_{\mathrm{e}} + \bar{\bar{\hat{H}}}_{\mathrm{B}} + \bar{\bar{V}}_{\mathrm{eB}}, \end{aligned} \tag{5.92}$$

where none of the similarity transformations effected $\hat{H}_{\mathrm{e}}$ because $\hat{T}$ and $\hat{Z}$ are purely (quasi)bosonic operators.

Working out $\bar{\bar{\hat{V}}}_{\mathrm{eB}}$

Using the first line of the footnote[1], the first similarity transformation gives

$$\begin{aligned}
\bar{\hat{V}}_{\mathrm{eB}} &= e^{-\hat{T}}\hat{V}_{\mathrm{eB}}e^{\hat{T}} \\
&= \sum_{pq}\sum_{\nu} V_{pq,\nu}\hat{a}_p^\dagger\hat{a}_q e^{-\hat{T}}\left(\hat{b}_\nu^\dagger + \hat{b}_\nu\right)e^{\hat{T}} \\
&= \sum_{pq}\sum_{\nu} V_{pq,\nu}\hat{a}_p^\dagger\hat{a}_q\left(\hat{b}_\nu^\dagger + \hat{b}_\nu + \sum_{\lambda} t_{\nu\lambda}\hat{b}_\lambda^\dagger\right) \\
&= \sum_{pq\nu} V_{pq,\nu}\hat{a}_p^\dagger\hat{a}_q\left(\hat{b}_\nu^\dagger + \hat{b}_\nu\right) + \sum_{pq\nu}\sum_{\lambda} V_{pq,\lambda}t_{\lambda\nu}\hat{a}_p^\dagger\hat{a}_q\hat{b}_\nu^\dagger
\end{aligned} \tag{5.95}$$

Then, using the second line of 1, the second similarity transformation gives

$$\bar{\bar{V}}_{\mathrm{eB}} = \sum_{pq\nu} V_{pq,\nu}\hat{a}_p^\dagger\hat{a}_q e^{\hat{Z}}\left(\hat{b}_\nu^\dagger + \hat{b}_\nu\right)e^{-\hat{Z}} + \sum_{pq\nu}\sum_{\lambda} V_{pq,\lambda}t_{\lambda\nu}\hat{a}_p^\dagger\hat{a}_q e^{\hat{Z}}\hat{b}_\nu^\dagger e^{-\hat{Z}} \tag{5.96}$$

$$= \sum_{pq\nu} V_{pq,\nu}\hat{a}_p^\dagger\hat{a}_q\left(\hat{b}_\nu^\dagger + \hat{b}_\nu + \sum_{\mu} z_{\nu\mu}\hat{b}_\mu\right) + \sum_{pq\nu}\sum_{\lambda} V_{pq,\lambda}t_{\lambda\nu}\hat{a}_p^\dagger\hat{a}_q\left(\hat{b}_\nu^\dagger + \sum_{\mu} z_{\nu\mu}\hat{b}_\mu\right) \tag{5.97}$$

$$= \sum_{pq\nu} V_{pq,\nu}\hat{a}_p^\dagger\hat{a}_q(\hat{b}_\nu^\dagger + \hat{b}_\nu) + \sum_{pq\nu}\left(\sum_{\lambda} V_{pq,\lambda}t_{\lambda\nu}\right)\hat{a}_p^\dagger\hat{a}_q\hat{b}_\nu^\dagger + \sum_{pq\nu}\left[\sum_{\lambda} V_{pq,\lambda}z_{\lambda\nu} + \sum_{\mu\lambda} V_{pq,\lambda}t_{\lambda\mu}z_{\mu\nu}\right]\hat{a}_p^\dagger\hat{a}_q\hat{b}_\nu \tag{5.98}$$

Working out $\bar{\bar{\hat{H}}}_{\mathrm{B}}$

Because $\hat{H}_{\mathrm{B}}$ is quadratic, the BCH expansion associated with the first similarity transformation truncates exactly at second order, yielding

$$\bar{\hat{H}}_{\mathrm{B}} = \hat{H}_{\mathrm{B}} + [\hat{H}_{\mathrm{B}}, \hat{T}] + \frac{1}{2}[[\hat{H}_{\mathrm{B}}, \hat{T}], \hat{T}] \tag{5.99}$$

[1]

$$[\hat{b}_\nu, \hat{T}] = \sum_{\lambda} t_{\nu\lambda}\hat{b}_\lambda^\dagger; \quad [\hat{b}_\nu^\dagger, \hat{T}] = 0 \tag{5.93}$$

$$[\hat{Z}, \hat{b}_\nu^\dagger] = \sum_{\mu} z_{\nu\mu}\hat{b}_\mu; \quad [\hat{Z}, \hat{b}_\nu] = 0 \tag{5.94}$$

Recalling that $\hat{H}_{\rm B} = \sum_{\mu\nu} A_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}_\nu + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}\left(\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu + \hat{b}_\mu\hat{b}_\nu\right)$ and $\hat{T} = \frac{1}{2}\sum_{\mu\nu} t_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu$, we evaluate

$$[\hat{H}_{\rm B},\hat{T}] = \sum_{\mu\nu} A_{\mu\nu}\hat{b}^\dagger_\mu[\hat{b}_\nu,\hat{T}] + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}[\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu,\hat{T}] + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}[\hat{b}_\mu\hat{b}_\nu,\hat{T}] \tag{5.100}$$

$$= \sum_{\mu\nu\lambda} A_{\mu\nu}t_{\nu\lambda}\hat{b}^\dagger_\mu\hat{b}^\dagger_\lambda + 0 + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}(\hat{b}_\mu[\hat{b}_\nu,\hat{T}] + [\hat{b}_\mu,\hat{T}]\hat{b}_\nu) \tag{5.101}$$

$$= \sum_{\mu\nu\lambda} A_{\mu\nu}t_{\nu\lambda}\hat{b}^\dagger_\mu\hat{b}^\dagger_\lambda + \frac{1}{2}\sum_{\mu\nu\lambda} B_{\mu\nu}t_{\nu\lambda}\hat{b}_\mu\hat{b}^\dagger_\lambda + \frac{1}{2}\sum_{\mu\nu\lambda} B_{\mu\nu}t_{\mu\lambda}\hat{b}^\dagger_\lambda\hat{b}_\nu \tag{5.102}$$

$$= \frac{1}{2}\sum_{\mu\lambda}(\mathbf{At}+\mathbf{tA})_{\mu\lambda}\hat{b}^\dagger_\mu\hat{b}^\dagger_\lambda + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}t_{\nu\mu} + \frac{1}{2}\sum_{\mu\nu}(\sum_\lambda t_{\nu\lambda}B_{\lambda\mu})\hat{b}^\dagger_\mu\hat{b}_\nu + \frac{1}{2}\sum_{\mu\nu\lambda} B_{\mu\nu}t_{\mu\lambda}\hat{b}^\dagger_\lambda\hat{b}_\nu \tag{5.103}$$

$$= \frac{1}{2}\sum_{\mu\lambda}(\mathbf{At}+\mathbf{tA})_{\mu\lambda}\hat{b}^\dagger_\mu\hat{b}^\dagger_\lambda + \frac{1}{2}\,{\rm Tr}(\mathbf{Bt}) + \sum_{\mu\nu}(\mathbf{Bt})_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}_\nu \tag{5.104}$$

Next, consider that only the term containing $\hat{b}^\dagger\hat{b}$ in the first commutator will have a non-vanishing commutator with $\hat{T}$, so

$$\frac{1}{2}[[\hat{H}_{\rm B},\hat{T}],\hat{T}] = \frac{1}{2}\sum_{\mu\nu}(\mathbf{Bt})_{\mu\nu}[\hat{b}^\dagger_\mu\hat{b}_\nu,\hat{T}] \tag{5.105}$$

$$= \frac{1}{2}\sum_{\mu\nu\lambda}(\mathbf{Bt})_{\mu\nu}t_{\nu\lambda}\hat{b}^\dagger_\mu\hat{b}^\dagger_\lambda \tag{5.106}$$

Note that any nested commutators of higher order must involve $[\hat{b}^\dagger\hat{b}^\dagger,\hat{T}] = 0$*, which is why the BCH expansion truncates at second order for quadratic Hamiltonians.* We can express the final result as

$$\bar{\hat{H}}_{\rm B} = \frac{1}{2}\sum_{\lambda\sigma} B_{\lambda\sigma}t_{\lambda\sigma} + \sum_{\mu\nu}\bar{A}_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}_\nu + \frac{1}{2}\sum_{\mu\nu}\bar{B}_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu + \frac{1}{2}\sum_{\mu\nu} B_{\mu\nu}\hat{b}_\mu\hat{b}_\nu \tag{5.107}$$

with

$$\bar{A}_{\mu\nu} = A_{\mu\nu} + \sum_\lambda t_{\nu\lambda}B_{\lambda\mu} \tag{5.108a}$$

$$\bar{B}_{\mu\nu} = B_{\mu\nu} + \sum_\lambda A_{\mu\lambda}t_{\lambda\nu} + \sum_\lambda t_{\mu\lambda}A_{\lambda\nu} + \sum_{\lambda\sigma} t_{\mu\lambda}B_{\lambda\sigma}t_{\sigma\nu} \tag{5.108b}$$

The second similarity transformation will also truncate at second order, but we will not work it out and instead just present the final result which reads

$$\begin{aligned}\bar{\bar{\hat{H}}}_{\rm B} &= \bar{\hat{H}}_{\rm B} + [\hat{Z},\bar{\hat{H}}_{\rm B}] + \frac{1}{2}[\hat{Z},[\hat{Z},\bar{\hat{H}}_{\rm B}]] \\ &\boxed{= \frac{1}{2}\sum_{\lambda\sigma}\bar{B}_{\lambda\sigma}z_{\lambda\sigma} + \sum_{\mu\nu}\bar{\bar{A}}_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}_\nu + \frac{1}{2}\sum_{\mu\nu}\bar{B}_{\mu\nu}\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu + \frac{1}{2}\sum_{\mu\nu}\bar{\bar{B}}_{\mu\nu}\hat{b}_\mu\hat{b}_\nu}\end{aligned} \tag{5.109}$$

where

$$\bar{\bar{A}}_{\mu\nu} = \bar{A}_{\mu\nu} + \sum_{\lambda} z_{\nu\lambda}\bar{B}_{\lambda\mu} \tag{5.110a}$$

$$\bar{\bar{B}}_{\mu\nu} = B_{\mu\nu} + \sum_{\lambda} \bar{A}_{\mu\lambda}z_{\lambda\nu} + \sum_{\lambda} z_{\mu\lambda}\bar{A}_{\lambda\nu} + \sum_{\lambda\sigma} z_{\mu\lambda}\bar{B}_{\lambda\sigma}z_{\sigma\nu} \tag{5.110b}$$

Left amplitude equations

Now, if we return to the left amplitude equation of 5.76, we recover the Riccati equation:

$$0 = \langle 0_{\mathrm{B}}|\hat{b}_\mu\hat{b}_\nu\bar{\bar{H}}_{\mathrm{B}}|0_{\mathrm{B}}\rangle \tag{5.111}$$

$$= \frac{1}{2}\sum_{\mu\nu}\bar{B}_{\mu\nu}\langle 0_{\mathrm{B}}|\hat{b}_\mu\hat{b}_\nu\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu|0_{\mathrm{B}}\rangle \tag{5.112}$$

$$= B_{\mu\nu} + \sum_{\lambda} A_{\mu\lambda}t_{\lambda\nu} + \sum_{\lambda} t_{\mu\lambda}A_{\lambda\nu} + \sum_{\lambda\sigma} t_{\mu\lambda}B_{\lambda\sigma}t_{\sigma\nu} \tag{5.113}$$

$$= \boldsymbol{B} + \mathbf{A}\cdot\mathbf{t} + \mathbf{t}\cdot\mathbf{A} + \mathbf{t}\cdot\mathbf{B}\cdot\mathbf{t} \tag{5.114}$$

with the solution given by $\mathbf{t} = \mathbf{Y}\cdot\mathbf{X}^{-1}$ where $\mathbf{X}$ and $\mathbf{Y}$ are the eigenvector matrices of the dRPA problem.

Right amplitude equations

Now, if we consider the right amplitude equation of 5.82, we have

$$0 = \langle 0_{\mathrm{B}}|\bar{\bar{H}}_{\mathrm{B}}\hat{b}_\mu\hat{b}_\nu|0_{\mathrm{B}}\rangle \tag{5.115}$$

$$= \frac{1}{2}\sum_{\mu\nu}\bar{B}_{\mu\nu}\langle 0_{\mathrm{B}}|\hat{b}_\mu\hat{b}_\nu\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu|0_{\mathrm{B}}\rangle \tag{5.116}$$

$$= B_{\mu\nu} + \sum_{\lambda} A_{\mu\lambda}t_{\lambda\nu} + \sum_{\lambda} t_{\mu\lambda}A_{\lambda\nu} + \sum_{\lambda\sigma} t_{\mu\lambda}B_{\lambda\sigma}t_{\sigma\nu} \tag{5.117}$$

$$= \boldsymbol{B} + \mathbf{A}\cdot\mathbf{t} + \mathbf{t}\cdot\mathbf{A} + \mathbf{t}\cdot\mathbf{B}\cdot\mathbf{t} \tag{5.118}$$

2. The Left Amplitude Equations (Stationarity wrt $t_{\mu\nu}$)

Setting $\frac{\partial E_{\mathrm{ECC}}}{\partial t_{\mu\nu}} = 0$ yields the adjoint equation:

$$\langle 0_{\mathrm{B}}|e^{\hat{Z}}\bar{\hat{H}}_{\mathrm{B}}\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu|0_{\mathrm{B}}\rangle = 0 \tag{5.119}$$

Using the property $e^{-\hat{Z}}|0_{\mathrm{B}}\rangle = |0_{\mathrm{B}}\rangle$, we can insert $e^{-\hat{Z}}e^{\hat{Z}}$ to write this in terms of the double-transformed Hamiltonian:

$$\langle 0_{\mathrm{B}}|\bar{\bar{\hat{H}}}_{\mathrm{B}}e^{\hat{Z}}\hat{b}^\dagger_\mu\hat{b}^\dagger_\nu e^{-\hat{Z}}|0_{\mathrm{B}}\rangle = 0 \tag{5.120}$$

Projecting this onto the vacuum extracts the coefficient of the annihilation part ($\hat{b}\hat{b}$) of the double-transformed Hamiltonian $\bar{\bar{\hat{H}}}_{\mathrm{B}}$. Looking at Eq. (5.109), this coefficient is $\bar{\bar{B}}_{\mu\nu}$. Therefore:

$$\bar{\bar{B}}_{\mu\nu} = 0 \tag{5.121}$$

3. Simplification and Decoupling

The full expression for $\bar{\bar{B}}_{\mu\nu}$ is:

$$\bar{\bar{B}}_{\mu\nu} = B_{\mu\nu} + \sum_{\lambda} \bar{A}_{\mu\lambda} z_{\lambda\nu} + \sum_{\lambda} z_{\mu\lambda} \bar{A}_{\lambda\nu} + \sum_{\lambda\sigma} z_{\mu\lambda} \bar{B}_{\lambda\sigma} z_{\sigma\nu} = 0 \tag{5.122}$$

In general ECC, the t and z equations are coupled because $\bar{B}$ appears in the z equation. However, we have already solved the t equations by setting $\bar{B}_{\mu\nu} = 0$.

Substituting $\bar{B}_{\mu\nu} = 0$ into the equation for $\bar{\bar{B}}_{\mu\nu}$ causes the quadratic term in z to vanish:

$$B_{\mu\nu} + \sum_{\lambda} \bar{A}_{\mu\lambda} z_{\lambda\nu} + \sum_{\lambda} z_{\mu\lambda} \bar{A}_{\lambda\nu} + 0 = 0 \tag{5.123}$$

This results in a linear equation for the z amplitudes:

$$B_{\mu\nu} + \sum_{\lambda} \bar{A}_{\mu\lambda} z_{\lambda\nu} + \sum_{\lambda} z_{\mu\lambda} \bar{A}_{\lambda\nu} = 0 \tag{5.124}$$

This proves that for the bosonic Hamiltonian, the t amplitudes can be determined first via a Riccati equation, and the z amplitudes follow by solving a subsequent linear system, effectively decoupling the two sets of equations. Let $|0_{\mathrm{B}}\rangle$ denote the bosonic reference vacuum. The right amplitude equations, obtained as $\langle 0_{\mathrm{B}}|\hat{b}_\mu \hat{b}_\nu \bar{\bar{H}}_{\mathrm{B}}|0_{\mathrm{B}}\rangle = 0$, yield the condition $\bar{B}_{\mu\nu} = 0$, which is equivalent to the (quadratic) Riccati equation in Eq. (**??**). The left amplitude equations, $\langle 0_{\mathrm{B}}|\bar{\bar{H}}_{\mathrm{B}} \hat{b}^\dagger_\mu \hat{b}^\dagger_\nu|0_{\mathrm{B}}\rangle = 0$, give $\bar{\bar{B}}_{\mu\nu} = 0$, which reduces to a set of linear equations in $z_{\mu\nu}$ once $\bar{B}_{\mu\nu} = 0$:

$$B_{\mu\nu} + \sum_{\lambda} \bar{A}_{\mu\lambda} z_{\lambda\nu} + \sum_{\lambda} z_{\mu\lambda} \bar{A}_{\lambda\nu} = 0 \tag{5.125}$$

Note that, although left- and right-amplitude equations are generally coupled, they are completely decoupled in the present case.

Once these two conditions are satisfied, the double similarity-transformed bosonic Hamiltonian simplifies to

$$\bar{\bar{H}}_{\mathrm{B}} = \sum_{\mu\nu} \bar{A}_{\mu\nu} \hat{b}^\dagger_\nu \hat{b}_\mu + \frac{1}{2} \sum_{\lambda\sigma} B_{\lambda\sigma} t_{\lambda\sigma} \tag{5.126}$$

where all non-number-conserving terms have been eliminated.

Having established the ECC treatment of the bosonic Hamiltonian, we can now turn to the full electron-boson Hamiltonian $\hat{H}_{\mathrm{eB}}$ given in Eq. (5.88). The double similarity-transformed electron-boson Hamiltonian reads

5.4.3 EOM on the ECC electron-boson Hamiltonian

Building on the ECC electron-boson Hamiltonian introduced in Sec. 5.4.2, we next derive the EOM formulation for charged excitations. Within the present ECC-based formalism,

the excitation operator $\bar{\bar{R}}^{(m)}$ for the mth excited state is defined as

$$\begin{aligned}\bar{\bar{R}}^{(m)} &= e^{\hat{T}}e^{-\hat{Z}}\hat{R}^{(m)}e^{\hat{Z}}e^{-\hat{T}} \\ &= \sum_M R_M^{(m)}\bar{\bar{c}}_M = \sum_M R_M^{(m)}e^{\hat{T}}e^{-\hat{Z}}\hat{c}_M e^{\hat{Z}}e^{-\hat{T}} \\ &= \sum_i r_i^{(m)}\hat{a}_i + \sum_{i\nu} r_{i\nu}^{(m)}\bar{\bar{b}}_\nu^\dagger\hat{a}_i + \sum_a r_a^{(m)}\hat{a}_a + \sum_{a\nu} r_{\nu a}^{(m)}\bar{\bar{b}}_\nu\hat{a}_a\end{aligned} \tag{5.127}$$

where $\hat{c}_M$ denotes an excitation operator associated with the Mth excitation process [see Eq. (**??**)]. The various excitation channels are expressed in terms of double similarity-transformed bosonic creation and annihilation operators,

$$\bar{\bar{b}}_\nu^\dagger = e^{\hat{T}}e^{-\hat{Z}}\hat{b}_\nu^\dagger e^{\hat{Z}}e^{-\hat{T}} \tag{5.128a}$$

$$\bar{\bar{b}}_\nu = e^{\hat{T}}e^{-\hat{Z}}\hat{b}_\nu e^{\hat{Z}}e^{-\hat{T}} \tag{5.128b}$$

as well as the corresponding left and right reference states in the combined fermionic-bosonic space,

$$|0_\mathrm{e}\bar{\bar{0}}_\mathrm{B}\rangle = e^{\hat{T}}e^{-\hat{Z}}|0_\mathrm{e}0_\mathrm{B}\rangle \tag{5.129a}$$

$$\langle 0_\mathrm{e}\bar{\bar{0}}_\mathrm{B}| = \langle 0_\mathrm{e}0_\mathrm{B}|e^{\hat{Z}}e^{-\hat{T}} \tag{5.129b}$$

The use of these transformed creation and annihilation operators is essential to ensure that the EOM formalism satisfies the required "killer" conditions.

The EOM eigenvalue problem becomes

$$\langle 0_\mathrm{e}\bar{\bar{0}}_\mathrm{B}|[\bar{\bar{c}}_M^\dagger,[\hat{H}_\mathrm{eB},\bar{\bar{R}}^{(m)}]]|0_\mathrm{e}\bar{\bar{0}}_\mathrm{B}\rangle = \langle 0_\mathrm{e}\bar{\bar{0}}_\mathrm{B}|[\bar{\bar{c}}_M^\dagger,\bar{\bar{R}}^{(m)}]|0_\mathrm{e}\bar{\bar{0}}_\mathrm{B}\rangle E^{(m)} \tag{5.130}$$

with $E^{(m)}$ denoting the energy of the mth charged excited state, and $\bar{\bar{c}}_M^\dagger$ to the Mth excitation/deexcitation process [see Eq. (5.127)], respectively. Equation (5.130) can be simplified to

$$\langle 0_\mathrm{e}0_\mathrm{B}|[\hat{c}_M^\dagger,[\bar{\bar{H}}_\mathrm{eB},\hat{R}^{(m)}]]|0_\mathrm{e}0_\mathrm{B}\rangle = \langle 0_\mathrm{e}0_\mathrm{B}|[\hat{c}_M^\dagger,\hat{R}^{(m)}]|0_\mathrm{e}0_\mathrm{B}\rangle E^{(m)} \tag{5.131}$$

and the resulting EOM eigenvalue problem can be written as

$$\mathbf{H}^\mathrm{EOM}\cdot\mathbf{R} = \mathbf{R}\cdot\mathbf{E} \tag{5.132}$$

The equivalence with the G_0W_0 quasiparticle energies is established analytically in Sec. **??**, identifying the matrix elements of the effective Hamiltonian $\mathbf{H}^\mathrm{EOM}$ with those of the GW supermatrix $\mathbf{H}^{GW}$ defined in Eq. (**??**).

Details regarding the determination of analytic properties within the ECC treatment of the electron-boson Hamiltonian can be found in Ref.

Chapter 6

Cumulant expansion

6.1 Background

6.1.1 The qualitative interpretation

The cumulant has the ability to enhance spectral features with respect to GW. The prototypical example that is given to motivate its utility is the analysis of the self-consistent GW spectral function by von Barth and Holm for the uniform electron gas. There they show that scGW predicts an unphysical peak, known as the plasmaron, which is off from the expected plasmon frequency, as the only feature in the satellite spectrum. Cumulant plus GW, on the other hand, gives a more accurate answer. The way I think about this is as follows. GW (with or without self-consistency) yields similar quasiparticle energies, while self-consistency mostly affects spectral weight distribution. However, GW often produces an artificial plasmaron satellite. The cumulant expansion not only redistributes weight but also corrects the satellite structure, eliminating the spurious plasmaron and aligning the features with the physical plasmon frequency.

As we know, the ansatz for the cumulant $C(t)$ is

$$G(t) = G_0(t)e^{C(t)} \tag{6.1}$$

where we have the non-interacting and fully interacting Green's functions G_0 and G, respectively. After some derivation, we arrive at the Landau form for the cumulant

$$C(t) = \int d\omega \frac{\beta(\omega)}{\omega^2} \left[e^{-i\omega t} + i\omega t - 1\right] \tag{6.2}$$

where the cumulant kernel is defined as

$$\beta(\omega) = -\frac{1}{\pi} \operatorname{Im} \Sigma^{\mathrm{c}}(\omega) \tag{6.3}$$

This form is amenable to physical interpretation if we partition into

$$C(t) = -a + i\Delta t + \tilde{C}(t) \tag{6.4}$$

where $a = \int d\omega \beta(\omega)/\omega^2$ is the net satellite strength, $\Delta = \int d\omega \beta(\omega)/\omega$ is the quasiparticle shift, or core-level "relaxation energy", and $\tilde{C}(t)$ is the remainder of the cumulant, which contains the information about the satellites.

6.1.2 Alternate derivation starting from Dyson equation

We can start from where Schwinger's fictitious potential u has already been introduced:

$$G_u(1,1') = G^0(1,1') + G^0(1,\overline{2})\left\{\left[u(\overline{2}) + v_{Hu}(\overline{2})\right] G_u\left(\overline{2},1'\right) + iv_c(\overline{2},\overline{3})\frac{\delta G_u\left(\overline{2},1'\right)}{\delta u\left(\overline{3}^+\right)}\right\} \quad (6.5)$$

where G^0 is the non-interacting Green's function and v_c is the bare Coulomb interaction. v_{Hu} is the Hartree potential built with the density $n_u(\mathbf{x}) = -iG_u(\mathbf{x},\mathbf{x},t,t^+)$. One of the complications of the equations is the fact that the density n_u in the Hartree potential introduces a term that is quadratic in G_u for the first term in the above equation. To overcome this problem, we introduce the total classical potential

$$u_{\rm cl}(1) = u(1) + v_{Hu}(1) \quad (6.6)$$

which allows us to rewrite the equation for $G_u \equiv G_u^{\rm cl}$ as

$$G_u(1,1') = G^0(1,1') + G^0(1,\overline{2})u_{\rm cl}(\overline{2})G_u\left(\overline{2},1'\right) + iG^0(1,\overline{2})v_c(\overline{2},\overline{3})\frac{\delta G_u\left(\overline{2},1'\right)}{\delta u\left(\overline{3}^+\right)} \quad (6.7)$$

$$= G^0(1,1') + G^0(1,\overline{2})u_{\rm cl}(\overline{2})G_u\left(\overline{2},1'\right) + iG^0(1,\overline{2})W_u(\overline{2},\overline{3})\frac{\delta G_u\left(\overline{2},1'\right)}{\delta u_{\rm cl}\left(\overline{3}^+\right)} \quad (6.8)$$

where we have defined the screened Coulomb interaction $W_u = \epsilon_u^{-1} v_c$ with the time-ordered inverse dielectric function $\epsilon_u^{-1} = \delta u_{\rm cl}/\delta u$. Note that ϵ_u^{-1} is not the usual linear response dielectric function, since it depends on the perturbing potential. However, since we are interested in the solution for vanishing u, a reasonable approximation is to evaluate the equation using $\epsilon_u^{-1} \approx \epsilon^{-1}$ at $u = 0$. *This corresponds to a linear-response approximation.* Written in a basis, the resulting equation reads

$$G_{ij}^{u_{cl}}(t_{12}) = G_{ij}^0(t_{12}) + G_{im}^0(t_{13})\,u_{{\rm cl},mk}(t_3)\,G_{kj}^{u_{cl}}(t_{32}) + iG_{ik}^0(t_{13})\,W_{klmn}(t_{34})\,\frac{\partial G_{nj}^{u_{cl}}(t_{32})}{\partial u_{{\rm cl},lm}(t_4)} \quad (6.9)$$

where $t_{12} \equiv (t_1, t_2)$ or $(t_1 - t_2)$ in equilibrium, W_{klmn} is a matrix element of the screened Coulomb interaction W, and we have replaced functional derivatives by partial derivatives, supposing the basis to be discrete, which corresponds to calculations in practice. Repeated indices are summed over.

GW approximation

The GW approximation sets

$$\frac{\partial G_{nj}^u(t_{32})}{\partial u_{{\rm cl},lm}(t_4)} \approx G_{nl}(t_{34})\,G_{mj}(t_{42}) \quad (6.10)$$

At $u = 0$, this yields the Dyson equation $G = G^0 + G^0\Sigma^{GW}G$ with the GW approximation for the self-energy,

$$\Sigma_{im}^{GW}(t_{34}) = v_{H,im} + iG_{nl}(t_{34})\,W_{ilmn}(t_{34})\,. \quad (6.11)$$

But we will take a different route here, in order to obtain the cumulant expression for the Green's function.

Cumulant expansion

The basic idea is to introduce a quasi-particle Green's function $G^{QP,u}$, defined as

$$\left[G^{QP,u}\right]^{-1}_{ij} = \left[G_0^{-1}\right]_{ij} - \left[u_{\rm cl}\right]_{ij} - \Sigma^{GW}_{ij}\left(\frac{\varepsilon_i+\varepsilon_j}{2}\right) \tag{6.12}$$

$$\implies \left[G_0\right]_{ij} = G^{QP,u}_{ij} - G^{QP,u}_{im}\left(u_{{\rm cl},mk} + \Sigma^{GW}_{mk}\left(\frac{\varepsilon_m+\varepsilon_k}{2}\right)\right)\left[G_0\right]_{kj} \tag{6.13}$$

Plugging this into 6.9 (suppressing the time and orbital indices for ease of notation) gives

$$\begin{aligned} G^u &= G^{QP,u} - G^{QP,u}\left(u_{\rm cl}+\Sigma\right)G_0 \\ &+ \left(G^{QP,u} - G^{QP,u}\left(u_{\rm cl}+\Sigma\right)G_0\right)u_{\rm cl}G^u \\ &+ i\left(G^{QP,u} - G^{QP,u}\left(u_{\rm cl}+\Sigma\right)G_0\right)W\frac{\partial G^u}{\partial u_{\rm cl}} \end{aligned} \tag{6.14}$$

$$\begin{aligned} &= G^{QP,u} + G^{QP,u}u_{\rm cl}G^u + iG^{QP,u}W\frac{\partial G^u}{\partial u_{\rm cl}} \\ &- G^{QP,u}\left(u_{\rm cl}+\Sigma\right)\left[\underbrace{G_0 + G_0u_{\rm cl}G^u + iG_0W\frac{\partial G^u}{\partial u_{\rm cl}}}_{G^u}\right] \\ &= G^{QP,u} + iG^{QP,u}W\frac{\partial G^u}{\partial u_{\rm cl}} - G^{QP,u}\Sigma G^u \end{aligned} \tag{6.15}$$

$$\implies G^u_{ij}(t_{12}) = G^{QP,u}_{ij}(t_{12}) + iG^{QP,u}_{ik}(t_{13})W_{klmn}(t_{34})\frac{\partial G^u_{mj}(t_{32})}{\partial u_{{\rm cl},lm}(t_4)} - G^{QP,u}_{ik}(t_{13})\Sigma^{GW}_{kl}\left(\frac{\varepsilon_k+\varepsilon_l}{2}\right)G^u_{lj}(t_{32}) \tag{6.16}$$

We have made use of the *GW* self-energy. It is known that in the *GW* approximation, because of a fortuitous cancellation of errors, it is most wise to use the RPA approximation for the dielectric function that goes into the screened Coloumb interaction. I wonder if this is also the smartest decision for the cumulant expansion; we could try using a more accurate approximation to the dielectric function and see what the effect is. Now one decouples the equations by supposing that G^u and $G^{QP,u}$ are diagonal in the same u-independent basis. So we take the diagonal components of all operators in the above equation, so $G^u_{ij} \to G^u_{ii} \equiv \mathcal{G}^u$, $G^{QP,u}_{ij} \to G^{QP,u}_{ii} \equiv \mathcal{G}^{QP,u}$, $\Sigma^{GW}_{ij}\left(\frac{\varepsilon_i+\varepsilon_j}{2}\right) \to \Sigma^{GW}_{ii}\left(\varepsilon_i\right) \equiv \tilde{\Sigma}^{GW}$, $W_{ijkl} \to W_{iiii} \equiv \mathcal{W}$, and $u_{{\rm cl},ij} \to u_{{\rm cl},ii} \equiv u$. This is a strong assumption, but nevertheless, it allows us to get to the conventional form for the cumulant expansion. The resulting equation is

$$\mathcal{G}^u(t_{12}) = \mathcal{G}^{QP,u}(t_{12}) + i\mathcal{G}^{QP,u}(t_{13})\mathcal{W}(t_{34})\frac{\partial \mathcal{G}^u(t_{32})}{\partial u(t_4)} - \mathcal{G}^{QP,u}(t_{13})\tilde{\Sigma}^{GW}\mathcal{G}^u(t_{32}) \tag{6.17}$$

In the paper they claim that the solution for $u \to 0$ is

$$\mathcal{G}\left(t_{12}\right) = \mathcal{G}^0_{QP}\left(t_{12}\right)e^{i(t_1-t_2)\Sigma^{GW}_{ii}(\varepsilon_i)}\exp\left[-i\int_{t_1}^{t_2}dt'\int_{t'}^{t_2}dt''\mathcal{W}\left(t'-t''\right)\right]. \tag{6.18}$$

I have not been able to derive, but my attempts are below, and we will just continue for now. The double integral can be evaluated as

$$-i\int_{t_1}^{t_2}dt'\int_{t'}^{t_2}dt''\mathcal{W}\left(t'-t''\right)=-\frac{i}{2\pi}\int d\omega\,\mathcal{W}(\omega)\int_{t_1}^{t_2}dt'\int_{t'}^{t_2}dt''\,e^{-i\omega(t'-t'')} \tag{6.19}$$

$$=-\frac{1}{2\pi}\int d\omega\,\frac{\mathcal{W}(\omega)}{\omega}\int_{t_1}^{t_2}dt'\Big(e^{i\omega(t_2-t')}-1\Big) \tag{6.20}$$

$$=-\frac{1}{2\pi}\int d\omega\,\frac{\mathcal{W}(\omega)}{\omega}\left[\frac{e^{i\omega(t_2-t_1)}-1}{i\omega}-(t_2-t_1)\right] \tag{6.21}$$

$$=-\left(t_1-t_2\right)\frac{1}{2\pi}\int d\omega\frac{\mathcal{W}(\omega)}{\omega}+\frac{i}{2\pi}\int d\omega\frac{\mathcal{W}(\omega)}{\omega^2}\left(e^{-i\omega(t_1-t_2)}-1\right) \tag{6.22}$$

where in going from Eq. (6.19) to Eq. (6.20) we have used $\int_{t'}^{t_2}dt''\,e^{-i\omega(t'-t'')}=\frac{e^{i\omega(t_2-t')}-1}{i\omega}$, and in going from Eq. (6.20) to Eq. (6.21) we have used $\int_{t_1}^{t_2}dt'\big(e^{i\omega(t_2-t')}-1\big)=\frac{e^{i\omega\Delta}-1}{i\omega}-\Delta$ with $\Delta=t_2-t_1$. Let us first examine the term proportional to (t_1-t_2) in Eq. (6.22), by comparing it to a GW quasi-particle shift. In the decoupling approximation, $\Sigma^{GW}_{kk}\approx iG_{kk}W_{kkkk}$. Evaluated at the quasi-particle energy, this yields exactly the term we are interested in. This means that $e^{-(t_1-t_2)\frac{1}{2\pi}\int d\omega\frac{\mathcal{W}(\omega)}{\omega}}$ approximately cancels with the GW shift in $e^{i(t_1-t_2)\Sigma^{GW}_{ii}(\varepsilon_i)}$, and we are left with

$$\mathcal{G}\left(t_{12}\right)=\mathcal{G}^0_{QP}\left(t_{12}\right)\exp\left[\frac{i}{2\pi}\int d\omega\frac{\mathcal{W}(\omega)}{\omega^2}\left(e^{i\omega(t_1-t_2)}-1\right)\right] \tag{17}$$

Using the functional derivative identity $\frac{\partial \mathcal{G}^u(t_{32})}{\partial u(t_4)}|_{u=0} = \mathcal{G}(t_{34})\mathcal{G}(t_{42})$ and correspondingly setting $\mathcal{G}^u \to \mathcal{G}$, $\mathcal{G}^{QP,u} \to \mathcal{G}^{QP}$ gives

$$\mathcal{G}(t_{12}) = \mathcal{G}^{QP}(t_{12}) + i\mathcal{G}^{QP}(t_{13})\mathcal{W}(t_{34})\mathcal{G}(t_{34})\mathcal{G}(t_{42}) - \mathcal{G}^{QP}(t_{13})\tilde{\Sigma}^{GW}\mathcal{G}(t_{32}) \tag{6.23}$$

Now define $\tilde{\mathcal{G}}(t_{12}) = e^{i(t_1-t_2)\tilde{\Sigma}^{GW}}\mathcal{G}(t_{12}) \implies \mathcal{G}(t_{12}) = e^{-i(t_1-t_2)\tilde{\Sigma}^{GW}}\tilde{\mathcal{G}}(t_{12})$. Plugging this in gives

$$\begin{aligned}\tilde{\mathcal{G}}(t_{12}) = {} & e^{i(t_1-t_2)\tilde{\Sigma}^{GW}}\mathcal{G}^{QP}(t_{12}) + ie^{i(t_1-t_2)\tilde{\Sigma}^{GW}}\mathcal{G}^{QP}(t_{13})\mathcal{W}(t_{34})e^{-i(t_3-t_4)\tilde{\Sigma}^{GW}}\tilde{\mathcal{G}}(t_{34})e^{-i(t_4-t_2)\tilde{\Sigma}^{GW}}\tilde{\mathcal{G}}(t_{42}) \\ & - e^{i(t_1-t_2)\tilde{\Sigma}^{GW}}\mathcal{G}^{QP}(t_{13})\tilde{\Sigma}^{GW}e^{-i(t_3-t_2)\tilde{\Sigma}^{GW}}\tilde{\mathcal{G}}(t_{32})\end{aligned} \tag{6.24}$$

Inegration factor method

Define $A(u) = -\mathcal{G}^{QP,u}(t_{13})\,\tilde{\Sigma}^{GW}, \quad B(u) = \mathcal{G}^{QP,u}(t_{12}), \quad C(u) = i\mathcal{G}^{QP,u}(t_{13})\,\mathcal{W}(t_{34})$. Then the equation can be written as

$$C(u)\frac{\partial \mathcal{G}^u}{\partial u} + A(u)\mathcal{G}^u = B(u). \tag{6.25}$$

Divide through by $C(u)$, we get

$$\frac{\partial \mathcal{G}^u}{\partial u} + \frac{A(u)}{C(u)}\mathcal{G}^u = \frac{B(u)}{C(u)}. \tag{6.26}$$

So our integrating factor is

$$\mu(u) = \exp\left(\int^u \frac{A(s)}{C(s)}ds\right) = \exp\left(+i\int^u ds\frac{\tilde{\Sigma}^{GW}}{\mathcal{W}}\right) \tag{6.27}$$

and the solution is

$$\mathcal{G}^u = \frac{1}{\mu(u)}\left[C_0 + \int^u \mu(s)\frac{B(s)}{C(s)}ds\right] \tag{6.28}$$

$$= \exp\left(-i\int^u ds\frac{\tilde{\Sigma}^{GW}}{\mathcal{W}}\right)\left[C_0 + \int^u ds\exp\left(+i\int^s ds'\frac{\tilde{\Sigma}^{GW}}{\mathcal{W}}\right)\frac{\mathcal{G}^{QP,s}}{i\mathcal{G}^{QP,s}\mathcal{W}}\right] \tag{6.29}$$

$$= \exp\left(-i\int^u ds\frac{\tilde{\Sigma}^{GW}}{\mathcal{W}}\right)\left[C_0 + \int^u ds\exp\left(+i\int^s ds'\frac{\tilde{\Sigma}^{GW}}{\mathcal{W}}\right)\frac{-i}{\mathcal{W}}\right] \tag{6.30}$$

$$\tag{6.31}$$

6.2 Cumulant expansion for electronic structure

We know that $\mathrm{Re}\Sigma = \Sigma_{\mathrm{SEX}} + \Sigma_{\mathrm{COH}}$. Assuming that $\boldsymbol{G}$ and $\boldsymbol{G_n}$ represent valid plane wave indices with the indices $\boldsymbol{Q_1}, \boldsymbol{Q_2}$ running over all possible momentum transfers, we can write

$$\Sigma^{\mathrm{SEX}}_{GG}(\omega) = -\frac{1}{\Omega}\sum_{G_n}^{\mathrm{occ}}\sum_{Q_1Q_2}\psi_{G_n}(G-Q_1)\psi^*_{G_n}(G-Q_2)\left(1+\frac{\Omega^2_{Q_1Q_2}}{\left(\omega-\varepsilon_{G_n}\right)^2-\tilde{\omega}^2_{Q_1Q_2}}\right)v\left(Q_2\right) \tag{6.32}$$

$$= -\frac{1}{\Omega}\sum_{G_n}^{\mathrm{occ}}\sum_{Q_1Q_2}\delta_{G_n,G-Q_1}\delta_{G_n,G-Q_2}\left(1+\frac{\Omega^2_{Q_1Q_2}}{\left(\omega-\varepsilon_{G_n}\right)^2-\tilde{\omega}^2_{Q_1Q_2}}\right)v\left(Q_2\right) \tag{6.33}$$

$$= -\frac{1}{\Omega}\sum_{G_n}^{\mathrm{occ}}\left(1+\frac{\Omega^2_{G-G_n,G-G_n}}{\left(\omega-\varepsilon_{G_n}\right)^2-\tilde{\omega}^2_{G-G_n,G-G_n}}\right)v\left(G-G_n\right) \tag{6.34}$$

since the delta functions require that $G-Q_1 = G_n = G-Q_2 \implies Q_1 = Q_2 = G-G_n$. Next,

$$\Sigma^{\mathrm{COH}}_{GG}(\omega) = \frac{1}{\Omega}\sum_{G_n}\sum_{Q_1Q_2}\psi_{G_n}(G-Q_1)\psi^*_{G_n}(G-Q_2)\frac{1}{2}\frac{\Omega^2_{Q_1Q_2}}{\tilde{\omega}_{Q_1Q_2}\left[\omega-\varepsilon_{G_n}-\tilde{\omega}_{Q_1Q_2}\right]}v\left(Q_2\right) \tag{6.35}$$

$$= \frac{1}{2\Omega}\sum_{G_n}\frac{\Omega^2_{G-G_n,G-G_n}}{\tilde{\omega}_{G-G_n,G-G_n}\left[\omega-\varepsilon_{G_n}-\tilde{\omega}_{G-G_n,G-G_n}\right]}v\left(G-G_n\right) \tag{6.36}$$

So the procedure will be to generate a set of $\Omega_{Q_1Q_1}$ and $\tilde{\omega}_{Q_1Q_1}$ for each momentum transfer vector Q_1 via the PPM and then to insert these into the above expressions for the self-energy.

6.2.1 LDA+GW+C

Aryasetiwan's derivation

They postulate that the hole Green's function is given for negative times by

$$G(k,t) = ie^{-i\varepsilon_k t + C(k,t)} \tag{6.37}$$

where ε_k is the LDA single-particle energy of the hole state k. It is shown that the cumulant can be written as the sum $C(k,t) = C^{\mathrm{QP}}(k,t) + C^{\mathrm{sat}}(k,t)$ where

$$C^{\mathrm{QP}}(k,t) = -\left(i\alpha_k+\gamma_k\right) + \left(-i\Delta\varepsilon_k+\Gamma_k\right)t \tag{6.38}$$

with

$$i\alpha_k + \gamma_k = -\left[\frac{\partial \Sigma^h(k,\omega)}{\partial \omega}\right]_{\omega=\varepsilon_k} = \sum_{i\nu} \frac{\left|M^?_{ki\nu}\right|^2 \Theta\left(\mu - \varepsilon^?_i + \Omega^?_\nu\right)}{\left(\varepsilon_k - \varepsilon^?_i + \Omega^?_\nu - i\delta\right)^2} - \sum_{a\nu} \frac{\left|M^?_{ka\nu}\right|^2 \Theta\left(\mu - \varepsilon^?_a - \Omega^?_\nu\right)}{\left(\varepsilon_k - \varepsilon^?_a - \Omega^?_\nu - i\delta\right)^2}$$

(6.39)

$$\Gamma_k = \operatorname{Im}\Sigma\left(k,\varepsilon_k\right) = \operatorname{Im}\Sigma^c(k,\varepsilon_k) \tag{6.40}$$

$$\Sigma^h(k,\omega) = \frac{1}{\pi}\int_{-\infty}^{\mu} d\omega' \frac{\operatorname{Im}\Sigma\left(k,\omega'\right)}{\omega - \omega' - i\delta} = \sum_{i\nu} \frac{\left|M^?_{ki\nu}\right|^2 \Theta\left(\mu - \varepsilon^?_i + \Omega^?_\nu\right)}{\omega - \varepsilon^?_i + \Omega^?_\nu - i\delta} - \underbrace{\sum_{a\nu} \frac{\left|M^?_{ka\nu}\right|^2 \Theta\left(\mu - \varepsilon^?_a - \Omega^?_\nu\right)}{\omega - \varepsilon^?_a - \Omega^?_\nu - i\delta}}_{0?}$$

(6.41)

and

$$\Delta\varepsilon_k = \operatorname{Re}\Sigma\left(k,\varepsilon_k\right) - v_{\mathrm{xc}}(k) = \Sigma^x(k) + \operatorname{Re}\Sigma^c(k,\varepsilon_k) - v_{\mathrm{xc}}(k). \tag{6.42}$$

Note that the self-energy can be written as $\Sigma(k,\omega) = \Sigma^x(k) + \Sigma^c(k,\omega)$, where Σ^x is the bare exchange contribution and $\Sigma^c(k,\omega) = \sum_{i\nu} \frac{\left(M^?_{ki\nu}\right)^2}{\omega - \varepsilon^?_i + \Omega^?_\nu + i\eta} + \sum_{a\nu} \frac{\left(M^?_{ka\nu}\right)^2}{\omega - \varepsilon^?_a - \Omega^?_\nu + i\eta}$, where $\varepsilon^?$, $\Omega^?$, and $M^?$ are the single-particle energies, excitation energies, and transition densities, respectively, whose form depends on the level of self-consistency chosen. While

$$C^S(k,t) = \frac{1}{\pi}\int_{\varepsilon_k - \mu}^{\infty} d\omega \frac{\operatorname{Im}\Sigma\left(k,\varepsilon_k - \omega\right)}{(\omega - i\delta)^2} e^{(i\omega+\delta)t} \tag{6.43}$$

$$= \sum_{i\nu} \frac{\left|M_{ki\nu}\right|^2 \Theta\left(\mu - \varepsilon_i + \Omega_\nu\right)}{\left(\Delta^h_{i\nu} - i\delta\right)^2} e^{\left(i\Delta^h_{i\nu}+\delta\right)t} - \sum_{a\nu} \frac{\left|M_{ka\nu}\right|^2 \Theta\left(\mu - \varepsilon_a - \Omega_\nu\right)}{\left(\Delta^h_{a\nu} - i\delta\right)^2} e^{\left(i\Delta^h_{a\nu}+\delta\right)t} \tag{6.44}$$

with $\Delta^h_{i\nu} = \varepsilon_k - \varepsilon_i + \Omega_\nu$ and $\Delta^h_{a\nu} = \varepsilon_k - \varepsilon_a - \Omega_\nu$.

A QP

They write

$$A^{\mathrm{QP}}(k,\omega) = \frac{n_k}{\pi} e^{-\gamma_k} \frac{\Gamma_k \cos\alpha_k - (\omega - E_k)\sin\alpha_k}{\left(\omega - E_k\right)^2 + \Gamma_k^2} \tag{6.45}$$

where $E_k = \varepsilon_k + \Delta\varepsilon_k$ is the quasiparticle energy.

My derivation

$$G^0_k(\omega) = \frac{1}{\omega - \epsilon_k \pm i\eta} \tag{6.46}$$

(6.47)

where ϵ_k are the single particle energies either due to LDA or LDA+GW. The upper sign corresponds to the retarded formalism, while the lower sign corresponds to the time-ordered

formalism for the hole state k. Plugging this into what we get by equating the Dyson and cumulant expansions both to first order in the screened interaction, we have

$$C_k(t) = \pm ie^{i\epsilon_k t}\int \frac{d\omega}{2\pi}e^{-i\omega t}G_k^0(\omega)\left[\Sigma_k^x + \Sigma_k^c(\omega) - V_k^{xc}\right]G_k^0(\omega) \tag{6.48}$$

$$= \pm ie^{i\epsilon_k t}\int \frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Sigma_k^x + \Sigma_k^c(\omega) - V_k^{xc}}{(\omega - \varepsilon_k \pm i\eta)^2}, \tag{6.49}$$

where $V^{xc} \equiv V^{xc}_{\text{LDA}}$ in the LDA and $V^{xc} \equiv \Sigma^x + \Sigma^c(\epsilon_k)$ for LDA+GW. Seperating the static contributions $\Delta_k = \Sigma_k^x - V_k^{xc}$ from the dynamical $\Sigma_k^c(\omega)$ gives

$$C_k(t) = \pm ie^{i\epsilon_k t}\int \frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Delta_k}{(\omega - \epsilon_k \pm i\eta)^2} \pm i\int \frac{d\omega}{2\pi}e^{-i(\omega-\epsilon_k)t}\frac{\Sigma_k^c(\omega)}{(\omega - \epsilon_k \pm i\eta)^2} \tag{6.50}$$

$$= -i\Delta_k t\Theta(\pm t) \pm i\int \frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Sigma_k^c(\omega+\epsilon_k)}{(\omega \pm i\eta)^2}. \tag{6.51}$$

At this point, we can absorb the static contribution into the pole giving

$$G_k(t) = \mp i\Theta(\pm t)\, e^{-i\tilde{\epsilon}_k t}e^{C_k^{\text{dyn}}(t)}, \tag{6.52}$$

where $\tilde{\epsilon}_k = \epsilon_k + \Delta_k$ is the shifted pole position and $C_k^{\text{dyn}}(t) = \pm i\int \frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Sigma_k^c(\omega+\epsilon_k)}{(\omega\pm i\eta)^2}$ is the dynamical contribution to the cumulant.

Retarded formalism

Here we have $\Sigma_k^c(\omega) \equiv \sum_{i\nu}\frac{\left(M^?_{ki\nu}\right)^2}{\omega-\varepsilon_i^?+\Omega_\nu^?+i\eta} + \sum_{a\nu}\frac{\left(M^?_{ka\nu}\right)^2}{\omega-\varepsilon_a^?-\Omega_\nu^?+i\eta}$, where $\varepsilon^?$, $\Omega^?$, and $M^?$ are the single-particle energies, excitation energies, and transition densities, respectively, whose form depends on the level of self-consistency chosen. Plugging this in leads to

$$C_k^{R,\text{dyn}}(t) = i\int \frac{d\omega}{2\pi}\frac{e^{-i\omega t}}{(\omega+i\eta)^2}\left[\sum_{i\nu}\frac{\left(M^?_{ki\nu}\right)^2}{\omega + \underbrace{\epsilon_k - \varepsilon_i^? + \Omega_\nu^? + i\eta}_{-\delta_{ki\nu}}} + \sum_{a\nu}\frac{\left(M^?_{ka\nu}\right)^2}{\omega + \underbrace{\epsilon_k - \varepsilon_a^? - \Omega_\nu^? + i\eta}_{-\delta_{ka\nu}}}\right] \tag{6.53}$$

$$= \mathrm{i}\sum_{iv}\left(M^?_{kiv}\right)^2\int\frac{\mathrm{d}\omega}{2\pi}e^{-\mathrm{i}\omega t}\frac{1}{[\omega+\mathrm{i}\eta]^2}\frac{1}{\omega-\delta_{kiv}} + \mathrm{i}\sum_{av}\left(M^?_{kav}\right)^2\int\frac{\mathrm{d}\omega}{2\pi}e^{-\mathrm{i}\omega t}\frac{1}{[\omega+\mathrm{i}\eta]^2}\frac{1}{\omega-\delta_{kav}} \tag{6.54}$$

$$= \Theta(t)\left[\sum_{i\nu}\zeta_{ki\nu}\left[e^{-i\delta_{ki\nu}t} - 1 + i\delta_{ki\nu}t\right] + \sum_{a\nu}\zeta_{ka\nu}\left[e^{-i\delta_{ka\nu}t} - 1 + i\delta_{ka\nu}t\right]\right], \tag{6.55}$$

where $\zeta_{kiv} = \frac{\left(M^?_{kiv}\right)^2}{\delta^2_{kiv}}$ and $\zeta_{kav} = \frac{\left(M^?_{kav}\right)^2}{\delta^2_{kav}}$. So the full Green's function in the retarded cumulant formalism, to first order in the screened interaction, is

$$G_k^R(t) = -i\Theta(t)e^{-i\tilde{\epsilon}_k t}\exp\left[\sum_{i\nu}\zeta_{ki\nu}\left[e^{-i\delta_{ki\nu}t} - 1 + i\delta_{ki\nu}t\right] + \sum_{a\nu}\zeta_{ka\nu}\left[e^{-i\delta_{ka\nu}t} - 1 + i\delta_{ka\nu}t\right]\right]. \tag{6.56}$$

Time-ordered formalism

Here, the 1h2p term drops out [1], and we are left with

$$\Sigma_k^c(\omega) \equiv \sum_{i\nu} \frac{\left(M_{ki\nu}^?\right)^2}{\omega - \varepsilon_i^? + \Omega_\nu^? - i\eta}. \tag{6.65}$$

Thus,

$$C_k^{T,\mathrm{dyn}}(t) = -i \int \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{(\omega - i\eta)^2} \sum_{i\nu} \frac{\left(M_{ki\nu}^?\right)^2}{\omega + \underbrace{\epsilon_k - \varepsilon_i^? + \Omega_\nu^? - i\eta}_{-\delta_{ki\nu}}} \tag{6.66}$$

$$= -\mathrm{i} \sum_{iv} \left(M_{kiv}^?\right)^2 \int \frac{\mathrm{d}\omega}{2\pi} e^{-\mathrm{i}\omega t} \frac{1}{[\omega - \mathrm{i}\eta]^2} \frac{1}{\omega - \delta_{kiv}} \tag{6.67}$$

$$= \Theta(-t) \sum_{i\nu} \zeta_{ki\nu} \left[e^{-i\delta_{ki\nu}t} - 1 - i\delta_{ki\nu}t\right]. \tag{6.68}$$

Getting the cumulant directly from the PPM

Consider the PPM correlation self-energy,

$$\Sigma_k^{c,R}(\omega) = \Sigma_k^{\mathrm{COH,R}}(\omega) + \widetilde{\Sigma}_k^{\mathrm{SEX,R}}(\omega), \tag{6.69}$$

[1] Reining was able to show that

$$C_{ii}^{T,\mathrm{dyn}}(t) = \int d\omega \frac{\beta_{ii}^-(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.57}$$

with

$$\beta_{ii}^-(\omega) = \left|\mathrm{Im}\, \Sigma_{ii}^R\left(\omega + \epsilon_i^{\mathrm{MF}}\right)\right| \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \omega\right) \tag{6.58}$$

$$= \left|\mathrm{Im}\left[\sum_{j\nu} \frac{M_{ij\nu}^2}{\omega + \epsilon_i^{\mathrm{MF}} - \epsilon_j^{\mathrm{MF}} + \Omega_\nu + i\eta} + \sum_{a\nu} \frac{M_{ia\nu}^2}{\omega + \epsilon_i^{\mathrm{MF}} - \epsilon_a^{\mathrm{MF}} - \Omega_\nu + i\eta}\right]\right| \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \omega\right) \tag{6.59}$$

$$= \pi \left[\sum_{j\nu} M_{ij\nu}^2 \delta\left(\omega + \epsilon_i^{\mathrm{MF}} - \epsilon_j^{\mathrm{MF}} + \Omega_\nu\right) + \sum_{a\nu} M_{ia\nu}^2 \delta\left(\omega + \epsilon_i^{\mathrm{MF}} - \epsilon_a^{\mathrm{MF}} - \Omega_\nu\right)\right] \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \omega\right) \tag{6.60}$$

$$= \pi \left[\sum_{j\nu} M_{ij\nu}^2 \delta\left(\omega - \omega_{ij\nu}^-\right) + \sum_{a\nu} M_{ia\nu}^2 \delta\left(\omega - \omega_{ia\nu}^+\right)\right] \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \omega\right). \tag{6.61}$$

with $\omega_{ij\nu}^- = \epsilon_j^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \Omega_\nu$ and $\omega_{ia\nu}^+ = \epsilon_a^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} + \Omega_\nu$. We can assume that the second term does drop out.

$$\delta\left(\omega - \omega_{ia\nu}^+\right) \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \omega\right) = \delta\left(\omega - \omega_{ia\nu}^+\right) \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_i^{\mathrm{MF}} - \omega_{ia\nu}^+\right) \tag{6.62}$$

$$= \delta\left(\omega - \omega_{ia\nu}^+\right) \theta\left(\epsilon_{k_F}^{\mathrm{MF}} - \epsilon_a^{\mathrm{MF}} - \Omega_\nu\right) \tag{6.63}$$

$$= 0, \tag{6.64}$$

because the RPA excitation energy Ω_ν is positive and $\epsilon_a^{\mathrm{MF}} > \epsilon_{k_F}^{\mathrm{MF}}$.

with

$$\widetilde{\Sigma}_k^{\mathrm{SEX,R}}(\omega) = -\frac{1}{V}\sum_{\boldsymbol{G}_i}^{\mathrm{occ}} \frac{\Omega_{\boldsymbol{q}}^2}{(\omega + i\eta - \varepsilon_i)^2 - \widetilde{\omega}_{\boldsymbol{q}}^2} v(\boldsymbol{q}), \tag{6.70}$$

$$\Sigma_k^{\mathrm{COH,R}}(\omega) = \frac{1}{2V}\sum_{\boldsymbol{G}_i} \frac{\Omega_{\boldsymbol{q}}^2}{\widetilde{\omega}_{\boldsymbol{q}}[\omega + i\eta - \varepsilon_i - \widetilde{\omega}_{\boldsymbol{q}}]} v(\boldsymbol{q}), \tag{6.71}$$

where

$$\boldsymbol{q} = \boldsymbol{G}_k - \boldsymbol{G}_i. \tag{6.72}$$

Define

$$A_{ki}^{\mathrm{PPM}} = \frac{1}{2V}\frac{\Omega_{\boldsymbol{q}}^2}{\widetilde{\omega}_{\boldsymbol{q}}} v(\boldsymbol{q}). \tag{6.73}$$

The SEX denominator can be decomposed as

$$\frac{1}{(\omega + i\eta - \varepsilon_i)^2 - \widetilde{\omega}_{\boldsymbol{q}}^2} = \frac{1}{2\widetilde{\omega}_{\boldsymbol{q}}}\left[\frac{1}{\omega + i\eta - \varepsilon_i - \widetilde{\omega}_{\boldsymbol{q}}} - \frac{1}{\omega + i\eta - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}}}\right]. \tag{6.74}$$

Therefore,

$$\widetilde{\Sigma}_k^{\mathrm{SEX,R}}(\omega) = -\sum_i^{\mathrm{occ}} A_{ki}^{\mathrm{PPM}}\left[\frac{1}{\omega + i\eta - \varepsilon_i - \widetilde{\omega}_{\boldsymbol{q}}} - \frac{1}{\omega + i\eta - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}}}\right]. \tag{6.75}$$

The COH term is

$$\Sigma_k^{\mathrm{COH,R}}(\omega) = \sum_i A_{ki}^{\mathrm{PPM}} \frac{1}{\omega + i\eta - \varepsilon_i - \widetilde{\omega}_{\boldsymbol{q}}}. \tag{6.76}$$

For occupied states, the first SEX pole cancels the COH pole. For unoccupied states, there is no SEX contribution. Thus the total PPM correlation self-energy becomes

$$\Sigma_k^{c,R}(\omega) = \sum_i^{\mathrm{occ}} A_{ki}^{\mathrm{PPM}} \frac{1}{\omega - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}} + i\eta} + \sum_a^{\mathrm{unocc}} A_{ka}^{\mathrm{PPM}} \frac{1}{\omega - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{q}} + i\eta}. \tag{6.77}$$

This has exactly the same pole structure as the general spectral representation of the GW correlation self-energy.

Now shift the integration variable,

$$\omega \to \omega + \varepsilon_k. \tag{6.78}$$

Then

$$C_k^{\mathrm{dyn}}(t) = i\int \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{(\omega + i\eta)^2} \Sigma_k^{c,R}(\omega + \varepsilon_k). \tag{6.79}$$

For the occupied contribution, define

$$\Delta_{ki}^{\mathrm{PPM}} = \varepsilon_i - \varepsilon_k - \widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_i} - i\eta. \tag{6.80}$$

For the unoccupied contribution, define

$$\Delta_{ka}^{\mathrm{PPM}} = \varepsilon_a + \widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_a} - \varepsilon_k - i\eta. \tag{6.81}$$

Then

$$C_k^{\mathrm{dyn}}(t) = i \sum_i^{\mathrm{occ}} A_{ki}^{\mathrm{PPM}} \int \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{(\omega + i\eta)^2(\omega - \Delta_{ki}^{\mathrm{PPM}} + i\eta)} \tag{6.82}$$

$$+ i \sum_a^{\mathrm{unocc}} A_{ka}^{\mathrm{PPM}} \int \frac{d\omega}{2\pi} \frac{e^{-i\omega t}}{(\omega + i\eta)^2(\omega - \Delta_{ka}^{\mathrm{PPM}} + i\eta)} \tag{6.83}$$

$$= \sum_i^{\mathrm{occ}} \zeta_{ki}^{\mathrm{PPM}} \left[e^{-i\Delta_{ki}^{\mathrm{PPM}} t} - 1 + i\Delta_{ki}^{\mathrm{PPM}} t \right] \tag{6.84}$$

$$+ \sum_a^{\mathrm{unocc}} \zeta_{ka}^{\mathrm{PPM}} \left[e^{-i\Delta_{ka}^{\mathrm{PPM}} t} - 1 + i\Delta_{ka}^{\mathrm{PPM}} t \right], \tag{6.85}$$

where

$$\zeta_{ki}^{\mathrm{PPM}} = \frac{A_{ki}^{\mathrm{PPM}}}{(\Delta_{ki}^{\mathrm{PPM}})^2}, \qquad \zeta_{ka}^{\mathrm{PPM}} = \frac{A_{ka}^{\mathrm{PPM}}}{(\Delta_{ka}^{\mathrm{PPM}})^2}. \tag{6.86}$$

Analytic derivation of Hedin's result from the PPM

Assuming negative times for a hole state k, Hedin's cumulant form gives

$$A_k(\omega) = \frac{1}{2\pi} \int dt\, e^{i(\omega - \epsilon_k)t} e^{S_k(t)}, \tag{6.87}$$

where

$$S_k(t) = \int d\omega\, \beta_k(\omega) \frac{e^{i\omega t} - i\omega t - 1}{\omega^2}, \tag{6.88}$$

and

$$\beta_k(\omega) = \frac{1}{\pi} \operatorname{Im} \Sigma^{GW}(k, \epsilon_k - \omega) \theta(\epsilon_F - \epsilon_k + \omega). \tag{6.89}$$

We use the time-ordered PPM correlation self-energy

$$\Sigma^{GW}(k, \omega) = \sum_i^{\mathrm{occ}} A_{ki}^{\mathrm{PPM}} \frac{1}{\omega - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}} - i\eta} + \sum_a^{\mathrm{unocc}} A_{ka}^{\mathrm{PPM}} \frac{1}{\omega - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{q}} + i\eta}, \tag{6.90}$$

where

$$\boldsymbol{q} = \boldsymbol{G}_k - \boldsymbol{G}_i \tag{6.91}$$

for the occupied contribution, and similarly

$$\boldsymbol{q} = \boldsymbol{G}_k - \boldsymbol{G}_a \tag{6.92}$$

for the unoccupied contribution. The PPM coupling weight is

$$A_{kn}^{\mathrm{PPM}} = \frac{1}{2V} \frac{\Omega^2_{\boldsymbol{G}_k - \boldsymbol{G}_n}}{\widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_n}} v(\boldsymbol{G}_k - \boldsymbol{G}_n). \tag{6.93}$$

We now evaluate the self-energy at $\epsilon_k - \omega$. For the occupied part,

$$\epsilon_k - \omega - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}} - i\eta = -\left[\omega - (\epsilon_k - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}}) + i\eta\right]. \tag{6.94}$$

Therefore,

$$\frac{1}{\epsilon_k - \omega - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}} - i\eta} = -\frac{1}{\omega - \Delta^h_{ki} + i\eta}, \tag{6.95}$$

where

$$\Delta^h_{ki} = \epsilon_k - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_i}. \tag{6.96}$$

Using

$$\mathrm{Im}\, \frac{1}{x + i\eta} = -\pi\delta(x), \tag{6.97}$$

we obtain

$$\mathrm{Im}\left[-\frac{1}{\omega - \Delta^h_{ki} + i\eta}\right] = \pi\delta(\omega - \Delta^h_{ki}). \tag{6.98}$$

Hence the occupied contribution to $\beta_k(\omega)$ is

$$\sum_i^{\mathrm{occ}} A_{ki}^{\mathrm{PPM}} \delta(\omega - \Delta^h_{ki}). \tag{6.99}$$

For the unoccupied part, define

$$\Delta^u_{ka} = \epsilon_k - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_a}. \tag{6.100}$$

Then

$$\epsilon_k - \omega - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{q}} + i\eta = -\left[\omega - \Delta^u_{ka} - i\eta\right], \tag{6.101}$$

and therefore

$$\mathrm{Im}\left[-\frac{1}{\omega - \Delta^u_{ka} - i\eta}\right] = -\pi\delta(\omega - \Delta^u_{ka}). \tag{6.102}$$

This gives a possible contribution

$$-\sum_a^{\mathrm{unocc}} A_{ka}^{\mathrm{PPM}} \delta(\omega - \Delta^u_{ka}). \tag{6.103}$$

However, for a hole state k, this contribution is removed by the step function in $\beta_k(\omega)$. At the unoccupied pole,

$$\epsilon_F - \epsilon_k + \Delta^u_{ka} = \epsilon_F - \epsilon_k + \epsilon_k - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{q}} \tag{6.104}$$

$$= \epsilon_F - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{q}} < 0, \tag{6.105}$$

because $\varepsilon_a > \epsilon_F$ and $\widetilde{\omega}_{\boldsymbol{q}} > 0$. Thus

$$\theta(\epsilon_F - \epsilon_k + \Delta^u_{ka}) = 0. \tag{6.106}$$

Therefore, for a hole state, Hedin's spectral density reduces to the discrete PPM form

$$\beta_k(\omega) = \sum_i^{\text{occ}} A^{\text{PPM}}_{ki} \delta\left(\omega - \Delta^h_{ki}\right). \tag{6.107}$$

Substituting this into the cumulant gives

$$S_k(t) = \int d\omega \sum_i^{\text{occ}} A^{\text{PPM}}_{ki} \delta(\omega - \Delta^h_{ki}) \frac{e^{i\omega t} - i\omega t - 1}{\omega^2} \tag{6.108}$$

$$= \sum_i^{\text{occ}} A^{\text{PPM}}_{ki} \frac{e^{i\Delta^h_{ki} t} - i\Delta^h_{ki} t - 1}{(\Delta^h_{ki})^2}. \tag{6.109}$$

Equivalently,

$$S_k(t) = \sum_i^{\text{occ}} \zeta^{h,\text{PPM}}_{ki} \left[e^{i\Delta^h_{ki} t} - i\Delta^h_{ki} t - 1\right], \tag{6.110}$$

with

$$\zeta^{h,\text{PPM}}_{ki} = \frac{A^{\text{PPM}}_{ki}}{(\Delta^h_{ki})^2}. \tag{6.111}$$

Thus the PPM version of Hedin's hole cumulant spectral function is

$$A_k(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt\, e^{i(\omega - \epsilon_k)t} \exp\left[\sum_i^{\text{occ}} \zeta^{h,\text{PPM}}_{ki} \left(e^{i\Delta^h_{ki} t} - i\Delta^h_{ki} t - 1\right)\right] \tag{6.112}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dt\, e^{i\omega t} G_k(t) \tag{6.113}$$

where

$$G_k(t) = i\theta(-t) e^{-i\epsilon_k t} e^{S_k(t)}. \tag{6.114}$$

Defining the renormalized quasiparticle energy

$$\tilde{\epsilon}_k = \epsilon_k + \Delta^{\text{static}}_k = \epsilon_k + \Sigma^x_k - V^{xc}_k, \tag{6.115}$$

we arrive at

$$G_k^R(t) = -i\theta(t)\, e^{-i\tilde{\epsilon}_k t} e^{C_k^{\rm dyn}(t)} \tag{6.116}$$

$$= -i\theta(t)\, e^{-i\tilde{\epsilon}_k t} \exp\left\{ \sum_{i\nu} \zeta_{ki\nu} \left[e^{-i\Delta_{ki\nu} t} - 1 + i\Delta_{ki\nu} t \right] + \sum_{a\nu} \zeta_{ka\nu} \left[e^{-i\Delta_{ka\nu} t} - 1 + i\Delta_{ka\nu} t \right] \right\} \tag{6.117}$$

$$= -i\theta(t)\, \underbrace{\exp\left\{ -\sum_{i\nu} \zeta_{ki\nu} - \sum_{a\nu} \zeta_{ka\nu} \right\}}_{Z_k} \exp\left\{ -i \underbrace{\left(\tilde{\epsilon}_k - \sum_{i\nu} \zeta_{ki\nu} \Delta_{ki\nu} - \sum_{a\nu} \zeta_{ka\nu} \Delta_{ka\nu} \right)}_{\epsilon_k^{QP}} t \right\} \tag{6.118}$$

$$\times \exp\left\{ \sum_{i\nu} \zeta_{ki\nu} e^{-i\Delta_{ki\nu} t} + \sum_{a\nu} \zeta_{ka\nu} e^{-i\Delta_{ka\nu} t} \right\} \tag{6.119}$$

$$= -i\theta(t)\, Z_k e^{-i\epsilon_k^{QP} t} \exp\left\{ \sum_{i\nu} \zeta_{ki\nu} e^{-i\Delta_{ki\nu} t} + \sum_{a\nu} \zeta_{ka\nu} e^{-i\Delta_{ka\nu} t} \right\} \tag{6.120}$$

Now we can perform the analytic Fourier transform:

$$G_k^R(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} G_k^R(t) \tag{6.121}$$

$$= -iZ_k \int_0^{\infty} dt\, e^{i(\omega - \epsilon_k^{QP})t} \exp\left\{ \sum_{i\nu} \zeta_{ki\nu} e^{-i\Delta_{ki\nu} t} + \sum_{a\nu} \zeta_{ka\nu} e^{-i\Delta_{ka\nu} t} \right\} \tag{6.122}$$

$$\approx -iZ_k \int_0^{\infty} dt\, e^{i(\omega - \epsilon_k^{QP})t} \left[1 + \sum_{i\nu} \zeta_{ki\nu} e^{-i\Delta_{ki\nu} t} + \sum_{a\nu} \zeta_{ka\nu} e^{-i\Delta_{ka\nu} t} \right] \tag{6.123}$$

$$= -iZ_k \int_0^{\infty} dt\, e^{-i\eta t + i(\omega - \epsilon_k^{QP})t} - iZ_k \sum_{i\nu} \zeta_{ki\nu} \int_0^{\infty} dt\, e^{-i\eta t + i(\omega - \epsilon_k^{QP} - \Delta_{ki\nu})t} \tag{6.124}$$

$$- iZ_k \sum_{a\nu} \zeta_{ka\nu} \int_0^{\infty} dt\, e^{-i\eta t + i(\omega - \epsilon_k^{QP} - \Delta_{ka\nu})t} \tag{6.125}$$

$$= \frac{Z_k}{\omega - \epsilon_k^{QP} + i\eta} + \sum_{i\nu} \frac{\overbrace{Z_k \zeta_{ki\nu}}^{Z^{\rm sat}_{ki\nu}}}{\omega - \underbrace{\left(\epsilon_k^{QP} + \Delta_{ki\nu} \right)}_{\epsilon^{\rm sat}_{ki\nu}} + i\eta} + \sum_{a\nu} \frac{\overbrace{Z_k \zeta_{ka\nu}}^{Z^{\rm sat}_{ka\nu}}}{\omega - \underbrace{\left(\epsilon_k^{QP} + \Delta_{ka\nu} \right)}_{\epsilon^{\rm sat}_{ka\nu}} + i\eta} \tag{6.126}$$

Then,

$$A_{pp}^{GW+C}(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G_k^R(\omega) \tag{6.127}$$

$$= -\frac{1}{\pi}\,\mathrm{Im}\left[\frac{Z_k}{\omega-\epsilon_k^{QP}+i\eta}+\sum_{i\nu}\frac{Z_{ki\nu}^{\mathrm{sat}}}{\omega-\epsilon_{ki\nu}^{\mathrm{sat}}+i\eta}+\sum_{a\nu}\frac{Z_{ka\nu}^{\mathrm{sat}}}{\omega-\epsilon_{ka\nu}^{\mathrm{sat}}+i\eta}\right] \tag{6.128}$$

$$= -\frac{1}{\pi}\,\mathrm{Im}\left[\frac{\mathrm{Re}\,Z_k+i\,\mathrm{Im}\,Z_k}{\omega-\mathrm{Re}\,\epsilon_k^{QP}+i(\eta-\mathrm{Im}\,\epsilon_k^{QP})}+\sum_{i\nu}\frac{\mathrm{Re}\,Z_{ki\nu}^{\mathrm{sat}}+i\,\mathrm{Im}\,Z_{ki\nu}^{\mathrm{sat}}}{\omega-\mathrm{Re}\,\epsilon_{ki\nu}^{\mathrm{sat}}+i(\eta-\mathrm{Im}\,\epsilon_{ki\nu}^{\mathrm{sat}})}+\ldots\right] \tag{6.129}$$

$$= -\frac{1}{\pi}\,\mathrm{Im}\left[\frac{(\mathrm{Re}\,Z_k+i\,\mathrm{Im}\,Z_k)\left(\omega-\mathrm{Re}\,\epsilon_k^{QP}-i(\eta-\mathrm{Im}\,\epsilon_k^{QP})\right)}{\left(\omega-\mathrm{Re}\,\epsilon_k^{QP}\right)^2+(\eta-\mathrm{Im}\,\epsilon_k^{QP})^2}\right.$$

$$\left.+\sum_{i\nu}\frac{(\mathrm{Re}\,Z_{ki\nu}^{\mathrm{sat}}+i\,\mathrm{Im}\,Z_{ki\nu}^{\mathrm{sat}})\,(\omega-\mathrm{Re}\,\epsilon_{ki\nu}^{\mathrm{sat}}-i(\eta-\mathrm{Im}\,\epsilon_{ki\nu}^{\mathrm{sat}}))}{(\omega-\mathrm{Re}\,\epsilon_{ki\nu}^{\mathrm{sat}})^2+(\eta-\mathrm{Im}\,\epsilon_{ki\nu}^{\mathrm{sat}})^2}+\ldots\right] \tag{6.130}$$

$$= -\frac{1}{\pi}\left[\frac{(\mathrm{Re}\,Z_k)\left(\mathrm{Im}\,\epsilon_k^{QP}\right)+\left(\mathrm{Im}\,Z_k^{QP}\right)\left(\omega-\mathrm{Re}\,\epsilon_k^{QP}\right)}{\left(\omega-\mathrm{Re}\,\epsilon_k^{QP}\right)^2+\left(\mathrm{Im}\,\epsilon_k^{QP}\right)^2}\right. \tag{6.131}$$

$$\left.+\sum_{i\nu}\frac{(\mathrm{Re}\,Z_{ki\nu}^{sat})\,(\mathrm{Im}\,\epsilon_{ki\nu}^{sat})+(\mathrm{Im}\,Z_{ki\nu}^{sat})\,(\omega-\mathrm{Re}\,\epsilon_{ki\nu}^{sat})}{(\omega-\mathrm{Re}\,\epsilon_{ki\nu}^{sat})^2+(\mathrm{Im}\,\epsilon_{ki\nu}^{sat})^2}+\ldots\right] \tag{6.132}$$

It is conventional to denote the dynamical cumulant by $\tilde{C}_k^R(t)$:

$$G_k^R(t) = -i\theta(t)\,e^{-i\tilde{\epsilon}_k t}e^{\tilde{C}_k^R(t)}. \tag{6.133}$$

6.2.2 How we do the discrete Fourier transform

Retarded case

To do the DFT, we need a periodic signal. However, the retarded and time-ordered Green's function for hole states are only non-zero on one side of the time axis. For the retarded case we had

$$G^R(t) = -i\Theta(t)\int d\omega A(\omega)e^{-i\omega t} \tag{6.134}$$

$$\implies G^R(t) = -i\int d\omega A(\omega)e^{-i\omega t} \quad \forall t>0. \tag{6.135}$$

Since $A(\omega)$ is real, we can write

$$G^R(t)^* = +i\int d\omega A(\omega)e^{+i\omega t} \tag{6.136}$$

$$\implies -G^R(-t)^* = -i\int d\omega A(\omega)e^{-i\omega t} \tag{6.137}$$

Now we have a definition for $G^R(t)$ for all t, so defining $G_{\text{wrapped}}(t) = G^R(t)$ for $t > 0$ and $G_{\text{wrapped}}(t) = -G^R(-t)^*$ for $t < 0$, we can generate the periodic signal as

$$[G^R(0), G^R(dt), \ldots, G^R(Ndt), -G^R(-Ndt)^*, \ldots, -G^R(-dt)^*]. \tag{6.138}$$

Then we can Fourier transform as

$$G_{\text{wrapped}}(t) = -i \int d\omega A(\omega) e^{-i\omega t}. \tag{6.139}$$

with the spectral function computed as $A(\omega) = -\frac{1}{\pi} \operatorname{Im} G^R(\omega)$.

Time-Ordered case

For the time-ordered case, with $\tau = -t$ for a hole state, we have

$$G^T(-\tau) = i\Theta(\tau) \int d\omega A(\omega) e^{i\omega\tau} \tag{6.140}$$

$$\implies G^T(-\tau) = i \int d\omega A(\omega) e^{i\omega\tau} \quad \forall \tau > 0. \tag{6.141}$$

Since $A(\omega)$ is real, we can write

$$G^T(-\tau)^* = -i \int d\omega A(\omega) e^{-i\omega\tau} \tag{6.142}$$

$$\implies -G^T(\tau)^* = i \int d\omega A(\omega) e^{i\omega\tau} \tag{6.143}$$

Now we have a definition for $G^T(\tau)$ for all τ, so defining $G_{\text{wrapped}}(t) = G^T(-\tau)$ for $\tau > 0$ and $G_{\text{wrapped}}(t) = -G^T(\tau)^*$ for $\tau < 0$, we can generate the periodic signal as

$$[G^T(0), G^T(d\tau), \ldots, G^T(Nd\tau), -G^T(-Nd\tau)^*, \ldots, -G^T(-d\tau)^*]. \tag{6.144}$$

Then we can Fourier transform as

$$G_{\text{wrapped}}(t) = i \int d\omega A(\omega) e^{i\omega\tau}. \tag{6.145}$$

with the spectral function computed as $A(\omega) = +\frac{1}{\pi} \operatorname{Im} G^T(\omega)$.

6.2.3 Considerations for the broadening

-C

We always do

$$A_k(\omega) \propto \frac{X}{(\omega - \epsilon_k - \operatorname{Re}\Sigma)^2 + X^2}. \tag{6.146}$$

v2 has $X = \operatorname{Im}\Sigma + \eta$, while v1 has just $X = \operatorname{Im}\Sigma$. Note that to match Loos's result for HF+GW, I need to use v1.

+C

To compute the Green's function in the time domain, we need

$$G_k(t) = -i\theta(t)e^{-(i\epsilon_k+\eta_1)t}e^{C_k(t)}. \tag{6.147}$$

where

$$C_k(t) = i\int \frac{d\omega}{2\pi}e^{-i\omega t}\frac{|\operatorname{Im}\Sigma(\omega+\epsilon_k)[\eta_2]|}{(\omega^2)}\left[e^{-i\omega t} - i\omega t - 1\right]. \tag{6.148}$$

I have verified numerically that one if one chooses $\eta_1 = 0$, then at long enough times, we recover the "first order" answer. Choosing $\eta_1 \neq 0$ broadens too much. In my LDA+GW+C curves, I choose $\eta_1 = 0$.

6.2.4 Annotation of Loos paper

Getting a density matrix from the equal-time limit of the cumulant Green's function

We can start with

$$G^\sigma\left(\mathbf{r}_1t_1,\mathbf{r}_2t_2\right) = G_0^\sigma\left(\mathbf{r}_1t_1,\mathbf{r}_2t_2\right)\exp\left[C^\sigma\left(\mathbf{r}_1t_1,\mathbf{r}_2t_2\right)\right]. \tag{6.149}$$

By relating the Dyson equation to the Taylor series expansion of the exponential (both to first order), we can write:

$$G_0^\sigma\left(\mathbf{r}_1t_1,\mathbf{r}_2t_2\right)C^\sigma\left(\mathbf{r}_1t_1,\mathbf{r}_2t_2\right) = \iint dt_3dt_4G_0^\sigma\left(\mathbf{r}_1t_1,\mathbf{r}_3t_3\right)\Sigma^{c,\sigma}\left(\mathbf{r}_3t_3,\mathbf{r}_4t_4\right)G_0^\sigma\left(\mathbf{r}_4t_4,\mathbf{r}_2t_2\right). \tag{6.150}$$

Projecting to the spin-orbital basis and inserting the resolution of the identity, we get:

$$G_{pp}^{0,\sigma}\left(t_1,t_2\right)C_{pq}^\sigma\left(t_1,t_2\right) = \iint dt_3dt_4G_{pp}^{0,\sigma}\left(t_1,t_3\right)\Sigma_{pq}^{c,\sigma}\left(t_3,t_4\right)G_{qq}^{0,\sigma}\left(t_4,t_2\right) \tag{6.151}$$

$$\implies C_{pq}^\sigma\left(t_1,t_2\right) = \frac{1}{G_{pp}^{0,\sigma}\left(t_1,t_2\right)}\int\frac{d\omega}{2\pi}e^{-i\omega(t_1-t_2)}G_{pp}^{0,\sigma}(\omega)\Sigma_{pq}^{c,\sigma}(\omega)G_{qq}^{0,\sigma}(\omega) \tag{6.152}$$

The definition of the cumulant ansatz for the retarded Green's function is given by:

$$\boldsymbol{G}(t) = \boldsymbol{G}^0(t)e^{\boldsymbol{C}(t)} \tag{6.153}$$

where $\boldsymbol{C}(t)$ is the retarded cumulant and $\boldsymbol{G}^0(t)$ is the retarded HF Green's function. By relating the Dyson equation to the Taylor series expansion of the exponential (both to first order), we can write:

$$\boldsymbol{G}^0(t)\boldsymbol{C}(t) = \iint \mathrm{d}t_1\mathrm{d}t_2\boldsymbol{G}^0(t-t_1)\boldsymbol{\Sigma}^c(t_1-t_2)\boldsymbol{G}^0(t_2) \tag{6.154}$$

This first order expansion is exact up to the first order in the screened Coulomb interaction W. If we choose instead to use a second order self-energy, now

the cumulant will be exact to second order in the bare Coulomb interaction v. Projecting to the spin-orbital basis and inserting the resolution of the identity, we get:

$$\sum_r \langle p| \boldsymbol{G}^0(t) |r\rangle \langle r| \boldsymbol{C}(t) |q\rangle = \sum_{rs} \iint \mathrm{d}t_1 \mathrm{d}t_2 \langle p| \boldsymbol{G}^0(t-t_1) |r\rangle \langle r| \boldsymbol{\Sigma}^c(t_1-t_2) |s\rangle \langle s| \boldsymbol{G}^0(t_2) |q\rangle \tag{6.155}$$

$$\sum_r \boldsymbol{G}^0_{pr}(t)\boldsymbol{C}_{rq}(t) = \sum_{rs} \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{ps}(t-t_1)\boldsymbol{\Sigma}^c_{sr}(t_1-t_2)\boldsymbol{G}^0_{rq}(t_2) \tag{6.156}$$

$$\boldsymbol{G}^0_{pp}(t)\boldsymbol{C}_{pq}(t) = \underbrace{\iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{pp}(t-t_1)\boldsymbol{\Sigma}^c_{pq}(t_1-t_2)\boldsymbol{G}^0_{qq}(t_2)}_{*} \tag{6.157}$$

where $\boldsymbol{G}^0(t)$ is the retarded HF Green's function, which is diagonal in the spin-orbital basis, specifically $\boldsymbol{G}^0_{pp}(t) = -i\Theta(t)e^{-i\epsilon_p t}$, where ϵ_p is the HF energy of the p-th spin-orbital. The formula for the inverse Fourier transform is given by:

$$f(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} f(\omega) \tag{6.158}$$

which implies that

$$\boldsymbol{G}^0_{pp}(t-t_1) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega) \tag{6.159}$$

$$\boldsymbol{\Sigma}^c_{pq}(t_1-t_2) = \int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega') \tag{6.160}$$

$$\boldsymbol{G}^0_{qq}(t_2) = \int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'') \tag{6.161}$$

and plugging into the double time integral *, we get:

$$* = \iint \mathrm{d}t_1 \mathrm{d}t_2 \left[\int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega)\right] \left[\int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega')\right] \left[\int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'')\right] \tag{6.162}$$

$$= \underbrace{\int \mathrm{d}t_1 e^{-i(\omega'-\omega)t_1} \int \mathrm{d}t_2 e^{-i(\omega''-\omega')t_2}}_{4\pi^2\delta(\omega'-\omega)\delta(\omega''-\omega')} \iiint \mathrm{d}\omega \mathrm{d}\omega' \mathrm{d}\omega'' \frac{e^{-i\omega t}}{8\pi^3} \boldsymbol{G}^0_{pp}(\omega)\boldsymbol{\Sigma}^c_{pq}(\omega')\boldsymbol{G}^0_{qq}(\omega'') \tag{6.163}$$

$$= \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} \boldsymbol{G}^0_{pp}(\omega)\boldsymbol{\Sigma}^c_{pq}(\omega)\boldsymbol{G}^0_{qq}(\omega) \tag{6.164}$$

Multiplying both sides of eqn. 6.1376 by $ie^{\epsilon_p t}$ and recalling that we are dealing with retarded quantities that vanish for negative times, we get:

6.2.5 Notes for BWs2+C

The definition of the cumulant ansatz for the retarded Green's function is given by:

$$\boldsymbol{G}(t) = \boldsymbol{G}^0(t)e^{\boldsymbol{C}(t)} \tag{6.165}$$

where $\boldsymbol{C}(t)$ is the retarded cumulant and $\boldsymbol{G}^0(t)$ is the retarded BWs2 Green's function. By relating the Dyson equation to the Taylor series expansion of the exponential, we can write:

$$\boldsymbol{G}^0(t) + \boldsymbol{G}^0(t)\boldsymbol{C}(t) + O(v^4) = \boldsymbol{G}^0(t) + \iint \mathrm{d}t_1\mathrm{d}t_2 \boldsymbol{G}^0(t-t_1)\boldsymbol{\Sigma}^c(t_1-t_2)\boldsymbol{G}^0(t_2) + O(v^4) \tag{6.166}$$

$$\implies \boldsymbol{G}^0(t)\boldsymbol{C}(t) = \iint \mathrm{d}t_1\mathrm{d}t_2 \boldsymbol{G}^0(t-t_1)\boldsymbol{\Sigma}^c(t_1-t_2)\boldsymbol{G}^0(t_2) \quad \text{(up to second order in } v) \tag{6.167}$$

If we use the second order self-energy, the cumulant will be exact to second order in the bare Coulomb interaction v. **Will this also be the case for BWs2?** Projecting to the spin-orbital basis and inserting the resolution of the identity, we get:

$$\sum_r \langle p|\, \boldsymbol{G}^0(t)\, |r\rangle \langle r|\, \boldsymbol{C}(t)\, |q\rangle = \sum_{rs} \iint \mathrm{d}t_1\mathrm{d}t_2 \langle p|\, \boldsymbol{G}^0(t-t_1)\, |r\rangle \langle r|\, \boldsymbol{\Sigma}^c(t_1-t_2)\, |s\rangle \langle s|\, \boldsymbol{G}^0(t_2)\, |q\rangle \tag{6.168}$$

$$\implies \sum_r \boldsymbol{G}^0_{pr}(t)\boldsymbol{C}_{rq}(t) = \sum_{rs} \iint \mathrm{d}t_1\mathrm{d}t_2 \boldsymbol{G}^0_{ps}(t-t_1)\boldsymbol{\Sigma}^c_{sr}(t_1-t_2)\boldsymbol{G}^0_{rq}(t_2) \tag{6.169}$$

$$\implies \boldsymbol{G}^0_{pp}(t)\boldsymbol{C}_{pq}(t) = \underbrace{\iint \mathrm{d}t_1\mathrm{d}t_2 \boldsymbol{G}^0_{pp}(t-t_1)\boldsymbol{\Sigma}^c_{pq}(t_1-t_2)\boldsymbol{G}^0_{qq}(t_2)}_{*} \tag{6.170}$$

where $\boldsymbol{G}^0(t)$ is the retarded BWs2 Green's function, which is diagonal in the spin-orbital basis, as $\boldsymbol{G}^0_{pp}(t) = -i\Theta(t)e^{-i\epsilon_p t}$, where ϵ_p is the dressed, single-particle energy of the p-th spin-orbital. The formula for the inverse Fourier transform is given by:

$$f(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} f(\omega) \tag{6.171}$$

which implies that

$$\boldsymbol{G}^0_{pp}(t-t_1) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega) \tag{6.172}$$

$$\boldsymbol{\Sigma}^c_{pq}(t_1-t_2) = \int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega') \tag{6.173}$$

$$\boldsymbol{G}^0_{qq}(t_2) = \int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'') \tag{6.174}$$

and plugging into the double time integral *, we get:

$$* = \iint \mathrm{d}t_1 \mathrm{d}t_2 \left[\int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega)\right] \left[\int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega')\right] \left[\int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'')\right] \tag{6.175}$$

$$= \underbrace{\int \mathrm{d}t_1 e^{-i(\omega'-\omega)t_1} \int \mathrm{d}t_2 e^{-i(\omega''-\omega')t_2}}_{4\pi^2\delta(\omega'-\omega)\delta(\omega''-\omega')} \iiint \mathrm{d}\omega \mathrm{d}\omega' \mathrm{d}\omega'' \frac{e^{-i\omega t}}{8\pi^3} \boldsymbol{G}^0_{pp}(\omega) \boldsymbol{\Sigma}^c_{pq}(\omega') \boldsymbol{G}^0_{qq}(\omega'') \tag{6.176}$$

$$= \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} \boldsymbol{G}^0_{pp}(\omega) \boldsymbol{\Sigma}^c_{pq}(\omega) \boldsymbol{G}^0_{qq}(\omega) \tag{6.177}$$

With $\eta \to 0^+$, we get

$$-i\Theta(t) e^{-i\epsilon_p t} \boldsymbol{C}_{pq}(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} \boldsymbol{G}^0_{pp}(\omega) \boldsymbol{\Sigma}^c_{pq}(\omega) \boldsymbol{G}^0_{qq}(\omega) \tag{6.178}$$

$$\implies C_{pp}(t) = i \int \frac{\mathrm{d}\omega}{2\pi} e^{-i(\omega-\epsilon_p)t} \frac{\Sigma^c_{pp}(\omega)}{(\omega - \epsilon_p + i\eta)^2} \quad \forall t > 0 \tag{6.179}$$

$$= i \int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega' t} \frac{\Sigma^c_{pp}(\omega' + \epsilon_p)}{(\omega' + i\eta)^2} \tag{6.180}$$

The second order self-energy, which uses a HF reference, can be written as

$$\Sigma^{(2)}_{pq}(\omega) = \frac{1}{2} \sum_{iab} \frac{\langle pi||ab\rangle\langle ab||qi\rangle}{\omega + \epsilon_i - \epsilon_a - \epsilon_b} + \frac{1}{2} \sum_{ija} \frac{\langle pa||ij\rangle\langle ij||qa\rangle}{\omega + \epsilon_a - \epsilon_i - \epsilon_j} \tag{6.181}$$

$$\implies \Sigma^{(2)}_{pp}(\omega + \epsilon_p) = \frac{1}{2} \sum_{iab} \frac{\langle pi \,||\, ab\rangle^2}{\omega - \epsilon^{ab}_{pi}} + \frac{1}{2} \sum_{ija} \frac{\langle pa \,||\, ij\rangle^2}{\omega - \epsilon^{ij}_{pa}} \tag{6.182}$$

where $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$. So we can write the diagonal cumulant[2] as

$$C_{pp}(t) \equiv i \int \frac{d\omega}{2\pi} \frac{\Sigma_{pp}^{c}(\omega + \epsilon_p)}{(\omega + i\eta)^2} e^{-i\omega t} \tag{6.183}$$

$$\implies C_{pp}^{(2)}(t) = \frac{1}{2} \sum_{iab} \langle pi \,||\, ab \rangle^2 \int \frac{d\omega}{2\pi} \frac{i e^{-i\omega t}}{\omega^2 \left(\omega - \epsilon_{pi}^{ab}\right)} + \frac{1}{2} \sum_{ija} \langle pa \,||\, ij \rangle^2 \int \frac{d\omega}{2\pi} \frac{i e^{-i\omega t}}{\omega^2 \left(\omega - \epsilon_{pa}^{ij}\right)} \tag{6.184}$$

$$= \frac{1}{2} \sum_{iab} \frac{\langle pi \,||\, ab \rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2} \left(e^{-i\epsilon_{pi}^{ab} t} + i\epsilon_{pi}^{ab} t - 1 \right) + \frac{1}{2} \sum_{ija} \frac{\langle pa \,||\, ij \rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2} \left(e^{-i\epsilon_{pa}^{ij} t} + i\epsilon_{pa}^{ij} t - 1 \right) \tag{6.185}$$

$$= \frac{1}{2} \sum_{iab} \langle pi \| ab \rangle^2 f\left(\epsilon_{pi}^{ab}\right) + \frac{1}{2} \sum_{ija} \langle pa \| ij \rangle^2 f\left(\epsilon_{pa}^{ij}\right) \tag{6.186}$$

$$= \int d\omega \, \beta(\omega) f(\omega) \tag{6.187}$$

where $f(\omega) \equiv \frac{e^{-i\omega t} + i\omega t - 1}{\omega^2}$ and we identify the cumulant kernel as [3]

$$\beta(\omega) = -\frac{1}{\pi} \operatorname{Im} \Sigma_{pp}^{(2)}(\omega + \epsilon_p) \tag{6.188}$$

$$\tag{6.189}$$

$$= \frac{1}{2} \sum_{iab} \langle pi \| ab \rangle^2 \delta\left(\omega - \epsilon_{pi}^{ab}\right) + \frac{1}{2} \sum_{ija} \langle pa \| ij \rangle^2 \delta\left(\omega - \epsilon_{pa}^{ij}\right) \tag{6.190}$$

A connection can be made between the cumulant kernel $\beta(\omega) = \sum_q g_q^2 \delta(\omega - \omega_q)$ for the 2nd-order self energy to that for electrons coupled to bosonic excitations at the frequencies $\omega_q \equiv \epsilon_{pq}^{rs}$ in the quasi-boson approximation (QBA) with coupling coefficients $g_q \equiv \langle pq \| rs \rangle$. **I feel that everything up to 8.101 holds generally. It needs to be determined how to proceed through 6.804 with BWs2.**

[2]Note that in order to get from eqn. 6.798 to eqn. 6.799 we have used the identity $\int \frac{d\omega}{2\pi} \frac{i e^{-i\omega t}}{\omega^2(\omega - \epsilon)} = \frac{1}{\epsilon^2}\left(e^{-i\epsilon t} + i\epsilon t - 1\right) \operatorname{sgn}(t)$, and since we are dealing with retarded quantities, we only care about $t > 0$ so $\operatorname{sgn}(t) = 1$.

[3]To get from eqn. 6.802 to eqn. 6.804, we have used the identity $\operatorname{Im} \frac{1}{x + i\eta} = -\pi\delta(x)$ as $\eta \to 0^+$.

6.2.6 Standard route: diagonal cumulant with GW self-energy

$$C_{pp}(t) = i\int \frac{d\omega}{2\pi}\frac{\Sigma^c_{pp}\left(\omega+\epsilon_p^{HF}\right)}{(\omega+i\eta)^2}e^{-i\omega t} \tag{6.191}$$

$$= i\int \frac{d\omega}{2\pi}\frac{1}{(\omega+i\eta)^2}e^{-i\omega t}\left[\sum_{i\nu}\frac{M^2_{pi\nu}}{\omega+\underbrace{\epsilon_p^{HF}-\epsilon_i+\Omega_\nu+i\eta}_{-\Delta_{pi\nu}}}+\sum_{a\nu}\frac{M^2_{pa\nu}}{\omega+\underbrace{\epsilon_p^{HF}-\epsilon_a-\Omega_\nu+i\eta}_{-\Delta_{pa\nu}}}\right] \tag{6.192}$$

$$= \mathrm{i}\sum_{iv} M^2_{piv}\int\frac{\mathrm{d}\omega}{2\pi}e^{-\mathrm{i}\omega t}\frac{1}{[\omega+\mathrm{i}\eta]^2}\frac{1}{\omega-\Delta_{piv}}+\mathrm{i}\sum_{av}M^2_{pav}\int\frac{\mathrm{d}\omega}{2\pi}e^{-\mathrm{i}\omega t}\frac{1}{[\omega+\mathrm{i}\eta]^2}\frac{1}{\omega-\Delta_{pav}} \tag{6.193}$$

$$= \sum_{i\nu}\zeta_{pi\nu}\left[e^{-i\Delta_{pi\nu}t}-1+i\Delta_{pi\nu}t\right]+\sum_{a\nu}\zeta_{pa\nu}\left[e^{-i\Delta_{pa\nu}t}-1+i\Delta_{pa\nu}t\right] \tag{6.194}$$

where in going from eqn. 6.569 to eqn. 8.101, where we made a diagonal approximation for the self-energy and introduced the frequency shift $\omega \to \omega+\epsilon_p^{HF}$, and then from eqn. 6.193 to eqn. 8.102, we have evaluated a contour integral. For the final expression, we have defined $\zeta_{pi\nu} = \left(\frac{M_{pi\nu}}{\Delta_{pi\nu}}\right)^2$ and $\zeta_{pa\nu} = \left(\frac{M_{pa\nu}}{\Delta_{pa\nu}}\right)^2$. This allows us to arrive at the something similar to the Landau form of the cumulant.

A few notes on how to evaluate this integral: there is a double pole at $\omega_1 = -i\eta$ and a simple pole at $\omega_2 = -\Delta - i\eta$. Closing the contour in the lower half-plane because $\mathrm{Im}\left(\omega_1\right), \mathrm{Im}\left(\omega_2\right) < 0$, and applying Cauchy's residue theorem, leads to

$$\int\frac{\mathrm{d}\omega}{2\pi}e^{-\mathrm{i}\omega t}\frac{1}{(\omega-\omega_1)^2}\frac{1}{\omega-\omega_2} = (-\mathrm{i})\left\{\left[\partial_\omega\left(\frac{e^{-\mathrm{i}\omega t}}{\omega-\omega_2}\right)\right]_{\omega=\omega_1}+\left[\frac{e^{-\mathrm{i}\omega t}}{(\omega-\omega_1)^2}\right]_{\omega=\omega_2}\right\} \tag{6.195}$$

$$= \frac{(-\mathrm{i})}{(\omega_1-\omega_2)^2}\left\{\left[(-\mathrm{i}t)\left(\omega_1-\omega_2\right)-1\right]e^{-\mathrm{i}\omega_1 t}+e^{-\mathrm{i}\omega_2 t}\right\} \tag{6.196}$$

$$\implies \int\frac{\mathrm{d}\omega}{2\pi}e^{-i\omega t}\frac{1}{[\omega-(0-\mathrm{i}\eta)]^2}\frac{1}{\omega-\Delta} = \frac{-\mathrm{i}}{\Delta^2}\left(e^{-\mathrm{i}\Delta t}+\mathrm{i}\Delta t-1\right) \tag{6.197}$$

Now we plug in our derived expression for $C_{pp}(t)$ into the cumulant ansatz for the retarded

Green's function:

$$G_{pp}(t) = G_{pp}^{HF}(t)e^{C_{pp}(t)} \tag{6.198}$$

$$= -i\Theta(t)e^{-i\epsilon_p^{HF}t + C_{pp}(t)} \tag{6.199}$$

$$= -i\Theta(t)e^{-i\epsilon_p^{HF}t + \sum_{i\nu}\zeta_{pi\nu}\left(e^{-i\Delta_{pi\nu}t} + i\Delta_{pi\nu}t - 1\right) + \sum_{a\nu}\zeta_{pa\nu}\left(e^{-i\Delta_{pa\nu}t} + i\Delta_{pa\nu}t - 1\right)} \tag{6.200}$$

$$= -i\Theta(t)\underbrace{e^{-\sum_{i\nu}\zeta_{pi\nu} - \sum_{a\nu}\zeta_{pa\nu}}}_{Z_p^{QP}} e^{-i\overbrace{\left(\epsilon_p^{HF} - \sum_{i\nu}\zeta_{pi\nu}\Delta_{pi\nu} - \sum_{a\nu}\zeta_{pa\nu}\Delta_{pa\nu}\right)}^{\epsilon_p^{QP}}t} e^{\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi\nu}t} + \sum_{a\nu}\zeta_{pa\nu}e^{-i\Delta_{pa\nu}t}} \tag{6.201}$$

$$= -i\Theta(t)Z_p^{QP}e^{-i\epsilon_p^{QP}t}e^{\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi\nu}t} + \sum_{a\nu}\zeta_{pa\nu}e^{-i\Delta_{pa\nu}t}} \tag{6.202}$$

$$\tag{6.203}$$

where we have the weight of the quasiparticle peak $Z_p^{QP} = \exp\left(-\sum_{i\nu}\zeta_{pi\nu} - \sum_{a\nu}\zeta_{pa\nu}\right)$ and the quasiparticle energy $\epsilon_p^{QP} = \epsilon_p^{HF} - \left(\sum_{i\nu}\zeta_{pi\nu}\Delta_{pi\nu} + \sum_{a\nu}\zeta_{pa\nu}\Delta_{pa\nu}\right)$.

We pause to make some important connections. Notice

$$\epsilon_p^{QP} = \epsilon_p^{HF} - \left(\sum_{i\nu}\zeta_{pi\nu}\Delta_{pi\nu} + \sum_{a\nu}\zeta_{pa\nu}\Delta_{pa\nu}\right) \tag{6.204}$$

$$= \epsilon_p^{HF} - \left(\sum_{i\nu}\frac{M_{pi\nu}^2}{\Delta_{pi\nu}} + \sum_{a\nu}\frac{M_{pa\nu}^2}{\Delta_{pa\nu}}\right) \tag{6.205}$$

$$= \epsilon_p^{HF} + \Sigma_{pp}^c\left(\epsilon_p^{HF}\right) \tag{6.206}$$

and

$$Z_p^{QP} = \exp\left(-\sum_{i\nu}\zeta_{pi\nu} - \sum_{a\nu}\zeta_{pa\nu}\right) \tag{6.207}$$

$$= \exp\left(-\sum_{i\nu}\left(\frac{M_{pi\nu}}{\Delta_{pi\nu}}\right)^2 - \sum_{a\nu}\left(\frac{M_{pa\nu}}{\Delta_{pa\nu}}\right)^2\right) \tag{6.208}$$

$$= \exp\left(\left[\frac{\partial\Sigma_{pp}^c(\omega)}{\partial\omega}\right]_{\omega=\epsilon_p^{HF}}\right) \tag{6.209}$$

$$\tag{6.210}$$

where we have used the fact that $\Sigma_{pp}^c(\omega) = \sum_{i\nu}\frac{M_{pi\nu}^2}{\omega - \epsilon_i + \Omega_\nu} + \sum_{a\nu}\frac{M_{pa\nu}^2}{\omega - \epsilon_a - \Omega_\nu} \implies \left[\frac{\partial\Sigma_{pp}^c(\omega)}{\partial\omega}\right]_{\omega=\epsilon_p^{HF}} = -\sum_{i\nu}\left(\frac{M_{pi\nu}}{\Delta_{pi\nu}}\right)^2 - \sum_{a\nu}\left(\frac{M_{pa\nu}}{\Delta_{pa\nu}}\right)^2$.

Next, we want to perform a Fourier transform.

$$G_{pp}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} G_{pp}(t) \tag{6.211}$$

$$= -iZ_p^{QP} \int_0^{\infty} dt e^{i(\omega-\epsilon_p^{QP})t} e^{\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi\nu}t}+\sum_{a\nu}\zeta_{pa\nu}e^{-i\Delta_{pa\nu}t}} \tag{6.212}$$

$$= -iZ_p^{QP} \int_0^{\infty} dt\, e^{i(\omega-\epsilon_p^{QP})t} \left(1 + \sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi\nu}t} + \sum_{a\nu}\zeta_{pa\nu}e^{-i\Delta_{pa\nu}t}4\right) \tag{6.213}$$

$$= -iZ_p^{QP} \int_0^{\infty} dt\, e^{[-\eta+i(\omega-\epsilon_p^{QP})t]} \tag{6.214}$$

$$- iZ_p^{QP} \sum_{i\nu}\zeta_{pi\nu} \int_0^{\infty} dt\, e^{[-\eta+i(\omega-\epsilon_p^{QP}-\Delta_{pi\nu})t]} \tag{6.215}$$

$$- iZ_p^{QP} \sum_{a\nu}\zeta_{pa\nu} \int_0^{\infty} dt\, e^{[-\eta+i(\omega-\epsilon_p^{QP}-\Delta_{pa\nu})t]} + \ldots \tag{6.216}$$

$$= \frac{Z_p^{QP}}{\omega-\epsilon_p^{QP}+i\eta} + \sum_{i\nu}\frac{Z_p^{QP}\zeta_{pi\nu}}{\omega-\epsilon_p^{QP}-\Delta_{pi\nu}+i\eta} + \sum_{a\nu}\frac{Z_p^{QP}\zeta_{pa\nu}}{\omega-\epsilon_p^{QP}-\Delta_{pa\nu}+i\eta} + \ldots \tag{6.217}$$

$$= \frac{Z_p^{QP}}{\omega-\epsilon_p^{QP}+i\eta} + \sum_{i\nu}\frac{Z_{pi\nu}^{sat}}{\omega-\epsilon_{pi\nu}^{sat}+i\eta} + \sum_{a\nu}\frac{Z_{pa\nu}^{sat}}{\omega-\epsilon_{pa\nu}^{sat}+i\eta} + \ldots \tag{6.218}$$

In going from eqn. 8.96 to eqn. 8.97, we used the step function to restrict the lower bound of the integral to 0 and then we end by defining the satellite energies $\epsilon_{pi\nu}^{sat} = \epsilon_p^{QP} + \Delta_{pi\nu}$ and $\epsilon_{pa\nu}^{sat} = \epsilon_p^{QP} + \Delta_{pa\nu}$, as well as the satellite weights $Z_{pi\nu}^{sat} = Z_p^{QP}\zeta_{pi\nu}$ and $Z_{pa\nu}^{sat} = Z_p^{QP}\zeta_{pa\nu}$. *What happens when we treat the exponential more than just up to the first order? Is this useful?*

Spectral function

The diagonal elements of the spectral function are obtained as (the virtual satellites spectral function, whose derivation will mirror that of the occupied satellites, are omitted for the

sake of brevity)

$$A_{pp}^{GW+C}(\omega) = -\frac{1}{\pi}\operatorname{Im} G_{pp}(\omega) \tag{6.219}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{Z_p^{QP}}{\omega - \epsilon_p^{QP} + i\eta} + \sum_{i\nu}\frac{Z_{pi\nu}^{sat}}{\omega - \epsilon_{pi\nu}^{sat} + i\eta} + \sum_{a\nu}\frac{Z_{pa\nu}^{sat}}{\omega - \epsilon_{pa\nu}^{sat} + i\eta}\right] \tag{6.220}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\operatorname{Re} Z_p^{QP} + i\operatorname{Im} Z_p^{QP}}{\omega - \operatorname{Re}\epsilon_p^{QP} + i\left(\eta - \operatorname{Im}\epsilon_p^{QP}\right)} + \sum_{i\nu}\frac{\operatorname{Re} Z_{pi\nu}^{sat} + i\operatorname{Im} Z_{pi\nu}^{sat}}{\omega - \operatorname{Re}\epsilon_{pi\nu}^{sat} + i\left(\eta - \operatorname{Im}\epsilon_{pi\nu}^{sat}\right)} + \dots\right] \tag{6.221}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\left(\operatorname{Re} Z_p^{QP} + i\operatorname{Im} Z_p^{QP}\right)\left(\omega - \operatorname{Re}\epsilon_p^{QP} - i\left(\eta - \operatorname{Im}\epsilon_p^{QP}\right)\right)}{\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)^2 + \left(\operatorname{Im}\epsilon_p^{QP}\right)^2}\right. \tag{6.222}$$

$$\left. + \sum_{i\nu}\frac{\left(\operatorname{Re} Z_{pi\nu}^{sat} + i\operatorname{Im} Z_{pi\nu}^{sat}\right)\left(\omega - \operatorname{Re}\epsilon_{pi\nu}^{sat} - i\left(\eta - \operatorname{Im}\epsilon_{pi\nu}^{sat}\right)\right)}{\left(\omega - \operatorname{Re}\epsilon_{pi\nu}^{sat}\right)^2 + \left(\operatorname{Im}\epsilon_{pi\nu}^{sat}\right)^2} + \dots\right] \tag{6.223}$$

$$= -\frac{1}{\pi}\left[\frac{\left(\operatorname{Re} Z_p^{QP}\right)\left(\operatorname{Im}\epsilon_p^{QP}\right) + \left(\operatorname{Im} Z_p^{QP}\right)\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)}{\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)^2 + \left(\operatorname{Im}\epsilon_p^{QP}\right)^2}\right. \tag{6.224}$$

$$\left. + \sum_{i\nu}\frac{\left(\operatorname{Re} Z_{pi\nu}^{sat}\right)\left(\operatorname{Im}\epsilon_{pi\nu}^{sat}\right) + \left(\operatorname{Im} Z_{pi\nu}^{sat}\right)\left(\omega - \operatorname{Re}\epsilon_{pi\nu}^{sat}\right)}{\left(\omega - \operatorname{Re}\epsilon_{pi\nu}^{sat}\right)^2 + \left(\operatorname{Im}\epsilon_{pi\nu}^{sat}\right)^2} + \dots\right] \tag{6.225}$$

6.2.7 Annotation of CC Cumulant Green's function paper

Cumulant from second order self-energy

The second order self-energy, which uses a HF reference, can be written as

$$\Sigma_{pq}^{(2)}(\omega) = \frac{1}{2}\sum_{iab}\frac{\langle pi||ab\rangle\langle ab||qi\rangle}{\omega + \epsilon_i - \epsilon_a - \epsilon_b} + \frac{1}{2}\sum_{ija}\frac{\langle pa||ij\rangle\langle ij||qa\rangle}{\omega + \epsilon_a - \epsilon_i - \epsilon_j} \tag{6.226}$$

$$\implies \Sigma_{pp}^{(2)}(\omega + \epsilon_p) = \frac{1}{2}\sum_{iab}\frac{\langle pi\,||\,ab\rangle^2}{\omega - \epsilon_{pi}^{ab}} + \frac{1}{2}\sum_{ija}\frac{\langle pa\,||\,ij\rangle^2}{\omega - \epsilon_{pa}^{ij}} \tag{6.227}$$

where $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$. So we can write the diagonal cumulant[4] as

$$C_{pp}(t) \equiv i \int \frac{d\omega}{2\pi} \frac{\Sigma_{pp}^{c}(\omega + \epsilon_p)}{(\omega + i\eta)^2} e^{-i\omega t} \tag{6.228}$$

$$\implies C_{pp}^{(2)}(t) = \frac{1}{2} \sum_{iab} \langle pi \,||\, ab \rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2 \left(\omega - \epsilon_{pi}^{ab}\right)} + \frac{1}{2} \sum_{ija} \langle pa \,||\, ij \rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2 \left(\omega - \epsilon_{pa}^{ij}\right)} \tag{6.229}$$

$$= \frac{1}{2} \sum_{iab} \frac{\langle pi \,||\, ab \rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2} \left(e^{-i\epsilon_{pi}^{ab} t} + i\epsilon_{pi}^{ab} t - 1\right) + \frac{1}{2} \sum_{ija} \frac{\langle pa \,||\, ij \rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2} \left(e^{-i\epsilon_{pa}^{ij} t} + i\epsilon_{pa}^{ij} t - 1\right) \tag{6.230}$$

$$= \frac{1}{2} \sum_{iab} \langle pi \| ab \rangle^2 f\left(\epsilon_{pi}^{ab}\right) + \frac{1}{2} \sum_{ija} \langle pa \| ij \rangle^2 f\left(\epsilon_{pa}^{ij}\right) \tag{6.231}$$

$$= \int d\omega \, \beta(\omega) f(\omega) \tag{6.232}$$

where $f(\omega) \equiv \frac{e^{-i\omega t} + i\omega t - 1}{\omega^2}$ and we identify the cumulant kernel as [5]

$$\beta(\omega) = -\frac{1}{\pi} \operatorname{Im} \Sigma_{pp}^{(2)}(\omega + \epsilon_p) \tag{6.233}$$

$$\tag{6.234}$$

$$= \frac{1}{2} \sum_{iab} \langle pi \| ab \rangle^2 \delta\left(\omega - \epsilon_{pi}^{ab}\right) + \frac{1}{2} \sum_{ija} \langle pa \| ij \rangle^2 \delta\left(\omega - \epsilon_{pa}^{ij}\right) \tag{6.235}$$

A connection can be made between the cumulant kernel $\beta(\omega) = \sum_q g_q^2 \delta(\omega - \omega_q)$ for the 2nd-order self energy to that for electrons coupled to bosonic excitations at the frequencies $\omega_q \equiv \epsilon_{pq}^{rs}$ in the quasi-boson approximation with coupling coefficients $g_q \equiv \langle pq \| rs \rangle$.

Equivalence between the second order self-energy and the CC retarded Green's function

When both the IP and EA branches are included, the CC GF can be written in frequency space as [6]

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[4] Note that in order to get from eqn. 6.798 to eqn. 6.799 we have used the identity $\int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2(\omega - \epsilon)} = \frac{1}{\epsilon^2}\left(e^{-i\epsilon t} + i\epsilon t - 1\right) \operatorname{sgn}(t)$, and since we are dealing with retarded quantities, we only care about $t > 0$ so $\operatorname{sgn}(t) = 1$.

[5] To get from eqn. 6.802 to eqn. 6.804, we have used the identity $\operatorname{Im} \frac{1}{x + i\eta} = -\pi\delta(x)$ as $\eta \to 0^+$.

[6] A diagrammatic analysis shows that only three non-zero terms when $p, q \in occ$ survive, which allows us to move from eqn. 6.243 to eqn. 6.244.

The retarded single-particle Green's function in the time domain is defined as

$$G^R_{pq}(t-t') = -i\,\Theta(t-t')\langle 0|\{a_p(t), a_q^\dagger(t')\}|0\rangle. \tag{6.236}$$

Fourier transforming and inserting a resolution of identity over $(N\pm1)$-electron eigenstates gives the Lehmann representation for both the IP and EA parts:

$$G^R_{pq}(\omega) = \sum_n \frac{\langle 0|a_p(t)|n^{N+1}\rangle\langle n^{N+1}|a_q^\dagger(t')|0\rangle}{\omega - (E_n^{N+1} - E_0^N) + i\delta} + \sum_m \frac{\langle 0|a_q^\dagger(t')|m^{N-1}\rangle\langle m^{N-1}|a_p(t)|0\rangle}{\omega + (E_m^{N-1} - E_0^N) + i\delta}. \tag{6.237}$$

In coupled-cluster theory, the exact, correlated ground state can be written as $|\Psi_0\rangle = e^T|\Phi\rangle$ and $\langle\Psi_0| = \langle\Phi|(1+\Lambda)e^{-T}$ and defining the similarity-transformed operators

$$\bar{H}_N = e^{-T}H_N e^T, \qquad \bar{a}_p(t) = e^{-T}a_p(t)e^T, \qquad \bar{a}_q^\dagger(t') = e^{-T}a_q^\dagger(t')e^T. \tag{6.238}$$

we can write

$$G^R_{pq}(\omega) = \langle\Phi|(1+\Lambda)\,\bar{a}_q^\dagger(t')\,\frac{1}{\omega + i\delta + \bar{H}_N}\,\bar{a}_p(t)|\Phi\rangle + \langle\Phi|(1+\Lambda)\,\bar{a}_p(t)\,\frac{1}{\omega + i\delta - \bar{H}_N}\,\bar{a}_q^\dagger(t')|\Phi\rangle. \tag{6.239}$$

Now, if we start again with Eqn. 6.236 and decide just to consider the retarded Green's function for one core orbital c, it is just diagonal and reads

$$G^R_c(t) = -i\Theta(t)e^{iE_0t}\left\langle 0\left|a_c e^{-iHt}a_c^\dagger\right|0\right\rangle - i\Theta(t)e^{-iE_0t}\left\langle 0\left|a_c^\dagger e^{iHt}a_c\right|0\right\rangle \tag{6.240}$$

$$= -i\Theta(t)e^{-iE_0t}\left\langle N-1\left|e^{iHt}\right|N-1\right\rangle \tag{6.241}$$

Note that we could only do this because we were able to make the separable approximation to the ground state $|0\rangle \simeq a_c^\dagger|N-1\rangle$, where $|N-1\rangle$ is the exact $N-1$ wavefunction with the core electron separated from it. We are just able to make this approximation for core states because they are localized and thus weakly correlated, so their removal won't relax the exact wavefunction too much. This is not the case for valence states, which are highly correlated, so the separable approximation would not make sense.

$$G^R_{pq}(\omega) = \left\langle \Phi \left| (1+\Lambda)\bar{a^\dagger_q}(\omega + \bar{H}_N + i\delta)^{-1}\bar{a_p} \right| \Phi \right\rangle + \left\langle \Phi \left| (1+\Lambda)\bar{a_p}(\omega - \bar{H}_N + i\delta)^{-1}\bar{a^\dagger_q} \right| \Phi \right\rangle \tag{6.242}$$

$$= \langle\Phi| (1+\Lambda_2) \left(a^\dagger_q + \left(a^\dagger_q T_2\right)_C \right) X_p(\omega)|\Phi\rangle + \langle\Phi| (1+\Lambda_2) \left(a_p + (a_p T_2)_C \right) Y_q(\omega)|\Phi\rangle \tag{6.243}$$

$$= \langle\Phi|a^\dagger_q X_{1,p}(\omega)|\Phi\rangle + \langle\Phi|\Lambda_2 \left(a^\dagger_q T_2\right)_C X_{1,p}(\omega)|\Phi\rangle + \langle\Phi|\Lambda_2 a_p Y_{2,q}(\omega)|\Phi\rangle \tag{6.244}$$

$$= x^q(\omega)_p - \frac{1}{2}\sum_{ijab} \lambda^{ab}_{ij} t^{qj}_{ab} x^i(\omega)_p - \frac{1}{2}\sum_{iab} \lambda^{ab}_{pi} y^i_{ab}(\omega)_q \tag{6.245}$$

where to get eqn. 6.243 we limited the CC expansion to doubles, i.e. $T \approx T_2 = \frac{1}{4}\sum_{ijab} t^{ab}_{ij} a^\dagger_a a^\dagger_b a_j a_i$ and $\Lambda \approx \Lambda_2 = \frac{1}{4}\sum_{ijab} \lambda^{ab}_{ij} a^\dagger_i a^\dagger_j a_b a_a$, and expanded the $\bar{a}_p = a_p + [a_p, T_2]$ and $\bar{a}^\dagger_q = a^\dagger_q + [a^\dagger_q, T_2]$ operators into their connected forms. We also defined the $X_p(\omega) \equiv (\omega + \bar{H}_N + i\delta)^{-1}\bar{a_p}$ and $Y_q(\omega) \equiv (\omega - \bar{H}_N + i\delta)^{-1}\bar{a_q}^\dagger$ operators that have the components

$$X_p(\omega) = \sum_i x^i(\omega)_p a_i + \frac{1}{2!}\sum_{ij,a} x^{ij}_a(\omega)_p a^\dagger_a a_j a_i = X_{1,p}(\omega) + X_{2,p}(\omega) \tag{6.246}$$

$$Y_q(\omega) = \sum_a y_a(\omega)_q a^\dagger_a + \frac{1}{2!}\sum_{i,ab} y^i_{ab}(\omega)_q a^\dagger_a a^\dagger_b a_i = Y_{1,q}(\omega) + Y_{2,q}(\omega) \tag{6.247}$$

This seems similar to second RPA theory, probably because it is. At this point, we can form a perturbation series defined by a perturbation parameter ξ, e.g. $t^{rs}_{pq} = t^{(0)rs}_{pq} + \xi t^{(1)rs}_{pq} + \xi^2 t^{(2)rs}_{pq}$, etc. By keeping only terms up to second order we get:

$$G^{(2)R}_{pq}(\omega) = x^{(2)q}(\omega)_p - \frac{1}{2}\sum_{ijab} \lambda^{(1)ab}_{ij} t^{(1)qj}_{ab} x^{(0)i}(\omega)_p - \frac{1}{2}\sum_{iab} \left(\lambda^{(1)ab}_{pi} y^{(1)i}_{ab}(\omega)_q + \lambda^{(2)ab}_{pi} y^{(0)i}_{ab}(\omega)_q \right) \tag{6.248}$$

$$= x^{(0)p}(\omega)_p \delta_{pq} + x^{(2)q}(\omega)_p - \frac{1}{2}\sum_{iab} \lambda^{(1)ab}_{pi} \left(t^{(1)qi}_{ab} x^{(0)p}(\omega)_p + y^{(1)i}_{ab}(\omega)_q \right) \tag{6.249}$$

$$= \frac{\delta_{pq}}{(\omega - \epsilon_p)} + \frac{1}{(\omega - \epsilon_p)} \left[\frac{1}{2}\sum_{ija} \frac{v^{qa}_{ij} v^{pa}_{ij}}{(\omega + \epsilon_a - \epsilon_i - \epsilon_j)} + \frac{1}{2}\sum_{iab} \frac{v^{pi}_{ab} v^{qi}_{ab}}{(\omega + \epsilon_i - \epsilon_a - \epsilon_b)} \right] \frac{1}{(\omega - \epsilon_q)} \tag{6.250}$$

$$\equiv G^{R(0)}_{pq}(\omega) + G^{R(0)}_{pq}(\omega)\Sigma^{(2)}_{pq}(\omega)G^{R(0)}_{pq}(\omega) \tag{6.251}$$

where we used the simplifications $x^{(1)q}(\omega)_p = 0$ and $x^{(0)q}(\omega)_p = x^{(0)p}(\omega)_p \delta_{pq}$ and in the perturbation analysis, we identified that $G^{(0)R}_{pq}(\omega) = x^{(0)q}(\omega)_p = \frac{1}{(\omega - \epsilon_q)}$.

Real-time EOM-CC Cumulant GF

We restrict the discussion to the retarded core-hole Green's function for a given deep core level $p = c$, $G_c^R = G_{cc}$ given by

$$G_c^R(t) = -i\Theta(t-t') \left\langle 0 \left| \left\{ a_c(t), a_c^\dagger(t') \right\} \right| 0 \right\rangle \tag{6.252}$$

$$= -i\Theta(t) e^{iE_0 t} \left\langle 0 \left| a_c e^{-iHt} a_c^\dagger \right| 0 \right\rangle + -i\Theta(t) e^{-iE_0 t} \left\langle 0 \left| a_c^\dagger e^{iHt} a_c \right| 0 \right\rangle \tag{6.253}$$

$$= -i\Theta(t) e^{iE_0 t} \left\langle N-1 \left| a_c a_c e^{-iHt} a_c^\dagger a_c^\dagger \right| N-1 \right\rangle + -i\Theta(t) e^{-iE_0 t} \left\langle N-1 \left| a_c a_c^\dagger e^{iHt} a_c a_c^\dagger \right| N-1 \right\rangle \tag{6.254}$$

$$= -i\Theta(t) e^{-iE_0 t} \left\langle N-1 \left| e^{iHt} \right| N-1 \right\rangle \tag{6.255}$$

$$= -i\Theta(t) e^{-iE_0 t} \left\langle N-1 | N-1, t \right\rangle \tag{6.256}$$

where we have used the separable approximation to the ground state $|0\rangle \simeq a_c^\dagger |N-1\rangle$, which is reasonable for core excitations. And then at the end we have defined $|N-1,t\rangle = e^{iHt}|N-1\rangle$, which is a solution to $-i\frac{d|N-1,t\rangle}{dt} = H|N-1,t\rangle$. The next step is to assume a time-dependent, CC ansatz for $|N-1,t\rangle = N(t)e^{T(t)}|\phi\rangle$, where $|\phi\rangle = a_c|N\rangle$ is the reference determinant with a core hole, $N(t)$ is a *scalar* normalization factor, and $T(t)$ is the time-dependent cluster operator acting only on the $N-1$ electron Fock space. Inserting this ansatz into the differential equation for $|N-1,t\rangle$ and left multiplying by $e^{-T(t)}$, we obtain the coupled EOM [7]

$$-i\frac{d\left[N(t)e^{T(t)}|\phi\rangle\right]}{dt} = HN(t)e^{T(t)}|\phi\rangle \tag{6.257}$$

$$-ie^{-T(t)}\frac{d\left[N(t)e^{T(t)}\right]}{dt}|\phi\rangle = N(t)\underbrace{e^{-T(t)}He^{T(t)}}_{\bar{H}(t)}|\phi\rangle \tag{6.258}$$

$$-iN(t)\left(\frac{d\ln N(t)}{dt} + \frac{dT(t)}{dt}\right)|\phi\rangle = N(t)\bar{H}(t)|\phi\rangle \tag{6.259}$$

$$-i\left(\frac{d\ln N(t)}{dt} + \frac{dT(t)}{dt}\right)|\phi\rangle = \left(\bar{H}_N(t) + E^{N-1}\right)|\phi\rangle \tag{6.260}$$

where we have defined a normal-order, similarity transformed Hamiltonian as $\bar{H}_N(t) = \bar{H}(t) - E^{N-1}$, where $E^{N-1} = \langle\phi|\bar{H}|\phi\rangle$. Projecting on the reference $\langle\phi|$ and excited de-

[7] In going eqn. 6.258 to eqn. 6.259, we used:

$$\begin{aligned}
e^{-T(t)}\frac{d}{dt}\left(N(t)e^{T(t)}\right) &= e^{-T(t)}\left(\dot{N}(t)e^{T(t)} + N(t)\dot{T}(t)e^{T(t)}\right) \\
&= \dot{N}(t)e^{-T(t)}e^{T(t)} + N(t)\left(\dot{T}(t) + \frac{1}{2!}[\dot{T}(t),T(t)] + \frac{1}{3!}[[\dot{T}(t),T(t)],T(t)] + \ldots\right) \\
&\approx \dot{N}(t) + N(t)\dot{T}(t) \\
&= N(t)\left(\frac{\dot{N}(t)}{N(t)} + \dot{T}(t)\right) \\
&= N(t)\left(\frac{d}{dt}\ln N(t) + \dot{T}(t)\right)
\end{aligned}$$

where we have used the Baker-Campbell-Hausdorff expansion and the truncation is consistent with CCSD.

terminants $\left\langle \phi_{ij}^{ab} \right|$, respectively, we obtain the coupled EOMs for the normalization factor and the cluster amplitudes, respectively:

$$-i\frac{d\ln N(t)}{dt} = \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle + E^{N-1} \tag{6.261}$$

$$\implies N(t) = e^{iE^{N-1}t} e^{i\int_0^t \left\langle \phi \left| \bar{H}_N(t') \right| \phi \right\rangle dt'} \tag{6.262}$$

$$-i\left\langle \phi_{ij\ldots}^{ab\ldots} \left| \frac{dT(t)}{dt} \right| \phi \right\rangle = \langle \phi_{ij\ldots}^{ab\ldots} \left| \bar{H}_N(t) \right| \phi \rangle. \tag{6.263}$$

In order to further evaluate, we need to introduce some additional approximations. First, we assume that the ground state is uncorrelated, i.e. $|N-1\rangle \simeq a_c |\Phi\rangle = |\phi\rangle$, so that

$$\begin{aligned}
\langle N-1| \, N-1,t\rangle =& N(t) \left\langle N-1 \left| e^{iHt} \right| \phi \right\rangle \\
=& N(t) \left\langle \phi \left| \left(1 + iHt + \frac{(iHt)^2}{2!} + \ldots \right) \right| \phi \right\rangle \\
=& N(t) \left(1 + \left\langle \phi \left| R(t) \right| \phi \right\rangle \right) \\
=& N(t).
\end{aligned} \tag{6.264}$$

where $R(t)$ is the excitation operator that collects all excited terms in the series expansion, so it has expectation value $\langle \phi | R(t) | \phi \rangle = 0$. Now, if we insert eqn. 6.262 into eqn. 6.256, we see that the Green's function is given by

$$G_c^R(t) = -i\Theta(t) e^{-iE_0 t} N(t) \tag{6.265}$$

$$= -i\Theta(t) e^{-i(E_0 - E^{N-1})t} e^{i\int_0^t \left\langle \phi \left| \bar{H}_N(t') \right| \phi \right\rangle dt'} \tag{6.266}$$

$$= -i\Theta(t) e^{-i\epsilon_c t} e^{C_c^R(t)} \tag{6.267}$$

In the last line, we take $E_0 \equiv E_{HF}$ as the reference energy of the N electron system and so we have $E_0 - E^{N-1} \simeq \epsilon_c$, as expected from Koopmans' theorem. And importantly, we have identified the cumulant as

$$C_c^R(t) = i\int_0^t \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle dt'. \tag{6.268}$$

Now, the known Landau form for the cumulant is

$$C_L(t) = \int d\omega \frac{\beta(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.269}$$

$$\implies C_L''(t) = -\int d\omega \beta(\omega) e^{-i\omega t} \tag{6.270}$$

$$\tag{6.271}$$

But now if we do the same thing by differentiating eqn. 6.268 twice, we have

$$C_c''(t) = i\frac{d}{dt} \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle \tag{6.272}$$

$$\tag{6.273}$$

So we can identify that

$$\int d\omega \beta(\omega) e^{-i\omega t} = -i\frac{d}{dt} \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle \tag{6.274}$$

Taking the inverse Fourier transform, we find a form for the CC cumulant kernel as

$$\beta(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt e^{i\omega t} \left[-i\frac{d}{dt} \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle \right] \tag{6.275}$$

$$= \frac{1}{\pi} \mathrm{Re} \int_{0}^{\infty} dt e^{-i\omega t} \left[-i\frac{d}{dt} \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle \right] \tag{6.276}$$

The above relation is not derived yet. Recall, that in the GW+C approach, we had $\beta(\omega) = \frac{1}{\pi} \left| \mathrm{Im}\, \Sigma_{pp}(\omega + \epsilon_p) \right|$. *Therefore, we can identify that there is a connection, but it is unclear currently how to explore it.*

So given the differential equation for the logarithm of the normalization factor in eqn. 6.261 and comparing with eqn. 6.267, we see that the cumulant obeys the differential equation

$$\begin{aligned} -i\frac{dC_c^R(t)}{dt} &= \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle \\ &= \sum_{ia} f_{ia} t_i^a + \frac{1}{2} \sum_{ijab} v_{ij}^{ab} t_j^b t_i^a, \end{aligned} \tag{6.277}$$

The above relation is not derived yet. Supposedly it requires some tedious algebra and diagrammatic analysis. We can also use eqn. 6.263 to write the equation of motion for the cluster amplitudes, where for the singles, it is given by

$$-i\dot{t}_i^a(t) \equiv -i \left\langle \phi_i^a \left| \frac{dT(t)}{dt} \right| \phi \right\rangle = \left\langle \phi_i^a \right| \bar{H}_N(t) \left| \phi \right\rangle . \tag{6.278}$$

I choose to not continue further for now.

6.2.8 Annotating Noojien DSO-GF paper

In the supermatrix formulation, MP2 tells us that we have to diagonalize:

$$A_{pq}^{\mathrm{MP2}}(\omega) = \epsilon_p \delta_{pq} + \Sigma_{pq}^{(2)}(\omega) \tag{6.279}$$

$$= \epsilon_p \delta_{pq} + \frac{1}{2} \sum_{l,c,d} \frac{V_{pl[cd]} V_{cd[ql]}}{\omega + \epsilon_l - \epsilon_c - \epsilon_d} - \frac{1}{2} \sum_{k,l,d} \frac{V_{pd[kl]} V_{kl[qd]}}{\epsilon_k + \epsilon_l - \epsilon_d - \omega} \tag{6.280}$$

whereas in their diagonal second-order Green's function (DSO-GF) approach, they diagonalize:

$$A_{ij}^{\mathrm{DSO-GF}}(\omega) = U_{ij} - \frac{1}{2} \sum_{k,l,d} \frac{W_{id[kl]} W_{kl[jd]}}{\epsilon_k + \epsilon_l - \epsilon_d - \omega} \tag{6.281}$$

$$= \epsilon_i \delta_{ij} + \sum_{c,d,l} V_{ilcd} \left(2t_{jl}^{cd} - t_{jl}^{dc} \right) - \frac{1}{2} \sum_{k,l,d} \frac{W_{id[kl]} V_{kl[jd]}}{\epsilon_k + \epsilon_l - \epsilon_d - \omega} \tag{6.282}$$

$$= \epsilon_i \delta_{ij} + \frac{1}{2} \sum_{l,c,d} V_{il[cd]} t_{jl}^{[cd]} - \frac{1}{2} \sum_{k,l,d} \frac{W_{id[kl]} V_{kl[jd]}}{\epsilon_k + \epsilon_l - \epsilon_d - \omega} \tag{6.283}$$

$$= \epsilon_i \delta_{ij} + \frac{1}{2} \sum_{l,c,d} V_{il[cd]} \frac{V_{cd[jl]}}{\epsilon_j + \epsilon_l - \epsilon_c - \epsilon_d} - \frac{1}{2} \sum_{k,l,d} \frac{W_{id[kl]} V_{kl[jd]}}{\epsilon_k + \epsilon_l - \epsilon_d - \omega} \tag{6.284}$$

where the CC doubles amplitudes are replaced by MBPT(2) values:

$$t_{ij}^{ab} = \frac{V_{abij}}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}$$

and V_{abij} denotes a nonantisymmetrized two-electron integral in "1212" notation and the ϵ_p denote canonical Hartree-Fock orbital eigenvalues. The brackets indicate antisymmetrization.

6.3 Cumulant expansion for electron-phonon interactions

6.3.1 Annotating PJ's SC-CE paper

We can define the electron boson Hamiltonian as

$$\hat{H} = \sum_k \epsilon_k a_k^\dagger a_k + \sum_q \omega_q b_q^\dagger b_q + \sum_{qk} g_{qk} a_{k+q}^\dagger a_k \left(b_q + b_{-q}^\dagger \right) \tag{6.285}$$

where a_k and $a_k^\dagger$ are the annihilation and creation operators for an electron of momentum k and energy ϵ_k, and b_q and $b_q^\dagger$ are the annihilation and creation operators for a phonon of momentum q and frequency ω_q. The electron-phonon coupling is given by g_{qk}. We can split the Hamiltonian into a non-interacting part $\hat{H}_0 = \sum_k \epsilon_k a_k^\dagger a_k + \sum_q \omega_q b_q^\dagger b_q$ and an interaction part $V = \sum_{qk} g_{qk} a_{k+q}^\dagger a_k \left(b_q + b_{-q}^\dagger \right)$. In the interaction picture, we have $V_I(t) = e^{i\hat{H}_0 t} V e^{-i\hat{H}_0 t}$.

The time-ordered form of the electron Green's function for one electron can be written as the power series expansion of an exponential ansatz

$$\mathcal{G}_k(t) = -i\theta(t)\left\langle a_k e^{-i\hat{H}t} a_k^\dagger \right\rangle \tag{6.286}$$

$$= -i\theta(t)\left\langle a_k e^{-i(\hat{H}_0+V)t} a_k^\dagger \right\rangle \tag{6.287}$$

$$= -i\theta(t)\left\langle a_k e^{-i\hat{H}_0 t}\mathcal{T} e^{-i\int_0^t d\tau e^{i\hat{H}_0\tau} V e^{-i\hat{H}_0\tau}} a_k^\dagger \right\rangle \tag{6.288}$$

$$= \mathcal{G}_k^{(0)}(t)\left\langle a_k T e^{-i\int_0^t d\tau V_I(\tau)} a_k^\dagger \right\rangle \tag{6.289}$$

$$= \mathcal{G}_k^{(0)}(t)\left\langle a_k \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_0^t dt_1 \cdots \int_0^t dt_n T\left[V_I(t_1)\cdots V_I(t_n)\right] a_k^\dagger \right\rangle \tag{6.290}$$

$$= \mathcal{G}_k^{(0)}(t)\left\langle a_k \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_0^t dt_1 \cdots \int_0^t dt_n T\left[\prod_{j=1}^{n}\sum_{q_j} g_{q_j,k_j} a^\dagger_{k_j+q_j}(t_j)\, a_{k_j}(t_j)\, A_{q_j}(t_j)\right] a_k^\dagger \right\rangle \tag{6.291}$$

$$= \mathcal{G}_k^{(0)}(t)\left\langle a_k \sum_{n=0}^{\infty} T\left[\frac{(-i)^n}{n!} \int_0^t dt_1 \cdots \int_0^t dt_n \prod_{j=1}^{n}\sum_{q_j} g_{q_j,k_j} a^\dagger_{k_j+q_j} a_{k_j} e^{-it_j(\epsilon_{k_j}-\epsilon_{k_j+q_j})} A_{q_j}(t_j)\right] a_k^\dagger \right\rangle \tag{6.292}$$

$$= \mathcal{G}_k^{(0)}(t)\left\langle \sum_{n=0}^{\infty} T\left[\frac{(-i)^n}{n!} \int_0^t dt_1 \cdots \int_0^t dt_n \prod_{j=1}^{n}\sum_{q_j} g_{q_j,k_j} e^{-it_j(\epsilon_{k_j}-\epsilon_{k_j+q_j})}\left(b_{q_j} e^{-i\omega_{q_j}t} + b^\dagger_{-q_j} e^{i\omega_{q_j}t}\right)\right]\right\rangle \tag{6.293}$$

where $k_j = k + \sum_{i=j+1}^{n} q_i$, $\sum_i q_i = 0$, and $[b_q, b^\dagger_{q'}] = \delta_{qq'}$. The interpretation of the above expression is that the electron starts in state k at time 0, and then scatters off n phonons, each with momentum q_j, before returning to state k at time t. Also, $A_q(t) = b_q e^{-i\omega_q t} + b^\dagger_{-q} e^{i\omega_q t}$, where b_q and $b_q^\dagger$ are the annihilation and creation operators for a phonon of momentum q and frequency ω_q. If we define the vertex operator $\Gamma_{qk}(t) = g_{qk} e^{-it\epsilon_k} e^{q\cdot\frac{d}{dk}} e^{it\epsilon_k}$ we can rewrite the electron Green's function again as an exponential,

$$\mathcal{G}_k(t) = \mathcal{G}_k^{(0)}(t)\left\langle T e^{-i\sum_q \int_0^t d\tau \Gamma_{qk}(\tau) A_q(\tau)} \right\rangle \tag{6.294}$$

where the remaining trace is only over the bosonic degrees of freedom. Assuming harmonic bosons, we can use $\left\langle e^B \right\rangle = e^{\left\langle B^2 \right\rangle/2}$ with the definition

$$B = -i\sum_q \int_0^t d\tau \Gamma_{qk}(\tau) A_q(\tau) \tag{6.295}$$

where our B is linear in the bosonic operators b_q, so the identity applies. Then,

$$\left\langle Te^{B}\right\rangle \equiv \left\langle e^{T[B]}\right\rangle = \exp\left(\frac{1}{2}\left\langle TB^{2}\right\rangle\right) \tag{6.296}$$

$$\implies S = \frac{1}{2}\left\langle TB^{2}\right\rangle \tag{6.297}$$

$$= \frac{1}{2}(-i)^{2}\sum_{qq'}\int_{0}^{t}d\tau\int_{0}^{t}d\tau'\Gamma_{qk}(\tau)\Gamma_{q'k}\left(\tau'\right)\left\langle TA_{q}(\tau)A_{q'}\left(\tau'\right)\right\rangle \tag{6.298}$$

$$= -\frac{1}{2}\sum_{qq'}\int_{0}^{t}d\tau\int_{0}^{t}d\tau'\Gamma_{qk}(\tau)\Gamma_{q'k}\left(\tau'\right)\left[iD_{q}^{(0)}\left(\tau-\tau'\right)\delta_{q,-q'}\right] \tag{6.299}$$

$$= -i\sum_{q}\int_{0}^{t}d\tau\int_{0}^{t}d\tau' D_{q}^{(0)}\left(\tau-\tau'\right)\Gamma_{-qk}(\tau)\Gamma_{qk}\left(\tau'\right) \tag{6.300}$$

where the non-interacting bosonic Green's function is $D_{q}^{(0)}\left(\tau-\tau'\right) = -i\left[(1+N_{q})e^{-i\omega_{q}|\tau-\tau'|}+N_{q}e^{i\omega_{q}|\tau-\tau'|}\right]$ with N_q the Bose-Einstein distribution $N_{q}=\frac{1}{e^{\beta\omega_{q}}-1}$. To find a self consistent expression for S, we use the Feynman operator ordering theorem, which states that given a functional of time-dependent operators $F[\hat{A}(\tau),\hat{B}(\tau),\ldots]$ with $\tau\in[0,t]$, which in our case is $e^{S[\Gamma(\tau)]}$, and with a unitary operator of the form $\hat{U}\left(\tau'\right)=T\exp\left[\int_{0}^{\tau'}d\tau\hat{P}(\tau)\right]$, then

$$\hat{U}(t)F[\hat{A}(\tau),\hat{B}(\tau),\ldots] = T\left[\exp\left(\int_{0}^{t}d\tau\hat{P}(\tau)\right)F\left[\hat{U}(\tau)\hat{A}(\tau)\hat{U}^{-1}(\tau),\ldots\right]\right]. \tag{6.301}$$

We choose $\hat{P}(\tau)=-i\left[\epsilon_{k}+\frac{d}{d\tau}\phi_{k}(\tau)\right]$ and so the transformed S is given by

$$\bar{S} = -i\int_{0}^{t}d\tau\left(\epsilon_{k}+\frac{d}{d\tau}\phi_{k}(\tau)\right)-i\sum_{q}\int_{0}^{t}d\tau\int_{0}^{\tau}d\tau' D_{q}^{(0)}\left(\tau-\tau'\right)\bar{\Gamma}_{-qk}(\tau)\bar{\Gamma}_{qk}\left(\tau'\right) \tag{6.302}$$

with the transformed vertex operators $\bar{\Gamma}_{qk}(\tau)\equiv\hat{U}(\tau)\Gamma_{qk}(\tau)\hat{U}^{-1}(\tau)=g_{qk}e^{-i\phi_{k}(\tau)}e^{q\cdot\frac{d}{dk}}e^{+i\phi_{k}(\tau)}$. The Green's function can then be written as

$$\mathcal{G}_{k}(t) = i\Theta(t)e^{i\phi_{k}(t)}T\left[\exp\left(-i\int_{0}^{t}d\tau(\epsilon_{k}+\frac{d}{d\tau}\phi_{k}(\tau))\right)\right. \tag{6.303}$$

$$\left.\times\exp\left(-i\sum_{q}\int_{0}^{t}d\tau\int_{0}^{\tau}d\tau' D_{q}^{(0)}\left(\tau-\tau'\right)\bar{\Gamma}_{-qk}(\tau)\bar{\Gamma}_{qk}\left(\tau'\right)\right)\right]$$

$$= i\Theta(t)e^{i\phi_{k}(t)}\exp\left(\sum_{n=1}^{\infty}\frac{1}{n!}T\left[\bar{S}^{n}\right]_{c}\right) \tag{6.304}$$

$$= i\Theta(t)e^{C_{k}(t)} \tag{6.305}$$

where

$$C_{k}(t) = i\phi_{k}(t)+\sum_{n=1}^{\infty}\frac{1}{n!}T\left[\bar{S}^{n}\right]_{c} \tag{6.306}$$

where the notation $[\cdots]_c$ denotes the cumulant of an operator, and $T[\bar{S}]_c = T[\bar{S}]$ with subsequent terms given by $T\left[\bar{S}^n\right]_c = T\left[\bar{S}^n\right] - \sum_{m=1}^{n-1} \frac{(n-1)!}{m!(n-m-1)!} T\left[\bar{S}^m\right] T\left[\bar{S}^{n-m}\right]_c$. Then, evaluating 6.381 with $\phi_k(t) = C_k(t)$ to leading order produces

$$C_k(t) = -i\epsilon_k t - i\frac{g^2}{N}\sum_q \int_0^t d\tau \int_0^\tau d\tau' D_q^{(0)}\left(\tau - \tau'\right) e^{-C_k(\tau)+C_{k-q}(\tau)-C_{k-q}(\tau')+C_k(\tau')} \tag{6.307}$$

$$\implies \dot{C}_k(t) = -i\epsilon_k - i\frac{g^2}{N}\sum_q \int_0^t d\tau D_q^{(0)}(t-\tau) e^{-C_k(t)+C_{k-q}(t)-C_{k-q}(\tau)+C_k(\tau)} \tag{6.308}$$

with the bare phonon propagator $D_q^{(0)}(t) = -i\left[\left(1+N_q\right)e^{-i\omega_q|l|} + N_q e^{|\omega_q|l|}\right]$ and the Bose occupation factor $N_q = \left[\exp\left(\beta\omega_q\right) - 1\right]^{-1}$. Eqn 6.307 can be recast as a self-consistent

equation for the interacting Green's function by considering

$$\frac{d\mathcal{G}_k(t)}{dt} = \frac{d}{dt}\left[-i\Theta(t)e^{C_k(t)}\right] \tag{6.309}$$

$$= \dot{C}_k(t)\mathcal{G}_k(t) \tag{6.310}$$

$$= -i\epsilon_k\mathcal{G}_k(t) - i\frac{g^2}{N}\sum_q\int_0^t d\tau\mathcal{G}_k(t)D_q^{(0)}(t-\tau)e^{-C_k(t)+C_{k-q}(t)-C_{k-q}(\tau)+C_k(\tau)} \tag{6.311}$$

$$= -i\epsilon_k\mathcal{G}_k(t) - i\frac{g^2}{N}\sum_q\int_0^t d\tau\mathcal{G}_k(t)D_q^{(0)}(t-\tau)\frac{\mathcal{G}_{k-q}(t)\mathcal{G}_k(\tau)}{\mathcal{G}_{k-q}(\tau)\mathcal{G}_k(t)} \tag{6.312}$$

$$= -i\epsilon_k\mathcal{G}_k(t) - i\frac{g^2}{N}\sum_q\int_0^t d\tau D_q^{(0)}(t-\tau)\frac{\mathcal{G}_k(\tau)\mathcal{G}_{k-q}(t)}{\mathcal{G}_{k-q}(\tau)} \tag{6.313}$$

$$\implies \frac{1}{\mathcal{G}_k(t)}\frac{d\mathcal{G}_k(t)}{dt} = -i\epsilon_k - i\frac{g^2}{N}\sum_q\int_0^t d\tau D_q^{(0)}(t-\tau)\frac{\mathcal{G}_{k-q}(t)}{\mathcal{G}_{k-q}(\tau)}\frac{\mathcal{G}_k(\tau)}{\mathcal{G}_k(t)} \tag{6.314}$$

$$\implies \frac{d\ln\mathcal{G}_k(t)}{dt} - \frac{d\ln\mathcal{G}_k^{(0)}(t)}{dt} = -i\frac{g^2}{N}\sum_q\int_0^t d\tau D_q^{(0)}(t-\tau)\frac{\mathcal{G}_{k-q}(t)}{\mathcal{G}_{k-q}(\tau)}\frac{\mathcal{G}_k(\tau)}{\mathcal{G}_k(t)} \tag{6.315}$$

$$\implies \frac{d\ln\left[\mathcal{G}_k(t)/\mathcal{G}_k^{(0)}(t)\right]}{dt} = -i\frac{g^2}{N}\sum_q\int_0^t d\tau D_q^{(0)}(t-\tau)\frac{\mathcal{G}_{k-q}(t)}{\mathcal{G}_{k-q}(\tau)}\frac{\mathcal{G}_k(\tau)}{\mathcal{G}_k(t)} \tag{6.316}$$

$$\implies ln\left[\mathcal{G}_k(t)/\mathcal{G}_k^{(0)}(t)\right] = -i\frac{g^2}{N}\sum_q\int_0^t d\sigma\int_0^\sigma d\tau D_q^{(0)}(\sigma-\tau)\frac{\mathcal{G}_{k-q}(\sigma)}{\mathcal{G}_{k-q}(\tau)}\frac{\mathcal{G}_k(\tau)}{\mathcal{G}_k(\sigma)} \tag{6.317}$$

$$\implies \mathcal{G}_k(t) = \mathcal{G}_k^{(0)}(t)\exp\left[i\sum_q\int_0^t d\sigma\int_0^\sigma d\tau\left|g_{qk}\right|^2 D_q^{(0)}(\sigma-\tau)\frac{\mathcal{G}_{k-q}(\sigma)}{\mathcal{G}_{k-q}(\tau)}\frac{\mathcal{G}_k(\tau)}{\mathcal{G}_k(\sigma)}\right] \tag{6.318}$$

$$\implies \mathcal{G}_k^{(n+1)}(t) = \mathcal{G}_k^{(0)}(t)\exp\left[i\sum_q\int_0^t d\sigma\int_0^\sigma d\tau\ \left|g_{qk}\right|^2 D_q^{(0)}(\sigma-\tau)\frac{\mathcal{G}_{k-q}^{(n)}(\sigma)}{\mathcal{G}_{k-q}^{(n)}(\tau)}\frac{\mathcal{G}_k^{(n)}(\tau)}{\mathcal{G}_k^{(n)}(\sigma)}\right] \tag{6.319}$$

where the last equation admits a form for an iterative solution of this self-consistent equation. For $n = 0$, we insert the bare Green's function, and get the second order cumulant,

$$\mathcal{G}_k^{(1)}(t) = \mathcal{G}_k^{(0)}(t)\exp\left[i\sum_q\int_0^t d\sigma\int_0^\sigma d\tau\left|g_{qk}\right|^2 D_q^{(0)}(\sigma-\tau)\frac{\mathcal{G}_{k-q}^{(0)}(\sigma)}{\mathcal{G}_{k-q}^{(0)}(\tau)}\frac{\mathcal{G}_k^{(0)}(\tau)}{\mathcal{G}_k^{(0)}(\sigma)}\right] \tag{6.320}$$

$$= \mathcal{G}_k^{(0)}(t)\exp\left[i\sum_q\int_0^t d\sigma\int_0^\sigma d\tau\left|g_{qk}\right|^2 D_q^{(0)}(\sigma-\tau)e^{-i\left(\epsilon_{k-q}-\epsilon_k\right)(\sigma-\tau)}\right] \tag{6.321}$$

$$\equiv \mathcal{G}_k^{(0)}(t)\exp\left[C_2(k,t)\right] \tag{6.322}$$

For $n = 1$, we have

$$\mathcal{G}_k^{(2)}(t) = \mathcal{G}_k^{(0)}(t) \exp\left[i \sum_q \int_0^t d\sigma \int_0^\sigma d\tau \, |g_{qk}|^2 \, D_q^{(0)}(\sigma - \tau) \frac{\mathcal{G}_{k-q}^{(1)}(\sigma)}{\mathcal{G}_{k-q}^{(1)}(\tau)} \frac{\mathcal{G}_k^{(1)}(\tau)}{\mathcal{G}_k^{(1)}(\sigma)} \right] \tag{6.323}$$

$$= \mathcal{G}_k^{(0)}(t) \exp\left[i \sum_q \int_0^t d\sigma \int_0^\sigma d\tau \, |g_{qk}|^2 \, D_q^{(0)}(\sigma - \tau) \frac{\mathcal{G}_{k-q}^{(0)}(\sigma)}{\mathcal{G}_{k-q}^{(0)}(\tau)} \frac{\mathcal{G}_k^{(0)}(\tau)}{\mathcal{G}_k^{(0)}(\sigma)} \right.$$

$$\left. \times e^{C_2(k,\tau) + C_2(k-q,\sigma) - C_2(k,\sigma) - C_2(k-q,\tau)} \right] \tag{6.324}$$

$$= \mathcal{G}_k^{(0)}(t) \exp\left[\sum_{n=0}^\infty \frac{-i}{n!} \sum_q \int_0^t d\sigma \int_0^\sigma d\tau \, |g_{qk}|^2 \, D_q^{(0)}(\sigma - \tau) \mathcal{G}_{k-q}^{(0)}(\sigma - \tau) \mathcal{G}_k^{(0)}(\tau - \sigma) \right. \tag{6.325}$$

$$\left. \times \left[C_2(k,\tau) + C_2(k-q,\sigma) - C_2(k,\sigma) - C_2(k-q,\tau) \right]^n \right]$$

$$= \mathcal{G}_k^{(0)}(t) \exp\left(\sum_{m=1}^\infty C_{2n}^{(2)}(k,t) \right) \tag{6.326}$$

where $C_{2n}^{(2)}(k,t)$ is the approximation to the $2n$th cumulant produced by the second iteration. Because the second iteration produces cumulants of all orders, further iterations require expanding the exponent into moments. For iterations $n \geq 2$, we use

$$\mathcal{G}_{\mathbf{k}}^{(n+1)}(t) = \mathcal{G}_{\mathbf{k}}^{(0)}(t) \exp\left(-i \sum_{\mathbf{q}} \int_0^t d\sigma \int_0^\sigma d\tau \, |g_{qk}|^2 \right. \tag{6.327}$$

$$\left. \times D_{\mathbf{q}}^0(\sigma - \tau) \mathcal{G}_{\mathbf{k}}^{(0)}(\tau - \sigma) \mathcal{G}_{\mathbf{k}-\mathbf{q}}^{(0)}(\sigma - \tau) e^{F^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau)} \right)$$

introducing the notation

$$F^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau) = F_2^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau) + F_4^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau) + \cdots \tag{6.328}$$

where

$$F_i^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau) = C_i^{(n)}(\mathbf{k}-\mathbf{q},\sigma) - C_i^{(n)}(\mathbf{k}-\mathbf{q},\tau) - C_i^{(n)}(\mathbf{k},\sigma) + C_i^{(n)}(\mathbf{k},\tau) \tag{6.329}$$

The procedure for constructing the approximation to the nth iteration of the mth cumulant is given by the moment expansion

$$\sum_{m=0}^\infty \lambda^{2m} W_{2m}^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau) = \exp\left(\sum_{m=1}^\infty \lambda^{2m} F_{2m}^{(n)}(\mathbf{k},\mathbf{q},\sigma,\tau) \right) \tag{6.330}$$

where we can get explicit expressions by matching the orders of λ on both sides. Our experience with equation (6.325) suggests that each iteration introduces another power of g^2 to all of the approximated cumulants, and since the nth-order cumulant is defined as being proportional to g^n, the nth iteration cannot alter any of the approximated cumulants of order lower than n. Because of this, and the fact that the approximate cumulant for a

given order is an integral over lower-order approximations, the iterative method is actually recursive. So alternatively, we can write

$$C_{2n}^{\mathrm{SC-CE}}(\mathbf{k},t) = -i\sum_{\mathbf{q}}\int_0^t d\sigma\int_0^\sigma d\tau\,|g_{qk}|^2\,D_{\mathbf{q}}^{(0)}(\sigma-\tau)\mathcal{G}_{\mathbf{k}}^{(0)}(\tau-\sigma)\mathcal{G}_{\mathbf{k}-\mathbf{q}}^{(0)}(\sigma-\tau)W_{2n-2}(\mathbf{k},\mathbf{q},\sigma,\tau) \tag{6.331}$$

To determine what is actually solved, at this point consider again equation (6.313), where, if we define $y_k(t) = e^{C_k(t)}$, we can rewrite it as

$$\frac{dy_k(t)}{dt} = -\,i\sum_q\int_0^t d\tau\,|g_{qk}|^2\,D_q^{(0)}(t-\tau)e^{i(\epsilon_k-\epsilon_{k-q})(t-\tau)}\frac{y_k(\tau)y_{k-q}(t)}{y_{k-q}(\tau)} \tag{6.332}$$

Numerical solutions to equation (6.332) are carried out with a multi-step predictor corrector method as derived by Linz. His method is general for first order Volterra Integro-Differential Equations (VIDEs) of the form

$$y'(x) = F\left(x, y(x), \int_0^x dtK(x,t,y(t))\right). \tag{6.333}$$

Equation (6.332) implies that F can be written purely as a function of $\int_0^x dtK(x,t,y(t))$. An Adams-Bashforth second-order predictor step and a third-order Adams-Moulton corrector step are used. Following Linz, for the starting procedure of the Holstein model we applied the self-consistent Simpson's method iterated five times. *Do we do the same thing for the starting procedure of the finite-basis UEG calculations?*

6.3.2 Annotation of Littlewood's CE paper

Power series ansatze

They use a power series expansion of the cumulant, instead of the usual exponential ansatz. This has the function of making sure that the term involving the g^{2n} EPI constant can only be found in the nth-order cumulant. So the retarded Green's function is given by

$$G(n,t) = G_o(n,t)\mathcal{P}(n,t) = G_o(n,t)\sum_{k=0}^{\infty} g^{2k}C_k(n,t) \tag{6.334}$$

To properly construct the electron self energy, they replace it by power series ansatz in order to re-introduce self-consistency from the start.

$$\begin{aligned} -i\Sigma(t) &= g^2\sum_{n=\pm}\sum_{N=\pm}\mathcal{D}(N,t)\underbrace{G_0(n,t)\mathcal{P}(n,t)}_{G(n,t)} \\ &= g^2\sum_{n=\pm} -i\Sigma_0(n,t)\mathcal{P}(n,t) \end{aligned} \tag{6.335}$$

where $\Sigma_0(n,t) \equiv -i\mathcal{D}(N,t)G_0(n,t)$.

Plugging into the Dyson equation

Now, we can plug into the temporal Dyson equation:

$$G(m,t-t_0) = G_0(m,t-t_0) + \iint dt_1 dt_2\, G_0(m,t-t_2)\,\Sigma(t_2-t_1)\,G(m,t_1-t_0) \tag{6.336}$$

$$G_0(m,t-t_0)\mathcal{P}(m,t-t_0) = G_0(m,t-t_0) - ig^2 \sum_{n=\pm} \iint dt_1 dt_2\, G_0(m,t-t_2)\,\Sigma_0(n,t_2-t_1) \tag{6.337}$$

$$\times\, \mathcal{P}(n,t_2-t_1)\,G_0(m,t_1-t_0)\,\mathcal{P}(m,t_1-t_0)$$

$$G_0(m,t)\mathcal{P}(m,t) = G_0(m,t) - ig^2 \sum_{n=\pm} \iint dt_1 dt_2\, G_0\,(m,t)\, e^{i\varepsilon_m(t_2-t_1)}\,\Sigma_0(n,t_2-t_1) \tag{6.338}$$

$$\times\, \mathcal{P}(n,t_2-t_1)\mathcal{P}(m,t_1-t_0)$$

$$\mathcal{P}(m,t) = 1 - ig^2 \sum_{n=\pm} \int_0^t dt_2 \int_0^{t_2} d\tau e^{i\varepsilon_m\tau}\,\Sigma_0(n,\tau)\mathcal{P}(n,\tau)\mathcal{P}(m,t_2-\tau) \tag{6.339}$$

where in the final line we have defined $\tau = t_2 - t_1$ and set $t_0 = 0$. Also, the temporal contraction property due to the boundary value dependence on time is used, which is only valid when the diagonal approximation is made. There are two distinct terms in this equation: a self correction term P_{SC} that occurs when $n = m$ and an inter-band scattering term P_{IC} that occurs when $n \neq m$. The contraction property is valid for the former but not the latter. So

$$\mathcal{P}(m,t) = 1 + P_{SC} + P_{IC} \tag{6.340}$$

with

$$P_{SC} = -ig^2 \int_0^t dt_2 \int_0^{t_2} d\tau\, e^{i\varepsilon_m\tau}\Sigma_0(m,\tau)\mathcal{P}(m,t_2) \tag{6.341}$$

$$P_{IC} = -ig^2 \int_0^t dt_2 \int_0^{t_2} d\tau\, e^{i\varepsilon_m\tau}\,\Sigma_0(n,\tau)\mathcal{P}(n,\tau)\mathcal{P}(m,t_2-\tau) \tag{6.342}$$

For numerical solution, they start with an initial guess of $\mathcal{P} = 1$ on the right and self consistently compute better values for $\mathcal{P}$ on the left until it converges. It also might be useful to know that we can write

$$C_k(m,t) = -i \int_0^t dt_2 \int_0^{t_2} d\tau \left[e^{i\varepsilon_m\tau}\,\Sigma_o(m,\tau)C_{k-1}(m,t_2) + \sum_{n\neq m}\sum_{l=0}^{k} e^{i\varepsilon_m\tau}\,\Sigma_o(n,\tau)\,C_l(n,\tau)\,C_{k-1-l}(m,t_2-\tau) \right] \tag{6.343}$$

6.3.3 Annotation of Dunn paper

Exact identities

The given Hamiltonian is

$$H = \sum_k \epsilon_k a_k^\dagger a_k + \frac{1}{2}\sum_q (P_q^\dagger P_q + \omega_q^2 Q_q^\dagger Q_q) + \sum_q \gamma_q Q_q \rho_q^\dagger \tag{6.344}$$

$$= H_{\text{el}} + H_{\text{ph}} + H_{\text{int}} \tag{6.345}$$

$$= \sum_k \epsilon_k a_k^\dagger a_k + \sum_q \omega_q \left(b_q^\dagger b_q + \frac{1}{2}\right) + \sum_{qk} \gamma_q \sqrt{\frac{1}{2\omega_q}} (b_q + b_{-q}^\dagger) a_k^\dagger a_{k+q} \langle k|e^{-iq\cdot r}|k+q\rangle \tag{6.346}$$

$$= \sum_k \epsilon_k a_k^\dagger a_k + \sum_q \omega_q b_q^\dagger b_q + \sum_{qk} g_{qk}(b_q + b_{-q}^\dagger) a_k^\dagger a_{k+q} \tag{6.347}$$

where we have introduced bosonic second quantization operators $Q_q = \sqrt{\frac{1}{2\Omega_q}}\left(b_q + b_{-q}^\dagger\right)$ and $P_q = i\sqrt{\frac{\Omega_q}{2}}\left(b_{-q}^\dagger - b_q\right)$, which correspond to the position and momentum, respectively, of a harmonic oscillator with frequency Ω_q. The electron density operator is given by $\rho_q = \sum_k a_{k+q}^\dagger a_k \langle k+q|e^{iq\cdot r}|k\rangle$, where the matrix element is between Bloch states. The electron-boson coupling constant is then given by $g_{qk} = \gamma_q \sqrt{\frac{1}{2\Omega_q}} \langle k|e^{-iq\cdot r}|k+q\rangle$. *Garnet started with such an electron boson Hamiltonian to start his AB theory, so maybe we want to do something similar.* Without any interaction, the electron annihilation operator is

$$a_k(t) = e^{-i\epsilon_k t} a_k \tag{6.348}$$

In the presence of this interaction, the EOM for the electron annihilation operator is given by

$$\frac{da_k(t)}{dt} = -i\epsilon_k a_k(t) - i\sum_q g_{qk} Q_q(t) a_{k+q}(t) \tag{6.349}$$

$$\implies a_k(t) = e^{-i\epsilon_k t} T \exp\left[-i\int_0^t d\tau \sum_q Q_q(\tau)\Gamma_{qk}(\tau)\right] a_k \tag{6.350}$$

$$\tag{6.351}$$

So the idea is that as time evolves when we are including the interaction, the electronic and bosonic processes become inseparable.

The above is derived. Start by introducing a rotated operator.

$$\tilde{a}_k(t) = e^{i\epsilon_k t} a_k(t) \tag{6.352}$$

$$\implies \frac{d\tilde{a}_k(t)}{dt} = i\epsilon_k e^{i\epsilon_k t} a_k(t) + e^{i\epsilon_k t} \frac{da_k(t)}{dt} \tag{6.353}$$

$$= i\epsilon_k e^{i\epsilon_k t} a_k(t) + e^{i\epsilon_k t} \left(-i\epsilon_k a_k(t) - i \sum_q g_{qk} Q_q(t) a_{k+q}(t) \right) \tag{6.354}$$

$$= -i \sum_q g_{qk} e^{i\epsilon_k t} Q_q(t) a_{k+q}(t) \tag{6.355}$$

$$= -i \sum_q g_{qk} Q_q(t) e^{i\left(\epsilon_k - \epsilon_{k+q}\right)t} \tilde{a}_{k+q}(t) \tag{6.356}$$

$$= -i \sum_q g_{qk} Q_q(t) e^{i\left(\epsilon_k - \epsilon_{k+q}\right)t} e^{q \cdot \frac{d}{dk}} \tilde{a}_k(t) \tag{6.357}$$

$$= -i \sum_q Q_q(t) \Gamma_{qk}(t) \tilde{a}_k(t) \tag{6.358}$$

$$\implies \tilde{a}_k(t) = T \exp\left[-i \int_0^t d\tau \sum_q Q_q(\tau) \Gamma_{qk}(\tau) \right] \tilde{a}_k \tag{6.359}$$

$$\implies a_k(t) = e^{-i\epsilon_k t} T \exp\left[-i \int_0^t d\tau \sum_q Q_q(\tau) \Gamma_{qk}(\tau) \right] a_k \tag{6.360}$$

where we defined the vertex operator $\Gamma_{qk}(t) = g_{qk} e^{i\epsilon_k t} e^{q \cdot \frac{d}{dk}} e^{-i\epsilon_k t} = g_{qk} e^{i\left(\epsilon_k - \epsilon_{k+q}\right)t} e^{q \cdot \frac{d}{dk}}$, which uses $f(k+q) = \sum_{n=0}^{\infty} \frac{(q \cdot \nabla_k)^n}{n!} f(k) = e^{q \cdot \nabla_k} f(k)$ so thatt $e^{q \cdot \frac{d}{dk}}$ can be understood as a translation operator in k-space.

The causal Green's function for a insulator, where the probability of an electron conduc-

tion state being occupied is negligible, is

$$G_k(t) = i\,\mathrm{Tr}\left[\rho T\left(a_k(t)a_k^\dagger\right)\right] \tag{6.361}$$

$$= i\Theta(t)\,\mathrm{Tr}\left[\rho a_k(t)a_k^\dagger\right] + i\Theta(-t)\,\mathrm{Tr}\left[\rho a_k^\dagger a_k(t)\right] \tag{6.362}$$

$$= i\Theta(t)\,\mathrm{Tr}\left[\rho a_k(t)a_k^\dagger\right] \tag{6.363}$$

$$= i\Theta(t)e^{-i\epsilon_k t}\,\mathrm{Tr}\left[\rho T\exp\left(-i\sum_q\int_0^t d\tau Q_q(\tau)\Gamma_{qk}(\tau)\right)\right]_0 \quad \text{trace over zero electron states} \tag{6.364}$$

$$= i\Theta(t)e^{-i\epsilon_k t}\,\mathrm{Tr}\left[\rho T\left(\exp\left(-i\sum_q\int_0^t d\tau\frac{b_q(\tau)+b_{-q}^\dagger(\tau)}{\sqrt{2\Omega_q}}\Gamma_{qk}(\tau)\right)\right)\right]_0 \tag{6.365}$$

$$= i\Theta(t)e^{-i\epsilon_k t}\,\mathrm{Tr}\left[\rho T\left(\exp\left(\underbrace{-i\sum_q\int_0^t d\tau\frac{b_q(\tau)\Gamma_{qk}(\tau)}{\sqrt{2\Omega_q}}}_{A}\right)\right.\right. \tag{6.366}$$

$$\left.\left.\times\exp\left(\underbrace{-i\sum_q\int_0^t d\tau\frac{b_{-q}^\dagger(\tau)\Gamma_{-qk}(\tau)}{\sqrt{2\Omega_q}}}_{B}\right)\right)\right]_0 \tag{6.367}$$

$$= i\Theta(t)e^{-i\epsilon_k t}\left\langle 1 + T(A+B) + \frac{1}{2}T(A^2+2AB+B^2) + \cdots\right\rangle_\rho \quad \text{with} \quad \rho\propto e^{-\beta H_{\mathrm{ph}}} \tag{6.368}$$

$$= i\Theta(t)e^{-i\epsilon_k t}\left\langle 1 + TAB + \ldots\right\rangle_\rho \quad \text{(vacuum buubles and disconnected terms vanish)} \tag{6.369}$$

$$= i\Theta(t)e^{-i\epsilon_k t}T\exp\left[i\sum_q\int_0^t d\tau\int_0^\tau d\tau' D_q(\tau-\tau')\Gamma_{qk}(\tau)\Gamma_{-qk}(\tau')\right] \tag{6.370}$$

where the free phonon Green function is

$$D_q(\tau-\tau') = \frac{i}{2\Omega_q}\left((1+N(\Omega_q))e^{-i\Omega_q|\tau-\tau'|} + N(\Omega_q)e^{+i\Omega_q|\tau-\tau'|}\right) \tag{6.371}$$

and $N(\Omega_q)$ is the phonon occupation number

$$N(\Omega_q) = \frac{1}{\exp\left(\frac{\Omega_q}{kT}\right)-1} \tag{6.372}$$

Approximations

The exact result for the Green function is

$$G_k(t) = i\Theta(t)e^{-i\epsilon_k t}T\exp\left[S\right] \tag{6.373}$$

$$\equiv i\Theta(t)\exp\left(iA_k(t)\right) \tag{6.374}$$

$$= i\Theta(t)\exp\left(-i\epsilon_k t - i\sum_{n=1}^{\infty}\frac{1}{n!}T\left[S^n\right]_c\right) \tag{6.375}$$

where we found $S = i\sum_q \int_0^t d\tau \int_0^\tau d\tau' D_q(\tau-\tau')\Gamma_{qk}(\tau)\Gamma_{-qk}(\tau')$, so the action function

$$A_k(t) = -\epsilon_k t - i\log[T\exp(S)] \tag{6.376}$$

$$= -\epsilon_k t - i\sum_{n=1}^{\infty}\frac{1}{n!}T\left[S^n\right]_c \tag{6.377}$$

with $T\left[S^n\right]_c = T\left[S^n\right] - \sum_{m=1}^{n-1}\frac{(n-1)!}{m!(n-m-1)!}T\left[S^m\right]T\left[S^{n-m}\right]_c$, where $T[S]_c = T[S]$. We showed earlier that only the connected terms survive, i.e. this is why we have $[TS^n]_c$; this final form is amenable to approximation. So to Nth order, we can write

$$G_k(t) = i\Theta(t)e^{-i\epsilon_k t}\prod_{n=1}^{N}e^{T[S^n]_c} \tag{6.378}$$

A telling comparison with 6.373 shows that we managed to replace the time-ordered exponential by a product of exponentials, where the time-ordering is done in the arguments and only involves connected terms. [8]

[8]Note that in PJ's paper, he pursued the fully self consistent approach where $A_k(t) = -iC_k(t)$ and

$$C_k(t) = i\phi_k(t) + \sum_{n=1}^{\infty}\frac{1}{n!}T\left[\bar{S}^n\right]_c \tag{6.379}$$

$$= -i\epsilon_k t - i\frac{g^2}{N}\sum_q\int_0^t d\tau\int_0^\tau d\tau' D_q^{(0)}\left(\tau-\tau'\right)e^{-C_k(\tau)+C_{k-q}(\tau)-C_{k-q}(\tau')+C_k(\tau')} + O(g^4) \tag{6.380}$$

$$\tag{6.381}$$

The diagrammatic explanation of what this cluster expansion does is that unlike usual perturbation theory, which sums a finite subset of diagrams exactly and does not treat the rest, this cluster expansion treats an infinite subset of diagrams approximately. And illustrative example of this phenomenon is that in this procedure, we are effectively substituting the renormalized energy for a multi-boson process as $\epsilon_{k+q_1+q_2} \to \epsilon_{k+q_1} + \epsilon_{k+q_2} - \epsilon_k$, which incurs the error of $(q_1 \cdot q_2)/m$ for a parabolic band with mass m because

$$\epsilon_{k+q_1+q_2} = \frac{|k+q_1+q_2|^2}{2m} = \frac{|k|^2+|q_1|^2+|q_2|^2+2k\cdot q_1+2k\cdot q_2+2q_1\cdot q_2}{2m} \tag{6.382}$$

$$\epsilon_{k+q_1}+\epsilon_{k+q_2}-\epsilon_k = \frac{|k|^2+|q_1|^2+2k\cdot q_1}{2m}+\frac{|k|^2+|q_2|^2+2k\cdot q_2}{2m}-\frac{|k|^2}{2m} \tag{6.383}$$

$$\implies \epsilon_{k+q_1+q_2}-(\epsilon_{k+q_1}+\epsilon_{k+q_2}-\epsilon_k) = \frac{q_1\cdot q_2}{m} \tag{6.384}$$

So the procedure is valid for either small q values (which corresponds to low energy or high wavelength excitations as is typical in large polarons) or large band mass m (remember that the single particle energies are given by $\epsilon_k = \frac{|k|^2}{2m}$, so a large m means a flat band). Also, obviously it works when Γ_k is large, which is when the peaks are very broad, so fine resolution is not needed. It can also be considered to be applicable when ($2N$)th order perturbation theory is good for ϵ_k and Γ_k. In any case, $G_{\mathrm{k}}(\omega)$ should be determined accurately by the cluster expansion if ω is far from the region of the renormalized energy.

Examples

Now, we can write down the first and second order corrections in the cluster expansion for the action function as

$$A_k^{(1)}(t) = -iT[S] \tag{6.385}$$

$$= -i^2\sum_q \int_0^t d\tau \int_0^\tau d\tau' D_q(\tau-\tau')T\left[\Gamma_{qk}(\tau)\Gamma_{-qk}(\tau')\right] \tag{6.386}$$

$$= \sum_q \int_0^t d\tau \int_0^\tau d\tau' \left|g_{qk}\right|^2 D_q(\tau-\tau')e^{i(\epsilon_k-\epsilon_{k+q})(\tau-\tau')} \tag{6.387}$$

and then

$$A_k^{(2)}(t) = -\frac{i}{2}T\left[S^2\right]_c \tag{6.388}$$

$$= -\frac{i}{2}\left(T\left[S^2\right] - T\left[S\right]T\left[S\right]\right) \tag{6.389}$$

$$= i\sum_{q_1}\sum_{q_2}\int_0^t d\tau_1\int_0^{\tau_1}d\tau_2\int_0^{\tau_2}d\tau_3\int_0^{\tau_3}d\tau_4\left[D_{q_1}(\tau_1-\tau_3)D_{q_2}(\tau_2-\tau_4)\right. \tag{6.390}$$

$$\times\left(g_{q_1k}g_{q_2k+q_1}g^*_{q_2k}g^*_{q_1k+q_2}e^{i\left(\left\{\epsilon_k(\tau_1-\tau_4)-\epsilon_{k+q_1}(\tau_1-\tau_2)-\epsilon_{k+q_1+q_2}(\tau_2-\tau_3)-\epsilon_{k+q_2}(\tau_3-\tau_4)\right\}\right.}\right. \tag{6.391}$$

$$\left.\left.-|g_{q_1k}g_{q_2k}|^2e^{i\left((\epsilon_k-\epsilon_{k+q_1})(\tau_1-\tau_3)+(\epsilon_k-\epsilon_{k+q_2})(\tau_2-\tau_4)\right)}\right) + D_{q_1}(\tau_1-\tau_4)D_{q_2}(\tau_2-\tau_3)\right. \tag{6.392}$$

$$\times\left(|g_{q_1k}g_{q_2k+q_1}|^2e^{i\left(\epsilon_k(\tau_1-\tau_4)-\epsilon_{k+q_1}(\tau_1-\tau_2)-\epsilon_{k+q_1+q_2}(\tau_2-\tau_3)-\epsilon_{k+q_2}(\tau_3-\tau_4)\right)}\right. \tag{6.393}$$

$$\left.\left.-|g_{q_1k}g_{q_2k}|^2e^{i\left((\epsilon_k-\epsilon_{k+q_2})(\tau_1-\tau_4)+(\epsilon_k-\epsilon_{k+q_1})(\tau_2-\tau_3)\right)}\right)\right] \tag{6.394}$$

so in the second order cluster approximation $G_k(\omega)$ is given by

$$G_k(\omega) = i\int_0^\infty dt e^{i\omega t}\exp\left[i\left(-\epsilon_k t + A_k^{(1)}(t) + A_k^{(2)}(t)\right)\right] \tag{6.395}$$

The cluster expansion for the action function works well except for the long time limit, but we want it to be well resolved for all times in order to capture the full frequency range. If we use the Feynman operator ordering theorem as in section 6.3.1, the action function can now be expressed as

$$A_k(t) = \phi_k(t) - i\log[T\exp\left(\bar{S}\right)] \tag{6.396}$$

$$= \phi_k(t) - i\sum_{n=1}^\infty\frac{1}{n!}T\left[\bar{S}^n\right]_c \tag{6.397}$$

where $\phi_k(t)$ can be chosen so as to improve the approximation with

$$\bar{S} = -i\int_0^t d\tau\left(\epsilon_k + \frac{d}{d\tau}\phi_k(\tau)\right) - i\sum_q\int_0^t d\tau\int_0^\tau d\tau' D_q^{(0)}\left(\tau-\tau'\right)\bar{\Gamma}_{-qk}(\tau)\bar{\Gamma}_{qk}\left(\tau'\right) \tag{6.398}$$

using the transformed vertex operators

$$\bar{\Gamma}_{qk}(\tau) \equiv \hat{U}(\tau)\Gamma_{qk}(\tau)\hat{U}^{-1}(\tau) \tag{6.399}$$

$$= g_{qk}e^{-i\phi_k(\tau)}e^{q\cdot\frac{d}{dk}}e^{+i\phi_k(\tau)} \tag{6.400}$$

and the bare bosonic propagator

$$D_q^{(0)}(t) = -i\left[\left(1+N_q\right)e^{-i\omega_q|l|} + N_q e^{|\omega_q|l|}\right] \tag{6.401}$$

with the Bose occupation factor $N_q = \left[\exp\left(\beta\omega_q\right)-1\right]^{-1}$. . But Dunn chooses $\phi_k(t) = -\epsilon_k t$, where ϵ_k is a (as yet undetermined) renormalized energy. With $\dot{\phi}_k = -\epsilon_k \implies \epsilon_k + \dot{\phi}_k = 0$,

the action function to first order in the cluster expansion becomes

$$A_k(t) \approx \phi_k(t) - iT[\bar{S}]_c \tag{6.402}$$

$$= -\epsilon_k t - i\langle \bar{S} \rangle \tag{6.403}$$

$$= -\epsilon_k t - \sum_q \int_0^t d\tau \int_0^\tau d\tau' D_q^{(0)}(\tau - \tau') \langle \bar{\Gamma}_{-qk}(\tau) \bar{\Gamma}_{qk}(\tau') \rangle \tag{6.404}$$

$$= -\epsilon_k t - \sum_q \int_0^t d\tau \int_0^\tau d\tau' D_q^{(0)}(\tau - \tau') \langle g_{-qk} g_{qk} e^{-i\phi_k(\tau)} e^{-q\cdot\frac{d}{dk}} e^{+i\phi_k(\tau)} e^{-i\phi_k(\tau')} e^{+q\cdot\frac{d}{dk}} e^{+i\phi_k(\tau')} \rangle \tag{6.405}$$

$$= -\epsilon_k t - \sum_q |g_{qk}|^2 \int_0^t d\tau \int_0^\tau d\tau' D_q^{(0)}(\tau - \tau') e^{i\epsilon_k \tau} e^{-i\epsilon_{k-q}\tau} e^{i\epsilon_{k-q}\tau'} e^{-i\epsilon_k \tau'} \tag{6.406}$$

$$= -\epsilon_k t - \sum_q |g_{qk}|^2 \int_0^t d\tau \int_0^\tau d\tau' D_q^{(0)}(\tau - \tau') e^{i\left(\epsilon_k - \epsilon_{k-q}\right)(\tau - \tau')} \tag{6.407}$$

$$= -\epsilon_k t + i \sum_q |g_{qk}|^2 \int_0^t d\Delta (t - \Delta) \left[(1 + N_q) e^{i\left(\epsilon_k - \epsilon_{k-q} - \Omega_q\right)\Delta} + N_q e^{i\left(\epsilon_k - \epsilon_{k-q} + \Omega_q\right)\Delta} \right] \tag{6.408}$$

$$= -\epsilon_k t + i \sum_q |g_{qk}|^2 \left[(1 + N_q) \frac{e^{itb_-} - 1 - itb_-}{b_-^2} + N_q \frac{e^{itb_+} - 1 - itb_+}{b_+^2} \right] \tag{6.409}$$

where we have set $\Delta \equiv \tau - \tau' \in [0, t]$ and $b_\mp = \epsilon_k - \epsilon_{k-q} \mp \Omega_q$. The solution resembles the Landau form of the cumulant of $C(t) = \int d\omega \frac{\beta(\omega)}{\omega^2} [e^{-i\omega t} + i\omega t - 1]$. This is not a coincidence because the action function is supposed to contain the cumulant by design. Then, ϵ_k can be identified from the long time behavior

$$\epsilon_k \equiv -\lim_{t\to\infty} \frac{d}{dt} \mathrm{Re}\left(A_k(t)\right) \tag{6.410}$$

$$= -\lim_{t\to\infty} \mathrm{Re}\left(\frac{d}{dt} \left[-\epsilon_k t + i \sum_q |g_{qk}|^2 \left[(1 + N_q) \frac{e^{itb_-} - 1 - itb_-}{b_-^2} + N_q \frac{e^{itb_+} - 1 - itb_+}{b_+^2} \right] \right] \right) \tag{6.411}$$

$$= -\lim_{t\to\infty} \left[-\epsilon_k^{(0)} - \sum_q |g_{qk}|^2 \left[(1 + N_q) \mathrm{Re}\left(\frac{e^{itb_-} - 1}{b_-} \right) + N_q \mathrm{Re}\left(\frac{e^{itb_+} - 1}{b_+} \right) \right] \right] \tag{6.412}$$

$$= \epsilon_k^{(0)} - \sum_q |g_{qk}|^2 \left[(1 + N_q) \mathcal{P}\left(\frac{1}{b_-} \right) + N_q \mathcal{P}\left(\frac{1}{b_+} \right) \right] \tag{6.413}$$

$$= \epsilon_k^{(0)} - \mathcal{P} \sum_q |g_{qk}|^2 \left[\frac{1 + N_q}{\epsilon_k - \epsilon_{k+q} - \Omega_q} + \frac{N_q}{\epsilon_k - \epsilon_{k+q} + \Omega_q} \right] \tag{6.414}$$

which gives a self-consistent condition for ϵ_k. [9] We could do the penultimate step because $\lim_{t\to\infty} \frac{e^{ibt} - 1}{b} = i \int_0^\infty d\tau e^{i\tau b} = i\pi\delta(b) - \mathcal{P}\left(\frac{1}{b}\right)$, where $\mathcal{P}$ denotes the Cauchy principal value.

[9] This is only self consistent in the energy, which PJ did not like. The self consistent choice was $\phi_k(t) = -iC_k(t)$ in PJ's paper.

6.4 My derivations

6.4.1 Derivation of second-order self-energy using RHF reference

To derive this, we can start with the definition (we will just deal with the 2p1h channel for simplicity)

$$\Sigma^{(2)}_{pp}(\omega) = \frac{1}{2}\sum_{iab}\frac{\langle pi||ab\rangle\langle ab||pi\rangle}{\omega+\epsilon_i-\epsilon_a-\epsilon_b}+\dots \tag{6.415}$$

$$= \frac{1}{2}\sum_{iab}\frac{\left(\langle pi|ab\rangle-\langle pi|ba\rangle\right)\left(\langle ab|pi\rangle-\langle ab|ip\rangle\right)}{\omega+\epsilon_i-\epsilon_a-\epsilon_b}+\dots \tag{6.416}$$

$$= \frac{1}{2}\sum_{iab}\frac{\langle pi|ab\rangle\langle ab|pi\rangle-\langle pi|ab\rangle\langle ab|ip\rangle-\langle pi|ba\rangle\langle ab|pi\rangle+\langle pi|ba\rangle\langle ab|ip\rangle}{\omega+\epsilon_i-\epsilon_a-\epsilon_b}+\dots \tag{6.417}$$

But if we swap the dummy indices for the third term, we see that $\langle pi|ba\rangle\langle ab|pi\rangle = \langle pi|ab\rangle\langle ba|pi\rangle = \langle pi|ab\rangle\langle ab|ip\rangle$, which we can group together with the second term. Similarly, for the fourth term, we can swap $a \leftrightarrow b$ to get $\langle pi|ba\rangle\langle ab|ip\rangle = \langle pi|ab\rangle\langle ba|ip\rangle = \langle pi|ab\rangle\langle ab|pi\rangle$, which we can group together with the first term. Now assigning a spin variable to each center, this gives

$$\Sigma^{(2)}_{p^+p^+}(\omega) = \frac{1}{2}\sum_{iab}\sum_{\beta\gamma\sigma}\frac{2\langle p^+i^\beta|a^\gamma b^\sigma\rangle\langle a^\gamma b^\sigma|p^+i^\beta\rangle-2\langle p^+i^\beta|a^\gamma b^\sigma\rangle\langle b^\sigma a^\gamma|p^+i^\beta\rangle}{\omega+\epsilon_i-\epsilon_a-\epsilon_b}+\dots \tag{6.418}$$

$$= \sum_{iab}\frac{2\langle pi|ab\rangle\langle ab|pi\rangle-\langle pi|ab\rangle\langle ba|pi\rangle}{\omega+\epsilon_i-\epsilon_a-\epsilon_b}+\dots \tag{6.419}$$

$$= \sum_{iab}\frac{\langle pi|ab\rangle\left(2\langle ab|pi\rangle-\langle ba|pi\rangle\right)}{\omega+\epsilon_i-\epsilon_a-\epsilon_b}+\dots \tag{6.420}$$

where in the first expression, because to start with the index p is already constrained to have spin $+$, so a must be $+$ as well, and we can accept 2 combinations of spin for the two-electron integrals ($\beta,\gamma = +,-$), while in the second expression, we can only accept 1 combinations of $\beta,\gamma = +$. For the expression of the off-diagonal self-energy, we can just accept Szabo's form in 3.79. In Joonho's paper, they say that

$$\langle pq|rs\rangle = \frac{1}{\Omega}\frac{4\pi}{|\mathbf{K}_p-\mathbf{K}_r|^2}\delta_{\mathbf{K}_p-\mathbf{K}_r,\mathbf{K}_q-\mathbf{K}_s}\times\left(1-\delta_{\mathbf{K}_p,\mathbf{K}_r}\right) \tag{6.421}$$

but I notice that I get the "better" results for the GF2 poles (i.e. closer to the exact result for $r_s = 4$), if instead it is

$$\langle pq|rs\rangle = \frac{1}{\Omega}\frac{4\pi}{|\mathbf{K}_p-\mathbf{K}_r|^2}\delta_{\mathbf{K}_p-\mathbf{K}_r,\mathbf{K}_s-\mathbf{K}_q}\times\left(1-\delta_{\mathbf{K}_p,\mathbf{K}_r}\right). \tag{6.422}$$

Making this change, also makes the MP2 energy match that obtained from the sine-cosine basis. I also notice that this is the momentum conservation condition that iPie suggests in the below function for the two-electron integrals in the plane wave basis. This is also the condition that seems suggested in the review.

$$i + j = a + b \implies i - a = b - j \tag{6.423}$$

To make things simpler, instead of working in the plan wave bases, we can transform to the sine-cosine basis. I need to make sure that I am consistent with this basis choice, though. Below, I go through the logic, to ensure consistency.

Second-order self-energy in sine-cosine basis

The retarded, closed shell expression given by Szabo is

$$\Sigma^{(2)}_{pq}(\omega) = \sum_{iab} \frac{\langle ab|pi\rangle \Big(2\langle qi|ab\rangle - \langle iq|ab\rangle\Big)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{\langle ij|pa\rangle \Big(2\langle qa|ij\rangle - \langle aq|ij\rangle\Big)}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{6.424}$$

$$= \sum_{iab} \frac{(pa|ib)\Big(2(qa|ib) - (qb|ia)\Big)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{((pi|aj)(2(qi|aj) - (qj|ai)))}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{6.425}$$

Now, for the purposes of the diagonal cumulant, we want to be able to compute $\Sigma^{(2)}(\tilde{p}, \omega)$, where $\tilde{p}$ is the state index in the plane wave basis. Because we want to compute the above expressions using the standard quantum chemistry methods, instead of directly computing $\Sigma^{(2)}_{\tilde{p}\tilde{p}}(\omega)$, we can transform to the sine-cosine basis as

$$\Sigma^{(2),R}_{pq}(\omega) = \sum_{mn} U_{pm} \Sigma^{(2),R}_{mn}(\omega) U^{\dagger}_{nq} \tag{6.426}$$

$$\tag{6.427}$$

But we must take into account the correct mapping between the plane wave and sine-cosine basis, leading to

$$\Sigma^{(2),R}(p, \omega) \equiv \Sigma^{(2),R}_{pp}(\omega) = \frac{1}{2}\left(\Sigma^{(2),R}_{cc}(\omega) + \Sigma^{(2),R}_{ss}(\omega)\right) + \frac{i}{2}\left(\Sigma^{(2),R}_{sc}(\omega) - \Sigma^{(2),R}_{cs}(\omega)\right) \tag{6.428}$$

where c and s refer to the cosine and sine functions, respectively. It will be more convenient to work in the sine-cosine basis, here. The transformation that allows us to move between the plane wave and sine-cosine basis is given by

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \implies \begin{bmatrix} |\mathbf{K}, s\rangle \\ |\mathbf{K}, c\rangle \end{bmatrix} = \mathbf{U}^{\dagger} \begin{bmatrix} |\mathbf{K}\rangle \\ |-\mathbf{K}\rangle \end{bmatrix} \tag{6.429}$$

for a $\mathbf{K} \neq 0$ state from the plane wave basis. But we are just interested in computing the self energy for the state $\Sigma(\mathbf{K}, \omega)$, so we just care about

$$\begin{bmatrix} s \\ c \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} \tag{6.430}$$

After some algebra, we are left with the fact that

$$\Sigma^{c,(2),R}(p, \omega) = \frac{1}{2}\left(\Sigma'_{cc} + \Sigma'_{ss}\right) + \frac{i}{2}\left(\Sigma'_{sc} - \Sigma'_{cs}\right) \tag{6.431}$$

6.4.2 Derivation of the analytic form for GF2+C

We can start with

$$C_{pp}^{(2)}(t) = \frac{1}{2}\sum_{iab}\frac{\langle pi\,||\,ab\rangle^2}{\left(\Delta_{pa}^{\mu}\right)^2}\left(e^{-i\Delta_{pa}^{\mu}t} + i\Delta_{pa}^{\mu}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa\,||\,ij\rangle^2}{\left(\Delta_{pi}^{\nu}\right)^2}\left(e^{-i\Delta_{pi}^{\nu}t} + i\Delta_{pi}^{\nu}t - 1\right) \tag{6.432}$$

We can define $\zeta_{pa\mu} = \frac{\langle pi||ab\rangle^2}{\left(\Delta_{pa}^{\mu}\right)^2}$ and $\zeta_{pi\nu} = \frac{\langle pa||ij\rangle^2}{\left(\Delta_{pi}^{\nu}\right)^2}$ with $\Delta_{pa}^{\mu} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i - i\eta$ and $\Delta_{pi}^{\nu} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a - i\eta$, where μ and ν describe the OV pairs $i \to b$ and $a \to j$, respectively. Then, we can write the retarded Green's function as

$$G(p,t) = -i\Theta(t)e^{-i\epsilon_p t}\exp\left[\frac{1}{2}\sum_{i\nu}\zeta_{pi\nu}\left(e^{-i\Delta_{pi}^{\nu}t} + i\Delta_{pi}^{\nu}t - 1\right) + \frac{1}{2}\sum_{a\mu}\zeta_{pa\mu}\left(e^{-i\Delta_{pa}^{\mu}t} + i\Delta_{pa}^{\mu}t - 1\right)\right] \tag{6.433}$$

$$= -i\Theta(t)\underbrace{\exp\left\{-\frac{1}{2}\left(\sum_{i\nu}\zeta_{pi\nu} + \sum_{a\mu}\zeta_{pa\mu}\right)\right\}}_{Z_p^{QP}} \times \exp\left\{-i\overbrace{\left(\epsilon_p - \frac{1}{2}\sum_{i\nu}\zeta_{pi\nu}\Delta_{pi}^{\nu} - \frac{1}{2}\sum_{a\mu}\zeta_{pa\mu}\Delta_{pa}^{\mu}\right)}^{\epsilon_p^{QP}}t\right\} \tag{6.434}$$

$$\times \exp\left\{\frac{1}{2}\left(\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi}^{\nu}t} + \sum_{a\mu}\zeta_{pa\mu}e^{-i\Delta_{pa}^{\mu}t}\right)\right\} \tag{6.435}$$

$$= -i\Theta(t)Z_p^{QP}e^{-i\epsilon_p^{QP}t}\exp\left\{\frac{1}{2}\left(\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi}^{\nu}t} + \sum_{a\mu}\zeta_{pa\mu}e^{-i\Delta_{pa}^{\mu}t}\right)\right\} \tag{6.436}$$

where we have defined the quasiparticle weight Z_p^{QP} and the quasiparticle energy ϵ_p^{QP}.

It is worth checking that we satisfy

$$\epsilon_p^{QP} = \epsilon_p - \frac{1}{2}\left(\sum_{i\nu}\zeta_{pi\nu}\Delta^{\nu}_{pi} + \sum_{a\mu}\zeta_{pa\mu}\Delta^{\mu}_{pa}\right) \tag{6.437}$$

$$= \epsilon_p - \frac{1}{2}\left(\sum_{i\nu}\frac{\langle pa\,||\,ij\rangle^2}{\Delta^{\nu}_{pi}} + \sum_{a\mu}\frac{\langle pi\,||\,ab\rangle^2}{\Delta^{\mu}_{pa}}\right) \tag{6.438}$$

$$= \epsilon_p + \Sigma^{(2)}_{pp}(\epsilon_p) \tag{6.439}$$

and

$$Z_p^{QP} = \exp\left(-\frac{1}{2}\left[\sum_{i\nu}\zeta_{pi\nu} - \sum_{a\nu}\zeta_{pa\nu}\right]\right) \tag{6.440}$$

$$= \exp\left(\left[\frac{\partial\Sigma^{(2)}_{pp}(\omega)}{\partial\omega}\right]_{\omega=\epsilon_p^{HF}}\right) \tag{6.441}$$

$$\tag{6.442}$$

where we have used the fact that $\Sigma^{(2)}_{pp}(\omega) = \frac{1}{2}\left(\sum_{i\nu}\frac{\langle pa||ij\rangle^2}{\omega-\Delta^{\nu}_{pi}} + \sum_{a\mu}\frac{\langle pi||ab\rangle^2}{\omega-\Delta^{\mu}_{pa}}\right) \implies \left[\frac{\partial\Sigma^{(2)}_{pp}(\omega)}{\partial\omega}\right]_{\omega=\epsilon_p^{HF}} = -\frac{1}{2}\left(\sum_{i\nu}\frac{\langle pa||ij\rangle^2}{\left(\Delta^{\nu}_{pi}\right)^2} + \sum_{a\mu}\frac{\langle pi||ab\rangle^2}{\left(\Delta^{\mu}_{pa}\right)^2}\right)$.

Next, we perform the Fourier transform analytically, inserting the positive infinitesimal where appropriate:

$$G(p,\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t}G(p,t) \tag{6.443}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{i(\omega-\epsilon_p^{QP})t}\exp\left\{\frac{1}{2}\left(\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta^{\nu}_{pi}t} + \sum_{a\mu}\zeta_{pa\mu}e^{-i\Delta^{\mu}_{pa}t}\right)\right\} \tag{6.444}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{i(\omega-\epsilon_p^{QP})t}\left(1 + \frac{1}{2}\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta^{\nu}_{pi}t} + \frac{1}{2}\sum_{a\mu}\zeta_{pa\mu}e^{-i\Delta^{\mu}_{pa}t} + \dots\right) \tag{6.445}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{-\eta+i(\omega-\epsilon_p^{QP})t} \tag{6.446}$$

$$-\frac{i}{2}Z_p^{QP}\sum_{i\nu}\zeta_{pi\nu}\int_0^{\infty} dt\, e^{-\eta+i(\omega-\epsilon_p^{QP}-\Delta^{\nu}_{pi})t} \tag{6.447}$$

$$-\frac{i}{2}Z_p^{QP}\sum_{a\mu}\zeta_{pa\mu}\int_0^{\infty} dt\, e^{-\eta+i(\omega-\epsilon_p^{QP}-\Delta^{\mu}_{pa})t} + \dots \tag{6.448}$$

$$= \frac{Z_p^{QP}}{\omega-\epsilon_p^{QP}+i\eta} + \frac{1}{2}Z_p^{QP}\sum_{i\nu}\frac{\zeta_{pi\nu}}{\omega-\epsilon_p^{QP}-\Delta^{\nu}_{pi}+i\eta} + \frac{1}{2}Z_p^{QP}\sum_{a\mu}\frac{\zeta_{pa\mu}}{\omega-\epsilon_p^{QP}-\Delta^{\mu}_{pa}+i\eta} + \dots \tag{6.449}$$

Then, proceeding as in the GW+C case, we can write down the spectral function as

$$A(p,\omega) = -\frac{1}{\pi}\left[\frac{\left(\mathrm{Re}\, Z_p^{QP}\right)\left(\mathrm{Im}\,\epsilon_p^{QP}\right) + \left(\mathrm{Im}\, Z_p^{QP}\right)\left(\omega - \mathrm{Re}\,\epsilon_p^{QP}\right)}{\left(\omega - \mathrm{Re}\,\epsilon_p^{QP}\right)^2 + \left(\mathrm{Im}\,\epsilon_p^{QP}\right)^2}\right. \tag{6.450}$$

$$\left. + \sum_{i\nu} \frac{\left(\mathrm{Re}\, Z_{pi\nu}^{sat}\right)\left(\mathrm{Im}\,\epsilon_{pi\nu}^{sat}\right) + \left(\mathrm{Im}\, Z_{pi\nu}^{sat}\right)\left(\omega - \mathrm{Re}\,\epsilon_{pi\nu}^{sat}\right)}{\left(\omega - \mathrm{Re}\,\epsilon_{pi\nu}^{sat}\right)^2 + \left(\mathrm{Im}\,\epsilon_{pi\nu}^{sat}\right)^2} + \ldots\right] \tag{6.451}$$

where we have defined the satellite weights and energies as

$$Z_{pi\nu}^{sat} = \frac{1}{2} Z_p^{QP} \zeta_{pi\nu} \tag{6.452}$$

$$\epsilon_{pi\nu}^{sat} = \epsilon_p^{QP} + \Delta_{pi}^{\nu} \tag{6.453}$$

and similarly for the $a\mu$ terms. To understand how the FFT comes about, we can analytically carry it out. We started with the non-interacting Green's function in the time domain as

$$G_{pp}^0(t) = -i\Theta(t) e^{-t(i\epsilon_p + \eta)} \tag{6.454}$$

Then, the FFT provides a map between t_n and ω_m as

$$G_{pp}^0(\omega_m) = -i \sum_{n=0}^{N_T - 1} e^{i\omega_m t_n} e^{-t_n(i\epsilon_p + \eta)} \tag{6.455}$$

$$= -i \sum_{n=0}^{N_T - 1} e^{t_n(i\omega_m - i\epsilon_p - \eta)} \tag{6.456}$$

with $t_n = n\Delta t$ with $\Delta t = np.float64(1.0000150001500014)$ and $N_T = 100000$. Then, $\omega_m =$ np.fft.fftshift(np.fft.fftfreq(len(times), d=dt) * 2.0 * np.pi). I get $\Delta\omega = 2\pi/(N_T \Delta t) = 6.283139999628673e{-}05$. How is this connected to the standard result for the non-interacting Green's function in frequency space, which is

$$G_{pp}^0(\omega) = \frac{1}{\omega - \epsilon_p + i\eta} \tag{6.457}$$

6.4.3 Taking advantage of connection between EOMs

We use the formula $\mathcal{L}\{(f * g)(t)\}(s) = \mathcal{L}\{f(t)\}(s)\mathcal{L}\{g(t)\}(s)$ for the Laplace transform of a convolution in the below.

EOM with the retarded self energy

We know the retarded Green's function satisfies the equation of motion in the time domain:

$$(i\partial_t - \epsilon_0)G^R(t) - \int_0^t d\tau\, \Sigma^R(t-\tau)\, G^R(\tau) = 0 \tag{6.458}$$

$$\implies i\mathcal{L}\{\partial_t G^R(t)\}(s) - \epsilon_0\mathcal{L}\{G^R(t)\}(s) - \mathcal{L}\left\{\int_0^t d\tau\, \Sigma^R(t-\tau)\, G^R(\tau)\right\}(s) = 0 \tag{6.459}$$

$$i\left(s\mathcal{L}\{G^R(t)\}(s) - G^R(0)\right) - \epsilon_0\mathcal{L}\{G^R(t)\}(s) - \mathcal{L}\{\Sigma^R(t)\}(s)\mathcal{L}\{G^R(t)\}(s) = 0 \tag{6.460}$$

$$\left[s + i\epsilon_0 + i\mathcal{L}\{\Sigma^R(t)\}(s)\right]\mathcal{L}\{G^R(t)\}(s) = -i \tag{6.461}$$

$$\tag{6.462}$$

where we used the boundary condition $G^R(0) = -i$. This implies that

$$\mathcal{L}\{G^R(t)\}(s) = \frac{-i}{s + i\epsilon_0 + i\mathcal{L}\{\Sigma^R(t)\}(s)}. \tag{6.463}$$

With the standard analytic continuation $s \to -i\omega + \eta$, we find

$$G^R(-i\omega+\eta) = \frac{-i}{-i\omega + \eta + i\epsilon_0 + i\mathcal{L}\{\Sigma^R(t)\}(-i\omega+\eta)} \times \frac{i}{i} \tag{6.464}$$

$$= \frac{1}{\omega - \epsilon_0 - \mathcal{L}\{\Sigma^R(t)\}(-i\omega+\eta) + i\eta} \tag{6.465}$$

$$\implies G^R(\omega) = \frac{1}{\omega - \epsilon_0 - \Sigma^R(\omega) + i0^+} \tag{6.466}$$

EOM with the retarded cumulant

It is also valid to replace the convolution term in (6.458) with the time derivative of the cumulant. Considering the Laplace transform of just this term, we have

$$\mathcal{L}\left\{\dot{C}^R(t)G^R(t)\right\}(s) = \int_0^\infty dt\, e^{-st}\dot{C}^R(t)\left(-ie^{-i\epsilon_0 t}e^{C^R(t)}\right) \tag{6.467}$$

$$= -i\int_0^\infty e^{-t(s+i\epsilon_0)}\frac{d}{dt}\left(e^{C^R(t)}\right)dt \tag{6.468}$$

$$= i - i(s+i\epsilon_0)\int_0^\infty e^{t(s+i\epsilon_0)}e^{C^R(t)}\,dt \tag{6.469}$$

$$= i - i(s+i\epsilon_0)\mathcal{L}\{G^R(t)\}(s) \tag{6.470}$$

where we used integration by parts. [10]

[10]

$$\int_0^\infty e^{-t(s+i\epsilon_0)}\frac{d}{dt}e^{C^R(t)}\,dt = \left[e^{-t(s+i\epsilon_0)}e^{C^R(t)}\right]_0^\infty + (s+i\epsilon_0)\int_0^\infty e^{-t(s+i\epsilon_0)}e^{C^R(t)}\,dt \tag{6.471}$$

$$= -1 + (s+i\epsilon_0)\int_0^\infty e^{-t(s+i\epsilon_0)}e^{C^R(t)}\,dt \tag{6.472}$$

EOM from Nakajima-Zwanzig

The Nakajima-Zwanzig equation for a correlation function $\mathcal{A}(t)$ is given by

$$\dot{\mathcal{A}}(t) = \mathcal{A}(t)\Omega_1 - \int_0^t d\tau\, \mathcal{A}(t-\tau)\mathcal{K}_1(\tau) + D(t) \tag{6.473}$$

So it is useful to consider the term involving the memory kernel. Its Laplace transform is

$$\mathcal{L}\left\{\mathcal{A}(t) * \mathcal{K}_1(t)\right\}(s) = \mathcal{L}\{\mathcal{A}(t)\}(s)\mathcal{L}\{\mathcal{K}_1(t)\}(s) \tag{6.474}$$

In order to make more progress, we would want to equate

$$\mathcal{L}\left\{\mathcal{A}(t) * \mathcal{K}_1(t)\right\}(s) = \mathcal{L}\left\{\dot{C}^R(t)G^R(t)\right\}(s) \tag{6.475}$$

$$\implies \mathcal{L}\{\mathcal{A}(t)\}(s)\mathcal{L}\{\mathcal{K}_1(t)\}(s) = i - i(s+i\epsilon_0)\mathcal{L}\{G^R(t)\}(s) \tag{6.476}$$

It would be nice to solve for $\mathcal{L}\{G^R(t)\}(s)$ in terms of $\mathcal{L}\{\mathcal{K}_1(t)\}(s)$ now, but to do so, we would need to establish a relationship between $\mathcal{L}\{G^R(t)\}(s)$ and $\mathcal{L}\{\mathcal{A}(t)\}(s)$. We know that the relationship between the fermionic retarded Green function $G^R(t) = -i\theta(t)\langle\{\hat{\mu}(t), \hat{\mu}(0)\}\rangle$ and the autocorrelation function $\mathcal{A}(t) = \langle\hat{\mu}(t)\hat{\mu}(0)\rangle$ of Nakajima-Zwanzig is given by

$$G^R(t) = -i\theta(t)\left[\mathcal{A}(t) + \mathcal{A}(-t)\right] \tag{6.477}$$

$$\implies \mathcal{L}\{G^R(t)\}(s) = -i\mathcal{L}\{[\mathcal{A}(t) + \mathcal{A}(-t)]\}(s) \tag{6.478}$$

$$= -i\left[\mathcal{L}\{\mathcal{A}(t)\}(s) + \mathcal{L}\{\mathcal{A}(-t)\}(s)\right] \tag{6.479}$$

$$\stackrel{?}{=} -i\mathcal{L}\{[\mathcal{A}(t) + \mathcal{A}^*(t)]\}(s) \tag{6.480}$$

$$= -2i\mathcal{L}\{\mathrm{Re}[\mathcal{A}(t)]\}(s) \tag{6.481}$$

The last two equalities only hold if the operators $\hat{\mu}(t)$ that make up the autocorrelation function $\mathcal{A}(t)$ are Hermitian. The fermionic second quantization operators are not Hermitian though. Only observables are Hermitian, which is why they were able to use the property leading to equation (6.480) in Eq. 35-36 of *Any type of spectroscopy can be efficiently simulated on a quantum computer.*

6.4.4 SC-CE

My derivation

The cumulant ansatz for the retarded Green's function is:

$$G^R(t) = \underbrace{-i\theta(t)e^{-i\epsilon_0 t}}_{G_0^R(t)} e^{C^R(t)}. \tag{6.482}$$

Note that because this is for the retarded Green's function, we can assume that $t > 0$ when differentiating, so $\theta(t) = 1$. Differentiating with respect to time gives:

$$\partial_t G^R(t) = -i\partial_t\left(e^{-i\epsilon_0 t}e^{C^R(t)}\right) \tag{6.483}$$

$$= -i\epsilon_0 G^R(t) - i\dot{C}^R(t)G^R(t) \tag{6.484}$$

Now, the equation of motion [11] for the retarded Green's function in the Dyson formulation with the self energy Σ is given by

$$\partial_t G^R(t) = -i\epsilon_0 G^R(t) - i\int_0^t d\tau G^R(t-\tau)\Sigma^R(\tau) \tag{6.486}$$

Equating the two gives an expression for the cumulant derivative:

$$\dot{C}^R(t)G^R(t) = \int_0^t d\tau G^R(t-\tau)\Sigma^R(\tau) \tag{6.487}$$

From here there are two directions I have in mind. Firstly, the Kowalski paper gave the EOM for the retarded core-hole cumulant as $-i\frac{dC_c^R(t)}{dt} = \left\langle \phi \left| \bar{H}_N(t) \right| \phi \right\rangle = \sum_{ia} f_{ia} t_i^a + \frac{1}{2}\sum_{ijab} v_{ij}^{ab} t_j^b t_i^a$. I could plug this in and see.

Secondly, we also know that

$$i\dot{C}^R(t)G^R(t) = \int_0^t d\tau\, \mathcal{A}(t-\tau)\mathcal{K}_1(\tau) \tag{6.488}$$

$$\implies \mathcal{L}\left\{ i\dot{C}^R(t)G^R(t) \right\}(s) = \mathcal{L}\left\{ \int_0^t d\tau\, \mathcal{A}(t-\tau)\mathcal{K}_1(\tau) \right\}(s) \tag{6.489}$$

where the RHS comes from the Nakajima-Zwanzig equation (6.485) and $\mathcal{L}\left\{ i\dot{C}^R(t)G^R(t) \right\}(s)$ has a known analytical form. [12] . This Laplace transform might be promising because the RHS is a convolution ($(f*g)(t) = \int_0^t d\tau f(t-\tau)g(\tau)$, with $f(t) = G^R(t)$ and $g(t) = \Sigma^R(t)$)

[11] The Nakajima-Zwanzig equation for a correlation function $\mathcal{A}(t)$ is given by

$$\dot{\mathcal{A}}(t) = \mathcal{A}(t)\Omega_1 - \int_0^t d\tau\, \mathcal{A}(t-\tau)\mathcal{K}_1(\tau) + D(t) \tag{6.485}$$

Since we know the connection $\int_0^t d\tau\, \mathcal{A}(t-\tau)\mathcal{K}_1(\tau) = i\dot{C}^R(t)G^R(t) = \int_0^t d\tau G^R(t-\tau)\Sigma^R(\tau)$ and the outer definitions involve convolutions, we might be able to make some progress by demanding equality of their Laplace transforms.

[12]

$$\mathcal{L}\left\{ \dot{C}^R(t)G^R(t) \right\}(s) = \int_0^\infty dt\, e^{-st}\dot{C}^R(t)\left(-ie^{-i\epsilon_0 t}e^{C^R(t)} \right) \tag{6.490}$$

$$= -i\int_0^\infty e^{-t(s+i\epsilon_0)}\frac{d}{dt}\left(e^{C^R(t)}\right)dt \tag{6.491}$$

$$= i - i(s+i\epsilon_0)\int_0^\infty e^{-t(s+i\epsilon_0)}e^{C^R(t)}\,dt \tag{6.492}$$

$$= i - i(s+i\epsilon_0)\mathcal{L}\{G^R(t)\}(s) \tag{6.493}$$

where to proceed from (6.491) we take $u' = \frac{d}{dt}\left(e^{C^R(t)}\right) \implies u = e^{C^R(t)}$ and $v = e^{-t(s+i\epsilon_0)} \implies v' = -(s+i\epsilon_0)e^{-t(s+i\epsilon_0)}$, so that

$$\int_0^\infty e^{-t(s+i\epsilon_0)}\frac{d}{dt}e^{C^R(t)}\,dt = \left[e^{-t(s+i\epsilon_0)}e^{C^R(t)} \right]_0^\infty + (s+i\epsilon_0)\int_0^\infty e^{-t(s+i\epsilon_0)}e^{C^R(t)}\,dt \tag{6.494}$$

$$= -1 + (s+i\epsilon_0)\int_0^\infty e^{-t(s+i\epsilon_0)}e^{C^R(t)}\,dt \tag{6.495}$$

and we know that there is a nice formula for the Laplace transform of a convolution as $\mathcal{L}\{(f*g)(t)\}(s) = \mathcal{L}\{f(t)\}(s)\mathcal{L}\{g(t)\}(s)$. From here on out, we will interchange with the notation that $\mathcal{L}\{f(t)\}(s) = \tilde{f}(s)$. Then

$$\mathcal{L}\left\{\text{RHS}\right\}(s) = \tilde{G}^R(s)\tilde{\Sigma}^R(s). \tag{6.498}$$

We already know the form of $\tilde{G}^R(s)$, but we need to find $\tilde{\Sigma}^R(s)$. We can start by taking the Laplace transform of $\Sigma^R(t)$:

$$\mathcal{L}\{\Sigma^R(t)\}(s) = \int_0^\infty dt\, e^{-st}\Sigma^R(t) \tag{6.499}$$

$$= \int_0^\infty dt\, e^{-st}\left[\left(G_0^R(t)\right)^{-1} - \left(G^R(t)\right)^{-1}\right] \tag{6.500}$$

Given that it is not simple to invert the Green's function analytically in the time domain, we can stop here. But perhaps you can think of certain limits where this inversion is possible to do numerically? Setting $\mathcal{L}\left\{\text{LHS}\right\}(s) = \mathcal{L}\left\{\text{RHS}\right\}(s)$, we have

$$i - i(s+i\epsilon_0)\tilde{G}^R(s) = \tilde{G}^R(s)\tilde{\Sigma}^R(s) \tag{6.501}$$

$$\implies \tilde{\Sigma}^R(s) = \frac{i}{\tilde{G}^R(s)} - i(s+i\epsilon_0) \quad\iff\quad \tilde{G}^R(s) = \frac{i}{\tilde{\Sigma}^R(s) + i(s+i\epsilon_0)}. \tag{6.502}$$

So the relationship of $\tilde{G}^R(s)$ to the $C^R(t)$ becomes clear if we define

$$F(z) \equiv \mathcal{L}\{e^{C^R(t)}\}(z) = \int_0^\infty e^{-zt}\, e^{C^R(t)}\, dt \tag{6.503}$$

$$\implies \tilde{G}^R(s) = -i\int_0^\infty dt\, e^{-t(s+i\epsilon_0)}e^{C^R(t)} = -iF(s+i\epsilon_0) \tag{6.504}$$

So with $z = s + i\epsilon_0$ in (6.502), we have

$$-i\,F(z) = \frac{i}{\tilde{\Sigma}^R(z-i\epsilon_0) + iz} \tag{6.505}$$

$$\implies F(z) = \mathcal{L}\{e^{C^R(t)}\}(z) = -\frac{1}{\tilde{\Sigma}^R(z-i\epsilon_0) + iz} \tag{6.506}$$

$$\implies C^R(t) = \ln\left[\mathcal{L}_z^{-1}\left\{-\frac{1}{\tilde{\Sigma}^R(z-i\epsilon_0) + iz}\right\}(t)\right], \tag{6.507}$$

where $\mathcal{L}_z^{-1}$ is the inverse Laplace transform with respect to z.

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and then in (6.492) we recognize

$$\tilde{G}^R(s) = \int_0^\infty dt\, e^{-st}\left(-ie^{-i\epsilon_0 t}e^{C^R(t)}\right) \tag{6.496}$$

$$= -i\int_0^\infty dt\, e^{-t(s+i\epsilon_0)}e^{C^R(t)} \tag{6.497}$$

If we pass to the frequency domain by setting $z = \eta - i\omega$ (with $\eta \to 0^+$), the boxed relation becomes equivalent to the familiar Dyson form:

$$G^R(\omega) = \frac{1}{\omega - \epsilon_0 - \Sigma^R(\omega)}. \tag{6.508}$$

Thus, the Laplace-space identity is the causal/Laplace version of Dyson's equation combined with the cumulant ansatz.

Under the usual GW+cumulant approximations, one expands around the quasiparticle pole and rewrites $C^R(t)$ in terms of the satellite spectral weight:

$$C^R(t) = \int_{-\infty}^{\infty} \frac{d\omega}{\pi} \frac{\text{Im } \Sigma^R(\epsilon_0 + \omega)}{\omega^2} \big(e^{-i\omega t} + i\omega t - 1\big), \tag{6.509}$$

which can be recovered by (i) expressing the inverse Laplace kernel via a spectral representation of Σ^R, and (ii) performing the Bromwich integral—this uses analyticity and retarded boundary conditions. The result is the well-known closed form used for plasmon satellites and shake-up structures.

Practical use

The compact identity

$$F(z) = -\frac{1}{\tilde{\Sigma}^R(z - i\epsilon_0) + iz} \tag{6.510}$$

is a convenient starting point for controlled approximations:

- **Low-t expansion:** expand the denominator for large z to obtain short-time moments of $C^R(t)$.
- **Long-t (satellite) behavior:** analyze the singularity structure of the denominator (branch cuts from Σ^R) to extract asymptotics and satellite weights.

Maybe the best thing now would be to start with the Landau form of the cumulant and to simplify the notation it is assumed that we deal with retarded quantities only.

$$C(t) = \int d\omega \frac{\beta(\omega)}{\omega^2} \left[e^{-i\omega t} + i\omega t - 1\right] \tag{6.511}$$

where the cumulant kernel is defined as

$$\beta(\omega) = -\frac{1}{\pi} \text{Im}\, \Sigma(\omega) \tag{6.512}$$

$$= \frac{1}{\pi} \text{Im} \left(G_0^{-1}(\omega) - G^{-1}(\omega)\right) \tag{6.513}$$

$$\tag{6.514}$$

If we partition into a physical and auxiliary space, denoted by $\mathcal{S}$ and $\mathcal{A}$, respectively, $\boldsymbol{G}_0(\omega) = \begin{pmatrix} \boldsymbol{G}^0_{\mathcal{SS}}(\omega) & 0 \\ 0 & \boldsymbol{G}^0_{\mathcal{AA}}(\omega) \end{pmatrix}$ and $\boldsymbol{G}(\omega) = \begin{pmatrix} \boldsymbol{G}_{\mathcal{SS}}(\omega) & \boldsymbol{G}_{\mathcal{SA}}(\omega) \\ \boldsymbol{G}_{\mathcal{AS}}(\omega) & \boldsymbol{G}_{\mathcal{AA}}(\omega) \end{pmatrix}$, and plug in to get an expression for the cumulant kernel in the partitioned space:

$$\beta_{SS}(\omega) = \frac{1}{\pi} \operatorname{Im} \left[\mathbf{G}^{0,-1}_{SS}(\omega) - \left(\mathbf{G}_{SS}(\omega) - \mathbf{G}_{S\mathcal{A}}(\omega) \mathbf{G}^{-1}_{\mathcal{AA}}(\omega) \mathbf{G}_{\mathcal{A}S}(\omega) \right)^{-1} \right] \tag{6.515}$$

I don't know if this form is useful.

One-shot cumulant

In the first iteration, we set the exponential factor to 1, which gives the time-derivative of the one-shot cumulant:

$$\dot{C}^R_0(t) = \int_0^t d\tau \Sigma^R(\tau) e^{i\epsilon_0 \tau}. \tag{6.516}$$

Note that we can express the second-derivative of the one-shot retarded cumulant as

$$\ddot{C}^R_0 = \Sigma^R(t) e^{i\epsilon_0 t}. \tag{6.517}$$

We integrate with respect to time to get the cumulant itself:

$$C^R_0(t) = \int_0^t dt' \int_0^{t'} d\tau \Sigma^R(\tau) e^{i\epsilon_0 \tau} = \int_0^t d\tau \Sigma^R(\tau) e^{i\epsilon_0 \tau} (t - \tau). \tag{6.518}$$

Now, we perform a Fourier transform to get the frequency-domain cumulant:

$$C^R_0(\omega) = \int_0^\infty dt\, e^{i\omega t} \int_0^t d\tau\, \Sigma^R(\tau) e^{i\epsilon_0 \tau} (t - \tau) \tag{6.519}$$

$$= \int_0^\infty d\tau\, \Sigma^R(\tau) e^{i\epsilon_0 \tau} \int_\tau^\infty dt\, e^{i\omega t} (t - \tau) \tag{6.520}$$

Let $t' = t - \tau$, so $dt = dt'$, and when $t = \tau$, $t' = 0$, and as $t \to \infty$, $t' \to \infty$:

$$\int_\tau^\infty dt\, e^{i\omega t} (t - \tau) = \int_0^\infty dt'\, e^{i\omega (t' + \tau)} t' \tag{6.521}$$

$$= e^{i\omega\tau} \int_0^\infty dt'\, e^{i\omega t'} t' \tag{6.522}$$

We can evaluate the integral over t' as

$$\int_0^\infty dt'\, e^{i\omega t'} t' = -\frac{1}{(\omega + i\eta)^2} \tag{6.523}$$

where where η is the positive infinitesimal convergence factor.

Putting it all together,

$$C^R_0(\omega) = \int_0^\infty d\tau\, \Sigma^R(\tau) e^{i(\omega + \epsilon_0)\tau} \left(-\frac{1}{(\omega + i\eta)^2} \right) \tag{6.524}$$

$$= -\frac{1}{(\omega + i\eta)^2} \int_0^\infty d\tau\, \Sigma^R(\tau) e^{i(\omega + \epsilon_0)\tau} \tag{6.525}$$

$$= -\frac{\Sigma^R(\omega + \epsilon_0)}{(\omega + i\eta)^2} \tag{6.526}$$

Now, consider the inverse Fourier transform to get back to the time domain:

$$C_0^R(t) = \int_{-\infty}^{\infty} d\omega\, e^{-i\omega t} C_0^R(\omega) \tag{6.527}$$

$$= \int_{-\infty}^{\infty} d\omega\, e^{-i\omega t} \left(-\frac{\Sigma^R(\omega + \epsilon_0)}{(\omega + i\eta)^2} \right) \tag{6.528}$$

6.4.5 Fundamental properties of the cumulant

The interacting Green's function can be written as either a Dyson series

$$G = G_0 + G_0 \Sigma G_0 + G_0 \Sigma G_0 \Sigma G_0 + G_0 \Sigma G_0 \Sigma G_0 \Sigma G_0 + \dots \tag{6.529}$$

or using the cumulant ansatz $G = G_0 e^C$, where we can formally expand the cumulant C in powers of the interaction strength as $C = \lambda C + \lambda^2 C + \lambda^3 C + \mathcal{O}(\lambda^4) \equiv C^{(1)} + C^{(2)} + C^{(3)} + \mathcal{O}(\lambda^4)$. Expanding the exponential gives

$$G = G_0 \left(1 + C + \frac{C^2}{2!} + \frac{C^3}{3!} + \dots \right) \tag{6.530}$$

$$= G_0 \left(1 + C^{(1)} + \frac{1}{2!} \left[\left(C^{(1)} \right)^2 + 2C^{(2)} \right] + \frac{1}{3!} \left[\left(C^{(1)} \right)^3 + 3C^{(1)} C^{(2)} + 3C^{(2)} C^{(1)} + 6C^{(3)} \right] + \mathcal{O}(\lambda^4) \right) \tag{6.531}$$

To form a relationship between the two, we can connect them order by order in powers of λ, where the standard PT convention tells us that a Σ carries with it a λ, to get

Zeroth order:

$$G^{(0)}_{\text{Dyson}} = G^{(0)}_{\text{Cumulant}} \tag{6.532}$$

$$\implies M^{(0)} \equiv G_0 = G_0 \tag{6.533}$$

$$\implies C^{(0)} = 0 \tag{6.534}$$

First order:

$$G^{(1)}_{\text{Dyson}} = G^{(1)}_{\text{Cumulant}} \tag{6.535}$$

$$\implies G_0 \Sigma G_0 = G_0 C^{(1)} \tag{6.536}$$

$$\implies C^{(1)} = \Sigma G_0 \equiv M^{(1)} \tag{6.537}$$

Beyond the first order, we will have to consider products of lower-order terms as well.

Second order:

$$G^{(2)}_{\text{Dyson}} = G^{(2)}_{\text{Cumulant}} \tag{6.538}$$

$$\implies G_0\Sigma G_0\Sigma G_0 = G_0\left(C^{(2)} + \frac{1}{2!}(C^{(1)})^2\right) \tag{6.539}$$

$$= G_0\left(C^{(2)} + \frac{1}{2!}M^{(1)}\right) \tag{6.540}$$

$$\implies C^{(2)} = \underbrace{\Sigma G_0\Sigma G_0}_{M^{(2)}} - \frac{1}{2}(M^{(1)})^2 \tag{6.541}$$

Third order:

$$G^{(3)}_{\text{Dyson}} = G^{(3)}_{\text{Cumulant}} \tag{6.542}$$

$$\implies G_0\Sigma G_0\Sigma G_0\Sigma G_0 = G_0\left(C^{(3)} + \frac{1}{2}C^{(1)}C^{(2)} + \frac{1}{2}C^{(2)}C^{(1)} + \frac{1}{6}(C^{(1)})^3\right) \tag{6.543}$$

$$= G_0\left(C^{(3)} + \frac{1}{2}M^{(1)}\left(M^{(2)} - \frac{1}{2}(M^{(1)})^2\right) + \frac{1}{2}\left(M^{(2)} - \frac{1}{2}(M^{(1)})^2\right)M^{(1)} + \frac{1}{6}(M^{(1)})^3\right) \tag{6.544}$$

$$\implies C^{(3)} = \underbrace{\Sigma G_0\Sigma G_0\Sigma G_0}_{M^{(3)}} - \frac{1}{2}\left(M^{(1)}M^{(2)} + M^{(2)}M^{(1)}\right) + \frac{1}{3}(M^{(1)})^3 \tag{6.545}$$

This mirrors the third cumulant in statistics:

$$\kappa_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3.$$

6.4.6 GW+C

Using exact self-energy

If you give me the exact self-energy, like $\Sigma = G_0^{-1} - G^{-1}$, this means all my terms in the Dyson expansion are exact; i.e. $G = G_0 + G_0\Sigma G_0 + G_0\Sigma G_0\Sigma G_0 + \dots$ will give the exact G. In other words, I have the exact moments $\mu_{n,k}$ of the interacting Green's function, which in the retarded framework can be written as

$$G^R_k(t) = -i\theta(t)\left\langle\left\{c_k(t), c_k^\dagger(0)\right\}\right\rangle \tag{6.546}$$

$$= -i\theta(t)\left\langle\left\{e^{iHt}c_k e^{-iHt}, c_k^\dagger\right\}\right\rangle \tag{6.547}$$

$$= -i\theta(t)\left\langle\left\{\sum_{n=0}^{\infty}\frac{(-it)^n}{n!}\mathcal{L}^n c_k, c_k^\dagger\right\}\right\rangle \tag{6.548}$$

$$= -i\theta(t)\left(\mu_{0,k} - t\mu_{1,k} + \frac{t^2}{2}\mu_{2,k} + O(t^3)\right) \tag{6.549}$$

where we have defined the moments $\mu_{n,k} = \left\langle\left\{(i\mathcal{L})^n c_k, c_k^\dagger\right\}\right\rangle$ and the Liouvillian superoperator $\mathcal{L}O = [H, O]$. Then, it becomes the mathematical exercise of determining the

cumulant $C_k(t)$ in terms of the moments, such that the Taylor expansion of the exponential in $G_k^R(t) = G_{0,k}^R(t)e^{C_k(t)}$ matches the Dyson expansion of the exact $G_k^R(t)$ up to order n. Given that the exact cumulant can be expanded as $C_k^R(t) = \kappa_{1,k}t + \kappa_{2,k}t^2 + O(t^3)$, we can Taylor expand the exponential $e^{C_k^R(t)}$ and then match coefficients with the above expression for $R_k(t)$ to plug into the usual relations between the cumulants $\kappa_{n,k}$ and moments $\alpha_{n,k}$, to get

$$\kappa_{1,k} = \alpha_{1,k} = -\left(\mu_{1,k} - \mu_{1,k}^0\right) \tag{6.550}$$

$$\kappa_{2,k} = \alpha_{2,k} - \frac{\alpha_{1,k}^2}{2} = \frac{1}{2}\left(\mu_{2,k} - \mu_{2,k}^0\right) - \mu_{1,k}\mu_{1,k}^0 + \left(\mu_{1,k}^0\right)^2 - \frac{1}{2}\left[-\left(\mu_{1,k} - \mu_{1,k}^0\right)\right]^2 \tag{6.551}$$

$$= \frac{1}{2}\left(\mu_{2,k} - \mu_{2,k}^0\right) - \mu_{1,k}\mu_{1,k}^0 + \left(\mu_{1,k}^0\right)^2 - \frac{1}{2}\left(\mu_{1,k} - \mu_{1,k}^0\right)^2 \tag{6.552}$$

$$= \frac{1}{2}\left[\left(\mu_{2,k} - \mu_{2,k}^0\right) + \left(\mu_{1,k}\right)^2 + \left(\mu_{1,k}^0\right)^2\right] \tag{6.553}$$

We know that $\Sigma = G_0^{-1} - G^{-1}$, and inserting this gives
$G_0(\omega)\Sigma(\omega)G_0(\omega) = G_0(\omega)\left(G_0^{-1}(\omega) - G^{-1}(\omega)\right)G_0(\omega) = G_0(\omega) - G_0(\omega)G^{-1}(\omega)G_0(\omega)$

$$C_{pq}(t) = i\int \frac{d\omega}{2\pi} e^{-i(\omega-\epsilon_p^{HF})t}\left[G_0(\omega)\Sigma(\omega)G_0(\omega)\right]_{pq} \tag{6.554}$$

$$= i\int \frac{d\omega}{2\pi} e^{-i(\omega-\epsilon_p^{HF})t}\left[G_0(\omega) - G_0(\omega)G^{-1}(\omega)G_0(\omega)\right]_{pq} \tag{6.555}$$

$$= i\int \frac{d\omega}{2\pi} e^{-i(\omega-\epsilon_p^{HF})t}\left[\frac{1}{\omega - \epsilon_p^{HF} + i\eta} - \left[G_0(\omega)G^{-1}(\omega)G_0(\omega)\right]_{pq}\right] \tag{6.556}$$

$$= \theta(t) - i\int \frac{d\omega}{2\pi} e^{-i(\omega-\epsilon_p^{HF})t}\left[G_0(\omega)G^{-1}(\omega)G_0(\omega)\right]_{pq} \tag{6.557}$$

$$= i\int \frac{d\omega}{2\pi}\left[\frac{e^{-i(\omega-\epsilon_p^{HF})t}}{\omega - \epsilon_p^{HF} + i\eta} - \frac{e^{-i(\omega-\epsilon_p^{HF})t}G_{pq}^{-1}(\omega)}{(\omega - \epsilon_p^{HF} + i\eta)(\omega - \epsilon_q^{HF} + i\eta)}\right] \tag{6.558}$$

$$= \theta(t) - i\int \frac{d\omega}{2\pi}\frac{e^{-i(\omega-\epsilon_p^{HF})t}G_{pq}^{-1}(\omega)}{(\omega - \epsilon_p^{HF} + i\eta)(\omega - \epsilon_q^{HF} + i\eta)} \tag{6.559}$$

Now we plug this into the ansatz for the retarded Green's function:

$$G_{pq}(t) = G_{pp}^{HF}(t)e^{C_{pq}(t)} \tag{6.560}$$

$$= -i\theta(t)e^{-i\epsilon_p^{HF}t}\exp\left[\theta(t) - i\int \frac{d\omega}{2\pi}\frac{e^{-i(\omega-\epsilon_p^{HF})t}G_{pq}^{-1}(\omega)}{(\omega - \epsilon_p^{HF} + i\eta)(\omega - \epsilon_q^{HF} + i\eta)}\right] \tag{6.561}$$

$$\approx -i\theta(t)e^{-i\epsilon_p^{HF}t}\left[1 - i\int \frac{d\omega}{2\pi}\frac{e^{-i(\omega-\epsilon_p^{HF})t}G_{pq}^{-1}(\omega)}{(\omega - \epsilon_p^{HF} + i\eta)(\omega - \epsilon_q^{HF} + i\eta)}\right] \tag{6.562}$$

This doesn't seem useful, so we can start again with eqn. 6.557. We can use a projection technique and split into a HF space where the projector is defined as $\hat{P} = \sum_p^{\text{HF}} |p\rangle\langle p|$, which we can identify with $\begin{pmatrix}1\\0\end{pmatrix}$ and the rest space with projector $\hat{Q} = 1 - \hat{P}$, which we can identify

with $\begin{pmatrix}0\\1\end{pmatrix}$. Then, using the fact that $G^{-1} = G_0^{-1} - \Sigma$ and assuming the HF reference, we can write:

$$G_{pq}^{-1}(\omega) = \left[\hat{P}G^{-1}(\omega)\hat{P} + \hat{P}G^{-1}(\omega)\hat{Q} + \hat{Q}G^{-1}(\omega)\hat{P} + \hat{Q}G^{-1}(\omega)\hat{Q}\right]_{pq} \tag{6.563}$$

$$= \left(\left[\omega - \epsilon_p^{HF}\right]\delta_{pq} - \Sigma_{pq}(\omega)\right)\hat{P}\otimes\hat{P} - \left(\Sigma_{pQ}^c\right)\hat{P}\otimes\hat{Q} - \left(\Sigma_{Pq}^c\right)\hat{Q}\otimes\hat{P} \tag{6.564}$$

$$+ \left((\omega - \epsilon_P^{HF})\delta_{PQ} - \Sigma_{PQ}(\omega)\right)\hat{Q}\otimes\hat{Q} \tag{6.565}$$

$$= \begin{pmatrix}\left[\omega - \epsilon_p^{HF}\right]\delta_{pq} - \Sigma_{pq}(\omega) & -\Sigma_{pQ}^c \\ -\Sigma_{Pq}^c & (\omega - \epsilon_P^{HF})\delta_{PQ} - \Sigma_{PQ}(\omega)\end{pmatrix} \tag{6.566}$$

$$\implies G_{pq}(\omega) = \tag{6.567}$$

where lowercase letters indicate indices in the physical space, while uppercase letters indicate indices in the rest space.

Off-diagonal cumulant with GW self-energy

We can start with

$$C_{pq}(t) = i\int\frac{d\omega}{2\pi}e^{-i(\omega-\epsilon_p^{HF})t}G_{pp}^{HF}(\omega)\Sigma_{pq}^c(\omega)G_{qq}^{HF}(\omega) \tag{6.568}$$

$$= i\int\frac{d\omega}{2\pi}e^{-i(\omega-\epsilon_p^{HF})t}\frac{\Sigma_{pq}^c(\omega)}{(\omega-\epsilon_p^{HF}+i\eta)(\omega-\epsilon_q^{HF}+i\eta)} \tag{6.569}$$

$$\tag{6.570}$$

At this point we make the frequency shift $\omega \to \omega + \epsilon_p^{HF}$, and then we can write

$$C_{pq}(t) = i\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Sigma_{pq}^c(\omega+\epsilon_p^{HF})}{(\omega+i\eta)(\omega+\underbrace{\epsilon_p^{HF}-\epsilon_q^{HF}}_{\Delta_{pq}}+i\eta)} \tag{6.571}$$

Now using the partial fractions we can write $\frac{1}{(\omega+i\eta)(\omega+\Delta_{pq}+i\eta)} = \frac{1}{\Delta_{pq}}\left(\frac{1}{\omega+i\eta} - \frac{1}{\omega+\Delta_{pq}+i\eta}\right)$ and we can plug in the full form for the retarded GW self-energy as

$$\Sigma_{pq}^{c,\mathrm{G_0\ W_0}}(\omega) = \sum_{i\nu}\left[W_{pi\nu}\frac{1}{\omega-(\epsilon_i-\Omega_\nu)+\mathrm{i}\eta}W_{qi\nu}\right] + \sum_{a\nu}\left[W_{pa\nu}\frac{1}{\omega-(\epsilon_a+\Omega_\nu)+\mathrm{i}\eta}W_{qa\nu}\right] \tag{6.572}$$

to get

$$C_{pq}(t) = \frac{i}{\Delta_{pq}} \int \frac{d\omega}{2\pi} e^{-i\omega t} \left[\sum_{i\nu} \frac{W_{pi\nu} W_{qi\nu}}{\omega + \epsilon_p - (\epsilon_i - \Omega_\nu) + \mathrm{i}\eta} + \sum_{a\nu} \frac{W_{pa\nu} W_{qa\nu}}{\omega + \epsilon_p - (\epsilon_a + \Omega_\nu) + \mathrm{i}\eta} \right] \tag{6.573}$$

$$\left(\frac{1}{\omega + i\eta} - \frac{1}{\omega + (\epsilon_p - \epsilon_q) + i\eta} \right)$$

$$= i \sum_{i\nu} \frac{W_{pi\nu} W_{qi\nu}}{\Delta_{pq}} \int \frac{d\omega}{2\pi} e^{-i\omega t} \times \tag{6.574}$$

$$\left(\frac{1}{(\omega + \epsilon_p - (\epsilon_i - \Omega_\nu) + \mathrm{i}\eta)(\omega + i\eta)} - \frac{1}{(\omega + \epsilon_p - (\epsilon_i - \Omega_\nu) + \mathrm{i}\eta)(\omega + (\epsilon_p - \epsilon_q) + i\eta)} \right)$$

$$+ i \sum_{a\nu} \frac{W_{pa\nu} W_{qa\nu}}{\Delta_{pq}} \int \frac{d\omega}{2\pi} e^{-i\omega t} \times$$

$$\left(\frac{1}{(\omega + \epsilon_p - (\epsilon_a + \Omega_\nu) + \mathrm{i}\eta)(\omega + i\eta)} - \frac{1}{(\omega + \epsilon_p - (\epsilon_a + \Omega_\nu) + \mathrm{i}\eta)(\omega + (\epsilon_p - \epsilon_q) + i\eta)} \right)$$

To evaluate these contour integrals, recall that $\oint d\omega f(\omega) = \int_{-\infty}^{\infty} d\omega f(\omega) + \int_{\mathrm{arc}} d\omega f(\omega)$, so in order to equate the real frequency integral to the contour integral, which will then allow us to use the residue theorem, we need to ensure that the integral over the arc vanishes. This only will happen (due to Jordan's Lemma) if the numerator $e^{-i\omega t}$ vanishes for $\omega \to \infty$ and $\omega \to -\infty$; because we have $t > 0$, this is only true if $\mathrm{Im}(\omega) < 0$, and so we must close the contour in the lower half plane.

$$\int \frac{d\omega}{2\pi} e^{-i\omega t} \frac{1}{(\omega + \epsilon_p - (\epsilon_i - \Omega_\nu) + \mathrm{i}\eta)(\omega + i\eta)} = i \left(\frac{e^{-i(\epsilon_i - \Omega_\nu - \epsilon_p)t} - 1}{\epsilon_i - \Omega_\nu - \epsilon_p} \right) \tag{6.575}$$

$$\int \frac{d\omega}{2\pi} e^{-i\omega t} \frac{1}{(\omega + \epsilon_p - (\epsilon_i - \Omega_\nu) + \mathrm{i}\eta)(\omega + \Delta_{pq} + i\eta)} = i \left(\frac{e^{i(\Delta_{pq})t} - e^{-i(\epsilon_i - \Omega_\nu - \epsilon_p)t}}{-\epsilon_i + \Omega_\nu + \epsilon_p - \Delta_{pq}} \right) \tag{6.576}$$

$$= i \left(\frac{e^{i(\epsilon_p - \epsilon_q)t} - e^{-i(\epsilon_i - \Omega_\nu - \epsilon_p)t}}{-\epsilon_i + \Omega_\nu + \epsilon_q} \right)$$

$$\int \frac{d\omega}{2\pi} e^{-i\omega t} \frac{1}{(\omega + \epsilon_p - (\epsilon_a + \Omega_\nu) + \mathrm{i}\eta)(\omega + i\eta)} = i \left(\frac{e^{-i(\epsilon_a + \Omega_\nu - \epsilon_p)t} - 1}{\epsilon_a + \Omega_\nu - \epsilon_p} \right) \tag{6.577}$$

$$\int \frac{d\omega}{2\pi} e^{-i\omega t} \frac{1}{(\omega + \epsilon_p - (\epsilon_a + \Omega_\nu) + \mathrm{i}\eta)(\omega + \Delta_{pq} + i\eta)} = i \left(\frac{e^{i(\Delta_{pq})t} - e^{-i(\epsilon_a + \Omega_\nu - \epsilon_p)t}}{-\epsilon_a - \Omega_\nu + \epsilon_p - \Delta_{pq}} \right) \tag{6.578}$$

$$= i \left(\frac{e^{i(\epsilon_p - \epsilon_q)t} - e^{-i(\epsilon_a + \Omega_\nu - \epsilon_p)t}}{-\epsilon_a - \Omega_\nu + \epsilon_q} \right)$$

To ease the notation, we can introduce $\Xi_{i\nu} \equiv \epsilon_i - \Omega_\nu$, $\Xi_{a\nu} \equiv \epsilon_a + \Omega_\nu$. Then we can write

As a sanity check at this point, we can consider $\lim_{q\to p} T_{i\nu}(\Delta)$ and see if we recover the diagonal result. We can start by defining $f(\Delta) = \frac{e^{-i(\Xi_{i\nu}-\epsilon_p)t}-1}{\Xi_{i\nu}-\epsilon_p} + \frac{e^{i\Delta t}-e^{-i(\Xi_{i\nu}-\epsilon_p)t}}{\Xi_{i\nu}-\epsilon_q}$ and $g(\Delta) = \Delta$, then $\lim_{q\to p} T_{i\nu}(\Delta) = \lim_{q\to p} \frac{f(\Delta)}{g(\Delta)} = \frac{0}{0}$. So we can use L'Hôpital's rule to get $\lim_{q\to p} T_{i\nu}(\Delta) = \lim_{q\to p} \frac{f'(\Delta)}{g'(\Delta)} = \lim_{\Delta\to 0} \frac{f'(\Delta)}{1} = \lim_{\Delta\to 0} \frac{\left(ite^{i\Delta t}\right)(\Xi_{i\nu}-\epsilon_p+\Delta)-\left(e^{i\Delta t}-e^{-i(\Xi_{i\nu}-\epsilon_p)t}\right)}{(\Xi_{i\nu}-\epsilon_p+\Delta)^2} = \frac{i(\Xi_{i\nu}-\epsilon_p)t-1+e^{-i(\Xi_{i\nu}-\epsilon_p)t}}{(\Xi_{i\nu}-\epsilon_p)^2}$, which is indeed the diagonal result.

$$C_{pq}(t) = i^2 \left[\sum_{i\nu} W_{pi\nu} W_{qi\nu} \left(\underbrace{\frac{1}{\Delta}\left[\frac{e^{-i(\Xi_{i\nu}-\epsilon_p)t}-1}{\Xi_{i\nu}-\epsilon_p} + \frac{e^{i\Delta t}-e^{-i(\Xi_{i\nu}-\epsilon_p)t}}{\Xi_{i\nu}-\epsilon_q} \right]}_{T_{i\nu}(t)} \right) \right. \tag{6.579}$$

$$\left. + \sum_{a\nu} W_{pa\nu} W_{qa\nu} \left(\underbrace{\frac{1}{\Delta}\left[\frac{e^{-i(\Xi_{a\nu}-\epsilon_p)t}-1}{\Xi_{a\nu}-\epsilon_p} + \frac{e^{i\Delta t}-e^{-i(\Xi_{a\nu}-\epsilon_p)t}}{\Xi_{a\nu}-\epsilon_q} \right]}_{T_{a\nu}(t)} \right) \right] \tag{6.580}$$

Now we just work with

$$T_{i\nu}(t) = \frac{(\Xi_{i\nu}-\epsilon_q)\left[e^{-i(\Xi_{i\nu}-\epsilon_p)t}-1\right] + (\Xi_{i\nu}-\epsilon_p)\left[e^{i(\epsilon_p-\epsilon_q)t}-e^{-i(\Xi_{i\nu}-\epsilon_p)t}\right]}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} \tag{6.581}$$

$$= \frac{-(\Xi_{i\nu}-\epsilon_q) + (\Xi_{i\nu}-\epsilon_p)e^{i(\epsilon_p-\epsilon_q)t} + (\epsilon_p-\epsilon_q-2\Xi_{i\nu})e^{-i(\Xi_{i\nu}-\epsilon_p)t}}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} \tag{6.582}$$

$$= -\frac{1}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)} + \frac{e^{i(\epsilon_p-\epsilon_q)t}}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_q)} + \frac{e^{-i(\Xi_{i\nu}-\epsilon_p)t}}{(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} - \frac{2\Xi_{i\nu}e^{-i(\Xi_{i\nu}-\epsilon_p)t}}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)}. \tag{6.583}$$

and we can do the analogous computation to get $T_{a\nu}(t)$. Now we can plug into the expression for the retarded Green's function to get

$$G_{pq}(t) = G^{HF}_{pq}(t)e^{C_{pq}(t)} \tag{6.584}$$

$$= -i\Theta(t)e^{-i\epsilon_p t + C_{pq}(t)} \tag{6.585}$$

$$= -i\Theta(t)\exp\left[-i\epsilon_p t - \sum_{i\nu} W_{pi\nu}W_{qi\nu}T_{i\nu}(t) - \sum_{a\nu} W_{pa\nu}W_{qa\nu}T_{a\nu}(t)\right] \tag{6.586}$$

$$\tag{6.587}$$

The Fourier transform to the frequency domain is given by

$$G_{pq}(\omega) = \int dt e^{i\omega t} G_{pq}(t) \tag{6.588}$$

$$= -i \int_0^\infty dt e^{i(\omega-\epsilon_p)t} \exp\left\{ -\sum_{i\nu} W_{pi\nu} W_{qi\nu} T_{i\nu}(t) - \sum_{a\nu} W_{pa\nu} W_{qa\nu} T_{a\nu}(t) \right\} \tag{6.589}$$

$$\approx -i \int_0^\infty dt e^{i(\omega-\epsilon_p)t} \left[1 - \sum_{i\nu} W_{pi\nu} W_{qi\nu} T_{i\nu}(t) - \sum_{a\nu} W_{pa\nu} W_{qa\nu} T_{a\nu}(t) \right] \tag{6.590}$$

$$\tag{6.591}$$

where in the last step we have made a Taylor expansion of the exponential, keeping just the 0th and 1st order terms. This is the right thing to do because in one of our first steps we chose to make the expression exact up to the first order in the screened Coulomb interaction W. If we were to expand beyond the first order, we would be including terms beyond first order in W. So the form for the off-diagonal GW+C spectral function would be

$$A_{pq}(\omega) = -\frac{1}{\pi} \mathrm{Im} G_{pq}(\omega) \tag{6.592}$$

$$\tag{6.593}$$

so we would need to determine

$$\mathrm{Im}\, G_{pq}(\omega) \approx -\,\mathrm{Re} \int_0^\infty dt \left[e^{i(\omega-\epsilon_p)t} \left(1 - \sum_{i\nu} W_{pi\nu} W_{qi\nu} T_{i\nu}(t) - \sum_{a\nu} W_{pa\nu} W_{qa\nu} T_{a\nu}(t) \right) \right] \tag{6.594}$$

$$= -\pi\delta(\omega-\epsilon_p) + \sum_{i\nu} W_{pi\nu} W_{qi\nu}\, \mathrm{Re} \int_0^\infty dt e^{i(\omega-\epsilon_p)t} T_{i\nu}(t) + \sum_{a\nu} W_{pa\nu} W_{qa\nu}\, \mathrm{Re} \int_0^\infty dt e^{i(\omega-\epsilon_p)t} T_{a\nu}(t) \tag{6.595}$$

which leaves us with the task of computing

$$\mathrm{Re} \int_0^\infty dt e^{i(\omega-\epsilon_p)t} T_{i\nu}(t) = -\frac{\pi\delta(\omega-\epsilon_p)}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)} + \frac{\pi\delta(\omega-\epsilon_q)}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_q)} \tag{6.596}$$

$$+ \frac{\pi\delta(\omega-\Xi_{i\nu})}{(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} - \frac{\pi 2\Xi_{i\nu}\delta(\omega-\Xi_{i\nu})}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} \tag{6.597}$$

and similarly for $T_{a\nu}(t)$. So

$$A_{pq}(\omega) \approx \delta(\omega-\epsilon_p) - \sum_{i\nu} W_{pi\nu} W_{qi\nu} \left[-\frac{\delta(\omega-\epsilon_p)}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)} + \frac{\delta(\omega-\epsilon_q)}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_q)} + \frac{\delta(\omega-\Xi_{i\nu})}{(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} \right. \tag{6.598}$$

$$\left. - \frac{2\Xi_{i\nu}\delta(\omega-\Xi_{i\nu})}{(\epsilon_p-\epsilon_q)(\Xi_{i\nu}-\epsilon_p)(\Xi_{i\nu}-\epsilon_q)} \right] - \sum_{a\nu} W_{pa\nu} W_{qa\nu} \left[-\frac{\delta(\omega-\epsilon_p)}{(\epsilon_p-\epsilon_q)(\Xi_{a\nu}-\epsilon_p)} + \frac{\delta(\omega-\epsilon_q)}{(\epsilon_p-\epsilon_q)(\Xi_{a\nu}-\epsilon_q)} \right. \tag{6.599}$$

$$\left. + \frac{\delta(\omega-\Xi_{a\nu})}{(\Xi_{a\nu}-\epsilon_p)(\Xi_{a\nu}-\epsilon_q)} - \frac{2\Xi_{a\nu}\delta(\omega-\Xi_{a\nu})}{(\epsilon_p-\epsilon_q)(\Xi_{a\nu}-\epsilon_p)(\Xi_{a\nu}-\epsilon_q)} \right]$$

We can replace all of the delta functions with Lorentzians in practice.

6.4.7 Assorted

MKCT

$$\Omega_n = \frac{\left((\mathrm{i}\mathcal{L})^n \hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})}, \tag{6.600}$$

with the corresponding auxiliary kernels

$$K_n(t) = \frac{\left((\mathrm{i}\mathcal{L})^n \hat{f}(t), \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.601}$$

So we can show the main result of their first paper that the higher-order kernels satisfy the following coupled ordinary differential equation (ODE):

$$\dot{K}_n(t) = \frac{\left((\mathrm{i}\mathcal{L})^n \dot{\hat{f}}(t), \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.602}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^n \mathrm{i}\mathcal{Q}\mathcal{L}\hat{f}(t), \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.603}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^{n+1} \hat{f}(t), \hat{A}\right)}{(\hat{A}, \hat{A})} - \frac{\left((\mathrm{i}\mathcal{L})^n \mathrm{i}\mathcal{P}\mathcal{L}\hat{f}(t), \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.604}$$

$$= K_{n+1}(t) - \frac{(\mathrm{i}\mathcal{L}\hat{f}(t), \hat{A})}{(\hat{A}, \hat{A})} \times \frac{\left((\mathrm{i}\mathcal{L})^n \hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.605}$$

$$= K_{n+1}(t) - K_1(t)\Omega_n, \tag{6.606}$$

where we have used the fact that the random fluctuation operator is $\hat{f}(t) = e^{\mathrm{i}t\mathrm{QC}}\mathcal{Q}\mathrm{i}\mathcal{L}\hat{A} \Longrightarrow \dot{\hat{f}}(t) = \mathrm{i}\mathcal{Q}\mathcal{L}\hat{f}(t)$ and we can deduce the initial conditions in a similar fashion with $\hat{f}(0) = \mathcal{Q}i\mathcal{L}\hat{A}$, so

$$K_n(0) = \frac{\left((\mathrm{i}\mathcal{L})^n \mathrm{Q}\mathrm{i}\mathcal{L}\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.607}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^n (\mathrm{i}\mathcal{L}\hat{A} - \mathrm{P}\mathrm{i}\mathcal{L}\hat{A}), \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.608}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^{n+1} \hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} - \frac{\left((\mathrm{i}\mathcal{L})^n \mathrm{P}\mathrm{i}\mathcal{L}\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.609}$$

$$= \Omega_{n+1} - \Omega_1\Omega_n, \tag{6.610}$$

suggesting that the central quantity to compute is the $\{\Omega_n\}$. But an issue with this ODE is that it extends to infinite order, and hard truncation to finite order can lead to numerical instabilities. To this end, they introduce a truncation scheme with Padé approximants. First, they noticed that the m-th derivative of kernel $K_n(t)$ evaluated at $t = 0$ is

$$K_n^{(m)} = \frac{\left((\mathrm{i}\mathcal{L})^n(\mathcal{Q}\mathrm{i}\mathcal{L})^{m+1}\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.611}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^n\mathcal{Q}\mathrm{i}\mathcal{L}(\mathcal{Q}\mathrm{i}\mathcal{L})^m\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.612}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^n\left(1-\mathcal{P}\right)\mathrm{i}\mathcal{L}(\mathcal{Q}\mathrm{i}\mathcal{L})^m\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.613}$$

$$= \frac{\left((\mathrm{i}\mathcal{L})^{n+1}(\mathcal{Q}\mathrm{i}\mathcal{L})^m\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} - \frac{\left((\mathrm{i}\mathcal{L})^n\mathcal{P}\mathrm{i}\mathcal{L}(\mathcal{Q}\mathrm{i}\mathcal{L})^m\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.614}$$

$$= K_{n+1}^{(m-1)} - \frac{\left(\mathrm{i}\mathcal{L}(\mathcal{Q}\mathrm{i}\mathcal{L})^m\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \times \frac{\left((\mathrm{i}\mathcal{L})^n\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} \tag{6.615}$$

$$= K_{n+1}^{(m-1)} - \bar{\Omega}_m\Omega_n \tag{6.616}$$

where we introduced the auxiliary moment $\bar{\Omega}_m = \frac{\left(\mathrm{i}\mathcal{L}(\mathcal{Q}\mathrm{i}\mathcal{L})^m\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})}$. So recursively applying this expression leads to the relation:

$$K_n^{(m)} = K_{n+2}^{(m-2)} - \bar{\Omega}_{m-1}\Omega_{n+1} - \bar{\Omega}_m\Omega_n \tag{6.617}$$

$$= K_{n+m}^{(0)} - \sum_{j=0}^{m-1}\bar{\Omega}_{m-j}\Omega_{n+j} \tag{6.618}$$

$$= \Omega_{n+m+1} - \Omega_1\Omega_{n+m} - \sum_{j=0}^{m-1}\bar{\Omega}_{m-j}\Omega_{n+j} \tag{6.619}$$

$$= \Omega_{n+m+1} - \bar{\Omega}_0\Omega_{n+m} - \sum_{j=0}^{m-1}\bar{\Omega}_{m-j}\Omega_{n+j} \tag{6.620}$$

$$= \Omega_{n+m+1} - \sum_{j=0}^{m}\bar{\Omega}_{m-j}\Omega_{n+j} \tag{6.621}$$

$$\tag{6.622}$$

that $K_n^{(m)}$ can be expressed with moments and auxiliary moments. Similarly, the auxiliary moments themselves have recursions

$$\begin{aligned}\tilde{\Omega}_m &= \frac{\left((\mathrm{i}\mathcal{L}\mathcal{Q})^m \mathrm{i}\mathcal{L}\hat{A}, \hat{A}\right)}{(\hat{A}, \hat{A})} - \tilde{\Omega}_{m-1}\Omega_1 \\ &= \Omega_{m+1} - \sum_{j=0}^{m-1} \tilde{\Omega}_j \Omega_{m-j}\end{aligned} \tag{6.623}$$

which means the auxiliary moments $\{\tilde{\Omega}_m\}$ can be obtained by the moments $\{\Omega_n\}$; it becomes a problem of efficiently computing the moments. The series expansion for the n-th order auxiliary kernel can then be given by Padé approximant. Initially, $K_n(t)$ is expressed as a truncated Taylor series:

$$K_n(t) \approx \sum_{j=0}^{M} \frac{K_n^{(j)}(0)}{j!} t^j, \tag{6.624}$$

which is a good local approximation but lacks accuracy over a broader range of t. A more reliable approximation can be achieved using the Padé approximant:

$$K_n(t) \approx \frac{p_{M_1}(t)}{q_{M_2}(t)} = \frac{a_0 + a_1 t + \cdots + a_{M_1} t^{M_1}}{1 + b_1 t + \cdots + b_{M_2} t^{M_2}}, \tag{6.625}$$

where $p_{M_1}(t)$ and $q_{M_2}(t)$ are polynomials of degrees M_1 and M_2, respectively. The coefficients $\{a_i\}$ and $\{b_i\}$ are computed using the python library SciPy, which implements the standard Padé approximant procedure as described in Ref. [**?**]. Overall, Eq. 6.625 provides a numerically stable truncation for the MKCT Eq. **??**, where *all* coefficients can be evaluated with higher-order moments $\{\Omega_n\}$.

Relation of improper self energy to cumulant

The Dyson equation for the one particle greens function can be written using an improper Σ^I self energy as

$$G_k(t,t') = G_k^0(t,t') + G_k^0(t,t')\Sigma_k^I(t,t')G_k^0(t,t'), \tag{6.626}$$

$$\tag{6.627}$$

instead of using the proper self energy Σ^*, which satisfies $G_k(t,t') = G_k^0(t,t') + G_k^0(t,t')\Sigma_k^*(t,t')G_k(t,t')$. Now, the retarded cumulant ansatz can be written as

$$G_k^R(t,t') = G_k^{0,R}(t,t') e^{C_k^R(t,t')} \tag{6.628}$$

$$\implies \frac{G_k^R(t,t')}{G_k^{0,R}(t,t')} = e^{C_k^R(t,t')} \tag{6.629}$$

$$\tag{6.630}$$

Let's deal directly with this ratio instead of needing to introduce a self energy. The retarded Green's function can be written as

$$G_k^R(t) = -i\theta(t)\left\langle\left\{c_k(t), c_k^\dagger(0)\right\}\right\rangle \tag{6.631}$$

$$= -i\theta(t)\left\langle\left\{e^{iHt}c_k e^{-iHt}, c_k^\dagger\right\}\right\rangle \tag{6.632}$$

$$= -i\theta(t)\left\langle\left\{\sum_{n=0}^{\infty}\frac{(-it)^n}{n!}\mathcal{L}^n c_k, c_k^\dagger\right\}\right\rangle \tag{6.633}$$

$$= -i\theta(t)\left(\mu_{0,k} - t\mu_{1,k} + \frac{t^2}{2}\mu_{2,k} + O(t^3)\right) \tag{6.634}$$

where we have defined the moments $\mu_{n,k} = \left\langle\left\{(i\mathcal{L})^n c_k, c_k^\dagger\right\}\right\rangle$ and the Liouvillian superoperator $\mathcal{L}O = [H, O]$. The non-interacting retarded Green's function is given by

$$G_k^{R,0}(t) = -i\theta(t)\left(\mu_{0,k}^0 - t\mu_{1,k}^0 + \frac{t^2}{2}\mu_{2,k}^0 + O(t^3)\right) \tag{6.635}$$

where the non-interacting moments are $\mu_{n,k}^0 = \left\langle\left\{(i\mathcal{L}_0)^n c_k, c_k^\dagger\right\}\right\rangle$ and $\mathcal{L}_0 O = [H_0, O]$. Defining $A(t) = 1 - t\mu_{1,k} + \frac{t^2}{2}\mu_{2,k} + O(t^3)$ and $B(t) = 1 - t\mu_{1,k}^0 + \frac{t^2}{2}\mu_{2,k}^0 + O(t^3)$ we can use a geometric series to write

$$\frac{1}{B(t)} = 1 + t\mu_{1,k}^0 + t^2\left[\left(\mu_{1,k}^0\right)^2 - \frac{1}{2}\mu_{2,k}^0\right] + O\left(t^3\right) \tag{6.636}$$

Note that the inverse power series coefficients can be computed by Wronski's formula (if one is interested in still higher orders of t). So, we can write the ratio and then group terms by order in t, to get

$$R_k(t) = \frac{G_k^R(t)}{G_k^{R,0}(t)} = \frac{-i\theta(t)}{-i\theta(t)} \times \frac{A(t)}{B(t)} \tag{6.637}$$

$$= \left(1 - t\mu_{1,k} + \frac{t^2}{2}\mu_{2,k} + O(t^3)\right)\left(1 + t\mu_{1,k}^0 + t^2\left[\left(\mu_{1,k}^0\right)^2 - \frac{1}{2}\mu_{2,k}^0\right] + O\left(t^3\right)\right) \tag{6.638}$$

$$= \left(1 + t\underbrace{\left[-\left(\mu_{1,k} - \mu_{1,k}^0\right)\right]}_{\alpha_{1,k}} + t^2\underbrace{\left[\frac{1}{2}\left(\mu_{2,k} - \mu_{2,k}^0\right) - \mu_{1,k}\mu_{1,k}^0 + \left(\mu_{1,k}^0\right)^2\right]}_{\alpha_{2,k}} + O(t^3)\right) \tag{6.639}$$

Given that the cumulant can be expanded as $C_k^R(t) = \kappa_{1,k}t + \kappa_{2,k}t^2 + O(t^3)$, we can Taylor expand the exponential $e^{C_k^R(t)}$ and then match coefficients with the above expression for

$R_k(t)$ to plug into the usual relations between the cumulants $\kappa_{n,k}$ and moments $\alpha_{n,k}$, to get

$$\kappa_{1,k} = \alpha_{1,k} = -\left(\mu_{1,k} - \mu_{1,k}^0\right) \tag{6.640}$$

$$\kappa_{2,k} = \alpha_{2,k} - \frac{\alpha_{1,k}^2}{2} = \frac{1}{2}\left(\mu_{2,k} - \mu_{2,k}^0\right) - \mu_{1,k}\mu_{1,k}^0 + \left(\mu_{1,k}^0\right)^2 - \frac{1}{2}\left[-\left(\mu_{1,k} - \mu_{1,k}^0\right)\right]^2 \tag{6.641}$$

$$= \frac{1}{2}\left(\mu_{2,k} - \mu_{2,k}^0\right) - \mu_{1,k}\mu_{1,k}^0 + \left(\mu_{1,k}^0\right)^2 - \frac{1}{2}\left(\mu_{1,k} - \mu_{1,k}^0\right)^2 \tag{6.642}$$

$$= \frac{1}{2}\left[\left(\mu_{2,k} - \mu_{2,k}^0\right) + \left(\mu_{1,k}\right)^2 + \left(\mu_{1,k}^0\right)^2\right] \tag{6.643}$$

So now we need to determine how to compute the moments for this cumulant approach and potentially for the MKCT approach. But let's begin by discerning the form for the 2/2 Padé approximant to the cumulant for the memory kernel at n=1:

$$K_1(t) \approx \frac{a_0 + a_1 t + a_2 t^2}{1 + b_1 t + b_2 t^2} \tag{6.644}$$

$$\approx \left(a_0 + a_1 t + a_2 t^2\right)\left(1 - b_1 t + \left(b_1^2 - b_2\right) t^2 + O(t^3)\right) \tag{6.645}$$

$$\approx a_0 + t\left(a_1 - a_0 b_1\right) + t^2\left(a_2 - a_1 b_1 + a_0\left(b_1^2 - b_2\right)\right) + O(t^3) \tag{6.646}$$

with the derivatives given by $c_0 = K_1(0)$, $c_1 = K_1^{(1)}(0)$, and $c_2 = \frac{K_1^{(2)}(0)}{2}$. Matching coefficients of t^0, t^1, and t^2 gives us the following system of equations:

$$c_0 = a_0 \tag{6.647}$$

$$c_1 = a_1 - a_0 b_1 \implies a_1 = c_1 + a_0 b_1 \tag{6.648}$$

$$c_2 = a_2 - a_1 b_1 + a_0\left(b_1^2 - b_2\right) \implies a_2 = c_2 + a_1 b_1 - a_0\left(b_1^2 - b_2\right) \tag{6.649}$$

We can find that $a_0 = K_1(0)$, $a_1 = K_1^{(1)}(0) - \frac{K_1(0)K_1^{(2)}(0)}{2K_1^{(1)}(0)}$, and $b_1 = -\frac{K_1^{(2)}(0)/2}{K_1^{(1)}(0)}$. We start by computing

$$K_1(0) = \Omega_2 - \Omega_1^2 = \mu_{2,k} - \mu_{1,k}^2 \tag{6.650}$$

$$K_1^{(1)}(0) = K_2(0) - \bar{\Omega}_1\Omega_1 = \Omega_3 - \Omega_1\Omega_2 - \bar{\Omega}_1\Omega_1 = \mu_{3,k} - \mu_{1,k}\mu_{2,k} - \bar{\Omega}_1\mu_{1,k} \tag{6.651}$$

$$\tag{6.652}$$

Fundamental starting point

We start with the definition of the retarded Green's function in the time domain:

$$G_k^R(t) \equiv -i\theta(t)\left\langle\left\{c_k(t), c_k^\dagger(0)\right\}\right\rangle. \tag{6.653}$$

Immediately, we can Fourier transform to the frequency domain:

$$G_k^R(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} G_k^R(t). \tag{6.654}$$

But because $G_k^R(t)$ is already zero for $t < 0$ (by the $\theta(t)$), this integral reduces to

$$G_k^R(\omega) = -i \int_0^\infty dt e^{i\omega t} \left\langle \left\{ c_k(t), c_k^\dagger(0) \right\} \right\rangle \tag{6.655}$$

$$= -i \int_0^\infty dt e^{i\omega t} \sum_{n=0}^\infty \frac{(it)^n}{n!} \mu_{n,k} \tag{6.656}$$

$$= -i \sum_{n=0}^\infty \frac{i^n}{n!} \mu_{n,k} \int_0^\infty dt e^{i\omega t} t^n \tag{6.657}$$

$$= -i \sum_{n=0}^\infty \frac{i^n}{n!} \mu_{n,k} \frac{n!}{(i\omega)^{n+1}} \tag{6.658}$$

$$= -\sum_{n=0}^\infty \frac{\mu_{n,k}}{\omega^{n+1}}, \tag{6.659}$$

where we have defined the moments $\mu_{n,k} = \left\langle \left\{ \mathcal{L}^n c_k, c_k^\dagger \right\} \right\rangle$. This works because $c_k(t) = e^{iHt} c_k e^{-iHt} = e^{i\mathcal{L}t} c_k$, where $\mathcal{L}$ is the Liouvillian superoperator, so the time evolution operator can be expanded in a Taylor series, as $e^{i\mathcal{L}t} = \sum_{n=0}^\infty \frac{(it)^n}{n!} \mathcal{L}^n \implies \left\langle \left\{ c_k(t), c_k^\dagger(0) \right\} \right\rangle = \sum_{n=0}^\infty \frac{(it)^n}{n!} \mu_{n,k}$. The complementary projection operator is $\mathcal{Q} = 1 - \mathcal{P}$. The Liouvillian superoperator acts on an arbitrary operator X as $\mathcal{L}X = [H, X]$. Since we know that Mori's is formally a projector method with $\mathcal{P}X = \frac{(X, c_k)}{(c_k, c_k)} c_k$, where $(A, B) = \langle \{A, B^\dagger\} \rangle$ is the fermionic Mori inner product, so $(c_k, c_k) = \langle \{c_k, c_k^\dagger\} \rangle = 1$. Lanczos is also formally a projection onto the Krylov subspace, so we can try to exploit this by devising a Lanczos procedure. Then we can initiate a Lanczos sequence with $|f_0\rangle = c_k$ and then for $n = 0$, we have

$$|f_1\rangle = \mathcal{L}|f_0\rangle - a_0 |f_0\rangle, \tag{6.660}$$

where $a_0 = \frac{(\mathcal{L} f_0, f_0)}{(f_0, f_0)} = \mu_{1,k}$. Then for $n \geq 1$ define

$$|f_{n+1}\rangle = \mathcal{L}|f_n\rangle - a_n |f_n\rangle - b_n^2 |f_{n-1}\rangle, \tag{6.661}$$

where the coefficients are given by $a_n = \frac{(\mathcal{L} f_n, f_n)}{(f_n, f_n)}$ and $b_{n+1}^2 = \frac{(f_{n+1}, f_{n+1})}{(f_n, f_n)}$. Let us try to determine the action of the Liouvillian onto a given state $|f_n\rangle$. We can write

$$\mathcal{L}|f_n\rangle = [H_0 + V, |f_n\rangle] \tag{6.662}$$

$$= \mathcal{L}_0 |f_n\rangle + [V, |f_n\rangle] \tag{6.663}$$

$$\tag{6.664}$$

where H_0 is the non-interacting Hamiltonian and V is the interaction. We can also define a non-interacting Liouvillian $\mathcal{L}_0$ such that $\mathcal{L}_0 X = [H_0, X]$. So we can write

6.5 Derivation of the cumulant starting from the electron-boson Hamiltonian

The retarded one-particle Green's function for a state p can be expressed as

$$G_{pp}^{R}(t) = -i\theta(t) \langle N | \hat{c}_p e^{-i\hat{H}t} \hat{c}_p^{\dagger} | N \rangle \tag{6.665}$$

where $|N\rangle$ is the exact ground state and $\hat{H}$ is the Hamiltonian, both of the N-electron system.

Core hole approximation

The exact ground state $|N\rangle \equiv |N_F\rangle \otimes |0_B\rangle$, where $|N_F\rangle$ is the exact fermionic ground state and $|0_B\rangle$ is the bosonic vacuum. $\hat{c}_p$ and $\hat{c}_p^{\dagger}$ will be the annihilation and creation operators for the fermionic hole state p. So, we can write

$$\hat{c}_p^{\dagger} |N\rangle = \hat{c}_p^{\dagger} \Big(|N_F\rangle \otimes |0_B\rangle \Big) \tag{6.666}$$

$$\approx \hat{c}_p^{\dagger} \Big(|\Phi_0^N\rangle \otimes |0_B\rangle \Big) \tag{6.667}$$

$$= |\Phi_0^{N-1}\rangle \otimes |0_B\rangle \tag{6.668}$$

$$\equiv |\Psi_p\rangle \tag{6.669}$$

where in 6.667 we approximated the exact vacuum by the Fermi vacuum.

6.5.1 Specifying the effective Hamiltonian

We can choose the electronic-bosonic Hamiltonian $\hat{H}_p^{\mathrm{eB}}$ for the N-1 electron system with a hole in core state p. ϵ_p is the energy of the state p, Ω_ν are the bosonic excitation energies, and $W_{p\nu}$ are the coupling matrix elements between the fermionic hole state and the bosonic excitations

$$\hat{H}_{\mathrm{eB}} = \epsilon_p \hat{n}_p + \sum_{\nu} \Omega_\nu \hat{b}_\nu^{\dagger} \hat{b}_\nu + \sum_{\nu} W_{p\nu} \hat{n}_p \left(\hat{b}_\nu^{\dagger} + \hat{b}_\nu \right) \tag{6.670}$$

$$= \hat{H}_0 + \hat{V}, \tag{6.671}$$

defining $\hat{n}_p = \hat{c}_p^{\dagger} \hat{c}_p$ as the number operator for the core hole state p. We can partition into a non-interacting part $\hat{H}_0$ and an interaction $\hat{V}$ as

$$\hat{H}_0 = \epsilon_p \hat{n}_p + \sum_{\nu} \Omega_\nu \hat{b}_\nu^{\dagger} \hat{b}_\nu \tag{6.672}$$

$$\hat{V} = \sum_{\nu} W_{p\nu} \hat{n}_p \left(\hat{b}_\nu^{\dagger} + \hat{b}_\nu \right). \tag{6.673}$$

Plugging this into the definition of the retarded Green's function gives us

$$G_{pp}^{R}(t) \approx -i\theta(t) \langle \Psi_p | e^{-i\hat{H}_{\mathrm{eB}} t} | \Psi_p \rangle \tag{6.674}$$

6.5.2 Connecting to the CCGF

We can start with the definition of the cumulant retarded Green's function, which derives from the core-hole approximation, and we will call it $G_p^{R,CE}(t)$. Now, performing a double similarity transformation of the (electron-boson) Hamiltonian, we can also define an XCC retarded Green's function in the core-hole approximation $G_p^{R,XCC}(t)$:

$$G_p^{R,CE}(t) \equiv -i\theta(t)\left\langle \Psi_p \right| \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}\left|\Psi_p\right\rangle \tag{6.675}$$

$$\implies G_p^{R,XCC}(t) \equiv -i\theta(t)\left\langle \Psi_p \right| \exp\Big\{-i\bar{\bar{H}}_{\mathrm{eB}}t\Big\}\left|\Psi_p\right\rangle \tag{6.676}$$

$$= -i\theta(t)\left\langle \Psi_p \right| \exp\Big\{-ie^{\hat{Z}}e^{-\hat{T}}\hat{H}_{\mathrm{eB}}e^{\hat{T}}e^{-\hat{Z}}t\Big\}\left|\Psi_p\right\rangle \tag{6.677}$$

$$= -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)\hat{c}_p^\dagger e^{\hat{Z}}e^{-\hat{T}} \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}e^{\hat{T}}e^{-\hat{Z}}\hat{c}_p\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.678}$$

At this point, we can notice that all similarity trasnformations are (quasi)bosonic, so they commute with the fermionic creation and annihilation operators. Hence, we can write

$$G_p^{R,XCC}(t) = -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)e^{\hat{Z}}e^{-\hat{T}}\hat{c}_p^\dagger \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}\hat{c}_p e^{\hat{T}}e^{-\hat{Z}}\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.679}$$

$$= -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)e^{\hat{Z}}e^{-\hat{T}}\hat{c}_p^\dagger \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}\hat{c}_p e^{\hat{T}}\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.680}$$

$$\tag{6.681}$$

where we used the fact that $e^{-\hat{Z}} = 1-\hat{Z}+\mathcal{O}(\hat{Z}^2)$, but because a bosonic deexcitation operator annihilates the bosonic vacuum, we just keep the leading identity term in the expansion of $e^{-\hat{Z}}$. Now, we can insert the identity $\hat{1} = e^{\hat{T}}e^{-\hat{T}}$ twice to get

$$G_p^{R,XCC}(t) = -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)e^{\hat{Z}}e^{-\hat{T}}\hat{c}_p^\dagger e^{\hat{T}}e^{-\hat{T}} \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}e^{\hat{T}}e^{-\hat{T}}\hat{c}_p e^{\hat{T}}\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.682}$$

$$= -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)e^{\hat{Z}}\bar{\hat{c}}_p^\dagger \exp\Big\{-i\bar{\hat{H}}_{\mathrm{eB}}t\Big\}\bar{\hat{c}}_p\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.683}$$

Now, the CCGF is given by

$$G_p^{R,CC}(t) = i\left\langle \Phi \right| (1+\Lambda)\bar{a}_p^\dagger e^{i\bar{H}_N t}\bar{a}_p \Big|\Phi\Big\rangle \tag{6.684}$$

$$\overset{CC}{\approx} -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)c_p^\dagger\Big[1+\hat{\Lambda}\Big]e^{-\hat{T}} \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}e^{\hat{T}}\Big[1-\hat{\Lambda}\Big]c_p\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.685}$$

$$= -i\theta(t)\Big(\left\langle \Phi_N^0\right| \otimes \left\langle 0_B\right|\Big)c_p^\dagger\Big[1+\hat{\Lambda}\Big]e^{-\hat{T}} \exp\Big\{-i\hat{H}_{\mathrm{eB}}t\Big\}e^{\hat{T}}c_p\Big(\left|\Phi_N^0\right\rangle \otimes \left|0_B\right\rangle\Big) \tag{6.686}$$

Plugging this into the definition of the retarded Green's function gives us

$$G^R_{pp}(t) \approx -i\theta(t)\,\langle\Psi_p|\,e^{-i\hat{H}_{\rm eB}t}\,|\Psi_p\rangle \tag{6.687}$$

$$= -i\theta(t)\,\langle\Psi_p|\,e^{-i\left(\hat{H}_0+\hat{V}\right)t}\,|\Psi_p\rangle \tag{6.688}$$

$$= -i\theta(t)\,\langle\Psi_p|\,e^{-i\hat{H}_0t}\mathcal{T}\exp\left(-i\int_0^t dt' e^{i\hat{H}_0t'}\hat{V}e^{-i\hat{H}_0t'}\right)|\Psi_p\rangle \tag{6.689}$$

$$= -i\theta(t)e^{-i\epsilon_p t}\underbrace{\langle\Psi_p|\,\mathcal{T}\exp\left(-i\int_0^t dt' V_I(t')\right)|\Psi_p\rangle}_{*} \tag{6.690}$$

where $V_I(t) = e^{i\hat{H}_0t}\hat{V}e^{-i\hat{H}_0t}$ is the interaction picture representation of the interaction $\hat{V}$. *Is there a relationship between this interaction picture representation and the CC similarity transformation?* Now, we can work out * by a Taylor expansion of the time-ordered exponential:

$$* = \langle\Psi_p|\sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int_0^t dt_1\cdots\int_0^t dt_n\mathcal{T}\left[V_I(t_1)\cdots V_I(t_n)\right]|\Psi_p\rangle \tag{6.691}$$

$$= 1 + (-i)\int_0^t dt'\,\langle\Psi_p|\,V_I(t')\,|\Psi_p\rangle + \frac{(-i)^2}{2!}\int_0^t dt_1\int_0^t dt_2\,\langle\Psi_p|\,\mathcal{T}\left[V_I(t_1)V_I(t_2)\right]|\Psi_p\rangle + O(W^3) \tag{6.692}$$

$$= 1 - \frac{1}{2}\int_0^t dt_1\int_0^t dt_2\,\langle\Psi_p|\,\mathcal{T}\left[V_I(t_1)V_I(t_2)\right]|\Psi_p\rangle + O(W^4) \tag{6.693}$$

where in 6.727 we have used the fact that terms with an odd number of V_I's will vanish because they do not conserve the number of bosons. Because $[\hat{H}_0, \hat{n}_p] = 0$, we can evaluate $V_I(t)$ as

$$\hat{V}_I(t) = e^{i\hat{H}_0t}\Big[\sum_\nu W_{p\nu}\,\hat{n}_p(\hat{b}^\dagger_\nu + \hat{b}_\nu)\Big]e^{-i\hat{H}_0t} \tag{6.694}$$

$$= \sum_\nu W_{p\nu}\,\hat{n}_p\Big(e^{i\hat{H}_0t}\hat{b}^\dagger_\nu e^{-i\hat{H}_0t} + e^{i\hat{H}_0t}\hat{b}_\nu e^{-i\hat{H}_0t}\Big) \tag{6.695}$$

$$= \sum_\nu W_{p\nu}\,\hat{n}_p\Big(\hat{b}^\dagger_\nu(t) + \hat{b}_\nu(t)\Big) \tag{6.696}$$

$$= \sum_\nu W_{p\nu}\,\hat{n}_p\big(e^{+i\Omega_\nu t}\hat{b}^\dagger_\nu + e^{-i\Omega_\nu t}\hat{b}_\nu\big). \tag{6.697}$$

Now the two-time expectation value:

$$\begin{aligned}\langle\Psi_p|\hat{V}_I(t_1)\hat{V}_I(t_2)|\Psi_p\rangle &= \sum_{\nu,\nu'} W_{p\nu}W_{p\nu'}\langle\Psi_p|\,\hat{n}_p\hat{n}_p\left(e^{+i\Omega_\nu t_1}\hat{b}^\dagger_\nu + e^{-i\Omega_\nu t_1}\hat{b}_\nu\right)\\ &\qquad\qquad\times\left(e^{+i\Omega_{\nu'} t_2}\hat{b}^\dagger_{\nu'} + e^{-i\Omega_{\nu'} t_2}\hat{b}_{\nu'}\right)|\Psi_p\rangle\\ &= \sum_{\nu,\nu'} W_{p\nu}W_{p\nu'}\; e^{-i\Omega_\nu t_1}e^{+i\Omega_{\nu'} t_2}\langle 0|\hat{b}_\nu\hat{b}^\dagger_{\nu'}|0\rangle\\ &= \sum_\nu |W_{p\nu}|^2\; e^{-i\Omega_\nu(t_1-t_2)}. \end{aligned} \tag{6.698}$$

So, recognizing

$$* = 1 - \frac{1}{2}\int_0^t dt_1\int_0^t dt_2\;\langle\Psi_p|\mathcal{T}\left[\hat{V}_I(t_1)\hat{V}_I(t_2)\right]|\Psi_p\rangle + O(W^4) \tag{6.699}$$

$$= 1 - \int_0^t dt_1\int_0^{t_1} dt_2\;\langle\Psi_p|\hat{V}_I(t_1)\hat{V}_I(t_2)|\Psi_p\rangle + O(W^4) \tag{6.700}$$

$$= 1 - \sum_\nu |W_{p\nu}|^2\int_0^t dt_1\int_0^{t_1} dt_2\; e^{-i\Omega_\nu(t_1-t_2)} + O(W^4) \tag{6.701}$$

$$= 1 - \sum_\nu |W_{p\nu}|^2\left(\frac{t}{i\Omega_\nu} + \frac{1-e^{-i\Omega_\nu t}}{\Omega_\nu^2}\right) + O(W^4) \tag{6.702}$$

where the integration is shown explicitly below for clarity. [13] Plugging this back into the expression for the retarded Green's function gives us

$$G^R_{pp}(t) = -i\,\theta(t)\,e^{-i\epsilon_p t}\left[1 - \sum_\nu |W_{p\nu}|^2\left(\frac{t}{i\Omega_\nu} + \frac{1-e^{-i\Omega_\nu t}}{\Omega_\nu^2}\right) + O(W^4)\right] \tag{6.705}$$

$$\approx -i\,\theta(t)\,e^{-i\epsilon_p t}e^{C_2(t)} \tag{6.706}$$

where the second-order cumulant, which is correct up to the second order in the interaction, is given by

$$C_2(t) = -\sum_\nu |W_{p\nu}|^2\left(\frac{t}{i\Omega_\nu} + \frac{1-e^{-i\Omega_\nu t}}{\Omega_\nu^2}\right). \tag{6.707}$$

6.5.3 Pursuing the alternate definition for the electronic-bosonic Hamiltonian

Now, we can choose the exact ground state to be $|N\rangle \equiv |N_F\rangle \otimes |0_{RPA}\rangle$, where $|N_F\rangle$ is the exact fermionic ground state and $|0_{RPA}\rangle$ is the RPA ground state. $\hat{c}_p$ and $\hat{c}^\dagger_p$ will be the

[13]

$$\int_0^{t_1} dt_2\, e^{-i\Omega(t_1-t_2)} = \int_0^{t_1} du\, e^{-i\Omega u} = \frac{1-e^{-i\Omega t_1}}{i\Omega}. \tag{6.703}$$

$$\int_0^t dt_1\,\frac{1-e^{-i\Omega t_1}}{i\Omega} = \frac{t}{i\Omega} + \frac{1-e^{-i\Omega t}}{\Omega^2}. \tag{6.704}$$

annihilation and creation operators for the fermionic hole state p. So, we can write

$$\hat{c}_p^\dagger |N\rangle = \hat{c}_p^\dagger \Big(|N_F\rangle \otimes |0_{RPA}\rangle \Big) \tag{6.708}$$

$$\approx \hat{c}_p^\dagger \Big(|\Phi_0^N\rangle \otimes |0_{RPA}\rangle \Big) \tag{6.709}$$

$$\approx |\Phi_0^{N-1}\rangle \otimes |0_{RPA}\rangle \tag{6.710}$$

$$\equiv |\Psi_p\rangle \,. \tag{6.711}$$

where in 6.709 we approximated the exact fermionic ground state by the Fermi vacuum, and in 6.710 we approximated the RPA ground state as unaffected by the removal of the core electron. A more revealing choice for the electron-boson Hamiltonian $\hat{H}_p^{\mathrm{eB}}$ of the N-1 electron system in the quasi-boson approximation with a hole in core state p

$$\hat{H}_p^{\mathrm{eB}} = \underbrace{\epsilon_p}_{\hat{H}^e} + \underbrace{\sum_{\mu\nu} \left(A_{\mu\nu} \hat{b}_\mu^\dagger \hat{b}_\nu + \frac{1}{2} B_{\mu\nu} \left(\hat{b}_\mu^\dagger \hat{b}_\nu^\dagger + \hat{b}_\mu \hat{b}_\nu \right) \right)}_{\hat{H}^B} + \underbrace{\sum_\nu V_{p\nu} \left(\hat{b}_\nu^\dagger + \hat{b}_\nu \right)}_{\hat{V}}, \tag{6.712}$$

where ϵ_p is the energy of the state p, $\boldsymbol{A}$ and $\boldsymbol{B}$ are the RPA matrices defined as

$$A_{ia,jb} = (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + \langle ib||aj\rangle \tag{6.713}$$

$$B_{ia,jb} = \langle ia||jb\rangle, \tag{6.714}$$

and $V_{p\nu} \equiv V_{pp,ia} = \langle pa||pi\rangle$ is the two-electron integral between the core hole state p and the quasi-bosonic excitation $i \to a$. The first term of 6.712 describes the single, fermionic core hole, the second term describes the bosonic bath of plasmons, while the third term describes the interaction between the core hole and the plasmonic bath. The bosonic Hamiltonian can also be expressed as

$$\hat{H}^B = -\frac{1}{2} \operatorname{Tr} \boldsymbol{A} + \frac{1}{2} \begin{pmatrix} \hat{\boldsymbol{b}}^\dagger & \hat{\boldsymbol{b}} \end{pmatrix} \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{B} & \boldsymbol{A} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{b}} \\ \hat{\boldsymbol{b}}^\dagger \end{pmatrix} \tag{6.715}$$

and by introducing the quasiparticle (Bogoliubov) operators

$$\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\beta}}^\dagger \end{pmatrix} = \begin{pmatrix} \boldsymbol{X} & -\boldsymbol{Y} \\ -\boldsymbol{Y} & \boldsymbol{X} \end{pmatrix}^\dagger \begin{pmatrix} \hat{\boldsymbol{b}} \\ \hat{\boldsymbol{b}}^\dagger \end{pmatrix} \tag{6.716}$$

defined by a unitary transformation of the RPA eigenvectors, just the bosonic and interaction parts of the Hamiltonian change to become

$$\hat{H}^B = \underbrace{\frac{1}{2} \operatorname{Tr}(\boldsymbol{\Omega} - \boldsymbol{A})}_{E_c^{\mathrm{RPA}}} + \sum_\mu \Omega_\mu \hat{\beta}_\mu^\dagger \hat{\beta}_\mu \tag{6.717}$$

$$\hat{V} = \sum_\nu M_{p\nu} \left(\hat{\beta}_\nu^\dagger + \hat{\beta}_\nu \right) \tag{6.718}$$

where $\mathbf{\Omega}$ is the diagonal matrix of RPA excitation energies and $M_{p\nu} = \sum_{ia} V_{pp,ia}(X_{ia,\nu}+Y_{ia,\nu})$ are the coupling matrix elements between the core hole state and the RPA excitations. With the manipulations complete, as before, we can further partition into a non-interacting part $\hat{H}_0$ and an interaction $\hat{W}$ as

$$\hat{H}_0 = \epsilon_p + E_c^{\mathrm{RPA}} + \sum_\mu \Omega_\mu \hat{\beta}_\mu^\dagger \hat{\beta}_\mu \tag{6.719}$$

$$\hat{W} = \sum_\nu M_{p\nu} \left(\hat{\beta}_\nu^\dagger + \hat{\beta}_\nu \right). \tag{6.720}$$

We can plug this into the definition of the retarded Green's function.

$$G_{pp}^R(t) \approx -i\theta(t) \langle \Psi_p | \, e^{-i\hat{H}_{\mathrm{eB}}t} \, | \Psi_p \rangle \tag{6.721}$$

$$= -i\theta(t) \langle \Psi_p | \, e^{-i\left(\hat{H}_0+\hat{V}\right)t} \, | \Psi_p \rangle \tag{6.722}$$

$$= -i\theta(t) \langle \Psi_p | \, e^{-i\hat{H}_0 t} \mathcal{T} \exp\left(-i \int_0^t dt' e^{i\hat{H}_0 t'} \hat{V} e^{-i\hat{H}_0 t'} \right) | \Psi_p \rangle \tag{6.723}$$

$$= -i\theta(t) e^{-i(\epsilon_p + E_c^{\mathrm{RPA}})t} \underbrace{\langle \Psi_p | \, \mathcal{T} \exp\left(-i \int_0^t dt' V_I(t') \right) | \Psi_p \rangle}_{*} \tag{6.724}$$

where we were able to get to 6.724 as before because by definition $\beta|0_{RPA}\rangle = 0$, but notice that now . Again, we can work out * by a Taylor expansion of the time-ordered exponential:

$$* = \langle \Psi_p | \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_0^t dt_1 \cdots \int_0^t dt_n \mathcal{T} \left[V_I(t_1) \cdots V_I(t_n) \right] | \Psi_p \rangle \tag{6.725}$$

$$= 1 + (-i) \int_0^t dt' \langle \Psi_p | \, V_I(t') \, | \Psi_p \rangle + \frac{(-i)^2}{2!} \int_0^t dt_1 \int_0^t dt_2 \langle \Psi_p | \, \mathcal{T} \left[V_I(t_1) V_I(t_2) \right] | \Psi_p \rangle + O(M^3) \tag{6.726}$$

$$= 1 - \frac{1}{2} \int_0^t dt_1 \int_0^t dt_2 \langle \Psi_p | \, \mathcal{T} \left[V_I(t_1) V_I(t_2) \right] | \Psi_p \rangle + O(M^4) \tag{6.727}$$

where in 6.727 we have used the fact that terms with an odd number of V_I's will vanish because they do not conserve the number of bosons. We can evaluate $V_I(t)$ as

$$\hat{V}_I(t) = e^{i\hat{H}_0 t} \Big[\sum_\nu M_{p\nu} \left(\hat{\beta}_\nu^\dagger + \hat{\beta}_\nu \right) \Big] e^{-i\hat{H}_0 t} \tag{6.728}$$

$$= \sum_\nu M_{p\nu} \Big(e^{i\hat{H}_0 t} \hat{\beta}_\nu^\dagger e^{-i\hat{H}_0 t} + e^{i\hat{H}_0 t} \hat{\beta}_\nu e^{-i\hat{H}_0 t} \Big) \tag{6.729}$$

$$= \sum_\nu M_{p\nu} \Big(\hat{\beta}_\nu^\dagger(t) + \hat{\beta}_\nu(t) \Big) \tag{6.730}$$

$$= \sum_\nu M_{p\nu} \big(e^{+i\Omega_\nu t} \hat{\beta}_\nu^\dagger + e^{-i\Omega_\nu t} \hat{\beta}_\nu \big). \tag{6.731}$$

so the two-time expectation value is

$$\langle\Psi_p|\hat{V}_I(t_1)\hat{V}_I(t_2)|\Psi_p\rangle = \sum_{\mu\nu} M_{p\mu}M_{p\nu}\langle\Psi_p|\left(e^{+i\Omega_\mu t_1}\hat{\beta}^\dagger_\mu + e^{-i\Omega_\mu t_1}\hat{\beta}_\mu\right)\left(e^{+i\Omega_\nu t_2}\hat{\beta}^\dagger_\nu + e^{-i\Omega_\nu t_2}\hat{\beta}_\nu\right)|\Psi_p\rangle \tag{6.732}$$

$$= \sum_{\mu\nu} M_{p\mu}M_{p\nu}\, e^{-i\Omega_\mu t_1}e^{+i\Omega_\nu t_2}\langle 0_{\mathrm{RPA}}|\hat{\beta}_\mu\hat{\beta}^\dagger_\nu|0_{\mathrm{RPA}}\rangle \tag{6.733}$$

$$= \sum_{\nu} |M_{p\nu}|^2\, e^{-i\Omega_\nu(t_1-t_2)}. \tag{6.734}$$

So, recognizing

$$* = 1 - \frac{1}{2}\int_0^t dt_1 \int_0^t dt_2\, \langle\Psi_p|\,\mathcal{T}\left[V_I(t_1)V_I(t_2)\right]|\Psi_p\rangle + O(M^4) \tag{6.735}$$

$$= 1 - \int_0^t dt_1 \int_0^{t_1} dt_2\, \langle\Psi_p|\,V_I(t_1)V_I(t_2)\,|\Psi_p\rangle + O(M^4) \tag{6.736}$$

$$= 1 - \sum_\nu |M_{p\nu}|^2 \int_0^t dt_1 \int_0^{t_1} dt_2 e^{-i\Omega_\nu(t_1-t_2)} + O(M^4) \tag{6.737}$$

$$= 1 - \sum_\nu |M_{p\nu}|^2 \left(\frac{t}{i\Omega_\nu} + \frac{1-e^{-i\Omega_\nu t}}{\Omega_\nu^2}\right) + O(M^4), \tag{6.738}$$

we can plug back into the expression for the retarded Green's function, yielding

$$G^{R,GW+C}_{pp}(t) = -i\theta(t)e^{-i(\epsilon_p + E_c^{\mathrm{RPA}})t}\left[1 - \sum_\nu |M_{p\nu}|^2\left(\frac{t}{i\Omega_\nu} + \frac{1-e^{-i\Omega_\nu t}}{\Omega_\nu^2}\right) + O(M^4)\right] \tag{6.739}$$

$$\approx -i\theta(t)e^{-i(\epsilon_p + E_c^{\mathrm{RPA}})t}e^{C_2(t)} \tag{6.740}$$

where the second-order cumulant, which is correct up to the second order in the screened interaction M, is given by

$$C_2(t) = -\sum_\nu |M_{p\nu}|^2\left(\frac{t}{i\Omega_\nu} + \frac{1-e^{-i\Omega_\nu t}}{\Omega_\nu^2}\right). \tag{6.741}$$

This shows that one recovers the same cumulant irregardless if we chose the bosonic ground state to be the RPA ground state or the non-interacting bosonic vacuum. It also provides a rigorous justification why our G_0 can be from a correlated single particle reference and not just the MF one. In other words, it is defining an algebraic procedure for how to obtain an improved Green's function apart from how we think diagrammatically as

$$G^{GW} = G^{HF} + G^{HF}\Sigma^{GW}G^{GW} \quad \Rightarrow \quad G^{GW+C} = G^{GW}e^{C}. \tag{6.742}$$

Here we were able to get to G^{GW+C} directly without any Dyson iteration.

Comparison to the Landau formula for the cumulant

The Landau formula for the cumulant is given by

$$C(t) = \int d\omega \frac{\beta(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.743}$$

where $\beta(\omega) = \frac{1}{\pi}|\mathrm{Im}\Sigma(\omega + \epsilon_p)|$ is proportional to the GW self-energy. Comparing this to the expression we have derived for $C_2(t)$, we see that the two expressions are consistent *only in the TDL* if we identify $\beta(\omega) = \pi \sum_\nu |M_{p\nu}|^2 \delta(\omega - \Omega_\nu)$, which is precisely the spectral density of the self-energy in the GW approximation. This shows that the

$$\hat{H}_{\mathrm{eB}} = \hat{H}_{\mathrm{e}} + \hat{H}_{\mathrm{B}} + \hat{V}_{\mathrm{eB}} \tag{6.744}$$

where

$$\hat{H}_{\mathrm{e}} \equiv \sum_{pq} f_{pq} c_p^\dagger c_q = \sum_p \epsilon_p c_p^\dagger c_p \tag{6.745}$$

$$\hat{H}_{\mathrm{B}} = \sum_{\mu\nu} A_{\mu\nu} b_\mu^\dagger b_\nu + \frac{1}{2} \sum_{\mu\nu} B_{\mu\nu} \left(b_\mu^\dagger b_\nu^\dagger + b_\mu b_\nu\right) \tag{6.746}$$

$$\hat{V}_{\mathrm{eB}} \equiv \sum_{pq\nu} V_{pq,\nu} c_p^\dagger c_q \left(b_\nu^\dagger + b_\nu\right) = \sum_{p\nu} W_{p\nu} c_p^\dagger c_p \left(b_\nu^\dagger + b_\nu\right) \tag{6.747}$$

respectively describe the fermionic subsystem, the bosonic bath, and their mutual coupling.

In the cumulant approximation, this effective Hamiltonian can be expressed as a supermatrix in the space of fermionic and bosonic excitations. Tolle claimsthat any vertex corrections can be added by modifying certain quantities in the EOM supermatrix:

$$\boldsymbol{H}^{\mathrm{EOM}} = \begin{bmatrix} \boldsymbol{F} & \boldsymbol{W}^< & \boldsymbol{W}^> \\ (\boldsymbol{W}^<)^\dagger & \boldsymbol{d}^< & \boldsymbol{0} \\ (\boldsymbol{W}^>)^\dagger & \boldsymbol{0} & \boldsymbol{d}^> \end{bmatrix} \tag{6.748}$$

where $\boldsymbol{F}$ represents the fermionic space, $\boldsymbol{d}^<$ and $\boldsymbol{d}^>$ reprsent the bosonic deexcitation and excitation spaces, respectively, while $\boldsymbol{W}^<$ and $\boldsymbol{W}^>$ represent the coupling matrices between the fermionic and bosonic sectors.

6.5.4 Examples

A good place to start in an effort to understand this abstract framework is to consider what we have encountered in the past.

GW

$\boldsymbol{F}$ is a generalized Fock matrix,

$$\boldsymbol{W}^<_{pk\nu} = \sum_{ia} (pk|ia) \left(X^\nu_{ia} + Y^\nu_{ia}\right) \quad \text{and} \quad \boldsymbol{W}^>_{pc\nu} = \sum_{ia} (pc|ia) \left(X^\nu_{ia} + Y^\nu_{ia}\right), \tag{6.749}$$

and

$$\boldsymbol{d}^<_{k\nu,l\nu'} = \left(\epsilon_k - \Omega_\nu\right) \delta_{k,l} \delta_{\nu,\nu'} \quad \text{and} \quad \boldsymbol{d}^>_{c\nu,d\nu'} = \left(\epsilon_c + \Omega_\nu\right) \delta_{c,d} \delta_{\nu,\nu'}, \tag{6.750}$$

where the bosonic excitation energies Ω_ν and amplitudes X^ν_{ia} and Y^ν_{ia} are due to the RPA.

MP2

$\boldsymbol{F}$ is the HF Fock matrix,

$$\boldsymbol{W}^{<}_{pk\nu} \equiv W^{<}_{pkia} = \frac{1}{\sqrt{2}}\langle pa||ik\rangle \quad \text{and} \quad \boldsymbol{W}^{>}_{pc\nu} \equiv W^{>}_{pcia} = \frac{1}{\sqrt{2}}\langle pi||ca\rangle, \tag{6.751}$$

and

$$\boldsymbol{d}^{<}_{kia,ljb} = \left(-\epsilon_a + \epsilon_i + \epsilon_k\right)\delta_{kl}\delta_{ij}\delta_{ab} \quad \text{and} \quad \boldsymbol{d}^{>}_{cai,dbj} = \left(-\epsilon_i + \epsilon_c + \epsilon_a\right)\delta_{cd}\delta_{ab}\delta_{ij}. \tag{6.752}$$

It has been shown that MP2 contains diagrams (vertex corrections) that are absent in GW.

ECC

Tolle's argument is that ECC can also be placed in this framework, which is able to systematically treat vertex corrections, albeit at a high computational cost.

6.5.5 Why this framework doesn't apply to the cumulant approximation

The cumulant approximation considers a single fermionic state p coupled to a bosonic bath of excitations and deexcitations. With this simplification, we can replace the fermionic space $\boldsymbol{F}$ with the single particle energy of a state p as ϵ_p. The coupling matrices lose one dimension, becoming $\boldsymbol{W}^{<} \equiv \delta_{pk}W^{<}_{k\nu}$ and $\boldsymbol{W}^{>} \equiv \delta_{pc}W^{>}_{c\nu}$, while the bosonic deexcitation and excitation spaces remain unchanged. The EOM supermatrix then takes the form

$$\boldsymbol{H}^{\text{EOM}} = \begin{bmatrix} \epsilon_p & 0 & \boldsymbol{W}^{>} \\ 0 & 0 & \boldsymbol{0} \\ (\boldsymbol{W}^{>})^{\dagger} & \boldsymbol{0} & \boldsymbol{d}^{>} \end{bmatrix} \tag{6.753}$$

MP2

We can consider a concrete example such as MP2, where $W^{<}_{k\nu} = \frac{1}{\sqrt{2}}\langle pa||ip\rangle$, expanded as

$$W^{<}_{k\nu} = \frac{1}{\sqrt{2}}\langle pa||ip\rangle = \frac{1}{\sqrt{2}}\left(\langle pa|ip\rangle - \langle pa|pi\rangle\right) = 0. \tag{6.754}$$

We reached the final equality because we know that for periodic systems the two-electron integrals must vanish due to momentum conservation, since we must have $\boldsymbol{k}_p + \boldsymbol{k}_a = \boldsymbol{k}_i + \boldsymbol{k}_p \implies \boldsymbol{k}_a = \boldsymbol{k}_i$, but this is impossible because a is a virtual index and i is an occupied index. So, the coupling matrix $\boldsymbol{W}^{<}$ vanishes and likewise $\boldsymbol{W}^{>}$ also vanishes. This means that the EOM supermatrix in the single fermion approximation is block diagonal, written as

$$\boldsymbol{H}^{\text{EOM}} = \begin{bmatrix} \epsilon_p & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{d}^{<} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{d}^{>} \end{bmatrix} \tag{6.755}$$

We can now apply the Lowdin partitioning technique to this matrix. We first define an excitation/deexcitation vector $\boldsymbol{R} = \begin{bmatrix} R_p & \boldsymbol{R}^< & \boldsymbol{R}^> \end{bmatrix}^T$. We can then write the following system of equations:

$$\epsilon_p R_p = \omega R_p \tag{6.756}$$
$$\boldsymbol{d}^< \boldsymbol{R}^< = \omega \boldsymbol{R}^< \tag{6.757}$$
$$\boldsymbol{d}^> \boldsymbol{R}^> = \omega \boldsymbol{R}^> \tag{6.758}$$

This implies that we got a complete decoupling of the fermionic and bosonic sectors.

6.6 PT2

6.6.1 SRG-MP2

We are interested in finding the self consistent solution to

$$\boldsymbol{F}^{QP} = \boldsymbol{F}^{(0)} + \boldsymbol{F}^{(2)}[\boldsymbol{F}^{QP}] \tag{6.759}$$

where $\boldsymbol{F}^{(0)}$ is the canonical HF Fock matrix and $\boldsymbol{F}^{(2)}$ is the second order correction to the Fock matrix, which we take to be diagonal with elements given by

$$F^{(2)}_{pp}(s) = \frac{1}{2}\sum_{ija} \frac{|\langle pa||ij\rangle|^2}{\Delta^{pa}_{ij}} \left[1 - e^{-2\left(\Delta^{pa}_{ij}\right)^2 s}\right] + \frac{1}{2}\sum_{abi} \frac{|\langle pi||ab\rangle|^2}{\Delta^{pi}_{ab}} \left[1 - e^{-2\left(\Delta^{pi}_{ab}\right)^2 s}\right] \tag{6.760}$$
$$= F^{2h1p}_{pp}(s) + F^{2p1h}_{pp}(s) \tag{6.761}$$

where $F^{2h1p}_{pp}(s)$ and $F^{2p1h}_{pp}(s)$ are the contributions from the 2h1p and 2p1h channels, and $\Delta^{pa}_{ij} = \epsilon_p + \epsilon_a - \epsilon_i - \epsilon_j$ and $\Delta^{pi}_{ab} = \epsilon_p + \epsilon_i - \epsilon_a - \epsilon_b$. We always follow the same procedure when applying the one-shot cumulant.

Deriving the analytic gradient

Since our objective function to minimize is

$$\mathcal{L} = \frac{1}{2}\sum_p (\Delta\epsilon_p)^2 \tag{6.762}$$

where at iteration n,

$$\Delta\epsilon^n_p = \epsilon^0_p + F_{pp}[\left\{\epsilon^{n-1}_q\right\}] - \epsilon^{n-1}_p, \tag{6.763}$$

Taking the derivative of the Lagrangian with respect to ϵ_p, we get

$$\frac{\partial \mathcal{L}}{\partial \epsilon_p} = \sum_q \Delta\epsilon_q \frac{\partial \Delta\epsilon_q}{\partial \epsilon_p} \tag{6.764}$$
$$= \sum_q \Delta\epsilon_q \left(\frac{\partial F_{qq}}{\partial \epsilon_p} - \delta_{pq}\right) \tag{6.765}$$
$$\tag{6.766}$$

Partition $\frac{\partial F_{qq}}{\partial \epsilon_p}$ into its 2h1p and 2p1h contributions: Working out $\frac{\partial F_{qq}^{2h1p}}{\partial \epsilon_p}$, we get

$$\frac{\partial F_{qq}^{2h1p}}{\partial \epsilon_p} = \frac{1}{2}\sum_{ija} |\langle qa||ij\rangle|^2 \partial_{\epsilon_p}\left[\frac{1}{\Delta_{ij}^{qa}}\left(1 - e^{-2\left(\Delta_{ij}^{qa}\right)^2 s}\right)\right] \tag{6.767}$$

$$= \frac{1}{2}\sum_{ija} |\langle qa||ij\rangle|^2 \left[\partial_{\epsilon_p}\left[\frac{1}{\Delta_{ij}^{qa}}\right]\left[1 - e^{-2\left(\Delta_{ij}^{qa}\right)^2 s}\right] + \frac{1}{\Delta_{ij}^{qa}}\partial_{\epsilon_p}\left[1 - e^{-2\left(\Delta_{ij}^{qa}\right)^2 s}\right]\right] \tag{6.768}$$

and

$$\partial_{\epsilon_p}\left[\frac{1}{\Delta_{ij}^{qa}}\right] = \frac{\partial}{\partial \Delta_{ij}^{qa}}\frac{\partial \Delta_{ij}^{qa}}{\partial \epsilon_p}\left[\frac{1}{\Delta_{ij}^{qa}}\right] = -\frac{\partial_{\epsilon_p}[\Delta_{ij}^{qa}]}{\left(\Delta_{ij}^{qa}\right)^2} \tag{6.769}$$

Consider

$$\partial_{\epsilon_p}\left[\Delta_{ij}^{qa}\right] = \partial_{\epsilon_p}\left[\epsilon_q + \epsilon_a - \epsilon_i - \epsilon_j\right] \tag{6.770}$$

$$= \delta_{pq} + \delta_{pa} - \delta_{pi} - \delta_{pj} \tag{6.771}$$

$$\implies \partial_{\epsilon_p}\left[\frac{1}{\Delta_{ij}^{qa}}\right] = -\frac{\delta_{pq} + \delta_{pa} - \delta_{pi} - \delta_{pj}}{\left(\Delta_{ij}^{qa}\right)^2} \tag{6.772}$$

Next, do

$$\partial_{\epsilon_p}\left[1 - e^{-2\left(\Delta_{ij}^{qa}\right)^2 s}\right] = \frac{\partial}{\partial \Delta_{ij}^{qa}}\frac{\partial \Delta_{ij}^{qa}}{\partial \epsilon_p}\left[1 - e^{-2\left(\Delta_{ij}^{qa}\right)^2 s}\right] \tag{6.773}$$

$$= \left(\delta_{pq} + \delta_{pa} - \delta_{pi} - \delta_{pj}\right)\left(-\exp\left[-2\left(\Delta_{ij}^{qa}\right)^2 s\right] \cdot \left(-4s\Delta_{ij}^{qa}\right)\right) \tag{6.774}$$

So, we found

$$\frac{\partial F_{qq}^{2h1p}}{\partial \epsilon_p} = \frac{1}{2}\sum_{ija} |\langle qa||ij\rangle|^2 \left(\delta_{pq} + \delta_{pa} - \delta_{pi} - \delta_{pj}\right)\left[-\frac{1 - e^{-2(\Delta_{ij}^{qa})^2 s}}{(\Delta_{ij}^{qa})^2} + 4s\, e^{-2(\Delta_{ij}^{qa})^2 s}\right]. \tag{6.775}$$

Analogously, we can work out $\frac{\partial F_{qq}^{2p1h}}{\partial \epsilon_p}$, yielding

$$\frac{\partial F_{qq}^{2p1h}}{\partial \epsilon_p} = \frac{1}{2}\sum_{abi} |\langle qi||ab\rangle|^2 \left(\delta_{pq} + \delta_{pi} - \delta_{pa} - \delta_{pb}\right)\left[-\frac{1 - e^{-2(\Delta_{ab}^{qi})^2 s}}{(\Delta_{ab}^{qi})^2} + 4s\, e^{-2(\Delta_{ab}^{qi})^2 s}\right]. \tag{6.776}$$

Thus, the formula for the gradient is given by

$$\frac{\partial F_{qq}}{\partial \epsilon_p} = \frac{\partial F_{qq}^{2h1p}}{\partial \epsilon_p} + \frac{\partial F_{qq}^{2p1h}}{\partial \epsilon_p}$$

$$= \frac{1}{2}\sum_{ija} |\langle qa||ij\rangle|^2 \left(\delta_{pq} + \delta_{pa} - \delta_{pi} - \delta_{pj}\right)\left[-\frac{1 - e^{-2(\Delta_{ij}^{qa})^2 s}}{(\Delta_{ij}^{qa})^2} + 4s\, e^{-2(\Delta_{ij}^{qa})^2 s}\right]$$

$$+ \frac{1}{2}\sum_{abi} |\langle qi||ab\rangle|^2 \left(\delta_{pq} + \delta_{pi} - \delta_{pa} - \delta_{pb}\right)\left[-\frac{1 - e^{-2(\Delta_{ab}^{qi})^2 s}}{(\Delta_{ab}^{qi})^2} + 4s\, e^{-2(\Delta_{ab}^{qi})^2 s}\right] \tag{6.777}$$

The cumulant ansatz for the interacting, retarded Green's function is given by:

$$\boldsymbol{G}(t) = \boldsymbol{G}^0(t) e^{\boldsymbol{C}(t)} \tag{6.778}$$

where $\boldsymbol{C}(t)$ is the retarded cumulant and $\boldsymbol{G}^0(t)$ is the noninteracting, retarded Green's function. By relating the Dyson equation to the Taylor series expansion of the exponential, we can write:

$$\boldsymbol{G}^0(t) + \boldsymbol{G}^0(t)\boldsymbol{C}(t) + O(v^4) = \boldsymbol{G}^0(t) + \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0(t-t_1)\boldsymbol{\Sigma}^c(t_1-t_2)\boldsymbol{G}^0(t_2) + O(v^4) \tag{6.779}$$

$$\implies \boldsymbol{G}^0(t)\boldsymbol{C}(t) = \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0(t-t_1)\boldsymbol{\Sigma}^c(t_1-t_2)\boldsymbol{G}^0(t_2) \quad \text{(up to second order in } v) \tag{6.780}$$

As indicated, when we use a second order self-energy, the cumulant should be exact to second order in the bare Coulomb interaction v. Projecting to the spin-orbital basis and inserting the resolution of the identity, we get:

$$\sum_r \langle p| \boldsymbol{G}^0(t) |r\rangle \langle r| \boldsymbol{C}(t) |q\rangle = \sum_{rs} \iint \mathrm{d}t_1 \mathrm{d}t_2 \langle p| \boldsymbol{G}^0(t-t_1) |r\rangle \langle r| \boldsymbol{\Sigma}^c(t_1-t_2) |s\rangle \langle s| \boldsymbol{G}^0(t_2) |q\rangle \tag{6.781}$$

$$\implies \sum_r \boldsymbol{G}^0_{pr}(t)\boldsymbol{C}_{rq}(t) = \sum_{rs} \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{ps}(t-t_1)\boldsymbol{\Sigma}^c_{sr}(t_1-t_2)\boldsymbol{G}^0_{rq}(t_2) \tag{6.782}$$

$$\implies \boldsymbol{G}^0_{pp}(t)\boldsymbol{C}_{pq}(t) = \underbrace{\iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0_{pp}(t-t_1)\boldsymbol{\Sigma}^c_{pq}(t_1-t_2)\boldsymbol{G}^0_{qq}(t_2)}_{*} \tag{6.783}$$

where $\boldsymbol{G}^0(t)$ is diagonal in the spin-orbital basis, as $\boldsymbol{G}^0_{pp}(t) = -i\Theta(t)e^{-i\epsilon_p t}$, where ϵ_p is the single-particle energy of the p-th spin-orbital. The formula for the inverse Fourier transform is given by:

$$f(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} f(\omega) \tag{6.784}$$

which implies that

$$\boldsymbol{G}^0_{pp}(t-t_1) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega) \tag{6.785}$$

$$\boldsymbol{\Sigma}^c_{pq}(t_1-t_2) = \int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega') \tag{6.786}$$

$$\boldsymbol{G}^0_{qq}(t_2) = \int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'') \tag{6.787}$$

and plugging into the double time integral *, we get:

$$* = \iint \mathrm{d}t_1 \mathrm{d}t_2 \left[\int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega(t-t_1)} \boldsymbol{G}^0_{pp}(\omega)\right] \left[\int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega'(t_1-t_2)} \boldsymbol{\Sigma}^c_{pq}(\omega')\right] \left[\int \frac{\mathrm{d}\omega''}{2\pi} e^{-i\omega'' t_2} \boldsymbol{G}^0_{qq}(\omega'')\right] \tag{6.788}$$

$$= \underbrace{\int \mathrm{d}t_1 e^{-i(\omega'-\omega)t_1} \int \mathrm{d}t_2 e^{-i(\omega''-\omega')t_2}}_{4\pi^2\delta(\omega'-\omega)\delta(\omega''-\omega')} \iiint \mathrm{d}\omega \mathrm{d}\omega' \mathrm{d}\omega'' \frac{e^{-i\omega t}}{8\pi^3} \boldsymbol{G}^0_{pp}(\omega) \boldsymbol{\Sigma}^c_{pq}(\omega') \boldsymbol{G}^0_{qq}(\omega'') \tag{6.789}$$

$$= \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} \boldsymbol{G}^0_{pp}(\omega) \boldsymbol{\Sigma}^c_{pq}(\omega) \boldsymbol{G}^0_{qq}(\omega) \tag{6.790}$$

With $\eta \to 0^+$, we get

$$-i\Theta(t) e^{-i\epsilon_p t} \boldsymbol{C}_{pq}(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t} \boldsymbol{G}^0_{pp}(\omega) \boldsymbol{\Sigma}^c_{pq}(\omega) \boldsymbol{G}^0_{qq}(\omega) \tag{6.791}$$

$$\implies C_{pp}(t) = i \int \frac{\mathrm{d}\omega}{2\pi} e^{-i(\omega-\epsilon_p)t} \frac{\Sigma^c_{pp}(\omega)}{(\omega - \epsilon_p + i\eta)^2} \quad \forall t > 0 \tag{6.792}$$

$$= i \int \frac{\mathrm{d}\omega'}{2\pi} e^{-i\omega' t} \frac{\Sigma^c_{pp}(\omega' + \epsilon_p)}{(\omega' + i\eta)^2} \tag{6.793}$$

A retarded second order self-energy can be written as

$$\Sigma^{(2)}_{pq}(\omega) = \frac{1}{2} \sum_{iab} \frac{\langle pi||ab\rangle\langle ab||qi\rangle}{\omega + \epsilon_i - \epsilon_a - \epsilon_b + i\eta} + \frac{1}{2} \sum_{ija} \frac{\langle pa||ij\rangle\langle ij||qa\rangle}{\omega + \epsilon_a - \epsilon_i - \epsilon_j + i\eta} \tag{6.794}$$

$$\implies \Sigma^{(2)}_{pp}(\omega + \epsilon_p) = \frac{1}{2} \sum_{iab} \frac{\langle pi \,||\, ab\rangle^2}{\omega - \epsilon^{ab}_{pi}} + \frac{1}{2} \sum_{ija} \frac{\langle pa \,||\, ij\rangle^2}{\omega - \epsilon^{ij}_{pa}} \tag{6.795}$$

where

$$\epsilon^{ab}_{pi} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i - i\eta \quad \text{and} \quad \epsilon^{ij}_{pa} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a - i\eta \tag{6.796}$$

are generated from combinations of the single particle energies of G_0. So we can write the diagonal cumulant[14] as

$$C_{pp}(t) \equiv i \int \frac{d\omega}{2\pi} \frac{\Sigma^c_{pp}(\omega + \epsilon_p)}{(\omega + i\eta)^2} e^{-i\omega t} \tag{6.797}$$

$$\implies C^{(2)}_{pp}(t) = \frac{1}{2} \sum_{iab} \langle pi \,||\, ab\rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2 \left(\omega - \epsilon^{ab}_{pi}\right)} + \frac{1}{2} \sum_{ija} \langle pa \,||\, ij\rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2 \left(\omega - \epsilon^{ij}_{pa}\right)} \tag{6.798}$$

$$= \frac{1}{2} \sum_{iab} \frac{\langle pi \,||\, ab\rangle^2}{\left(\epsilon^{ab}_{pi}\right)^2} \left(e^{-i\epsilon^{ab}_{pi} t} + i\epsilon^{ab}_{pi} t - 1\right) + \frac{1}{2} \sum_{ija} \frac{\langle pa \,||\, ij\rangle^2}{\left(\epsilon^{ij}_{pa}\right)^2} \left(e^{-i\epsilon^{ij}_{pa} t} + i\epsilon^{ij}_{pa} t - 1\right) \tag{6.799}$$

$$= \frac{1}{2} \sum_{iab} \langle pi \| ab\rangle^2 f\left(\epsilon^{ab}_{pi}\right) + \frac{1}{2} \sum_{ija} \langle pa \| ij\rangle^2 f\left(\epsilon^{ij}_{pa}\right) \tag{6.800}$$

$$= \int d\omega \, \beta(\omega) f(\omega) \tag{6.801}$$

where $f(\omega) \equiv \frac{e^{-i\omega t} + i\omega t - 1}{\omega^2}$ and we identify the cumulant kernel as [15]

$$\beta(\omega) = -\frac{1}{\pi} \operatorname{Im} \Sigma^{(2)}_{pp}(\omega + \epsilon_p) \tag{6.802}$$

$$\tag{6.803}$$

$$= \frac{1}{2} \sum_{iab} \langle pi \| ab\rangle^2 \delta\left(\omega - \epsilon^{ab}_{pi}\right) + \frac{1}{2} \sum_{ija} \langle pa \| ij\rangle^2 \delta\left(\omega - \epsilon^{ij}_{pa}\right) \tag{6.804}$$

A connection can be made between the cumulant kernel $\beta(\omega) = \sum_q g_q^2 \delta(\omega - \omega_q)$ for a 2nd-order self energy to that for electrons coupled to bosonic excitations at the frequencies $\omega_q \equiv \epsilon^{rs}_{pq}$ in the quasi-boson approximation (QBA) with coupling coefficients $g_q \equiv \langle pq \| rs\rangle$.

Why the cumulant regularizes irregardless of an empirical parameter

Whenever $\epsilon^{ab}_{pi} \to 0$ (or ϵ^{ij}_{pa} by the same argument), due to L'Hopital's rule

$$\lim_{\epsilon^{ab}_{pi} \to 0} \frac{e^{-i\epsilon^{ab}_{pi} t} + i\epsilon^{ab}_{pi} t - 1}{\left(\epsilon^{ab}_{pi}\right)^2} = \lim_{\epsilon^{ab}_{pi} \to 0} \frac{-ite^{-i\epsilon^{ab}_{pi} t} + it}{2\epsilon^{ab}_{pi}} = \lim_{\epsilon^{ab}_{pi} \to 0} \frac{-t^2 e^{-i\epsilon^{ab}_{pi} t}}{2} = -\frac{t^2}{2}. \tag{6.805}$$

This is why the cumulant spectral function should stay stable even in the case of a vanishing gap.

[14] Note that in order to get from eqn. 6.798 to eqn. 6.799 we have used the identity $\int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2(\omega - \epsilon)} = \frac{1}{\epsilon^2}\left(e^{-i\epsilon t} + i\epsilon t - 1\right) \operatorname{sgn}(t)$, and since we are dealing with retarded quantities, we only care about $t > 0$ so $\operatorname{sgn}(t) = 1$.

[15] To get from eqn. 6.802 to eqn. 6.804, we have used the identity $\operatorname{Im} \frac{1}{x + i\eta} = -\pi\delta(x)$ as $\eta \to 0^+$.

6.6.2 Comment on plots

First order

When I say that I am computing the cumulant spectral function *to first order*, it means that I directly compute the spectral function in frequency domain as

$$A_{pp}(\omega)\frac{Z_p^{QP}}{\omega-\epsilon_p^{QP}+i\eta}+\sum_{iab}\frac{Z_{piab}^{sat}}{\omega-\epsilon_{piab}^{sat}+i\eta}+\sum_{ija}\frac{Z_{pija}^{sat}}{\omega-\epsilon_{pija}^{sat}+i\eta} \tag{6.806}$$

where we defined the satellite energies $\epsilon_{piab}^{sat}=\epsilon_p^{QP}+\epsilon_{pi}^{ab}$ and $\epsilon_{pija}^{sat}=\epsilon_p^{QP}+\epsilon_{pa}^{ij}$, and the corresponding spectral weights $Z_{piab}^{sat}=Z_p^{QP}\zeta_{pi}^{ab}$ and $Z_{pija}^{sat}=Z_p^{QP}\zeta_{pa}^{ij}$. These have the definitions $Z_p^{QP}=e^{-\sum_{iab}\zeta_{pi}^{ab}-\sum_{ija}\zeta_{pa}^{ij}}$, $\epsilon_p^{QP}=\epsilon_p-\sum_{iab}\zeta_{pi}^{ab}\epsilon_{pi}^{ab}-\sum_{ija}\zeta_{pa}^{ij}\epsilon_{pa}^{ij}$, $\zeta_{pi}^{ab}=\frac{1}{2}\frac{\langle pi\|ab\rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2}$, and $\zeta_{pa}^{ij}=\frac{1}{2}\frac{\langle pa\|ij\rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2}$.

Infinite order

When I say that I am computing the cumulant spectral function *to infinite order*, it means that I compute the spectral function as $A_{pp}(\omega)=-\frac{1}{\pi}\operatorname{Im}G_{pp}(\omega)$, where I get $G_{pp}(\omega)$ from the DFT of $G_{pp}(t)=G_{pp}^0(t)e^{C_{pp}(t)}$ using

$$C_{pp}(t)=\frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t}+i\epsilon_{pi}^{ab}t-1\right)+\frac{1}{2}\sum_{ija}\frac{\langle pa\|ij\rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t}+i\epsilon_{pa}^{ij}t-1\right) \tag{6.807}$$

and $G_{pp}^0(t)=-i\Theta(t)e^{-i\epsilon_p t}$. $\epsilon_{pi}^{ab}=\epsilon_a+\epsilon_b-\epsilon_p-\epsilon_i-i\eta$ and $\epsilon_{pa}^{ij}=\epsilon_i+\epsilon_j-\epsilon_p-\epsilon_a-i\eta$.

Where these equations come from

Loos showed that first order and infinite order share the same features for GW+C; all that I have done is to replace the GW quantites by their MP2 counterparts. All ϵ's above are the single particle energies due to SRG-MP2 since I took this as my noninteracting reference.

Frequency domain version

Now, if we plug in our derived expression for $C_{pp}(t)$ into the cumulant ansatz for the retarded Green's function:

$$G_{pp}(t) = G_{pp}^{HF}(t)e^{C_{pp}(t)} \tag{6.808}$$

$$= -i\Theta(t)e^{-i\epsilon_p t + C_{pp}(t)} \tag{6.809}$$

$$= -i\Theta(t)e^{-i\epsilon_p t}\exp\left\{\frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t} + i\epsilon_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa\|ij\rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t} + i\epsilon_{pa}^{ij}t - 1\right)\right\} \tag{6.810}$$

$$= -i\Theta(t)\underbrace{e^{-\sum_{iab}\zeta_{pi}^{ab} - \sum_{ija}\zeta_{pa}^{ij}}}_{Z_p^{QP}} e^{-i\overbrace{\left(\epsilon_p - \sum_{iab}\zeta_{pi}^{ab}\epsilon_{pi}^{ab} - \sum_{ija}\zeta_{pa}^{ij}\epsilon_{pa}^{ij}\right)}^{\epsilon_p^{QP}}t} e^{\sum_{iab}\zeta_{pi}^{ab}e^{-i\epsilon_{pi}^{ab}t} + \sum_{ija}\zeta_{pa}^{ij}e^{-i\epsilon_{pa}^{ij}t}} \tag{6.811}$$

where $\zeta_{pi}^{ab} = \frac{1}{2}\frac{\langle pi\|ab\rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2}$ and $\zeta_{pa}^{ij} = \frac{1}{2}\frac{\langle pa\|ij\rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2}$ are the bosonic coupling coefficients. Next, we want to perform a Fourier transform.

$$G_{pp}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t}G_{pp}(t) \tag{6.812}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt e^{i(\omega-\epsilon_p^{QP})t}\exp\left\{\sum_{iab}\zeta_{pi}^{ab}e^{-i\epsilon_{pi}^{ab}t} + \sum_{ija}\zeta_{pa}^{ij}e^{-i\epsilon_{pa}^{ij}t}\right\} \tag{6.813}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{i(\omega-\epsilon_p^{QP})t}\left(1 + \sum_{iab}\zeta_{pi}^{ab}e^{-i\epsilon_{pi}^{ab}t} + \sum_{ija}\zeta_{pa}^{ij}e^{-i\epsilon_{pa}^{ij}t} + \ldots\right) \tag{6.814}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{[-\eta + i(\omega-\epsilon_p^{QP})t]} \tag{6.815}$$

$$- iZ_p^{QP}\sum_{iab}\zeta_{pi}^{ab}\int_0^{\infty} dt\, e^{[-\eta + i(\omega-\epsilon_p^{QP}-\epsilon_{pi}^{ab})t]} \tag{6.816}$$

$$- iZ_p^{QP}\sum_{a\nu}\zeta_{pa}^{ij}\int_0^{\infty} dt\, e^{[-\eta + i(\omega-\epsilon_p^{QP}-\epsilon_{pa}^{ij})t]} + \ldots \tag{6.817}$$

$$= \frac{Z_p^{QP}}{\omega - \epsilon_p^{QP} + i\eta} + \sum_{iab}\frac{Z_{piab}^{sat}}{\omega - \epsilon_{piab}^{sat} + i\eta} + \sum_{ija}\frac{Z_{pija}^{sat}}{\omega - \epsilon_{pija}^{sat} + i\eta} + \ldots \tag{6.818}$$

where we defined the satellite energies $\epsilon_{piab}^{sat} = \epsilon_p^{QP} + \epsilon_{pi}^{ab}$ and $\epsilon_{pija}^{sat} = \epsilon_p^{QP} + \epsilon_{pa}^{ij}$, and the corresponding spectral weights $Z_{piab}^{sat} = Z_p^{QP}\zeta_{pi}^{ab}$ and $Z_{pija}^{sat} = Z_p^{QP}\zeta_{pa}^{ij}$. The diagonal elements of the spectral function are obtained as (the virtual satellites spectral function, whose derivation

will mirror that of the occupied satellites, are omitted for the sake of brevity):

$$A_{pp}^{MP2+C}(\omega) = -\frac{1}{\pi}\operatorname{Im} G_{pp}(\omega) \tag{6.819}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{Z_p^{QP}}{\omega - \epsilon_p^{QP} + i\eta} + \sum_{iab}\frac{Z_{pi}^{ab}}{\omega - \epsilon_{pi}^{ab} + i\eta} + \sum_{ija}\frac{Z_{pa}^{ij}}{\omega - \epsilon_{pa}^{ij} + i\eta} + \ldots\right] \tag{6.820}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\operatorname{Re} Z_p^{QP} + i\operatorname{Im} Z_p^{QP}}{\omega - \operatorname{Re}\epsilon_p^{QP} + i\left(\eta - \operatorname{Im}\epsilon_p^{QP}\right)} + \sum_{iab}\frac{\operatorname{Re} Z_{pi}^{ab} + i\operatorname{Im} Z_{pi}^{ab}}{\omega - \operatorname{Re}\epsilon_{pi}^{ab} + i\left(\eta - \operatorname{Im}\epsilon_{pi}^{ab}\right)} + \ldots\right] \tag{6.821}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\left(\operatorname{Re} Z_p^{QP} + i\operatorname{Im} Z_p^{QP}\right)\left(\omega - \operatorname{Re}\epsilon_p^{QP} - i\left(\eta - \operatorname{Im}\epsilon_p^{QP}\right)\right)}{\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)^2 + \left(\operatorname{Im}\epsilon_p^{QP}\right)^2}\right. \tag{6.822}$$

$$\left. + \sum_{iab}\frac{\left(\operatorname{Re} Z_{pi}^{ab} + i\operatorname{Im} Z_{pi}^{ab}\right)\left(\omega - \operatorname{Re}\epsilon_{pi}^{ab} - i\left(\eta - \operatorname{Im}\epsilon_{pi}^{ab}\right)\right)}{\left(\omega - \operatorname{Re}\epsilon_{pi}^{ab}\right)^2 + \left(\operatorname{Im}\epsilon_{pi}^{ab}\right)^2} + \ldots\right] \tag{6.823}$$

$$= -\frac{1}{\pi}\left[\frac{\left(\operatorname{Re} Z_p^{QP}\right)\left(\operatorname{Im}\epsilon_p^{QP}\right) + \left(\operatorname{Im} Z_p^{QP}\right)\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)}{\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)^2 + \left(\operatorname{Im}\epsilon_p^{QP}\right)^2}\right. \tag{6.824}$$

$$\left. + \sum_{iab}\frac{\left(\operatorname{Re} Z_{pi}^{ab}\right)\left(\operatorname{Im}\epsilon_{pi}^{ab}\right) + \left(\operatorname{Im} Z_{pi}^{ab}\right)\left(\omega - \operatorname{Re}\epsilon_{pi}^{ab}\right)}{\left(\omega - \operatorname{Re}\epsilon_{pi}^{ab}\right)^2 + \left(\operatorname{Im}\epsilon_{pi}^{ab}\right)^2} + \ldots\right] \tag{6.825}$$

6.6.3 MP2+C

Here we start with $G_0 = G^{HF}$. To summarize, we are solving

$$G^{MP2+C} = G^{HF} e^{C^{MP2}} \tag{6.826}$$

where C^{MP2} is given by eqn. 6.800 with the single particle energies and two-electron integrals evaluated at the HF level. However, we have seen that doing it this way introduces issues for higher system sizes in metals, where MP2 is known to diverge.

6.6.4 SRG-MP2+C

Here, I tried $G_0 = G^{SRG-MP2}$. That is, I was solving

$$G^{SRG-MP2+C} = G^{SRG-MP2} e^{C^{SRG-MP2}} \tag{6.827}$$

where $C^{SRG-MP2}$ was determined as

$$C_{pp}^{SRG-MP2}(t) = \frac{1}{2}\sum_{iab}\frac{\langle pi \,||\, ab\rangle^2}{\left(\varepsilon_{pi}^{ab}\right)^2}\left(e^{-i\varepsilon_{pi}^{ab}t} + i\varepsilon_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa \,||\, ij\rangle^2}{\left(\varepsilon_{pa}^{ij}\right)^2}\left(e^{-i\varepsilon_{pa}^{ij}t} + i\varepsilon_{pa}^{ij}t - 1\right) \tag{6.828}$$

where $\varepsilon_{pi}^{ab} = \varepsilon_a + \varepsilon_b - \varepsilon_p - \varepsilon_i - i\eta$ and $\varepsilon_{pa}^{ij} = \varepsilon_i + \varepsilon_j - \varepsilon_p - \varepsilon_a - i\eta$ was generated from combinations of the dressed SRG-MP2 single particle energies. This formulation does *not* have firm theoretical grounds. To see why, consider that we derived the second-order cumulant by defining

$$G(t) = G_0(t) + \iint dt_1 dt_2 G_0(t - t_1)\Sigma(t_1 - t_2)G(t_2) \tag{6.829}$$

$$= G_0(t) + \iint dt_1 dt_2 G_0(t - t_1)\Sigma(t_1 - t_2)G_0(t_2) + \ldots \tag{6.830}$$

and

$$G(t) = G_0(t)e^{C(t)} \tag{6.831}$$

$$= G_0(t)\left(1 + C(t) + \ldots\right) \tag{6.832}$$

And then we equated the terms linear in Σ and C to get the expression for C in terms of Σ. If we wanted to use $G_0 = G^{SRG-MP2}$, then the self energy that we would be using is the one generated from $G^{SRG-MP2}$, which is not the same as the second order self energy generated from G^{HF}. That is, it is no longer true that

$$G_0(t)C(t) = \iint dt_1 dt_2 G_0(t - t_1)\Sigma^{(2)}[G^{SRG-MP2}](t_1 - t_2)G_0(t_2) \tag{6.833}$$

up to second order in the bare Coulomb interaction. BWs2+C is more rigorously defined, as we will see below.

6.7 BWs2+C

6.7.1 Electronic structure Hamiltonian

In the below, it is assumed that we are dealing with retarded quantities and in the spin-restricted formalism. The BWs2 partition chooses

$$\hat{H} = \hat{H}_0' + \hat{H}_1', \tag{6.834}$$

with

$$\mathbf{H}_0' = \begin{pmatrix} \mathbf{F}_{oo} + \alpha_1 \mathbf{W}^O & 0 \\ 0 & \mathbf{F}_{vv} + \alpha_2 \mathbf{W}^V \end{pmatrix}, \tag{6.835}$$

where we assume a canonical Hartree-Fock matrix and define

$$W_{ii}^O = \frac{1}{2}\mathbf{\Sigma}_{ii}^{(2),2p1h}(\omega = \tilde{\varepsilon}_i) = \frac{1}{2}\sum_{abj} \frac{\mathcal{K}_{ij}^{ab}}{\tilde{\varepsilon}_i + \tilde{\varepsilon}_j - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b}, \tag{6.836}$$

$$\mathbf{W}_{aa}^V = \frac{1}{2}\mathbf{\Sigma}_{aa}^{(2),2h1p}(\omega = \tilde{\varepsilon}_a) = \frac{1}{2}\sum_{ijb} \frac{\mathcal{K}_{ij}^{ab}}{\tilde{\varepsilon}_a + \tilde{\varepsilon}_b - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j}, \tag{6.837}$$

where $\tilde{\varepsilon}$ is the BWs2 ground-state MO energy and we defined $\mathcal{K}_{pq}^{rs} = (pr|qs)\,[2(pr|qs) - (ps|qr)]$. Note that in this spin-restricted formalism, we must assert that for general α_1, $\mathrm{Tr}(\alpha_1 \mathbf{W}^O) = \frac{\alpha_1}{2} E_{\mathrm{MP2}}$. As usual,

$$\hat{H}_1' = \hat{H} - \hat{H}_0'. \tag{6.838}$$

We define the noninteracting Green's function as

$$G^{(0)'}(\omega) = \big(\omega - (\mathbf{F} + \underbrace{\alpha_1 \mathbf{W}^O \oplus \alpha_2 \mathbf{W}^V}_{\mathbf{W}})\big)^{-1}. \tag{6.839}$$

The Dyson equation must remain unchanged under any partition, so we have

$$\begin{aligned} G^{-1}(\omega) &= \omega - \mathbf{F} - \mathbf{\Sigma}(\omega) && (6.840)\\ &= \omega - (\mathbf{F} + \mathbf{W}) - \big[\mathbf{\Sigma}(\omega) - \mathbf{W}\big] && (6.841)\\ &= \big(G^{(0)'}(\omega)\big)^{-1} - \mathbf{\Sigma}_{\mathrm{eff}}(\omega). && (6.842) \end{aligned}$$

This gives us the Dyson equation for the BWs2 partition:

$$\begin{aligned} G(\omega) &= G^{(0)'}(\omega) + G^{(0)'}(\omega)\mathbf{\Sigma}_{\mathrm{eff}}(\omega)G(\omega) && (6.843)\\ &\approx G^{(0)'}(\omega) + G^{(0)'}(\omega)\mathbf{\Sigma}_{\mathrm{eff}}(\omega)G^{(0)'}(\omega) && (6.844)\\ \implies G(t) &\approx G^{(0)'}(t) + \int dt'dt''\, G^{(0)'}(t-t')\mathbf{\Sigma}_{\mathrm{eff}}(t'-t'')G^{(0)'}(t''). && (6.845) \end{aligned}$$

Note that 6.845 is exact up to second order in the bare Coulomb interaction since

$$\begin{aligned} \mathbf{\Sigma}_{\mathrm{eff}}(\omega) &\equiv \mathbf{\Sigma}(\omega) - \mathbf{W} && (6.846)\\ &\approx \mathbf{\Sigma}^{(2)}\left[G^{(0)'}\right](\omega) - \mathbf{W} && (6.847) \end{aligned}$$

and these quantities are exact to second order in the bare Coulomb interaction. Alternatively, the cumulant ansatz for the Green's function is

$$G(t) = G^{(0)}(t)e^{C(t)}.$$

Thus, the BWs2 cumulant ansatz for state p in the diagonal approximation is

$$\begin{aligned} G_{pp}^{\mathrm{BWs2+C}}(t) &= G_{pp}^{(0)'}(t)e^{C_{pp}^{\mathrm{BWs2}}(t)} && (6.848)\\ &\approx G_{pp}^{(0)'}(t) + G_{pp}^{(0)'}(t)C_{pp}^{\mathrm{BWs2}}(t). && (6.849) \end{aligned}$$

where we have retained terms in the Taylor expansion of the exponential up to first order in the cumulant. Equating (6.845) and (6.849) gives

$$G_{pp}^{(0)'}(t)C_{pp}^{\mathrm{BWs2}}(t) = \int dt'dt''\, G_{pp}^{(0)'}(t-t')\Sigma_{\mathrm{eff},pp}(t'-t'')G_{pp}^{(0)'}(t''), \tag{6.850}$$

which implies

$$\begin{aligned} C_{pp}^{\mathrm{BWs2}}(t) &= i\int \frac{d\omega}{2\pi} e^{-i(\omega-\tilde{\varepsilon}_p)t}\frac{\Sigma_{\mathrm{eff},pp}(\omega)}{(\omega-\tilde{\varepsilon}_p)^2} && (6.851)\\ &= i\int \frac{d\omega'}{2\pi} e^{-i\omega' t}\frac{\Sigma_{\mathrm{eff},pp}(\omega'+\tilde{\varepsilon}_p)}{(\omega')^2}. && (6.852) \end{aligned}$$

6.7.2 Occupied p

$$\Sigma_{\text{eff},pp}(\omega) = \Sigma^{(2)}_{pp}\left[G^{(0)\prime}\right](\omega) - W_{pp} \tag{6.853}$$

$$= \Sigma^{(2)}_{pp}\left[G^{(0)\prime}\right](\omega) - \frac{\alpha_1}{2}\Sigma^{(2),2p1h}_{pp}\left[G^{(0)\prime}\right](\tilde{\varepsilon}_p) \tag{6.854}$$

$$\implies \Sigma_{\text{eff},pp}(\omega+\tilde{\varepsilon}_p) = \Sigma^{(2),2h1p+2p1h}_{pp}\left[G^{(0)\prime}\right](\omega+\tilde{\varepsilon}_p) - \frac{\alpha_1}{2}\Sigma^{(2),2p1h}_{pp}\left[G^{(0)\prime}\right](\tilde{\varepsilon}_p) \tag{6.855}$$

$$= \sum_{ija} \mathcal{K}^{ij}_{pa}\left(\frac{1}{\omega+\tilde{\varepsilon}_p+\tilde{\varepsilon}_a-\tilde{\varepsilon}_i-\tilde{\varepsilon}_j+i\eta}\right) \tag{6.856}$$

$$+ \sum_{iab} \mathcal{K}^{ab}_{pi}\left(\frac{1}{\omega+\tilde{\varepsilon}_p+\tilde{\varepsilon}_i-\tilde{\varepsilon}_a-\tilde{\varepsilon}_b+i\eta} - \frac{\frac{\alpha_1}{2}}{\tilde{\varepsilon}_p+\tilde{\varepsilon}_i-\tilde{\varepsilon}_a-\tilde{\varepsilon}_b+i\eta}\right), \tag{6.857}$$

where we identify $\tilde{\epsilon^{ab}_{pi}} = \tilde{\varepsilon}_a+\tilde{\varepsilon}_b-\tilde{\varepsilon}_p-\tilde{\varepsilon}_i-i\eta$ and $\tilde{\epsilon}^{ij}_{pa} = \tilde{\varepsilon}_i+\tilde{\varepsilon}_j-\tilde{\varepsilon}_p-\tilde{\varepsilon}_a-i\eta$ as the bosonic frequencies in the quasi-boson approximation due to the BWs2 partition. Using the partial fraction decomposition[16] and the time-domain integral identities[17] we find for the 2p1h term:

$$I^{\alpha_1}_{2p1h}(t) = i\int\frac{d\omega}{2\pi}\frac{e^{-i\omega t}}{\omega^2}\left(\frac{1}{\omega-\tilde{\epsilon}^{ab}_{pi}} + \frac{\alpha_1/2}{\tilde{\epsilon}^{ab}_{pi}}\right) \tag{6.862}$$

$$= \frac{e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\left(1-\alpha_1/2\right)\tilde{\epsilon}^{ab}_{pi}t - 1}{(\tilde{\epsilon}^{ab}_{pi})^2} \tag{6.863}$$

The 2h1p term can be integrated as:

$$I_{2h1p}(t) = i\int\frac{d\omega}{2\pi}\frac{e^{-i\omega t}}{\omega^2}\frac{1}{\omega-\tilde{\epsilon}^{ij}_{pa}} \tag{6.864}$$

$$= \frac{e^{-i\tilde{\epsilon}^{ij}_{pa}t} + i\tilde{\epsilon}^{ij}_{pa}t - 1}{(\tilde{\epsilon}^{ij}_{pa})^2} \tag{6.865}$$

[16]
$$\frac{1}{\omega^2(\omega-\epsilon)} = \frac{1}{\epsilon^2}\frac{1}{\omega-\epsilon} - \frac{1}{\epsilon^2}\frac{1}{\omega} - \frac{1}{\epsilon}\frac{1}{\omega^2} \tag{6.858}$$

[17]
$$i\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{1}{\omega-\epsilon} = e^{-i\epsilon t} \tag{6.859}$$

$$i\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{1}{\omega} = 1 \tag{6.860}$$

$$i\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{1}{\omega^2} = -it, \tag{6.861}$$

where in the intermediate steps, we use the partial fraction decomposition[18] and the time-domain integral identities mentioned above. Combining the two yields the final form:

$$C_{pp}^{(2;\alpha_1)}(t) = \sum_{iab} \frac{\mathcal{K}_{pi}^{ab}}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2} \left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\left(1-\alpha_1/2\right)\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \sum_{ija} \frac{\mathcal{K}_{ij}^{pa}}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2} \left(e^{-i\tilde{\epsilon}_{pa}^{ij}t} + i\tilde{\epsilon}_{pa}^{ij}t - 1\right) \tag{6.867}$$

Thus, the retarded Green's function in the BWs2+C method will read

$$G_{pp}^{(2;\alpha_1)+\mathrm{C}}(t) = -i\theta(t)e^{-i\tilde{\varepsilon}_p t}\exp\left\{C_{pp}^{(2;\alpha_1)}(t)\right\}. \tag{6.868}$$

How this reduces to a HF poled Green's function

We can rewrite as

$$G_{pp}^{(2;\alpha_1)+\mathrm{C}}(t) = -i\theta(t)e^{-i\tilde{\varepsilon}_p t}\exp\left\{C_{pp}^{(2;\alpha_1)}(t)\right\} \tag{6.869}$$

$$= -i\theta(t)e^{-i\epsilon_p^{\mathrm{HF}}t}\exp\left\{-i\underbrace{W_{pp}}_{\frac{\alpha_1}{2}\mathbf{\Sigma}_{pp}^{(2),2p1h}[G^{(0)\prime}](\omega=\tilde{\varepsilon}_p)}t + \sum_{iab}\frac{\mathcal{K}_{pi}^{ab}}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2}\left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\left(1-\alpha_1/2\right)\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \ldots\right. \tag{6.870}$$

$$= -i\theta(t)e^{-i\epsilon_p^{\mathrm{HF}}t}\exp\left\{\sum_{iab}\frac{\mathcal{K}_{pi}^{ab}}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2}\left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \sum_{ija}\frac{\mathcal{K}_{ij}^{pa}}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2}\left(e^{-i\tilde{\epsilon}_{pa}^{ij}t} + i\tilde{\epsilon}_{pa}^{ij}t - 1\right)\right\} \tag{6.871}$$

6.7.3 Virtual p

Going through the same procedures will get us to

$$C_{pp}^{(2;\alpha_2)}(t) = \sum_{iab} \frac{\mathcal{K}_{pi}^{ab}}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2} \left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \sum_{ija} \frac{\mathcal{K}_{ij}^{pa}}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2} \left(e^{-i\tilde{\epsilon}_{pa}^{ij}t} + i\left(1-\alpha_2/2\right)\tilde{\epsilon}_{pa}^{ij}t - 1\right) \tag{6.872}$$

with the retarded Green's function reading

$$G_{pp}^{(2;\alpha_2)+\mathrm{C}}(t) = -i\theta(t)e^{-i\tilde{\varepsilon}_p t}\exp\left\{C_{pp}^{(2;\alpha_2)}(t)\right\}. \tag{6.873}$$

This is to be compared with the usual second order cumulant, to which the above reduces when $\alpha_1, \alpha_2 = 0$:

$$C_{pp}^{(2)}(t) = \sum_{iab} \frac{\mathcal{K}_{pi}^{ab}}{\left(\epsilon_{pi}^{ab}\right)^2} \left(e^{-i\epsilon_{pi}^{ab}t} + i\epsilon_{pi}^{ab}t - 1\right) + \sum_{ija} \frac{\mathcal{K}_{ij}^{pa}}{\left(\epsilon_{pa}^{ij}\right)^2} \left(e^{-i\epsilon_{pa}^{ij}t} + i\epsilon_{pa}^{ij}t - 1\right) \tag{6.874}$$

The quasi-boson frequencies are potentially different due to the different choice of partition.

[18]
$$\frac{1}{\omega(\omega-\epsilon)} = \frac{1}{\epsilon}\left(\frac{1}{\omega-\epsilon} - \frac{1}{\omega}\right) \tag{6.866}$$

6.7.4 Self-consistent BWs2

At iteration $n > 0$, the chemical potential is determined from

$$\frac{N}{2} = \sum_p \int_{-\infty}^{\mu^{(n)}} A_p^{(n)}(\omega)\, d\omega, \tag{6.875}$$

where $\mu^{(0)} = \frac{\epsilon^{\rm HF}_{\rm HOMO} + \epsilon^{\rm HF}_{\rm LUMO}}{2}$ and $A_p^{(0)}(\omega)$ is a delta function at the BW single-particle energy. Then, the second-Born spectral kernel is

$$\begin{aligned} B_p^{(n)}(\Omega) = \sum_{qrs} \mathcal{K}_{pq}^{rs} \int d\omega_q\, d\omega_r\, d\omega_s\, A_q^{(n)}(\omega_q) A_r^{(n)}(\omega_r) A_s^{(n)}(\omega_s) \\ \times \left[f_q^{(n)}(1 - f_r^{(n)})(1 - f_s^{(n)}) + (1 - f_q^{(n)}) f_r^{(n)} f_s^{(n)} \right] \\ \times \delta(\Omega - \omega_r - \omega_s + \omega_q) \end{aligned} \tag{6.876}$$

and so we can identify

$$\begin{aligned} B_{p,2p1h}^{(n)}(\Omega) = \sum_{qrs} \mathcal{K}_{pq}^{rs} \int d\omega_q\, d\omega_r\, d\omega_s\, A_q^{(n)}(\omega_q) A_r^{(n)}(\omega_r) A_s^{(n)}(\omega_s) \\ \times f_q^{(n)} \left(1 - f_r^{(n)}\right) \left(1 - f_s^{(n)}\right) \delta(\Omega - \omega_r - \omega_s + \omega_q)\,, \end{aligned} \tag{6.877}$$

$$\begin{aligned} B_{p,2h1p}^{(n)}(\Omega) = \sum_{qrs} \mathcal{K}_{pq}^{rs} \int d\omega_q\, d\omega_r\, d\omega_s\, A_q^{(n)}(\omega_q) A_r^{(n)}(\omega_r) A_s^{(n)}(\omega_s) \\ \times \left(1 - f_q^{(n)}\right) f_r^{(n)} f_s^{(n)} \delta(\Omega - \omega_r - \omega_s + \omega_q)\,, \end{aligned} \tag{6.878}$$

$$B_p^{(n)}(\Omega) = B_{p,2p1h}^{(n)}(\Omega) + B_{p,2h1p}^{(n)}(\Omega), \tag{6.879}$$

where $f_x^{(n)} \equiv f^{(n)}(\omega_x)$ with $f^{(n)}(\omega) = \Theta\big(\mu^{(n)} - \omega\big)$ at zero temperature. The associated retarded correlation self-energy is

$$\Sigma_{p,c}^{R,(n)}(\omega) = \int d\Omega\, \frac{B_p^{(n)}(\Omega)}{\omega - \Omega + i0^+}. \tag{6.880}$$

In the initial iteration, this takes on the one-shot form

$$\Sigma_{p,c}^{R,(0)}(\omega) = \int d\Omega\, \frac{B_p^{(0)}(\Omega)}{\omega - \Omega + i0^+} \tag{6.881}$$

$$= \int d\Omega\, \frac{1}{\omega - \Omega + i0^+} \left[\sum_{iab} \mathcal{K}_{pi}^{ab}\, \delta(\Omega - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b + \tilde{\varepsilon}_i) + \sum_{ija} \mathcal{K}_{ij}^{pa}\, \delta(\Omega - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j + \tilde{\varepsilon}_a) \right] \tag{6.882}$$

$$= \sum_{iab} \frac{\mathcal{K}_{pi}^{ab}}{\omega + \tilde{\varepsilon}_i - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b + i0^+} + \sum_{ija} \frac{\mathcal{K}_{ij}^{pa}}{\omega + \tilde{\varepsilon}_a - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j + i0^+} \tag{6.883}$$

$$\equiv \Sigma_{p,c}^{(2),2p1h}\left[G^{(0)\prime}\right](\omega) + \Sigma_{p,c}^{(2),2h1p}\left[G^{(0)\prime}\right](\omega). \tag{6.884}$$

We are always taking

$$A_p^{(n)}(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G_p^{R,(n)}(\omega) \tag{6.885}$$

and at each iteration, we have

$$\Sigma_k^{x,(n)} = n_k^{(n)} v_m - \sum_{\boldsymbol{q}\neq\boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}}^{(n)} v(\boldsymbol{q}), \tag{6.886}$$

where v_m is the Madelung constant and $v(\boldsymbol{q})$ is the Fourier transform of the Coulomb potential. We have $n_k^{(n)} = \int_{-\infty}^{\mu^{(n)}} A_k^{(n)}(\omega)\,d\omega$.

Dyson formalism

Here we can write out

$$A_p^{(n+1)}(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G_p^{R,(n+1)}(\omega) \tag{6.887}$$

$$= -\frac{1}{\pi}\frac{\mathrm{Im}\,\Sigma_{p,c}^{R,(n)}(\omega)}{\left[\omega - p^2/2 - \Sigma_p^{x,(n)} - \mathrm{Re}\,\Sigma_{p,c}^{R,(n)}(\omega)\right]^2 + \left[\mathrm{Im}\,\Sigma_{p,c}^{R,(n)}(\omega)\right]^2}. \tag{6.888}$$

Cumulant formalism

Define

$$\xi_p^{(n)} = p^2/2 + \Sigma_p^{x,(n)} + \Lambda_p^{(n)}, \tag{6.889}$$

with

$$\Lambda_p^{(n)} = \begin{cases} \dfrac{\alpha_1}{2}\,\mathrm{Re}\,\Sigma_{p,c}^{R,(n),2p1h}\left(\xi_p^{(n)}\right), & p \in \mathcal{O}, \\ \dfrac{\alpha_2}{2}\,\mathrm{Re}\,\Sigma_{p,c}^{R,(n),2h1p}\left(\xi_p^{(n)}\right), & p \in \mathcal{V}, \end{cases} \tag{6.890}$$

where

$$\mathrm{Re}\,\Sigma_{p,c}^{R,(n),2p1h}(\omega) = \mathcal{P}\int d\Omega\,\frac{B_{p,2p1h}^{(n)}(\Omega)}{\omega - \Omega}, \tag{6.891}$$

$$\mathrm{Re}\,\Sigma_{p,c}^{R,(n),2h1p}(\omega) = \mathcal{P}\int d\Omega\,\frac{B_{p,2h1p}^{(n)}(\Omega)}{\omega - \Omega}. \tag{6.892}$$

This gives

$$C_p^{(2),(n)}(t) = \int d\Omega\,B_p^{(n)}(\Omega)\frac{e^{-i(\Omega-\xi_p^{(n)})t} + i(\Omega - \xi_p^{(n)})t - 1}{(\Omega - \xi_p^{(n)})^2} \tag{6.893}$$

$$= \int d\Omega\,\frac{B_p^{(n)}(\Omega + \xi_p^{(n)})}{\Omega^2}\left(e^{-i\Omega t} + i\Omega t - 1\right). \tag{6.894}$$

Then, we get the Green's function via

$$G_p^{R,(n+1)}(t) = -i\Theta(t)e^{-i(p^2/2+\Sigma_p^{x,(n)})t}e^{C_p^{(2),(n)}(t)} \tag{6.895}$$

In the initial iteration, this takes on the one-shot form of

$$C_p^{(2),(0)}(t) = \int d\Omega \, \frac{B_p^{(0)}(\Omega + \xi_p^{(0)})}{\Omega^2} \left(e^{-i\Omega t} + i\Omega t - 1\right) \tag{6.896}$$

$$= \sum_{iab} \frac{\mathcal{K}_{pi}^{ab}}{(\tilde{\epsilon}_{pi}^{ab})^2} \left(e^{-i\tilde{\epsilon}_{pi}^{ab} t} + i\tilde{\epsilon}_{pi}^{ab} t - 1\right) + \sum_{ija} \frac{\mathcal{K}_{ij}^{pa}}{(\tilde{\epsilon}_{pa}^{ij})^2} \left(e^{-i\tilde{\epsilon}_{pa}^{ij} t} + i\tilde{\epsilon}_{pa}^{ij} t - 1\right), \tag{6.897}$$

where $\tilde{\epsilon}_{pq}^{rs} = \tilde{\varepsilon}_r + \tilde{\varepsilon}_s - \tilde{\varepsilon}_p - \tilde{\varepsilon}_q - i0^+$ and $\mathcal{K}_{pq}^{rs} = (pr|qs)\,[2(pr|qs) - (ps|qr)]$.

6.7.5 Why Garnet's HF+GW is wrong

I was able to match Garnet's HF+GW and I think that was wrong. An ingredient of HF+GW is dRPA@HF, in which we use

$$A_{ia,jb}(\mathbf{q}) = \delta_{ij}\delta_{ab}(\varepsilon_a^{\mathrm{HF}} - \varepsilon_i^{\mathrm{HF}}) + 2(ia|bj) \tag{6.898}$$

By Koopmans', $\varepsilon_a^{\mathrm{HF}}$ is the EA of the a-th virtual orbital, and $\varepsilon_i^{\mathrm{HF}}$ is the negative IP of the i-th occupied orbital.

The first thing you write

Your paper writes that "inclusion of the Madelung constant in the Hamiltonian means the IP gets shifted by $-\frac{\xi}{2}$ and the EA gets shifted by $+\frac{\xi}{2}$". This then implies that in this shifted Ewald scheme the difference between virtual and occupied Hartree-Fock single-particle energies in equation (6.898) can be written as

$$\varepsilon_a^{\mathrm{HF}} - \varepsilon_i^{\mathrm{HF}} = \bar{\varepsilon}_a^{\mathrm{HF}} + \frac{\xi}{2} - \left(\bar{\varepsilon}_i^{\mathrm{HF}} + \frac{\xi}{2}\right) = \bar{\varepsilon}_a^{\mathrm{HF}} - \bar{\varepsilon}_i^{\mathrm{HF}}, \tag{6.899}$$

where $\bar{\varepsilon}_a^{\mathrm{HF}}$ and $\bar{\varepsilon}_i^{\mathrm{HF}}$ are the single-particle energies that one would get from standard Ewald summation, where all Madelung contribution is neglected. So, computation of screening via dRPA@HF should never see a Madelung contribution.

The second thing you write

However, we know that the HF poles on the spectral function are not at $\bar{\varepsilon}_i^{\mathrm{HF}} + \frac{\xi}{2}$ and $\bar{\varepsilon}_a^{\mathrm{HF}} + \frac{\xi}{2}$, respectively, but rather at $\bar{\varepsilon}_i^{\mathrm{HF}} + \xi$ and $\bar{\varepsilon}_a^{\mathrm{HF}}$. This is because you then go on to write that "regardless of whether the Madelung constant is included in the Hamiltonian, there is an additional correction of $-\frac{\xi}{2}$ for both the IP and EA to remove spurious image interactions coming from the excess of charge created".

Why both things can be true

The dRPA calculation computes neutral excitation energies, which never see an excess of charge. So it must deal with $\bar{\varepsilon}_i^{\mathrm{HF}} + \frac{\xi}{2}$ and $\bar{\varepsilon}_a^{\mathrm{HF}} + \frac{\xi}{2}$. Meanwhile, the HF spectral pole occurs

at the frequency at which the excess of charge is created. So it must deal with $\bar{\varepsilon}_i^{\mathrm{HF}} + \xi$ and $\bar{\varepsilon}_a^{\mathrm{HF}}$. *So the IPs and EAs of the HF reference are not the same as the image-corrected HF poles of the physical spectral function.*

6.7.6 Insights from the UEG Hamiltonian

We begin from

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \underbrace{\frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{k}\mathbf{k}'} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}'-\mathbf{q}}^\dagger a_{\mathbf{k}'} a_{\mathbf{k}}}_{\hat{V}'}, \tag{6.900}$$

where $\mathbf{k}, \mathbf{k}'$ run over all valid momenta, and $\mathbf{q}$ runs over all nonzero momentum transfers. $\epsilon_{\mathbf{k}}^{(0)} = \frac{k^2}{2} + \frac{v_M}{2}$ is the bare single-particle energy with $v_M = -2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_s^{-1}$ the Madelung constant for the cubic simulation cell. We now normal order the two-body

interaction $\hat{V}'$ with respect to the HF reference $|\Phi_0\rangle$. Wick's theorem gives

$$a^\dagger a^\dagger a a =: a^\dagger a^\dagger a a : + \sum_{\text{single contractions}} (\text{contraction}) :a^\dagger a: + \sum_{\text{complete contractions}} (\text{contractions})$$

(6.901)

$$\implies a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}} \tag{6.902}$$

$$\begin{aligned}
&=: a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}} : \\
&\quad + \overbrace{a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}}} : a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} : - \overbrace{a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}}} : a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k}} : \\
&\quad - a^\dagger_{\mathbf{k+q}} \overbrace{a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}}} : a^\dagger_{\mathbf{k+q}} a_{\mathbf{k'}} : + a^\dagger_{\mathbf{k+q}} \overbrace{a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}}} : a^\dagger_{\mathbf{k+q}} a_{\mathbf{k}} : \\
&\quad + \overbrace{a^\dagger_{\mathbf{k+q}} \overbrace{a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}}}} - \overbrace{a^\dagger_{\mathbf{k+q}} \overbrace{a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}}}}
\end{aligned} \tag{6.903}$$

$$\begin{aligned}
&=: a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}} : \\
&\quad + \delta_{\mathbf{q},\mathbf{0}} n_{\mathbf{k}} : a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} : - \delta_{\mathbf{k+q},\mathbf{k'}} n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k}} : - \delta_{\mathbf{k'-q},\mathbf{k}} n_{\mathbf{k}} : a^\dagger_{\mathbf{k+q}} a_{\mathbf{k'}} : + \delta_{\mathbf{q},\mathbf{0}} n_{\mathbf{k'}} : a^\dagger_{\mathbf{k+q}} a_{\mathbf{k}} : \\
&\quad + \delta_{\mathbf{q},\mathbf{0}} \delta_{\mathbf{q},\mathbf{0}} n_{\mathbf{k}} n_{\mathbf{k'}} - \delta_{\mathbf{k+q},\mathbf{k'}} \delta_{\mathbf{k'-q},\mathbf{k}} n_{\mathbf{k+q}} n_{\mathbf{k}}
\end{aligned} \tag{6.904}$$

$$\overset{\mathbf{q}\neq\mathbf{0}}{=}: a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}} : - \delta_{\mathbf{k+q},\mathbf{k'}} n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k}} : - \delta_{\mathbf{k'-q},\mathbf{k}} n_{\mathbf{k}} : a^\dagger_{\mathbf{k+q}} a_{\mathbf{k'}} : - \delta_{\mathbf{k+q},\mathbf{k'}} \delta_{\mathbf{k'-q},\mathbf{k}} n_{\mathbf{k+q}} n_{\mathbf{k}} \tag{6.905}$$

$$\implies \hat{V}' = \frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{kk'}} \Big[: a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}} : - \delta_{\mathbf{k},\mathbf{k'}} n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k}} : \tag{6.906}$$

$$- \delta_{\mathbf{k},\mathbf{k'}} n_{\mathbf{k}} : a^\dagger_{\mathbf{k+q}} a_{\mathbf{k'}} : - \delta_{\mathbf{k+q},\mathbf{k'}} \delta_{\mathbf{k'-q},\mathbf{k}} n_{\mathbf{k+q}} n_{\mathbf{k}} \Big] \tag{6.907}$$

$$= \underbrace{\frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{kk'}} : a^\dagger_{\mathbf{k+q}} a^\dagger_{\mathbf{k'-q}} a_{\mathbf{k'}} a_{\mathbf{k}} :}_{\hat{V}_{\text{res}}} \underbrace{- \frac{1}{\Omega} \sum_{\mathbf{k}} \sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k}} a_{\mathbf{k}} :}_{\sum_{\mathbf{k}} \Sigma^x_{\mathbf{k}} :\hat{n}_{\mathbf{k}}:} \underbrace{- \frac{1}{2\Omega} \sum_{\mathbf{k}} \sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k}} n_{\mathbf{k+q}}}_{E_x} \tag{6.908}$$

$$\implies \hat{H} = \sum_{\mathbf{k}} \epsilon^{(0)}_{\mathbf{k}} \left(n_{\mathbf{k}} + : a^\dagger_{\mathbf{k}} a_{\mathbf{k}} : \right) + E_{\text{x}} + \sum_{\mathbf{k}} \Sigma^x_{\mathbf{k}} : \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} : + \hat{V}_{\text{res}} \tag{6.909}$$

$$\equiv E_{\text{HF}} + \underbrace{\sum_{\mathbf{k}} \epsilon^{\text{HF}}_{\mathbf{k}} : \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} :}_{\hat{F}} + \hat{V}_{\text{res}} \tag{6.910}$$

$$\implies \hat{H}_N = \hat{H} - E_{\text{HF}} = \hat{F} + \hat{V}_{\text{res}}. \tag{6.911}$$

with $\epsilon^{\text{HF}}_{\mathbf{k}} = \epsilon^{(0)}_{\mathbf{k}} + \Sigma^x_{\mathbf{k}} = \frac{k^2}{2} + \frac{v_M}{2} - \frac{1}{\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k+q}}$ and $E_{\text{HF}} = \sum_{\mathbf{k}} \epsilon^{(0)}_{\mathbf{k}} n_{\mathbf{k}} + E_{\text{x}}$.

MP partitioning

Partition the normal-ordered Hamiltonian into a zero-order part $\hat{F}$ and a perturbation $\hat{V}_{\text{res}}$. Also, define the Madelung-free Fock operator $\hat{F}' = \hat{F} - \frac{v_M}{2}\hat{N}$ with eigenvalues $\epsilon'^{\text{HF}}_{\mathbf{k}} = \frac{k^2}{2} -$

$\frac{1}{\Omega}\sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k}+\mathbf{q}}$. RSPT gives the second-order correlation energy as

$$E^{(2)} = \frac{1}{2}\sum_{ijab}\frac{|\langle\Phi_{ij}^{ab}|\hat{V}_{\rm res}|\Phi_0\rangle|^2}{E_0^{(0)} - E_{ij,ab}^{(0)}} \tag{6.912}$$

$$\overset{\rm RHF}{=} \sum_{ijab}\frac{\mathcal{K}_{ij}^{ab}}{E_0'^{(0)} - E_{ij,ab}'^{(0)}} \tag{6.913}$$

$$= E'^{(2)}, \tag{6.914}$$

since, in the restricted formalism $\frac{1}{2}|\langle\Phi_{ij}^{ab}|\hat{V}_{\rm res}|\Phi_0\rangle|^2 \equiv \mathcal{K}_{ij}^{ab} = (ia|jb)\left[2(ia|jb) - (ib|ja)\right]$ and

$$E_{ij,ab}^{(0)} - E_0^{(0)} = \langle\Phi_{ij}^{ab}|\hat{F}|\Phi_{ij}^{ab}\rangle - \langle\Phi_0|\hat{F}|\Phi_0\rangle \tag{6.915}$$

$$= \langle\Phi_{ij}^{ab}|\hat{F}'|\Phi_{ij}^{ab}\rangle - \langle\Phi_0|\hat{F}'|\Phi_0\rangle + \frac{\upsilon_M}{2}\langle\Phi_{ij}^{ab}|\hat{N} - N|\Phi_{ij}^{ab}\rangle \tag{6.916}$$

$$= \underbrace{\langle\Phi_{ij}^{ab}|\hat{F}'|\Phi_{ij}^{ab}\rangle}_{E_{ij,ab}'^{(0)}} - \underbrace{\langle\Phi_0|\hat{F}'|\Phi_0\rangle}_{E_0'^{(0)}} \tag{6.917}$$

So, we have shown that the MP2 correlation energy does not depend on the Madelung potential.

Second-order self-energy

The first iteration of BWs2 must satisfy the constraint $\mathrm{Tr}(\mathbf{W}^O) - \mathrm{Tr}(\mathbf{W}^V) = E_N^{(2)}$. To see this, consider the definitions

$$W_{ii}^O \equiv \frac{1}{2}\mathbf{\Sigma}_{ii}^{(2),2p1h}[\hat{F}](\omega = \epsilon_i^{\rm HF}) = \frac{1}{2}\sum_{abj}\frac{\mathcal{K}_{ij}^{ab}}{\epsilon_i^{\rm HF} + \epsilon_j^{\rm HF} - \epsilon_a^{\rm HF} - \epsilon_b^{\rm HF}} \tag{6.918}$$

$$= \frac{1}{2}\mathbf{\Sigma}_{ii}^{(2),2p1h}[\hat{F}'](\omega = \epsilon_i'^{\rm HF}) = \frac{1}{2}\sum_{abj}\frac{\mathcal{K}_{ij}^{ab}}{\epsilon_i'^{\rm HF} + \epsilon_j'^{\rm HF} - \epsilon_a'^{\rm HF} - \epsilon_b'^{\rm HF}}, \tag{6.919}$$

$$W_{aa}^V \equiv \frac{1}{2}\mathbf{\Sigma}_{aa}^{(2),2h1p}[\hat{F}](\omega = \epsilon_a^{\rm HF}) = \frac{1}{2}\sum_{ijb}\frac{\mathcal{K}_{ij}^{ab}}{\epsilon_a^{\rm HF} + \epsilon_b^{\rm HF} - \epsilon_i^{\rm HF} - \epsilon_j^{\rm HF}} \tag{6.920}$$

$$= \frac{1}{2}\mathbf{\Sigma}_{aa}^{(2),2h1p}[\hat{F}'](\omega = \epsilon_a'^{\rm HF}) = \frac{1}{2}\sum_{ijb}\frac{\mathcal{K}_{ij}^{ab}}{\epsilon_a'^{\rm HF} + \epsilon_b'^{\rm HF} - \epsilon_i'^{\rm HF} - \epsilon_j'^{\rm HF}}. \tag{6.921}$$

This implies that the second-order self-energy can be written as

$$\Sigma^{(2)}[F](\omega) \equiv \mathbf{\Sigma}^{(2),2p1h}[F](\omega) + \mathbf{\Sigma}^{(2),2h1p}[F](\omega) \tag{6.922}$$

$$= \sum_{iab}\frac{\mathcal{K}_{pi}^{ab}}{\omega - \epsilon_a^{\rm HF} - (\epsilon_b^{\rm HF} - \epsilon_i^{\rm HF})} + \sum_{ija}\frac{\mathcal{K}_{ij}^{pa}}{\omega - \epsilon_i^{\rm HF} + (\epsilon_a^{\rm HF} - \epsilon_j^{\rm HF})} \tag{6.923}$$

$$= \sum_{iab}\frac{\mathcal{K}_{pi}^{ab}}{\omega' - \epsilon_a'^{\rm HF} - (\epsilon_b'^{\rm HF} - \epsilon_i'^{\rm HF})} + \sum_{ija}\frac{\mathcal{K}_{ij}^{pa}}{\omega' - \epsilon_i'^{\rm HF} + (\epsilon_a'^{\rm HF} - \epsilon_j'^{\rm HF})} \tag{6.924}$$

$$\equiv \mathbf{\Sigma}^{(2),2p1h}[F'](\omega') + \mathbf{\Sigma}^{(2),2h1p}[F'](\omega'), \tag{6.925}$$

where $\omega' = \omega - \frac{v_M}{2}$. Then, the following quasiparticle equations are the same:

$$\omega - \epsilon_p^{\mathrm{HF}} - \mathbf{\Sigma}_{pp}^{(2)}[\hat{F}](\omega) = 0 \tag{6.926}$$

$$\omega' - \epsilon_p'^{\mathrm{HF}} - \mathbf{\Sigma}_{pp}^{(2)}[\hat{F}'](\omega') = 0 \tag{6.927}$$

The same expressions should hold when solving for HF+G0W0 quasiparticle energies, where one just makes the substitution $\mathbf{\Sigma}^{(2)} \to \mathbf{\Sigma}^{\mathrm{GW}}$, where the spectral representation is

$$\mathbf{\Sigma}_{pp}^{\mathrm{G0W0}}(\omega) = \sum_{i\nu_{\boldsymbol{q}}} \frac{|g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})|^2}{\omega - \epsilon_i^{\mathrm{HF}} - \omega_{\nu_{\boldsymbol{q}}}^{\mathrm{pl}} + i0^+} + \sum_{a\nu_{\boldsymbol{q}}} \frac{|g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})|^2}{\omega - \epsilon_a^{\mathrm{HF}} + \omega_{\nu_{\boldsymbol{q}}}^{\mathrm{pl}} - i0^+}, \tag{6.928}$$

with the plasmonic frequencies $\omega_{\nu_{\boldsymbol{q}}}^{\mathrm{pl}}$ and couplings $g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})$ are computed from a Madelung-free polarizability for reasons discussed below.

6.8 Deriving the electron-boson Hamiltonian

We start from

$$\hat{H} = E_{\mathrm{HF}} + \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} : \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} : + \hat{V}_{\mathrm{res}}, \tag{6.929}$$

where $\hat{V}_{\mathrm{res}} = \frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{k}\mathbf{k}'} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}'-\mathbf{q}}^\dagger a_{\mathbf{k}'} a_{\mathbf{k}}$.

Quadratic bosonic bath from orbital rotations

Define the particle–hole excitation and de-excitation operators $E_{i\mathbf{q}}^\dagger = a_{\mathbf{i}+\mathbf{q}}^\dagger a_{\mathbf{i}}$ and $E_{i-q} = a_{\mathbf{i}}^\dagger a_{\mathbf{i}-\mathbf{q}}$, respectively, where $\mathbf{i} \in \mathcal{O}$ and $\mathbf{i}\pm\mathbf{q} \in \mathcal{V}$. Define a vector of orbital-rotation amplitudes $\boldsymbol{\kappa}_{\mathbf{q}} = \begin{pmatrix} \kappa_{1\mathbf{q}} \\ \kappa_{2\mathbf{q}} \\ \vdots \end{pmatrix}$ and the doubled coordinate vector $\mathbf{z}_{\mathbf{q}} = \begin{pmatrix} \boldsymbol{\kappa}_{\mathbf{q}} \\ \boldsymbol{\kappa}_{-\mathbf{q}}^* \end{pmatrix}$. Then, the anti-Hermitian orbital-rotation generator is

$$\hat{\kappa}[\mathbf{z}] = \sum_{i\mathbf{q}} \left[\kappa_{i\mathbf{q}} E_{i\mathbf{q}}^\dagger - \kappa_{i\mathbf{q}}^* E_{i\mathbf{q}} \right] \implies \hat{\kappa}^\dagger = -\hat{\kappa}. \tag{6.930}$$

Taking the single best Slater determinant in the HF $|\Phi_0\rangle$ as our equilibrium point, nearby Slater determinants are parameterized as

$$|\Phi(\mathbf{z})\rangle = e^{\hat{\kappa}[\mathbf{z}]} |\Phi_0\rangle, \tag{6.931}$$

where $|\Phi_0\rangle$ is the Hartree–Fock determinant. The corresponding energy functional is

$$E(\mathbf{z}) = \langle\Phi(\mathbf{z})|\hat{H}|\Phi(\mathbf{z})\rangle \tag{6.932}$$

$$= \langle\Phi_0|e^{-\hat{\kappa}[\mathbf{z}]} \hat{H} e^{\hat{\kappa}[\mathbf{z}]}|\Phi_0\rangle \tag{6.933}$$

$$= E_{\mathrm{HF}} + \langle\Phi_0|[\hat{H}, \hat{\kappa}[\mathbf{z}]]|\Phi_0\rangle + \frac{1}{2}\langle\Phi_0|[[\hat{H}, \hat{\kappa}[\mathbf{z}]], \hat{\kappa}[\mathbf{z}]]|\Phi_0\rangle + \mathcal{O}(\kappa^3[\mathbf{z}]), \tag{6.934}$$

where we used the BCH expansion $e^{-\hat{\kappa}[\mathbf{z}]}\hat{H}e^{\hat{\kappa}[\mathbf{z}]} = \hat{H} + [\hat{H},\hat{\kappa}[\mathbf{z}]] + \frac{1}{2}[[\hat{H},\hat{\kappa}[\mathbf{z}]],\hat{\kappa}[\mathbf{z}]] + \mathcal{O}(\kappa^3[\mathbf{z}])$. The generalized gradient of the energy is

$$\nabla_{\mathbf{z}} E(\mathbf{z}) = \begin{pmatrix} \dfrac{\partial E(\mathbf{z})}{\partial \boldsymbol{\kappa}} \\ \\ \dfrac{\partial E(\mathbf{z})}{\partial \boldsymbol{\kappa}^*} \end{pmatrix} \implies \nabla_{\mathbf{z}} E(\mathbf{z})\big|_{\mathbf{z}=0} = 0. \tag{6.935}$$

since for all $i\mathbf{q}$, $\frac{\partial E(\mathbf{z})}{\partial \kappa_{i\mathbf{q}}}\Big|_{\mathbf{z}=0} = \langle\Phi_0|[\hat{H}, E^\dagger_{i\mathbf{q}}]|\Phi_0\rangle = 0$ and $\frac{\partial E(\mathbf{z})}{\partial \kappa^*_{i,-\mathbf{q}}}\Big|_{\mathbf{z}=0} = -\langle\Phi_0|[\hat{H}, E_{i,-\mathbf{q}}]|\Phi_0\rangle = 0$ by the Brillouin theorem. The Hessian is

$$\mathcal{H}_{\mathrm{HF}} = \left.\frac{\partial^2 E(\mathbf{z})}{\partial \begin{pmatrix} \boldsymbol{\kappa}^* \\ \boldsymbol{\kappa} \end{pmatrix} \partial \begin{pmatrix} \boldsymbol{\kappa} & \boldsymbol{\kappa}^* \end{pmatrix}}\right|_{\mathbf{z}=0} = \begin{pmatrix} \dfrac{\partial^2 E}{\partial \boldsymbol{\kappa}^* \partial \boldsymbol{\kappa}} & \dfrac{\partial^2 E}{\partial \boldsymbol{\kappa}^* \partial \boldsymbol{\kappa}^*} \\ \dfrac{\partial^2 E}{\partial \boldsymbol{\kappa} \partial \boldsymbol{\kappa}} & \dfrac{\partial^2 E}{\partial \boldsymbol{\kappa} \partial \boldsymbol{\kappa}^*} \end{pmatrix}_{\mathbf{z}=0} \tag{6.936}$$

$$= \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix}. \tag{6.937}$$

The energy functional then becomes

$$E(\kappa,\kappa^*) = E_{\mathrm{HF}} + \frac{1}{2}\begin{pmatrix} \boldsymbol{\kappa}^*_{\mathbf{q}} & \boldsymbol{\kappa}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} \boldsymbol{\kappa}_{\mathbf{q}} \\ \boldsymbol{\kappa}^*_{-\mathbf{q}} \end{pmatrix} + \mathcal{O}(\kappa^3) \tag{6.938}$$

$$\implies E^2(\kappa,\kappa^*) = \frac{1}{2}\begin{pmatrix} \boldsymbol{\kappa}^*_{\mathbf{q}} & \boldsymbol{\kappa}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} \boldsymbol{\kappa}_{\mathbf{q}} \\ \boldsymbol{\kappa}^*_{-\mathbf{q}} \end{pmatrix} \tag{6.939}$$

Quantizing the space as $\boldsymbol{\kappa}^*_{\mathbf{q}} \to \hat{\boldsymbol{b}}^\dagger_{\mathbf{q}}$ and $\boldsymbol{\kappa}_{-\mathbf{q}} \to \hat{\boldsymbol{b}}_{-\mathbf{q}}$, where $\hat{\boldsymbol{b}}^\dagger_{\mathbf{q}}$ and $\hat{\boldsymbol{b}}_{-\mathbf{q}}$ are bosonic creation and annihilation operators, respectively, leads to

$$\hat{H}^{(2)} = \frac{1}{2}\begin{pmatrix} \hat{\boldsymbol{b}}^\dagger_{\mathbf{q}} & \hat{\boldsymbol{b}}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{b}}_{\mathbf{q}} \\ \hat{\boldsymbol{b}}^\dagger_{-\mathbf{q}} \end{pmatrix} \tag{6.940}$$

Redefining the bosonic operators according to the Bogoliubov transformation as

$$\begin{pmatrix} \hat{\boldsymbol{b}}_{\mathbf{q}} \\ \hat{\boldsymbol{b}}^\dagger_{-\mathbf{q}} \end{pmatrix} = \begin{pmatrix} \mathbf{X}_{\mathbf{q}} & \mathbf{Y}_{\mathbf{q}} \\ \mathbf{Y}^*_{-\mathbf{q}} & \mathbf{X}^*_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_{\mathbf{q}} \\ \hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{q}} \end{pmatrix}, \quad \begin{pmatrix} \hat{\boldsymbol{\beta}}^\dagger_{\mathbf{q}} & \hat{\boldsymbol{\beta}}_{-\mathbf{q}} \end{pmatrix} = \begin{pmatrix} \mathbf{b}^\dagger_{\mathbf{q}} & \mathbf{b}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \mathbf{X}^\dagger_{\mathbf{q}} & \mathbf{Y}^{\mathrm{T}}_{-\mathbf{q}} \\ \mathbf{Y}^\dagger_{\mathbf{q}} & \mathbf{X}^{\mathrm{T}}_{-\mathbf{q}} \end{pmatrix}, \tag{6.941}$$

yields

$$\hat{H}^{(2)} = \frac{1}{2}\begin{pmatrix} \hat{\boldsymbol{\beta}}^\dagger_{\mathbf{q}} & \hat{\boldsymbol{\beta}}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \mathbf{X}^\dagger_{\mathbf{q}} & \mathbf{Y}^{\mathrm{T}}_{-\mathbf{q}} \\ \mathbf{Y}^\dagger_{\mathbf{q}} & \mathbf{X}^{\mathrm{T}}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \begin{pmatrix} \mathbf{X}_{\mathbf{q}} & \mathbf{Y}_{\mathbf{q}} \\ \mathbf{Y}^*_{-\mathbf{q}} & \mathbf{X}^*_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_{\mathbf{q}} \\ \hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{q}} \end{pmatrix} \tag{6.942}$$

$$= \frac{1}{2}\begin{pmatrix} \hat{\boldsymbol{\beta}}^\dagger_{\mathbf{q}} & \hat{\boldsymbol{\beta}}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \boldsymbol{\Omega}_{\mathbf{q}} & 0 \\ 0 & \boldsymbol{\Omega}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}}_{\mathbf{q}} \\ \hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{q}} \end{pmatrix} \tag{6.943}$$

$$= \frac{1}{2}\begin{pmatrix} \hat{\boldsymbol{\beta}}^\dagger_{\mathbf{q}} & \hat{\boldsymbol{\beta}}_{-\mathbf{q}} \end{pmatrix} \begin{pmatrix} \boldsymbol{\Omega}_{\mathbf{q}}\hat{\boldsymbol{\beta}}_{\mathbf{q}} \\ \boldsymbol{\Omega}_{-\mathbf{q}}\hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{q}} \end{pmatrix} \tag{6.944}$$

$$= \hat{\boldsymbol{\beta}}^\dagger_{\mathbf{q}}\boldsymbol{\Omega}_{\mathbf{q}}\hat{\boldsymbol{\beta}}_{\mathbf{q}} + \frac{1}{2}\operatorname{Tr}(\boldsymbol{\Omega}_{\mathbf{q}} - A) + \frac{1}{2}\operatorname{Tr}(A) \tag{6.945}$$

$$= \sum_{\nu_{\mathbf{q}}} \omega_{\nu_{\mathbf{q}}} \hat{\beta}^\dagger_{\nu_{\mathbf{q}}}\hat{\beta}_{\nu_{\mathbf{q}}} + E_{\mathrm{RPA}} + \frac{1}{2}\operatorname{Tr}(A) \tag{6.946}$$

where $\mathbf{\Omega_q} = \mathbf{\Omega_{-q}}$ are diagonal matrices of the RPA excitation energies and we once again used the fact that normal ordering of the string leads to $\hat{\boldsymbol{\beta}}_{-\mathbf{q}}\mathbf{\Omega}_{-\mathbf{q}}\hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{q}} = \mathrm{Tr}(\mathbf{\Omega_q}) + \hat{\boldsymbol{\beta}}^\dagger_{-\mathbf{q}}\mathbf{\Omega}_{-\mathbf{q}}\hat{\boldsymbol{\beta}}_{-\mathbf{q}}$.

Fermionic one-hole Hamiltonian

We can write

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) = \sum_{\mathbf{k},\mathbf{k}'\in\mathcal{O}} h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}(\boldsymbol{\kappa}) h^\dagger_{\mathbf{k}'} h_{\mathbf{k}}. \tag{6.947}$$

Expanding to first order about the Hartree–Fock reference gives

$$h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}(\boldsymbol{\kappa}) = h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}(0) + \sum_{i\mathbf{q}} \left.\frac{\partial h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \kappa_{i\mathbf{q}} + \sum_{i\mathbf{q}} \left.\frac{\partial h^{\mathrm{h}}_{\mathbf{k}'\mathbf{k}}}{\partial \kappa^*_{i\mathbf{q}}}\right|_0 \kappa^*_{i\mathbf{q}} + \mathcal{O}(\kappa^2), \tag{6.948}$$

Consider

$$\left.\frac{\partial h^{\mathrm{h}}_{\mathbf{k+q},\mathbf{k}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 = -\left.\frac{\partial F_{\mathbf{k},\mathbf{k+q}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \tag{6.949}$$

$$= -\sum_{\mathbf{l}} \frac{\partial F_{\mathbf{k},\mathbf{k+q}}}{\partial P_{\mathbf{l+q},\mathbf{l}}} \left.\frac{\partial P_{\mathbf{l+q},\mathbf{l}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \tag{6.950}$$

$$= -\frac{v(\mathbf{q})}{\Omega} \sum_{\mathbf{l}} \left.\frac{\partial P_{\mathbf{l+q},\mathbf{l}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 \tag{6.951}$$

$$= -\frac{v(\mathbf{q})}{\Omega} \tag{6.952}$$

because, by definition

$$P(\boldsymbol{\kappa}) = P_0 + \sum_{j\mathbf{Q}} \left[\kappa_{j\mathbf{Q}} |j+\mathbf{Q}\rangle\langle j| + \kappa^*_{j\mathbf{Q}} |j\rangle\langle j+\mathbf{Q}|\right] + \mathcal{O}(\kappa^2) \tag{6.953}$$

$$\implies P_{\mathbf{l+q},\mathbf{l}} \equiv \langle \mathbf{l}+\mathbf{q}|P(\boldsymbol{\kappa})|\mathbf{l}\rangle \tag{6.954}$$

$$= \langle \mathbf{l}+\mathbf{q}|P_0|\mathbf{l}\rangle + \sum_{j\mathbf{Q}} \kappa_{j\mathbf{Q}} \langle \mathbf{l}+\mathbf{q}|\mathbf{j}+\mathbf{Q}\rangle\langle \mathbf{j}|\mathbf{l}\rangle + \sum_{j\mathbf{Q}} \kappa^*_{j\mathbf{Q}} \langle \mathbf{l}+\mathbf{q}|\mathbf{j}\rangle\langle \mathbf{j}+\mathbf{Q}|\mathbf{l}\rangle + \mathcal{O}(\kappa^2) \tag{6.955}$$

$$\implies \left.\frac{\partial P_{\mathbf{l+q},\mathbf{l}}}{\partial \kappa_{i\mathbf{q}}}\right|_0 = \delta_{\mathbf{l},i} + \mathcal{O}(\kappa). \tag{6.956}$$

where we truncated at linear order and $P_0 = \sum_{i\in\mathcal{O}} |i\rangle\langle i|$ is the Hartree–Fock density matrix. Similar arguments will hold for the other derivative. Thus, keeping only terms linear in the particle–hole displacement yields

$$\hat{H}_{\mathrm{h}}(\boldsymbol{\kappa}) = -\sum_{\mathbf{k}\in\mathcal{O}} \epsilon^{\mathrm{HF}}_{\mathbf{k}} h^\dagger_{\mathbf{k}} h_{\mathbf{k}} - \sum_{\mathbf{kq}} \sum_i \frac{v(\mathbf{q})}{\Omega} h^\dagger_{\mathbf{k+q}} h_{\mathbf{k}} \left(\kappa_{i\mathbf{q}} + \kappa^*_{i,-\mathbf{q}}\right) + \mathcal{O}(\kappa^2). \tag{6.957}$$

Again quantizing as $\kappa_{i\mathbf{q}} \to \hat{b}_{i\mathbf{q}}$ and $\kappa^*_{i,-\mathbf{q}} \to \hat{b}^\dagger_{i,-\mathbf{q}}$, the linear coupling becomes

$$\hat{H}_{\mathrm{h-pl}} = -\frac{1}{\Omega} \sum_{\mathbf{kq}} \sum_i v(\mathbf{q}) h^\dagger_{\mathbf{k+q}} h_{\mathbf{k}} \left(\hat{b}_{i\mathbf{q}} + \hat{b}^\dagger_{i,-\mathbf{q}}\right). \tag{6.958}$$

Applying the same Bogoliubov transformation as above yields

$$\hat{\mathbf{b}}_{\mathbf{q}} + \hat{\mathbf{b}}^{\dagger}_{-\mathbf{q}} = (X_{\mathbf{q}} + Y_{\mathbf{q}}) \left(\hat{\boldsymbol{\beta}}_{\mathbf{q}} + \hat{\boldsymbol{\beta}}^{\dagger}_{-\mathbf{q}} \right). \tag{6.959}$$

Substitution into $\hat{H}_{\mathrm{h-ph}}$ gives

$$\hat{H}_{\mathrm{h-pl}} = \sum_{\mathbf{kq}} \sum_{\nu_{\mathbf{q}}} g^{\mathrm{pl}}_{\nu}(\mathbf{k}, \mathbf{q}) h^{\dagger}_{\mathbf{k+q}} h_{\mathbf{k}} \left(\hat{\beta}_{\nu_{\mathbf{q}}} + \hat{\beta}^{\dagger}_{\nu_{-\mathbf{q}}} \right), \tag{6.960}$$

where the coupling to the collective RPA mode is

$$g^{\mathrm{pl}}_{\nu}(\mathbf{k}, \mathbf{q}) = -\frac{v(\mathbf{q})}{\Omega} \sum_{\mathbf{i}} \left[X_{i\nu}(\mathbf{q}) + Y_{i\nu}(\mathbf{q}) \right]. \tag{6.961}$$

So overall, we have obtained (omitting constant terms)

$$\hat{H}_{\mathrm{eb}} = -\sum_{\mathbf{k} \in \mathcal{O}} \epsilon^{\mathrm{HF}}_{\mathbf{k}} h^{\dagger}_{\mathbf{k}} h_{\mathbf{k}} + \sum_{\nu_{\mathbf{q}}} \omega^{\mathrm{pl}}_{\nu_{\mathbf{q}}} \hat{\beta}^{\dagger}_{\nu_{\mathbf{q}}} \hat{\beta}_{\nu_{\mathbf{q}}} + \sum_{\mathbf{kq}} \sum_{\nu_{\mathbf{q}}} g^{\mathrm{pl}}_{\nu}(\mathbf{k}, \mathbf{q}) h^{\dagger}_{\mathbf{k+q}} h_{\mathbf{k}} \left(\hat{\beta}_{\nu_{\mathbf{q}}} + \hat{\beta}^{\dagger}_{\nu_{-\mathbf{q}}} \right). \tag{6.962}$$

A single charge carrier

The ionization of a single electron has turned the previously neutral UEG into one with a single hole, which is our fermionic charge carrier. The hole may be created in any occupied state, so the fermionic contribution to the total Hamiltonian is

$$\hat{H}_{\mathrm{h}} = \sum_{\boldsymbol{k} \in \mathcal{O}} \epsilon^{h}_{\boldsymbol{k}} h^{\dagger}_{\boldsymbol{k}} h_{\boldsymbol{k}}, \tag{6.963}$$

where $\epsilon^{h}_{\boldsymbol{k}}$ is the hole energy and $h^{\dagger}_{\boldsymbol{k}}$ ($\mathrm{h}_{\boldsymbol{k}}$) creates (annihilates) a hole in the occupied state $\boldsymbol{k}$.

Charge-neutral particle-hole fluctuations up to quadratic order

A second contribution to the total Hamiltonian is through a bath of neutral particle-hole fluctuations. A way to think about this is that at each wavevector accessible to the hole, we place a harmonic oscillator in the N-dimensional Hilbert space, whose equilibrium is at the single best Slater determinant, or the HF state. We want to consider fluctuations about this equilibrium, as generated by OV rotation, to lowest non-trivial order, which is quadratic. Specifically, with Φ_0 the HF reference, we define a particle-hole excited state as

$$|\Phi(\kappa)\rangle = e^{\sum_{ia} \kappa_{ia} a^{\dagger}_{a} a_i} |\Phi_0\rangle \tag{6.964}$$

$$\approx \left(1 + \sum_{ia} \kappa_{ia} a^{\dagger}_{a} a_i + \frac{1}{2} \sum_{ia,jb} \kappa_{ia} \kappa_{jb} a^{\dagger}_{a} a_i a^{\dagger}_{b} a_j \right) |\Phi_0\rangle, \tag{6.965}$$

$$\hat{H}^{(2)}_{\mathrm{ph}} = E_{\mathrm{n}} + \sum_{\mathbf{q}} \sum_{ij} A_{ij}(\mathbf{q}) E^{\dagger}_{i\mathbf{q}} E_{j\mathbf{q}} + \frac{1}{2} \sum_{\mathbf{q}} \sum_{ij} \left[B_{ij}(\mathbf{q}) E^{\dagger}_{i\mathbf{q}} E^{\dagger}_{j,-\mathbf{q}} + B^{*}_{ij}(\mathbf{q}) E_{j,-\mathbf{q}} E_{i\mathbf{q}} \right], \tag{6.966}$$

Define the particle–hole excitation and de-excitation operators $E^\dagger_{i\mathbf{q}} = a^\dagger_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}}$ and $E_{i,-\mathbf{q}} = a^\dagger_{\mathbf{i}} a_{\mathbf{i}-\mathbf{q}}$, respectively, where $\mathbf{i} \in \mathcal{O}$ and $\mathbf{i} \pm \mathbf{q} \in \mathcal{V}$. The space defined by these operators is $\mathcal{S}_{\mathbf{q}} = \text{span}\{E^\dagger_{i\mathbf{q}}, E_{i,-\mathbf{q}}\}$. Introduce $\mathcal{P}$, which is a projection onto $\mathcal{S}_{\mathbf{q}}$, and $\mathcal{Q} = 1 - \mathcal{P}$. The EOM for the particle–hole operators is

$$-i\frac{d}{dt}\begin{pmatrix} E^\dagger_{i\mathbf{q}} \\ E_{i,-\mathbf{q}} \end{pmatrix} = \begin{pmatrix} \left[\hat{H}_N, E^\dagger_{i\mathbf{q}}\right] \\ \left[\hat{H}_N, E_{i,-\mathbf{q}}\right] \end{pmatrix} \tag{6.967}$$

$$= \begin{pmatrix} \mathcal{P}\left[\hat{H}_N, E^\dagger_{i\mathbf{q}}\right] \\ \mathcal{P}\left[\hat{H}_N, E_{i,-\mathbf{q}}\right] \end{pmatrix} + \begin{pmatrix} \mathcal{Q}\left[\hat{H}_N, E^\dagger_{i\mathbf{q}}\right] \\ \mathcal{Q}\left[\hat{H}_N, E_{i,-\mathbf{q}}\right] \end{pmatrix} \tag{6.968}$$

$$\approx \begin{pmatrix} \mathcal{P}\left[\hat{H}_N, E^\dagger_{i\mathbf{q}}\right] \\ \mathcal{P}\left[\hat{H}_N, E_{i,-\mathbf{q}}\right] \end{pmatrix} \tag{6.969}$$

$$= \begin{pmatrix} \sum_j \left[A_{ji}(\mathbf{q}) E^\dagger_{j\mathbf{q}} + B_{ji}(\mathbf{q}) E_{j,-\mathbf{q}}\right] \\ -\sum_j \left[A^*_{ji}(\mathbf{q}) E_{j,-\mathbf{q}} + B^*_{ji}(\mathbf{q}) E^\dagger_{j\mathbf{q}}\right] \end{pmatrix} \tag{6.970}$$

We can always write equation (6.970) because the projected commutators belong to $\mathcal{S}_{\mathbf{q}}$, so they must be expressible as linear combinations of the basis operators spanning that space. The second equation follows from Hermitian conjugation of the first.

Since for $\mathbf{q} \neq 0$, $\left\langle \Phi_0 \middle| \left[E_{i\mathbf{q}}, E^\dagger_{j\mathbf{q}}\right] \middle| \Phi_0 \right\rangle = \delta_{ij}$ [19] and $\langle \Phi_0 | [E_{i\mathbf{q}}, E_{j,-\mathbf{q}}] | \Phi_0 \rangle = 0$,

$$A_{ij}(\mathbf{q}) = \sum_k A_{kj}(\mathbf{q}) \delta_{ik} \tag{6.973}$$

$$= \sum_k A_{kj}(\mathbf{q}) \left\langle \Phi_0 \middle| \left[E_{i\mathbf{q}}, E^\dagger_{k\mathbf{q}}\right] \middle| \Phi_0 \right\rangle + \sum_k B_{kj}(\mathbf{q}) \langle \Phi_0 | [E_{i\mathbf{q}}, E_{k,-\mathbf{q}}] | \Phi_0 \rangle \tag{6.974}$$

$$= \left\langle \Phi_0 \middle| \left[E_{i\mathbf{q}}, \mathcal{P}[\hat{H}_N, E^\dagger_{j\mathbf{q}}]\right] \middle| \Phi_0 \right\rangle \tag{6.975}$$

$$= \left\langle \Phi_0 \middle| \left[E_{i\mathbf{q}}, [\hat{F}, E^\dagger_{j\mathbf{q}}]\right] \middle| \Phi_0 \right\rangle + \left\langle \Phi_0 \middle| \left[E_{i\mathbf{q}}, [\hat{V}_{\text{res}}, E^\dagger_{j\mathbf{q}}]\right] \middle| \Phi_0 \right\rangle \tag{6.976}$$

$$= \left(\epsilon^{\text{HF}}_{\mathbf{i}+\mathbf{q}} - \epsilon^{\text{HF}}_{\mathbf{i}}\right) \delta_{ij} + \langle aj || ib \rangle \tag{6.977}$$

$$= \left(\epsilon^{\text{HF}}_{\mathbf{i}+\mathbf{q}} - \epsilon^{\text{HF}}_{\mathbf{i}}\right) \delta_{ij} + \frac{1}{\Omega}\Big[v_q - (1 - \delta_{ij})\, v_{|\mathbf{i}-\mathbf{j}|}\Big] \tag{6.978}$$

where $a = \mathbf{i}+\mathbf{q}$ and $b = \mathbf{j}+\mathbf{q}$ and in the penultimate step, we simplified the Fock operator con-

[19]

$$\left[E_{i\mathbf{q}}, E^\dagger_{j\mathbf{q}}\right] = \left[a^\dagger_{\mathbf{i}} a_{\mathbf{i}+\mathbf{q}}, a^\dagger_{\mathbf{j}+\mathbf{q}} a_{\mathbf{j}}\right] \tag{6.971}$$

$$= \delta_{ij}\left(a^\dagger_{\mathbf{i}} a_{\mathbf{i}} - a^\dagger_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}+\mathbf{q}}\right) \tag{6.972}$$

because $\mathbf{i}$ is occupied while $\mathbf{i}+\mathbf{q}$ is virtual.

tribution [20] , and we used the Slater-Condon rule. Similarly, since $\left\langle \Phi_0 \left| \left[E^\dagger_{i,-\mathbf{q}}, E_{k,-\mathbf{q}} \right] \right| \Phi_0 \right\rangle =$

[20]

$$[\hat{F}, E^\dagger_{i\mathbf{q}}] = \left[\sum_{\mathbf{p}} \epsilon^{\mathrm{HF}}_{\mathbf{p}} :a^\dagger_{\mathbf{p}} a_{\mathbf{p}}:, a^\dagger_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}} \right] \tag{6.979}$$

$$= \sum_{\mathbf{p}} \epsilon^{\mathrm{HF}}_{\mathbf{p}} \left(\left[a^\dagger_{\mathbf{p}} a_{\mathbf{p}}, a^\dagger_{\mathbf{i}+\mathbf{q}} \right] a_{\mathbf{i}} + a^\dagger_{\mathbf{i}+\mathbf{q}} \left[a^\dagger_{\mathbf{p}} a_{\mathbf{p}}, a_{\mathbf{i}} \right] \right) \tag{6.980}$$

$$= \sum_{\mathbf{p}} \epsilon^{\mathrm{HF}}_{\mathbf{p}} \left(\delta_{\mathbf{p},\mathbf{i}+\mathbf{q}} a^\dagger_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}} - \delta_{\mathbf{p},\mathbf{i}} a^\dagger_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}} \right) \tag{6.981}$$

$$= \left(\epsilon^{\mathrm{HF}}_{\mathbf{i}+\mathbf{q}} - \epsilon^{\mathrm{HF}}_{\mathbf{i}} \right) a^\dagger_{\mathbf{i}+\mathbf{q}} a_{\mathbf{i}}. \tag{6.982}$$

$-\delta_{ik}$ and $\langle\Phi_0|\left[E^\dagger_{i,-\mathbf{q}}, E^\dagger_{k\mathbf{q}}\right]|\Phi_0\rangle = 0$,

$$B_{ij}(\mathbf{q}) = -\sum_k B_{kj}(\mathbf{q})\left\langle\Phi_0\left|\left[E^\dagger_{i,-\mathbf{q}}, E_{k,-\mathbf{q}}\right]\right|\Phi_0\right\rangle - \sum_k A_{kj}(\mathbf{q})\left\langle\Phi_0\left|\left[E^\dagger_{i,-\mathbf{q}}, E^\dagger_{k\mathbf{q}}\right]\right|\Phi_0\right\rangle \tag{6.983}$$

$$= -\left\langle\Phi_0\left|\left[E^\dagger_{i,-\mathbf{q}}, \mathcal{P}[\hat{H}_N, E^\dagger_{j\mathbf{q}}]\right]\right|\Phi_0\right\rangle \tag{6.984}$$

$$= -\left(\epsilon^{\mathrm{HF}}_{\mathbf{j+q}} - \epsilon^{\mathrm{HF}}_{\mathbf{j}}\right)\left\langle\Phi_0\left|\left[E^\dagger_{i,-\mathbf{q}}, E^\dagger_{j\mathbf{q}}\right]\right|\Phi_0\right\rangle - \left\langle\Phi_0\left|\left[E^\dagger_{i,-\mathbf{q}}, [\hat{V}_{\mathrm{res}}, E^\dagger_{j\mathbf{q}}]\right]\right|\Phi_0\right\rangle \tag{6.985}$$

$$= -\left\langle\Phi_0\left|E^\dagger_{i,-\mathbf{q}}\hat{V}_{\mathrm{res}}E^\dagger_{j\mathbf{q}} - E^\dagger_{i,-\mathbf{q}}E^\dagger_{j\mathbf{q}}\hat{V}_{\mathrm{res}} - \hat{V}_{\mathrm{res}}E^\dagger_{j\mathbf{q}}E^\dagger_{i,-\mathbf{q}} + E^\dagger_{j\mathbf{q}}\hat{V}_{\mathrm{res}}E^\dagger_{i,-\mathbf{q}}\right|\Phi_0\right\rangle \tag{6.986}$$

$$= \left\langle\Phi_0\left|\hat{V}_{\mathrm{res}}\,E^\dagger_{j\mathbf{q}}\,E^\dagger_{i,-\mathbf{q}}\right|\Phi_0\right\rangle \tag{6.987}$$

$$= \frac{1}{2\Omega}\sum_{\mathbf{p}\neq\mathbf{0}} v_p \sum_{\mathbf{kk'}}\left\langle\Phi_0\left|: a^\dagger_{\mathbf{k+p}}a^\dagger_{\mathbf{k'-p}}a_{\mathbf{k'}}a_{\mathbf{k}} : a^\dagger_{\mathbf{j+q}}a_{\mathbf{j}}a^\dagger_{\mathbf{i-q}}a_{\mathbf{i}}\right|\Phi_0\right\rangle \tag{6.988}$$

$$= \frac{1}{2\Omega}\sum_{\mathbf{p}\neq\mathbf{0}} v_p \sum_{\mathbf{kk'}}\Bigg[a^\dagger_{\mathbf{k+p}}a^\dagger_{\mathbf{k'-p}}a_{\mathbf{k'}}a_{\mathbf{k}}a^\dagger_{\mathbf{j+q}}a_{\mathbf{j}}a^\dagger_{\mathbf{i-q}}a_{\mathbf{i}} + a^\dagger_{\mathbf{k+p}}a^\dagger_{\mathbf{k'-p}}a_{\mathbf{k'}}a_{\mathbf{k}}a^\dagger_{\mathbf{j+q}}a_{\mathbf{j}}a^\dagger_{\mathbf{i-q}}a_{\mathbf{i}}$$
$$- a^\dagger_{\mathbf{k+p}}a^\dagger_{\mathbf{k'-p}}a_{\mathbf{k'}}a_{\mathbf{k}}a^\dagger_{\mathbf{j+q}}a_{\mathbf{j}}a^\dagger_{\mathbf{i-q}}a_{\mathbf{i}} - a^\dagger_{\mathbf{k+p}}a^\dagger_{\mathbf{k'-p}}a_{\mathbf{k'}}a_{\mathbf{k}}a^\dagger_{\mathbf{j+q}}a_{\mathbf{j}}a^\dagger_{\mathbf{i-q}}a_{\mathbf{i}}\Bigg] \tag{6.989}$$

$$= \frac{1}{2\Omega}\sum_{\mathbf{p}\neq\mathbf{0}} v_p \sum_{\mathbf{kk'}}\Big[\delta_{\mathbf{k+p,i}}\,n_{\mathbf{i}}\;\delta_{\mathbf{k'-p,j}}\,n_{\mathbf{j}}\;\delta_{\mathbf{k',j+q}}(1-n_{\mathbf{j+q}})\;\delta_{\mathbf{k,i-q}}(1-n_{\mathbf{i-q}})$$
$$+ \delta_{\mathbf{k+p,j}}\,n_{\mathbf{j}}\;\delta_{\mathbf{k'-p,i}}\,n_{\mathbf{i}}\;\delta_{\mathbf{k',i-q}}(1-n_{\mathbf{i-q}})\;\delta_{\mathbf{k,j+q}}(1-n_{\mathbf{j+q}})$$
$$- \delta_{\mathbf{k+p,i}}\,n_{\mathbf{i}}\;\delta_{\mathbf{k'-p,j}}\,n_{\mathbf{j}}\;\delta_{\mathbf{k',i-q}}(1-n_{\mathbf{i-q}})\;\delta_{\mathbf{k,j+q}}(1-n_{\mathbf{j+q}})$$
$$- \delta_{\mathbf{k+p,j}}\,n_{\mathbf{j}}\;\delta_{\mathbf{k'-p,i}}\,n_{\mathbf{i}}\;\delta_{\mathbf{k',j+q}}(1-n_{\mathbf{j+q}})\;\delta_{\mathbf{k,i-q}}(1-n_{\mathbf{i-q}})\Big] \tag{6.990}$$

$$\overset{\substack{\mathbf{i,j}\in\mathcal{O}\\ \mathbf{i-q,j+q}\in\mathcal{V}}}{=} \frac{1}{2\Omega}\Big[\underbrace{v_{\mathbf{q}}}_{\mathbf{p=q}} + \underbrace{v_{-\mathbf{q}}}_{\mathbf{p=-q}} - \underbrace{v_{\mathbf{i-j-q}}}_{\mathbf{p=i-j-q}} - \underbrace{v_{\mathbf{j-i+q}}}_{\mathbf{p=j-i+q}}\Big] \tag{6.991}$$

$$= \frac{1}{\Omega}\Big[v_q - v_{|\mathbf{i-j-q}|}\Big]. \tag{6.992}$$

Coupling of the hole to particle-hole fluctuations up to linear order

6.8.1 Electron-boson Hamiltonian

Given a set of hole energies $\{\epsilon^h_{\boldsymbol{k}} = -\epsilon^{\mathrm{HF}}_{\boldsymbol{k}}\}$, as well as plasmonic frequencies $\{\omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}\}$ and couplings $\{g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\}$, we can write the effective electron–boson Hamiltonian as

$$\begin{aligned}\hat{H}_{\mathrm{eb}} &= \hat{H}_{\mathrm{el}} + \hat{H}_{\mathrm{b}} + \hat{H}_{\mathrm{eb,int}} \qquad (6.993)\\ &= \sum_{\boldsymbol{k}\in\mathcal{O}} \epsilon^h_{\boldsymbol{k}}\, h^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}} + \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}} b^\dagger_{\nu_{\boldsymbol{q}}} b_{\nu_{\boldsymbol{q}}} \\ &\quad + \sum_{\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}} \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q}) h^\dagger_{\boldsymbol{k}+\boldsymbol{q}} h_{\boldsymbol{k}} \left(b^\dagger_{\nu_{-\boldsymbol{q}}} + b_{\nu_{\boldsymbol{q}}}\right). \qquad (6.994)\end{aligned}$$

The Brillouin–Wigner construction chooses to repartition

$$\hat{H}_{\mathrm{eb}} = \hat{H}^{(0)}_{\mathrm{BW}} + \hat{H}^{(1)}_{\mathrm{BW}} \qquad (6.995)$$

with

$$\hat{H}^{(0)}_{\mathrm{BW}} = \sum_{\boldsymbol{k}\in\mathcal{O}} \underbrace{\left(\epsilon^h_{\boldsymbol{k}} + W_{\boldsymbol{k}\boldsymbol{k}}\right)}_{\tilde{\varepsilon}_{\boldsymbol{k}}} h^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}} + \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}} b^\dagger_{\nu_{\boldsymbol{q}}} b_{\nu_{\boldsymbol{q}}}, \qquad (6.996)$$

where our fermionic space just consists of occupied states so

$$W_{\boldsymbol{k}\boldsymbol{k}} = \alpha \Sigma^{(2)}_{\boldsymbol{k}\boldsymbol{k}}(\omega = \tilde{\varepsilon}_{\boldsymbol{k}}) = \alpha \sum_{\substack{\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}\\ \nu_{\boldsymbol{q}}\in\mathcal{M}}} \frac{\left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2}{\tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}} - \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}}. \qquad (6.997)$$

The upshot is that, before we do anything, we need to solve for this modified BW reference. This implies that we must start by finding a self-consistent solution to the matrix equation

$$\left(\mathbf{F}^{\mathrm{HF}}_h + \alpha \mathbf{W}[\tilde{\boldsymbol{\varepsilon}}_h]\right) \boldsymbol{U} = \tilde{\boldsymbol{\varepsilon}}_h \boldsymbol{U}, \qquad (6.998)$$

where $\mathbf{F}^{\mathrm{HF}}_h = \begin{pmatrix} \epsilon^h_{\boldsymbol{k}_1} & 0 & \cdots & 0 \\ 0 & \epsilon^h_{\boldsymbol{k}_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \epsilon^h_{\boldsymbol{k}_N} \end{pmatrix}$ is the Fock matrix in the occupied space, and $\tilde{\boldsymbol{\varepsilon}}_h = \begin{pmatrix} \tilde{\varepsilon}_{\boldsymbol{k}_1} & 0 & \cdots & 0 \\ 0 & \tilde{\varepsilon}_{\boldsymbol{k}_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \tilde{\varepsilon}_{\boldsymbol{k}_N} \end{pmatrix}$ is the diagonal matrix of the modified BW reference energies. To leave the full Hamiltonian unchanged, the perturbation is defined as

$$\hat{H}^{(1)}_{\mathrm{BW}} = \sum_{\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}} \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q}) h^\dagger_{\boldsymbol{k}+\boldsymbol{q}} h_{\boldsymbol{k}} \left(b^\dagger_{\nu_{-\boldsymbol{q}}} + b_{\nu_{\boldsymbol{q}}}\right) - \sum_{\boldsymbol{k}\in\mathcal{O}} W_{\boldsymbol{k}\boldsymbol{k}} h^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}}. \qquad (6.999)$$

BWs2+C for the electron-boson Hamiltonian

The derivation remains the same until the step

$$C_{\boldsymbol{kk}}^{\mathrm{BWs2}}(t) = i\int \frac{d\omega}{2\pi} e^{-i(\omega-\tilde{\varepsilon}_{\boldsymbol{k}})t} \frac{\Sigma_{\boldsymbol{kk}}^{\mathrm{eff}}(\omega)}{(\omega-\tilde{\varepsilon}_{\boldsymbol{k}})^2} \tag{6.1000}$$

$$= i\int \frac{d\omega'}{2\pi} e^{-i\omega' t} \frac{\Sigma_{\boldsymbol{kk}}^{\mathrm{eff}}(\omega'+\tilde{\varepsilon}_{\boldsymbol{k}})}{(\omega')^2}. \tag{6.1001}$$

Now, we have

$$\Sigma_{\boldsymbol{kk}}^{\mathrm{eff}}(\omega) = \Sigma_{\boldsymbol{kk}}^{(2)}\left[G^{(0)'}\right](\omega) - W_{\boldsymbol{kk}} \tag{6.1002}$$

$$= \Sigma_{\boldsymbol{kk}}^{(2)}\left[G^{(0)'}\right](\omega) - \alpha\Sigma_{\boldsymbol{kk}}^{(2)}\left[G^{(0)'}\right](\tilde{\varepsilon}_{\boldsymbol{k}}) \tag{6.1003}$$

$$\implies \Sigma_{\boldsymbol{kk}}^{\mathrm{eff}}(\omega+\tilde{\varepsilon}_{\boldsymbol{k}}) = \Sigma_{\boldsymbol{kk}}^{(2)}\left[G^{(0)'}\right](\omega+\tilde{\varepsilon}_{\boldsymbol{k}}) - \alpha\Sigma_{\boldsymbol{kk}}^{(2)}\left[G^{(0)'}\right](\tilde{\varepsilon}_{\boldsymbol{k}}) \tag{6.1004}$$

$$= \sum_{\substack{\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}\\ \nu_{\boldsymbol{q}}\in\mathcal{M}}} \frac{\left|g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})\right|^2}{\omega+\tilde{\varepsilon}_{\boldsymbol{k}}-\tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}}-\omega_{\nu_{\boldsymbol{q}}}^{\mathrm{pl}}} - \alpha\sum_{\substack{\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}\\ \nu_{\boldsymbol{q}}\in\mathcal{M}}} \frac{\left|g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})\right|^2}{\tilde{\varepsilon}_{\boldsymbol{k}}-\tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}}-\omega_{\nu_{\boldsymbol{q}}}^{\mathrm{pl}}}. \tag{6.1005}$$

where we identify $\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q}) = -\tilde{\varepsilon}_{\boldsymbol{k}} + \tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}} + \omega_{\nu_{\boldsymbol{q}}}^{\mathrm{pl}}$. Using the partial fraction decomposition[21] and the time-domain integral identities[22] we find:

$$I^\alpha(t) = i\int \frac{d\omega}{2\pi}\frac{e^{-i\omega t}}{\omega^2}\left(\frac{1}{\omega-\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q})} + \frac{\alpha}{\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q})}\right) \tag{6.1010}$$

$$= \frac{e^{-i\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q})t} + i\,(1-\alpha)\,\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q})t - 1}{(\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q}))^2}. \tag{6.1011}$$

This yields the final form:

$$C_{\boldsymbol{kk}}^{(2,\alpha)}(t) = \sum_{\substack{\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}\\ \nu_{\boldsymbol{q}}\in\mathcal{M}}} \frac{\left|g_\nu^{\mathrm{pl}}(\boldsymbol{k},\boldsymbol{q})\right|^2}{(\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q}))^2}\left(e^{-i\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q})t} + i\,(1-\alpha)\,\tilde{\epsilon}^\nu(\boldsymbol{k},\boldsymbol{q})t - 1\right) \tag{6.1012}$$

Thus, the retarded Green's function in the BWs2+C method will read

$$G_{\boldsymbol{kk}}^{(2,\alpha)+\mathrm{C}}(t) = -i\theta(t)e^{-i\tilde{\varepsilon}_{\boldsymbol{k}}t}\exp\left\{C_{\boldsymbol{kk}}^{(2,\alpha)}(t)\right\}. \tag{6.1013}$$

[21]
$$\frac{1}{\omega^2(\omega-\epsilon)} = \frac{1}{\epsilon^2}\frac{1}{\omega-\epsilon} - \frac{1}{\epsilon^2}\frac{1}{\omega} - \frac{1}{\epsilon}\frac{1}{\omega^2} \tag{6.1006}$$

[22]
$$i\int \frac{d\omega}{2\pi} e^{-i\omega t}\frac{1}{\omega-\epsilon} = e^{-i\epsilon t} \tag{6.1007}$$

$$i\int \frac{d\omega}{2\pi} e^{-i\omega t}\frac{1}{\omega} = 1 \tag{6.1008}$$

$$i\int \frac{d\omega}{2\pi} e^{-i\omega t}\frac{1}{\omega^2} = -it, \tag{6.1009}$$

Why BWs2 doesn't diverge in the TDL of the UEG

To show this property, it is sufficient to show that the BWs2 nonlinear eigenvalue equation of

$$\left(\mathbf{F}_h^{\mathrm{HF}} + \alpha \mathbf{W}[\tilde{\boldsymbol{\varepsilon}}_h]\right) \boldsymbol{U} = \tilde{\boldsymbol{\varepsilon}}_h \boldsymbol{U} \tag{6.1014}$$

has a well-defined solution in the TDL.Since we know that Hartree-Fock does not diverge in the TDL, we just need to show this same property for the regularizer. For the arbitrary hole state $\boldsymbol{k}$, the regularizer $\boldsymbol{W} \propto \boldsymbol{\Sigma}^{(2)}$ has the form

$$\Sigma^{(2)}_{\boldsymbol{k}\boldsymbol{k}}(\omega = \tilde{\varepsilon}_{\boldsymbol{k}}) = \int_0^\infty \frac{d^3\boldsymbol{q}}{(2\pi)^3} \frac{|g(\boldsymbol{q})|^2}{\tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}} - \omega_{\mathrm{pl}}(\boldsymbol{q})} \tag{6.1015}$$

$$= \underbrace{\int_0^{\boldsymbol{q}_c} \frac{d^3\boldsymbol{q}}{(2\pi)^3} \frac{|g(\boldsymbol{q})|^2}{\tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}} - \omega_{\mathrm{pl}}(\boldsymbol{q})}}_{\boldsymbol{W}^{\mathrm{IR}}_{\boldsymbol{k}}} + \int_{\boldsymbol{q}_c}^\infty \frac{d^3\boldsymbol{q}}{(2\pi)^3} \frac{|g(\boldsymbol{q})|^2}{\tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}} - \omega_{\mathrm{pl}}(\boldsymbol{q})}, \tag{6.1016}$$

where we have introduced the infitesimally small plasma frequency cutoff $\boldsymbol{q}_c$. We are confident that the second term does not diverge, so it suffices to show that $\boldsymbol{W}^{\mathrm{IR}}_{\boldsymbol{k}}$ does not diverge. For a fixed hole momentum $\boldsymbol{k}$, define

$$D_{\boldsymbol{k}}(\boldsymbol{q}) = \tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_{\boldsymbol{k}+\boldsymbol{q}} - \omega_{\mathrm{pl}}(q). \tag{6.1017}$$

We consider the contribution to the regularizer from an arbitrarily small neighborhood $|\boldsymbol{q}_c| < \delta$ of the origin,

$$W^{\mathrm{IR}}_{\boldsymbol{k}}(\delta) = \int_{\substack{|\boldsymbol{q}_c|<\delta \\ \boldsymbol{k}+\boldsymbol{q}_c\in\mathcal{O}}} \frac{d^3q}{(2\pi)^3} \frac{|g(q)|^2}{D_{\boldsymbol{k}}(\boldsymbol{q})}. \tag{6.1018}$$

In the long-wavelength limit of the 3D UEG, the plasmon-pole representation, which is exact in the TDL, gives

$$|g(q)|^2 = \frac{2\pi e^2 \omega_p}{q^2}, \qquad q \to 0, \tag{6.1019}$$

while continuity of the fermionic dispersion and $\omega_{\mathrm{pl}}(q) \to \omega_p$ imply

$$\lim_{\boldsymbol{q}\to 0} D_{\boldsymbol{k}}(\boldsymbol{q}) = -\omega_p. \tag{6.1020}$$

Since the bulk Sommerfeld plasma frequency $\omega_p > 0$, the definition of this limit guarantees the existence of a $\delta_0 > 0$ such that, for $0 < q < \delta_0$,

$$|D_{\boldsymbol{k}}(\boldsymbol{q}) + \omega_p| < \frac{\omega_p}{2} \tag{6.1021}$$

$$\implies |D_{\boldsymbol{k}}(\boldsymbol{q})| > \frac{\omega_p}{2} \tag{6.1022}$$

$$\implies \frac{1}{|D_{\boldsymbol{k}}(\boldsymbol{q})|} < \frac{2}{\omega_p}. \tag{6.1023}$$

For any $\delta < \delta_0$, it therefore follows that

$$\left|W^{\mathrm{IR}}_{\boldsymbol{k}}(\delta)\right| \leq \int_{|\boldsymbol{q}_c|<\delta} \frac{d^3q}{(2\pi)^3} \frac{|g(q)|^2}{|D_{\boldsymbol{k}}(\boldsymbol{q})|} \tag{6.1024}$$

$$< \frac{2}{\omega_p} \int_{|\boldsymbol{q}_c|<\delta} \frac{d^3q}{(2\pi)^3} \frac{2\pi e^2 \omega_p}{q^2}. \tag{6.1025}$$

Using $d^3q = 4\pi q^2 dq$ gives

$$\left|W_{\boldsymbol{k}}^{\mathrm{IR}}(\delta)\right| < \frac{2e^2}{\pi}\int_0^\delta dq \tag{6.1026}$$

$$= \frac{2e^2}{\pi}\delta. \tag{6.1027}$$

Therefore,

$$\lim_{\delta\to 0^+} W_{\boldsymbol{k}}^{\mathrm{IR}}(\delta) = 0. \tag{6.1028}$$

and there is no infrared divergence of the Brillouin–Wigner regularizer in the TDL of the 3D UEG.

Numerator

The plasmon-pole representation of the correlation part of the screened Coulomb interaction, which is exact in the TDL, is

$$W_c(q,\omega) = \frac{|g(q)|^2}{\omega - \omega_p + i0^+} - \frac{|g(q)|^2}{\omega + \omega_p - i0^+} \tag{6.1029}$$

$$= \frac{2\omega_p|g(q)|^2}{\omega^2 - \omega_p^2}. \tag{6.1030}$$

Now, formally, the correlation part of the screened Coulomb interaction is defined as

$$W_c(q,\omega) = \frac{v(q)}{\epsilon(q,\omega)} - v(q)$$

$$= v(q)\frac{\omega_p^2}{\omega^2 - \omega_p^2} \qquad (q\to 0), \tag{6.1031}$$

because the long-wavelength dielectric function satisfies $\lim_{q\to 0}\epsilon(q,\omega) = 1 - \frac{\omega_p^2}{\omega^2}$. Setting Eqs. (6.1030) and (6.1031) equal to each other gives

$$2\omega_p|g(q)|^2 = v(q)\omega_p^2 \tag{6.1032}$$

$$\implies |g(q)|^2 = \frac{2\pi e^2\omega_p}{q^2}, \tag{6.1033}$$

since $v(q) = 4\pi e^2/q^2$ in three dimensions.

Comparison with the ordinary spectral function

Defining

$$\tilde{\Sigma}_p^{(2),R}(\omega) = \frac{1}{2}\sum_{iab}\frac{|\langle pi\|ab\rangle|^2}{\omega + \tilde{\varepsilon}_i - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b + i\eta_\Sigma} + \frac{1}{2}\sum_{ija}\frac{|\langle pa\|ij\rangle|^2}{\omega + \tilde{\varepsilon}_a - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j + i\eta_\Sigma} \tag{6.1034}$$

where the single-particle energies in the denominator are due to the ground state BWs2 calculation. The spectral function is then given by

$$A_p(\omega) = -\frac{1}{\pi}\operatorname{Im} G_p^R(\omega) \tag{6.1035}$$

$$= -\frac{1}{\pi}\operatorname{Im}\frac{1}{\omega + i\eta_A - \varepsilon_p^{\mathrm{HF}} - \tilde{\Sigma}_p^{(2),R}(\omega)} \tag{6.1036}$$

$$= \frac{1}{\pi}\frac{\eta_A + \Gamma_p(\omega)}{\left[\omega - \varepsilon_p^{\mathrm{HF}} - \operatorname{Re}\tilde{\Sigma}_p^{(2),R}(\omega)\right]^2 + \left[\eta_A + \Gamma_p(\omega)\right]^2} \tag{6.1037}$$

where $\Gamma_p(\omega) = |\operatorname{Im}\tilde{\Sigma}_p^{(2),R}(\omega)|$. We can take $\eta_A = \eta_\Sigma = 0.2\mathrm{eV}$ to facilitate comparison with my BWs2+C spectral function. Let us also put a vertical line for the frequency of the quasi-particle energy of

$$\varepsilon_p^{\mathrm{BWs2}} = \varepsilon_p^{\mathrm{HF}} + Z_p(\tilde{\varepsilon}_p)\operatorname{Re}\tilde{\Sigma}_p^{(2),R}(\varepsilon_p^{\mathrm{BWs2}}) \tag{6.1038}$$

where $Z_p(\tilde{\varepsilon}_p) = \left(1 - \frac{\partial \operatorname{Re}\tilde{\Sigma}_p^{(2),R}(\omega)}{\partial\omega}\big|_{\omega=\tilde{\varepsilon}_p}\right)^{-1}$ is the renormalization factor.

Double check on my previous results

An equivalent way to compute the BWs2 cumulant Greens function should be via

$$\bar{G}_{pp}^{\mathrm{BWs2+C}}(t) = -i\theta(t)e^{-i\varepsilon_p^{HF}t}\exp\left\{\bar{C}_{pp}^{\mathrm{BWs2}}(t)\right\}. \tag{6.1039}$$

where

$$\bar{C}_{pp}^{\mathrm{BWs2}}(t) = \frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2}\left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa\|ij\rangle^2}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2}\left(e^{-i\tilde{\epsilon}_{pa}^{ij}t} - 1\right). \tag{6.1040}$$

We should verify numerically for some small system size (maybe 14e19o) that $G_{pp}^{\mathrm{BWs2+C}}(t) = \bar{G}_{pp}^{\mathrm{BWs2+C}}(t)$ To numerical precision, given that they should be the same.

Frequency domain version

Now, if we plug in our derived expression for $C_{pp}^{\mathrm{BWs2}}(t)$ into the cumulant ansatz for the retarded Green's function:

$$G_{pp}(t) = G_{pp}^{0,BWs2}(t)e^{C_{pp}^{BWs2}(t)} \tag{6.1041}$$

$$= -i\Theta(t)e^{-i\tilde{\epsilon}_p t + C_{pp}^{BWs2}(t)} \tag{6.1042}$$

$$= -i\Theta(t)e^{-i\tilde{\epsilon}_p t}\exp\left\{\frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2}\left(e^{-i\tilde{\epsilon}_{pi}^{ab}t} + i\tilde{\epsilon}_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa\|ij\rangle^2}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2}\left(e^{-i\tilde{\epsilon}_{pa}^{ij}t} - 1\right)\right\} \tag{6.1043}$$

$$= -i\Theta(t)\underbrace{e^{-\sum_{iab}\zeta_{pi}^{ab} - \sum_{ija}\zeta_{pa}^{ij}}}_{Z_p^{QP}} e^{-i\overbrace{\left(\tilde{\epsilon}_p - \sum_{iab}\zeta_{pi}^{ab}\tilde{\epsilon}_{pi}^{ab}\right)}^{\epsilon_p^{QP}}t}\exp\left\{\sum_{iab}\zeta_{pi}^{ab}e^{-i\tilde{\epsilon}_{pi}^{ab}t} + \sum_{ija}\zeta_{pa}^{ij}e^{-i\tilde{\epsilon}_{pa}^{ij}t}\right\} \tag{6.1044}$$

where $\zeta_{pi}^{ab} = \frac{1}{2}\frac{\langle pi\|ab\rangle^2}{\left(\tilde{\epsilon}_{pi}^{ab}\right)^2}$ and $\zeta_{pa}^{ij} = \frac{1}{2}\frac{\langle pa\|ij\rangle^2}{\left(\tilde{\epsilon}_{pa}^{ij}\right)^2}$ are the bosonic coupling coefficients. Next, we want to perform a Fourier transform.

$$G_{pp}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t}G_{pp}(t) \tag{6.1045}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt e^{i(\omega-\epsilon_p^{QP})t}\exp\left\{\sum_{iab}\zeta_{pi}^{ab}e^{-i\tilde{\epsilon}_{pi}^{ab}t} + \sum_{ija}\zeta_{pa}^{ij}e^{-i\tilde{\epsilon}_{pa}^{ij}t}\right\} \tag{6.1046}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{i(\omega-\epsilon_p^{QP})t}\left(1 + \sum_{iab}\zeta_{pi}^{ab}e^{-i\tilde{\epsilon}_{pi}^{ab}t} + \sum_{ija}\zeta_{pa}^{ij}e^{-i\tilde{\epsilon}_{pa}^{ij}t} + \ldots\right) \tag{6.1047}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{[-\eta + i(\omega-\epsilon_p^{QP})t]} \tag{6.1048}$$

$$- iZ_p^{QP}\sum_{iab}\zeta_{pi}^{ab}\int_0^{\infty} dt\, e^{[-\eta + i(\omega-\epsilon_p^{QP}-\tilde{\epsilon}_{pi}^{ab})t]} \tag{6.1049}$$

$$- iZ_p^{QP}\sum_{a\nu}\zeta_{pa}^{ij}\int_0^{\infty} dt\, e^{[-\eta + i(\omega-\epsilon_p^{QP}-\tilde{\epsilon}_{pa}^{ij})t]} + \ldots \tag{6.1050}$$

$$= \frac{Z_p^{QP}}{\omega - \epsilon_p^{QP} + i\eta} + \sum_{iab}\frac{Z_{piab}^{sat}}{\omega - \epsilon_{piab}^{sat} + i\eta} + \sum_{ija}\frac{Z_{pija}^{sat}}{\omega - \epsilon_{pija}^{sat} + i\eta} + \ldots \tag{6.1051}$$

where we defined the satellite energies $\epsilon_{piab}^{sat} = \epsilon_p^{QP} + \tilde{\epsilon}_{pi}^{ab}$ and $\epsilon_{pija}^{sat} = \epsilon_p^{QP} + \tilde{\epsilon}_{pa}^{ij}$, and the corresponding spectral weights $Z_{piab}^{sat} = Z_p^{QP}\zeta_{pi}^{ab}$ and $Z_{pija}^{sat} = Z_p^{QP}\zeta_{pa}^{ij}$. The diagonal elements of the spectral function are obtained as (the virtual satellites spectral function, whose derivation

will mirror that of the occupied satellites, are omitted for the sake of brevity):

$$A_{pp}^{MP2+C}(\omega) = -\frac{1}{\pi}\operatorname{Im} G_{pp}(\omega) \tag{6.1052}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{Z_p^{QP}}{\omega - \epsilon_p^{QP} + i\eta} + \sum_{iab}\frac{Z_{piab}^{sat}}{\omega - \tilde{\epsilon}_{pi}^{ab} + i\eta} + \sum_{ija}\frac{Z_{pija}^{sat}}{\omega - \tilde{\epsilon}_{pa}^{ij} + i\eta} + \ldots\right] \tag{6.1053}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\operatorname{Re} Z_p^{QP} + i\operatorname{Im} Z_p^{QP}}{\omega - \operatorname{Re}\epsilon_p^{QP} + i\left(\eta - \operatorname{Im}\epsilon_p^{QP}\right)} + \sum_{iab}\frac{\operatorname{Re} Z_{piab}^{sat} + i\operatorname{Im} Z_{piab}^{sat}}{\omega - \operatorname{Re}\epsilon_{piab}^{sat} + i\left(\eta - \operatorname{Im}\epsilon_{piab}^{sat}\right)} + \ldots\right] \tag{6.1054}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\left(\operatorname{Re} Z_p^{QP} + i\operatorname{Im} Z_p^{QP}\right)\left(\omega - \operatorname{Re}\epsilon_p^{QP} - i\left(\eta - \operatorname{Im}\epsilon_p^{QP}\right)\right)}{\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)^2 + \left(\operatorname{Im}\epsilon_p^{QP}\right)^2}\right. \tag{6.1055}$$

$$\left. + \sum_{iab}\frac{\left(\operatorname{Re} Z_{piab}^{sat} + i\operatorname{Im} Z_{piab}^{sat}\right)\left(\omega - \operatorname{Re}\epsilon_{piab}^{sat} - i\left(\eta - \operatorname{Im}\epsilon_{piab}^{sat}\right)\right)}{\left(\omega - \operatorname{Re}\epsilon_{piab}^{sat}\right)^2 + \left(\operatorname{Im}\epsilon_{piab}^{sat}\right)^2} + \ldots\right] \tag{6.1056}$$

$$= -\frac{1}{\pi}\left[\frac{\left(\operatorname{Re} Z_p^{QP}\right)\left(\operatorname{Im}\epsilon_p^{QP}\right) + \left(\operatorname{Im} Z_p^{QP}\right)\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)}{\left(\omega - \operatorname{Re}\epsilon_p^{QP}\right)^2 + \left(\operatorname{Im}\epsilon_p^{QP}\right)^2}\right. \tag{6.1057}$$

$$\left. + \sum_{iab}\frac{\left(\operatorname{Re} Z_{piab}^{sat}\right)\left(\operatorname{Im}\epsilon_{piab}^{sat}\right) + \left(\operatorname{Im} Z_{piab}^{sat}\right)\left(\omega - \operatorname{Re}\epsilon_{piab}^{sat}\right)}{\left(\omega - \operatorname{Re}\epsilon_{piab}^{sat}\right)^2 + \left(\operatorname{Im}\epsilon_{piab}^{sat}\right)^2} + \ldots\right] \tag{6.1058}$$

Everything I present in the plots is one-shot. When I speak of GF2+C with analytic frequency integration, I am building the cumulant as

$$C_{pp}(t) = \sum_{iab}\frac{(pa|ib)\left(2(pa|ib) - (pb|ia)\right)}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t} + i\epsilon_{pi}^{ab}t - 1\right) \tag{6.1059}$$

$$+ \sum_{ija}\frac{(pi|aj)\left(2(pi|aj) - (pj|ai)\right)}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t} + i\epsilon_{pa}^{ij}t - 1\right)$$

with $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$. As can be seen, this gives qualitatively the same result as GF2+C with numerical frequency integration, where I instead build the cumulant as

$$C_{pp}(t) = \int d\omega \frac{\beta_{pp}(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.1060}$$

with $\beta_{pp}(\omega) = \frac{1}{\pi}|\operatorname{Im}\Sigma_{pp}(\omega + \epsilon_p)|$ with the self-energy defined in chemists notation as

$$\Sigma_{pq}^{(2)}(\omega) = \sum_{iab}\frac{(pa|ib)\Big(2(qa|ib) - (qb|ia)\Big)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija}\frac{((pi|aj)(2(qi|aj) - (qj|ai)))}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{6.1061}$$

or in physicists notation as

$$\Sigma^{(2)}_{pq}(\omega) = \sum_{iab} \frac{\langle pi|ab\rangle \left(2\langle qi|ab\rangle - \langle qi|ba\rangle\right)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{\langle pa|ij\rangle \left(2\langle qa|ij\rangle - \langle qa|ji\rangle\right)}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{6.1062}$$

If I remember correctly, our idea for a self consistent scheme was to solve equation 6.1377 using

$$\Sigma_{pp}(\omega) = \frac{1}{G^0_{pp}(\omega)} - \frac{1}{G_{pp}(\omega)} \tag{6.1063}$$

In the initial trials, I have not been getting any useful results with the above. However, I notice that I only use under 1e4 frequency grid points. Should I try to get this scheme to work with something like 1e6 grid points or were you thinking of something different?

6.8.2 BW-PEOM(2)+C

The second-order self-energy can be written as

$$\begin{aligned}\Sigma^{(2)}_{pq}(\omega) =& \frac{1}{2}\sum_{iab} \frac{\langle pi||ab\rangle\langle ab||qi\rangle}{\omega + \epsilon_i - \epsilon_a - \epsilon_b + i\eta} \\ &+ \frac{1}{2}\sum_{ija} \frac{\langle pa||ij\rangle\langle ij||qa\rangle}{\omega + \epsilon_a - \epsilon_i - \epsilon_j + i\eta}.\end{aligned} \tag{6.1064}$$

The P-EOM-MBPT(2) self-energy may instead be written as

$$\Sigma^{\mathrm{PEOM(2)}}_{ij}(\omega) = \underbrace{\frac{1}{2}\sum_{l,c,d} \frac{\langle il||cd\rangle\langle cd||jl\rangle}{\epsilon_j + \epsilon_l - \epsilon_c - \epsilon_d}}_{2p1h} - \underbrace{\frac{1}{2}\sum_{k,l,d} \frac{\tilde{V}_{id[kl]}\langle kl||jd\rangle}{\epsilon_k + \epsilon_l - \epsilon_d - \omega}}_{2h1p}. \tag{6.1065}$$

Here,

$$\begin{aligned}\tilde{V}_{kb[ij]} =& \langle kb||ij\rangle + \sum_{l,d}\langle kl||jd\rangle t^{bd}_{il} - \sum_{l,d}\langle lk||jd\rangle t^{bd}_{il} \\ &+ \sum_{c,d}\langle bk||dc\rangle t^{dc}_{ji} - \sum_{l,c}\langle lk||jc\rangle t^{bc}_{li},\end{aligned} \tag{6.1066}$$

with

$$t^{ab}_{ij} = \frac{\langle ab||ij\rangle}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}. \tag{6.1067}$$

Defining

$$\mathbf{S} = \mathbf{\Sigma}^{\mathrm{PEOM(2)},2p1h}, \tag{6.1068}$$

the Dyson equation may be partitioned as

$$G^{-1}(\omega) \approx \omega - (\mathbf{F} + \alpha\mathbf{S}) - \left[\mathbf{\Sigma}^{\mathrm{PEOM(2)}}[G^{(0)'}](\omega) - \alpha\mathbf{S}\right] \tag{6.1069}$$

$$= \left(G^{(0)'}(\omega)\right)^{-1} - \mathbf{\Sigma}_{\mathrm{eff}}(\omega), \tag{6.1070}$$

where

$$G^{(0)'}(\omega) = [\omega - (\mathbf{F} + \alpha\mathbf{S})]^{-1} \tag{6.1071}$$

is the BWS2 reference Green's function. Expanding the Dyson equation to first order in the residual self-energy gives

$$G(t) \approx G^{(0)'}(t) + \int dt' dt'' \, G^{(0)'}(t - t') \mathbf{\Sigma}_{\text{eff}}(t' - t'') G^{(0)'}(t''). \tag{6.1072}$$

The BW-PEOM(2) cumulant ansatz for state p, within the diagonal approximation, is

$$G_{pp}^{\text{BW}-\text{PEOM(2)}+C}(t) = G_{pp}^{(0)'}(t) \exp\left[C_{pp}^{\text{BW}-\text{PEOM(2)}}(t)\right] \tag{6.1073}$$

$$\approx G_{pp}^{(0)'}(t) + G_{pp}^{(0)'}(t) C_{pp}^{\text{BW}-\text{PEOM(2)}}(t). \tag{6.1074}$$

Equating Eqs. (6.1072) and (6.1074) gives

$$G_{pp}^{(0)'}(t) C_{pp}^{\text{BW}-\text{PEOM(2)}}(t) = \int dt' dt'' \, G_{pp}^{(0)'}(t - t') \Sigma_{\text{eff},pp}(t' - t'') G_{pp}^{(0)'}(t''), \tag{6.1075}$$

which implies

$$C_{pp}^{\text{BW}-\text{PEOM(2)}}(t) = i \int \frac{d\omega}{2\pi} e^{-i(\omega - \tilde{\varepsilon}_p)t} \frac{\Sigma_{\text{eff},pp}(\omega)}{(\omega - \tilde{\varepsilon}_p)^2} \tag{6.1076}$$

$$= i \int \frac{d\omega}{2\pi} e^{-i\omega t} \frac{\Sigma_{\text{eff},pp}(\omega + \tilde{\varepsilon}_p)}{\omega^2}. \tag{6.1077}$$

Since

$$\Sigma_{pp}^{\text{PEOM(2)}}(\omega) = S_p + \Sigma_{pp}^{\text{PEOM(2)},2h1p}(\omega), \tag{6.1078}$$

the effective self-energy becomes

$$\Sigma_{\text{eff},pp}(\omega + \tilde{\varepsilon}_p) = (1 - \alpha) S_p + \Sigma_{pp}^{\text{PEOM(2)},2h1p}(\omega + \tilde{\varepsilon}_p) \tag{6.1079}$$

$$\overset{\alpha=1}{=} \Sigma_{pp}^{\text{PEOM(2)},2h1p}(\omega + \tilde{\varepsilon}_p). \tag{6.1080}$$

Introducing

$$A_{p;kld} \equiv \tilde{V}_{pd[kl]} \langle kl \| pd \rangle \tag{6.1081}$$

and

$$\tilde{\Omega}_{p;kld} \equiv \tilde{\varepsilon}_k + \tilde{\varepsilon}_l - \tilde{\varepsilon}_p - \tilde{\varepsilon}_d - i\eta, \tag{6.1082}$$

the $2h1p$ self-energy may be written as

$$\Sigma_{pp}^{\text{PEOM(2)},2h1p}(\omega + \tilde{\varepsilon}_p) = -\frac{1}{2} \sum_{kld} \frac{A_{p;kld}}{\tilde{\Omega}_{p;kld} - \omega}. \tag{6.1083}$$

Substituting into Eq. (6.1077) gives

$$C_{pp}^{\text{BW}-\text{PEOM(2)}}(t) \overset{\alpha=1}{=} \frac{1}{2} \sum_{kld} \frac{\tilde{V}_{pd[kl]} \langle kl \| pd \rangle}{\tilde{\Omega}_{p;kld}^2} \left[e^{-i\tilde{\Omega}_{p;kld} t} + i\tilde{\Omega}_{p;kld} t - 1 \right]. \tag{6.1084}$$

6.9 Self-consistent cumulant

The scheme involves multiple junctures at which we are free to make different choices. A few parameters should always stay the same. Specifically, we need to use a consistent set of $\{\omega_i, t_j, \Delta t, \eta, k\}$, where ω_i and t_j are the frequency and time grids related by the discrete Fourier transform. η is the broadening parameter. We should experiment with a range of different values for this. What should we try, and what kind of dependence should this have on my time grid? , and k is the momentum index. We are most in the deep core state. The other parameters can be varied to explore different self-consistent cumulant schemes.

6.9.1 Solving for the retarded Green's function initially

We solve for the Green's function via

$$G_k^R(t) = G_{0,k}^R(t) e^{C_k^R(t)} \tag{6.1085}$$

where the non-interacting cumulant takes the form

$$G_{0,k}^R(t) = -i\Theta(t) e^{-i\epsilon_k t}. \tag{6.1086}$$

1. If we use a HF reference, $\epsilon_k = \epsilon_k^{\mathrm{HF}}$.
2. If we use an LDA reference, $\epsilon_k = \epsilon_k^{\mathrm{LDA}} + (\Sigma^x - V_{xc})_{kk}$, where Σ^x is the Fock exchange operator and V_{xc} is the LDA exchange-correlation potential.

The Landau formula for the cumulant is given by

$$C_k^R(t) = \int d\omega \frac{\beta_k(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1 \right) \tag{6.1087}$$

where

$$\beta_k(\omega) = \frac{1}{\pi} |\mathrm{Im}\Sigma_k^R(\omega + \tilde{\epsilon}_k)| \tag{6.1088}$$

is proportional to the retarded self-energy. In the past, we have always used the proper form, But given that document showed the different exact relationship between the improper self-energy and the retarded cumulant, we also try using that. In the initial iteration, the improper prescription is

1. If we are using a HF or LDA reference , $\tilde{\epsilon}_k = \epsilon_k^{\mathrm{HF}}$ or $\tilde{\epsilon}_k = \epsilon_k^{\mathrm{LDA}}$, respectively.

We have already investigated the one-shot form where we utilize the proper self-energy. Relative to this, the improper self-energy is defined as

Then, do the one-sided causal transform:

$$G_k^R(\omega) = \int_0^\infty dt e^{i\omega t} G_k^R(t). \tag{6.1089}$$

We can do this via the discrete Fourier transform. Here we should use the trick that forms the periodic signal.

Frequency Grid

So, we have

6.9.2 Updating the self-energy

We can try one of two schemes to update the self-energy after the initial iteration $n = 0$.

1. $\Sigma^{P,R}_{k,n>0}(\omega) = \frac{1}{G^R_{0,k}(\omega)} - \frac{1}{G^R_{k,n-1}(\omega)}$ is the appropriate choice for a proper self-energy.

2. $\Sigma^{I,R}_{k,n>0}(\omega) = \frac{G^R_{k,n-1}(\omega) - G^R_{0,k}(\omega)}{\left[G^R_{0,k}(\omega)\right]^2}$ is the appropriate choice for an improper self-energy.

In both schemes, we deal with the complex scalar at the frequency point ω. $G^R_{0,k}(\omega) = \frac{1}{\omega - \epsilon_k + i\eta}$ is the noninteracting Green's function, so $\left[G^R_{0,k}(\omega)\right]^{-1} = \omega - \epsilon_k + i\eta$.

Seeing if we can get an analytic expression for Hartree-Fock reference

After G0W0+C, the resulting Green's function can be shown to be

$$G^R_k(\omega) = -iZ_k \int_0^\infty dt\, e^{i(\omega - \epsilon^{QP}_k)t} \exp\left\{ \sum_{i\nu} \zeta_{ki\nu} e^{-i\Delta_{ki\nu}t} + \sum_{a\nu} \zeta_{ka\nu} e^{-i\Delta_{ka\nu}t} \right\} \tag{6.1090}$$

$$\approx \frac{Z_k}{\omega - \epsilon^{QP}_k + i\eta} + \sum_{i\nu} \frac{\overbrace{Z_k \zeta_{ki\nu}}^{Z^{\text{sat}}{}_{ki\nu}}}{\omega - \underbrace{\left(\epsilon^{QP}_k + \Delta_{ki\nu}\right)}_{\epsilon^{\text{sat}}_{ki\nu}} + i\eta} + \sum_{a\nu} \frac{\overbrace{Z_k \zeta_{ka\nu}}^{Z^{\text{sat}}_{ka\nu}}}{\omega - \underbrace{\left(\epsilon^{QP}_k + \Delta_{ka\nu}\right)}_{\epsilon^{\text{sat}}_{ka\nu}} + i\eta}. \tag{6.1091}$$

where

$$Z_k = \exp\left\{ -\sum_{i\nu} \zeta_{ki\nu} - \sum_{a\nu} \zeta_{ka\nu} \right\} \tag{6.1092}$$

$$\epsilon^{QP}_k = \epsilon^{\text{HF}}_k - \sum_{i\nu} \zeta_{ki\nu} \Delta_{ki\nu} - \sum_{a\nu} \zeta_{ka\nu} \Delta_{ka\nu}. \tag{6.1093}$$

with the definitions $\zeta_{ki\nu} = \left(\frac{M_{ki\nu}}{\Delta_{ki\nu}}\right)^2$ and $\zeta_{ka\nu} = \left(\frac{M_{ka\nu}}{\Delta_{ka\nu}}\right)^2$, M are transition densities, and $\Delta_{ki\nu} = \epsilon_i - \epsilon^{HF}_k - \Omega_\nu - i\eta$ and $\Delta_{ka\nu} = \epsilon_a - \epsilon^{HF}_k + \Omega_\nu - i\eta$. The approximation that we have made to get the final expression, which comes from performing a Taylor expansion of the outer exponential of the first expression, has been shown to generate the main quasiparticle peak along with the first two satellites. After some algebra, we can derive

$$\Sigma^{I,R}_k(\omega) \approx \frac{Z_k \left(x_k - \epsilon_k\right)^2}{x_k - \epsilon^{QP}_k} + \sum_{i\nu} \frac{Z^{\text{sat}}_{ki\nu} \left(x_k - \epsilon_k\right)^2}{x_k - \epsilon^{\text{sat}}_{ki\nu}} + \sum_{a\nu} \frac{Z^{\text{sat}}_{ka\nu} \left(x_k - \epsilon_k\right)^2}{x_k - \epsilon^{\text{sat}}_{ka\nu}} - \left(x_k - \epsilon_k\right) \tag{6.1094}$$

and

$$\Sigma_k^{P,R}(\omega) \approx x_k - \epsilon_k - \left[\frac{Z_k}{x_k - \epsilon_k^{QP}} + \sum_{i\nu} \frac{Z_{ki\nu}^{\text{sat}}}{x_k - \epsilon_{ki\nu}^{\text{sat}}} + \sum_{a\nu} \frac{Z_{ka\nu}^{\text{sat}}}{x_k - \epsilon_{ka\nu}^{\text{sat}}} \right]^{-1} \tag{6.1095}$$

where $x_k = \omega + i\eta$.

Starting from LDA reference

For LDA, we have

$$G^{-1}(\omega) = \left[G_0^{\text{LDA}}(\omega)\right]^{-1} - \left[\Sigma(\omega) - v_{xc}^{\text{LDA}}\right] \tag{6.1096}$$

$$\implies \Sigma^c(\omega) = \left[G_0^{\text{LDA}}(\omega)\right]^{-1} - G^{-1}(\omega) + v_{xc}^{\text{LDA}} - \Sigma^x \tag{6.1097}$$

where $G_0^{\text{LDA}}(\omega) = \frac{1}{\omega - \epsilon_k^{\text{LDA}} + i\eta}$ is the noninteracting Green's function.

6.9.3 Self-consistency condition

We can monitor the Green's function norm at a given iteration $n > 0$ as

$$\Delta_G^{(n)} = \frac{\sqrt{\sum_i w_i |G_k^{(n)}(\omega_i) - G_k^{(n-1)}(\omega_i)|^2}}{\sqrt{\sum_i w_i |G_k^{(n-1)}(\omega_i)|^2}} \tag{6.1098}$$

where w_i are the quadrature weights for the frequency grid chosen above, stopping once $\Delta_G^{(n)} < \epsilon$ for some small threshold ϵ.

1. Experiment with how different parameters, including the broadening η, effect what choice of ϵ is realistic.

6.9.4 Problems with this formulation

To arrive at the Landau form in the first place, we expanded the cumulant ansatz alongside the Dyson equation as

$$G_0 e^{C(t)} \equiv G_0[1 + C(t) + \frac{C(t)^2}{2} + \ldots] = G_0 + G_0\Sigma_{GW}G_0 + G_0\Sigma_{GW}G_0\Sigma_{GW}G_0 + \ldots \equiv G_0 + G_0\Sigma_{GW}G. \tag{6.1099}$$

Then, to the first order in the screened interaction, with which we will associate the parameter λ, we have

$$G_0 C(t) = G_0 \Sigma_{GW} G_0 \tag{6.1100}$$

$$\implies C_k^R(t) = \frac{1}{\pi} \int d\omega \frac{|\Sigma_{GW}^R(\omega + \epsilon_k)|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.1101}$$

Exponentiating introduces additional powers of λ:

$$G_k^R(t) = G_{0,k}^R(t) e^{C_k^R(t)} \propto \lambda + \lambda^2 + \lambda^3 + \dots \tag{6.1102}$$

$$\implies \Sigma_k^R(\omega) = \left[G_0^R(\omega)\right]^{-1} - \left[G^R(\omega)\right]^{-1} \propto \lambda + \lambda^2 + \lambda^3 + \dots \tag{6.1103}$$

where (6.1103) is the proposed self-energy update. But then, we shouldn't plug back into (6.1101), because deriving this expression relied on Σ being correct up to first order in λ, which was true in the initial G0W0 case but is no longer true during our update. If instead we follow Holms' formulation, where they just update G self-consistently, this would also be some sort of approximate formulation since technically we would be including powers of λ into G, but at least now it would always be the case that W stays controlled at the RPA level. To understand if this scheme is actually effective would require numerical experimentation.

6.9.5

In this scheme, we envisioned we would plug the Σ from equation (6.1103) into equation (6.1101) to get a new cumulant. But this would violate the order matching of equation (6.1100) that gave us the Landau form in the first place. Recall that in order to derive the Landau form for the cumulant of

$$C_k^R(t) = \frac{1}{\pi} \int d\omega \frac{|\Sigma_{GW}^R(\omega + \epsilon_k)|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right), \tag{6.1104}$$

We can start by performing the expansion in powers of the screened Coulomb interaction, associating with it the parameter λ. It is clear that at the one-shot level

$$C_k^R(t) = \frac{1}{\pi} \int d\omega \frac{|\Sigma_{GW}^R(\omega + \epsilon_k)|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \propto GW \propto \lambda \tag{6.1105}$$

So then

$$G_{CE,k}^R(t) = G_{0,k}^R(t) e^{C_k^R(t)} \propto \lambda + \lambda^2 + \lambda^3 + \dots. \tag{6.1106}$$

Then, the self-energy update is

$$\Sigma_k^R(\omega) = \left[G_0^R(\omega)\right]^{-1} - \left[G^R(\omega)\right]^{-1} \propto \lambda^{-1} + \lambda^{-2} + \lambda^{-3} + \dots. \tag{6.1107}$$

To arrive at the Landau form in the first place, we expanded the cumulant ansatz alongside the Dyson equation as

$$G_0 e^{C(t)} \equiv G_0[1 + C(t) + \frac{C(t)^2}{2} + \dots] = G_0 + G_0\Sigma G_0 + G_0\Sigma G_0\Sigma G_0 + \dots \equiv G_0 + G_0\Sigma G. \tag{6.1108}$$

Then, to the first order in λ, it is true that

$$G_0 C(t) = G_0 \Sigma G_0 \tag{6.1109}$$

$$\implies C_k^R(t) = \frac{1}{\pi} \int d\omega \frac{|\Sigma_{GW}^R(\omega + \epsilon_k)|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.1110}$$

In this scheme, we envisioned we would plug the Σ from equation (6.1103) into equation (6.1101) to get a new cumulant. But this would violate the order matching of equation (6.1100) that gave us the Landau form in the first place. Recall that in order to derive the Landau form for the cumulant of

$$C_k^R(t) = \frac{1}{\pi}\int d\omega \frac{|\Sigma_{GW}^R(\omega+\epsilon_k)|}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right), \tag{6.1111}$$

which we have used successfully for one-shot calculations, we equated the cumulant ansatz to the Dyson equation in a way that was correct up to first-order (second-order) in the screened (bare) Coulomb interaction, using

$$G_0 C = G_0 \Sigma G_0, \tag{6.1112}$$

for a Σ due to GW (BWs2). But when we go on to form

$$G_{CE,k}^R(t) = G_{0,k}^R(t) e^{C_k^R(t)}, \tag{6.1113}$$

it is known that the resulting interacting Green's function can no longer be understood as just being just correct up to a finite order of the interaction strength (this the same reason why we say that G_{CE} contains an approximate infinity of diagrams). In the current self-consistent scheme, when we want to update Σ, we use the *exact* relation

$$\Sigma_k^R(\omega) = \left[G_0^R(\omega)\right]^{-1} - \left[G^R(\omega)\right]^{-1}, \tag{6.1114}$$

where now it is unclear to which order of interaction strength $G^R(\omega)$ is correct. Therefore it no longer makes sense to utilize equation (6.1101) again, since the derivation of this relation relied on the assumption that we were going to stay at a certain order in the interaction strength.

6.9.6 Results from our scheme

As before, we can do either

1. $\beta_k(\omega) = \left|\mathrm{Im}\,\Sigma^{c,R}(\mathbf{k}, \omega + E_k^{\mathrm{GW}})\right|$ with E_k^{GW} the GW quasiparticle energy, which I call epep@qp.
2. $\beta_k(\omega) = \left|\mathrm{Im}\,\Sigma^{c,R}(\mathbf{k}, \omega + \varepsilon_{\mathbf{k}}^{\mathrm{MF}})\right|$ with $\varepsilon_{\mathbf{k}}^{\mathrm{MF}}$ the non-interacting single-particle energy, which I call epep@mf.

I just present results for epep@mf, as that is what Loos suggested, seems more theoretically sound, and falls closer to the CCSD curve for 114e485o. I show the results for the initial iteration (labeled as LDA+G0W0+C) and then, for the two following iterations after that, which gives a good idea as to what the solution is converging to.

Proper self-energy

Here we start with the standard one-shot retarded cumulant (obtained from the retarded proper G0W0 self-energy Σ^R_{GW}) I have presented before, and then we update the self energy via

$$\Sigma^R_k(\omega) = \left[G^R_0(\omega)\right]^{-1} - \left[G^R(\omega)\right]^{-1}. \tag{6.1115}$$

I have plotted both the spectral function and the self-energy obtained from subsequent iterations. *As we can see, after the one-shot cumulant, the spectral function just becomes featureless because the self-energy loses its features in this update scheme.*

Improper self-energy

Here we start with the retarded G0W0 improper self-energy $\Sigma^R_{\rm imp}$, which is related to the proper one by

$$\Sigma^R_{\rm imp} = \Sigma^R_{GW}\left(1 - G^R_0\Sigma^R_{GW}\right)^{-1}. \tag{6.1116}$$

Consequently, our self-energy update scheme instead uses

$$\Sigma^R_k(\omega) = \frac{G^R(\omega) - G^R_0(\omega)}{\left[G^R_0(\omega)\right]^2}. \tag{6.1117}$$

I have plotted both the spectral function and the self-energy obtained from subsequent iterations. Clearly, the scheme isn't working for the GW case. *Recall that Reichman used this improper retarded, cumulant scheme to get one-shot spectral functions in the TDL, but the results weren't that good there either.*

6.9.7 Results from the Holm-inspired SC-CE(GW0)

As before, we start with the standard one-shot cumulant (obtained from the proper G0W0 self-energy Σ^R_{GW}). But then to do our self-energy update we build

$$\Sigma^R_i(t) = i\sum_j \left[G^R_j(t)W^{0,>}_{ij}(t) + G^<_j(t)W^{0,R}_{ij}(t)\right]. \tag{6.1118}$$

$G^R_j(t)$ and $G^<_j(t)$ are the retarded and lesser cumulant Green's functions for the state j. $W^{0,R}_{ij}(t)$ and $W^{0,>}_{ij}(t)$ are the retarded and greater matrix elements of the bare RPA screened interaction in the time domain. I obtain $W(t)$ by first forming the frequency domain sum-over-states expression for the screened interaction via a Casida diagonalization and then doing a DFT to get it into the time domain. Once we have built $\Sigma^R_i(t)$, we can then Fourier transform to get $\Sigma^R_i(\omega)$, then form a new cumulant via the Landau form, and iterate until convergence.

6.9.8 Choosing the correct pole for our self-consistent cumulant

In a self-consistent cumulant, we update the pole along with the cumulant. But this is nuanced and to see why, we juxtapose how we do the one-shot cumulant starting from Hartree-Fock versus LDA references. We always have $G^R_{CE,k}(t) = G^R_{0,k}(t)e^{C^R_k(t)} = -i\Theta(t)e^{-i\tilde{\epsilon}_k t}e^{C^R_k(t)}$, where $C^R_k(t) = \frac{1}{\pi}\int d\omega \frac{|\Sigma^{c,R}_{GW}(\omega+\epsilon_k)|}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right)$.

1. HF chooses $\tilde{\epsilon}_k = \epsilon_k = \frac{k^2}{2} + \Sigma^x_{kk} \equiv \epsilon^{HF}_k$

2. LDA chooses $\tilde{\epsilon}_k = \epsilon^{LDA}_k + \Sigma^x_{kk} - v^{xc}_{kk} = \frac{k^2}{2} + \Sigma^x_{kk}$ and $\epsilon_k = \epsilon^{LDA}_k$

Therefore, it becomes clear that one always builds the Green function from three main ingredients. The first two are static, constituting the completely non-interacting piece and the bare exchange self-energy. The third is the dynamic correlation piece, which is taken care of by the cumulant. A self-consistent scheme must respect the same split. Therefore, if one started from a quasiparticle G_0, the correlation component would have to be contained in the cumulant, so we would do $\tilde{\epsilon}_k = \epsilon^{QP}_k - \Sigma^c_{kk}(\epsilon^{QP}_k) = \frac{k^2}{2} + \Sigma^x_{kk}$ and $\epsilon_k = \epsilon^{QP}_k$. So SC-CE(GW0) should use $\tilde{\epsilon}_k = \frac{k^2}{2} + \Sigma^x_{kk}$ and $\epsilon_k = \epsilon^{QP}_k$.

6.9.9 A New Scheme for Self-Consistent Cumulant

We can start with a one-shot cumulant computation as

$$C^R_k(t) = \frac{1}{\pi}\int d\omega \frac{|\Sigma^R_{GW}(\omega + \epsilon_k)|}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right), \tag{6.1119}$$

leading to a time-domain Green's function

$$G^R_{CE,k}(t) = G^R_{0,k}(t)e^{C^R_k(t)}. \tag{6.1120}$$

Keeping the screened interaction constant at the RPA level, we can then iterate and build a new self-energy using

$$\Sigma^R_i(t) = i\sum_j \left[G^R_j(t)W^{0,>}_{ij}(t) + G^<_j(t)W^{0,R}_{ij}(t)\right]. \tag{6.1121}$$

In order to do this, note that we have to get the cumulant Green's function for each state j. Now, since we are working with the electron gas, we know that there are abundant degeneracies for states with the same K-vector, so we can exploit that to do fewer computations of G. Also, we already have the spectral representation of the screened interaction $W^{0,R}_{ij}(\omega)$ from the initial Casida diagonalization, so we can just Fourier transform that initially to get to what we need, which is the screened interaction in the time domain. With these quantities, we can form a new GW self-energy, plug it into equation (6.1119) to get a new cumulant, and iterate until convergence.

A scheme for doing self-consistency in G and W

Hedin's equations tell us that we can form the polarizability as

$$P_{ij}(t) = -i\sum_k G_{ik}(t)G_{kj}(-t). \tag{6.1122}$$

Then, we perform a Fourier transform to get this into frequency space yielding $P_{ij}(\omega)$. Then, we can form the screened interaction as

$$W_{ij}(\omega) = v_{ij} + \sum_{kl} v_{ik}P_{kl}(\omega)W_{lj}(\omega), \tag{6.1123}$$

where v_{ij} is the bare Coulomb interaction. Then, we can Fourier transform this back into the time domain to get $W_{ij}(t)$, which we can then use to form a new self-energy as

$$\Sigma_i^R(t) = i \sum_j \left[G_j^R(t) W_{ij}^>(t) + G_j^<(t) W_{ij}^R(t) \right]. \tag{6.1124}$$

Self-consistency condition

We can monitor the Green's function norm at a given iteration $n > 0$ as

$$\Delta_G^{(n)} = \frac{\sqrt{\sum_i |G_k^{(n)}(\omega_i) - G_k^{(n-1)}(\omega_i)|^2}}{\sqrt{\sum_i |G_k^{(n-1)}(\omega_i)|^2}} \tag{6.1125}$$

where ω_i are the frequency grid points chosen, stopping once $\Delta_G^{(n)} < \epsilon$ for some small threshold ϵ.

1. Experiment with how different parameters, including the broadening η, effect what choice of ϵ is realistic.

6.9.10 Parameters of interest

We can obtain the spectral function $A_k(\omega)$ by generating a periodic signal from the converged retarded Green's function. Then, we can compute the total energies E from Galitskii-Migdal as:

$$E = \sum_k \int_{-\infty}^{\mu} (\omega + \epsilon_k) A_k(\omega) d\omega. \tag{6.1126}$$

where μ is the chemical potential, which we can take to be equivalent to the Fermi energy ϵ_F, and $\epsilon_k = \frac{k^2}{2}$ is the bare single-particle energy. This allows us to compute the correlation energy per particle as

$$\epsilon_{\text{corr}} = \frac{E - E_{\text{HF}}}{N} \tag{6.1127}$$

where N is the number of particles. Finally, we can obtain occupation numbers as

$$n_k = \int_{-\infty}^{\mu} A_k(\omega) d\omega. \tag{6.1128}$$

6.10 Update notes

I compute the LDA+G0W0 spectral functions via

$$A(\mathbf{k}, \omega) = \frac{1}{\pi} |\operatorname{Im} G^R(\mathbf{k}, \omega)| = \frac{1}{\pi} \left| \frac{\operatorname{Im} \Sigma^{c,R}(\mathbf{k}, \omega)[\eta_s] + \eta_P}{\left[\omega - (\varepsilon_{\mathbf{k}}^0 + \Sigma_k^{\text{x}} + \operatorname{Re} \Sigma^{c,R}(\mathbf{k}, \omega))\right]^2 + \left[\operatorname{Im} \Sigma^{c,R}(\mathbf{k}, \omega)[\eta_s] + \eta_P\right]^2} \right|, \tag{6.1129}$$

where I used $\eta_s = \eta_P = 0.8$ eV , computing $\mathrm{Im}\,\Sigma^{c,R}(\mathbf{k},\omega)[\eta_s]$ via the exact sum over states expression. In computing LDA+G0W0+C, I use

$$C_k(t) = \pi^{-1} \int d\omega \, |\mathrm{Im}\,\beta_k(\omega)| \, \frac{e^{-i\omega t} + i\omega t - 1}{\omega^2} \tag{6.1130}$$

and then build the Green's function as

$$G^R(\mathbf{k},t) = -i\theta(t) e^{-i\varepsilon^{\mathrm{eff}}_{\mathbf{k}} t} e^{C_k(t)}, \tag{6.1131}$$

where $\varepsilon^{\mathrm{eff}}_{\mathbf{k}} = \varepsilon^{\mathrm{LDA}}_{\mathbf{k}} + \Sigma^{\mathbf{x}}_k - V^{xc}_{\mathbf{k}} = \varepsilon^0_{\mathbf{k}} + \Sigma^{\mathbf{x}}_k$ with $\varepsilon^0_{\mathbf{k}} = \frac{k^2}{2}$. Two variants have been explored.

1. $\beta_k(\omega) = \left|\mathrm{Im}\,\Sigma^{c,R}(\mathbf{k}, \omega + E^{\mathrm{GW}}_k)[\eta_s]\right|$, which I call epep@qp.
2. $\beta_k(\omega) = \left|\mathrm{Im}\,\Sigma^{c,R}(\mathbf{k}, \omega + \varepsilon^{\mathrm{LDA}}_{\mathbf{k}})[\eta_s]\right|$, which I call epep@mf.

6.11 Insights from the independent boson model

Here we work with the independent Boson model, which has the Hamiltonian:

$$H = \epsilon_c c^\dagger c + \sum_\nu \Omega_{c\nu} b^\dagger_\nu b_\nu + c^\dagger c \sum_\nu V_\nu \left(b_\nu + b^\dagger_\nu\right). \tag{6.1132}$$

with $\Omega_{c\nu} = \tilde{\epsilon_c} - \epsilon_c + \Omega_\nu$, where Ω_ν and V_ν are the excitation energies and couplings for the QB mode ν. ϵ_c and $\tilde{\epsilon_c}$ are noninteracting single-particle energies of the core state at $\mathbf{k}_c = [0,0,0]$. Note that for $G_{c,0}$@HF, $\epsilon_c = \epsilon^{\mathrm{HF}}_c = \tilde{\epsilon_c}$; for $G_{c,0}$@LDA, $\epsilon_c = \epsilon^{\mathrm{LDA}}_c + \Sigma^x_{cc} - v^{xc}_{cc} = \epsilon^{\mathrm{HF}}_c$ while $\tilde{\epsilon_c} = \epsilon^{\mathrm{LDA}}_c$; and for $G_{c,0}$@BWs2, $\epsilon_c = \epsilon^{\mathrm{BWs2}}_c = \tilde{\epsilon_c}$. We have been able to compute QB quantities for dRPA@HF, dRPA@LDA, and BWs2 with $\alpha_1 = 2$ and $\alpha_2 = -1, 0, 1$ for a variety of system sizes; they are all stored in

- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_hf_rpa_rs4_2949212`
- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_lda_rpa_rs4_294925`
- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_hf_bws2_a1_2_a2_-1_`
- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_hf_bws2_a1_2_a2_0_r`
- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_hf_bws2_a1_2_a2_1_r`

The single-particle energies are Exposed immediately within the h5s, whereas the QB quantities for a given momentum transfer vector $\mathbf{q}$ are nested, where each momentum transfer vector $\boldsymbol{q}$ belies a set of modes ν. Should I just enumerate all possible modes ν in all possible $\boldsymbol{q}$ and store them in a long dictionary along with their couplings? Because right now the layout is such that we have separate groups for each $\boldsymbol{q}$ with the modes ν belonging to that $\boldsymbol{q}$ inside. Or is it imperative that once I have this long dictionary, I then sort it, placing degenerate modes together? The latter is more faithful to the equations, but would take an extra step to implement. I guess a useful check that we could do is once we have our long dictionary of modes ν with their excitation energies and couplings, we could then scan

the excitation energies to see which modes are degenerate, and then we can demand the couplings should be identical. For the 114e485o system, there are 3911 $\boldsymbol{q}$ vectors, each with potentially multiple modes ν. Will the performance of the code suffer greatly if we don't organize before we run? The exact retarded Green's function for this model is given by

$$G^R_{c,\text{exact}}(\omega) = \sum_{\{n_\nu\}} \frac{e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega - \epsilon_c + \sum_\nu (\zeta_{c\nu} - n_\nu)\Omega_{c\nu} + i\eta}, \tag{6.1133}$$

with $\zeta_{c\nu} = \frac{|V_\nu|^2}{\Omega^2_{c\nu}}$ and n_ν is the occupation number of the boson mode ν.

6.11.1 Landau cumulant

The Landau cumulant is given by

$$C^R_c(t) = \frac{1}{\pi} \int d\omega \frac{|\operatorname{Im} \Sigma^R(\omega + \tilde{\epsilon_c})|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1 \right), \tag{6.1134}$$

where for $G_{c,0}$@HF, $\tilde{\epsilon_c} = \epsilon_c^{\text{HF}}$; for $G_{c,0}$@LDA, $\tilde{\epsilon_c} = \epsilon_c^{\text{LDA}}$; and for $G_{c,0}$@BWs2, $\tilde{\epsilon_c} = \epsilon_c^{\text{BWs2}}$. Then we plug this into the cumulant ansatz for the Green's function as

$$G^R_c(t) = G^R_{c,0}(t) e^{C^R_c(t)}, \tag{6.1135}$$

where $G^R_{c,0}(t) = -i\Theta(t) e^{-i\epsilon_c t}$.

Putting the GW self-energy into the Landau cumulant

It is known that plugging in the GW form of the self-energy into the Landau cumulant and then using it in the cumulant ansatz for the Green's function reproduces the exact solution

of 6.1163. The GW self-energy for this model is given by

$$\Sigma^R_{c,GW}(\omega) = \sum_\nu \frac{|V_\nu|^2}{\omega - \epsilon_c + \Omega_\nu + i\eta} \tag{6.1136}$$

$$\implies \Sigma^R_{c,GW}(\omega + \tilde{\epsilon_c}) = \sum_\nu \frac{|V_\nu|^2}{\omega + \Omega_{c\nu} + i\eta} \tag{6.1137}$$

$$\implies \operatorname{Im}\Sigma^R_{c,GW}(\omega + \tilde{\epsilon_c}) = -\pi \sum_\nu |V_\nu|^2 \frac{\eta}{(\omega + \Omega_{c\nu})^2 + \eta^2} \tag{6.1138}$$

$$\implies C^R_{c,GW}(t) = \int d\omega \frac{\sum_\nu |V_\nu|^2 \frac{\eta}{(\omega+\Omega_{c\nu})^2+\eta^2}}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{6.1139}$$

$$= \sum_\nu \zeta_{c\nu} \left[e^{-i\Omega_{c\nu}t} - 1 + i\Omega_{c\nu}t\right] \tag{6.1140}$$

$$\implies G^R_{c,GW}(t) = G^R_{c,0}(t) e^{C^R_{c,GW}(t)} \tag{6.1141}$$

$$= -i\Theta(t) e^{-i\epsilon_c t} e^{\sum_\nu \zeta_{c\nu}\left[e^{-i\Omega_{c\nu}t} - 1 + i\Omega_{c\nu}t\right]} \tag{6.1142}$$

$$= -i\Theta(t) e^{-i\epsilon_c t} e^{-\sum_\nu \zeta_{c\nu}} e^{+i\sum_\nu \zeta_{c\nu}\Omega_{c\nu}t} \prod_\nu \exp\left[\zeta_{c\nu} e^{-i\Omega_{c\nu}t}\right] \tag{6.1143}$$

$$= -i\Theta(t) e^{-i\epsilon_c t} e^{-\sum_\nu \zeta_{c\nu}} e^{+i\sum_\nu \zeta_{c\nu}\Omega_{c\nu}t} \prod_\nu \sum_{n_\nu=0}^{\infty} \frac{\left(\zeta_{c\nu} e^{-i\Omega_{c\nu}t}\right)^{n_\nu}}{n_\nu!} \tag{6.1144}$$

$$= -i\Theta(t) e^{-i\epsilon_c t} e^{-\sum_\nu \zeta_{c\nu}} e^{+i\sum_\nu \zeta_{c\nu}\Omega_{c\nu}t} \sum_{\{n_\nu\}} \prod_\nu \frac{\zeta_{c\nu}^{n_\nu}}{n_\nu!} e^{-i\sum_\nu n_\nu \Omega_{c\nu} t} \tag{6.1145}$$

$$= -i\Theta(t) \sum_{\{n_\nu\}} e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \frac{\zeta_{c\nu}^{n_\nu}}{n_\nu!} e^{-i\left[\epsilon_c - \sum_\nu \zeta_{c\nu}\Omega_{c\nu} + \sum_\nu n_\nu \Omega_{c\nu}\right]t} \tag{6.1146}$$

$$\implies G^R_{c,GW}(\omega) = \sum_{\{n_\nu\}} \frac{e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega - \epsilon_c + \sum_\nu (\zeta_{c\nu} - n_\nu)\Omega_{c\nu} + i\eta} \tag{6.1147}$$

Putting the exact self-energy into the Landau cumulant

In order to observe the ramifications of our SCCE procedure, we can compute the exact self-energy from the exact Green's function as

$$\Sigma^R_{c,\text{Dyson}}(\omega) = \left[G^R_{c,0}(\omega)\right]^{-1} - \left[G^R_{c,\text{exact}}(\omega)\right]^{-1} \tag{6.1148}$$

$$\implies \Sigma^R_{c,\text{Dyson}}(\omega) = \omega - \epsilon_c + i\eta - \left[\sum_{\{n_\nu\}} \frac{e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega - \epsilon_c + \sum_\nu (\zeta_{c\nu} - n_\nu)\Omega_{c\nu} + i\eta}\right]^{-1} \tag{6.1149}$$

$$\implies \Sigma^R_{c,\text{Dyson}}(\omega + \tilde{\epsilon_c}) = \begin{cases} \omega + i\eta - \left[\sum_{\{n_\nu\}} \frac{e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega + \sum_\nu (\zeta_{c\nu} - n_\nu)\Omega_{c\nu} + i\eta}\right]^{-1}, & G_0\text{@HF or } G_0\text{@BW} \\ \omega + v^{xc}_{cc} - \Sigma^x_{cc} + i\eta - \left[\sum_{\{n_\nu\}} \frac{e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega + v^{xc}_{cc} - \Sigma^x_{cc} + \sum_\nu (\zeta_{c\nu} - n_\nu)\Omega_{c\nu} + i\eta}\right]^{-1}, & G_0\text{@LDA} \end{cases} \tag{6.1150}$$

We should be able to back-out the value of $v_{cc}^{xc} - \Sigma_{cc}^{x}$ by taking $\epsilon_c^{\mathrm{LDA}} - \epsilon_c^{\mathrm{HF}}$. Then, we can plug this into the Landau cumulant to get a $C^R_{c,\mathrm{Dyson}}(t)$, which we can then use to get a $G^R_{c,\mathrm{Dyson}}(t)$ and then Fourier transform to get $G^R_{c,\mathrm{Dyson}}(\omega)$.

6.11.2 Results for the independent boson model

All results are at $r_s = 4$. The independent Boson model has the Hamiltonian:

$$H = \tilde{\epsilon_c} c^\dagger c + \sum_\nu \Omega_\nu b_\nu^\dagger b_\nu + c^\dagger c \sum_\nu V_\nu \left(b_\nu + b_\nu^\dagger\right). \tag{6.1151}$$

where Ω_ν and V_ν are quasiboson (QB) frequencies and couplings, which can be taken from dRPA@HF, dRPA@LDA, and BWs2 ($\alpha_1 = 2, \alpha_2 = 0$). The latter two are the ones which were found to give competitive spectral functions with respect to CCSD for 114e485o. $\tilde{\epsilon_c}$ is the noninteracting single-particle energy of the core state at $\mathbf{k}_c = [0, 0, 0]$. This is a model for which we can get the exact retarded Green's function, which is given by

$$G^R_{c,\mathrm{exact}}(\omega) = \frac{e^{-\sum_\nu \zeta_\nu} \prod_\nu \zeta_\nu}{\omega - \tilde{\epsilon_c} + \sum_\nu (\zeta_\nu)\Omega_\nu + i\eta}, \tag{6.1152}$$

with $\zeta_\nu = \frac{|V_\nu|^2}{\Omega_\nu^2}$.

Putting the GW self-energy into the Landau cumulant

It is known that one-shot GW+C reproduces 6.1163.

Putting the exact self-energy into the Landau cumulant

We can compute the exact self-energy from the exact Green's function as

$$\Sigma^R_{c,\mathrm{Dyson}}(\omega) = \left[G^R_{c,0}(\omega)\right]^{-1} - \left[G^R_{c,\mathrm{exact}}(\omega)\right]^{-1} \tag{6.1153}$$

$$\implies \Sigma^R_{c,\mathrm{Dyson}}(\omega + \tilde{\epsilon_c}) = \omega + i\eta - \left[\frac{e^{-\sum_\nu \zeta_\nu} \prod_\nu \zeta_\nu}{\omega + \sum_\nu (\zeta_\nu)\Omega_\nu + i\eta}\right]^{-1} \tag{6.1154}$$

Getting the self-energy this way defines our self-consistent cumulant scheme (SC-CE). The way to read the plots is that I am doing $\Sigma_{\mathrm{Dyson}}[G_{\mathrm{exact}}] \to$ SC-CE (Iter 1) $\to \Sigma_{\mathrm{Dyson}}[G_1] \to$ SC-CE (Iter 2). This is where I terminate my iteration because it has become clear that the results I will get past this point are not useful. The exact spectral function and the GW self-energy are shown for reference.

6.11.3 Dyson versus cumulant expansion for the independent boson model (IBM)

Let λ count powers of W. The exact cumulant C and exact self-energy Σ admit expansions as

$$C = \lambda C^{(1)} + \lambda^2 C^{(2)} + \cdots \tag{6.1155}$$

and

$$\Sigma = \lambda\Sigma^{(1)} + \lambda^2\Sigma^{(2)} + \cdots, \tag{6.1156}$$

respectively. Matching the cumulant and Dyson expansions of G to first order in W gives

$$G_0 C^{(1)} = G_0 \Sigma^{(1)} G_0, \tag{6.1157}$$

from which the Landau form for $C^{(1)}$ is derived. For the IBM, we can get the exact Green's function G via

$$G = G_0 \exp\big(C^{(1)}[\Sigma^{(1)}]\big), \quad \text{where } \Sigma^{(1)} \equiv \Sigma_{GW}. \tag{6.1158}$$

From this, we can get

$$\Sigma = G_0^{-1} - G^{-1} = \lambda\Sigma^{(1)} + \lambda^2\Sigma^{(2)} + \cdots. \tag{6.1159}$$

SCCE(GW)

Our self-consistent cumulant scheme then proposes to recumulate as

$$\tilde{G} = G_0 \exp\big(C^{(1)}[\Sigma^{(1)} + \Sigma^{(2)} + \cdots]\big). \tag{6.1160}$$

Assuming high-order terms of the self-energy are non-zero (this is reasonable because we expect the self energy to decay slowly), we then have $\tilde{G} \neq G$. In other words, our scheme incorpates vertex corrections in an uncontrolled manner. Since it fails for the simple model system of the IBM, it cannot be trusted to deliver reliable results for more complex systems, such as the electron gas.

SCCE(GW_0)

In Holms' scheme, at iteration $n \geq 1$ we update the self-energy for occupied state i as

$$\Sigma_i^{n,R}(t) = i \sum_j \left[G_j^{n-1,R}(t) W_{ij}^{0,>}(t) + G_j^{n-1,<}(t) W_{ij}^{0,R}(t) \right], \tag{6.1161}$$

where G_j^{n-1} is the cumulant Green's function of state j from the previous iteration. Then, we do a DFT to get Σ into the frequency domain and stick it into the Landau form. Since G_j^{n-1}, for all $n \geq 0$, contains all orders of W, the IBM spectral functions past one-shot cannot give the exact result, assuming $C \neq 0 \implies G \neq G_0 \implies \Sigma^n \neq \underbrace{\Sigma^{(1)}}_{\Sigma_{G_0W_0}}$, since we know that $C^{(1)}[\Sigma^{(1)}]$ gives the exact result for the IBM. My IBM plots support this conclusion.

Moving to more complex models

I understand that even if SCCE(GW_0) is bad for the IBM, it may still be useful for the UEG. However, theoretically speaking, in order to use the Landau form, the cumulant and Dyson expansions of G should be matched up to first order in W, but as argued above SCCE(GW_0) violates this for all $n \geq 1$.

Results for the UEG

My UEG plot for 114e485o supports this conclusion that SCCE(GW_0) doesn't improve results past one-shot.

Results for the finite-size Holstein model

PJ tried this scheme (which he called HA-SCCE $\equiv SCCE(GW_0)$) out for the finite-size Holstein model and the results didnt look good there as well. Specifically, at moderate coupling HA-SCCE is devoid of satellite features, which is not the case for the one-shot CE.

6.11.4 Holms' GW_0 self-consistent cumulant scheme for the independent boson model

Here we work with the independent Boson model, which has the Hamiltonian:

$$H = \epsilon_c c^\dagger c + \sum_\nu \Omega_{c\nu} b_\nu^\dagger b_\nu + c^\dagger c \sum_\nu V_\nu \left(b_\nu + b_\nu^\dagger\right). \tag{6.1162}$$

with $\Omega_{c\nu} = \tilde{\epsilon_c} - \epsilon_c + \Omega_\nu$, where Ω_ν and V_ν are the excitation energies and couplings for the QB mode ν. ϵ_c and $\tilde{\epsilon_c}$ are noninteracting single-particle energies of the core state at $\mathbf{k}_c = [0,0,0]$. Note that for $G_{c,0}$@HF, $\epsilon_c = \epsilon_c^{\mathrm{HF}} = \tilde{\epsilon_c}$; for $G_{c,0}$@LDA, $\epsilon_c = \epsilon_c^{\mathrm{LDA}} + \Sigma_{cc}^x - v_{cc}^{xc} = \epsilon_c^{\mathrm{HF}}$ while $\tilde{\epsilon_c} = \epsilon_c^{\mathrm{LDA}}$. We have been able to compute QB quantities for dRPA@HF, dRPA@LDA for a variety of system sizes; they are all stored in

- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_hf_rpa_rs4_2949212`
- `qcpbc/libpbc/gw_tests/useful_tests_all/ueg/7/quasiboson/quasiboson_lda_rpa_rs4_294925`

The single-particle energies are Exposed immediately within the h5s, whereas the QB quantities for a given momentum transfer vector $\mathbf{q}$ are nested, where each momentum transfer vector $\boldsymbol{q}$ belies a set of modes ν. Also note that for this independent Boson model, my h5 files give all of the possible momentum transfers starting from occupied states. Since in this model I am only considering the excitations that can arise after the core state, the subset of allowed excitations is indeed smaller. I think this is already taken into account in the pricing script, but I just want you to be aware of this issue. The exact retarded Green's function for this model is given by

$$G_{c,\mathrm{exact}}^R(\omega) = \sum_{\{n_\nu\}} \frac{e^{-\sum_\nu \zeta_{c\nu}} \prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega - \epsilon_c + \sum_\nu (\zeta_{c\nu} - n_\nu)\Omega_{c\nu} + i\eta}, \tag{6.1163}$$

with $\zeta_{c\nu} = \frac{|V_\nu|^2}{\Omega_{c\nu}^2}$ and n_ν is the occupation number of the boson mode ν.

6.11.5 Landau cumulant

The Landau cumulant is given by

$$C_c^R(t) = \frac{1}{\pi}\int d\omega \frac{|\operatorname{Im}\Sigma^R(\omega+\tilde{\epsilon_c})|}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right), \tag{6.1164}$$

where for $G_{c,0}$@HF, $\tilde{\epsilon_c}=\epsilon_c^{\mathrm{HF}}$; for $G_{c,0}$@LDA, $\tilde{\epsilon_c}=\epsilon_c^{\mathrm{LDA}}$. Then we plug this into the cumulant ansatz for the Green's function as

$$G_c^R(t) = G_{c,0}^R(t)e^{C_c^R(t)}, \tag{6.1165}$$

where $G_{c,0}^R(t) = -i\Theta(t)e^{-i\epsilon_c t}$.

Putting the GW self-energy into the Landau cumulant

It is known that plugging in the GW form of the self-energy into the Landau cumulant and then using it in the cumulant ansatz for the Green's function reproduces the exact solution of 6.1163. The GW self-energy for this model is given by

$$\Sigma_{c,GW}^R(\omega) = \sum_\nu \frac{|V_\nu|^2}{\omega-\epsilon_c+\Omega_\nu+i\eta} \tag{6.1166}$$

$$\implies \Sigma_{c,GW}^R(\omega+\tilde{\epsilon_c}) = \sum_\nu \frac{|V_\nu|^2}{\omega+\Omega_{c\nu}+i\eta} \tag{6.1167}$$

$$\implies \operatorname{Im}\Sigma_{c,GW}^R(\omega+\tilde{\epsilon_c}) = -\pi\sum_\nu |V_\nu|^2 \frac{\eta}{(\omega+\Omega_{c\nu})^2+\eta^2} \tag{6.1168}$$

$$\implies C_{c,GW}^R(t) = \int d\omega \frac{\sum_\nu |V_\nu|^2 \frac{\eta}{(\omega+\Omega_{c\nu})^2+\eta^2}}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right) \tag{6.1169}$$

$$= \sum_\nu \zeta_{c\nu}\left[e^{-i\Omega_{c\nu}t}-1+i\Omega_{c\nu}t\right] \tag{6.1170}$$

$$\implies G_{c,GW}^R(t) = G_{c,0}^R(t)e^{C_{c,GW}^R(t)} \tag{6.1171}$$

$$= -i\Theta(t)e^{-i\epsilon_c t}e^{\sum_\nu \zeta_{c\nu}\left[e^{-i\Omega_{c\nu}t}-1+i\Omega_{c\nu}t\right]} \tag{6.1172}$$

$$= -i\Theta(t)e^{-i\epsilon_c t}e^{-\sum_\nu \zeta_{c\nu}}e^{+i\sum_\nu \zeta_{c\nu}\Omega_{c\nu}t}\prod_\nu \exp\left[\zeta_{c\nu}e^{-i\Omega_{c\nu}t}\right] \tag{6.1173}$$

$$= -i\Theta(t)e^{-i\epsilon_c t}e^{-\sum_\nu \zeta_{c\nu}}e^{+i\sum_\nu \zeta_{c\nu}\Omega_{c\nu}t}\prod_\nu \sum_{n_\nu=0}^{\infty}\frac{\left(\zeta_{c\nu}e^{-i\Omega_{c\nu}t}\right)^{n_\nu}}{n_\nu!} \tag{6.1174}$$

$$= -i\Theta(t)e^{-i\epsilon_c t}e^{-\sum_\nu \zeta_{c\nu}}e^{+i\sum_\nu \zeta_{c\nu}\Omega_{c\nu}t}\sum_{\{n_\nu\}}\prod_\nu \frac{\zeta_{c\nu}^{n_\nu}}{n_\nu!}e^{-i\sum_\nu n_\nu\Omega_{c\nu}t} \tag{6.1175}$$

$$= -i\Theta(t)\sum_{\{n_\nu\}}e^{-\sum_\nu \zeta_{c\nu}}\prod_\nu \frac{\zeta_{c\nu}^{n_\nu}}{n_\nu!}e^{-i\left[\epsilon_c-\sum_\nu \zeta_{c\nu}\Omega_{c\nu}+\sum_\nu n_\nu\Omega_{c\nu}\right]t} \tag{6.1176}$$

$$\implies G_{c,GW}^R(\omega) = \sum_{\{n_\nu\}}\frac{e^{-\sum_\nu \zeta_{c\nu}}\prod_\nu \zeta_{c\nu}^{n_\nu}/n_\nu!}{\omega-\epsilon_c+\sum_\nu(\zeta_{c\nu}-n_\nu)\Omega_{c\nu}+i\eta} \tag{6.1177}$$

Putting the exact self-energy into the Landau cumulant

Note that we already have a script which performs full self-consistency in `/n/home03/pkozlowski/qcpbc/li`
Now, I want to pursue self-consistency in G but keep W at the bare RPA level. I have done something similar already in /n/home03/pkozlowski/qcpbc/libpbc/gw_tests/gw-ac/tests/sc_cumulant_pilot I was thinking the simplest thing would be to add this kind of functionality to my IBM firm app initial Python script. We can just run both variants in tandem with a flag on the config section of the script, controlling whether we want to run the full self-consistency or just the GW zero self-consistency. This next question that I have, and I want to hear your opinion on, is: what is the best way to form the GW0 self-energy in this game? Is it what I am doing right now, where I first formed the self-energy in the time domain by summing the retarded Green's function times a lesser screen interaction, plus some other combination, all in the time domain? At the very end, I Fourier transform this to the frequency domain. This requires me to first Fourier transform the screened interaction from the frequency domain into the time domain, only to be Fourier transformed back into the frequency domain later as the DFT of the self-energy proceeds.

I am thinking of a potential alternative to this, where we, of course, first form the cumulonand Green's function via a fast Fourier transform of the cumulonand Green's function in the time domain to get it into the frequency domain. We then convolve in frequency space this cumulonand Green's function with the bare Coulomb interaction already in frequency space to just get the self-energy in frequency space already. It would seem that there becomes a complicated pulse structure as we go along. I would need you to verify that the result that we get is not dependent on the size of the frequency grid. The smallest non-trivial system where we get a complicated pulse structure is for 14 electrons in 57 orbitals. For this system, I would want you to investigate if the results that we get for the self-energy are indeed dependent on what kind of frequency grid we choose. In order to observe the ramifications of our GW0 SCCE procedure, we can update the self-energy from the cumulant Green's function.

6.12 Reproducing Holms numbers for G0W0+C in the TDL

I am interested in reproducing Holms curves for the one-shot and self-consistent cumulant schemes he outlined in his paper (attached). However, they formulated their theory in the TDL and CBS, but I am interested in what happens with the finite basis. I think they are getting the spectral function from the time-ordered Green function at zero temperature. A way to do this is described in Reining's paper (attached). They instruct to build the Greens function as

$$G_k^T(t) = G_k^{T,0}(t) e^{\tilde{C}_k^T(t)} \tag{6.1178}$$

where the non-interacting part is $G_k^{T,0}(t) = \mp i\theta(\pm t)\theta\left(\pm\left(\epsilon_k - \epsilon_{kF}\right)\right) e^{-i\epsilon_k^{\text{eff}} t}$, where the upper (lower) sign refers to particles with $k > k_F$ (holes $k < k_F$), and θ is the unit step function. Our reference is an LDA calculation and we are at zero temperature, so we can take $\epsilon_k = \epsilon_k^{\text{LDA}}$, $\epsilon_k^{\text{eff}} = \epsilon_k^{\text{LDA}} + \Sigma_k^x - v_k^{xc}$, and $\epsilon_{k_F} = \epsilon_{k_F}^{\text{LDA}} = \frac{\epsilon_{k\equiv\text{HOMO}}^{\text{LDA}} + \epsilon_{k\equiv\text{LUMO}}^{\text{LDA}}}{2}$. The dynamical cumulant is defined

as

$$\tilde{C}_k^T(t) = \int d\omega \frac{\beta_k^{\pm}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right). \tag{6.1179}$$

Note that $\beta_k(\omega) = \beta_k^+(\omega) + \beta_k^-(\omega)$, where $\beta_k^{\pm}(\omega) = \frac{1}{\pi} \left|\operatorname{Im} \Sigma_k^R(\omega + \epsilon_k)\right| \theta\left(\pm\left(\epsilon_{k_F} - \epsilon_k - \omega\right)\right)$. Then we can Generate a periodic signal from the time-ordered Green's function like we did before at which point we get the 1shot spectral function as $A_k(\omega) = \frac{1}{\pi} |\operatorname{Im} G_k^T(\omega)|$.

6.13 Efficiently reaching the TDL with G0W0+C

In order to form the self-energy, which is proportional to the cumulant kernel, we need to build the retarded GW self-energy. The formal expression is

$$\Sigma_c^R(\mathbf{k}, \omega) = \sum_{\mathbf{q}\nu} \left|M_{\mathbf{k},\mathbf{k}-\mathbf{q}}^{\nu}\right|^2 \left[\frac{f_{\mathbf{k}-\mathbf{q}}}{\omega - \xi_{\mathbf{k}-\mathbf{q}} + \Omega_{\mathbf{q}\nu} + i\eta} + \frac{1 - f_{\mathbf{k}-\mathbf{q}}}{\omega - \xi_{\mathbf{k}-\mathbf{q}} - \Omega_{\mathbf{q}\nu} + i\eta}\right]. \tag{6.1180}$$

where $\xi_{\mathrm{k}} = \frac{k^2}{2} - \mu_0$ with the chemical potential $\mu_0 = \frac{\epsilon_{\mathrm{HOMO}} + \epsilon_{\mathrm{LUMO}}}{2}$. In order to evaluate this expression efficiently, we start by summing over all valid momentum transfer vectors $\boldsymbol{q}$. This step is embarrassingly parallel.

6.13.1 Casida eigenproblem for a momentum transfer

We can generate the RPA quantities at each momentum transfer vector $\boldsymbol{q}$ by solving its Casida eigenproblem. The Casida eigenproblem is given by

$$\begin{pmatrix} \boldsymbol{A}(\boldsymbol{q}) & \boldsymbol{B}(\boldsymbol{q}) \\ -\boldsymbol{B}(\boldsymbol{q}) & -\boldsymbol{A}(\boldsymbol{q}) \end{pmatrix} \begin{pmatrix} \boldsymbol{X}(\boldsymbol{q}) & \boldsymbol{Y}(\boldsymbol{q}) \\ \boldsymbol{Y}(\boldsymbol{q}) & \boldsymbol{X}(\boldsymbol{q}) \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Omega}(\boldsymbol{q}) & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{\Omega}(\boldsymbol{q}) \end{pmatrix} \begin{pmatrix} \boldsymbol{X}(\boldsymbol{q}) & \boldsymbol{Y}(\boldsymbol{q}) \\ \boldsymbol{Y}(\boldsymbol{q}) & \boldsymbol{X}(\boldsymbol{q}) \end{pmatrix}. \tag{6.1181}$$

In practice, we can solve an eigenproblem of half the size by exploiting the symmetric formulation of the Casida eigenproblem, which is already implemented. The $\boldsymbol{A}(\boldsymbol{q})$ and $\boldsymbol{B}(\boldsymbol{q})$ matrices are defined as

$$A_{\boldsymbol{k}_i\boldsymbol{k}_a,\,\boldsymbol{k}_b\boldsymbol{k}_j}(\boldsymbol{q}) = \left(\xi_{\boldsymbol{k}_a} - \xi_{\boldsymbol{k}_i}\right) \delta_{\boldsymbol{k}_i\boldsymbol{k}_j} \delta_{\boldsymbol{k}_a\boldsymbol{k}_b} + \frac{4\pi}{|\boldsymbol{q}|^2} \tag{6.1182}$$

$$B_{\boldsymbol{k}_i\boldsymbol{k}_a,\,\boldsymbol{k}_j\boldsymbol{k}_b}(\boldsymbol{q}) = \frac{4\pi}{|\boldsymbol{q}|^2} \tag{6.1183}$$

In which we consider all occupied and virtual wave vectors connected by the momentum transfer $\boldsymbol{q}$, i.e. $\boldsymbol{k}_a = \boldsymbol{k}_i + \boldsymbol{q}$ and $\boldsymbol{k}_b = \boldsymbol{k}_j + \boldsymbol{q}$. After solving the Casida eigenvalue problem, I perform the bioorthogonalization procedure, which ensures that

$$\left(\boldsymbol{X}(\boldsymbol{q}) - \boldsymbol{Y}(\boldsymbol{q})\right)^{\dagger} \left(\boldsymbol{X}(\boldsymbol{q}) + \boldsymbol{Y}(\boldsymbol{q})\right) = \boldsymbol{I} \tag{6.1184}$$

Then, I have a list of excitation energies $\boldsymbol{\Omega}$ and corresponding right eigenvectors $\boldsymbol{X}, \boldsymbol{Y}$ for each momentum transfer $\boldsymbol{q}$. But for the self energy, we actually need rather the transition densities $M_{\boldsymbol{k},\boldsymbol{k}-\boldsymbol{q}}^{\nu}$ for each mode ν at each momentum transfer $\boldsymbol{q}$. These are given by

$$M_{\mathrm{k},\mathrm{k}-\mathrm{q}}^{\nu} = \frac{v_q}{\mathcal{V}} \sum_{ia \in \mathrm{q}} \left(X_{ia}^{\nu} + Y_{ia}^{\nu}\right), \tag{6.1185}$$

where $v_q = 4\pi/|\boldsymbol{q}|^2$ is the bare Coulomb interaction and $\mathcal{V}$ is the volume of the simulation cell. $f_{\boldsymbol{k}}$ is the occupation number of state $\boldsymbol{k}$. We are at zero temperature, so this should be 1 for occupied states and 0 for unoccupied states. *Should we devote 1/32 CPUs in OpenMPI to each momentum transfer vector $\boldsymbol{q}$ to parallelize the self-energy evaluation?*

Verification

With these ingredients, we can construct the screened-interaction pole expansion directly:

$$W^R_{0,c}(\mathbf{q},\omega) = \sum_\nu |M_{\mathbf{q}\nu}|^2 \left[\frac{1}{\omega - \Omega_{\mathbf{q}\nu} + i\eta} - \frac{1}{\omega + \Omega_{\mathbf{q}\nu} + i\eta}\right],$$

and numerically verify that it reproduces

$$W^R_{0,c}(\mathbf{q},\omega) = v_q \left[\frac{1}{1 - v_q \chi^R_0(\mathbf{q},\omega)} - 1\right].$$

in order to ensure that I don't have any missing factors of $v_q, \mathcal{V}$, two, or spin degeneracy.

6.13.2 Cumulant spectral functions

Retarded cumulant spectral function

Here we can use

$$C^R_k(t) = \frac{?}{\pi}\int d\omega \frac{|\operatorname{Im}\Sigma^R_k(\omega+\epsilon_k)|}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right), \tag{6.1186}$$

and then

$$G^R_k(t) = G^R_{k,0}(t) e^{C^R_k(t)}, \tag{6.1187}$$

where $G^R_{k,0}(t) = -i\Theta(t)e^{-i(\epsilon_k+\Sigma^x_k)t}$, where we have already devised an efficient way to compute the exchange self-energy in ¡file¿.

Time-ordered cumulant spectral function

Below, the upper (lower) sign refers to particles with $k > k_F$ (holes $k < k_F$). *When I am comparing $\tilde{\epsilon}_k$ to $\tilde{\epsilon}_{k_F}$, should we take $\tilde{\epsilon}_{k_F} = \frac{\epsilon_{k\equiv HOMO} + \epsilon_{k\equiv LUMO}}{2}$? and $\tilde{\epsilon}_k = \epsilon_k = k^2/2$? Or do we want to solve the GW quasi-particle equation for each ϵ_k, and then whenever I am evaluating a step function that is comparing $\tilde{\epsilon}_k$ to $\tilde{\epsilon}_{k_F}$, actually insert E^{GW}_k and $E^{GW}_{k_F}$, respectively, where $E^{GW}_{k_F} = \frac{E^{GW}_{k\equiv HOMO} + E^{GW}_{k\equiv LUMO}}{2}$?* Here we can take

$$\tilde{C}^T_k(t) = \int d\omega \frac{\beta^\pm_k(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right). \tag{6.1188}$$

Note that $\beta_k(\omega) = \beta^+_k(\omega) + \beta^-_k(\omega)$, where $\beta^\pm_k(\omega) = \frac{1}{\pi}\left|\operatorname{Im}\Sigma^R_k(\omega+\epsilon_k)\right|\theta\left(\pm\left(\tilde{\epsilon}_{k_F} - \tilde{\epsilon}_k - \omega\right)\right)$. Then,

$$G^T_k(t) = G^{T,0}_k(t) e^{\tilde{C}^T_k(t)} \tag{6.1189}$$

where the non-interacting part is $G^{T,0}_k(t) = \mp i\theta(\pm t)\theta\left(\pm\left(\tilde{\epsilon}_k - \tilde{\epsilon}_{k_F}\right)\right)e^{-i(\epsilon_k+\Sigma^x_k)t}$.

Reproducing the reference result

I have tried to implement LDA+GW+C

$$G_c^R(t) = G_{c,0}^R(t) e^{C_c^R(t)}, \tag{6.1190}$$

with $G_{c,0}^R(t) = -i\Theta(t)e^{-i\epsilon_c t}$, $\epsilon_c = \epsilon_c^{\mathrm{LDA}} + \Sigma_{cc}^x - v_{cc}^{xc} = \epsilon_c^{\mathrm{HF}}$, and

$$C_c^R(t) = \frac{1}{\pi}\int d\omega \frac{|\operatorname{Im}\Sigma^R(\omega+\tilde{\epsilon_c})|}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right), \tag{6.1191}$$

for $\tilde{\epsilon_c} = \epsilon_c^{\mathrm{LDA}}$ or $\tilde{\epsilon_c} = E_c^{GW}$, Yielding the plots in respectively, but none of these permutations is giving me the reported TDL or finite basis results. I have seemed to verify though that my LDA+GW converges towards the reference in the TDL in ¡file¿. Either my formulas for LDA+GW+C are wrong or the reported TDL result may be wrong. Is this the same conclusion that you arrived at? The one thing that I am unsure about is that I am using the restricted formalism throughout. When I am building the RPA matrix, I tack on an extra factor of 2 to the column kernel in the DRP calculation. When I go to form the transition densities, I further multiply by a factor of sqrt of 2. To Reproduce the reported result, I wonder what happens if we try out different forms for Σ. For example, we could try one variation where we do not multiply the transition densities by sqrt of 2 or another variation where we do not multiply the column kernel by 2 or both. We can then see if any of these variations reproduce the reported TDL result. Then we plug this into the cumulant ansatz for the Green's function as

Frames

In both plots, we are constructing the retarded cumulant as

$$C_k^R(t) = \frac{1}{\pi}\int d\omega \frac{|\operatorname{Im}\Sigma_k^R(\omega+\epsilon_k)|}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right). \tag{6.1192}$$

In the $\beta(\omega+E_{QP})$ and $\beta(\omega+\epsilon_{\mathrm{LDA}})$ frames, we are using $\epsilon_k \equiv E_k^{GW}$ and $\epsilon_k \equiv \epsilon_k^{\mathrm{LDA}}$, respectively.

6.13.3 Choosing the appropriate broadening

The theory

We are always forming the retarded Green's function as

$$G_k^R(t) = G_{k,0}^R(t) e^{C_k^R(t)}, \tag{6.1193}$$

with $G_{k,0}^R(t) = -i\Theta(t)e^{-(i\epsilon_k+\eta_G)t}$, $\epsilon_k = \epsilon_k^{\mathrm{LDA}} + \Sigma_{kk}^x - v_{kk}^{xc} = \epsilon_k^{\mathrm{HF}}$, and

$$C_k^R(t) = \frac{1}{\pi}\int d\omega \frac{|\operatorname{Im}\Sigma^R[\eta_\Sigma](\omega+\tilde{\epsilon_k})|}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right), \tag{6.1194}$$

for $\tilde{\epsilon_k} = \epsilon_k^{\mathrm{LDA}}$ ($\beta(\omega+\epsilon_{\mathrm{LDA}})$ frame) or $\tilde{\epsilon_k} = E_k^{GW}$ ($\beta(\omega+E_{QP})$ frame).

Reproduction of Garnet's TDL curve

I was able to reproduce Garnet's reported TDL curve, using a continuum integration approach following Reining (the implementation uses the one-dimensional quadrature you had mentioned yesterday). To get this to work, I used $\eta_\Sigma \approx 0^+ =$ 2e-2 eV and $\eta_G = 0.8$eV.

What I was doing in what I showed you yesterday

Following Loos' convention, I was choosing $\eta_G = 0$ and $\eta_\Sigma = 0.8$eV, which was giving me the OLD plots.

What do I do in these updated plots

These plots are using $\eta_G = 0.8$eV and $\eta_\Sigma =$ 2e-2 eV. I chose the latter parameter for my initial test, because choosing smaller η_Σ did not make a qualitative difference to the 114e485o spectral functions. I submitted tests to check that this is indeed the case also for larger supercells. These plots are labeled NEW.

6.13.4 Quasiparticle equation

For a given wave vector $\boldsymbol{k}$, the linearized quasiparticle equation is

$$E_k^{GW} = \epsilon_k + Z_k \operatorname{Re}\Sigma_k^R(\epsilon_k) \tag{6.1195}$$

where $Z_k = \left(1 - \frac{\partial \Sigma_k^R}{\partial \omega}\big|_{\omega=\epsilon_k}\right)^{-1}$ is the renormalization factor and $\Sigma_k^R(\epsilon_k) = \Sigma_{\text{corr}}^R(\boldsymbol{k}, \epsilon_k) + \Sigma_k^x$.

6.14 SCCE in the finite basis

6.14.1 GW0

In the retarded formalism, the cumulant ansatz for the interacting Green's function at the iteration $n+1$ is

$$G_k^{n+1,R}(t) = G_{0,k}^{n,R}(t) e^{C_k^{n+1,R}(t)} \tag{6.1196}$$

where $G_{0,k}^{n,R}(t) = -i\Theta(t) e^{-i\left(k^2/2 + \Gamma_k^n\right)t}$ with, for $n \geq 0$, $\Gamma_k^{n+1} = n_k^{(n)} v_m + \Sigma_k^{n+1,x(E)}[G^{(n)}] = n_k^{(n)} v_m - \sum_{\boldsymbol{q}\neq\boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}}^{(n)} v(\boldsymbol{q})$ and $\Gamma_k^0 = V_k^{xc}$. $\Sigma_k^{n+1,x(E)}[G^{(n)}]$ is the exchange self-energy at iteration $n+1$ computed with Ewald summation (neglecting the $\boldsymbol{q} = 0$ term) and $n_k^{(n)} v_m$ is the Madelung potential at iteration n. Note that this may change during self-consistent iteration as the chemical potential and thus occupation numbers are updated. We are always expanding around the current pole $E_k^{(n)}$; at the one-shot level, $E_k^{(0)} = \varepsilon_k^{\text{LDA}}$, but for $n > 0$, $E_k^{(n)} = k^2/2 + \Gamma_k^n + \operatorname{Re}\Sigma_k^{n,c}[G^{(n-1)}](E_k^{(n-1)})$.

Intuition behind this decision

In any self-consistent flavor of GW based on iteration of the Dyson equation, the scheme is to compute the quasiparticle energies and then feed that back into the self-energy to close the self-consistency loop. Therefore, an interpretation is that this scheme is doing the exact same thing, but in the cumulant formalism.

Equating the Dyson and cumulant expansions to first order in the screened interaction yields,

$$C_k^{n+1,R}(t) = ie^{iE_k^{(n)}t}\int \frac{d\omega}{2\pi}e^{-i\omega t}\frac{k^2/2+\Gamma_k^{n+1}-(k^2/2+\Gamma_k^n)+\Sigma_k^{n+1,c}[G^{(n)}](\omega)}{(\omega-E_k^{(n)}+i\eta)^2} \tag{6.1197}$$

$$= -i\left(\Gamma_k^{n+1}-\Gamma_k^n\right)t\Theta(t) + \underbrace{i\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Sigma_k^{n+1,c}[G^{(n)}](\omega+E_k^{(n)})}{(\omega+i\eta)^2}}_{C_k^{n+1,\mathrm{dyn}}(t)}. \tag{6.1198}$$

where $\Sigma_k^{n+1,c}[G^{(n)}](\omega)$ is the correlation self-energy at iteration $n+1$. We can rewrite the dynamical cumulant as

$$C_k^{n+1,\mathrm{dyn}}(t) = \int d\omega\,\beta_k^{n+1,R}(\omega)\frac{e^{-i\omega t}-1}{\omega^2} \tag{6.1199}$$

$$= \underbrace{\int \mathrm{d}\omega\,\beta_k^{n+1,R}(\omega)\frac{e^{-i\omega t}+i\omega t-1}{\omega^2}}_{C_k^{\mathrm{Landau}}(t)} - i\,S_1[\beta_k]\,t \tag{6.1200}$$

where $\beta_{k,c}^{n+1,R}(\omega) = \frac{1}{\pi}\left|\mathrm{Im}\,\Sigma_k^{n+1,c}\left(\omega+E_k^{(n)}\right)\right|$ and $S_1[\beta_k] = \mathcal{P}\int\mathrm{d}\omega\,\beta_k^{n+1,R}(\omega)/\omega$. At a given iteration, the correlation self-energy takes the form:

$$\Sigma_{m,c}^{(n+1),R}[G^{(n)}](t) = -i\theta(t)\left[\sum_{i,I}\left|M_{m,i}^I\right|^2 G_i^{(n),<}(t)e^{-i\Omega_I t} + \sum_{a,I}\left|M_{m,a}^I\right|^2 G_a^{(n),>}(t)e^{-i\Omega_I t}\right], \tag{6.1201}$$

with M and Ω the transition densities and energies from the dRPA problem. This allows us to write the Green's function as

$$G_k^{n+1,R}(t) = -i\Theta(t)\,\exp\left[-i\left(k^2/2+\Gamma_k^{n+1}\right)t\right]e^{C_k^{n+1,\mathrm{dyn}}(t)}. \tag{6.1202}$$

For iterations $n \geq 1$: Assume that we have $A_k^{(n-1)}(\omega) \propto \mathrm{Im}\,G_k^{(n-1)}(\omega)$. This gives us a recipe to update the chemical potential $\mu^{(n+1)}$, where $\mu^{(1)} = \frac{\left(k^2/2+\Gamma_k^1\right)_{\mathrm{HOMO}}+\left(k^2/2+\Gamma_k^1\right)_{\mathrm{LUMO}}}{2}$ as

$$N/2 = \sum_k\int_{-\infty}^{\mu^{(n+1)}} A_k^{(n)}(\omega)d\omega, \tag{6.1203}$$

where we can take $N/2$ as the number of occupied bands in our spin-restricted UEG. Once we have the chemical potential, we determine the lesser and greater Green's functions as

$$G_k^{<,(n)}(t) = i \int_{-\infty}^{\mu^{(n+1)}} d\omega A_k^{(n)}(\omega) e^{-i\omega t} \tag{6.1204}$$

$$G_k^{>,(n)}(t) = -i \int_{\mu^{(n+1)}}^{\infty} d\omega A_k^{(n)}(\omega) e^{-i\omega t}. \tag{6.1205}$$

Then, we would do a DFT to get the real and imaginary parts of $\Sigma_{p,c}^{n+1,R}(\omega)$, which can then be used to update $G_k^{n+1,R}(t)$. This closes the self-consistency loop.

Doing it with forward time integration

The cumulant representation of the Green's function is

$$G_k(t) = G_k^{(0)}(t) e^{C_k(t)} = G_k^{(0)}(t) \left[1 + C_k(t) + \cdots\right]. \tag{6.1206}$$

where to determine the cumulant, we compare this expression with the Dyson expansion,

$$G_k(\omega) = G_k^{(0)}(\omega) + G_k^{(0)}(\omega)\Sigma_k(\omega)G_k^{(0)}(\omega) + \cdots. \tag{6.1207}$$

Equating the terms first order in the coupling and Fourier transforming the Dyson contribution to the time domain gives

$$C_k(t) = \frac{1}{G_k^{(0)}(t)} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \Sigma_k(\omega) \left[G_k^{(0)}(\omega)\right]^2. \tag{6.1208}$$

For the electron gas, the GW self-energy in the time domain is given as

$$\Sigma_k(t) = i \sum_q G_{k+q}(t) W_q^{(0)}(t). \tag{6.1209}$$

The time-domain cumulant can therefore be written as

$$\begin{aligned} C_k(t) = \frac{1}{G_k^{(0)}(t)} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \int_{-\infty}^{\infty} dt_1\, e^{it_1\omega} \int_{-\infty}^{\infty} dt_2\, e^{it_2\omega} \int_{-\infty}^{\infty} dt_3\, e^{it_3\omega} \\ \times \Sigma_k(t_1) G_k^{(0)}(t_2) G_k^{(0)}(t_3). \end{aligned} \tag{6.1210}$$

Collecting the exponential factors,

$$C_k(t) = \frac{1}{G_k^{(0)}(t)} \iiint \frac{d\omega}{2\pi}\, dt_1\, dt_2\, dt_3\, e^{i\omega(t_1+t_2+t_3-t)} \Sigma_k(t_1) G_k^{(0)}(t_2) G_k^{(0)}(t_3). \tag{6.1211}$$

The frequency integral generates a delta function,

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{i\omega(t_1+t_2+t_3-t)} = \delta(t_1 + t_2 + t_3 - t), \tag{6.1212}$$

which imposes

$$t_3 = t - t_1 - t_2. \tag{6.1213}$$

Thus,

$$C_k(t) = \frac{1}{G_k^{(0)}(t)} \iint dt_1\, dt_2\, \Sigma_k(t_1) G_k^{(0)}(t_2) G_k^{(0)}(t - t_1 - t_2). \tag{6.1214}$$

We now specialize to the retarded Green's function,

$$G_k^{(0)}(t) = -i\theta(t) e^{-i\epsilon_k^{\mathrm{HF}} t}. \tag{6.1215}$$

Substituting this into Eq. (6.1214) gives

$$\begin{aligned} C_k(t) = -ie^{i\epsilon_k t} \iint dt_1\, dt_2\, \Sigma_k(t_1)\theta(t_1)\theta(t_2)\theta(t - t_1 - t_2) \\ \times e^{-i\epsilon_k^{\mathrm{HF}} t_2} e^{-i\epsilon_k^{\mathrm{HF}}(t - t_1 - t_2)}. \end{aligned} \tag{6.1216}$$

The step functions restrict the integration domain, yielding

$$C_k(t) = -i \int_0^t dt_2 \int_0^{t_2} dt_1\, \Sigma_k(t_1) e^{i\epsilon_k^{\mathrm{HF}} t_1}. \tag{6.1217}$$

Since

$$G_k^{(0)}(t_1) = -i\theta(t_1) e^{-i\epsilon_k^{\mathrm{HF}} t_1}, \tag{6.1218}$$

we may equivalently write

$$C_k(t) = -\int_0^t dt_2 \int_0^{t_2} dt_1\, \frac{\Sigma_k(t_1)}{G_k^{(0)}(t_1)}. \tag{6.1219}$$

Using

$$G_k(t) = G_k^{(0)}(t) e^{C_k(t)}, \tag{6.1220}$$

we differentiate to obtain

$$\frac{\partial G_k(t)}{\partial t} = -i\epsilon_k G_k(t) - \int_0^t d\tau\, \frac{\Sigma_k(\tau) G_k(t)}{G_k^{(0)}(\tau)}. \tag{6.1221}$$

Substituting in the GW self-energy,

$$\Sigma_k(\tau) = i \sum_q G_{k+q}(\tau) W_q^{(0)}(\tau), \tag{6.1222}$$

and rearranging gives

$$0 = i\frac{\partial G_k(t)}{\partial t} - \epsilon_k G_k(t) + \sum_q \int_0^t d\tau\, W_q^{(0)}(\tau) \frac{G_{k+q}(\tau) G_k(t)}{G_k^{(0)}(\tau)}. \tag{6.1223}$$

To make contact with the time-domain Dyson equation, we may formally multiply and divide the kernel by $G_k(\tau)$,

$$0 = i\frac{\partial G_k(t)}{\partial t} - \epsilon_k G_k(t) + \sum_q \int_0^t d\tau\, W_q^{(0)}(\tau) \frac{G_{k+q}(\tau) G_k(t)}{G_k^{(0)}(\tau) G_k(\tau)} G_k(\tau). \tag{6.1224}$$

Hence, the HA-SCC construction may be interpreted as replacing the usual self-energy kernel by the effective kernel

$$\Sigma_k(t-\tau) \longrightarrow -W_q^{(0)}(\tau)\frac{G_{k+q}(\tau)G_k(t)}{G_k^{(0)}(\tau)G_k(\tau)}. \tag{6.1225}$$

My hypothesis is that we would recover the self-consistent scheme carried out in the frequency domain if we froze the exchange.

6.15 Fermionic self-consistent cumulants

6.15.1 GW+C formalism

One-shot formalism

Define the first-order cumulant as $C_p^{(1)}(t) \equiv -\mathrm{i}\,\delta_p t$, where $\delta_p \equiv \varepsilon_p^{\mathrm{HF}} - \varepsilon_p^{\mathrm{LDA}} = \Sigma_{pp}^{\mathrm{x}} - V_{pp}^{\mathrm{xc}}$. With screening parameters $W_{pq\nu}$ and Ω_ν computed from dRPA@LDA, the second-order cumulant is

$$C_p^{(2)}(t) = -\mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau'\, D_\nu^{(q)}(\tau-\tau')\, \exp\!\left[\mathrm{i}\varepsilon_p^{\mathrm{LDA}}\tau - \mathrm{i}\varepsilon_q^{\mathrm{LDA}}\tau + \mathrm{i}\varepsilon_q^{\mathrm{LDA}}\tau' - \mathrm{i}\varepsilon_p^{\mathrm{LDA}}\tau'\right], \tag{6.1226}$$

using $D_\nu^{(q)}(t) \equiv -\mathrm{i}\,\Theta(t)\, e^{-\mathrm{i}s_q\Omega_\nu t}$ and $s_q = \Theta(\varepsilon_q^{\mathrm{LDA}} - \varepsilon_F) - \Theta(\varepsilon_F - \varepsilon_q^{\mathrm{LDA}})$. The retarded Green's function is

$$G_p^R(t) = -\mathrm{i}\,\Theta(t)\, e^{C_p^{(0)}(t)+C_p^{(1)}(t)+C_p^{(2)}(t)} = -\mathrm{i}\,\Theta(t)\, e^{-\mathrm{i}\varepsilon_p^{\mathrm{HF}}t}\, e^{C_p^{(2)}(t)}, \tag{6.1227}$$

where we defined $C_p^{(0)}(t) \equiv -\mathrm{i}\,\varepsilon_p^{\mathrm{LDA}} t$. This is $\mathrm{G_0W_0}$+C.

Self-consistent resummation

Looking at 6.1226, we see that the phase in the memory kernel of $C_p^{(2)}(t)$ is $e^{-C_p^0(\tau)+C_q^0(\tau)-C_q^0(\tau')+C_p^0(\tau')}$. Taking $C^0 \to C$ affects the partial resummation:

$$C_p(t) = -\mathrm{i}\varepsilon_p^{\mathrm{LDA}} t - \mathrm{i}\delta_p t - \mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau \int_0^\tau \mathrm{d}\tau'\, D_\nu^{(q)}(\tau-\tau')\, e^{-C_p(\tau)+C_q(\tau)-C_q(\tau')+C_p(\tau')}. \tag{6.1228}$$

Equations of motion

Differentiating, and using $\mathrm{d}G_p^R/\mathrm{d}t = G_p^R\,\mathrm{d}C_p/\mathrm{d}t$ since $e^{C_p(t)} = \mathrm{i}\,G_p^R(t)$ for $t > 0$, gives

$$\frac{\mathrm{d}C_p(t)}{\mathrm{d}t} = -\mathrm{i}\varepsilon_p^{\mathrm{LDA}} - \mathrm{i}\delta_p - \mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau\, D_\nu^{(q)}(t-\tau)\, e^{-C_p(t)+C_q(t)-C_q(\tau)+C_p(\tau)} \tag{6.1229}$$

$$\implies \frac{\mathrm{d}G_p^R(t)}{\mathrm{d}t} = -\mathrm{i}\varepsilon_p^{\mathrm{HF}} G_p^R(t) - \mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau\, D_\nu^{(q)}(t-\tau)\, \frac{G_p^R(\tau)\, G_q^R(t)}{G_q^R(\tau)}, \qquad G_p^R(0^+) = -\mathrm{i}. \tag{6.1230}$$

Let $y_p(t) \equiv e^{C_p(t)+\mathrm{i}\varepsilon_p^{\mathrm{LDA}}t} \equiv e^{\tilde{C}_p(t)}$, so that $G_p^R(t) = -\mathrm{i}\,\Theta(t)\,e^{-\mathrm{i}\varepsilon_p^{\mathrm{LDA}}t}\,y_p(t)$. Then

$$\frac{\mathrm{d}G_p^R(t)}{\mathrm{d}t} = -\mathrm{i}\varepsilon_p^{\mathrm{LDA}} G_p^R(t) - \mathrm{i}\,e^{-\mathrm{i}\varepsilon_p^{\mathrm{LDA}}t}\,\dot{y}_p(t), \tag{6.1231}$$

$$\frac{G_p^R(\tau)\,G_q^R(t)}{G_q^R(\tau)} = -\mathrm{i}\,e^{-\mathrm{i}\varepsilon_p^{\mathrm{LDA}}\tau}\,e^{-\mathrm{i}\varepsilon_q^{\mathrm{LDA}}(t-\tau)}\,\frac{y_p(\tau)\,y_q(t)}{y_q(\tau)}. \tag{6.1232}$$

Inserting both identities into the equation of motion, using $\varepsilon_p^{\mathrm{HF}} - \varepsilon_p^{\mathrm{LDA}} = \delta_p$, and dividing through by $-\mathrm{i}\,e^{-\mathrm{i}\varepsilon_p^{\mathrm{LDA}}t}$, gives

$$\dot{y}_p(t) = -\mathrm{i}\delta_p\,y_p(t) - \mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau\, D_\nu^{(q)}(t-\tau)\,e^{\mathrm{i}(\varepsilon_p^{\mathrm{LDA}}-\varepsilon_q^{\mathrm{LDA}})(t-\tau)}\,\frac{y_p(\tau)\,y_q(t)}{y_q(\tau)}, \qquad y_p(0) = 1. \tag{6.1233}$$

Writing $\frac{y_p(\tau)\,y_q(t)}{y_q(\tau)} = y_p(t)\,e^{-\tilde{C}_p(t)+\tilde{C}_q(t)-\tilde{C}_q(\tau)+\tilde{C}_p(\tau)}$ and setting $\tilde{C} \to 0$ in the exponent,

$$\dot{y}_p(t) = y_p(t)\left[-\mathrm{i}\delta_p + \dot{C}_p^{(2)}(t)\right] \implies y_p(t) = e^{C_p^{(1)}(t)+C_p^{(2)}(t)}, \tag{6.1234}$$

which is G_0W_0+C@LDA.

The form to implement numerically

Define $\bar{y}_p(t) \equiv e^{\mathrm{i}\delta_p t}\,y_p(t) = e^{C_p(t)+\mathrm{i}\varepsilon_p^{\mathrm{HF}}t}$, so that $G_p^R(t) = -\mathrm{i}\,\Theta(t)\,e^{-\mathrm{i}\varepsilon_p^{\mathrm{HF}}t}\,\bar{y}_p(t)$ and

$$\begin{aligned}
\dot{\bar{y}}_p(t) &= e^{\mathrm{i}\delta_p t}\left[\dot{y}_p(t) + \mathrm{i}\delta_p y_p(t)\right] \\
&= -\mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau\, D_\nu^{(q)}(t-\tau)\,e^{\mathrm{i}\delta_p t}\,e^{\mathrm{i}(\varepsilon_p^{\mathrm{LDA}}-\varepsilon_q^{\mathrm{LDA}})(t-\tau)}\,\frac{y_p(\tau)\,y_q(t)}{y_q(\tau)} \\
&= -\mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau\, D_\nu^{(q)}(t-\tau)\,e^{\mathrm{i}(\varepsilon_p^{\mathrm{HF}}-\varepsilon_q^{\mathrm{HF}})(t-\tau)}\,e^{\mathrm{i}\delta_p\tau+\mathrm{i}\delta_q(t-\tau)}\,\frac{y_p(\tau)\,y_q(t)}{y_q(\tau)} \\
&= -\mathrm{i}\sum_{q\nu} W_{pq\nu}^2 \int_0^t \mathrm{d}\tau\, D_\nu^{(q)}(t-\tau)\,e^{\mathrm{i}(\varepsilon_p^{\mathrm{HF}}-\varepsilon_q^{\mathrm{HF}})(t-\tau)}\,\frac{\bar{y}_p(\tau)\,\bar{y}_q(t)}{\bar{y}_q(\tau)}, \qquad \bar{y}_p(0) = 1,
\end{aligned} \tag{6.1235}$$

where we used $\frac{y_p(\tau)\,y_q(t)}{y_q(\tau)} = e^{-\mathrm{i}\delta_p\tau-\mathrm{i}\delta_q(t-\tau)}\frac{\bar{y}_p(\tau)\,\bar{y}_q(t)}{\bar{y}_q(\tau)}$. We consider the electronic Hamiltonian with the traditional partitioning as

$$\hat{H} = \hat{H}_0 + \hat{V} \tag{6.1236}$$

$$= \sum_p \epsilon_p\,\underbrace{\{a_p^\dagger a_p\}}_{\hat{n}_p} + \frac{1}{4}\sum_{pqrs} \overline{v}_{pqrs}\,\{a_p^\dagger a_q^\dagger a_s a_r\}, \tag{6.1237}$$

with $\overline{v}_{pqrs} \equiv \langle pq\|rs\rangle$. Introducing a set of time-dependent scalar functions $\{\phi_p(t)\}$ satisfying $\phi_p(0) = 0$, we can repartition the electronic Hamiltonian according to

$$\hat{H} = \hat{H}_0^\phi(t) + \hat{V}^\phi(t), \tag{6.1238}$$

with $\hat{H}_0^\phi(t) = \hat{H}_0 + \sum_p \dot{\phi}_p(t)\,\hat{n}_p$ and $\hat{V}^\phi(t) = \hat{V} - \sum_p \dot{\phi}_p(t)\,\hat{n}_p$. The evolution operator generated by the new reference Hamiltonian is

$$\hat{U}_0^\phi(t,0) = T\exp\left[-i\int_0^t d\tau\,\hat{H}_0^\phi(\tau)\right] \tag{6.1239}$$

$$= \exp\left[-i\hat{H}_0 t - i\sum_p \phi_p(t)\hat{n}_p\right]$$

$$= e^{-i\hat{H}_0 t}e^{-i\hat{\Phi}(t)}. \tag{6.1240}$$

6.15.2 Particle Green's function in the interaction picture

The time-ordered Green's function for the addition of an electron into state p, with $t > 0$, is:

$$G_c(t) = -i\Theta(t)\frac{\mathrm{Tr}\left[\hat{\rho}_0\,T\left\{a_{c,I}^\phi(t)a_{c,I}^{\phi\dagger}(0)\exp\left[-i\int_0^t d\tau\,\hat{V}_{\phi,I}(\tau)\right]\right\}\right]}{\mathrm{Tr}\left[\hat{\rho}_0\,T\exp\left[-i\int_0^t d\tau\,\hat{V}_{\phi,I}(\tau)\right]\right]} \tag{6.1241}$$

$$\overset{T=0}{=} -i\Theta(t)\frac{\langle\Phi_0|a_{c,I}^\phi(t)T\left\{\exp\left[-i\int_0^t d\tau\,\hat{V}_{\phi,I}(\tau)\right]\right\}a_{c,I}^{\phi\dagger}(0)|\Phi_0\rangle}{\langle\Phi_0|T\exp\left[-i\int_0^t d\tau\,\hat{V}_{\phi,I}(\tau)\right]|\Phi_0\rangle} \tag{6.1242}$$

with the interaction-picture operators $\hat{O}_I^\phi(t) = \hat{U}_0^{\phi\dagger}(t,0)\,\hat{O}\,\hat{U}_0^\phi(t,0)$.

Interaction-picture fermionic operators

Since

$$a_{c,I}^\phi(t) = \hat{U}_0^{\phi\dagger}(t,0)a_c\hat{U}_0^\phi(t,0)$$

$$= e^{-i\epsilon_c t}e^{-i\phi_c(t)}a_c, \tag{6.1243}$$

while

$$a_{c,I}^{\phi\dagger}(t) = e^{+i\epsilon_c t}e^{+i\phi_c(t)}a_c^\dagger \tag{6.1244}$$

$$\implies a_{c,I}^{\phi\dagger}(0) = a_c^\dagger. \tag{6.1245}$$

Substituting in yields:

$$G_c(t) = -i\Theta(t)e^{-i\epsilon_c t}e^{-i\phi_c(t)}\underbrace{\frac{\langle\Phi_0|a_c T\left\{\exp\left[-i\int_0^t d\tau\,\hat{V}_{\phi,I}(\tau)\right]\right\}a_c^\dagger|\Phi_0\rangle}{\langle\Phi_0|T\exp\left[-i\int_0^t d\tau\,\hat{V}_{\phi,I}(\tau)\right]|\Phi_0\rangle}}_{y_c^\phi(t)} \tag{6.1246}$$

So defining the bare propagator $G_c^{(0)}(t) = -i\Theta(t)e^{-i\epsilon_c t}$, we can write

$$G_c(t) = G_c^{(0)}(t)e^{-i\phi_c(t)}y_c^\phi(t). \tag{6.1247}$$

Comparing to the traditional parametrization of the interacting Green's function by a cumulant,

$$G_c(t) = G_c^{(0)}(t)e^{C_c(t)} \tag{6.1248}$$

$$\implies e^{C_c(t)} = e^{-i\phi_c(t)}y_c^{\phi}(t) \tag{6.1249}$$

$$\implies C_c(t) = -i\phi_c(t) + \ln y_c^{\phi}(t). \tag{6.1250}$$

Expansion of the normalized interaction functional

Defining

$$\hat{S}_{\phi}(t) \equiv -i\int_0^t d\tau\, \hat{V}_{\phi,I}(\tau), \tag{6.1251}$$

we have

$$y_c^{\phi}(t) = \frac{\langle\Phi_0|a_c\, T e^{\hat{S}_{\phi}(t)} a_c^{\dagger}|\Phi_0\rangle}{\langle\Phi_0|T e^{\hat{S}_{\phi}(t)}|\Phi_0\rangle} \equiv \frac{\mathcal{N}_c^{\phi}(t)}{\mathcal{D}^{\phi}(t)} \tag{6.1252}$$

Associating the coupling parameter λ with the residual two-body correlation interaction (the first-order static contribution has already been absorbed into the HF reference Hamiltonian), the correlation cumulant begins at second order,

$$C_c(t) = \lambda^2 C_c^{(2)}(t) + \mathcal{O}(\lambda^3). \tag{6.1253}$$

Since the self-consistent choice in 6.1250 will take the phase to have the same perturbative order as the correlation cumulant, we assign $\phi_c(t) = \mathcal{O}(\lambda^2) \implies \dot{\phi}_c(t) = \mathcal{O}(\lambda^2)$.

The full residual interaction in the ϕ-interaction picture is

$$\begin{aligned}
\hat{V}_{\phi,I}(\tau) &= \hat{U}_0^{\phi\dagger}(\tau,0)\hat{V}^{\phi}(\tau)\hat{U}_0^{\phi}(\tau,0) \\
&= \lambda \hat{U}_0^{\phi\dagger}(\tau,0)\hat{V}\hat{U}_0^{\phi}(\tau,0) - \lambda^2 \sum_r \dot{\phi}_r(\tau)\hat{n}_r \\
&\equiv \lambda \hat{W}_{\phi,I}(\tau) - \lambda^2 \hat{K}_{\phi}(\tau). \qquad (6.1254)
\end{aligned}$$

$$\begin{aligned}
\implies \hat{S}_{\phi}(t) &= -i\int_0^t d\tau\, \hat{V}_{\phi,I}(\tau) \\
&= \lambda \underbrace{\left[-i\int_0^t d\tau\, \hat{W}_{\phi,I}(\tau)\right]}_{\hat{S}_W(t)} + \lambda^2 \underbrace{\left[+i\int_0^t d\tau\, \hat{K}_{\phi}(\tau)\right]}_{\hat{S}_K(t)} \qquad (6.1255)
\end{aligned}$$

$$\implies Te^{\hat{S}_{\phi}(t)} = 1 + \lambda \hat{S}_W(t) + \lambda^2 \left[\hat{S}_K(t) + \frac{1}{2}T\hat{S}_W^2(t)\right] + \mathcal{O}(\lambda^3) \qquad (6.1256)$$

$$\implies \mathcal{N}_c^{\phi}(t) = \langle a_c a_c^{\dagger}\rangle_0 + \lambda \underbrace{\langle a_c \hat{S}_W a_c^{\dagger}\rangle_0}_{N_1} + \lambda^2 \underbrace{\left[\langle a_c \hat{S}_K a_c^{\dagger}\rangle_0 + \frac{1}{2}\langle a_c T\hat{S}_W^2 a_c^{\dagger}\rangle_0\right]}_{N_2} + \mathcal{O}(\lambda^3) \qquad (6.1257)$$

$$\implies \mathcal{D}^{\phi}(t) = 1 + \lambda \underbrace{\langle \hat{S}_W\rangle_0}_{D_1} + \lambda^2 \underbrace{\left[\langle \hat{S}_K\rangle_0 + \frac{1}{2}\langle T\hat{S}_W^2\rangle_0\right]}_{D_2} + \mathcal{O}(\lambda^3) \qquad (6.1258)$$

$$\begin{aligned}
\implies \frac{1}{\mathcal{D}^{\phi}} &= \frac{1}{1+\lambda D_1 + \lambda^2 D_2} \\
&= 1 - \lambda D_1 + \lambda^2\left(D_1^2 - D_2\right) + \mathcal{O}(\lambda^3), \qquad (6.1259)
\end{aligned}$$

$$\begin{aligned}
\implies y_c^{\phi} &= \frac{\mathcal{N}_c^{\phi}}{\mathcal{D}^{\phi}} \\
&= \left(1 + \lambda N_1 + \lambda^2 N_2\right)\left[1 - \lambda D_1 + \lambda^2(D_1^2 - D_2)\right] + \mathcal{O}(\lambda^3) \\
&= 1 + \lambda(N_1 - D_1) + \lambda^2\left[N_2 - D_2 - N_1 D_1 + D_1^2\right] + \mathcal{O}(\lambda^3) \\
&= 1 + \lambda^2\left[\underbrace{\langle a_c \hat{S}_K a_c^{\dagger}\rangle_0 - \langle \hat{S}_K\rangle_0}_{\left[y_c^{\phi}(t)\right]_K^{(1)}} + \underbrace{\frac{1}{2}\Big(\langle a_c T\hat{S}_W^2 a_c^{\dagger}\rangle_0 - \langle T\hat{S}_W^2\rangle_0\Big)}_{\left[y_c^{\phi}(t)\right]_{WW}^{(2)}}\right] + \mathcal{O}(\lambda^3), \qquad (6.1260)
\end{aligned}$$

where we used that for a normal-ordered residual correlation interaction $\langle \hat{S}_W\rangle_0 = 0$, $\langle a_c \hat{S}_W a_c^{\dagger}\rangle_0 = 0$ (there is no contribution of order λ), and $\langle a_c a_c^{\dagger}\rangle_0 = 1$ for $c \in \mathcal{V}$. This implies that

$$\ln y_c^{\phi}(t) = \left[\ln y_c^{\phi}(t)\right]_K^{(1)} + \left[\ln y_c^{\phi}(t)\right]_{WW}^{(2)} + \mathcal{O}(\lambda^3) \qquad (6.1261)$$

with

$$\left[\ln y_c^{\phi}(t)\right]_K^{(1)} = i\int_0^t d\tau \sum_r \dot{\phi}_r(\tau)\Big[\langle a_c \hat{n}_r a_c^\dagger\rangle_0 - \langle a_c a_c^\dagger\rangle_0\langle \hat{n}_r\rangle_0\Big] \tag{6.1262}$$

$$\overset{c\in\mathcal{V}}{=} i\int_0^t d\tau\, \dot{\phi}_c(\tau) \tag{6.1263}$$

$$= i\phi_c(t), \tag{6.1264}$$

Thus, the explicit auxiliary phase cancels from our definition of the exact cumulant in equation (6.1250) and so, through second order,

$$C_c(t) = \left[\ln y_c^{\phi}(t)\right]_{WW}^{(2)} + \mathcal{O}(\lambda^3). \tag{6.1265}$$

So now we are left to evaluate

$$\left[\ln y_c^{\phi}(t)\right]_{WW}^{(2)} = \frac{(-i)^2}{2}\int_0^t d\tau_1 \int_0^t d\tau_2$$
$$\times \Big[\langle a_c T\{\hat{W}_{\phi,I}(\tau_1)\hat{W}_{\phi,I}(\tau_2)\}a_c^\dagger\rangle_0 - \langle a_c a_c^\dagger\rangle_0\langle T\{\hat{W}_{\phi,I}(\tau_1)\hat{W}_{\phi,I}(\tau_2)\}\rangle_0\Big] \tag{6.1266}$$

$$= -\int_0^t d\tau \int_0^\tau d\tau' \left\langle a_c \hat{W}_{\phi,I}(\tau)\hat{W}_{\phi,I}(\tau')a_c^\dagger\right\rangle_{0,c,c}. \tag{6.1267}$$

with 0,c,c standing for the connected contractions with respect to the reference state $|\Phi_0\rangle$ and with the external creation and annihilation operators $a_c^\dagger$ and a_c.

Second-order linked contractions

Simplifying the ϕ-interaction picture two-body interaction gives:

$$\hat{W}_{\phi,I}(\tau) = \hat{U}_0^{\phi\dagger}(\tau,0)\,\hat{V}\,\hat{U}_0^{\phi}(\tau,0) \tag{6.1268}$$

$$= e^{+i\hat{\Phi}(\tau)}e^{+i\hat{H}_0\tau}\left[\frac{1}{4}\sum_{pqrs}\overline{v}_{pqrs}\,a_p^\dagger a_q^\dagger a_s a_r\right]e^{-i\hat{H}_0\tau}e^{-i\hat{\Phi}(\tau)} \tag{6.1269}$$

$$= \frac{1}{4}\sum_{pqrs}\overline{v}_{pqrs}\,e^{+i\hat{\Phi}(\tau)}\left[e^{+i\hat{H}_0\tau}a_p^\dagger e^{-i\hat{H}_0\tau}\right]\left[e^{+i\hat{H}_0\tau}a_q^\dagger e^{-i\hat{H}_0\tau}\right]\left[e^{+i\hat{H}_0\tau}a_s e^{-i\hat{H}_0\tau}\right]\left[e^{+i\hat{H}_0\tau}a_r e^{-i\hat{H}_0\tau}\right]e^{-i\hat{\Phi}(\tau)} \tag{6.1270}$$

$$= \frac{1}{4}\sum_{pqrs}\overline{v}_{pqrs}\,e^{+i(\epsilon_p+\epsilon_q-\epsilon_s-\epsilon_r)\tau}e^{+i\hat{\Phi}(\tau)}a_p^\dagger a_q^\dagger a_s a_r e^{-i\hat{\Phi}(\tau)} \tag{6.1271}$$

$$= \frac{1}{4}\sum_{pqrs}\overline{v}_{pqrs}\,e^{+i(\epsilon_p+\epsilon_q-\epsilon_r-\epsilon_s)\tau}\left[e^{+i\hat{\Phi}(\tau)}a_p^\dagger e^{-i\hat{\Phi}(\tau)}\right]\left[e^{+i\hat{\Phi}(\tau)}a_q^\dagger e^{-i\hat{\Phi}(\tau)}\right]\left[e^{+i\hat{\Phi}(\tau)}a_s e^{-i\hat{\Phi}(\tau)}\right]\left[e^{+i\hat{\Phi}(\tau)}a_r e^{-i\hat{\Phi}(\tau)}\right] \tag{6.1272}$$

$$= \frac{1}{4}\sum_{pqrs}\overline{v}_{pqrs}\,e^{+i(\epsilon_p+\epsilon_q-\epsilon_r-\epsilon_s)\tau}\left[e^{+i\phi_p(\tau)}a_p^\dagger\right]\left[e^{+i\phi_q(\tau)}a_q^\dagger\right]\left[e^{-i\phi_s(\tau)}a_s\right]\left[e^{-i\phi_r(\tau)}a_r\right] \tag{6.1273}$$

$$= \frac{1}{4}\sum_{pqrs}\overline{v}_{pqrs}\,e^{+i(\epsilon_p+\epsilon_q-\epsilon_r-\epsilon_s)\tau}e^{+i[\phi_p(\tau)+\phi_q(\tau)-\phi_r(\tau)-\phi_s(\tau)]}a_p^\dagger a_q^\dagger a_s a_r, \tag{6.1274}$$

where we have used the result

$$e^{+i\hat{\Phi}(\tau)}a_p^\dagger e^{-i\hat{\Phi}(\tau)} = a_p^\dagger + \left[i\hat{\Phi}(\tau), a_p^\dagger\right] + \frac{1}{2!}\left[i\hat{\Phi}(\tau), \left[i\hat{\Phi}(\tau), a_p^\dagger\right]\right] + \cdots \tag{6.1275}$$

$$= a_p^\dagger + i\phi_p(\tau)a_p^\dagger + \frac{1}{2!}\left[i\phi_p(\tau)\right]^2 a_p^\dagger + \frac{1}{3!}\left[i\phi_p(\tau)\right]^3 a_p^\dagger + \cdots \tag{6.1276}$$

$$= \left[1 + i\phi_p(\tau) + \frac{\left[i\phi_p(\tau)\right]^2}{2!} + \frac{\left[i\phi_p(\tau)\right]^3}{3!} + \cdots\right] a_p^\dagger \tag{6.1277}$$

$$= e^{+i\phi_p(\tau)}a_p^\dagger \tag{6.1278}$$

with an analogous one holding for the annihilation operator. Since we are considering the particle-addition propagator, the connected contributions are obtained via propagation through the intermediate $2p1h$ configuration, so for $\tau > \tau'$, we have

$$\left\langle a_c T\left\{\hat{W}_{\phi,I}(\tau)\hat{W}_{\phi,I}(\tau')\right\} a_c^\dagger\right\rangle_0 \propto \frac{1}{16}\sum_{qrst}\sum_{uvwx}\bar{v}_{qrst}\bar{v}_{uvwx}\left\langle a_c a_q^\dagger a_r^\dagger a_t a_s a_u^\dagger a_v^\dagger a_x a_w a_c^\dagger\right\rangle_0 \tag{6.1279}$$

$$\overset{2p1h}{=} \frac{1}{16}\sum_{\substack{i,i'\in\mathcal{O}\\ a,a',b,b'\in\mathcal{V}}}\Bigg[\bar{v}_{ciab}\bar{v}_{a'b'ci'}\left\langle a_c a_c^\dagger a_i^\dagger a_b a_a a_{a'}^\dagger a_{b'}^\dagger a_{i'} a_c a_c^\dagger\right\rangle_0$$
$$+ \bar{v}_{icab}\bar{v}_{a'b'ci'}\left\langle a_c a_i^\dagger a_c^\dagger a_b a_a a_{a'}^\dagger a_{b'}^\dagger a_{i'} a_c a_c^\dagger\right\rangle_0$$
$$+ \bar{v}_{ciab}\bar{v}_{a'b'ic}\left\langle a_c a_c^\dagger a_i^\dagger a_b a_a a_{a'}^\dagger a_{b'}^\dagger a_c a_{i'} a_c^\dagger\right\rangle_0$$
$$+ \bar{v}_{icab}\bar{v}_{a'b'ic}\left\langle a_c a_i^\dagger a_c^\dagger a_b a_a a_{a'}^\dagger a_{b'}^\dagger a_c a_{i'} a_c^\dagger\right\rangle_0\Bigg] \tag{6.1280}$$

$$= \frac{4}{16}\sum_{\substack{i,i'\in\mathcal{O}\\ a,a',b,b'\in\mathcal{V}}}\bar{v}_{ciab}\bar{v}_{a'b'ci'}\Bigg[\overbrace{a_c a_c^\dagger a_i^\dagger a_b a_a a_{a'}^\dagger a_{b'}^\dagger a_{i'} a_c a_c^\dagger} + \overbrace{a_c a_c^\dagger a_i^\dagger a_b a_a a_{a'}^\dagger a_{b'}^\dagger a_{i'} a_c a_c^\dagger}\Bigg] \tag{6.1281}$$

$$\tag{6.1282}$$

$$= \frac{4}{16}\sum_{iab}\bar{v}_{ciab}\left(\bar{v}_{abci} - \bar{v}_{baci}\right) \tag{6.1283}$$

$$= \frac{8}{16}\sum_{iab}\bar{v}_{ciab}\bar{v}_{abci} \tag{6.1284}$$

$$= \frac{1}{2}\sum_{iab}\left|\langle ab\|ci\rangle\right|^2. \tag{6.1285}$$

In summary, we obtain

$$\left\langle a_c T\left\{\hat{W}_{\phi,I}(\tau)\hat{W}_{\phi,I}(\tau')\right\} a_c^\dagger\right\rangle_{0,c,c} = \frac{1}{2}\sum_{i\in\mathcal{O}}\sum_{a,b\in\mathcal{V}}\left|\langle ab\|ci\rangle\right|^2 \mathcal{P}_{cabi}(\tau,\tau'), \qquad \tau > \tau', \tag{6.1286}$$

where $\mathcal{P}_{cabi}$ contains the time-dependent one-particle phases generated by the two vertices.

Time-dependent one-particle phases

Based on equation 6.1274, we know that at the earlier time τ', the transition $p+i \longrightarrow a+b$ carries the phase

$$\exp\Big\{ +i[\epsilon_a+\epsilon_b-\epsilon_c-\epsilon_i]\tau' \Big\} \exp\Big\{ +i[\phi_a(\tau')+\phi_b(\tau')-\phi_c(\tau')-\phi_i(\tau')] \Big\}, \tag{6.1287}$$

while at the later time τ, the inverse transition $a+b \longrightarrow p+i$ carries

$$\exp\Big\{ -i[\epsilon_a+\epsilon_b-\epsilon_c-\epsilon_i]\tau \Big\} \exp\Big\{ -i[\phi_a(\tau)+\phi_b(\tau)-\phi_c(\tau)-\phi_i(\tau)] \Big\}. \tag{6.1288}$$

Their product is therefore

$$\begin{aligned}\mathcal{P}_{cabi}(\tau,\tau') = \exp\Big[-i\Delta^{(+)}_{cabi}(\tau-\tau')\Big] \exp\Big\{ &-i[\phi_a(\tau)-\phi_a(\tau')+\phi_b(\tau)-\phi_b(\tau') \\ &-\phi_c(\tau)+\phi_c(\tau')-\phi_i(\tau)+\phi_i(\tau')] \Big\},\end{aligned} \tag{6.1289}$$

where

$$\Delta^{(+)}_{cabi} \equiv \epsilon_a+\epsilon_b-\epsilon_i-\epsilon_c. \tag{6.1290}$$

So we are left with

$$\begin{aligned}C^{(2)}_c(t) = \ln y^\phi_c(t) = -\frac{1}{2}\sum_{i\in\mathcal{O}}\sum_{a,b\in\mathcal{V}} |\langle ab\|ci\rangle|^2 \int_0^t d\tau \int_0^\tau d\tau' e^{-i\Delta^{(+)}_{cabi}(\tau-\tau')} \\ \times \exp\Big\{ -i[\phi_a(\tau)-\phi_a(\tau')+\phi_b(\tau)-\phi_b(\tau')-\phi_c(\tau)+\phi_c(\tau')-\phi_i(\tau)+\phi_i(\tau')] \Big\}.\end{aligned} \tag{6.1291}$$

Looking back to the equation 6.1250, we see that the self-consistent choice is $C_c(t) = -i\phi_c(t)$.

6.15.3 Hole Green's function in the interaction picture

Here we are interested in solving for

$$G_k(t) = +i\Theta(t)\langle a^\dagger_k(t) a_k(0)\rangle_0 \tag{6.1292}$$

$$= G^{(0)}_k(t) e^{+i\phi_k(t)} y^\phi_k(t), \tag{6.1293}$$

where $G^{(0)}_k(t) = +i\Theta(t)e^{+i\epsilon_k t}$ is the bare hole propagator. Comparing with the cumulant ansatz yields

$$C_k(t) = +i\phi_k(t) + \ln y^\phi_k(t). \tag{6.1294}$$

The ensuing derivation is similar to the one we did previously, with the exception that now our intermediate configuration is $2h1p$. The final result is

$$\begin{aligned}C^{(2)}_k(t) = -\frac{1}{2}\sum_{a\in\mathcal{V}}\sum_{i,j\in\mathcal{O}} |\langle ka\|ij\rangle|^2 \int_0^t d\tau \int_0^\tau d\tau' e^{-i\Delta^{(-)}_{kija}(\tau-\tau')} \\ \times \exp\Big\{ -i[\phi_i(\tau)-\phi_i(\tau')+\phi_j(\tau)-\phi_j(\tau')-\phi_k(\tau)+\phi_k(\tau')-\phi_a(\tau)+\phi_a(\tau')] \Big\},\end{aligned} \tag{6.1295}$$

where $\Delta^{(-)}_{kija} = \epsilon_k+\epsilon_a-\epsilon_i-\epsilon_j$. Now, the self-consistent choice is $C_k(t) = +i\phi_k(t)$. $c \in \mathcal{V}$ and $k \in \mathcal{O}$ are particle and hole indices, respectively.

6.15.4 BWs2-partitioned self-consistent cumulant

In the spin-restricted formalism, we define the static BWs2 shift as

$$\Lambda_p = \begin{cases} \alpha_1 W_p^O, & p \in \mathcal{O}, \\ \alpha_2 W_p^V, & p \in \mathcal{V}, \end{cases} \qquad \tilde{\epsilon}_p = \epsilon_p + \Lambda_p. \tag{6.1296}$$

where

$$W_{ii}^O = \frac{1}{2}\mathbf{\Sigma}_{ii}^{(2),2p1h}(\omega = \tilde{\varepsilon}_i) = \frac{1}{2}\sum_{abj} \frac{\mathcal{K}_{ij}^{ab}}{\tilde{\varepsilon}_i + \tilde{\varepsilon}_j - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b}, \tag{6.1297}$$

$$\mathbf{W}_{aa}^V = \frac{1}{2}\mathbf{\Sigma}_{aa}^{(2),2h1p}(\omega = \tilde{\varepsilon}_a) = \frac{1}{2}\sum_{ijb} \frac{\mathcal{K}_{ij}^{ab}}{\tilde{\varepsilon}_a + \tilde{\varepsilon}_b - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j}, \tag{6.1298}$$

where $\tilde{\varepsilon}$ is the BWs2 ground-state MO energy and we defined $\mathcal{K}_{pq}^{rs} = (pr|qs)\,[2(pr|qs) - (ps|qr)]$. The Hamiltonian is partitioned as

$$\hat{H} = \hat{H}_0^{\mathrm{BW}} + \hat{V}^{\mathrm{BW}}, \tag{6.1299}$$

$$\hat{H}_0^{\mathrm{BW}} = \sum_p \tilde{\epsilon}_p \hat{n}_p, \tag{6.1300}$$

$$\hat{V}^{\mathrm{BW}} = \hat{V} - \sum_p \Lambda_p \hat{n}_p. \tag{6.1301}$$

Thus, the end result for the coupled VIDEs

$$\begin{aligned} \dot{C}_c^+(t) = +i\Lambda_c - \sum_{iab} \mathcal{K}_{ci}^{ab} \int_0^t d\tau\, e^{-i\widetilde{\Delta}_{cabi}^{(+)}(t-\tau)} \\ \times \exp\left[-\delta C_c^+ + \delta C_a^+ + \delta C_b^+ + \delta C_i^-\right], \end{aligned} \tag{6.1302}$$

$$\begin{aligned} \dot{C}_k^-(t) = -i\Lambda_k + \sum_{ija} \mathcal{K}_{ij}^{pa} \int_0^t d\tau\, e^{-i\widetilde{\Delta}_{kija}^{(-)}(t-\tau)} \\ \times \exp\left[-\delta C_k^- + \delta C_i^- + \delta C_j^- + \delta C_a^+\right], \end{aligned} \tag{6.1303}$$

with $\delta C_q \equiv C_q(t) - C_q(\tau)$, $\widetilde{\Delta}_{cabi}^{(+)} = \tilde{\epsilon}_a + \tilde{\epsilon}_b - \tilde{\epsilon}_c - \tilde{\epsilon}_i$, and $\widetilde{\Delta}_{kija}^{(-)} = \tilde{\epsilon}_k + \tilde{\epsilon}_a - \tilde{\epsilon}_i - \tilde{\epsilon}_j$. Note that the BWs2 shifts of Λ_c and Λ_k take the place of the auxiliary phases ϕ_c and ϕ_k in the VIDEs; this is to be expected, though, since we're just doing another repartitioning of the Hamiltonian, just as we did when we introduced the auxiliary phases in 6.1238.

6.15.5 Coupled VIDEs

Putting the results together and differentiating once gives

$$\dot{C}_c(t) = -\sum_{iab} |\langle ab\|ci\rangle|^2 \int_0^t d\tau e^{-i\Delta_{cabi}^+(t-\tau)} e^{-\delta C_c + \delta C_a + \delta C_b + \delta C_i} \tag{6.1304}$$

$$\dot{C}_k(t) = -\frac{1}{2}\sum_{ija} |\langle ka\|ij\rangle|^2 \int_0^t d\tau e^{-i\Delta_{kija}^-(t-\tau)} e^{-\delta C_k + \delta C_i + \delta C_j + \delta C_a} \tag{6.1305}$$

with $\delta C_q \equiv C_q(t) - C_q(\tau)$, $\Delta^{(+)}_{cabi} = \epsilon_a + \epsilon_b - \epsilon_c - \epsilon_i$, and $\Delta^{(-)}_{kija} = \epsilon_k + \epsilon_a - \epsilon_i - \epsilon_j$.

How to solve them?

Algorithm 1 Coupled propagation of particle and hole cumulants

1: Define the time grid

$$t_n = n\Delta t, \qquad n = 0, \ldots, N_t - 1,$$

and initialize

$$C^+_a(0) = 0, \qquad C^-_i(0) = 0.$$

2: **for** $n = 0, \ldots, N_t - 2$ **do**

3: Evaluate all particle and hole memory integrals from the common history

$$\mathbf{F}_n = \mathbf{F}[\mathbf{C}](t_n) = \begin{pmatrix} \{F^+_{a,n}\} \\ \{F^-_{i,n}\} \end{pmatrix},$$

where

$$F^+_{a,n} = \dot{C}^+_a(t_n), \qquad F^-_{i,n} = \dot{C}^-_i(t_n).$$

4: Form a simultaneous predictor

$$\mathbf{C}^{\mathrm{P}}_{n+1} = \mathbf{C}_n + \Delta t\, \mathbf{F}_n.$$

5: Evaluate the predicted derivative using the history augmented by $\mathbf{C}^{\mathrm{P}}_{n+1}$,

$$\mathbf{F}^{\mathrm{P}}_{n+1} = \mathbf{F}[\mathbf{C}_0, \ldots, \mathbf{C}_n, \mathbf{C}^{\mathrm{P}}_{n+1}](t_{n+1}).$$

6: Correct all particle and hole cumulants simultaneously,

$$\mathbf{C}_{n+1} = \mathbf{C}_n + \frac{\Delta t}{2}\left(\mathbf{F}_n + \mathbf{F}^{\mathrm{P}}_{n+1}\right).$$

7: **end for**

6.15.6 Comparing self-consistent cumulants on a GW reference

Initial definitions

We use the retarded convention throughout, so that the Green's function is

$$G_{0,k}(t) = -ie^{-i\epsilon^{\mathrm{LDA}}_k t}, \qquad G_k(t) = G_{0,k}(t)e^{C_k(t)}, \tag{6.1306}$$

which is applicable for the general state k. We assume that the chemical potential can be written as the perturbative correction $\mu = \mu_0 + \delta\mu$, where $\delta\mu$ is the correction from its initial value of $\mu_0 = \frac{\left(k^2/2+\Gamma^1_k\right)_{\mathrm{HOMO}} + \left(k^2/2+\Gamma^1_k\right)_{\mathrm{LUMO}}}{2}$. Defining the static correction

$$\Delta_k[\mu_0] = \Sigma^x_k[\mu_0] - V^{\mathrm{LDA}}_{xc,k}, \tag{6.1307}$$

with $\Sigma_k^x[\mu] = n_k[\mu]v_m - \sum_{q\neq 0} n_{k+q}[\mu]v(q)$, where $n_k[\mu] = \int_0^\mu A_k(\omega)d\omega \implies n_k[\mu_0] = \Theta(k_F - k)$, we can write the cumulant as

$$C_k(t) = -i\Delta_k[\mu_0]t + c_k(t) \tag{6.1308}$$

where we used $C_k(0) = 0$. Our initial formulation assumes a GW self-energy, so the quasibosonic frequencies Ω_ν and couplings $g_{pq\nu}$ are also held fixed. It can be shown that both schemes described below reduce to the one-shot G0W0+C result with $c_k(t) = 0$ in the initial iteration. Because we are linearizing the equations, this means that we can solve the forward time integration approximating $\mu \approx \mu_0$ and then just add a perturbative correction at the very end.

Holms' scheme

If we assume that the chemical potential stays fixed, which he does NOT do, then we can write EOMs in the time-domain. For the state $j \in \mathcal{O}$ below the Fermi level, we have

$$\dot{G}_j^{\mathrm{H}}(t) = -i\big(\epsilon_j^{\mathrm{LDA}} + \Delta_j[\mu_0]\big)G_j^{\mathrm{H}}(t) - \sum_{j\in\mathcal{O},\nu} |g_{ji\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_i^{\mathrm{H}}(s)G_j^{\mathrm{H}}(t)}{G_{0,j}(s)} \tag{6.1309}$$

$$- \sum_{a\in\mathcal{V},\nu} |g_{ja\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_a^{\mathrm{H}}(s)G_j^{\mathrm{H}}(t)}{G_{0,j}(s)}, \tag{6.1310}$$

while for the state $a \in \mathcal{V}$ above the Fermi level, we have

$$\dot{G}_a^{\mathrm{H}}(t) = -i\big(\epsilon_a^{\mathrm{LDA}} + \Delta_a[\mu_0]\big)G_a^{\mathrm{H}}(t) - \sum_{j\in\mathcal{O},\nu} |g_{aj\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_j^{\mathrm{H}}(s)G_a^{\mathrm{H}}(t)}{G_{0,a}(s)} \tag{6.1311}$$

$$- \sum_{b\in\mathcal{V},\nu} |g_{ab\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_b^{\mathrm{H}}(s)G_a^{\mathrm{H}}(t)}{G_{0,a}(s)}. \tag{6.1312}$$

I have demonstrated numerically that this is equivalent to carrying out self-consistency in the frequency domain, *only on the quasiboson Hamiltonian.*

Dunn's scheme

Dunn's EOM for holes is

$$\dot{G}_j^{\mathrm{D}}(t) = -i\big(\epsilon_j^{\mathrm{LDA}} + \Delta_j[\mu_0]\big)G_j^{\mathrm{D}}(t) - \sum_{i\in\mathcal{O},\nu} |g_{ji\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_i^{\mathrm{D}}(t)G_j^{\mathrm{D}}(t-s)}{G_i^{\mathrm{D}}(t-s)} \tag{6.1313}$$

$$- \sum_{a\in\mathcal{V},\nu} |g_{ja\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_a^{\mathrm{D}}(t)G_j^{\mathrm{D}}(t-s)}{G_a^{\mathrm{D}}(t-s)}, \tag{6.1314}$$

while for the particle states, we have

$$\dot{G}_a^{\mathrm{D}}(t) = -i\big(\epsilon_a^{\mathrm{LDA}} + \Delta_a[\mu_0]\big)G_a^{\mathrm{D}}(t) - \sum_{j\in\mathcal{O},\nu} |g_{aj\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_j^{\mathrm{D}}(t)G_a^{\mathrm{D}}(t-s)}{G_j^{\mathrm{D}}(t-s)} \tag{6.1315}$$

$$- \sum_{b\in\mathcal{V},\nu} |g_{ab\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_b^{\mathrm{D}}(t)G_a^{\mathrm{D}}(t-s)}{G_b^{\mathrm{D}}(t-s)}. \tag{6.1316}$$

If we solve these equations, then we get

$$G_j^D(t) = \exp\bigl(-i(\epsilon_j^{\mathrm{LDA}} + \Delta_j[\mu_0])t\bigr)\exp(c_j(t)) \tag{6.1317}$$

But we really want

$$G_j^D(t) = \exp\bigl(-i(\epsilon_j^{\mathrm{LDA}} + \Delta_j[\mu_0 + \delta\mu])t\bigr)\exp(c_j(t)) \tag{6.1318}$$

$$\approx \exp\left(-i(\epsilon_j^{\mathrm{LDA}} + \Delta_j[\mu_0] + \frac{\partial \Delta_j}{\partial \mu}\delta\mu)t\right)\exp(c_j(t)) \tag{6.1319}$$

$$\tag{6.1320}$$

where $G_{0,k}^{n,R}(t) = -i\Theta(t)e^{-i\left(k^2/2+\Gamma_k^n\right)t}$ with, for $n \geq 0$, $\Gamma_k^{n+1} = n_k^{(n)}v_m + \Sigma_k^{n+1,x(E)}[G^{(n)}] = n_k^{(n)}v_m - \sum_{\boldsymbol{q}\neq\boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}}^{(n)}v(\boldsymbol{q})$ and $\Gamma_k^0 = V_k^{xc}$. $\Sigma_k^{n+1,x(E)}[G^{(n)}]$ is the exchange self-energy at iteration $n+1$ computed with Ewald summation (neglecting the $\boldsymbol{q} = 0$ term) and $n_k^{(n)}v_m$ is the Madelung potential at iteration n. Note that this may change during self-consistent iteration as the chemical potential and thus occupation numbers are updated. We are always e

6.15.7 Comparing self-consistent cumulants on the hole-only quasiboson model

Holms' scheme

This comparison is most transparent if we assume that the exchange stays fixed. We use the retarded convention throughout, so that the Green's function is

$$G_{0,k}(t) = -ie^{-i\epsilon_k^{\mathrm{LDA}}t}, \qquad G_k(t) = G_{0,k}(t)e^{C_k(t)}, \tag{6.1321}$$

which is applicable for the general state k. Defining the static correction

$$\Delta_k = \Sigma_k^x - V_{xc,k}^{\mathrm{LDA}}, \tag{6.1322}$$

we can write the cumulant as

$$C_k(t) = -i\Delta_k t + c_k(t) \tag{6.1323}$$

$$\implies \dot{C}_k(0) = -i\Delta_k, \tag{6.1324}$$

where we used $C_k(0) = 0$. Our initial formulation assumes a GW self-energy, so the bosonic frequencies Ω_ν and couplings $g_{pq\nu}$ are also held fixed. Also, define

$$D_{ki\nu}^{\mathrm{LDA}} = \epsilon_k^{\mathrm{LDA}} - \epsilon_i^{\mathrm{LDA}} + \Omega_\nu. \tag{6.1325}$$

Matching the first Dyson correction to the cumulant expansion, and then evaluating the correlation self-energy with dressed propagators, Holm's self-consistent map is

$$C_k^{\mathrm{H}}(t) = -i\Delta_k t - \int_0^t ds\,(t-s)\frac{\Sigma_{k,c}[G^{\mathrm{H}}](s)}{G_{0,k}(s)} \tag{6.1326}$$

$$\implies \dot{C}_k^{\mathrm{H}}(t) = -i\Delta_k - \sum_{i\in\mathcal{O},\nu}|g_{ki\nu}|^2\int_0^t ds\,e^{+iD_{ki\nu}^{\mathrm{LDA}}s}e^{C_i^{\mathrm{H}}(s)} \tag{6.1327}$$

with

$$\Sigma_{k,c}[G^{\mathrm{H}}](s) = \sum_{i\in\mathcal{O},\nu} |g_{ki\nu}|^2 e^{+i\Omega_\nu s} G_i^{\mathrm{H}}(s), \qquad s > 0. \tag{6.1328}$$

Equivalently, the interacting Green's function obeys

$$\dot{G}_k^{\mathrm{H}}(t) = \dot{G}_{0,k}(t) e^{C_k^{\mathrm{H}}(t)} + G_k^{\mathrm{H}}(t)\dot{C}_k^{\mathrm{H}}(t) \tag{6.1329}$$

$$= -i\big(\epsilon_k^{\mathrm{LDA}} + \Delta_k\big) G_k^{\mathrm{H}}(t) - \sum_{i\in\mathcal{O},\nu} |g_{ki\nu}|^2 \int_0^t ds\, e^{iD_{ki\nu}^{\mathrm{LDA}} s} e^{C_i^{\mathrm{H}}(s)} G_k^{\mathrm{H}}(t) \tag{6.1330}$$

$$= -i\big(\epsilon_k^{\mathrm{LDA}} + \Delta_k\big) G_k^{\mathrm{H}}(t) - \sum_{i\in\mathcal{O},\nu} |g_{ki\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_i^{\mathrm{H}}(s) G_k^{\mathrm{H}}(t)}{G_{0,k}(s)}, \tag{6.1331}$$

where we used $e^{iD_{ki\nu}^{\mathrm{LDA}} s} e^{C_i^{\mathrm{H}}(s)} = e^{i\Omega_\nu s} e^{i(\epsilon_k^{\mathrm{LDA}} - \epsilon_i^{\mathrm{LDA}})s} e^{C_i^{\mathrm{H}}(s)} = e^{i\Omega_\nu s} \frac{G_i^{\mathrm{H}}(s)}{G_{0,k}(s)}$. For the same quasiboson model and static correction, Dunn's EOM is

$$\dot{G}_k^{\mathrm{D}}(t) = -i\big(\epsilon_k^{\mathrm{LDA}} + \Delta_k\big) G_k^{\mathrm{D}}(t) - \sum_{i\nu} |g_{ki\nu}|^2 \int_0^t ds\, e^{+i\Omega_\nu s} \frac{G_i^{\mathrm{D}}(t) G_k^{\mathrm{D}}(t-s)}{G_i^{\mathrm{D}}(t-s)} \tag{6.1332}$$

$$\implies \dot{C}_k^{\mathrm{D}}(t) = -i\Delta_k - \sum_{i\nu} |g_{ki\nu}|^2 \int_0^t ds\, e^{i\Omega_\nu s} \frac{G_i^{\mathrm{D}}(t) G_k^{\mathrm{D}}(t-s)}{G_i^{\mathrm{D}}(t-s) G_k^{\mathrm{D}}(t)} \tag{6.1333}$$

$$= -i\Delta_k - \sum_{i\nu} |g_{ki\nu}|^2 \int_0^t ds\, e^{+iD_{ki\nu}^{\mathrm{LDA}} s} \left[e^{C_i^{\mathrm{D}}(t) - C_i^{\mathrm{D}}(t-s) + C_k^{\mathrm{D}}(t-s) - C_k^{\mathrm{D}}(t)} \right], \tag{6.1334}$$

where we use the identity $\frac{G_i^{\mathrm{D}}(t) G_k^{\mathrm{D}}(t-s)}{G_i^{\mathrm{D}}(t-s) G_k^{\mathrm{D}}(t)} = e^{i(\epsilon_k^{\mathrm{LDA}} - \epsilon_i^{\mathrm{LDA}})s} e^{C_i^{\mathrm{D}}(t) - C_i^{\mathrm{D}}(t-s) + C_k^{\mathrm{D}}(t-s) - C_k^{\mathrm{D}}(t)}$.

6.15.8 $G\tilde{v}_0$

Define the noninteracting Green's function for state p as

$$G_p^n(\omega) = \big(\omega - E_p^n\big)^{-1} \tag{6.1335}$$

where

$$E_p^n = \big(\mathbf{F}[G^{(n-1)}] + \alpha \mathbf{W}[G^{(n)}]\big)_{pp} + (1 - \delta_{n0}) \operatorname{Re} \Sigma_{\mathrm{eff,pp}}[G^{(n-1)}](E_p^{n-1}). \tag{6.1336}$$

If we define $\tilde{\Omega}_{ia} = E_a^0 - E_i^0$, for the initial iteration $n = 0$, we have

$$E_p^0 = \big(\mathbf{F}^{\mathrm{HF}} + \alpha \mathbf{W}[G^{(0)}]\big)_{pp} \tag{6.1337}$$

$$= p^2/2 + \Gamma_p^n + \alpha \Sigma_{pp}^{2h1p}[G^{(0)}](E_p^0) \tag{6.1338}$$

$$= p^2/2 + \Gamma_p^n + \alpha \sum_{ij,a} \frac{\mathcal{K}_{paij}}{E_p^0 + \tilde{\Omega}_{ia} - E_j^0}. \tag{6.1339}$$

where $\mathcal{K}_{paij} = (pa|ij)\,(2(pa|ij) - (pi|aj))$ in the restricted formalism and the BW single-particle energies are determined self-consistently in the ground-state calculation, when they happen to take on the values

$$E_r^0 = \begin{cases} \varepsilon_r^{\mathrm{HF}}, & \text{if } |\boldsymbol{r}| > k_F, \\ \varepsilon_r^{\mathrm{HF}} + \alpha \sum_{ij,a} \frac{\mathcal{K}_{raij}}{E_r^0 + E_a^0 - E_i^0 - E_j^0}, & \text{if } |\boldsymbol{r}| \leq k_F. \end{cases} \tag{6.1340}$$

With the definition

$$\boldsymbol{\Sigma}_{\text{eff}}^{(n)}(\omega) \equiv \boldsymbol{\Sigma}(\omega) - \alpha\mathbf{W} \tag{6.1341}$$

$$\approx \tilde{\boldsymbol{\Sigma}}^{(2)}\left[G^{(n-1)}\right](\omega) - \alpha\mathbf{W}\left[G^{(n-1)}\right]. \tag{6.1342}$$

for iterations $n > 0$, we have

$$E_p^n = \left(\mathbf{F} + \alpha\mathbf{W}[G^{(n)}]\right)_{pp} + \Sigma_{\text{eff,pp}}^{(n)}(E_p^{n-1}) \tag{6.1343}$$

$$\approx \left(\mathbf{F} + \alpha\mathbf{W}[G^{(n)}]\right)_{pp} + \text{Re}\,\tilde{\Sigma}_{pp}^{(2)}\left[G^{(n-1)}\right](E_p^{n-1}) - \alpha\mathbf{W}[G^{(n)}]_{pp} \tag{6.1344}$$

$$= p^2/2 + \Gamma_p^n + \text{Re}\,\tilde{\Sigma}_{pp}^{(2)}\left[G^{(n-1)}\right](E_p^{n-1}) \tag{6.1345}$$

$$= p^2/2 + \Gamma_p^n + \sum_{ij,a} \frac{\mathcal{K}_{paij}}{E_p^{n-1} - E_j^{n-1} + \tilde{\Omega}_{ia}} + \sum_{i,ab} \frac{\mathcal{K}_{paib}}{E_p^{n-1} - E_b^{n-1} - \tilde{\Omega}_{ia}}. \tag{6.1346}$$

We saw previously that the following equation is approximately obeyed

$$G_{pp}^1(t) \approx G_{pp}^0(t) + \int dt'dt''\, G_{pp}^0(t-t')\boldsymbol{\Sigma}_{\text{eff}}(t'-t'')G_{pp}^0(t'') \tag{6.1347}$$

Then, for iterations $n > 0$, previous experience with GW suggests that we take the quasiparticle Green's function as our non-interacting reference.

How we should choose $W^{(n)}$

If we are supposed to freeze the interaction, this implies that the QB couplings and frequencies always stay at the non-interacting level as $\langle pa||ij\rangle$ and $\tilde{\Omega}_{ia} = E_a^0 - E_i^0$.

Getting the quasiparticle energies

What choice we take for the self-energy

For true self-consistency, I was thinking that we should do

$$\tilde{\Sigma}_{m,c}^{(2),n+1,R}[G^{(n)}](t) = -i\theta(t)\left[\sum_{ijb} \mathcal{K}_{pijb} G_i^{(n),<}(t) e^{-i\tilde{\Omega}_{jb}t} + \sum_{iab} \mathcal{K}_{paib} G_a^{(n),>}(t) e^{-i\tilde{\Omega}_{ib}t}\right] \tag{6.1348}$$

with $\mathcal{K}_{pijb} = (pi|jb)\,(2(pi|jb) - (pj|ib))$ and $\mathcal{K}_{paib} = (pa|bi)\,(2(pa|bi) - (pb|ai))$ in the restricted formalism. $\tilde{\Omega}_{kc} \equiv \tilde{\varepsilon}_c - \tilde{\varepsilon}_k = \varepsilon_c^{\text{HF}} - \tilde{\varepsilon}_k$ are the BW excitation energies frozen at the non-interacting level and $\tilde{\varepsilon}$ are the dressed BW single-particle energies.

How this reduces to the one-shot BWs2 self-energy

We can check that this reduces to the known form of the one-shot BWs2 self-energy when $n = 0$ and $G^{(0)}$ is the non-interacting Green's function. We have that

$$G_n^{(0),R}(t) = -i\Theta(t)e^{-i\tilde{\varepsilon}_n t} \tag{6.1349}$$

$$\implies G_i^{(0),<}(t) = ie^{-i\tilde{\varepsilon}_i t}, \quad G_a^{(0),>}(t) = -ie^{-i\tilde{\varepsilon}_a t} = -ie^{-i\varepsilon_a^{\mathrm{HF}} t} = \quad \forall t > 0 \tag{6.1350}$$

$$\implies \tilde{\Sigma}^{(2),1,R}_{m,c}[G^{(0)}](t) = -i\theta(t)\left[\sum_{ijb}\left(\mathcal{K}_{pijb}\right)e^{-i(\tilde{\varepsilon}_i - \tilde{\Omega}_{jb})t}\right. \tag{6.1351}$$

$$\left. + \sum_{iab}\left(\mathcal{K}_{paib}\right)e^{-i(\tilde{\varepsilon}_a + \tilde{\Omega}_{ib})t}\right], \tag{6.1352}$$

Since $F(\omega) = \int_{-\infty}^{\infty} dt e^{i(\omega t)}F(t)$ and $\int_0^{\infty}(-i)e^{i(\omega - E)t}dt = \frac{1}{\omega - E + i\eta}$, we have that

$$\tilde{\Sigma}^{(2),1,R}_{m,c}[G^{(0)}](\omega) = \sum_{ijb}\frac{\mathcal{K}_{pijb}}{\omega - \tilde{\varepsilon}_i + \tilde{\Omega}_I + i\eta} + \sum_{aij}\frac{\mathcal{K}_{paib}}{\omega - \tilde{\varepsilon}_a - \tilde{\Omega}_I + i\eta}, \tag{6.1353}$$

which is exactly the form of the second order self energy evaluated at a BW reference.

In all instances past the first iteration, we can do a discrete Fourier transform to get into the frequency domain, take the real part, and evaluate at E_p^{n-1} to get the correlation self-energy contribution to the quasiparticle energy. Equating the Dyson and cumulant expansions to first order in the screened interaction yields,

$$C_k^{n+1,R}(t) = ie^{iE_k^{(n)}t}\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{k^2/2 + \Gamma_k^{n+1} - (k^2/2 + \Gamma_k^n) + \Sigma_k^{n+1,c}[G^{(n)}](\omega)}{(\omega - E_k^{(n)} + i\eta)^2} \tag{6.1354}$$

$$= -i\left(\Gamma_k^{n+1} - \Gamma_k^n\right)t\Theta(t) + \underbrace{i\int\frac{d\omega}{2\pi}e^{-i\omega t}\frac{\Sigma_k^{n+1,c}[G^{(n)}](\omega + E_k^{(n)})}{(\omega + i\eta)^2}}_{C_k^{n+1,\mathrm{dyn}}(t)}. \tag{6.1355}$$

where $\Sigma_k^{n+1,c}[G^{(n)}](\omega)$ is the correlation self-energy at iteration $n+1$. We can rewrite the dynamical cumulant as

$$C_k^{n+1,\mathrm{dyn}}(t) = \int d\omega\,\beta_k^{n+1,R}(\omega)\frac{e^{-i\omega t} - 1}{\omega^2} \tag{6.1356}$$

$$= \underbrace{\int \mathrm{d}\omega\,\beta_k^{n+1,R}(\omega)\frac{e^{-i\omega t} + i\omega t - 1}{\omega^2}}_{C_k^{\mathrm{Landau}}(t)} - i\,S_1[\beta_k]\,t \tag{6.1357}$$

where $\beta_{k,c}^{n+1,R}(\omega) = \frac{1}{\pi}\left|\mathrm{Im}\,\Sigma_k^{n+1,c}\left(\omega + E_k^{(n)}\right)\right|$ and $S_1[\beta_k] = \mathcal{P}\int\mathrm{d}\omega\,\beta_k^{n+1,R}(\omega)/\omega$. This allows us to write the Green's function as

$$G_k^{n+1,R}(t) = -i\Theta(t)\,\exp\left[-i\left(k^2/2 + \Gamma_k^{n+1}\right)t\right]e^{C_k^{n+1,\mathrm{dyn}}(t)}. \tag{6.1358}$$

How this reduces to the MBPT self-energy with the MF Green's function

When we are iterating on BW, the correlation self-energy takes the form:

$$\Sigma_{m,c}^{(n+1),R}[G^{(n)}](t) = -i\theta(t)\left[\sum_{ijb}\mathcal{K}_{pijb}G_i^{(n),<}(t)e^{-i\tilde{\Omega}_{jb}t} + \sum_{iab}\mathcal{K}_{paib}G_a^{(n),>}(t)e^{-i\tilde{\Omega}_{ib}t}\right], \quad (6.1359)$$

with $\mathcal{K}_{pijb} = (pi|jb)\left(2(pi|jb) - (pj|ib)\right)$ and $\mathcal{K}_{paib} = (pa|bi)\left(2(pa|bi) - (pb|ai)\right)$ in the restricted formalism.

How this reduces to the BWs2 self-energy with a HF Green's function

For $n = 0$, we have that

$$G_n^{(0),R}(t) = -i\Theta(t)e^{-i\tilde{\varepsilon}_n t} \quad (6.1360)$$

$$\implies G_i^{(0),<}(t) = ie^{-i\tilde{\varepsilon}_i t}, \quad G_a^{(0),>}(t) = -ie^{-i\tilde{\varepsilon}_a t} \quad \forall t > 0 \quad (6.1361)$$

$$\implies \tilde{\Sigma}_{m,c}^{(2),1,R}[G^{(0)}](t) = -i\theta(t)\left[\sum_{ijb}\left(\mathcal{K}_{pijb}\right)e^{-i(\tilde{\varepsilon}_i - \tilde{\Omega}_{jb})t}\right. \quad (6.1362)$$

$$\left. + \sum_{iab}\left(\mathcal{K}_{paib}\right)e^{-i(\tilde{\varepsilon}_a + \tilde{\Omega}_{ib})t}\right], \quad (6.1363)$$

Since $F(\omega) = \int_{-\infty}^{\infty} dt e^{i(\omega t)}F(t)$ and $\int_0^{\infty}(-i)e^{i(\omega - E)t}dt = \frac{1}{\omega - E + i\eta}$, we have that

$$\tilde{\Sigma}_{m,c}^{(2),1,R}[G^{(0)}](\omega) = \sum_{ijb}\frac{\mathcal{K}_{pijb}}{\omega - \tilde{\varepsilon}_i + \tilde{\Omega}_I + i\eta} + \sum_{aij}\frac{\mathcal{K}_{paib}}{\omega - \tilde{\varepsilon}_a - \tilde{\Omega}_I + i\eta}, \quad (6.1364)$$

When we are iterating on GW, the correlation self-energy takes the form:

$$\Sigma_{m,c}^{(n+1),R}[G^{(n)}](t) = -i\theta(t)\left[\sum_{i,I}\left|M_{m,i}^I\right|^2 G_i^{(n),<}(t)e^{-i\Omega_I t} + \sum_{a,I}\left|M_{m,a}^I\right|^2 G_a^{(n),>}(t)e^{-i\Omega_I t}\right], \quad (6.1365)$$

with M and Ω the transition densities and energies from the dRPA problem.

How this reduces to G0W0 with a noninteracting Green's function

For $n = 0$, we have that

$$G^R_{0,n}(t) = -i\Theta(t)e^{-i\varepsilon^0_n t} \tag{6.1366}$$

$$\implies G^<_{0,i}(t) = ie^{-i\varepsilon^0_i t}, \quad G^>_{0,a}(t) = -ie^{-i\varepsilon^0_a t} \quad \forall t > 0 \tag{6.1367}$$

$$\implies \Sigma^R_{m,c}(t) = -i\theta(t)\left[\sum_{i,I}\left|M^I_{m,i}\right|^2 e^{-i(\varepsilon^0_i - \Omega_I)t} + \sum_{a,I}\left|M^I_{m,a}\right|^2 e^{-i(\varepsilon^0_a + \Omega_I)t}\right] \tag{6.1368}$$

Since $F(\omega) = \int_{-\infty}^{\infty} dt e^{i(\omega t)}F(t)$ and $\int_0^\infty (-i)e^{i(\omega - E)t}dt = \frac{1}{\omega - E + i\eta}$, we have that

$$\Sigma^R_{m,c}(\omega) = \sum_{i,I}\frac{\left|M^I_{m,i}\right|^2}{\omega - \varepsilon^0_i + \Omega_I + i\eta} + \sum_{a,I}\frac{\left|M^I_{m,a}\right|^2}{\omega - \varepsilon^0_a - \Omega_I + i\eta} \tag{6.1369}$$

For iterations $n \geq 1$: Assume that we have $A^{(n-1)}_k(\omega) \propto \operatorname{Im} G^{(n-1)}_k(\omega)$. This gives us a recipe to update the chemical potential $\mu^{(n+1)}$, where $\mu^{(1)} = \frac{\left(k^2/2 + \Gamma^1_k\right)_{\text{HOMO}} + \left(k^2/2 + \Gamma^1_k\right)_{\text{LUMO}}}{2}$ as

$$N/2 = \sum_k \int_{-\infty}^{\mu^{(n+1)}} A^{(n)}_k(\omega)d\omega, \tag{6.1370}$$

where we can take $N/2$ as the number of occupied bands in our spin-restricted UEG. Once we have the chemical potential, we determine the lesser and greater Green's functions as

$$G^{<,(n)}_k(t) = i\int_{-\infty}^{\mu^{(n+1)}} d\omega A^{(n)}_k(\omega)e^{-i\omega t} \tag{6.1371}$$

$$G^{>,(n)}_k(t) = -i\int_{\mu^{(n+1)}}^{\infty} d\omega A^{(n)}_k(\omega)e^{-i\omega t}. \tag{6.1372}$$

This gives a recipe for updating the correlation self-energy according to 6.1359 or 6.1365, for BWs2 and GW, respectively. Then, we would do a DFT to get the real and imaginary parts of $\Sigma^{n+1,R}_{p,c}(\omega)$, which can then be used to update $G^{n+1,R}_k(t)$. This closes the self-consistency loop.

6.15.9 Alternative expansion points

In the diagonal approximation, the cumulant ansatz is

$$\boldsymbol{G}_{pp}(t) = \boldsymbol{G}^0_{pp}(t)e^{\boldsymbol{C}_{pp}(t)}. \tag{6.1373}$$

In the retarded convention, $G^0_{pp}(t) = -i\Theta(t)e^{-i(p^2/2 + \Sigma^x_{pp})t}$, whereas in the time-ordered convention, $G^0_{pp}(t) = \mp i\theta(\pm t)\theta\left(\pm\left(\epsilon_p - \epsilon_{k_F}\right)\right)e^{-i(p^2/2 + \Sigma^x_{pp})t}$, where the upper (lower) sign refers to particles with $k > k_F$ (holes $k < k_F$), and θ is the unit step function. Equating the

Dyson expansion to the cumulant ansatz to first order in W yields

$$\boldsymbol{G}^0_{pp}(t)\boldsymbol{C}_{pp}(t) = \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t}\boldsymbol{G}^0_{pp}(\omega)\boldsymbol{\Sigma}^c_{pp}(\omega)\boldsymbol{G}^0_{pp}(\omega) \tag{6.1374}$$

$$\implies C_{pp}(t) = i\int \frac{d\omega}{2\pi}\frac{\Sigma^c_{pp}\left(\omega+\epsilon_p\right)}{(\omega+i\eta)^2}e^{-i\omega t} \tag{6.1375}$$

$$= \int d\omega \frac{\beta_{pp}(\omega)}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right), \tag{6.1376}$$

where $\beta_{pp}(\omega) = \frac{1}{\pi}|\mathrm{Im}\Sigma^c_{pp}(\omega+\epsilon_p)|$.

Bare non-interacting reference

Here, we take $\epsilon_p = p^2/2$. Correspondingly, $\epsilon_{k_F} = \frac{1}{2}\left(k^2_{\mathrm{nocc}} + k^2_{\mathrm{nocc}-1}\right)$, where k_{nocc} and $k_{\mathrm{nocc}-1}$ are the wave vectors of the LUMO and HOMO, respectively.

LDA non-interacting reference

Here, we take $\epsilon_p = \epsilon^{\mathrm{LDA}}_p$. Correspondigly, $\epsilon_{k_F} = \frac{1}{2}\left(\epsilon^{\mathrm{LDA}}_{\mathrm{nocc}} + \epsilon^{\mathrm{LDA}}_{\mathrm{nocc}-1}\right)$.

Quasiparticle non-interacting reference

Here, we take $\epsilon_p = E^{GW}_p$. Correspondingly, $\epsilon_{k_F} = \frac{1}{2}\left(E^{GW}_{\mathrm{nocc}} + E^{GW}_{\mathrm{nocc}-1}\right)$.

6.15.10 SCCE(GW0) on the LDA reference

Iteration on the retarded cumulant

Here,

$$C^R_k(t) = \int d\omega \frac{\beta_k(\omega)}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right) \tag{6.1377}$$

where $\beta_k(\omega) = \frac{1}{\pi}|\mathrm{Im}\Sigma^R_k[\eta_\Sigma](\omega+\varepsilon^{\mathrm{LDA}}_k)|$. Then,

$$G^R_k(t) = G^{R,0}_k(t)e^{\tilde{C}^R_k(t)}e^{-\eta_{\mathrm{X}}|t|} \tag{6.1378}$$

Iteration on the time-ordered cumulant

With the lower (upper) sign referring to particles (holes), here we take

$$\tilde{C}^T_k(t) = \int d\omega \frac{\beta^{\pm}_k(\omega)}{\omega^2}\left(e^{-i\omega t}+i\omega t-1\right). \tag{6.1379}$$

where $\beta^{\pm}_k(\omega) = \frac{1}{\pi}\left|\mathrm{Im}\,\Sigma^R_k[\eta_\Sigma]\left(\omega+\varepsilon^{\mathrm{LDA}}_k\right)\right|\theta\left(\pm\left(\varepsilon^{\mathrm{LDA}}_{k_F}-\varepsilon^{\mathrm{LDA}}_k-\omega\right)\right)$. Then,

$$G^T_k(t) = G^{T,0}_k(t)e^{\tilde{C}^T_k(t)}e^{-\eta_{\mathrm{X}}|t|} \tag{6.1380}$$

The self energy

At a given iteration n, including the one-shot level (I checked analytically), the self-energy is always constructed as

$$\Sigma^{n,R}_{k,c}[\eta_\Sigma](t) = \theta(t)\left[-\sum_{l<k_F,I}\left|M^I_{k,l}\right|^2 G^{<,(n)}_l(t)e^{+i\Omega_I t}e^{-\eta_\Sigma t} + \sum_{l>k_F,I}\left|M^I_{k,l}\right|^2 G^{>,(n)}_l(t)e^{-i\Omega_I t}e^{-\eta_\Sigma t}\right], \tag{6.1381}$$

with the quasiboson quantities fixed at the RPA level.

Broadening of the self-energy

Following the convention that allowed us to converge to the TDL, we use $\eta_\Sigma = 0.0272$ eV during self-consistent iteration. When plotting the self-energy at a given iteration, I instead use $\eta_\Sigma = 0.4$ eV to facilitate visualization.

Broadening of the Green's function

$\eta_X \equiv \eta_{\text{loop}} = 0$ eV, except in the final iteration where $\eta_X = 0.8$ eV, which is the final broadening applied to facilitate visualisation of the spectral function. We cannot set $\eta_X = 0.8$ eV during the self-consistent loop because this would smear out the self-energy; it is essential that the Landau cumulant can detect the fine structure of the self-energy in order to get a spectral function with features.

6.15.11 Three broadenings

Symbol	Enters	Value	Effect	Notes
η_Σ	$W_0(t)$ damping	0.0272 eV	smooths the boson poles	For small systems, I checked that going smaller doesn't change results and only makes the computation more expensive, but when we do the TDL ladder, we should experiment with different values for η_Σ.
η_{loop}	the fed-back loop G	0 eV	kept negligible, so that we don't smear out the Σ for the next iteration	
η_{G}	the final $t\to\omega$ transform	0.8 eV	pure Lorentzian convolution of $A(\omega)$; preserves m_0 and the centroid	

6.15.12 SCCE(GW0) in the TDL

Then the idea for the self-consistent scheme would be to build a new retarded self-energy from the time-ordered cumulant Green's function. I think we would need to do this for all $k = |\boldsymbol{k}|$ shells in the plane wave basis, but we wouldn't need to enumerate over all basis states? More precisely, we would only need to compute the Green's function for each state in a given shell once, but when we are summing over all possible Green's functions, we would need to sum over all states in the shell.

Constructing the Retarded Self Energy

We might want to use the form

$$\Sigma^R_{k,c}(t) = \theta(t) \sum_{mI} \left| M^I_{k,m} \right|^2 \left[G^>_{m,CE}(t) e^{-i\Omega_I t} - G^<_{m,CE}(t) e^{+i\Omega_I t} \right], \tag{6.1382}$$

where $M^I_{k,m}$ is the screened Coulomb matrix element between states k and m for the quasiboson mode I and Ω_I is the quasiboson frequency, as obtained from the preceding Casida diagonalization. $G^>_{m,CE}(t)$ and $G^<_{m,CE}(t)$ are the greater and lesser cumulant Green's functions, respectively. For a retarded self-energy, all that we need are

$$G^<_{i,\mathrm{CE}}(t) = -\left[G^T_{i,\mathrm{CE}}(-t) \right]^*, \quad t > 0,$$

and

$$G^>_{a,\mathrm{CE}}(t) = G^T_{a,\mathrm{CE}}(t), \quad t > 0.$$

, for occupied states i, and unoccupied states a, resulting in

$$\Sigma^R_{k,c}(t) = \theta(t) \left[-\sum_{m<k_F,I} \left| M^I_{k,m} \right|^2 G^<_{m,CE}(t) e^{+i\Omega_I t} + \sum_{m>k_F,I} \left| M^I_{k,m} \right|^2 G^>_{m,CE}(t) e^{-i\Omega_I t} \right] \tag{6.1383}$$

6.15.13 Getting the cumulant Green's function

We always have

$$G_k(t) = G^0_k(t) e^{\tilde{C}_k(t)} \tag{6.1384}$$

Retarded formalism

Here, the dynamical cumulant is always given as

$$\tilde{C}^R_k(t) = \frac{1}{\pi} \int d\omega \frac{|\operatorname{Im} \Sigma^R_k(\omega + \tilde{\epsilon_k})|}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1 \right), \tag{6.1385}$$

where for $G_{k,0}$@LDA, $\tilde{\epsilon_k} = \epsilon^{\mathrm{LDA}}_k$. Note that we should get equivalent results if we instead use $\tilde{\epsilon_k} = \epsilon^{\mathrm{LDA}}_k - v^{xc}_{kk} = k^2/2$. All that LDA really does is to introduce a non-dispersive exchange-correlation potential to each single-particle state. This does not change the RPA screening and we get a further cancellation of the exchange-correlation potential in the self-energy evaluations. The shifted argujment of the cumulant kernel introduces a positive

exchange-correlation contribution, while in the self-energy denominator, we have a negative exchange-correlation, and these cancel out. The non-interacting reference always has the form $G^R_{k,0}(t) = -i\Theta(t)e^{-i\epsilon_k t}$.

1. We can take $\epsilon_k = \epsilon_k^{\mathrm{LDA}}$.
2. We can take $\epsilon_k = \epsilon_k^{\mathrm{LDA}} - v^{xc}_{kk} + \Sigma^x_{kk} = k^2/2 + \Sigma^x_{kk}$.

Time-ordered formalism

Here, the dynamical cumulant is always given as

$$\tilde{C}^T_k(t) = \int d\omega \frac{\beta^{\pm}_k(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right), \tag{6.1386}$$

where $\beta^{\pm}_k(\omega) = \frac{1}{\pi}\left|\mathrm{Im}\,\Sigma^R_k(\omega+\epsilon_k)\right|\theta\left(\pm(\epsilon_{k_F} - \epsilon_k - \omega)\right)$. The non-interacting part is $G^{T,0}_k(t) = \mp i\theta(\pm t)\theta\left(\pm(\epsilon_k - \epsilon_{kF})\right)e^{-i\epsilon^{\mathrm{eff}}_k t}$, where the upper (lower) sign refers to particles with $k > k_F$ (holes $k < k_F$), and θ is the unit step function. Our reference is an LDA calculation and we are at zero temperature, so we can take $\epsilon_k = \epsilon_k^{\mathrm{LDA}}$, $\epsilon_k^{\mathrm{eff}} = \epsilon_k^{\mathrm{LDA}} + \Sigma^x_k - v^{xc}_k$, and $\epsilon_{k_F} = \epsilon^{\mathrm{LDA}}_{k_F} = \frac{\epsilon^{\mathrm{LDA}}_{k\equiv\mathrm{HOMO}} + \epsilon^{\mathrm{LDA}}_{\mathrm{LUMO}}}{2}$. Is this an appropriate way to define the Fermi energy, or should we instead take the midpoint between the quasiparticle energies of the HOMO and LUMO? Again, we choose to either take

1. We can take $\epsilon_k^{\mathrm{eff}} = \epsilon_k^{\mathrm{LDA}}$.
2. We can take $\epsilon_k^{\mathrm{eff}} = \epsilon_k^{\mathrm{LDA}} - v^{xc}_{kk} + \Sigma^x_{kk} = k^2/2 + \Sigma^x_{kk}$.

In total, this belies six different ways to do SCCE(GW0). We can do: - retarded first variant - retarded second variant - time-ordered first variant with fermi energy in terms of the LDA energies - time-ordered variant one with fermi energy defined in terms of quasiparticle energies - time-ordered variant two with both options Do you think this is complete? Which way do we currently do? Also, Holms synthesize s green functions at the chemical potential before embarking on self-consistency. Do we need to do this as well? Because right now my Greens functions, I think, are centered on the absolute frequency axis. I manually extracted both Garnet's and Holms' reported G0W0+C spectral functions in the TDL and ran my code towards the TDL.

My red curves

My red curves have the plasmon-pole structure; however, they differ from Garnet's. One reason is due to the choice of manual broadening; he used 0.8 eV, whereas I used 0.2 eV. Another difference originates from computation of the cumulant spectral function; recall that when I used the exact frequency integration scheme for GW in the finite basis, I was unable to reproduce his results. We attributed this to two reasons:

1. He was using a PPM to construct the GW self-energy when he wasn't supposed to; as I showed before, in the finite basis, the GW self-energy cannot be adequately represented by a single pole model.

2. He was constructing the cumulant kernel as $\beta_k(\omega) = 1/\pi |\operatorname{Im}\Sigma^R(\omega + E_k^{QP})|$, whereas we (and Loos) think it should be $\beta_k(\omega) = 1/\pi |\operatorname{Im}\Sigma^R(\omega + \epsilon_k^{\mathrm{LDA}})|$. This is because the latter choice better reproduces the location of the first satellite peak, as predicted by the wavefunction theories of CCSD and BWs2+C.

My green curves

In his paper, Holm states that all spectral functions are centered around the chemical potential. My green curves are an attempt to reproduce this idea. Specifically, we know that at zero temperature, $\mu = \epsilon_{k_F} \equiv \frac{\epsilon_{\mathrm{HOMO}}^{\mathrm{LDA}} + \epsilon_{\mathrm{LUMO}}^{\mathrm{LDA}}}{2}$. Clearly, I am not yet at the TDL. However, beyond this basis size, it becomes difficult to store the OO+OV Choleskies in memory.

Next steps

I could do now G0W0+C with PPMs at the TDL, which may allow me to reproduce Holm's SCCE(GW0) results, but I don't know if this would be transferable to a finite basis.

6.15.14 Reproducing the TDL G0W0+C using PPMs

We know that the plasmon-pole models are an exact way of doing the GW frequency integration at the TDL. So we can set an arbitrarily large number of electrons and basis functions (What should I choose specifically? It should be cheap even if we use a large number of electrons because the scaling is favorable.) If the total PPM correlation self-energy is

$$\Sigma_k^{c,R}(\omega) = \sum_i^{\mathrm{occ}} A_{ki}^{\mathrm{PPM}} \frac{1}{\omega - \varepsilon_i + \widetilde{\omega}_{\boldsymbol{q}} + i\eta} + \sum_a^{\mathrm{unocc}} A_{ka}^{\mathrm{PPM}} \frac{1}{\omega - \varepsilon_a - \widetilde{\omega}_{\boldsymbol{q}} + i\eta}. \tag{6.1387}$$

then if we define

$$\Delta_{ki}^{\mathrm{PPM}} = \varepsilon_i - \varepsilon_k - \widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_i} - i\eta. \tag{6.1388}$$

and

$$\Delta_{ka}^{\mathrm{PPM}} = \varepsilon_a + \widetilde{\omega}_{\boldsymbol{G}_k - \boldsymbol{G}_a} - \varepsilon_k - i\eta. \tag{6.1389}$$

Then

$$C_k^{\mathrm{dyn}}(t) = \sum_i^{\mathrm{occ}} \zeta_{ki}^{\mathrm{PPM}} \left[e^{-i\Delta_{ki}^{\mathrm{PPM}} t} - 1 + i\Delta_{ki}^{\mathrm{PPM}} t \right] \tag{6.1390}$$

$$+ \sum_a^{\mathrm{unocc}} \zeta_{ka}^{\mathrm{PPM}} \left[e^{-i\Delta_{ka}^{\mathrm{PPM}} t} - 1 + i\Delta_{ka}^{\mathrm{PPM}} t \right], \tag{6.1391}$$

where

$$\zeta_{ki}^{\mathrm{PPM}} = \frac{A_{ki}^{\mathrm{PPM}}}{(\Delta_{ki}^{\mathrm{PPM}})^2}, \qquad \zeta_{ka}^{\mathrm{PPM}} = \frac{A_{ka}^{\mathrm{PPM}}}{(\Delta_{ka}^{\mathrm{PPM}})^2}. \tag{6.1392}$$

with the bosonic frequency related to that of the Sommerfeld bulk plasma as

$$\Omega^2(\mathbf{q}) = \omega_p^2, \quad \omega_p = \sqrt{4\pi n} \tag{6.1393}$$

Then,

$$\tilde{\omega}^2(\mathbf{q}) = \frac{\Omega^2(\mathbf{q})}{1 - \varepsilon^{-1}(\mathbf{q}, \omega = 0)} \tag{6.1394}$$

$$\tag{6.1395}$$

where the RPA form for the dielectric function used is

$$\varepsilon(\mathbf{q}, \omega = 0) = 1 - v(\mathbf{q}) P(\mathbf{q}, \omega = 0) \tag{6.1396}$$

$$\approx 1 - v(\mathbf{q}) P_0(\mathbf{q}, \omega = 0). \tag{6.1397}$$

The static, non-interacting polarizability is given by

$$P_0(\mathbf{q}, \omega = 0) \overset{\text{UEG}}{=} \frac{1}{V} \sum_{\mathrm{k}} \frac{f_{\mathrm{k}+\mathbf{q}} - f_{\mathrm{k}}}{\epsilon_{\mathrm{k}+\mathbf{q}} - \epsilon_{\mathrm{k}}}, \tag{6.1398}$$

where we have a sum over the momentum transfers $\mathbf{k}$, and f is the Fermi-Dirac distribution, which is 1 for occupied states and 0 for unoccupied states at zero temperature. We can take the $\epsilon_{\mathrm{k}} = k^2/2$ for the single-particle energies. And finally,

$$A(\mathbf{q}) = -\frac{\pi}{2} \frac{\Omega^2(\mathbf{q})}{\tilde{\omega}(\mathbf{q})}. \tag{6.1399}$$

Then we can determine the static part of the cumulant, which is the bare exchange self-energy component as

$$\Sigma^x_{\boldsymbol{k}} = -\sum_{\mathbf{q}} v(\mathbf{k} + \mathbf{q}) f_{\mathbf{q}} \tag{6.1400}$$

In summary, we can obtain the cumulant Green's function as

$$G(\boldsymbol{k})^R = G_0(\boldsymbol{k})^R \exp\bigl(C_{\boldsymbol{k}(t)}\bigr) \tag{6.1401}$$

with

$$C_{\boldsymbol{k}}(t) = -i\Sigma^x_{\boldsymbol{k}} t + C^{\mathrm{dyn}}_{\boldsymbol{k}}(t). \tag{6.1402}$$

Check on the dielectric function

For a given momentum transfer vector $\mathbf{q}$, I was thinking we could solve

$$\begin{pmatrix} \boldsymbol{A}(\mathbf{q}) & \boldsymbol{B}(\mathbf{q}) \\ -\boldsymbol{B}(\mathbf{q}) & -\boldsymbol{A}(\mathbf{q}) \end{pmatrix} \begin{pmatrix} \boldsymbol{X}(\mathbf{q}) \\ \boldsymbol{Y}(\mathbf{q}) \end{pmatrix} = \boldsymbol{\Omega}(\mathbf{q}) \begin{pmatrix} \boldsymbol{X}(\mathbf{q}) \\ \boldsymbol{Y}(\mathbf{q}) \end{pmatrix} \tag{6.1403}$$

where The matrix elements are defined as $A_{ia,jb}(\mathbf{q}) = \delta_{ij}\delta_{ab}(\varepsilon_a - \varepsilon_i) + 2(ia|bj)$ and $B_{ia,jb}(\mathbf{q}) = 2(ia|jb)$ With the requirement that there be a momentum transfer of $\mathbf{q}$ to get from the occupied plane with State i to the unoccupied plane with state a. Correct me if I am wrong, but this should be cheap to do for a few of the low-energy momentum transfers because we know we are never diagonalizing a full matrix with dimension occupied and virtual. Rather, there were just a bunch of different momentum transfer sectors, and the amount of these sectors rose with this system size. I was thinking we just diagonalize for one of these low-energy momentum transfers. And then from the exact three w dielectric function as ¡insert¿ But then we would compare to the known analytic form for the dielectric function.

Constructing the Self-Energy in Frequency Space

Here the relation is that

$$\Sigma^R(\omega) = i \int \frac{d\omega'}{2\pi} \left[G^R(\omega - \omega') W^<(\omega') + G^<(\omega - \omega') W^R(\omega') \right] \tag{6.1404}$$

Or is it better to directly carry out in frequency space? Then we would Fourier transform this to the frequency domain and plug it into the Landau cumulant to get a new $\tilde{C}_k^T(t)$, Whose form clearly depends on whether we are considering a whole or particle state. We can use this to get a new $G_k^T(t)$, which we would then Fourier transform to get a new $G_k^T(\omega)$, which we would then use to get a new $\Sigma_k^T(t)$, and iterate until convergence. Note that I already have retarded the scheme using the retarded cumulant formulation, but that didn't give me satisfactory results in comparison with Holms values, so I want to try this time-ordered formulation. The research question that I am asking then is: Is distinguishing between particle and hole states important for this self-consistent cumulant scheme?

6.16 SCCE(GW0) in the finite basis

We are interested in obtaining the a self-consistent cumulant spectral function for $k = [0, 0, 0]$. In the first iteration (one-shot), we apply $\Sigma_m^{G_0W_0}$ to get the Landau cumulant, and then a cumulant Green function $G_{m,CE}(t)$ for each basis state m. For the following iterations, we form the retarded correlation self-energy as

$$\Sigma_m^{c,R}(t) = \theta(t) \sum_{nI} \left| M_{m,n}^I \right|^2 \left[G_{n,CE}^>(t) e^{-i\Omega_I t} - G_{n,CE}^<(t) e^{+i\Omega_I t} \right], \tag{6.1405}$$

where $M_{m,n}^I$ and Ω_I are the screened Coulomb matrix elements and frequencies of quasiboson (QB) mode I, respectively. We then conduct a DFT to get $\Sigma_m^{c,R}(\omega)$, which we evaluate at $\omega + \epsilon_m^{\text{LDA}}$ to form a new Landau cumulant. This closes the self-consistency loop, that we then iterate until convergence. This procedure is followed for both the independent boson model (IBM) and UEG, where the QB reference is always due to dRPA@LDA.

6.16.1 What Holm did

After the initial iteration, Holm updates the self-energy for target state k as

$$\Sigma^c(\mathbf{k}, \omega) = \int_C \frac{\Gamma(\mathbf{k}, \omega')}{\omega - \omega'} d\omega' \tag{6.1406}$$

where

$$\Gamma(\mathbf{k}, \omega) = \sum_{\mathbf{q}} \int_0^\omega d\omega' A(\mathbf{k} + \mathbf{q}, \omega - \omega') B_0(\mathbf{q}, \omega') \tag{6.1407}$$

with $B_0(\mathbf{q}, \omega)$ the spectral representation of the screened Coulomb interaction. They fit the spectral function to a sum of Gaussians as

$$A(k, \omega) = \sum_\nu \frac{W_\nu(k)}{\sqrt{2\pi\Gamma_\nu(k)}} \exp\left[-\frac{[\omega - E_\nu(k)]^2}{2\Gamma_\nu^2(k)} \right], \tag{6.1408}$$

where $E_\nu(k)$, $\Gamma_\nu(k)$, and $W_\nu(k)$ are the energy, width, and weight of the ν-th peak in the spectral function for state k. *Would I want to do the same thing in my finite basis?* Then they update the cumulant to first order in the interaction as

$$C(k,t) = -i \int_0^t dt' \int_{t'}^\infty d\tau \Sigma_{GW}(k,\tau) e^{i\varepsilon_k \tau} \tag{6.1409}$$

Why this doesn't work in a finite basis

I only present results that follow the retarded cumulant formalism, but I checked and the time-ordered results aren't any better.

6.16.2 Interpretation of the results for the IBM

Here, the one-shot cumulant is known to be exact. For both the minimal and intermediate system sizes of 14e19o and 38e123o, we see that SCCE(GW0) degrades particularly the satellite features in the spectrum.

6.16.3 Interpretation of the results for the UEG

Here I present results for 305 and 1021 orbitals. The one-shot cumulant yields a quasiparticle peak, with satellites occurring at regularly spaced intervals as we move to higher binding energies. In contrast, SCCE(GW0) yields a washed out spectrum, which loses the QP peak location and second satellite peak of the one-shot result.

6.16.4 Interpretation of PJ's results for the finite-size Holstein model

In the strong coupling regime (which is representative of the electron gas at $r_s = 4$), SCCE(GW0) fails to produce any satellite features with just a broad quasiparticle peak. This is similar to what I observe for the UEG. To interpret spectra in a finite basis, we need to apply a finite broadening.

$\nu \equiv$ QP

For the main quasiparticle peak ($\nu \equiv$ QP), we could determine these quantities as

$$E_{\mathrm{QP}}(k) = \epsilon_k + \mathrm{Re}\,\Sigma\left(k, E_{\mathrm{QP}}(k)\right) \equiv E_k \tag{6.1410}$$

and

$$W_{\mathrm{QP}}(k) = Z_k = \left[1 - \left.\frac{\partial\, \mathrm{Re}\,\Sigma(k,\omega)}{\partial\omega}\right|_{\omega=E_{\mathrm{QP}}}\right]^{-1} \tag{6.1411}$$

and

$$\Gamma_{\mathrm{QP}}(k) = \left|\mathrm{Im}\,\Sigma\left(k, E_{\mathrm{QP}}(k)\right)\right|. \tag{6.1412}$$

$\nu \equiv \text{sat}$

To determine the satellite positions, Holm $E_{h,1} = E_k - \omega_p$, $E_{h,2} = E_k - 2\omega_p$, $E_{p,1} = E_k + \omega_p$, and $E_{p,2} = E_k + 2\omega_p$, where $\omega_p = \sqrt{4\pi n}$ is taken as the bare Sommerfeld plasma frequency. We have seen before that this is no longer applicable in the finite basis as the plasma frequencies are dispersive; it was for this reason that we considered instead

$$\omega_p(\mathbf{q}) = \sqrt{\frac{4v_{\mathbf{q}}}{V} \sum_{(i,a)\in\mathcal{T}_{\mathbf{q}}} (\varepsilon_a - \varepsilon_i)}, \tag{6.1413}$$

where $\mathcal{T}_{\mathbf{q}}$ is the set of all particle-hole pairs (i, a) connected by the momentum transfer $\mathbf{q}$.

One-shot retarded formulation

Chapter 7

Self-consistent MP2

7.1 Self-consistent schemes for QPMP2

I can envision the following self-consistency schemes for this QPMP2 using

$$\Sigma^{(2)}_{pp}(\omega) = \sum_{iab} \frac{(pa|ib)\Big(2(pa|ib) - (pb|ia)\Big)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{(pi|aj)\Big(2(pi|aj) - (pj|ai)\Big)}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{7.1}$$

7.1.1 No self consistency (1shotQPMP2)

This just comprises of solving the QP equation using the self energy defined in 7.1 with the HF orbitals and MO energies in a one-shot fashion. So for each orbital p we solve

$$\epsilon_p^{\mathrm{QP}} = \epsilon_p^{\mathrm{HF}} + Z_p \Sigma^{(2)}_{pp}(\epsilon_p^{\mathrm{HF}}) \tag{7.2}$$

where $Z_p = \left(1 - \frac{\partial \Sigma^{(2)}_{pp}(\omega)}{\partial \omega}\Big|_{\omega=\epsilon_p^{\mathrm{HF}}}\right)^{-1}$ is the renormalization factor.

7.1.2 Eigenvalue self-consistency (evQPMP2)

1. Start with HF reference orbitals and energies.
2. Solve the QP equation 7.2 for each orbital using the MP2 self-energy, as given in equation 7.1.
3. Update all the orbital energies in 7.1, keeping the numerator fixed.
4. Repeat steps 2-3 until convergence in the orbital energies of interest.

7.2 Exploring the similarity renormalization group approach to quasi-particle self-consistency

The unrestricted form for the MP2 self-energy is

$$\Sigma^{(2)}_{pq}(\omega) = \frac{1}{2}\sum_{iab}\frac{\langle pi||ab\rangle\langle ab||qi\rangle}{\omega+\epsilon_i-\epsilon_a-\epsilon_b} + \frac{1}{2}\sum_{ija}\frac{\langle pa||ij\rangle\langle ij||qa\rangle}{\omega+\epsilon_a-\epsilon_i-\epsilon_j} \tag{7.3}$$

$$\tag{7.4}$$

To derive the restricted form, consider just the 2p1h sector and expand the antisymmetrized integrals. We can associate spin variables $\sigma_p, \sigma_q, \sigma_i, \sigma_a, \sigma_b$ to each of the orbitals p, q, i, a, b. Then, with $\Delta_{iab} \equiv \epsilon_a + \epsilon_b - \epsilon_i$, we have

$$\Sigma^{(2p1h)}_{pq}(\omega) = \frac{1}{2}\sum_{iab}\frac{\left(\langle pi|ab\rangle - \langle pi|ba\rangle\right)\left(\langle ab|qi\rangle - \langle ab|iq\rangle\right)}{\omega-\Delta_{iab}} \tag{7.5}$$

$$\implies \Sigma^{(2p1h)}_{p\sigma_p,q\sigma_q}(\omega) = \frac{1}{2}\sum_{\sigma_i,\sigma_a,\sigma_b}\sum_{iab}\frac{\left((p\sigma_p a\sigma_a|i\sigma_i b\sigma_b) - (p\sigma_p b\sigma_b|i\sigma_i a\sigma_a)\right)\left((a\sigma_a b\sigma_b|q\sigma_q i\sigma_i) - (a\sigma_a b\sigma_b|i\sigma_i q\sigma_q)\right)}{\omega-\Delta_{iab}} \tag{7.6}$$

$$\tag{7.7}$$

.

7.2.1 Downfolding the QPMP2 Hamiltonian

If we use that Lowdin partitioning we are able to recast the QPMP2 equation into a frequency independent Hamiltonian of the form

$$H^{\mathrm{QPMP2}} = \begin{pmatrix} \boldsymbol{F} & \boldsymbol{W}^{2h1p} & \boldsymbol{W}^{2p1h} \\ (\boldsymbol{W}^{2h1p})^\dagger & \boldsymbol{C}^{2h1p} & 0 \\ (\boldsymbol{W}^{2p1h})^\dagger & 0 & \boldsymbol{C}^{2p1h} \end{pmatrix} \tag{7.8}$$

where now the 2h1p and 2p1h blocks are defined as

$$C^{2h1p}_{ija,klb} = (\epsilon_k - \epsilon_l + \epsilon_b)\delta_{ik}\delta_{jl}\delta_{ab} \tag{7.9}$$

$$C^{2p1h}_{abi,cdj} = (\epsilon_c + \epsilon_d - \epsilon_j)\delta_{ac}\delta_{bd}\delta_{ij} \tag{7.10}$$

and the coupling blocks are

$$W^{2h1p}_{p,ija} = \frac{\sqrt{2}}{2}\langle pa||ij\rangle \tag{7.11}$$

$$W^{2p1h}_{p,abi} = \frac{\sqrt{2}}{2}\langle pi||ab\rangle \tag{7.12}$$

Now, we can multiply this matrix by a trial vector $\Psi = (\boldsymbol{X}^{1h1p}, \boldsymbol{X}^{2h1p}, \boldsymbol{X}^{2p1h})^T$ to obtain the following set of coupled equations

$$\boldsymbol{F}\boldsymbol{X}^{1h1p} + \boldsymbol{W}^{2h1p}\boldsymbol{X}^{2h1p} + \boldsymbol{W}^{2p1h}\boldsymbol{X}^{2p1h} = E\boldsymbol{X}^{1h1p} \tag{7.13}$$

$$(\boldsymbol{W}^{2h1p})^\dagger\boldsymbol{X}^{1h1p} + \boldsymbol{C}^{2h1p}\boldsymbol{X}^{2h1p} = E\boldsymbol{X}^{2h1p} \tag{7.14}$$

$$(\boldsymbol{W}^{2p1h})^\dagger\boldsymbol{X}^{1h1p} + \boldsymbol{C}^{2p1h}\boldsymbol{X}^{2p1h} = E\boldsymbol{X}^{2p1h} \tag{7.15}$$

From the last two equations we can solve for $\boldsymbol{X}^{2h1p}$ and $\boldsymbol{X}^{2p1h}$ as

$$\boldsymbol{X}^{2h1p} = (E\mathbf{1} - \boldsymbol{C}^{2h1p})^{-1}(\boldsymbol{W}^{2h1p})^\dagger\boldsymbol{X}^{1h1p} \tag{7.16}$$

$$\boldsymbol{X}^{2p1h} = (E\mathbf{1} - \boldsymbol{C}^{2p1h})^{-1}(\boldsymbol{W}^{2p1h})^\dagger\boldsymbol{X}^{1h1p} \tag{7.17}$$

Substituting these expressions back into the first equation we obtain an effective eigenvalue equation for the 1h1p space

$$\left[\boldsymbol{F} + \boldsymbol{W}^{2h1p}(E\mathbf{1} - \boldsymbol{C}^{2h1p})^{-1}(\boldsymbol{W}^{2h1p})^\dagger + \boldsymbol{W}^{2p1h}(E\mathbf{1} - \boldsymbol{C}^{2p1h})^{-1}(\boldsymbol{W}^{2p1h})^\dagger\right]\boldsymbol{X}^{1h1p} = E\boldsymbol{X}^{1h1p} \tag{7.18}$$

where we recognize the terms in square brackets as a sum of the Fock matrix and the MP2 self-energy evaluated at energy E.

7.2.2 SRG approach

Now, we define, within the SRG formalism, the diagonal and off-diagonal parts of the $\boldsymbol{H}^{\mathrm{QPMP2}}$ as

$$\boldsymbol{H}^{\mathrm{d}}(s) = \begin{pmatrix} \boldsymbol{F} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{C}^{\mathrm{2h1p}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{C}^{\mathrm{2p1h}} \end{pmatrix} \equiv \begin{pmatrix} \boldsymbol{F} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{C} \end{pmatrix}, \tag{7.19}$$

$$\boldsymbol{H}^{\mathrm{od}}(s) = \begin{pmatrix} \mathbf{0} & \boldsymbol{W}^{\mathrm{2h1p}} & \boldsymbol{W}^{\mathrm{2p1h}} \\ (\boldsymbol{W}^{\mathrm{2h1p}})^\dagger & \mathbf{0} & \mathbf{0} \\ (\boldsymbol{W}^{\mathrm{2p1h}})^\dagger & \mathbf{0} & \mathbf{0} \end{pmatrix} \equiv \begin{pmatrix} \mathbf{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \mathbf{0} \end{pmatrix} \tag{7.20}$$

where we omit the s dependence of the matrices for the sake of brevity. Then, our aim is to solve, order by order, the flow equation

$$\frac{\mathrm{d}\boldsymbol{H}(s)}{\mathrm{d}s} = [\boldsymbol{\eta}(s), \boldsymbol{H}(s)] \tag{7.21}$$

knowing that the initial conditions are

$$\boldsymbol{H}_d^{(0)}(0) = \begin{pmatrix} \boldsymbol{F} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{C} \end{pmatrix}, \boldsymbol{H}_{od}^{(0)}(0) = \mathbf{0}, \tag{7.22}$$

$$\boldsymbol{H}_d^{(1)}(0) = \mathbf{0}, \boldsymbol{H}_{od}^{(1)}(0) = \begin{pmatrix} \mathbf{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \mathbf{0} \end{pmatrix} \tag{7.23}$$

Once the closed-form expressions of the low-order perturbative expansions are known, they can be inserted into the downfolded self-energy to define a renormalized version of the quasiparticle equation.

7.2.3 Order by order expansion

Now, Wegner's generator is

$$\boldsymbol{\eta}(s) = [\boldsymbol{H}^{d}(s), \boldsymbol{H}^{od}(s)] \tag{7.24}$$

$$= [\boldsymbol{H}^{d(0)}(s) + \lambda\mathbf{0} + \lambda^2\boldsymbol{H}^{d(2)}(s) + \mathcal{O}(\lambda^3), \mathbf{0} + \lambda\boldsymbol{H}^{od(1)}(s) + \lambda^2\boldsymbol{H}^{od(2)}(s) + \mathcal{O}(\lambda^3)] \tag{7.25}$$

$$= \underbrace{\boldsymbol{H}_d^{(0)}(s)}_{\eta^{(0)}(s)} + \lambda \underbrace{[\boldsymbol{H}_d^{(0)}(s), \boldsymbol{H}_{od}^{(1)}(s)]}_{\eta^{(1)}(s)} + \lambda^2 \underbrace{\Big([\boldsymbol{H}_d^{(0)}(s), \boldsymbol{H}_{od}^{(2)}(s)]\Big)}_{\eta^{(2)}(s)} + \mathcal{O}(\lambda^3) \tag{7.26}$$

and the flow equation 7.21 can also be expanded order by order in λ as

$$\frac{\mathrm{d}\boldsymbol{H}(s)}{\mathrm{d}s} = [\boldsymbol{\eta}(s), \boldsymbol{H}(s)] \tag{7.27}$$

$$\text{LHS} = \frac{\mathrm{d}\Big(\boldsymbol{H}^{(0)}(s) + \lambda\boldsymbol{H}^{(1)}(s) + \lambda^2\boldsymbol{H}^{(2)}(s) + \mathcal{O}(\lambda^3)\Big)}{\mathrm{d}s} \tag{7.28}$$

$$\text{RHS} = \lambda[\boldsymbol{\eta}^{(1)}(s), \boldsymbol{H}^{(0)}(s)] + \lambda^2\Big([\boldsymbol{\eta}^{(1)}(s), \boldsymbol{H}^{(1)}(s)] + [\boldsymbol{\eta}^{(2)}(s), \boldsymbol{H}^{(0)}(s)]\Big) + \mathcal{O}(\lambda^3) \tag{7.29}$$

Equating to zeroth order

From the expansion of the flow equation, we get

$$\frac{\mathrm{d}\boldsymbol{H}^{(0)}(s)}{\mathrm{d}s} = \mathbf{0} \tag{7.30}$$

and then using our initial condition gives

$$\boldsymbol{H}^{(0)}(s) = \boldsymbol{H}^{(0)}(0) = \begin{pmatrix} \boldsymbol{F} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{C} \end{pmatrix} \tag{7.31}$$

Equating to first order

At first order we have

$$\frac{\mathrm{d}\boldsymbol{H}^{(1)}(s)}{\mathrm{d}s} = [\boldsymbol{\eta}^{(1)}(s), \boldsymbol{H}^{(0)}(s)] \tag{7.32}$$

$$\text{LHS} = \begin{pmatrix} \frac{\mathrm{d}\boldsymbol{F}^{(1)}(s)}{\mathrm{d}s} & \frac{\mathrm{d}\boldsymbol{W}^{(1)}(s)}{\mathrm{d}s} \\ \frac{\mathrm{d}(\boldsymbol{W}^{(1)}(s))^\dagger}{\mathrm{d}s} & \frac{\mathrm{d}\boldsymbol{C}^{(1)}(s)}{\mathrm{d}s} \end{pmatrix} \tag{7.33}$$

$$\text{RHS} = [[\boldsymbol{H}_d^{(0)}(s), \boldsymbol{H}_{od}^{(1)}(s)], \boldsymbol{H}_d^{(0)}(s)] \tag{7.34}$$

$$= [\boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s) - \boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s), \boldsymbol{H}_d^{(0)}(s)] \tag{7.35}$$

$$= \boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s) - \boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_d^{(0)}(s) - \boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s) + \boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s) \tag{7.36}$$

$$= \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} - \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \tag{7.37}$$

$$- \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} + \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \tag{7.38}$$

$$= \begin{pmatrix} \boldsymbol{0} & 2\boldsymbol{F}\boldsymbol{W}\boldsymbol{C} - \boldsymbol{F}^2\boldsymbol{W} - \boldsymbol{W}\boldsymbol{C}^2 \\ 2\boldsymbol{C}\boldsymbol{W}^\dagger\boldsymbol{F} - \boldsymbol{C}^2\boldsymbol{W}^\dagger - \boldsymbol{W}^\dagger\boldsymbol{F}^2 & \boldsymbol{0} \end{pmatrix} \tag{7.39}$$

which implies that the first-order flow equations are

$$\frac{\mathrm{d}\boldsymbol{F}^{(0)}(s)}{\mathrm{d}s} = \boldsymbol{0}, \frac{\mathrm{d}\boldsymbol{C}^{(0)}(s)}{\mathrm{d}s} = \boldsymbol{0} \tag{7.40}$$

$$\implies \boldsymbol{F}^{(0)}(s) = \boldsymbol{F}^{(0)}(0), \boldsymbol{C}^{(0)}(s) = \boldsymbol{C}^{(0)}(0) \tag{7.41}$$

$$\tag{7.42}$$

Now, we know that in the canonical HF basis both $\boldsymbol{F}$ and $\boldsymbol{C}$ are diagonal, so really we have found that

$$\frac{\mathrm{d}W_{pq,ia}^{(1)}(s)}{\mathrm{d}s} = 2F_{pp}W_{pq,ia}^{(1)}(s)C_{ia,ia} - F_{pp}^2W_{pq,ia}^{(1)}(s) - W_{pq,ia}^{(1)}(s)C_{ia,ia}^2 \tag{7.43}$$

$$= \left[2F_{pp}C_{ia,ia} - F_{pp}^2 - C_{ia,ia}^2\right] W_{pq,ia}^{(1)}(s) \tag{7.44}$$

$$= \left[2\epsilon_p(\epsilon_i - \epsilon_a) - \epsilon_p^2 - (\epsilon_i - \epsilon_a)^2\right] W_{pq,ia}^{(1)}(s) \tag{7.45}$$

$$= -(\epsilon_p - \epsilon_i + \epsilon_a)^2 W_{pq,ia}^{(1)}(s) \tag{7.46}$$

$$\implies W_{pq,ia}^{(1)}(s) = W_{pq,ia}^{(1)}(0)e^{-(\epsilon_p - \epsilon_i + \epsilon_a)^2 s} \tag{7.47}$$

so we get that $W_{pq,ia}^{(1)}(0) = W_{pq,ia}$ and $\lim_{s\to\infty} W_{pq,ia}^{(1)}(s) = 0$.

Equating to second order

At second order we have

$$\frac{\mathrm{d}\boldsymbol{H}^{(2)}(s)}{\mathrm{d}s} = [\boldsymbol{\eta}^{(1)}(s), \boldsymbol{H}^{(1)}(s)] + [\boldsymbol{\eta}^{(2)}(s), \boldsymbol{H}^{(0)}(s)] = [\boldsymbol{\eta}^{(1)}(s), \boldsymbol{H}^{(1)}(s)] \tag{7.48}$$

$$\text{LHS} = \begin{pmatrix} \frac{\mathrm{d}\boldsymbol{F}^{(2)}(s)}{\mathrm{d}s} & \frac{\mathrm{d}\boldsymbol{W}^{(2)}(s)}{\mathrm{d}s} \\ \frac{\mathrm{d}(\boldsymbol{W}^{(2)}(s))^\dagger}{\mathrm{d}s} & \frac{\mathrm{d}\boldsymbol{C}^{(2)}(s)}{\mathrm{d}s} \end{pmatrix} \tag{7.49}$$

$$\text{RHS} = [[\boldsymbol{H}_d^{(0)}(s), \boldsymbol{H}_{od}^{(1)}(s)], \boldsymbol{H}^{(1)}(s)] \tag{7.50}$$

$$= [\boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s) - \boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s), \boldsymbol{H}_{od}^{(1)}(s)] \tag{7.51}$$

$$= \boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_{od}^{(1)}(s) - \boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s) - \boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s)\boldsymbol{H}_{od}^{(1)}(s) + \boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_{od}^{(1)}(s)\boldsymbol{H}_d^{(0)}(s$$

(7.52)

$$= \begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} - \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} \tag{7.53}$$

$$- \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix} + \begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{W} \\ \boldsymbol{W}^\dagger & \boldsymbol{0} \end{pmatrix}\begin{pmatrix} \boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} \tag{7.54}$$

$$= \begin{pmatrix} \boldsymbol{F}\boldsymbol{W}\boldsymbol{W}^\dagger - 2\boldsymbol{W}\boldsymbol{C}\boldsymbol{W}^\dagger + \boldsymbol{W}\boldsymbol{W}^\dagger\boldsymbol{F} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{C}\boldsymbol{W}^\dagger\boldsymbol{W} - 2\boldsymbol{W}^\dagger\boldsymbol{F}\boldsymbol{W} + \boldsymbol{W}^\dagger\boldsymbol{W}\boldsymbol{C} \end{pmatrix} \tag{7.55}$$

which implies that the second-order flow equations are

$$\frac{\mathrm{d}\boldsymbol{F}^{(2)}(s)}{\mathrm{d}s} = \boldsymbol{F}\boldsymbol{W}\boldsymbol{W}^\dagger - 2\boldsymbol{W}\boldsymbol{C}\boldsymbol{W}^\dagger + \boldsymbol{W}\boldsymbol{W}^\dagger\boldsymbol{F} \tag{7.56}$$

$$\frac{\mathrm{d}\boldsymbol{C}^{(2)}(s)}{\mathrm{d}s} = \boldsymbol{C}\boldsymbol{W}^\dagger\boldsymbol{W} - 2\boldsymbol{W}^\dagger\boldsymbol{F}\boldsymbol{W} + \boldsymbol{W}^\dagger\boldsymbol{W}\boldsymbol{C} \tag{7.57}$$

Continuing with this approach yields the desired result.

7.2.4 Quasiparticle self-consistency (qpQPMP2)

We can follow what is done in the GW literature to define a quasiparticle self-consistent scheme for QPMP2. In GW, it involves implementing an off-diagonal self-energy

$$\Sigma_{pq}^{qsGW}(\omega) = \frac{1}{2}\Big(\Sigma_{pq}(\epsilon_p) + \Sigma_{pq}(\epsilon_q)\Big) \tag{7.58}$$

$$= \frac{1}{2}\sum_\nu \left[\sum_i \left[\frac{\Delta_{pi}^\nu}{(\Delta_{pi}^\nu)^2 + \eta^2} + \frac{\Delta_{qi}^\nu}{(\Delta_{qi}^\nu)^2 + \eta^2}\right] W_{pi}^\nu W_{qi}^\nu \right. \tag{7.59}$$

$$\left. + \sum_a \left[\frac{\Delta_{pa}^\nu}{(\Delta_{pa}^\nu)^2 + \eta^2} + \frac{\Delta_{qa}^\nu}{(\Delta_{qa}^\nu)^2 + \eta^2}\right] W_{pa}^\nu W_{qa}^\nu \right] \tag{7.60}$$

$$\tag{7.61}$$

with $\Delta_{pi}^\nu = \epsilon_p - \epsilon_i - \Omega_\nu$ and $\Delta_{pa}^\nu = \epsilon_p - \epsilon_a + \Omega_\nu$. In QPMP2, it will be analogous, and the expression will be

7.3 Why it is bad for the UEG

The VASP paper by Kresse and Grunes proves that scMP2 consists of terminating the Dyson equation for the interacting polarizability to lowest order, so setting $\chi(\omega) \approx \chi_0(\omega)$. This implies that the MP2 self-energy is only reliable if the polarizability of the system is small, so the neglect of higher order terms does not matter so much. I believe this is why the MP2 pole at $r_s = 4$ is bad. This can be proven analytically be examining the magnitude of the Linhard function.

7.4 Notes

David Tew's recent paper investigates the performance of this quasiparticle MP2 for finite systems, but they only do small molecules and the Hubbard model.

7.5 Decisions to make

7.5.1 Basis choice

I think def2-TZVPP-RIFIT would be good and then all-electron?

Chapter 8

UEG

Throughout, I am working in the retarded time formalism. We are working with a plane wave basis, where each state has its own wavevector.

8.1 Chemical potential

8.1.1 Thermodynamic definitions

We know that $E(N,V) = Ne(r_s)$, with $e(r_s)$ the energy per electron for the UEG. Note that r_s depends on N when the V is fixed.[1] So starting from the thermodynamic definition of

[1] Specifically (with $n = \frac{N}{V}$),

$$\frac{4\pi}{3} r_s^3 = \frac{V}{N} \tag{8.1}$$

$$\implies r_s = \left(\frac{3V}{4\pi N}\right)^{1/3} \tag{8.2}$$

$$\implies \left(\frac{\partial r_s}{\partial N}\right)_V = -\frac{1}{3}\left(\frac{3V}{4\pi}\right)^{1/3} N^{-4/3} \tag{8.3}$$

$$= -\frac{r_s}{3N} \tag{8.4}$$

the chemical potential at zero temperature, we find

$$\mu = \left(\frac{\partial E}{\partial N}\right)_V \tag{8.5}$$
$$= \left(\frac{\partial}{\partial N}\left[Ne(r_s)\right]\right)_V \tag{8.6}$$
$$= e(r_s) + N\left(\frac{\partial e}{\partial N}\right)_V \tag{8.7}$$
$$= e(r_s) + N\frac{\partial e}{\partial r_s}\left(\frac{\partial r_s}{\partial N}\right)_V \tag{8.8}$$
$$= e(r_s) + N\frac{de}{dr_s}\left(-\frac{r_s}{3N}\right) \tag{8.9}$$
$$= e(r_s) - \frac{r_s}{3}\frac{de}{dr_s}. \tag{8.10}$$

For the UEG, we can decompose the energy per particle as

$$e(r_s) = t_s(r_s) + e_x(r_s) + e_c(r_s) \tag{8.11}$$
$$\implies \mu = \underbrace{t_s(r_s) - \frac{r_s}{3}\left(\frac{dt_s}{dr_s}\right)}_{\mu_{\text{kin}}} + \underbrace{e_x(r_s) - \frac{r_s}{3}\left(\frac{de_x}{dr_s}\right)}_{\mu_x} + \underbrace{e_c(r_s) - \frac{r_s}{3}\left(\frac{de_c}{dr_s}\right)}_{\mu_c} \tag{8.12}$$
$$\tag{8.13}$$

Noninteracting Kinetic Contribution

For the 3D UEG, $E_F = k_F^2/2$, where $k_F = (9\pi/4)^{1/3} r_s^{-1}$. The kinetic energy per particle of the filled Fermi sphere is

$$t_s = \frac{3}{5}E_F \tag{8.14}$$
$$\implies \frac{dt_s}{dr_s} = -\frac{2t_s}{r_s} \tag{8.15}$$
$$\implies \mu_{\text{kin}} = t_s - \frac{r_s}{3}\left(-\frac{2t_s}{r_s}\right) \tag{8.16}$$
$$= t_s + \frac{2}{3}t_s \tag{8.17}$$
$$= \frac{5}{3}t_s \tag{8.18}$$
$$= E_F \tag{8.19}$$

This explains why the chemical potential occurs at the Fermi energy, for a *noninteracting* UEG at zero temperature.

Exchange Contribution

The exchange energy per electron is

$$e_x = -\frac{3k_F}{4\pi} \tag{8.20}$$

$$\implies \frac{de_x}{dr_s} = -\frac{e_x}{r_s} \tag{8.21}$$

$$\implies \mu_x = e_x - \frac{r_s}{3}\left(-\frac{e_x}{r_s}\right) \tag{8.22}$$

$$= e_x + \frac{1}{3}e_x \tag{8.23}$$

$$= \frac{4}{3}e_x \tag{8.24}$$

$$= \underbrace{-\frac{k_F}{\pi}}_{\Sigma_x^{\mathrm{HF}}(k_F)} \tag{8.25}$$

Correlation Contribution

For the correlation energy there is no simple power-law dependence on r_s. Instead, the field seems to use numbers that were tabulated for the given density with different methods. The full chemical potential is consequently

$$\boxed{\mu = E_F - \frac{k_F}{\pi} + e_c(r_s) - \frac{r_s}{3}\frac{de_c(r_s)}{dr_s}}. \tag{8.26}$$

8.1.2 Numerical Evaluation at $r_s = 4$

At $r_s = 4$, $\alpha = (9\pi/4)^{1/3} \approx 1.91916 \implies k_F \approx 0.47979$. Thus, $E_F = k_F^2/2 \approx 0.11510$ Ha and $\mu_x = -k_F/\pi \approx -0.15272$ Ha. So far, we have

$$\mu = 0.11510\text{Ha} - 0.15272\text{Ha} + e_c(4) - \frac{4}{3}\left.\frac{de_c(4)}{dr_s}\right|_{r_s=4} \tag{8.27}$$

Estimation using RC correlation energies

Kas reports RC correlation energies per particle near $r_s = 4$ as

$$e_c(3) = -0.041 \text{ Ha}, \tag{8.28}$$

$$e_c(4) = -0.035 \text{ Ha}, \tag{8.29}$$

$$e_c(5) = -0.030 \text{ Ha}. \tag{8.30}$$

We can estimate the derivative at $r_s = 4$ using a centered finite difference:

$$\left.\frac{de_c}{dr_s}\right|_{r_s=4} \approx \frac{e_c(5) - e_c(3)}{5-3} \tag{8.31}$$

$$= \frac{-0.030 - (-0.041)}{2} \tag{8.32}$$

$$= 0.0055 \text{ Ha.} \tag{8.33}$$

So the RC predicts

$$\mu_c = e_c(4) - \frac{4}{3}\left.\frac{de_c}{dr_s}\right|_{r_s=4} \tag{8.34}$$

$$= -0.035 - \frac{4}{3}(0.0055) \tag{8.35}$$

$$= -0.04233 \text{ Ha} \tag{8.36}$$

$$\implies \mu = E_F + \mu_x + \mu_c \tag{8.37}$$

$$= 0.11510 - 0.15272 - 0.04233 \tag{8.38}$$

$$\approx -0.07995 \text{ Ha} \tag{8.39}$$

$$\approx -2.18 \text{ eV} \tag{8.40}$$

$$\implies \frac{\mu}{\omega_p} \approx -0.369. \tag{8.41}$$

This approximately coincides with the chemical-potential reported in their Figure 1.

8.2 G0W0 quasiparticle

8.2.1 Energy

Holm defines the quasiparticle energy through

$$E_k^{GW} = \epsilon_k + \operatorname{Re}\Sigma\left(k, E_k\right). \tag{8.42}$$

He then postulates that E_k^{GW} is very close to the free-electron value, in order to justify replacing E_k by ϵ_k when evaluating the quasiparticle peak's structure. But let us check that this is true. Define

$$\xi_k^{GW} = \frac{k^2 - k_F^2}{2} + \operatorname{Re}\Sigma^{GW}\left(k, \xi_k^{GW}\right) - \operatorname{Re}\Sigma^{GW}\left(k_F, 0\right) \tag{8.43}$$

$$\implies \xi_0^{GW} = -E_F + \operatorname{Re}\Sigma^{GW}\left(0, \xi_0^{GW}\right) - \operatorname{Re}\Sigma^{GW}\left(k_F, 0\right) \tag{8.44}$$

$$= -E_F + \left[\Sigma_x(0) - \Sigma_x\left(k_F\right)\right] + \left[\operatorname{Re}\Sigma_c\left(0, \xi_0^{GW}\right) - \operatorname{Re}\Sigma_c\left(k_F, 0\right)\right] \tag{8.45}$$

Exchange part

The exchange part can be done completely analytically, since for the unpolarized 3D HEG, $\Sigma_x(k) = -\frac{k_F}{\pi}\left[1 + \frac{k_F^2-k^2}{2kk_F}\ln\left|\frac{k+k_F}{k-k_F}\right|\right]$. Taking the limits gives

$$\Sigma_x(0) = -\frac{2k_F}{\pi}, \tag{8.46}$$

$$\Sigma_x\left(k_F\right) = -\frac{k_F}{\pi} \tag{8.47}$$

$$\implies \Sigma_x(0) - \Sigma_x\left(k_F\right) = -\frac{k_F}{\pi}. \tag{8.48}$$

Since at $r_s = 4$, $k_F = \frac{(9\pi/4)^{1/3}}{4} \simeq 0.4798$Ha, we have

$$E_F = \frac{k_F^2}{2} \simeq 0.1151\text{Ha}, \tag{8.49}$$

and

$$-\frac{k_F}{\pi} \simeq -0.1527\text{Ha}. \tag{8.50}$$

So

$$\xi_0^{GW} \simeq -0.2678 + \left[\operatorname{Re}\Sigma_c\left(0, \xi_0^{GW}\right) - \operatorname{Re}\Sigma_c\left(k_F, 0\right)\right]. \tag{8.51}$$

What we need to compute numerically

We can either do it the way that is formally correct as was derived above, or we can do what Holm likely did of

$$\xi_0^{GW} \simeq -0.2678 + \left[\operatorname{Re}\Sigma_c\left(0, -E_F\right) - \operatorname{Re}\Sigma_c\left(k_F, 0\right)\right]. \tag{8.52}$$

This means that we need to evaluate $\operatorname{Re}\Sigma_c\left(0, -E_F\right)$, $\operatorname{Re}\Sigma_c\left(0, \xi_0^{GW}\right)$, and $\operatorname{Re}\Sigma_c\left(k_F, 0\right)$ numerically. For the evaluation of $\operatorname{Re}\Sigma_c\left(0, \xi_0^{GW}\right)$, we will need to calculate a nonlinear equation which implies a self-consistent evaluation. For this, we could implement Newton's method with an initial guess of $\xi_0^{GW} = -E_F$ and then iterate until convergence.

8.2.2 Weight

The weight is given by

$$W_{QP}(k) = Z_k = \left[1 - \frac{\partial \operatorname{Re}\Sigma(k,\omega)}{\partial\omega}\right]^{-1}_{\omega=\varepsilon_k} \tag{8.53}$$

In their paper, it is suggested that they may have used

$$W_{QP}(k) = 0.638 + 0.0595k^{1.4} \tag{8.54}$$

$$\overset{k=0}{=} 0.638 \tag{8.55}$$

8.2.3 Width

The width is given by

$$\Gamma_{QP}(k) = \operatorname{Im}\Sigma\left(k, \varepsilon_k\right). \tag{8.56}$$

In their paper, it is suggested that they may have used

$$\Gamma_{QP}(k) = \left(k^2 - 1\right)^2 / \left(9.8 + 7k^2\right) \tag{8.57}$$

$$\overset{k=0}{=} 1/9.8 \tag{8.58}$$

In any case, the QP peak is one of the Gaussian peaks in the spectral function as

$$A(k,\omega) = \sum_\nu \frac{W_\nu(k)}{\sqrt{2\pi\Gamma_\nu(k)}} \exp\left[-\frac{\left[\omega - E_\nu(k)\right]^2}{2\Gamma_\nu^2(k)}\right] \tag{8.59}$$

Visually, it seems that still they chose $E_0^{QP} = \varepsilon_0^0$.

8.3 Spectral function

The cumulant spectral function is defined as

$$A(\boldsymbol{k},\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dt e^{i\omega t} e^{-i\varepsilon_0 t + C(\boldsymbol{k},t)} \tag{8.60}$$

where

$$C(\boldsymbol{k},t) = -i\int_0^t dt' \int_{t'}^{\infty} d\tau \Sigma_{GW}(\boldsymbol{k},\tau) e^{i\varepsilon_0\tau}. \tag{8.61}$$

Now we have

$$\boldsymbol{\Sigma}(\boldsymbol{k},\omega) = \boldsymbol{\Sigma}_{HF}(\boldsymbol{k}) + \boldsymbol{\Sigma}^C(\boldsymbol{k},\omega). \tag{8.62}$$

The frequency-independent piece is given by

$$\boldsymbol{\Sigma}_{HF}(\boldsymbol{k}) = -\sum\nolimits_{\mathbf{q}} v(\boldsymbol{k}+\mathbf{q}) n_{\mathbf{q}} \tag{8.63}$$

$$= -\sum\nolimits_{\mathbf{q}} \frac{4\pi}{|\boldsymbol{k}+\mathbf{q}|^2} n_{\mathbf{q}} \tag{8.64}$$

with

$$n_{\mathbf{q}} = \int_{-\infty}^{\mu} A(\mathbf{q},\omega) d\omega. \tag{8.65}$$

The frequency-dependent piece is

$$\boldsymbol{\Sigma}^C(\boldsymbol{k},\omega) = \int_{\mathcal{C}} \frac{\Gamma\left(\boldsymbol{k},\omega'\right)}{\omega - \omega'} d\omega', \tag{8.66}$$

where

$$\Gamma(\boldsymbol{k},\omega) = \sum_{\mathbf{q}} \int_0^{\omega} d\omega' A\left(\boldsymbol{k}+\mathbf{q}, \omega-\omega'\right) B_0\left(\mathbf{q},\omega'\right). \tag{8.67}$$

Note that

$$B_0(\mathbf{q},\omega) = -\frac{1}{\pi}\frac{v^2(\mathbf{q})\,\mathrm{Im}\,\chi_0(\mathbf{q},\omega)}{\left|1-v(\mathbf{q})\chi_0(\mathbf{q},\omega)\right|^2}\,\mathrm{sgn}(\omega) \tag{8.68}$$

$$= \frac{v(\boldsymbol{q})}{\pi}\frac{\alpha(\boldsymbol{q})f_2(u,z)}{\left[1+\alpha(\boldsymbol{q})f_1(u,z)\right]^2+\left[\alpha(\boldsymbol{q})f_2(u,z)\right]^2}\,\mathrm{sgn}(\omega) \tag{8.69}$$

with $v(\mathbf{q}) = \frac{4\pi}{|\mathbf{q}|^2}$, $\alpha(\mathbf{q}) = \frac{3\omega_p^2}{|\mathbf{q}|^2 k_F^2}$, and $\omega_p = \sqrt{4\pi n}$ the plasma frequency. The functions f_1 and f_2 are defined as

$$f_1(u,z) = \frac{1}{2} + \frac{1}{8z}\left\{1-(z-u)^2\right\}\log\left|\frac{z-u+1}{z-u-1}\right| + \frac{1}{8z}\left\{1-(z+u)^2\right\}\log\left|\frac{z+u+1}{z+u-1}\right|, \tag{8.70}$$

$$f_2(u,z) = \left\{\begin{array}{l} \frac{\pi}{2}u, \text{ for } z+u<1, \\ \frac{\pi}{8z}\left\{1-(z-u)^2\right\}, \text{ for } |z-u|<1<z+u, \\ 0, \text{ for } |z-u|>1. \end{array}\right\} \tag{8.71}$$

with $z = \frac{|\mathbf{q}|}{2k_F}$ and $u = \frac{|\omega|}{|\mathbf{q}|k_F}$.

One-shot cumulant

Here we have $\boldsymbol{k} = 0$. Also, we have

$$A_0(\mathbf{q},\omega) = \delta\left(\omega - \epsilon_{\boldsymbol{q}}\right) \tag{8.72}$$

$$\implies n_{\mathbf{q}} = \Theta\left(k_F - |\mathbf{q}|\right). \tag{8.73}$$

I want us to save the value of this so we can plot it alongside the other methods.

Self-consistency with updates to the chemical potential

The idea will now be to compute the spectral functions $A(\mathbf{q},\omega)$ for all $\mathbf{q}$ (*it is different depending on whether* $\mathbf{q}$ *is inside or outside the Fermi surface*), and then use those to compute the self-energy $\Sigma(\boldsymbol{k},\omega)$, which will then be used to compute the cumulant $C(\boldsymbol{k},t)$, which will then be used to compute a new spectral function $A(\boldsymbol{k},\omega)$. This process will be iterated until convergence. Note that now since $A(\mathbf{q},\omega)$ is no longer a delta function, we will need to compute $n_{\mathbf{q}}$ numerically as well at each iteration. Now the occupation numbers are computed as

$$n_{\mathbf{q}} = \int_{-\infty}^{\mu} A(\mathbf{q},\omega)d\omega. \tag{8.74}$$

At each iteration of the self-consistent iteration, we should update the chemical potential by enforcing

$$N = \sum_{\mathbf{q}} n_{\mathbf{q}}. \tag{8.75}$$

Note that at the one-shot level, this is already enforced because

$$n_{\mathbf{q}} = \Theta\left(k_F - |\mathbf{q}|\right) \tag{8.76}$$

$$\implies N = \sum_{|\boldsymbol{q}|<k_F} 1 = N_{\mathrm{occ}}. \tag{8.77}$$

Self-consistency without updates to the chemical potential

I want to see the version that comes about when we update the spectral function that goes into the correlation self-energy, but we do not update the chemical potential at all, meaning that the Hartree-fock self-energy stays constant at the one-shot level.

8.3.1 Self-energy

I want us to store the self-energy as well so that I can see how the self-energy as a function of frequency changes for each of the methods we covered.

8.4 Holm's reported one-shot CE curve is wrong

Implementing Holm's equations gives a result analogous (I still need to figure out why they are not exactly the same) to Reining's, neither of which match Holm's reported result. Should I do self-consistent cumulant to match Holm's result even though his one-shot is wrong?

8.5 HF

8.5.1 Writing spectral functions for the UEG

Noninteracting case

We can start by writing down the form for the retarded non-interacting Green's function of a state p in time domain as

$$G^R_{0,p}(t) = -i\Theta(t)e^{-i(\epsilon_p - \varepsilon_F)t} \tag{8.78}$$

where ε_F is the Fermi energy. I am operating at zero temperature, so in my HF code, this means that ε_F is just the energy of the highest occupied molecular orbital; for this system of the UEG at $r_s = 4$ with 14 electrons in 19 orbitals it is just the energy of orbital 6 with orbitals numbered from 0). A broadening parameter of $\eta = 0.2$ eV is used (this coincides with what Joonho and Garnet used). The HF single particle energies I get from my code do not take the Madelung correction into account. In that case, Joonho's paper tells us that for the spectral function of these occupied orbitals, we need to apply a constant frequency shift of

$$\Delta_{IP} = -\frac{\xi}{2} \tag{8.79}$$

where $\xi = -2 \times 2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_{\mathrm{s}}^{-1}$ Hartrees. For my system, this means that $\frac{\xi}{2} = -4.97$ eV. In the restricted formalism, the retarded second order self-energy in the

frequency domain is given by

$$\Sigma^{c,(2),R}(p,\omega) = \frac{1}{2}\sum_{iab}\frac{\langle pi\,||\,ab\rangle^2}{\omega-\epsilon_a-\epsilon_b+\epsilon_i+i\eta} + \frac{1}{2}\sum_{ija}\frac{\langle pa\,||\,ij\rangle^2}{\omega-\epsilon_i-\epsilon_j+\epsilon_a+i\eta} \tag{8.80}$$

$$\overset{\text{RHF}}{=} \frac{1}{2}\sum_{iab}\frac{\left(2\langle pi|ab\rangle - \langle pi|ba\rangle\right)^2}{\omega-\epsilon_a-\epsilon_b+\epsilon_i+i\eta} + \frac{1}{2}\sum_{ija}\frac{\left(2\langle pa|ij\rangle - \langle pa|ji\rangle\right)^2}{\omega-\epsilon_j-\epsilon_i+\epsilon_a+i\eta} \tag{8.81}$$

$$= \frac{2\pi}{V}\left(\sum_{\mathbf{k}_i\in O;\,\mathbf{k}_a,\mathbf{k}_b\in V}\frac{\left[\frac{2\delta_{\mathbf{k}_p-\mathbf{k}_a,\mathbf{k}_b-\mathbf{k}_i}}{|\mathbf{k}_p-\mathbf{k}_a|^2} - \frac{\delta_{\mathbf{k}_p-\mathbf{k}_b,\mathbf{k}_a-\mathbf{k}_i}}{|\mathbf{k}_p-\mathbf{k}_b|^2}\right]^2}{\omega-\epsilon_{\mathbf{k}_a}-\epsilon_{\mathbf{k}_b}+\epsilon_{\mathbf{k}_i}+i\eta} + \sum_{\mathbf{k}_i,\mathbf{k}_j\in O;\,\mathbf{k}_a\in V}\frac{\left[\frac{2\delta_{\mathbf{k}_p-\mathbf{k}_i,\mathbf{k}_j-\mathbf{k}_a}}{|\mathbf{k}_p-\mathbf{k}_i|^2} - \frac{\delta_{\mathbf{k}_p-\mathbf{k}_j,\mathbf{k}_i-\mathbf{k}_a}}{|\mathbf{k}_p-\mathbf{k}_j|^2}\right]^2}{\omega-\epsilon_{\mathbf{k}_i}-\epsilon_{\mathbf{k}_j}+\epsilon_{\mathbf{k}_a}+i\eta}\right) \tag{8.82}$$

where $\mathbf{k}_b = \mathbf{k}_p + \mathbf{k}_i - \mathbf{k}_a$ in the first sum and $\mathbf{k}_j = \mathbf{k}_p - \mathbf{k}_i + \mathbf{k}_a$ in the second sum due to momentum conservation.

GW spectral function

Consider

$$\Sigma_k(\omega) = G_{0,k}^{-1}(\omega) - G_k^{-1}(\omega) \tag{8.83}$$

$$\implies G_k(\omega) = \frac{1}{G_{0,k}^{-1}(\omega)-\Sigma_k(\omega)} \tag{8.84}$$

$$= \frac{1}{\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)-i\operatorname{Im}\Sigma_k(\omega)} \times \frac{(\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)+i\operatorname{Im}\Sigma_k(\omega))}{(\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)+i\operatorname{Im}\Sigma_k(\omega))} \tag{8.85}$$

$$= \frac{\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)+i\operatorname{Im}\Sigma_k(\omega)}{\left(\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)\right)^2+\left(\operatorname{Im}\Sigma_k(\omega)\right)^2} \tag{8.86}$$

We want $-\frac{1}{\pi}\operatorname{Im}G_k(\omega)$, so

$$\operatorname{Im}G_k(\omega) = \frac{\operatorname{Im}\Sigma_k(\omega)}{\left(\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)\right)^2+\left(\operatorname{Im}\Sigma_k(\omega)\right)^2} \tag{8.87}$$

$$\tag{8.88}$$

Thus, the spectral function is

$$A(\omega) = -\frac{1}{\pi}\operatorname{Im}G_k(\omega) = -\frac{1}{\pi}\frac{\operatorname{Im}\Sigma_k(\omega)}{\left(\omega-\epsilon_k-\operatorname{Re}\Sigma_k(\omega)\right)^2+\left(\operatorname{Im}\Sigma_k(\omega)\right)^2} \tag{8.89}$$

This means we can write down the second order cumulant in the time domain as

$$C_{pp}^{(2)}(t) \equiv i \int \frac{d\omega}{2\pi} \frac{\tilde{\Sigma}_{pp}^{c,(2)}(\omega+\epsilon_p)}{(\omega+i\eta)^2} e^{-i\omega t} \tag{8.90}$$

$$= \frac{1}{2}\sum_{iab} \langle pi \,||\, ab\rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2\left(\omega-\epsilon_{pi}^{ab}\right)} + \frac{1}{2}\sum_{ija} \langle pa \,||\, ij\rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2\left(\omega-\epsilon_{pa}^{ij}\right)} \tag{8.91}$$

$$= \frac{1}{2}\sum_{iab} \frac{\langle pi \,||\, ab\rangle^2}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t} + i\epsilon_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija} \frac{\langle pa \,||\, ij\rangle^2}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t} + i\epsilon_{pa}^{ij}t - 1\right) \tag{8.92}$$

$$= \frac{2\pi}{V}\left(\sum_{\mathbf{k}_i\in O;\,\mathbf{k}_a,\mathbf{k}_b\in V} \frac{\left[\frac{2}{|\mathbf{k}_p-\mathbf{k}_a|^2} - \frac{1}{|\mathbf{k}_p-\mathbf{k}_b|^2}\right]^2}{\left(\varepsilon_{pi}^{ab}\right)^2}\left(e^{-i\varepsilon_{pi}^{ab}t} + i\varepsilon_{pi}^{ab}t - 1\right)\right. \tag{8.93}$$

$$\left. + \sum_{\mathbf{k}_i,\mathbf{k}_j\in O;\,\mathbf{k}_a\in V} \frac{\left[\frac{2}{|\mathbf{k}_p-\mathbf{k}_i|^2} - \frac{1}{|\mathbf{k}_p-\mathbf{k}_j|^2}\right]^2}{\left(\varepsilon_{pa}^{ij}\right)^2}\left(e^{-i\varepsilon_{pa}^{ij}t} + i\varepsilon_{pa}^{ij}t - 1\right)\right) \tag{8.94}$$

where $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$, but for the UEG, $\varepsilon_{pi}^{ab} = \epsilon_{\mathbf{k}_a} + \epsilon_{\mathbf{k}_b} - \epsilon_{\mathbf{k}_p} - \epsilon_{\mathbf{k}_i}$ and $\varepsilon_{pa}^{ij} = \epsilon_{\mathbf{k}_i} + \epsilon_{\mathbf{k}_j} - \epsilon_{\mathbf{k}_p} - \epsilon_{\mathbf{k}_a}$. When we took the single particle energies from HF, the Madelung correction was never included, so I need to find out if we need to include it here.

Interacting case

Now, the form for the retarded interacting Green's function in the time domain is

$$G_{pp}(t) = -i\Theta(t)e^{-i(\epsilon_p-\varepsilon_F)t}e^{C_{pp}(t)} \tag{8.95}$$

Explanation of one-shot plots

Recall

$$G_{pp}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} G_{pp}(t) \tag{8.96}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt e^{i(\omega-\epsilon_p^{QP})t} e^{\sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi\nu}t} + \sum_{a\nu}\zeta_{pa\nu}e^{-i\Delta_{pa\nu}t}} \tag{8.97}$$

$$= -iZ_p^{QP}\int_0^{\infty} dt\, e^{i(\omega-\epsilon_p^{QP})t}\left(1 + \sum_{i\nu}\zeta_{pi\nu}e^{-i\Delta_{pi\nu}t} + \sum_{a\nu}\zeta_{pa\nu}e^{-i\Delta_{pa\nu}t} + \ldots\right) \tag{8.98}$$

$$= \frac{Z_p^{QP}}{\omega-\epsilon_p^{QP}+i\eta} + \sum_{i\nu}\frac{Z_p^{QP}\zeta_{pi\nu}}{\omega-\epsilon_p^{QP}-\Delta_{pi\nu}+i\eta} + \sum_{a\nu}\frac{Z_p^{QP}\zeta_{pa\nu}}{\omega-\epsilon_p^{QP}-\Delta_{pa\nu}+i\eta} + \ldots \tag{8.99}$$

$$\approx \frac{Z_p^{QP}}{\omega-\epsilon_p^{QP}+i\eta} + \sum_{i\nu}\frac{Z_{pi\nu}^{sat}}{\omega-\epsilon_{pi\nu}^{sat}+i\eta} + \sum_{a\nu}\frac{Z_{pa\nu}^{sat}}{\omega-\epsilon_{pa\nu}^{sat}+i\eta} \tag{8.100}$$

$G_{pp}(\omega)$ is computed via 8.100 in the schemes which I call first order. This is what Loos does. On the other hand, $G_{pp}(\omega)$ is computed via 8.97 in the schemes which I call to infinite order.

As you can see, there are two different flavors of infinite order, which only differ in how the real-time cumulant is computed. Analytic to infinite order does

$$C_{pp}(t) = i \int \frac{d\omega}{2\pi} \frac{\Sigma^c_{pp}\left(\omega + \epsilon_p^{HF}\right)}{(\omega + i\eta)^2} e^{-i\omega t} \tag{8.101}$$

$$= \sum_{i\nu} \zeta_{pi\nu} \left[e^{-i\Delta_{pi\nu}t} - 1 + i\Delta_{pi\nu}t\right] + \sum_{a\nu} \zeta_{pa\nu} \left[e^{-i\Delta_{pa\nu}t} - 1 + i\Delta_{pa\nu}t\right] \tag{8.102}$$

whereas numerical to infinite order computes using the Landau form

$$C_{pp}(t) = \int d\omega \frac{\beta_{pp}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{8.103}$$

with $\beta_{pp}(\omega) = \frac{1}{\pi} |\operatorname{Im} \Sigma_{pp}(\omega + \epsilon_p^{HF})|$ using the basic Riemann sum for the frequency integration. In both cases, we then plug the cumulant into

$$G_{pp}(t) = -i\Theta(t) e^{-t(i\epsilon_p + \eta)t} e^{C_{pp}(t)} \tag{8.104}$$

and then use the DFT to get $A_{pp}(\omega)$.

Choosing the right shift for my spectral functions

My interpretation of your discussion is that when I have included the G=0 component in my HF reference, all I have to do is to plot $A_{pp}(\omega + \xi/2)$ for HF, MP2, and MP2+C alike, since I am only computing the spectral function for occupied states. Is that correct?

Updated results

As we discussed on Friday, now, I have a RHF reference which includes the G=0 component. I fixed the bug and now it matches with the G=0 excluded reference exactly. Therefore, now the MP2 correlation energy is the same, irrespective of whether G=0 is included or not. I am plotting $A_{pp}(\omega + \xi/2)$ for all my spectral functions. This is because I needed to plot $A_{pp}(\omega + \xi/2)$ for HF in order to match the reference, so I do the same for MP2 and MP2+C; other than this, nothing else gets shifted.

Update

If we choose

$$\xi = -2 \times 2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_{\mathrm{s}}^{-1} \tag{8.105}$$

then I implemented the direct term of my ERIs as

$$v(\mathbf{q}) = \begin{cases} \frac{4\pi}{\mathbf{q}^2 V} & \mathbf{q} \neq 0 \\ 0 & \mathbf{q} = 0 \end{cases} \tag{8.106}$$

and the exchange terms as

$$v(\mathbf{q}) = \begin{cases} \frac{4\pi}{\mathbf{q}^2 V} & \mathbf{q} \neq 0 \\ -\xi & \mathbf{q} = 0 \end{cases} \tag{8.107}$$

for use in the self energy as

$$\Sigma^{(2)}_{pp}(\omega) = \sum_{iab} \frac{\langle pi|ab\rangle \left(2\langle pi|ab\rangle - \langle pi|ba\rangle\right)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{\langle pa|ij\rangle \left(2\langle pa|ij\rangle - \langle pa|ji\rangle\right)}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{8.108}$$

Correct poles with G=0 included

I have changed the definition of ξ to

$$\xi = -1 \times 2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_{\mathrm{s}}^{-1} \tag{8.109}$$

so omitting the factor of 2 as was suggested. This produces the results that I provide with no manual shift ever done. I present two varieties of cumulants. The Landau form of the cumulant is

$$C_{pp}(t) \equiv i \int \frac{d\omega}{2\pi} \frac{\Sigma_{pp}\left(\omega + \varepsilon_p\right)}{(\omega + i\eta)^2} e^{-i\omega t} \tag{8.110}$$

$$\tag{8.111}$$

Then the cumulant ansatz for the time domain interacting Green's function is

$$G_{pp}(t) = -i\Theta(t) e^{-t(i\varepsilon_p + \eta)} e^{C_{pp}(t)} \tag{8.112}$$

When I present MP2+C; G_0 via HF, $\varepsilon_p \equiv \epsilon_p^{HF}$ in both 8.110 and 8.112, and $\Sigma_{pp}(\omega)$ is built from the set $\{\epsilon_p^{HF}\}$. When I present MP2+C; G_0 via SRG-MP2, $\varepsilon_p \equiv \epsilon_p^{SRG-MP2}$ in both 8.110 and 8.112, and $\Sigma_{pp}(\omega)$ is built from the set $\{\epsilon_p^{SRG-MP2}\}$. So the former does not solve Dyson's equation before adding the cumulant correction (it is a true one shot scheme), while the latter solves Dyson's equation at the SRG-MP2 level before adding the cumulant correction.

Why we need the G=0 component in the ERIs of the MP2 self-energy

It is instructive to start by understanding why having the G=0 component in the ERIs is irrelevant for computing the MP2 correlation energy. The MP2 correlation energy can be written as

$$E_c^{MP2} = \frac{1}{4} \sum_{ijab} \frac{|\langle \boldsymbol{k}_i \boldsymbol{k}_j | \boldsymbol{k}_a \boldsymbol{k}_b\rangle - \langle \boldsymbol{k}_i \boldsymbol{k}_j | \boldsymbol{k}_b \boldsymbol{k}_a\rangle|^2}{\epsilon_{\boldsymbol{k}_i} + \epsilon_{\boldsymbol{k}_j} - \epsilon_{\boldsymbol{k}_a} - \epsilon_{\boldsymbol{k}_b}} \tag{8.113}$$

Because i, j are occupied and a, b are virtual, and the correlation energy always has an OV pair at a ERI center, we will never encounter a zero momentum transfer at any center. Now, the MP2 self-energy is given by

$$\Sigma^{(2)}_{pq}(\omega) = \frac{1}{2} \sum_{iab} \frac{\langle \boldsymbol{k}_p \boldsymbol{k}_i | \boldsymbol{k}_a \boldsymbol{k}_b\rangle \langle \boldsymbol{k}_a \boldsymbol{k}_b | \boldsymbol{k}_q \boldsymbol{k}_i\rangle}{\omega + \epsilon_{\boldsymbol{k}_i} - \epsilon_{\boldsymbol{k}_a} - \epsilon_{\boldsymbol{k}_b}} + \frac{1}{2} \sum_{ija} \frac{\langle \boldsymbol{k}_p \boldsymbol{k}_a | \boldsymbol{k}_i \boldsymbol{k}_j\rangle \langle \boldsymbol{k}_i \boldsymbol{k}_j | \boldsymbol{k}_q \boldsymbol{k}_a\rangle}{\omega + \epsilon_{\boldsymbol{k}_a} - \epsilon_{\boldsymbol{k}_i} - \epsilon_{\boldsymbol{k}_j}} \tag{8.114}$$

If you take a look at the second summation (the 2h1p one), you can see that if p is occupied, then it is possible, for example, to have $\boldsymbol{k}_p = \boldsymbol{k}_i$, and hence a zero momentum transfer at the ERI center. Thus, if we do not include the G=0 component from the ouset in HF, we mistreat the situation where there might be a zero momentum transfer at an ERI center, and thus miss contributions to the self-energy.

Solution to the problem of the shifted poles

Tim's notes write that the many-body Hamiltonian based on the Ewald potential can be re-written as

$$H = -\sum_{i=1}^{N}\frac{1}{2}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M} z_A \left[v_{\mathrm{E}}\left(r_i, R_A\right) - v_{\mathrm{M}}\right] + \sum_{i=1}^{N}\sum_{j>i}^{N}\left[v_{\mathrm{E}}\left(r_i, r_j\right) - v_{\mathrm{M}}\right] \\ + \sum_{A=1}^{M}\sum_{B>A}^{M} z_A z_B \left[v_{\mathrm{E}}\left(R_A, R_B\right) - v_{\mathrm{M}}\right] + \frac{v_{\mathrm{M}}}{2}\left(N - \sum_A z_A\right)^2 \tag{8.115}$$

where the final term vanishes for a neutral system. The shifted Ewald potential $v_{\mathrm{s}}(r) \equiv v_{\mathrm{E}}(r) - v_{\mathrm{M}}$ has the Fourier components

$$v_{\mathrm{s}}(G) = \begin{cases} 4\pi/\Omega G^2 & G \neq 0 \\ -v_{\mathrm{M}} & G = 0 \end{cases}. \tag{8.116}$$

Because this is the homogeneous electron gas, there are no nuclei, so the second and fourth and last part of the final parentheses in (8.162) vanish. Thus, the Hamiltonian reduces to

$$H = -\sum_{i=1}^{N}\frac{1}{2}\nabla_i^2 + \sum_{\boldsymbol{G}} v_{\mathrm{s}}(G) + \frac{v_{\mathrm{M}}}{2}N^2 \tag{8.117}$$

Now, the IP is defined as the energy difference between the N and N-1 electron systems, so if the final term in 8.117 did not exist, we could say that the IPs are equivalent to the occupied MO energies by Koopmans' theorem. However, because of the final term, we have that

$$IP = -\left(E(N-1) - E(N)\right) = \epsilon_{\mathrm{i}} - \frac{v_{\mathrm{M}}}{2} \tag{8.118}$$

So the IPs should be shifted by $-v_{\mathrm{M}}/2$ to get the correct result. This is what I was doing to get the good looking results before this conversation all started.

8.5.2 What are the implications of including the G=0 component for spectral functions?

Deriving orbital energies

We know the form when G=0 is excluded, so with respect to that we don't modify the Coulomb matrix at all, but just the exchange matrix. We are at the GPT, so we can neglect

the $\vec{k}$ dependence of quantities. Thus, in the plane wave basis, we have

$$K_{pq}^{\sigma} = \int_{\Omega} d^2\vec{r}\phi_p^*(\vec{r})K^{\sigma}(\vec{r})\phi_q(\vec{r}) \tag{8.119}$$

$$= \sum_{ij} P_{ji}^{\sigma} \underbrace{\langle pi|jq\rangle}_{*} \tag{8.120}$$

Now,

$$* = \frac{1}{\Omega^2}\int_{\Omega} d^3\vec{r} e^{i\left(\vec{G}_j-\vec{G}_p\right)\cdot\vec{r}} \left\{e^{i\left(\vec{G}_q-\vec{G}_i\right)\cdot\vec{r}}\right\} \left\{\int d^3\vec{r}' \frac{1}{|\vec{r}'-\vec{r}|} e^{i\left(\vec{G}_q-\vec{G}_i\right)\cdot|\vec{r}'-\vec{r}|}\right\} \tag{8.121}$$

$$= \frac{1}{\Omega^2}\int_{\Omega} d^3\vec{r} \left\{e^{i\left(\vec{G}_j-\vec{G}_p\right)\cdot\vec{r}}\right\} \left\{e^{i\left(\vec{G}_q-\vec{G}_i\right)\cdot\vec{r}}\right\} \left(\frac{4\pi}{\left|\vec{G}_q-\vec{G}_i\right|^2}\left(1-\delta_{\vec{G}_q,\vec{G}_i}\right) - v_{\mathrm{M}}\Omega\delta_{\vec{G}_q,\vec{G}_i}\right) \tag{8.122}$$

$$= \frac{1}{\Omega}\left(\frac{4\pi}{\left|\vec{G}_q-\vec{G}_i\right|^2}\left(1-\delta_{\vec{G}_q,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_q,\vec{G}_i}\right)\delta\left[\left(\vec{G}_j-\vec{G}_p\right)+\left(\vec{G}_q-\vec{G}_i\right)\right] \tag{8.123}$$

$$\tag{8.124}$$

Thus, the exchange matrix elements become

$$K_{pq}^{\sigma} = \frac{1}{\Omega}\sum_{ij} P_{ji}^{\sigma}\left(\frac{4\pi}{\left|\vec{G}_q-\vec{G}_i\right|^2}\left(1-\delta_{\vec{G}_q,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_q,\vec{G}_i}\right)\delta\left[\left(\vec{G}_j-\vec{G}_p\right)+\left(\vec{G}_q-\vec{G}_i\right)\right] \tag{8.125}$$

$$= \frac{1}{\Omega}\sum_{ij} P_{p-(q-i),q-(q-i)}^{\sigma}\left(\frac{4\pi}{\left|\vec{G}_q-\vec{G}_i\right|^2}\left(1-\delta_{\vec{G}_q,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_q,\vec{G}_i}\right)\delta\left[\left(\vec{G}_j-\vec{G}_p\right)+\left(\vec{G}_q-\vec{G}_i\right)\right] \tag{8.126}$$

Given that the kinetic and Coulomb matrices are unchanged, we can write the Fock matrix for the UEG as

$$F_{pq}^{\sigma} = H_{pq}^{\mathrm{kin}} + J_{pq}^{\sigma} - K_{pq}^{\sigma} \tag{8.127}$$

$$= \delta_{pq}\frac{|\vec{G}_p|^2}{2} - \frac{\delta_{pq}}{\Omega}\sum_{q-i} n_{p-(q-i)}^{\sigma}\left(\frac{4\pi}{\left|\vec{G}_q-\vec{G}_i\right|^2}\left(1-\delta_{\vec{G}_q,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_q,\vec{G}_i}\right) \tag{8.128}$$

$$= \delta_{pq}\frac{|\vec{G}_p|^2}{2} - \frac{\delta_{pq}}{\Omega}\sum_{i\in occ}\left(\frac{4\pi}{\left|\vec{G}_p-\vec{G}_i\right|^2}\left(1-\delta_{\vec{G}_p,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_p,\vec{G}_i}\right) \tag{8.129}$$

where we have defined $n_p^\sigma \equiv P_{pp}^\sigma$ as the occupation number of orbital p with spin σ, but because we are at zero temperature, $n_p^\sigma = 0$ (unoccupied) or 1 (occupied). Because the Fock matrix is diagonal, we can read off the orbital energies as

$$\epsilon_p^\sigma = \frac{|\vec{G}_p|^2}{2} - \frac{1}{\Omega}\sum_{i\in occ}\left(\frac{4\pi}{\left|\vec{G}_p - \vec{G}_i\right|^2}\left(1 - \delta_{\vec{G}_p,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_p,\vec{G}_i}\right) \tag{8.130}$$

$$= \frac{|\vec{G}_p|^2}{2} - \frac{1}{\Omega}\sum_{i\in occ}\frac{4\pi}{\left|\vec{G}_p - \vec{G}_i\right|^2}\left(1 - \delta_{\vec{G}_p,\vec{G}_i}\right) + v_{\mathrm{M}} n_p^\sigma \tag{8.131}$$

Thus, occupied orbitals get a shift of v_{M}, while unoccupied orbitals are unchanged. In my plot I suggest two different definitions of v_{M}; they both are proportional to

$$\xi = -2 \times 2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_{\mathrm{s}}^{-1} \tag{8.132}$$

Derivation specific to RHF

Given that the kinetic and Coulomb matrices are unchanged, we can write the Fock matrix in RHF for the UEG as

$$F_{pq} = 2H_{pq}^{\mathrm{kin}} + 2J_{pq} - K_{pq} \tag{8.133}$$

$$= \delta_{pq}|\vec{G}_p|^2 - \frac{\delta_{pq}}{\Omega}\sum_{q-i} n_{p-(q-i)}\left(\frac{4\pi}{\left|\vec{G}_q - \vec{G}_i\right|^2}\left(1 - \delta_{\vec{G}_q,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_q,\vec{G}_i}\right) \tag{8.134}$$

$$= \delta_{pq}|\vec{G}_p|^2 - \frac{\delta_{pq}}{\Omega}\sum_{i\in occ}\left(\frac{4\pi}{\left|\vec{G}_p - \vec{G}_i\right|^2}\left(1 - \delta_{\vec{G}_p,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_p,\vec{G}_i}\right) \tag{8.135}$$

where we have defined $n_p \equiv P_{pp}$ as the occupation number of orbital p, but because we are at zero temperature, $n_p = 0$ (unoccupied) or 1 (occupied). Because the Fock matrix is diagonal, we can read off the orbital energies as

$$\epsilon_p = |\vec{G}_p|^2 - \frac{1}{\Omega}\sum_{i\in occ}\left(\frac{4\pi}{\left|\vec{G}_p - \vec{G}_i\right|^2}\left(1 - \delta_{\vec{G}_p,\vec{G}_i}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_p,\vec{G}_i}\right) \tag{8.136}$$

$$= |\vec{G}_p|^2 - \frac{1}{\Omega}\sum_{i\in occ}\frac{4\pi}{\left|\vec{G}_p - \vec{G}_i\right|^2}\left(1 - \delta_{\vec{G}_p,\vec{G}_i}\right) + v_{\mathrm{M}} n_p \tag{8.137}$$

Thus, occupied orbitals get a shift of v_{M}, while unoccupied orbitals are unchanged. In my plot I suggest two different definitions of v_{M}; they both are proportional to

$$\xi = -2 \times 2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_{\mathrm{s}}^{-1} \tag{8.138}$$

Deriving total energy

Earlier, we found that the value of the ERI is

$$\langle pq|rs\rangle = \frac{1}{\Omega}\left(\frac{4\pi}{\left|\vec{G}_s - \vec{G}_q\right|^2}\left(1-\delta_{\vec{G}_s,\vec{G}_q}\right) - \Omega v_{\mathrm{M}}\delta_{\vec{G}_s,\vec{G}_q}\right)\delta\left[\left(\vec{G}_r - \vec{G}_p\right) + \left(\vec{G}_s - \vec{G}_q\right)\right]. \tag{8.139}$$

Let us now determine the magnitude of the Madelung correction, given the the expression for the RHF energy in physicist's notation as

$$E^{(0)} = 2\sum_i h_{ii} + \sum_{ij}\left(2\langle ij|ij\rangle - \langle ij|ji\rangle\right) \tag{8.140}$$

$$= \ldots + \sum_i^{N/2}\sum_j^{N/2}\left(\delta_{ij}v_{\mathrm{M}}\right) \tag{8.141}$$

$$= \ldots + \frac{N}{2}v_{\mathrm{M}} \tag{8.142}$$

Since it is RHF, we don't need to worry about spin orthogonality. Then, we know that for the UEG, by defintion, the direct term vanishes. Finally, in the exchange term, a zero momentum transfer only happens when $i = j$, and we only get a v_{M} contribution in that case.

From Szabo, we know that the total HF energy is the sum of occupied orbital energies minus a correction term to avoid double counting the two-electron interactions:

$$E_{HF} = \sum_{i\in occ}\epsilon_i - \frac{1}{2}\sum_{i\in occ}\sum_{j\in occ}\left(\langle ij|ij\rangle - \langle ij|ji\rangle\right) \tag{8.143}$$

$$= \sum_{i\in occ}\left[\frac{|\vec{G}_i|^2}{2} - \frac{1}{\Omega}\sum_{j\in occ}\frac{4\pi}{\left|\vec{G}_i - \vec{G}_j\right|^2}\left(1-\delta_{\vec{G}_i,\vec{G}_j}\right)\right] + Nv_{\mathrm{M}} \tag{8.144}$$

$$-\frac{1}{2}\sum^{2N}(-v_{\mathrm{M}}) + \frac{1}{2}\sum_{i\neq j\in occ}\frac{1}{\Omega}\frac{4\pi}{\left|\vec{G}_i - \vec{G}_j\right|^2} + \frac{1}{2}\sum^{N}(-v_{\mathrm{M}}) \tag{8.145}$$

$$= \ldots + \frac{3Nv_{\mathrm{M}}}{2} \tag{8.146}$$

Deriving IPs

Now, the total HF energy for a general system can be written as a functional of occupation numbers $\{n_i\}$ for a fixed orthonormal orbital set $\{\phi_i\}$:

$$E[\{n\}] = \sum_i n_i\, h_{ii} + \frac{1}{2}\sum_i\sum_j n_i n_j\, G_{ij}, \tag{8.147}$$

where the one-body term and two-body terms are h_{ii} and G_{ij}, respectively, and the factor of $\frac{1}{2}$ avoids double counting of the two-body terms. We know that ionization consists of removing an electron from an occupied orbital p; we assume this happens via a vertical process, where other occupations and orbitals are frozen. In the N-electron reference state, $n_p = 1$, while in the $(N-1)$-electron state, $n_p = 0$. This motivates us to take a closer look at the double sum, separating terms involving orbital p,

$$\sum_{ij} n_i n_j G_{ij} = \sum_{i\neq p}\sum_{j\neq p} n_i n_j G_{ij} + 2n_p \sum_{j\neq p} n_j G_{pj} + n_p^2 G_{pp}. \tag{8.148}$$

Using the decomposition in 8.148, we compute the frozen-orbital energy difference as

$$E(N) - E(N-1) = h_{pp} + \frac{1}{2}\left(2\sum_{j\neq p} n_j G_{pj} + G_{pp}\right). \tag{8.149}$$

We found the HF orbital energy of orbital p to be

$$\epsilon_p = h_{pp} + \sum_j n_j G_{pj} = h_{pp} + \sum_{j\neq p} n_j G_{pj} + G_{pp} \tag{8.150}$$

$$\equiv \frac{|\vec{G}_p|^2}{2} + \frac{1}{\Omega}\sum_{i\in occ} \frac{4\pi}{\left|\vec{G}_p - \vec{G}_i\right|}\left(1 - \delta_{\vec{G}_p,\vec{G}_i}\right) + v_{\mathrm{M}} n_p^{\sigma} \tag{8.151}$$

where the first, second, and third terms are meant to be compared, respectively, and we can identify $G_{pp} \equiv v_{\mathrm{M}}$ for occupied orbitals. Comparing with 8.149 yields

$$E(N) - E(N-1) = \varepsilon_p - \frac{v_{\mathrm{M}}}{2} \tag{8.152}$$

$$\implies IP_p \equiv E(N-1) - E(N) = -\varepsilon_p + \frac{v_{\mathrm{M}}}{2}. \tag{8.153}$$

8.6 Determining IPs in simulation with periodic boundary conditions (PBCs)

8.6.1 Physical intuition when using PBCs

When we simulate with PBCs, we are repeating a N-electron unit cell across the entire lattice. Therefore, when we model the ionization process, PBCs force us to create a hole in every cell across the lattice. These periodically repeated holes interact with each other, but this is not physical, because the ionization process was just supposed to remove one electron from a single cell. This spurious interaction should not contribute to the IP.

How to resolve the issue of spurious interactions

Physically, N-1 and N+1 electrons systems both have an excess of one charge unit. But, as described before, PBCs are not able to place this excess of charge in just the unit cell where

it belongs. Instead, the PBCs force us to put this same excess of charge in *every* cell across the lattice, leading to spurious interactions between the periodically repeating charges. This spurious interaction has the function of raising the energy for the charged systems, relative to the N electron system, by $|\frac{v_{\mathrm{M}}}{2}|$. This is why it is said that the Madelung constant is twice the energy of a periodic array of unit point charges with a compensating background charge density.

8.6.2 The IP/EA problem

What are IPs?

Ionization involves the removal of an electron from a single location. The ionization potential (IP) quantifies the energy you have to pay to remove the electron. It is a positive quantity if the electron is bound or, in other words, if the electron removal has raised the energy. Thus,

$$\mathrm{IP} \equiv E(N-1) - E(N). \tag{8.154}$$

We know that the shifted Ewald approach predicts

What are EAs?

Electron attachment involves the addition of an electron to a single location. The electron affinity (EA) quantifies the energy released when an electron is added. It is a positive quantity if the extra electron is bound or, in other words, if the electron attachment has lowered the energy. Thus,

$$\mathrm{EA} \equiv E(N) - E(N+1). \tag{8.155}$$

Why Koopmans' theorem holds with PBCs

Koopmans' theorem tells us that IP$= -\epsilon_{\mathrm{HOMO}}$ and EA$= -\epsilon_{\mathrm{LUMO}}$. We know that the shifted Ewald approach, which is designed to be free from finite-size effects, predicts

$$\epsilon_{\mathrm{HOMO}} = (\ldots) + |v_{\mathrm{M}}| \tag{8.156}$$

and

$$\epsilon_{\mathrm{LUMO}} = (\ldots) \tag{8.157}$$

where $(\ldots)$ denotes uncorrected quantities. If we start with the fundamental definition of IP and EA, we should get the same answer.

Implication for the IP and EA

In the last section, we described why simulation with PBCs gives a positive shift of $|\frac{v_{\mathrm{M}}}{2}|$ for the total energy of both N-1 and N+1 systems. Comparing with the definitions 8.154 and 8.155, PBCs *artificially* subtract from the EA and add to the IP, in both situations by $|\frac{v_{\mathrm{M}}}{2}|$. We can correct for this spurious contribution *after* simulation with PBCs has been completed, now adding to the EA and subtracting from the IP by $|\frac{v_{\mathrm{M}}}{2}|$. The spectral function interrogates the density of states in a system, but there are no states at the PBC IPs, because

the PBC IPs include spurious interactions. This is why my PBC simulated IPs were shifted from the HF peaks of the spectral function (which plots $\epsilon_{\text{HOMO}} \equiv -\text{IP}$) by exactly $-|\frac{v_{\text{M}}}{2}|$. So we need to shift all spectral function peaks by $+|\frac{v_{\text{M}}}{2}|$.

8.6.3 Computing band gaps

What QCPBC does

The Coulomb kernel in QCPBC includes divergent contributions from the start, which is equivalent to shifted Ewald. Concretely, if $\{\epsilon_i\}$ and $\{\epsilon_a\}$ are the set of occupied and virtual MO energies, respectively, due to shifted Ewald and QCPBC alike, then

$$\epsilon_i = T_{ii} + V_{ii} + \sum_j [\overline{\langle ij|ij\rangle} - \overline{\langle ij|ja\rangle}] + v_{\text{M}} \tag{8.158}$$

$$\epsilon_a = T_{aa} + V_{aa} + \sum_j [\overline{\langle aj|aj\rangle} - \overline{\langle aj|ja\rangle}] \tag{8.159}$$

where the overline indicates that the ERIs exclude the divergent component. Now, QCPBC computes band gaps as $E_{\text{Gap}} = \epsilon_{\text{LUMO}} - \epsilon_{\text{HOMO}} + |v_{\text{M}}|$. **Note that this Madelung correction is completely separate from what we added to 8.158, where we were just trying to correct for the fact that we excluded the divergent component in the ERIs.** Because the promotion of an electron from the valence to the conduction band is localized to the unit cell and creates no excess charge, one might expect that no *a posteriori* correction should have been needed to obtain the band gap, in a seeming contradiction. However, this is okay because the MO energies are not observable quantities.

The IP/EA perspective

The fundamental definition of the band gap is

$$E_{\text{Gap}} = \text{IP} - \text{EA} \tag{8.160}$$

which comes from that fact that the IP/EA are observable quantities. As found in a previous section, we need to correct PBC computed EAs and IPs by a shift of $+|\frac{v_{\text{M}}}{2}|$ and $-|\frac{v_{\text{M}}}{2}|$, respectively. So if we do a HF simulation with PBCs and the IP (EA) that we get from Koopmans' is $-\epsilon_{\text{HOMO}}$ ($-\epsilon_{\text{LUMO}}$) this means we get $E_{\text{Gap}} = \epsilon_{\text{LUMO}} - \epsilon_{\text{HOMO}} + |v_{\text{M}}|$. This is consistent with what QCPBC does.

8.6.4 Implications for the numerics

What shifted Ewald can and can't do

We know that in PBC simulations, shifted Ewald summation is unique in the sense that unlike other approaches, such as standard Ewald summation, it does not neglect the divergent component of the Coulomb potential. Concretely, the shifted Ewald potential is

$$v_{\text{s}}(G) = \begin{cases} 4\pi/\Omega G^2 & G \neq 0 \\ -v_{\text{M}} & G = 0 \end{cases}, \tag{8.161}$$

where the Madelung potential $v_{\mathrm{M}} = -2 \times 2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_{\mathrm{s}}^{-1}$. Even though shifted Ewald treats the divergent component of the Coulomb potential, it is not able to remedy the unphysical interaction between the periodically repeated holes created during ionization.

When Koopmans' theorem does hold, but why we don't use this paradigm

Formally, we can define the Hamiltonian in the shifted Ewald formalism as

$$\begin{aligned} H = &- \sum_{i=1}^{\hat{N}} \frac{1}{2} \nabla_i^2 - \sum_{i=1}^{\hat{N}} \sum_{A=1}^{M} z_A \left[v_{\mathrm{E}} \left(r_i, R_A \right) - v_{\mathrm{M}} \right] + \sum_{i=1}^{\hat{N}} \sum_{j>i}^{\hat{N}} \left[v_{\mathrm{E}} \left(r_i, r_j \right) - v_{\mathrm{M}} \right] \\ &+ \sum_{A=1}^{M} \sum_{B>A}^{M} z_A z_B \left[v_{\mathrm{E}} \left(R_A, R_B \right) - v_{\mathrm{M}} \right] + \frac{v_{\mathrm{M}}}{2} \left(\hat{N} - \sum_A z_A \right)^2 \end{aligned} \tag{8.162}$$

As indicated, really $\hat{N}$ is a one-body operator; if we treated it as such, this would mitigate the issue described in the previous section and then Koopmans' theorem would hold. However, Tim showed that this approach leads to poor convergence of the orbital energies towards the TDL as L^{-1}. The other option is to approximate $\hat{N}$ as a constant N. Then Koopmans' theorem does not hold for the aforementioned reasons, but we get superior convergence to the TDL as L^{-3}. Since we are working in a finite basis, it is necessary have the latter feature.

Why this is compatible with the way to compute band gaps

Both the N-1 and N+1 systems have an excess of charge and so they both deal with unphysical interactions between the images of the periodically repeating charges. This interaction always has the function of raising the energy for the charged systems relative to the N electron system, which is charge neutral. This leads to a positive shift of $-\frac{v_{\mathrm{M}}}{2}$ both N-1 and N+1 systems. Therefore, the EA gets a shift of $-\frac{v_{\mathrm{M}}}{2}$, because $EA \equiv E(N+1) - E(N)$ and the IP gets a shift of $+\frac{v_{\mathrm{M}}}{2}$, because $IP \equiv -\left(E(N-1) - E(N)\right)$. If band gaps are the difference between the IP and EA, this means band gaps in periodic simulations should be $|v_{\mathrm{M}}|$ larger than if we were in a finite system. Nemo explained to me that QCPBC does compute band gaps by considering the difference between the lowest virtual and highest occupied orbitals. But then he also explained the Madelung shift they do in the K build has the function of shifting the occupied energies down by $|v_{\mathrm{M}}|$, but leaving the virtuals unchanged. This is consistent with what I described above.

Conclusion

So this is why we should always get that the calculated IPs are off from the poles by $-\frac{v_{\mathrm{M}}}{2}$, which is what I have observed. Because this is equivalent to always shifting by $+\frac{v_{\mathrm{M}}}{2}$ relative to the frequency variable, I hypothesize that we should continue doing this in the Landau form for the cumulant, where recall that inside of the integral, we are always computing some product between the self energy and frequency variable.

8.6.5 Simulating the quasi-particle with periodic boundary conditions

Definition of the Madelung potential

It is known that the potential arising

Mean-field theory

To start simple, let us start by considering how mean-field theories, such as HF, describe the quasi-particle. HF involves a self energy contribution, the exchange one, but this is static.

8.6.6 Self-consistent cumulant with second-order self-energy

Basis transformation

We want to evaluate the quantities in the sine-cosine basis, as this is simpler. But in the end, we will want the spectral function in the plane wave basis, and thus we need the Green's function in that basis. Now, the retarded Greens function in the time domain with the broadening factor η can be written as

$$G_{pp}(t) = -i\Theta(t)e^{-t(i\epsilon_p+\eta)}e^{C_{pp}(t)} \tag{8.163}$$

where the cumulant is defined as

$$C_{pp}(t) \equiv i\int \frac{d\omega}{2\pi}\frac{\Sigma_{pp}\left(\omega+\epsilon_p\right)}{(\omega+i\eta)^2}e^{-i\omega t} \tag{8.164}$$

$$= \frac{1}{2}\sum_{iab}\langle pi\,||\,ab\rangle^2\int\frac{d\omega}{2\pi}\frac{ie^{-i\omega t}}{\omega^2\left(\omega-\epsilon_{pi}^{ab}\right)} + \frac{1}{2}\sum_{ija}\langle pa\,||\,ij\rangle^2\int\frac{d\omega}{2\pi}\frac{ie^{-i\omega t}}{\omega^2\left(\omega-\epsilon_{pa}^{ij}\right)} \tag{8.165}$$

$$\overset{RHF}{=}\sum_{iab}\frac{(pa|ib)\left(2(pa|ib)-(pb|ia)\right)}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t}+i\epsilon_{pi}^{ab}t-1\right) \tag{8.166}$$

$$+\sum_{ija}\frac{(pi|aj)\left(2(pi|aj)-(pj|ai)\right)}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t}+i\epsilon_{pa}^{ij}t-1\right) \tag{8.167}$$

with $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$.

Self-consistency

Everything I present in the plots is one-shot. When I speak of GF2+C with analytic frequency integration, I am building the cumulant as

$$C_{pp}(t) = \sum_{iab}\frac{(pa|ib)\left(2(pa|ib)-(pb|ia)\right)}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t}+i\epsilon_{pi}^{ab}t-1\right) \tag{8.168}$$

$$+\sum_{ija}\frac{(pi|aj)\left(2(pi|aj)-(pj|ai)\right)}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t}+i\epsilon_{pa}^{ij}t-1\right)$$

with $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$. As can be seen, this gives qualitatively the same result as GF2+C with numerical frequency integration, where I instead build the cumulant as

$$C_{pp}(t) = \int d\omega \frac{\beta_{pp}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{8.169}$$

with $\beta_{pp}(\omega) = \frac{1}{\pi}|\operatorname{Im}\Sigma_{pp}(\omega + \epsilon_p)|$ with the self-energy defined in chemists notation as

$$\Sigma_{pq}^{(2)}(\omega) = \sum_{iab} \frac{(pa|ib)\Big(2(qa|ib) - (qb|ia)\Big)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{((pi|aj)(2(qi|aj) - (qj|ai)))}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{8.170}$$

or in physicists notation as

$$\Sigma_{pq}^{(2)}(\omega) = \sum_{iab} \frac{\langle pi|ab\rangle \left(2\langle qi|ab\rangle - \langle qi|ba\rangle\right)}{\omega - \epsilon_a - \epsilon_b + \epsilon_i + i\eta} + \sum_{ija} \frac{\langle pa|ij\rangle \left(2\langle qa|ij\rangle - \langle qa|ji\rangle\right)}{\omega - \epsilon_i - \epsilon_j + \epsilon_a + i\eta} \tag{8.171}$$

If I remember correctly, our idea for a self consistent scheme was to solve equation 6.1377 using

$$\Sigma_{pp}(\omega) = \frac{1}{G_{pp}^0(\omega)} - \frac{1}{G_{pp}(\omega)} \tag{8.172}$$

In the initial trials, I have not been getting any useful results with the above. However, I notice that I only use under 1e4 frequency grid points. Should I try to get this scheme to work with something like 1e6 grid points or were you thinking of something different?

This self-energy can then be used to compute a new cumulant $C_{pp}(t)$ as

$$C_{pp}(t) = i\int \frac{d\omega}{2\pi}\frac{\Sigma_{pp}\left(\omega+\epsilon_p\right)}{(\omega+i\eta)^2}e^{-i\omega t} \tag{8.173}$$

$$= i\int \frac{d\omega}{2\pi}\frac{\omega+i\eta-\frac{1}{G_{pp}(\omega+\epsilon_p)}}{(\omega+i\eta)^2}e^{-i\omega t} \tag{8.174}$$

$$= i\int \frac{d\omega}{2\pi}\frac{\omega+i\eta}{(\omega+i\eta)^2}e^{-i\omega t} - i\int \frac{d\omega}{2\pi}\frac{1}{G_{pp}(\omega+\epsilon_p)(\omega+i\eta)^2}e^{-i\omega t} \tag{8.175}$$

$$= i\int \frac{d\omega}{2\pi}\frac{e^{-i\omega t}}{\omega+i\eta} - i\int \frac{d\omega}{2\pi}\frac{1}{G_{pp}(\omega+\epsilon_p)(\omega+i\eta)^2}e^{-i\omega t} \tag{8.176}$$

$$= \Theta(t)e^{-\eta t} - i\int \frac{d\omega}{2\pi}\frac{1}{G_{pp}(\omega+\epsilon_p)(\omega+i\eta)^2}e^{-i\omega t} \tag{8.177}$$

where we can use some quadrature to perform the latter integral. This cumulant can then be used to compute $G_{pp}(t)$ using the usual recipe, and then Fourier transformed to obtain $G_{pp}(\omega)$, which can then be used to compute a new self-energy, and so on, until convergence is reached. We still need to determine what measure will be used to determine if self-consistency has been reached.

$$\begin{pmatrix}|\boldsymbol{K}_p\rangle\\|-\boldsymbol{K}_p\rangle\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & i\\1 & -i\end{bmatrix}\begin{pmatrix}|\alpha_p\rangle\\|\beta_p\rangle\end{pmatrix} \implies |\boldsymbol{K}_p\rangle = \frac{1}{\sqrt{2}}\left(|\alpha_p\rangle + i|\beta_p\rangle\right) \tag{8.178}$$

So the self energy (and also the cumulant as well as the Green's function) can be expressed as

$$\Sigma_{pp}(\omega) \equiv \langle p|\hat{\Sigma}(\omega)|p\rangle \tag{8.179}$$

$$= \frac{1}{2}\left(\langle\alpha_p| + -i\langle\beta_p|\right)\hat{\Sigma}(\omega)\left(|\alpha_p\rangle + i|\beta_p\rangle\right) \tag{8.180}$$

$$= \frac{1}{2}\left(\Sigma_{\alpha_p\alpha_p}(\omega) + \Sigma_{\beta_p\beta_p}(\omega) + i\Sigma_{\alpha_p\beta_p}(\omega) - i\Sigma_{\beta_p\alpha_p}(\omega)\right) \tag{8.181}$$

Suppose that p is the Fermi state, which in our case has wavevector $(1,0,0)$ and the complement is $(-1,0,0)$; in the plane wave basis, they may have non-consecutive indices, like 6 and 1, respectively. The sine-cosine basis is expressed as linear combinations of the two, so the sine and cosine states will now be in a consecutive order, say 5 and 6. I need to check that nothing changes if instead they occupy indices 1 and 2. Also, it is worth checking whether $\Sigma_{\alpha\beta} = \Sigma_{\beta\alpha}$ in the sine-cosin basis. If so, check if $\Sigma_{\alpha\alpha} = \Sigma_{\beta\beta}$ because, if so, that means the self-energy in the plane-wave basis might be equivalent to the self-energy in thee sine-cosine basis. To derive a form for $C^{(2)}_{pp}(t)$, start by equating the cumulant ansatz for the Green's

function the Dyson equation, which gives

$$C_{pp}^{(2)}(t) \equiv i \int \frac{d\omega}{2\pi} \frac{\tilde{\Sigma}_{pp}^{c,(2)}(\omega+\epsilon_p)}{(\omega+i\eta)^2} e^{-i\omega t} \tag{8.182}$$

$$= \frac{1}{2}\sum_{iab} \langle pi \,||\, ab\rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2\left(\omega-\epsilon_{pi}^{ab}\right)} + \frac{1}{2}\sum_{ija} \langle pa \,||\, ij\rangle^2 \int \frac{d\omega}{2\pi} \frac{ie^{-i\omega t}}{\omega^2\left(\omega-\epsilon_{pa}^{ij}\right)} \tag{8.183}$$

$$\overset{RHF}{=} \sum_{iab} \frac{(pa|ib)\left(2(pa|ib)-(pb|ia)\right)}{\left(\epsilon_{pi}^{ab}\right)^2}\left(e^{-i\epsilon_{pi}^{ab}t} + i\epsilon_{pi}^{ab}t - 1\right) \tag{8.184}$$

$$+ \sum_{ija} \frac{(pi|aj)\left(2(pi|aj)-(pj|ai)\right)}{\left(\epsilon_{pa}^{ij}\right)^2}\left(e^{-i\epsilon_{pa}^{ij}t} + i\epsilon_{pa}^{ij}t - 1\right) \tag{8.185}$$

$$\overset{SC}{=} \sum_{\tilde{p}}^{p_{\boldsymbol{K}},p_{-\boldsymbol{K}}} \left(\sum_{iab} \frac{(\tilde{p}i|ab)\left(2(\widetilde{p}i|ab)-(\widetilde{p}b|ia)\right)}{\left(\epsilon_{\widetilde{p}i}^{ab}\right)^2}\left(e^{-i\epsilon_{\widetilde{p}i}^{ab}t} + i\epsilon_{\widetilde{p}i}^{ab}t - 1\right) \right. \tag{8.186}$$

$$\left. + \sum_{ija} \frac{(\tilde{p}i|aj)\left(2(\widetilde{p}i|aj)-(\widetilde{p}j|ai)\right)}{\left(\epsilon_{\widetilde{p}a}^{ij}\right)^2}\left(e^{-i\epsilon_{\widetilde{p}a}^{ij}t} + i\epsilon_{\widetilde{p}a}^{ij}t - 1\right) \right) \tag{8.187}$$

where $\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i$ and $\epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a$. At this point we can stop and notice that if we want to determine use the sine-cosine basis for the evaluation of the self energy in the definition of the cumualant, we just need to do

$$\Sigma_{pp}(\omega) = \langle p|\hat{\Sigma}(\omega)|p\rangle = \sum_{\alpha,\beta} U_{p\alpha}\Sigma_{\alpha\beta}^{(\mathrm{sc})}(\omega)U_{p\beta}^*, \tag{8.188}$$

Now, we are told that this unitary transformation that allows us to move between the plane wave and sine-cosine basis is given by

$$\mathbf{U} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \implies \begin{bmatrix} |\mathbf{K},s\rangle \\ |\mathbf{K},c\rangle \end{bmatrix} = \mathbf{U}^\dagger \begin{bmatrix} |\mathbf{K}\rangle \\ |-\mathbf{K}\rangle \end{bmatrix} \tag{8.189}$$

for a $\mathbf{K} \neq 0$ state from the plane wave basis.

8.6.7 Basic relationships

The Wigner-Seitz radius r_s is defined as the radius of a sphere that contains one electron on average. In a finite simulation cell of length L containing N electrons, we can relate r_s to L and N via the following steps.

$$\frac{4\pi}{3}r_s^3 = \frac{1}{n} = \frac{L^3}{N} \tag{8.190}$$

$$\implies n = \frac{3}{4\pi r_s^3} \tag{8.191}$$

$$\tag{8.192}$$

We know that the Fermi wavevector of a uniform 3D gas has the definition:

$$k_F = \left(3\pi^2 n\right)^{1/3} \tag{8.193}$$

$$= \frac{(9\pi/4)^{1/3}}{r_s} \tag{8.194}$$

For periodic boundary conditions, the allowed single-particle wavevectors are:

$$\mathbf{k} = \frac{2\pi}{L}\mathbf{n}, \quad \mathbf{n} = (n_x, n_y, n_z), \quad n_i \in \mathbb{Z} \tag{8.195}$$

So the spacing of the k-grid is $\Delta k = 2\pi/L$. Using the definition that at $T = 0$, all states with $|\mathbf{k}| \leq k_F$ are occupied, with two electrons (spin up/down) per k-point, the number of electrons is

$$N = 2 \times \frac{L^3}{(2\pi)^3} \times \frac{4\pi}{3} k_F^3 \tag{8.196}$$

$$= \frac{L^3}{3\pi^2} k_F^3 \tag{8.197}$$

$$\implies \frac{L^3}{3\pi^2} k_F^3 = \frac{3L^3}{4\pi r_s^3} \tag{8.198}$$

$$\implies k_F = \left(\frac{9\pi}{4r_s^3}\right)^{1/3} \tag{8.199}$$

This implies that the following definition should hold in my code:

$$\frac{2\pi}{L} |\mathbf{n}_{\max}| = \frac{(9\pi/4)^{1/3}}{r_s}, \quad \mathbf{n}_{\max} = (n_{x,\max}, n_{y,\max}, n_{z,\max}) \quad n_{i,\max} \in \mathbb{Z} \tag{8.200}$$

where $\mathbf{n}_{\max}$ is the highest occupied wavevector in my plane wave basis.

This is the algebraic identity your assert is verifying. Even before doing HF, I generate my basis of plane waves, and their wave vectors in 3D are just a list of integers, three entries long. From HF, I get a set of MO coefficients $C_{\boldsymbol{k}_\mu \boldsymbol{k}_p}$ and orbital energies $\epsilon_{\boldsymbol{k}_p}$.

8.7 RPA

For the UEG, I am first looping over all momentum transfer vectors $\boldsymbol{k}_q$. To learn which excitations are valid for the given $\boldsymbol{k}_q$, we first loop over occupied states $\boldsymbol{k}_i$. Virtual states $\boldsymbol{k}_a$ are already determined by that point; we compute $\boldsymbol{k}_a \equiv \boldsymbol{k}_i + \boldsymbol{k}_q$, and based of of whether it is a valid virtual index, we either add it to our list or continue. The criterion for validity is that the wave vector must be outside the Fermi sphere, but still within our plane wave basis, i.e. within the KE cutoff. We do the same for the second pair of indices $\boldsymbol{k}_j, \boldsymbol{k}_b$. Then, we can form the Casida eigenproblem for each $\boldsymbol{k}_q$.

$$\begin{pmatrix} \boldsymbol{A}(\boldsymbol{k}_q) & \boldsymbol{B}(\boldsymbol{k}_q) \\ -\boldsymbol{B}(\boldsymbol{k}_q) & -\boldsymbol{A}(\boldsymbol{k}_q) \end{pmatrix} \begin{pmatrix} \boldsymbol{X}(\boldsymbol{k}_q) & \boldsymbol{Y}(\boldsymbol{k}_q) \\ \boldsymbol{Y}(\boldsymbol{k}_q) & \boldsymbol{X}(\boldsymbol{k}_q) \end{pmatrix} \begin{pmatrix} \boldsymbol{\Omega}(\boldsymbol{k}_q) & \mathbf{0} \\ \mathbf{0} & -\boldsymbol{\Omega}(\boldsymbol{k}_q) \end{pmatrix} = \begin{pmatrix} \boldsymbol{X}(\boldsymbol{k}_q) & \boldsymbol{Y}(\boldsymbol{k}_q) \\ \boldsymbol{Y}(\boldsymbol{k}_q) & \boldsymbol{X}(\boldsymbol{k}_q) \end{pmatrix} \tag{8.201}$$

The $\boldsymbol{A}(\boldsymbol{k}_q)$ and $\boldsymbol{B}(\boldsymbol{k}_q)$ matrices are defined as

$$A_{\boldsymbol{k}_i\boldsymbol{k}_a,\,\boldsymbol{k}_b\boldsymbol{k}_j}(\boldsymbol{k}_q) = (\epsilon_{\boldsymbol{k}_a} - \epsilon_{\boldsymbol{k}_i})\,\delta_{\boldsymbol{k}_i\boldsymbol{k}_j}\delta_{\boldsymbol{k}_a\boldsymbol{k}_b} + \frac{4\pi}{|\boldsymbol{k}_q|^2} \tag{8.202}$$

$$B_{\boldsymbol{k}_i\boldsymbol{k}_a,\,\boldsymbol{k}_j\boldsymbol{k}_b}(\boldsymbol{k}_q) = \frac{4\pi}{|\boldsymbol{k}_q|^2} \tag{8.203}$$

After solving the Casida eigenvalue problem, I perform the bioorthogonalization procedure, which ensures that

$$(\boldsymbol{X}(\boldsymbol{k}_q) - \boldsymbol{Y}(\boldsymbol{k}_q))^\dagger\,(\boldsymbol{X}(\boldsymbol{k}_q) + \boldsymbol{Y}(\boldsymbol{k}_q)) = \boldsymbol{I} \tag{8.204}$$

Now, I have a list of excitation energies Ω and corresponding right eigenvectors $\boldsymbol{X}, \boldsymbol{Y}$ for each momentum transfer $\boldsymbol{k}_q$. But for the self energy, we actually need rather the transition densities $\rho(\boldsymbol{k}_q)$, which are lists (with length μ) of scalars, defined as

$$\rho_\nu(\boldsymbol{k}_q) = \sum_{\boldsymbol{k}_i\boldsymbol{k}_a} \left(X^\nu_{\boldsymbol{k}_i\boldsymbol{k}_a}(\boldsymbol{k}_q) + Y^\nu_{\boldsymbol{k}_i\boldsymbol{k}_a}(\boldsymbol{k}_q)\right). \tag{8.205}$$

So to get them, I just need to contract over the indices $\boldsymbol{k}_i, \boldsymbol{k}_a$.

Analytic dielectric function

The expression is:

$$\epsilon(\mathbf{q},\omega) = 1 - v(\mathbf{q})\chi_0(\mathbf{q},\omega) \tag{8.206}$$

$$= 1 + \frac{\alpha r_s}{8\pi q^3}[H(q+u/q) - H(q-u/q)] \tag{8.207}$$

where $H(q) = 2q + (1-q^2)\ln[(q+1)/(q-1)], \alpha = (4/9\pi)^{1/3}$. Here we followed the notation of Ref. 9 where q denotes momenta in units of $2k_\mathrm{F}$ and u are energies in units of 4μ; thus, $q = |\mathbf{q}|/(2k_\mathrm{F})$ and $u = \omega/(4\mu)$. The Wigner-Seitz radius r_s is defined by $4\pi r_s^3/3 = 1/n$, where n is the electron density.

Verification

We can compute the RPA correlation energy via the plasmon formula:

$$E_c^{RPA} = \frac{1}{2}\,\mathrm{Tr}(\sum_{\boldsymbol{q}} \boldsymbol{\Omega}(\boldsymbol{q}) - \boldsymbol{A}(\boldsymbol{q})) \tag{8.208}$$

$$\tag{8.209}$$

and then we just have to divide by the number of electrons to get the correlation energy per electron.

8.8 GW

8.8.1 Correlation self energy

Derivation from scratch

We can start from the definition of the GW self energy, which is a convolution of the Green's function and the screened Coulomb interaction:

$$\Sigma(\mathbf{k},\omega) = \frac{i}{2\pi}\int d\omega' \frac{d^3q}{(2\pi)^3} G\left(\mathbf{k}-\mathbf{q},\omega-\omega'\right) W\left(\mathbf{q},\omega'\right) \tag{8.210}$$

The screened Coulomb interaction in the RPA is given by

$$\begin{aligned}
W(\mathbf{q},\omega) &= v(\mathbf{q}) + v(\mathbf{q})\chi_0(\mathbf{q},\omega)v(\mathbf{q}) + \ldots && (8.211)\\
&= \frac{v(\mathbf{q})}{1-v(\mathbf{q})\chi_0(\mathbf{q},\omega)} \equiv \frac{v(\mathbf{q})}{\varepsilon(\mathbf{q},\omega)} && (8.212)\\
&= v(\mathbf{q}) + v^2(\mathbf{q})\chi_{RPA}(\mathbf{q},\omega) && (8.213)\\
\implies W_p(\mathbf{q},\omega) \equiv W(\mathbf{q},\omega) - v(\mathbf{q}) &= v^2(\mathbf{q})\chi_{RPA}(\mathbf{q},\omega) && (8.214)\\
&= \sum_\nu \frac{2\omega_\nu v^2(\mathbf{q})|\rho_\nu(\mathbf{q})|^2}{\omega^2-\omega_\nu^2} && (8.215)
\end{aligned}$$

where it can be shown that $\varepsilon^{-1}(\mathbf{q},\omega) = 1 + v(\mathbf{q})\chi_{RPA}(\mathbf{q},\omega)$. At this point, we can just quote the final result of Hedin for a retarded self energy in the GW approximation:

$$\Sigma_{pol}\left(x,x';\omega\right) = \sum_{i,s\neq 0} \frac{V^s(\boldsymbol{r})V^s\left(\boldsymbol{r}'\right)\phi_i(x)\phi_i^*\left(x'\right)}{\omega+\omega_s\,\mathrm{sgn}\left(\mu-\varepsilon_i\right)-\varepsilon_i} \tag{8.216}$$

$$\implies \Sigma(\boldsymbol{k},\omega) = \sum_{q,\nu} \frac{v^2(\mathbf{q})|\rho_\nu(\mathbf{q})|^2}{\omega+\Omega_\nu\,\mathrm{sgn}(\mu-\epsilon_{\boldsymbol{k}-\boldsymbol{q}})-\epsilon_{\boldsymbol{k}-\boldsymbol{q}}+i\eta} \tag{8.217}$$

since we know that we will have a diagonal self energy in momentum space for the UEG. We defined

$$\begin{aligned}
V^s(\mathbf{r}) \to V^\nu(\mathbf{q}) &= v(\mathbf{q})\rho_\nu(\mathbf{q}), && (8.218)\\
\omega_s &\to \Omega_\nu && (8.219)
\end{aligned}$$

In relation to our Casida eigenproblem, we can identify that the excitation energies Ω_ν are just the eigenvalues from Eqn. 9.209, and the transition densities $\rho_\nu(\boldsymbol{q})$ are given by

$$\rho_\nu(\boldsymbol{q}) = \sum_{\boldsymbol{k}_i\boldsymbol{k}_a}(X^\nu_{\boldsymbol{k}_i\boldsymbol{k}_a}(\boldsymbol{q}) + Y^\nu_{\boldsymbol{k}_i\boldsymbol{k}_a}(\boldsymbol{q})). \tag{8.220}$$

So the expression I find has one more factor of the ERI then the one I have written below.

$$\Sigma^{\mathrm{corr}}(\boldsymbol{k}_p,\omega) = \sum_{\boldsymbol{k}_q}\sum_\nu^{\mathrm{RPA}} \frac{4\pi}{|\boldsymbol{k}_q|^2}|\rho_\nu(\boldsymbol{k}_q)|^2\left[\frac{f_{\boldsymbol{k}_p-\boldsymbol{k}_q}}{\omega+\Omega_\nu(\boldsymbol{k}_q)-\epsilon_{\boldsymbol{k}_p-\boldsymbol{k}_q}+i\eta} + \frac{1-f_{\boldsymbol{k}_p-\boldsymbol{k}_q}}{\omega-\Omega_\nu(\boldsymbol{k}_q)-\epsilon_{\boldsymbol{k}_p-\boldsymbol{k}_q}+i\eta}\right] \tag{8.221}$$

8.8.2 Spectral function

Finally, the spectral function is given by

$$A(\boldsymbol{k}_p, \omega) = \frac{1}{\pi} \frac{|\mathrm{Im}\, \Sigma(\boldsymbol{k}_p, \omega)|}{\left(\omega - \epsilon_{\boldsymbol{k}_p} - \mathrm{Re}\, \Sigma(\boldsymbol{k}_p, \omega)\right)^2 + \left(\mathrm{Im}\, \Sigma(\boldsymbol{k}_p, \omega)\right)^2} \tag{8.222}$$

8.8.3 Self-consistent GW

Self-energy

At a given iteration n, including the one-shot level, the correlation self-energy is always constructed as

$$\Sigma_{k,c}^{n,R}[\eta_\Sigma](t) = \theta(t) \left[-\sum_{l,I} \left| M_{k,l}^{I,(n)} \right|^2 G_l^{<,(n)}(t) e^{+i\Omega_I^{(n)} t} e^{-\eta_\Sigma t} + \sum_{l,I} \left| M_{k,l}^{I,(n)} \right|^2 G_l^{>,(n)}(t) e^{-i\Omega_I^{(n)} t} e^{-\eta_\Sigma t} \right]. \tag{8.223}$$

Spectral function

And we get the spectral function as

$$A_k^{(n)}(\omega) = \frac{1}{\pi} \frac{|\mathrm{Im}\, \Sigma_k^{n,R}[\eta_\Sigma](\omega)| + \eta_G}{\left(\omega - \epsilon_k^0 - \mathrm{Re}\, \Sigma_k^{n,R}[\eta_\Sigma](\omega)\right)^2 + \left(|\mathrm{Im}\, \Sigma_k^{n,R}[\eta_\Sigma](\omega)| + \eta_G\right)^2} \tag{8.224}$$

$$= \frac{1}{\pi} \frac{|\mathrm{Im}\, \Sigma_{k,c}^{n,R}[\eta_\Sigma](\omega)| + \eta_G}{\left(\omega - \epsilon_k^0 - \left(\mathrm{Re}\, \Sigma_{k,c}^{n,R}[\eta_\Sigma](\omega) + \Sigma_k^{x,(n)}\right)\right)^2 + \left(|\mathrm{Im}\, \Sigma_{k,c}^{n,R}[\eta_\Sigma](\omega)| + \eta_G\right)^2} \tag{8.225}$$

For $n \geq 0$, we have $\Sigma_k^{x,(n)} = n_k^{(n)} v_m - \sum_{\boldsymbol{q} \neq \boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}}^{(n)} v(\boldsymbol{q})$, where $v_m = -\frac{\alpha_0}{L}$. The interpretation is that the first term is Madelung potential at iteration n, and the second term is the exchange potential due to Ewald summation at iteration n. The spectral function is then used to compute the chemical potential $\mu^{(n)}$ and the lesser and greater Green's functions $G_k^{<,(n)}(t)$ and $G_k^{>,(n)}(t)$, which are then used to compute the correlation self-energy at iteration $n+1$.

Green's function

This gives us a recipe to update the chemical potential $\mu^{(n)}$ for $n > 0$, where $\mu^{(0)} = \frac{\left(k^2/2 + \Gamma_k^1\right)_{\mathrm{HOMO}} + \left(k^2/2 + \Gamma_k^1\right)_{\mathrm{LUMO}}}{2}$ as

$$N/2 = \sum_k \int_{-\infty}^{\mu^{(n)}} A_k^{(n)}(\omega) d\omega, \tag{8.226}$$

where we can take $N/2$ as the number of occupied bands in our spin-restricted UEG. Once we have the chemical potential, we determine the lesser and greater Green's functions as

$$G_k^{<,(n)}(t) = i\int_{-\infty}^{\mu^{(n)}} d\omega A_k^{(n)}(\omega)e^{-i\omega t} \tag{8.227}$$

$$G_k^{>,(n)}(t) = -i\int_{\mu^{(n)}}^{\infty} d\omega A_k^{(n)}(\omega)e^{-i\omega t}. \tag{8.228}$$

Recall that we determine the Green's functions at the zeroth iteration by

$$G_{0,n}^R(t) = -i\Theta(t)e^{-i\varepsilon_n^{\mathrm{LDA}}t} \tag{8.229}$$

$$\implies G_{0,i}^<(t) = ie^{-i\varepsilon_i^{\mathrm{LDA}}t}, \quad G_{0,a}^>(t) = -ie^{-i\varepsilon_a^{\mathrm{LDA}}t} \quad \forall t > 0 \tag{8.230}$$

$$\implies \Sigma_{m,c}^R(t) = -i\theta(t)\left[\sum_{i,I}\left|M_{m,i}^I\right|^2 e^{-i(\varepsilon_i^{\mathrm{LDA}}-\Omega_I)t} + \sum_{a,I}\left|M_{m,a}^I\right|^2 e^{-i(\varepsilon_a^{\mathrm{LDA}}+\Omega_I)t}\right] \tag{8.231}$$

$$\implies \Sigma_{m,c}^R(\omega) = \sum_{i,I}\frac{\left|M_{m,i}^I\right|^2}{\omega - \varepsilon_i^{\mathrm{LDA}} + \Omega_I + i\eta} + \sum_{a,I}\frac{\left|M_{m,a}^I\right|^2}{\omega - \varepsilon_a^{\mathrm{LDA}} - \Omega_I + i\eta}. \tag{8.232}$$

8.8.4 Screened interaction

For scGW0, we take $M^{(n)} = M^{(0)}$ and $\Omega^{(n)} = \Omega^{(0)}$ for all n. This has the interpretation of us freezing the interaction at the RPA level rather than updating it. For scGW, define

$$B_{\boldsymbol{q}}^{(n)}(\Omega) \equiv -\frac{1}{\pi}\operatorname{Im} W_c^{R,(n)}(\boldsymbol{q},\Omega) \tag{8.233}$$

$$= \frac{v_{\boldsymbol{q}}^2\, S_{\boldsymbol{q}}^{0,(n)}(\Omega)}{\left[1 - v_{\boldsymbol{q}}\operatorname{Re} P_{\boldsymbol{q}}^{(n)}(\Omega)\right]^2 + \left[\pi v_{\boldsymbol{q}} S_{\boldsymbol{q}}^{0,(n)}(\Omega)\right]^2} \tag{8.234}$$

with

$$S_{\boldsymbol{q}}^{0,(n)}(\Omega) \equiv -\frac{1}{\pi}\mathrm{Im}\, P^{R,(n)}(\boldsymbol{q},\Omega) = \frac{2}{V}\sum_{\boldsymbol{k}}\int d\epsilon\; A_{\boldsymbol{k}}^{<,(n)}(\epsilon)\, A_{\boldsymbol{k}+\boldsymbol{q}}^{>,(n)}(\epsilon+\Omega), \tag{8.235}$$

$$\operatorname{Re} P_{\boldsymbol{q}}^{(n)}(\Omega) = \mathcal{P}\int_{-\infty}^{\infty} d\Omega'\,\frac{S_{\boldsymbol{q}}^{0,(n)}(\Omega')}{\Omega - \Omega'}, \qquad S_{\boldsymbol{q}}^0(-\Omega) = -S_{\boldsymbol{q}}^0(\Omega). \tag{8.236}$$

Then,

$$\Sigma_{k,c}^{n,R}[\eta_\Sigma](t) \;=\; \frac{\theta(t)}{V}\int_0^\infty d\Omega \sum_{\boldsymbol{l}\neq\boldsymbol{k}} B_{\boldsymbol{k}-\boldsymbol{l}}^{(n)}(\Omega)\left[\; -\; G_l^{<,(n)}(t)\, e^{+i\Omega t} \;+\; G_l^{>,(n)}(t)\, e^{-i\Omega t}\right] e^{-\eta_\Sigma t}, \tag{8.237}$$

with

$$G_l^{<,(n)}(t) = i\int_{-\infty}^{\mu^{(n)}} d\omega\, A_l^{(n)}(\omega)e^{-i\omega t}, \qquad G_l^{>,(n)}(t) = -i\int_{\mu^{(n)}}^{\infty} d\omega\, A_l^{(n)}(\omega)e^{-i\omega t}. \tag{8.238}$$

Specification to scGW0

Plugging in the identity

$$\frac{1}{V}B^{(0)}_{\mathrm{k}-1}(\Omega) = \sum_I \left|M^{I,(0)}_{k,l}\right|^2 \delta\left(\Omega - \Omega^{(0)}_I\right), \tag{8.239}$$

we get

$$\Sigma^{n,R}_{k,c}[\eta_\Sigma](t) = \theta(t)\left[-\sum_{l,I}\left|M^{I,(n)}_{k,l}\right|^2 G^{<,(n)}_l(t)e^{+i\Omega^{(n)}_I t}e^{-\eta_\Sigma t} + \sum_{l,I}\left|M^{I,(n)}_{k,l}\right|^2 G^{>,(n)}_l(t)e^{-i\Omega^{(n)}_I t}e^{-\eta_\Sigma t}\right]. \tag{8.240}$$

8.8.5 scGW

8.8.6 Self-consistent GW

Formulation in the TDL

The full self-energy is

$$\Sigma^{(n)}(\mathbf{k},\omega) = \Sigma^{(n)}_{HF}(\mathbf{k}) + \Sigma^{(n),c}(\mathbf{k},\omega) \tag{8.241}$$

with

$$\mathbf{\Sigma}^{(n)}_{HF}(\mathbf{k}) = -\sum\!\!\!\!\!\!\int_{\mathbf{q}} v(\mathbf{k}+\mathbf{q})n^{(n)}_{\mathbf{q}} \tag{8.242}$$

where

$$n^{(n)}_{\mathrm{k}} = \int_{-\infty}^{\mu^{(n)}} A^{(n)}(\mathbf{k},\omega)d\omega. \tag{8.243}$$

But during the self-consistent cycle, we should update the chemical potential $\mu^{(n)}$, such that it always takes the form

$$N/2 = \sum_k \int_{-\infty}^{\mu^{(n)}} A^{(n)}_k(\omega)d\omega. \tag{8.244}$$

Note that in the initial iteration this reduces to $\mu^{(0)} = \frac{\left(k^2/2+\Sigma^{(0)}_{HF}\right)_{\mathrm{HOMO}}+\left(k^2/2+\Sigma^{(0)}_{HF}\right)_{\mathrm{LUMO}}}{2}$. The correlation self-energy gets computed as

$$\Sigma^{(n),c}(\mathbf{k},\omega) = \int_C \frac{\Gamma^{(n)}(\mathbf{k},\omega')}{\omega-\omega'}d\omega' \tag{8.245}$$

where

$$\Gamma^{(n)}(\mathbf{k},\omega) = \sum_{\mathbf{q}}\int_0^\omega d\omega' A^{(n)}(\mathbf{k}+\mathbf{q},\omega-\omega')B^{(n)}(\mathbf{q},\omega'). \tag{8.246}$$

If we were to do a calculation where we froze the interaction at the RPA level, we would actually have $B^{(n)} \equiv B_0$, where

$$B_0(q,\omega) = -\frac{1}{\pi}\frac{v^2(q)\,\mathrm{Im}\,\chi_0(q,\omega)}{|1-v(q)\chi_0(q,\omega)|^2}\,\mathrm{sgn}(\omega) \tag{8.247}$$

with χ_0 being the Lindhard function, whose value should we take from his original paper. If we want to self-consistently update the interaction, we would need to do

$$B^{(n)}(\mathbf{q},\omega) = \frac{v^2(\mathbf{q}) S_0^{(n)}(\mathbf{q},\omega)}{\left|1 - v(\mathbf{q}) P_0^{(n)}(\mathbf{q},\omega)\right|^2} \tag{8.248}$$

with

$$P_0^{(n)}(\mathbf{q},\omega) = \int_C \frac{S_0^{(n)}(\mathbf{q},\omega')}{\omega - \omega'} d\omega' \tag{8.249}$$

and

$$S_0^{(n)}(\mathbf{q},\omega) = 2\sum_{\mathbf{k}} \int_0^{\omega} A^{(n)}(\mathbf{k},\omega'-\omega) A^{(n)}(\mathbf{k}+\mathbf{q},\omega') d\omega'. \tag{8.250}$$

We should verify numerically that in our first iteration $B^{(0)} \equiv B_0$. This implies that $S_0^{(0)}(\mathbf{q},\omega) = -\frac{1}{\pi}\operatorname{Im}\chi_0(\mathbf{q},\omega)\operatorname{sgn}(\omega)$, which we should verify numerically. In the paper, they also suggest checking the identities $\int_{-\infty}^{\infty}\Gamma(\mathbf{k},\omega)d\omega = \sum_{\mathbf{q}}\int_0^{\infty} B_0(\mathbf{q},\omega)d\omega$ and $\int_0^{\infty} B(\mathbf{q},\omega)\omega d\omega = v^2(\mathbf{q})\int_0^{\infty} S_0(\mathbf{q},\omega)\omega d\omega$. Should we check these as well; I feel that the former one should hold only in the initial iteration, but the latter one may hold throughout the self-consistency cycle? Then we compute the spectral function via

$$A^{(n)}(\mathbf{k},\omega) = \frac{1}{\pi}\frac{\left|\operatorname{Im}\Sigma^{(n)}(\mathbf{k},\omega)\right|}{\left|\omega - \epsilon_{\mathbf{k}} - \operatorname{Re}\Sigma^{(n)}(\mathbf{k},\omega)\right|^2 + \left|\operatorname{Im}\Sigma^{(n)}(\mathbf{k},\omega)\right|^2}. \tag{8.251}$$

What value should we use for $\epsilon_{\mathbf{k}}$ during the self consistency cycle? Do we just take it to be the free electron value all the time?

Formulation in the finite basis

At iteration n :

1. Start with $G^{(n)}(\mathbf{k},\omega)$ and compute

$$A^{(n)}(\mathbf{k},\omega) = -\frac{1}{\pi}\operatorname{Im} G^{(n),R}(\mathbf{k},\omega). \tag{8.252}$$

2. Determine $\mu^{(n)}$ from

$$N/2 = \sum_{\mathbf{k}} \int_{-\infty}^{\mu^{(n)}} A^{(n)}(\mathbf{k},\omega) d\omega. \tag{8.253}$$

3. Update

$$\Sigma_x^{(n)}(\mathbf{k}) = -\sum_{\mathbf{q}} v(\mathbf{k}+\mathbf{q}) n_{\mathbf{q}}^{(n)}. \tag{8.254}$$

4. Construct the bubble

$$P^{(n)} = -iG^{(n)}G^{(n)} \tag{8.255}$$

$$\implies P^{(n),R}(\mathbf{q},\omega) = 2\sum_{\mathbf{k}} \int_{-\infty}^{\infty} d\epsilon \int_{-\infty}^{\infty} d\epsilon' A_{\mathbf{k}}^{(n)}(\epsilon) A_{\mathbf{k}+\mathbf{q}}^{(n)}(\epsilon') \frac{f^{(n)}(\epsilon) - f^{(n)}(\epsilon')}{\omega + \epsilon - \epsilon' + i\eta} \tag{8.256}$$

where at zero temperature $f^{(n)}(\epsilon) = \Theta\left(\mu^{(n)} - \epsilon\right)$.

5. Solve

$$W_c^{(n),R}(\mathbf{q},\omega) = \left[1 - v(\mathbf{q})P^{(n),R}(\mathbf{q},\omega)\right]^{-1} v(\mathbf{q}) - v(\mathbf{q}) \tag{8.257}$$

$$\implies B^{(n)}(\mathrm{q},\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,W_c^{(n),R}(\mathrm{q},\omega) \tag{8.258}$$

6. Compute

$$\Sigma_c^{(n+1),R}(\mathbf{k},\omega) = \sum_{\mathbf{q}} \int_{-\infty}^{\infty} d\epsilon\, A^{(n)}_{\mathbf{k}-\mathbf{q}}(\epsilon) \int_0^{\infty} d\Omega\, B^{(n)}(\mathbf{q},\Omega) \times \left[\frac{f^{(n)}(\epsilon)}{\omega - \epsilon + \Omega + i\eta} + \frac{1 - f^{(n)}(\epsilon)}{\omega - \epsilon - \Omega + i\eta}\right]. \tag{8.259}$$

8. Solve the Dyson equation

$$G^{(n+1),R}(\mathbf{k},\omega) = \left[G_0^{-1,R}(\mathbf{k},\omega) - \Sigma_x^{(n)}(\mathbf{k}) - \Sigma_c^{(n+1,R)}(\mathbf{k},\omega)\right]^{-1}. \tag{8.260}$$

where

$$\left[G_0^R(\mathbf{k},\omega)\right]^{-1} = \omega - \frac{k^2}{2} + i\eta. \tag{8.261}$$

or

$$\left[G_0^R(\mathbf{k},\omega)\right]^{-1} = \omega + \mu^{(n)} - \frac{k^2}{2} + i\eta. \tag{8.262}$$

I believe that we should use the first definition for the non-interacting Green's function if we are using an absolute energy axis.

9. Repeat until convergence. Then plot

$$A(\mathbf{k},\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G^R(\mathbf{k},\omega). \tag{8.263}$$

8.8.7 GW+2SOSEX-psd

I have also implemented GW+2SOSEX-psd. This is a vertex-corrected version (approximating G3W2) recently proposed by Bruneval et al. that is guaranteed to give a positive spectral function. The self-energy is given in HF MO basis, with an infinitesimal broadening η_Σ, by

$$\Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) = \sum_{ks} \frac{\left(w_{s,\mathrm{HF}}^{pk} + \tilde{w}_{s,\mathrm{HF}}^{pk}\right)^2}{\omega - \epsilon_k^{\bar{\mathrm{HF}}} + \Omega_s^{\mathrm{HF}} - \mathrm{i}\eta_\Sigma} + \sum_{cs} \frac{\left(w_{s,\mathrm{HF}}^{pc} + \tilde{w}_{s,\mathrm{HF}}^{pc}\right)^2}{\omega - \epsilon_c^{\bar{\mathrm{HF}}} - \Omega_s^{\mathrm{HF}} + \mathrm{i}\eta_\Sigma} \tag{8.264}$$

with

$$\tilde{w}_{s,\mathrm{HF}}^{pc} = \sum_{aj} w_{s,\mathrm{HF}}^{aj} \cdot \left[\frac{(pa|jc)}{\epsilon_a^{\bar{\mathrm{HF}}} - \epsilon_j^{\bar{\mathrm{HF}}} - \Omega_s^{\mathrm{HF}}} + \frac{(pj|ac)}{\epsilon_a^{\bar{\mathrm{HF}}} - \epsilon_j^{\bar{\mathrm{HF}}} + \Omega_s^{\mathrm{HF}}}\right]. \tag{8.265}$$

Note that the equations above assume dRPA@HF screening. *Should the $\bar{\epsilon}$ be the Madelung-corrected HF orbital energies or no? My suspicion is maybe not, as this would not be consistent with how we comppute the GW self-energy in the DFT case.* Then, with the finite broadening η_G, I obtain the spectral function as

$$A_p(\omega) = \frac{1}{\pi} \frac{\left|\mathrm{Im}\, \Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) + \eta_G\right|}{\left|\omega - \epsilon_p^0 - \mathrm{Re}\, \Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) - \Sigma_{pp}^x\right|^2 + \left|\mathrm{Im}\, \Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) + \eta_G\right|^2} \tag{8.266}$$

We should try this also with the LDA starting point. Note that if we are starting from LDA, we should be using dRPA@LDA screening and then

$$\Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) = \sum_{ks} \frac{\left(w_{s,\mathrm{LDA}}^{pk} + \tilde{w}_{s,\mathrm{LDA}}^{pk}\right)^2}{\omega - \epsilon_k^{\mathrm{LDA}} + \Omega_s^{\mathrm{LDA}} - \mathrm{i}\eta_\Sigma} + \sum_{cs} \frac{\left(w_{s,\mathrm{LDA}}^{pc} + \tilde{w}_{s,\mathrm{LDA}}^{pc}\right)^2}{\omega - \epsilon_c^{\mathrm{LDA}} - \Omega_s^{\mathrm{LDA}} + \mathrm{i}\eta_\Sigma} \tag{8.267}$$

with

$$\tilde{w}_{s,\mathrm{LDA}}^{pc} = \sum_{aj} w_{s,\mathrm{LDA}}^{aj} \cdot \left[\frac{(pa|jc)}{\epsilon_a^{\mathrm{LDA}} - \epsilon_j^{\mathrm{LDA}} - \Omega_s^{\mathrm{LDA}}} + \frac{(pj|ac)}{\epsilon_a^{\mathrm{LDA}} - \epsilon_j^{\mathrm{LDA}} + \Omega_s^{\mathrm{LDA}}}\right]. \tag{8.268}$$

Thus, with the finite broadening η_G, I obtain the spectral function as

$$A_p(\omega) = \frac{1}{\pi} \frac{\left|\mathrm{Im}\, \Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) + \eta_G\right|}{\left|\omega - \epsilon_p^0 - \mathrm{Re}\, \Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) - \Sigma_{pp}^x\right|^2 + \left|\mathrm{Im}\, \Sigma_{pp}^{GW+\mathrm{2SOSEX-psd}}(\omega) + \eta_G\right|^2} \tag{8.269}$$

The authors of this method say that "in the course of this study, we also evaluated the complete spectral functions, but we found that the satellite structures in the spectral functions were very sensitive to the technical details, basis set, mean-field starting point. In our opinion, they are not conclusive pieces of information."

8.9 GW+C

8.9.1 Cumulant

By relating the Dyson equation to the Taylor series expansion of the exponential (both to first order), we can write:

$$\boldsymbol{G}^0(t)\boldsymbol{C}(t) = \iint \mathrm{d}t_1 \mathrm{d}t_2 \boldsymbol{G}^0(t - t_1)\boldsymbol{\Sigma}^c(t_1 - t_2)\boldsymbol{G}^0(t_2) \tag{8.270}$$

We know that all of the operators are diagonal in momentum space for the UEG, so insertion of the resolution of the identity just gives us:

$$\boldsymbol{G}^0(\boldsymbol{p},t)\boldsymbol{C}(\boldsymbol{p},t) = \iint \mathrm{d}t_1\mathrm{d}t_2 \boldsymbol{G}^0(\boldsymbol{p},t-t_1)\boldsymbol{\Sigma}^c(\boldsymbol{p},t_1-t_2)\boldsymbol{G}^0(\boldsymbol{p},t_2) \tag{8.271}$$

$$= \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t}\boldsymbol{G}^0(\boldsymbol{p},\omega)\boldsymbol{\Sigma}^c(\boldsymbol{p},\omega)\boldsymbol{G}^0(\boldsymbol{p},\omega) \tag{8.272}$$

$$\implies \boldsymbol{C}(\boldsymbol{p},t) = i\int \frac{\mathrm{d}\omega}{2\pi} \frac{\boldsymbol{\Sigma}^c(\boldsymbol{p},\omega+\epsilon_{\boldsymbol{p}}^{HF})}{(\omega+i\eta)^2} e^{-i\omega t} \tag{8.273}$$

$$= i\int \frac{d\omega}{2\pi}\frac{1}{(\omega+i\eta)^2}e^{-i\omega t}\left\{\sum_{\boldsymbol{q}}\sum_{\nu}^{\mathrm{RPA}} \underbrace{\frac{4\pi}{|\boldsymbol{q}|^2}|\rho_\nu(\boldsymbol{q})|^2}_{M^2_{\boldsymbol{q}\nu}} \left[\frac{f_{\boldsymbol{p}-\boldsymbol{q}}}{\omega \underbrace{+\Omega_\nu(\boldsymbol{q})+(\epsilon_{\boldsymbol{p}}-\epsilon_{\boldsymbol{p}-\boldsymbol{q}})+i\eta}_{-\Delta^{\mathrm{occ}}_{\boldsymbol{q}\nu}}}\right.\right. \tag{8.274}$$

$$\left.\left. + \frac{1-f_{\boldsymbol{p}-\boldsymbol{q}}}{\omega \underbrace{-\Omega_\nu(\boldsymbol{q})+(\epsilon_{\boldsymbol{p}}-\epsilon_{\boldsymbol{p}-\boldsymbol{q}})+i\eta}_{-\Delta^{\mathrm{virt}}_{\boldsymbol{q}\nu}}}\right]\right\}$$

$$= i\sum_{\boldsymbol{q}\nu} M^2_{\boldsymbol{q}\nu}\int \frac{d\omega}{2\pi}e^{-i\omega t}\left[\frac{f_{\boldsymbol{p}-\boldsymbol{q}}}{(\omega+i\eta)^2(\omega-\Delta^{\mathrm{occ}}_{\boldsymbol{q}\nu})} + \frac{1-f_{\boldsymbol{p}-\boldsymbol{q}}}{(\omega+i\eta)^2(\omega-\Delta^{\mathrm{virt}}_{\boldsymbol{q}\nu})}\right] \tag{8.275}$$

$$= \sum_{\boldsymbol{q}\nu}\zeta^{\mathrm{occ}}_{\boldsymbol{p}\boldsymbol{q}\nu}\left[e^{-i\Delta^{\mathrm{occ}}_{\boldsymbol{q}\nu}t}-1+i\Delta^{\mathrm{occ}}_{\boldsymbol{q}\nu}t\right] + \sum_{\boldsymbol{q}\nu}\zeta^{\mathrm{virt}}_{\boldsymbol{p}\boldsymbol{q}\nu}\left[e^{-i\Delta^{\mathrm{virt}}_{\boldsymbol{q}\nu}t}-1+i\Delta^{\mathrm{virt}}_{\boldsymbol{q}\nu}t\right] \tag{8.276}$$

For the final expression, we have defined $\zeta^{\mathrm{occ}}_{\boldsymbol{p}\boldsymbol{q}\nu} = \left(\frac{f_{\boldsymbol{p}-\boldsymbol{q}}M_{\boldsymbol{q}\nu}}{\Delta^{\mathrm{occ}}_{\boldsymbol{q}\nu}}\right)^2$ and $\zeta^{\mathrm{virt}}_{\boldsymbol{p}\boldsymbol{q}\nu} = \left(\frac{(1-f_{\boldsymbol{p}-\boldsymbol{q}})M_{\boldsymbol{q}\nu}}{\Delta^{\mathrm{virt}}_{\boldsymbol{q}\nu}}\right)^2$. This allows us to arrive at the something similar to the Landau form of the cumulant.

A few notes on how to evaluate the contour integral: there is a double pole at $\omega_1 = -i\eta$ and a simple pole at $\omega_2 = -\Delta - i\eta$. Closing the contour in the lower half-plane because $\mathrm{Im}\left(\omega_1\right), \mathrm{Im}\left(\omega_2\right) < 0$, and applying Cauchy's residue theorem, leads to

$$\int \frac{\mathrm{d}\omega}{2\pi} e^{-\mathrm{i}\omega t}\frac{1}{\left(\omega-\omega_1\right)^2}\frac{1}{\omega-\omega_2} = (-\mathrm{i})\left\{\left[\partial_\omega\left(\frac{e^{-\mathrm{i}\omega t}}{\omega-\omega_2}\right)\right]_{\omega=\omega_1} + \left[\frac{e^{-\mathrm{i}\omega t}}{\left(\omega-\omega_1\right)^2}\right]_{\omega=\omega_2}\right\} \tag{8.277}$$

$$= \frac{(-\mathrm{i})}{\left(\omega_1-\omega_2\right)^2}\left\{\left[(-\mathrm{i}t)\left(\omega_1-\omega_2\right)-1\right]e^{-\mathrm{i}\omega_1 t} + e^{-\mathrm{i}\omega_2 t}\right\} \tag{8.278}$$

$$\implies \int \frac{\mathrm{d}\omega}{2\pi} e^{-i\omega t}\frac{1}{\left[\omega-(0-\mathrm{i}\eta)\right]^2}\frac{f}{\omega-\Delta} = \frac{-\mathrm{i}f}{\Delta^2}\left(e^{-\mathrm{i}\Delta t}+\mathrm{i}\Delta t-1\right) \tag{8.279}$$

8.9.2 Green's function

Now, we plug in our derived expression for $C(\boldsymbol{p}, t)$ into the cumulant ansatz for the retarded Green's function:

$$
\begin{aligned}
G^{GW+C}(\boldsymbol{p}, t) &= G^{HF}(\boldsymbol{p}, t) e^{C(\boldsymbol{p},t)} && (8.280)\\
&= -i\Theta(t) e^{-i\epsilon^{HF}_{\boldsymbol{p}} t + C(\boldsymbol{p},t)} && (8.281)\\
&= -i\Theta(t) e^{-i\epsilon^{HF}_{\boldsymbol{p}} t + \sum_{\boldsymbol{q}\nu} \zeta^{\text{occ}}_{\boldsymbol{pq}\nu} \left(e^{-i\Delta^{\text{occ}}_{\boldsymbol{q}\nu} t} + i\Delta^{\text{occ}}_{\boldsymbol{q}\nu} t - 1 \right) + \sum_{\boldsymbol{q}\nu} \zeta^{\text{virt}}_{\boldsymbol{pq}\nu} \left(e^{-i\Delta^{\text{virt}}_{\boldsymbol{q}\nu} t} + i\Delta^{\text{virt}}_{\boldsymbol{q}\nu} t - 1 \right)} && (8.282)\\
&= -i\Theta(t) Z^{QP}_{\boldsymbol{p}} e^{-i\epsilon^{QP}_{\boldsymbol{p}} t} e^{\sum_{\boldsymbol{q}\nu} \zeta^{\text{occ}}_{\boldsymbol{pq}\nu} e^{-i\Delta^{\text{occ}}_{\boldsymbol{q}\nu} t} + \sum_{\boldsymbol{q}\nu} \zeta^{\text{virt}}_{\boldsymbol{pq}\nu} e^{-i\Delta^{\text{virt}}_{\boldsymbol{q}\nu} t}} && (8.283)\\
& && (8.284)
\end{aligned}
$$

where we have the weight of the quasiparticle peak $Z^{QP}_{\boldsymbol{p}} = \exp\left(-\sum_{\boldsymbol{q}\nu} \zeta^{\text{occ}}_{\boldsymbol{pq}\nu} - \sum_{\boldsymbol{q}\nu} \zeta^{\text{virt}}_{\boldsymbol{pq}\nu} \right)$ and the quasiparticle energy $\epsilon^{QP}_{\boldsymbol{p}} = \epsilon^{HF}_{\boldsymbol{p}} - \left(\sum_{\boldsymbol{q}\nu} \zeta^{\text{occ}}_{\boldsymbol{pq}\nu} \Delta^{\text{occ}}_{\boldsymbol{q}\nu} + \sum_{\boldsymbol{q}\nu} \zeta^{\text{virt}}_{\boldsymbol{pq}\nu} \Delta^{\text{virt}}_{\boldsymbol{q}\nu} \right)$.

We pause to make some important connections. Notice

$$
\begin{aligned}
Z^{QP}_{\boldsymbol{p}} &= \exp\left(-\sum_{\boldsymbol{q}\nu} \zeta^{\text{occ}}_{\boldsymbol{pq}\nu} - \sum_{\boldsymbol{q}\nu} \zeta^{\text{virt}}_{\boldsymbol{pq}\nu} \right) && (8.285)\\
&= \exp\left(-\sum_{\boldsymbol{q}\nu} \left(\frac{f_{\boldsymbol{p}-\boldsymbol{q}} M_{\boldsymbol{q}\nu}}{\Delta^{\text{occ}}_{\boldsymbol{q}\nu}} \right)^2 - \sum_{\boldsymbol{q}\nu} \left(\frac{(1 - f_{\boldsymbol{p}-\boldsymbol{q}}) M_{\boldsymbol{q}\nu}}{\Delta^{\text{virt}}_{\boldsymbol{q}\nu}} \right)^2 \right) && (8.286)\\
&= \exp\left(\left[\frac{\partial \Sigma^c(\boldsymbol{p}, \omega)}{\partial \omega} \right]_{\omega = \epsilon^{HF}_{\boldsymbol{p}}} \right) && (8.287)
\end{aligned}
$$

and

$$
\begin{aligned}
\epsilon^{QP}_{\boldsymbol{p}} &= \epsilon^{HF}_{\boldsymbol{p}} - \left(\sum_{\boldsymbol{q}\nu} \zeta^{\text{occ}}_{\boldsymbol{pq}\nu} \Delta^{\text{occ}}_{\boldsymbol{q}\nu} + \sum_{\boldsymbol{q}\nu} \zeta^{\text{virt}}_{\boldsymbol{pq}\nu} \Delta^{\text{virt}}_{\boldsymbol{q}\nu} \right) && (8.288)\\
&= \epsilon^{HF}_{\boldsymbol{p}} - \left(\sum_{\boldsymbol{q}\nu} \frac{f_{\boldsymbol{p}-\boldsymbol{q}} M^2_{\boldsymbol{q}\nu}}{\Delta^{\text{occ}}_{\boldsymbol{q}\nu}} + \sum_{\boldsymbol{q}\nu} \frac{(1 - f_{\boldsymbol{p}-\boldsymbol{q}}) M^2_{\boldsymbol{q}\nu}}{\Delta^{\text{virt}}_{\boldsymbol{q}\nu}} \right) && (8.289)\\
&= \epsilon^{HF}_{\boldsymbol{p}} + \Sigma^c(\boldsymbol{p}, \epsilon^{HF}_{\boldsymbol{p}}) && (8.290)
\end{aligned}
$$

where we have used the fact that $\Sigma^c(\boldsymbol{p}, \omega) = \sum_{\boldsymbol{q}\nu} \frac{f_{\boldsymbol{p}-\boldsymbol{q}} M^2_{\boldsymbol{q}\nu}}{\omega + \Omega_\nu(\boldsymbol{q}) - \epsilon_{\boldsymbol{p}-\boldsymbol{q}}} + \sum_{\boldsymbol{q}\nu} \frac{(1 - f_{\boldsymbol{p}-\boldsymbol{q}}) M^2_{\boldsymbol{q}\nu}}{\omega - \Omega_\nu(\boldsymbol{q}) - \epsilon_{\boldsymbol{p}-\boldsymbol{q}}} \implies \left[\frac{\partial \Sigma^c(\boldsymbol{p}, \omega)}{\partial \omega} \right]_{\omega = \epsilon^{HF}_{\boldsymbol{p}}} = -\sum_{\boldsymbol{q}\nu} \left(\frac{f_{\boldsymbol{p}-\boldsymbol{q}} M_{\boldsymbol{q}\nu}}{\Delta^{\text{occ}}_{\boldsymbol{q}\nu}} \right)^2 - \sum_{\boldsymbol{q}\nu} \left(\frac{(1 - f_{\boldsymbol{p}-\boldsymbol{q}}) M_{\boldsymbol{q}\nu}}{\Delta^{\text{virt}}_{\boldsymbol{q}\nu}} \right)^2$.

Next, we want to perform a Fourier transform.

$$G^{GW+C}(\boldsymbol{p},\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} G^{GW+C}(\boldsymbol{p},t) \tag{8.291}$$

$$= -iZ_{\boldsymbol{p}}^{QP} \int_0^{\infty} dt\, e^{i(\omega-\epsilon_{\boldsymbol{p}}^{QP})t} e^{\sum_{\boldsymbol{q}\nu}\zeta_{\boldsymbol{pq}\nu}^{\text{occ}} e^{-i\Delta_{\boldsymbol{q}\nu}^{\text{occ}}t} + \sum_{\boldsymbol{q}\nu}\zeta_{\boldsymbol{pq}\nu}^{\text{virt}} e^{-i\Delta_{\boldsymbol{q}\nu}^{\text{virt}}t}} \tag{8.292}$$

$$= -iZ_{\boldsymbol{p}}^{QP} \int_0^{\infty} dt\, e^{i(\omega-\epsilon_{\boldsymbol{p}}^{QP})t} \left(1 + \sum_{\boldsymbol{q}\nu}\zeta_{\boldsymbol{pq}\nu}^{\text{occ}} e^{-i\Delta_{\boldsymbol{q}\nu}^{\text{occ}}t} + \sum_{\boldsymbol{q}\nu}\zeta_{\boldsymbol{pq}\nu}^{\text{virt}} e^{-i\Delta_{\boldsymbol{q}\nu}^{\text{virt}}t} + \dots\right) \tag{8.293}$$

$$= -iZ_{\boldsymbol{p}}^{QP} \int_0^{\infty} dt\, e^{[-\eta+i(\omega-\epsilon_{\boldsymbol{p}}^{QP})t]} \tag{8.294}$$

$$- iZ_{\boldsymbol{p}}^{QP} \sum_{\boldsymbol{q}\nu}\zeta_{\boldsymbol{pq}\nu}^{\text{occ}} \int_0^{\infty} dt\, e^{[-\eta+i(\omega-\epsilon_{\boldsymbol{p}}^{QP}-\Delta_{\boldsymbol{q}\nu}^{\text{occ}})t]}$$

$$- iZ_{\boldsymbol{p}}^{QP} \sum_{\boldsymbol{q}\nu}\zeta_{\boldsymbol{pq}\nu}^{\text{virt}} \int_0^{\infty} dt\, e^{[-\eta+i(\omega-\epsilon_{\boldsymbol{p}}^{QP}-\Delta_{\boldsymbol{q}\nu}^{\text{virt}})t]} + \dots$$

$$= \frac{Z_{\boldsymbol{p}}^{QP}}{\omega-\epsilon_{\boldsymbol{p}}^{QP}+i\eta} + \sum_{\boldsymbol{q}\nu}\frac{Z_{\boldsymbol{p}}^{QP}\zeta_{\boldsymbol{pq}\nu}^{\text{occ}}}{\omega-\epsilon_{\boldsymbol{p}}^{QP}-\Delta_{\boldsymbol{q}\nu}^{\text{occ}}+i\eta} + \sum_{\boldsymbol{q}\nu}\frac{Z_{\boldsymbol{p}}^{QP}\zeta_{\boldsymbol{pq}\nu}^{\text{virt}}}{\omega-\epsilon_{\boldsymbol{p}}^{QP}-\Delta_{\boldsymbol{q}\nu}^{\text{virt}}+i\eta} + \dots \tag{8.295}$$

$$= \frac{Z_{\boldsymbol{p}}^{QP}}{\omega-\epsilon_{\boldsymbol{p}}^{QP}+i\eta} + \sum_{\boldsymbol{q}\nu}\frac{Z_{\boldsymbol{pq}\nu}^{\text{occ-sat}}}{\omega-\epsilon_{\boldsymbol{pq}\nu}^{\text{occ-sat}}+i\eta} + \sum_{\boldsymbol{q}\nu}\frac{Z_{\boldsymbol{pq}\nu}^{\text{virt-sat}}}{\omega-\epsilon_{\boldsymbol{pq}\nu}^{\text{virt-sat}}+i\eta} + \dots \tag{8.296}$$

where we define the satellite energies $\epsilon_{\boldsymbol{pq}\nu}^{\text{occ-sat}} = \epsilon_{\boldsymbol{p}}^{QP} + \Delta_{\boldsymbol{q}\nu}^{\text{occ}}$ and $\epsilon_{\boldsymbol{pq}\nu}^{\text{virt-sat}} = \epsilon_{\boldsymbol{p}}^{QP} + \Delta_{\boldsymbol{q}\nu}^{\text{virt}}$, as well as the satellite weights $Z_{\boldsymbol{pq}\nu}^{\text{occ-sat}} = Z_{\boldsymbol{p}}^{QP}\zeta_{\boldsymbol{pq}\nu}^{\text{occ}}$ and $Z_{\boldsymbol{pq}\nu}^{\text{virt-sat}} = Z_{\boldsymbol{p}}^{QP}\zeta_{\boldsymbol{pq}\nu}^{\text{virt}}$.

8.9.3 Spectral function

The spectral function is obtained as (the virtual satellites spectral function, whose derivation will mirror that of the occupied satellites, are omitted for the sake of brevity)

$$A^{GW+C}(\boldsymbol{p},\omega) = -\frac{1}{\pi}\operatorname{Im} G^{GW+C}(\boldsymbol{p},\omega) \tag{8.297}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{Z^{QP}_{\boldsymbol{p}}}{\omega-\epsilon^{QP}_{\boldsymbol{p}}+i\eta}+\sum_{\boldsymbol{q}\nu}\frac{Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}}{\omega-\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}+i\eta}+\ldots\right] \tag{8.298}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\operatorname{Re} Z^{QP}_{\boldsymbol{p}}+i\operatorname{Im} Z^{QP}_{\boldsymbol{p}}}{\omega-\operatorname{Re}\epsilon^{QP}_{\boldsymbol{p}}+i\left(\eta-\operatorname{Im}\epsilon^{QP}_{\boldsymbol{p}}\right)}+\sum_{\boldsymbol{q}\nu}\frac{\operatorname{Re} Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}+i\operatorname{Im} Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}}{\omega-\operatorname{Re}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}+i\left(\eta-\operatorname{Im}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)}+\ldots\right] \tag{8.299}$$

$$= -\frac{1}{\pi}\operatorname{Im}\left[\frac{\left(\operatorname{Re} Z^{QP}_{\boldsymbol{p}}+i\operatorname{Im} Z^{QP}_{\boldsymbol{p}}\right)\left(\omega-\operatorname{Re}\epsilon^{QP}_{\boldsymbol{p}}-i\left(\eta-\operatorname{Im}\epsilon^{QP}_{\boldsymbol{p}}\right)\right)}{\left(\omega-\operatorname{Re}\epsilon^{QP}_{\boldsymbol{p}}\right)^2+\left(\operatorname{Im}\epsilon^{QP}_{\boldsymbol{p}}\right)^2}\right. \tag{8.300}$$

$$\left.+\sum_{\boldsymbol{q}\nu}\frac{\left(\operatorname{Re} Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}+i\operatorname{Im} Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)\left(\omega-\operatorname{Re}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}-i\left(\eta-\operatorname{Im}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)\right)}{\left(\omega-\operatorname{Re}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)^2+\left(\operatorname{Im}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)^2}+\ldots\right] \tag{8.301}$$

$$= -\frac{1}{\pi}\left[\frac{\left(\operatorname{Re} Z^{QP}_{\boldsymbol{p}}\right)\left(\operatorname{Im}\epsilon^{QP}_{\boldsymbol{p}}\right)+\left(\operatorname{Im} Z^{QP}_{\boldsymbol{p}}\right)\left(\omega-\operatorname{Re}\epsilon^{QP}_{\boldsymbol{p}}\right)}{\left(\omega-\operatorname{Re}\epsilon^{QP}_{\boldsymbol{p}}\right)^2+\left(\operatorname{Im}\epsilon^{QP}_{\boldsymbol{p}}\right)^2}\right. \tag{8.302}$$

$$\left.+\sum_{\boldsymbol{q}\nu}\frac{\left(\operatorname{Re} Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)\left(\operatorname{Im}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)+\left(\operatorname{Im} Z^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)\left(\omega-\operatorname{Re}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)}{\left(\omega-\operatorname{Re}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)^2+\left(\operatorname{Im}\epsilon^{\text{occ-sat}}_{\boldsymbol{pq}\nu}\right)^2}+\ldots\right] \tag{8.303}$$

8.10 LDA

Our presentation follows the restricted formalism where two electrons of opposite spin occupy each spatial orbital.

8.10.1 Analytic gradients

We first recall the definition of the density in terms of $P_{\mu\nu}$ and the basis functions $\phi_\mu(\mathbf{r})$:

$$\rho(\mathbf{r}) = \frac{1}{V}\sum_{\mu\nu} P_{\mu\nu}\operatorname{Re}\left[\phi^*_\mu(\mathbf{r})\phi_\nu(\mathbf{r})\right]. \tag{8.304}$$

Defining the real overlap factor

$$S_{\mu\nu}(\mathbf{r}) = \operatorname{Re}\left[\phi^*_\mu(\mathbf{r})\phi_\nu(\mathbf{r})\right], \tag{8.305}$$

the derivative of $\rho(\mathbf{r})$ with respect to a density-matrix element is

$$\frac{\partial \rho(\mathbf{r})}{\partial P_{\alpha\beta}} = \frac{1}{V} S_{\alpha\beta}(\mathbf{r}) = \frac{1}{V} \operatorname{Re}[\phi^*_\alpha(\mathbf{r})\phi_\beta(\mathbf{r})] \,. \tag{8.306}$$

Kinetic

The kinetic energy is given by

$$E_k = \sum_{\mu\nu} P_{\mu\nu} T_{\mu\nu} \tag{8.307}$$

Hartree

The Hartree energy is given by

$$E_H = \frac{1}{2} \int d\mathbf{r} \; v_H(\mathbf{r}) \, \rho(\mathbf{r}) \tag{8.308}$$

$$\tag{8.309}$$

Exchange

The LDA defines the exchange energy for a single particle as $\epsilon_x(\mathbf{r}) = -\frac{3}{4} \left(\frac{3}{\pi}\right)^{1/3} \rho(\mathbf{r})^{1/3}$. So the total exchange energy becomes

$$E_x[\rho] = \int d\mathbf{r} \; \epsilon_x(\mathbf{r}) \, \rho(\mathbf{r}) \tag{8.310}$$

$$= -\frac{3}{4} \left(\frac{3}{\pi}\right)^{1/3} \int d\mathbf{r} \; \rho(\mathbf{r})^{4/3} \tag{8.311}$$

$$= -\frac{3}{4} \left(\frac{3}{\pi}\right)^{1/3} \int d\mathbf{r} \; \left(\frac{1}{V} \sum_{\mu\nu} P_{\mu\nu} S_{\mu\nu}(\mathbf{r}) \right)^{4/3} \tag{8.312}$$

$$\approx -\frac{3}{4} \left(\frac{3}{\pi}\right)^{1/3} \frac{1}{V^{4/3}} \sum_g w_g \left(\sum_{\mu\nu} P_{\mu\nu} S_{\mu\nu}(\mathbf{r}_g) \right)^{4/3} . \tag{8.313}$$

Taking the derivative with respect to $P_{\alpha\beta}$ gives

$$\frac{\partial E_x}{\partial P_{\alpha\beta}} = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}\int d\mathbf{r}\ \frac{4}{3}\left(\frac{1}{V}\sum_{\mu\nu}P_{\mu\nu}S_{\mu\nu}(\mathbf{r})\right)^{1/3}\frac{\partial}{\partial P_{\alpha\beta}}\left(\frac{1}{V}\sum_{\mu\nu}P_{\mu\nu}S_{\mu\nu}(\mathbf{r})\right) \tag{8.314}$$

$$= -\left(\frac{3}{\pi}\right)^{1/3}\int d\mathbf{r}\ \left(\frac{1}{V}\sum_{\mu\nu}P_{\mu\nu}S_{\mu\nu}(\mathbf{r})\right)^{1/3}\left(\frac{1}{V}S_{\alpha\beta}(\mathbf{r})\right) \tag{8.315}$$

$$= -\left(\frac{3}{\pi}\right)^{1/3}\frac{1}{V^{4/3}}\int d\mathbf{r}\ \left(\sum_{\mu\nu}P_{\mu\nu}S_{\mu\nu}(\mathbf{r})\right)^{1/3}S_{\alpha\beta}(\mathbf{r}) \tag{8.316}$$

$$\approx -\left(\frac{3}{\pi}\right)^{1/3}\frac{1}{V^{4/3}}\sum_{g}w_g\left(\sum_{\mu\nu}P_{\mu\nu}S_{\mu\nu}(\mathbf{r}_g)\right)^{1/3}S_{\alpha\beta}(\mathbf{r}_g). \tag{8.317}$$

Correlation

Perdew defines the correlation energy as

$$E_c[\rho] = \int d\mathbf{r}\ \epsilon_c(\mathbf{r})\,\rho(\mathbf{r}), \tag{8.318}$$

with

$$\epsilon_c(\mathbf{r}) = \frac{\gamma}{1+\beta_1\sqrt{r_s(\mathbf{r})}+\beta_2 r_s(\mathbf{r})}, \tag{8.319}$$

$$r_s(\mathbf{r}) = \left(\frac{3}{4\pi\rho(\mathbf{r})}\right)^{1/3}. \tag{8.320}$$

Then

$$\frac{\delta E_c}{\delta\rho(\mathbf{r})} = \epsilon_c(\mathbf{r}) + \rho(\mathbf{r})\frac{d\epsilon_c}{d\rho}\Big|_{\rho(\mathbf{r})} \tag{8.321}$$

$$= v_c(\mathbf{r}) \tag{8.322}$$

with $v_c(\mathbf{r}) = \epsilon_c(\mathbf{r})\left(1+\frac{7}{6}\beta_1\sqrt{r_s(\mathbf{r})}+\frac{4}{3}\beta_2 r_s(\mathbf{r})\right)$. To obtain the gradient with respect to the density-matrix element $P_{\alpha\beta}$, we need:

$$\frac{\partial E_c}{\partial P_{\alpha\beta}} = \frac{\partial E_c}{\partial\rho(\mathbf{r})}\frac{\partial\rho(\mathbf{r})}{\partial P_{\alpha\beta}} \tag{8.323}$$

$$= \int d\mathbf{r}\ v_c(\mathbf{r})\frac{\partial\rho(\mathbf{r})}{\partial P_{\alpha\beta}} \tag{8.324}$$

$$= \int d\mathbf{r}\ v_c(\mathbf{r})\frac{1}{V}S_{\alpha\beta}(\mathbf{r}) \tag{8.325}$$

$$\approx \frac{1}{V}\sum_{g}w_g v_c(\mathbf{r}_g)S_{\alpha\beta}(\mathbf{r}_g). \tag{8.326}$$

8.10.2 Fock matrix

Density operator

I construct a real space cubic grid for the evaluation of my quantities on, which defines a set of grid points $\{\mathbf{r}_g\}$ with associated weights $\{w_g\}$. The density is then given by

$$\rho(\mathbf{r}) = \sum_{\mu\nu} P_{\mu\nu}\phi^*_\mu(\mathbf{r})\phi_\nu(\mathbf{r}) = \frac{1}{V}\sum_{\mu\nu} P_{\mu\nu} \tag{8.327}$$

8.11 Convergence study to TDL and CBS

I want to investigate convergence of my UEG spectral functions at rs=4 toward the TDL and CBS.

8.12 Applying the Dyson density matrix to the electron gas

We can Consider both Hartree-Fock and LDA references when using the GW Dyson Solver, whereas just the Hartree-Fock reference when using the Pt2 Dyson Solvers (MP2 and BWs2).
At the end of the day, we want to be able to plug everything into equation 10 of 10.1021/acs.jctc.9b00333 (attached).

Obtaining the quasar boson quantities

For each method, we need quasibosons frequencies, and fermionic couplings. For GW, These can be obtained from the excitation energies and transition densities taken from a Casida diagonalization of the dRPA problem. *What would be their specific forms?* For PT2 methods, these are The single-particle energy differences and the relevant ERIs between the occupied and virtual stattes. The best way to understand how these quantities are obtained is by considering the cumulant formulation for each method. Consider that for BWs2, I wrote down

$$\Sigma_{\text{eff},pp}(\omega + \tilde{\varepsilon}_p) = \Sigma^{(2),2p1h+2p1h}_{pp}\left[G^{(0)\prime}\right](\omega + \tilde{\varepsilon}_p) - \Sigma^{(2),2h1p}_{pp}\left[G^{(0)\prime}\right](\tilde{\varepsilon}_p) \tag{8.328}$$

$$= \frac{1}{2}\sum_{ija}\langle pa\|ij\rangle^2 \left(\frac{1}{\omega + \tilde{\varepsilon}_p + \tilde{\varepsilon}_a - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j + i\eta} - \frac{1}{\tilde{\varepsilon}_p + \tilde{\varepsilon}_a - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j + i\eta}\right) \tag{8.329}$$

$$+ \frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\omega + \tilde{\varepsilon}_p + \tilde{\varepsilon}_i - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b + i\eta} \tag{8.330}$$

$$= \frac{1}{2}\sum_{ija}\langle pa\|ij\rangle^2 \frac{\omega}{(\omega - \tilde{\epsilon}^{ij}_{pa})(\tilde{\epsilon}^{ij}_{pa})} + \frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\omega - \tilde{\epsilon}^{ab}_{pi}} \tag{8.331}$$

where we identify $\tilde{\epsilon}^{ab}_{pi} = \tilde{\varepsilon}_a + \tilde{\varepsilon}_b - \tilde{\varepsilon}_p - \tilde{\varepsilon}_i - i\eta$ and $\tilde{\epsilon}^{ij}_{pa} = \tilde{\varepsilon}_i + \tilde{\varepsilon}_j - \tilde{\varepsilon}_p - \tilde{\varepsilon}_a - i\eta$ as the bosonic frequencies in the quasi-boson approximation due to the BWs2 partition. I imagine that we should get something similar if we just do MP2. *Can you write that down as well?*

Obtaining the 1-RDM

The elements of this Dyson 1-RDM Can be computed via equation 10 For every method . *Tell me how you would do so in each case, paying, I guess, particular attention to what happens when we start from the DFT reference, because in that case some extra modification to the OV blocks of the Dyson 1-RDM are needed As can be seen by considering the appendage to the equation 12c of 10.48550/arXiv.2607.15695*

8.12.1 Properties to compute

Given that we can obtain the electronic density from the Dyson 1-RDM, I was thinking of doing this for multiple rs values; say 4-100; that AFQMC can't do but there is NQ reference data for somewhere? My collegaue said This might be found in https://journals.aps.org/prb/abstract/10.1103/P or https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.133.266504 or maybe some related papers. I guess the advantage that I am noticing is that NQS can only do supercellls Up to like 150 electrons, but I can run these many-body perturbation methods, which in the worst case exhibit a fifth-order scaling for thousands of electrons.

8.13 Madelung contributions to the self-energy

From here on out, quantities colored red include a Madelung contribution, while quantities colored green do not. We begin from

$$\hat{H} = \sum_{\mathbf{k}} \epsilon^{(0)}_{\mathbf{k}} \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \underbrace{\frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{k}\mathbf{k}'} a^\dagger_{\mathbf{k}+\mathbf{q}} a^\dagger_{\mathbf{k}'-\mathbf{q}} a_{\mathbf{k}'} a_{\mathbf{k}}}_{\hat{V}}, \tag{8.332}$$

where $\mathbf{k}, \mathbf{k}'$ run over all valid momenta, and $\mathbf{q}$ runs over all nonzero momentum transfers. $\epsilon^{(0)}_{\mathbf{k}} = \frac{k^2}{2} + \frac{v_M}{2}$ is the bare single-particle energy with $v_M = -2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_s^{-1}$ the Madelung constant for the cubic simulation cell. We will now normal order $\hat{V}$ with respect to the HF reference $|\Phi_0\rangle$. Wick's theorem, which says that $a^\dagger a^\dagger a a =: a^\dagger a^\dagger a a :$

$+\sum_{\text{single contractions}}(\text{contraction}) :a^\dagger a: +\sum_{\text{complete contractions}}(\text{contractions})$, gives

$$\implies a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}} \tag{8.333}$$

$$\begin{aligned}
&=: a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}} : \\
&\quad + \overbrace{a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}}} : a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}} : - \overbrace{a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}}a_{\mathbf{k'}}a_{\mathbf{k}} : a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k}} : \\
&\quad - a^\dagger_{\mathbf{k+q}}\overbrace{a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}}} : a^\dagger_{\mathbf{k+q}}a_{\mathbf{k'}} : + a^\dagger_{\mathbf{k+q}}\overbrace{a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}}a_{\mathbf{k}} : a^\dagger_{\mathbf{k+q}}a_{\mathbf{k}} : \\
&\quad + \overbrace{a^\dagger_{\mathbf{k+q}}\overbrace{a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}}a_{\mathbf{k}}} - \overbrace{a^\dagger_{\mathbf{k+q}}\overbrace{a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}}}}
\end{aligned} \tag{8.334}$$

$$\begin{aligned}
&=: a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}} : \\
&\quad + \delta_{\mathbf{q},\mathbf{0}}n_{\mathbf{k}} : a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}} : - \delta_{\mathbf{k+q},\mathbf{k'}}n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k}} : - \delta_{\mathbf{k'-q},\mathbf{k}}n_{\mathbf{k}} : a^\dagger_{\mathbf{k+q}}a_{\mathbf{k'}} : + \delta_{\mathbf{q},\mathbf{0}}n_{\mathbf{k'}} : a^\dagger_{\mathbf{k+q}}a_{\mathbf{k}} : \\
&\quad + \delta_{\mathbf{q},\mathbf{0}}\delta_{\mathbf{q},\mathbf{0}}n_{\mathbf{k}}n_{\mathbf{k'}} - \delta_{\mathbf{k+q},\mathbf{k'}}\delta_{\mathbf{k'-q},\mathbf{k}}n_{\mathbf{k+q}}n_{\mathbf{k}}
\end{aligned} \tag{8.335}$$

$$\overset{\mathbf{q}\neq\mathbf{0}}{=}: a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}} : -\delta_{\mathbf{k+q},\mathbf{k'}}n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k}} : -\delta_{\mathbf{k'-q},\mathbf{k}}n_{\mathbf{k}} : a^\dagger_{\mathbf{k+q}}a_{\mathbf{k'}} : -\delta_{\mathbf{k+q},\mathbf{k'}}\delta_{\mathbf{k'-q},\mathbf{k}}n_{\mathbf{k+q}}n_{\mathbf{k}} \tag{8.336}$$

$$\begin{aligned}
\implies \hat{V} &= \frac{1}{2\Omega}\sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{kk'}} \Big[: a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}} : -\delta_{\mathbf{k'},\mathbf{k+q}}\, n_{\mathbf{k+q}} \; : a^\dagger_{\mathbf{k}}a_{\mathbf{k}} : \\
&\qquad - \delta_{\mathbf{k'},\mathbf{k+q}}\, n_{\mathbf{k}} \; : a^\dagger_{\mathbf{k+q}}a_{\mathbf{k+q}} : -\delta_{\mathbf{k'},\mathbf{k+q}}\, n_{\mathbf{k+q}}n_{\mathbf{k}} \Big]
\end{aligned} \tag{8.337}$$

$$= \underbrace{\frac{1}{2\Omega}\sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{kk'}} : a^\dagger_{\mathbf{k+q}}a^\dagger_{\mathbf{k'-q}}a_{\mathbf{k'}}a_{\mathbf{k}} :}_{\hat{V}_{\text{res}}} \underbrace{-\frac{1}{\Omega}\sum_{\mathbf{k}}\sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k+q}} : a^\dagger_{\mathbf{k}}a_{\mathbf{k}} :}_{\sum_{\mathbf{k}}\Sigma^x_{\mathbf{k}}:\hat{n}_{\mathbf{k}}:} \underbrace{-\frac{1}{2\Omega}\sum_{\mathbf{k}}\sum_{\mathbf{q}\neq\mathbf{0}} v_q n_{\mathbf{k}}n_{\mathbf{k+q}}}_{E_x} \tag{8.338}$$

$$\implies \hat{H} = \sum_{\mathbf{k}} \epsilon^{(0)}_{\mathbf{k}} \Big(n_{\mathbf{k}} + : a^\dagger_{\mathbf{k}}a_{\mathbf{k}} : \Big) + E_{\text{x}} + \sum_{\mathbf{k}} \Sigma^x_{\mathbf{k}} : \hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} : + \hat{V}_{\text{res}} \tag{8.339}$$

$$\equiv E_{\text{HF}} + \underbrace{\sum_{\mathbf{k}} \epsilon^{\text{HF}}_{\mathbf{k}} : \hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} :}_{\hat{F}} + \hat{V}_{\text{res}} \tag{8.340}$$

$$\implies \hat{H}_N = \hat{H} - E_{\text{HF}}. \tag{8.341}$$

with $\epsilon^{\text{HF}}_{\mathbf{k}} = \epsilon^{(0)}_{\mathbf{k}} + \Sigma^x_{\mathbf{k}} = \frac{k^2}{2} + \frac{v_M}{2} + \Sigma^x_{\mathbf{k}}$ and $E_{\text{HF}} = \sum_{\mathbf{k}} \epsilon^{(0)}_{\mathbf{k}} n_{\mathbf{k}} + E_{\text{x}}$. At this point, we can pursue two different partitionings.

1. $\hat{H}_N = \underbrace{\hat{F}}_{\hat{H}_0} + \underbrace{\hat{V}_{\text{res}}}_{\hat{V}}$
2. $\hat{H}_N = \underbrace{\hat{F}_M}_{\hat{H}_0} + \underbrace{\hat{V}_{\text{res}} + \hat{S}}_{\hat{V}}$

Note that $\hat{F}_M = \sum_{\mathbf{k}} \left(\frac{k^2}{2} + \Sigma^x_{\mathbf{k}} + v_M n_{\mathbf{k}} \right) : \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} :$ and

$$\hat{S} = \hat{F} - \hat{F}_M \tag{8.342}$$

$$= \frac{v_M}{2} \sum_{\mathbf{k}} (1 - 2n_{\mathbf{k}}) : a^\dagger_{\mathbf{k}} a_{\mathbf{k}} : \tag{8.343}$$

8.13.1 Answering the question of RPA screening

First partitioning

Define $|\Phi^a_i\rangle = a^\dagger_a a_i |\Phi_0\rangle$. Then,

$$A_{ia,jb} = \langle \Phi^a_i | \hat{H}_N | \Phi^b_j \rangle = \langle \Phi^a_i | \hat{F} | \Phi^b_j \rangle + \langle \Phi^a_i | \hat{V}_{\text{res}} | \Phi^b_j \rangle, \tag{8.344}$$

with $\boldsymbol{AX} = \boldsymbol{X\omega}$, where $\boldsymbol{\omega} = \text{diag}(\omega_n)$, $\omega_n = E^{\text{CIS}}_n - E_{\text{HF}}$, and $\sum_{ia} \langle \Phi^b_j | \hat{H} | \Phi^a_i \rangle \, c^a_{i,n} = E^{\text{CIS}}_n \, c^b_{j,n}$. We know that the orbital energy differences come from

$$\langle \Phi^a_i | \hat{F} | \Phi^b_j \rangle = \delta_{ij} \delta_{ab} \left(\epsilon^{\text{HF}}_a - \epsilon^{\text{HF}}_i \right) \tag{8.345}$$

$$= \delta_{ij} \delta_{ab} \left[\frac{a^2 - i^2}{2} + \Sigma^x_a - \Sigma^x_i + \frac{v_M}{2} - \frac{v_M}{2} \right] \tag{8.346}$$

$$= \delta_{ij} \delta_{ab} \left[\frac{a^2 - i^2}{2} + \Sigma^x_a - \Sigma^x_i \right] \tag{8.347}$$

$$= \langle \Phi^a_i | \hat{F} | \Phi^b_j \rangle, \tag{8.348}$$

with $\hat{F} = \sum_{\mathbf{k}} \epsilon^{\text{HF}}_{\mathbf{k}} : \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} :$ and $\epsilon^{\text{HF}}_{\mathbf{k}} = \frac{k^2}{2} + \Sigma^x_{\mathbf{k}}$.

Second partitioning

Now,

$$A_{ia,jb} = \langle \Phi^a_i | \hat{F}_M | \Phi^b_j \rangle + \langle \Phi^a_i | \hat{V}_{\text{res}} | \Phi^b_j \rangle + \langle \Phi^a_i | \hat{S} | \Phi^b_j \rangle \tag{8.349}$$

$$= \delta_{ij} \delta_{ab} \left[\frac{a^2}{2} + \Sigma^x_a - \left(\frac{i^2}{2} + \Sigma^x_i + v_M \right) \right] + \langle \Phi^a_i | \hat{V}_{\text{res}} | \Phi^b_j \rangle + \delta_{ij} \delta_{ab} \, v_M \tag{8.350}$$

$$= \delta_{ij} \delta_{ab} \left[\frac{a^2 - i^2}{2} + \Sigma^x_a - \Sigma^x_i \right] + \langle \Phi^a_i | \hat{V}_{\text{res}} | \Phi^b_j \rangle \tag{8.351}$$

$$= \langle \Phi^a_i | \hat{F} | \Phi^b_j \rangle + \langle \Phi^a_i | \hat{V}_{\text{res}} | \Phi^b_j \rangle. \tag{8.352}$$

8.13.2 Answering the question of the GW self-energy

We take the non-interacting Green's function to be

$$G^0_p(\omega) = \frac{n_p}{\omega - \tilde{\epsilon}_p - i\eta} + \frac{1 - n_p}{\omega - \tilde{\epsilon}_p + i\eta}, \tag{8.353}$$

where $\tilde{\epsilon}_p$ is the single-particle energy of the reference. We then solve the Casida equation of

$$\begin{pmatrix} \mathrm{A} & \mathrm{B} \\ \mathrm{B} & \mathrm{A} \end{pmatrix} \begin{pmatrix} \mathrm{X} & \mathrm{Y} \\ \mathrm{Y} & \mathrm{X} \end{pmatrix} = \begin{pmatrix} \mathrm{X} & -\mathrm{Y} \\ -\mathrm{Y} & \mathrm{X} \end{pmatrix} \begin{pmatrix} \Omega & 0 \\ 0 & \Omega \end{pmatrix} \tag{8.354}$$

using these single-particle energies, since in the GW approximation, the non-interacting polarizability is written as $P_0 = -iG_0G_0$. Then we form the correlation self-energy as

$$\Sigma_{pp}^{c,G_0W_0}(\omega) = \sum_{j\nu} \frac{W_{pj\nu}W_{pj\nu}}{\omega - (\epsilon_j - \Omega_\nu) - \mathrm{i}\eta} + \sum_{b\nu} \frac{W_{pb\nu}W_{pb\nu}}{\omega - (\epsilon_b + \Omega_\nu) + \mathrm{i}\eta}, \tag{8.355}$$

where $W_{np\nu} = \sqrt{2}\sum_{ia} V_{npia}(\mathbf{X}+\mathbf{Y})_{ia,\nu}$ with $V_{npia} = (ni|pa)$ in the direct approximation.

First partitioning

Here, $\tilde{\epsilon}_p = \epsilon_p^{\mathrm{HF}} + \frac{v_M}{2}$. The Casida equation has

$$\begin{aligned} A_{ia,jb} &= \delta_{ij}\delta_{ab}\left(\epsilon_a^{\mathrm{HF}} + \frac{v_M}{2} - \epsilon_i^{\mathrm{HF}} - \frac{v_M}{2}\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle & (8.356)\\ &= \delta_{ij}\delta_{ab}\left(\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle & (8.357)\\ B_{ia,jb} &= \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle. & (8.358) \end{aligned}$$

The correlation self-energy is then

$$\Sigma_{pp}^{c,G_0W_0}(\omega) = \sum_{j\nu} \frac{W_{pj\nu}W_{pj\nu}}{\omega - \left(\epsilon_j^{\mathrm{HF}} + \frac{v_M}{2} - \Omega_\nu\right) - \mathrm{i}\eta} + \sum_{b\nu} \frac{W_{pb\nu}W_{pb\nu}}{\omega - \left(\epsilon_b^{\mathrm{HF}} + \frac{v_M}{2} + \Omega_\nu\right) + \mathrm{i}\eta} \tag{8.359}$$

with QP equation

$$\begin{aligned} E_p^{GW} &= \tilde{\epsilon}_p + \Sigma_{pp}^{c,G_0W_0}(E_p^{GW}) & (8.360)\\ &= \epsilon_p^{\mathrm{HF}} + \frac{v_M}{2} + \Sigma_{pp}^{c,G_0W_0}(E_p^{GW}). & (8.361) \end{aligned}$$

Second partitioning

Here, $\tilde{\epsilon}_p = \epsilon_p^{\mathrm{HF}} + v_M n_p$. The Casida equation has

$$\begin{aligned} A_{ia,jb} &= \delta_{ij}\delta_{ab}\left(\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - v_M\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle + \delta_{ij}\delta_{ab} v_M & (8.362)\\ &= \delta_{ij}\delta_{ab}\left(\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right) + \langle \Phi_i^a | \hat{V}_{\mathrm{res}} | \Phi_j^b \rangle & (8.363)\\ B_{ia,jb} &= \langle \Phi_{ij}^{ab} | \hat{V}_{\mathrm{res}} + S | \Phi_0 \rangle & (8.364)\\ &= \langle \Phi_{ij}^{ab} | \hat{V}_{\mathrm{res}} | \Phi_0 \rangle & (8.365)\\ & & (8.366) \end{aligned}$$

The correlation self-energy is then

$$\Sigma_{pp}^{c,G_0W_0}(\omega) = \sum_{j\nu} \frac{W_{pj\nu}W_{pj\nu}}{\omega - \left(\epsilon_j^{\mathrm{HF}} + v_M - \Omega_\nu\right) - \mathrm{i}\eta} + \sum_{b\nu} \frac{W_{pb\nu}W_{pb\nu}}{\omega - \left(\epsilon_b^{\mathrm{HF}} + \Omega_\nu\right) + \mathrm{i}\eta} \tag{8.367}$$

with QP equation for occupied states k

$$E_k^{GW} = \tilde{\epsilon}_k + \Sigma_{kk}^{c,G_0W_0}(E_k^{GW}) \tag{8.368}$$
$$= \epsilon_k^{\mathrm{HF}} + v_M + \Sigma_{kk}^{c,G_0W_0}(E_k^{GW}) \tag{8.369}$$

and for virtual states c

$$E_c^{GW} = \tilde{\epsilon}_c + \Sigma_{cc}^{c,G_0W_0}(E_c^{GW}) \tag{8.370}$$
$$= \epsilon_c^{\mathrm{HF}} + \Sigma_{cc}^{c,G_0W_0}(E_c^{GW}). \tag{8.371}$$

8.13.3 Answering the question of MP2

First partitioning

We know $E^{(1)} = \langle\Phi_0|\hat{V}_{\mathrm{res}}|\Phi_0\rangle = 0$. Then,

$$E^{(2)} = -\frac{1}{4}\sum_{ijab}\frac{\left|\langle\Phi_{ij}^{ab}|\hat{V}_{\mathrm{res}}|\Phi_0\rangle\right|^2}{E_{ij\to ab}^{(0)}} \tag{8.372}$$
$$\overset{RHF}{=} -\sum_{ijab}\frac{\mathcal{K}_{ij}^{ab}}{E_{ij\to ab}^{(0)}} \tag{8.373}$$

where $\mathcal{K}_{ij}^{ab} = (ia|bj)\left(2(ia|bj) - (ib|aj)\right)$, $\hat{F}|\Phi_{ij}^{ab}\rangle = E_{ij\to ab}^{(0)}|\Phi_{ij}^{ab}\rangle$, and

$$E_{ij\to ab}^{(0)} = \epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}} \tag{8.374}$$
$$= \epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}} + 2\frac{v_M}{2} - 2\frac{v_M}{2} \tag{8.375}$$
$$= \epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}} \tag{8.376}$$

Second partitioning

Because $\hat{S}$ is a normal-ordered, one-body operator, $\langle\Phi_0|\hat{S}|\Phi_0\rangle = 0$ and $\langle\Phi_{ij}^{ab}|\hat{S}|\Phi_0\rangle = 0$. So $E^{(1)} = \langle\Phi_0|\hat{V}_{\mathrm{res}}|\Phi_0\rangle + \langle\Phi_0|\hat{S}|\Phi_0\rangle = 0$. Then,

$$E_M^{(2)} = -\frac{1}{4}\sum_{ijab}\frac{\left|\langle\Phi_{ij}^{ab}|\hat{V}_{\mathrm{res}}|\Phi_0\rangle\right|^2}{\tilde{E}_{ij\to ab}^{(0)}} \tag{8.377}$$
$$\overset{RHF}{=} -\sum_{ijab}\frac{\mathcal{K}_{ij}^{ab}}{\tilde{E}_{ij\to ab}^{(0)}} \tag{8.378}$$

with $\hat{F}_M|\Phi_{ij}^{ab}\rangle = \tilde{E}_{ij\to ab}^{(0)}|\Phi_{ij}^{ab}\rangle$ and

$$\tilde{E}_{ij\to ab}^{(0)} = \epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \left(\epsilon_i^{\mathrm{HF}} + v_M\right) - \left(\epsilon_j^{\mathrm{HF}} + v_M\right) \tag{8.379}$$
$$= \epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}} - 2v_M \tag{8.380}$$
$$= E_{ij\to ab}^{(0)} - 2v_M. \tag{8.381}$$

8.13.4 Answering the question of the second-order self-energy

It is known that the first iteration of BWs2 must satisfy the constraint $\mathrm{Tr}(\mathbf{W}^O) - \mathrm{Tr}(\mathbf{W}^V) = E_N^{(2)}$. Consider the definitions

$$W_{ii}^O \equiv \frac{1}{2}\mathbf{\Sigma}_{ii}^{(2),2p1h}[\hat{F}](\omega = \epsilon_i^{\mathrm{HF}}) = \frac{1}{2}\sum_{abj} \frac{\mathcal{K}_{ij}^{ab}}{\epsilon_i^{\mathrm{HF}} + \epsilon_j^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \epsilon_b^{\mathrm{HF}}} \tag{8.382}$$

$$W_{aa}^V \equiv \frac{1}{2}\mathbf{\Sigma}_{aa}^{(2),2h1p}[\hat{F}](\omega = \epsilon_a^{\mathrm{HF}}) = \frac{1}{2}\sum_{ijb} \frac{\mathcal{K}_{ij}^{ab}}{\epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}}. \tag{8.383}$$

First partitioning

$$\mathrm{Tr}(\mathbf{W}^O) - \mathrm{Tr}(\mathbf{W}^V) = \frac{1}{2}\Sigma_{ii}^{(2),2p1h}[\hat{F}](\omega = \epsilon_i^{\mathrm{HF}}) - \frac{1}{2}\Sigma_{aa}^{(2),2h1p}[\hat{F}](\omega = \epsilon_a^{\mathrm{HF}}) \tag{8.384}$$

$$= \frac{1}{2}\sum_{ijab} \frac{\mathcal{K}_{ij}^{ab}}{\epsilon_i^{\mathrm{HF}} + \epsilon_j^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \epsilon_b^{\mathrm{HF}}} - \frac{1}{2}\sum_{ijab} \frac{\mathcal{K}_{ij}^{ab}}{\epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}} \tag{8.385}$$

$$= \sum_{ijab} \frac{\mathcal{K}_{ij}^{ab}}{\epsilon_i^{\mathrm{HF}} + \epsilon_j^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \epsilon_b^{\mathrm{HF}}} \tag{8.386}$$

$$= E^{(2)} \tag{8.387}$$

$$\implies \Sigma_{pp}^{(2)}[F](\omega) = \sum_{ija} \frac{\mathcal{K}_{ij}^{ap}}{\omega - \epsilon_i^{\mathrm{HF}} + \left(\epsilon_a^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}\right)} + \sum_{iab} \frac{\mathcal{K}_{ip}^{ab}}{\omega - \epsilon_a^{\mathrm{HF}} - \left(\epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right)} \tag{8.388}$$

$$= \sum_{ija} \frac{\mathcal{K}_{ij}^{ap}}{\omega - \epsilon_i^{\mathrm{HF}} + \left(\epsilon_a^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}\right) + \frac{|v_M|}{2}} + \sum_{iab} \frac{\mathcal{K}_{ip}^{ab}}{\omega - \epsilon_a^{\mathrm{HF}} - \left(\epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right) + \frac{|v_M|}{2}} \tag{8.389}$$

The quasiparticle equation reads

$$E_p^{(2)} = \epsilon_p^{\mathrm{HF}} + \Sigma_{pp}^{(2)}[F](\omega = E_p^{(2)}) \tag{8.390}$$

$$\approx \epsilon_p^{\mathrm{HF}} + Z_p[F]\Sigma_{pp}^{(2)}[F](\omega = \epsilon_p^{\mathrm{HF}}) \tag{8.391}$$

$$= \epsilon_p^{\mathrm{HF}} + \left(1 - \frac{\partial \Sigma_{pp}^{(2)}[F](\omega)}{\partial \omega}\Big|_{\omega=\epsilon_p^{\mathrm{HF}}}\right)^{-1} \Sigma_{pp}^{(2)}[F](\omega = \epsilon_p^{\mathrm{HF}}) \tag{8.392}$$

where

$$Z_p[F] = \left(1 - \frac{\partial \Sigma^{(2)}_{pp}[F](\omega)}{\partial \omega}\Big|_{\omega=\epsilon_p^{\mathrm{HF}}}\right)^{-1} \tag{8.393}$$

$$= \left(1 + \left[\sum_{ija} \frac{\mathcal{K}^{ap}_{ij}}{\left(\epsilon_p^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} + \left(\epsilon_a^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}\right)\right)^2} + \sum_{iab} \frac{\mathcal{K}^{ab}_{ip}}{\left(\epsilon_p^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \left(\epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right)\right)^2}\right]\right)^{-1} \tag{8.394}$$

$$= \left(1 + \left[\sum_{ija} \frac{\mathcal{K}^{ap}_{ij}}{\left(\epsilon_p^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} + \left(\epsilon_a^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}\right)\right)^2} + \sum_{iab} \frac{\mathcal{K}^{ab}_{ip}}{\left(\epsilon_p^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \left(\epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right)\right)^2}\right]\right)^{-1}. \tag{8.395}$$

Second partitioning

$$\mathrm{Tr}(\mathbf{W}^O) - \mathrm{Tr}(\mathbf{W}^V) = \frac{1}{2}\Sigma^{(2),2p1h}_{ii}[\hat{F}_M](\omega = \epsilon_i^{\mathrm{HF}} + v_M) - \frac{1}{2}\Sigma^{(2),2h1p}_{aa}[\hat{F}_M](\omega = \epsilon_a^{\mathrm{HF}}) \tag{8.396}$$

$$= \frac{1}{2}\sum_{ijab} \frac{\mathcal{K}^{ab}_{ij}}{\epsilon_i^{\mathrm{HF}} + \epsilon_j^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \epsilon_b^{\mathrm{HF}} + 2v_M} - \frac{1}{2}\sum_{ijab} \frac{\mathcal{K}^{ab}_{ij}}{\epsilon_a^{\mathrm{HF}} + \epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}} - 2v_M} \tag{8.397}$$

$$= \sum_{ijab} \frac{\mathcal{K}^{ab}_{ij}}{\epsilon_i^{\mathrm{HF}} + \epsilon_j^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \epsilon_b^{\mathrm{HF}} + 2v_M} \tag{8.398}$$

$$= E_M^{(2)} \tag{8.399}$$

$$\implies \Sigma^{(2)}_{pp}[F_M](\omega) = \sum_{ija} \frac{\mathcal{K}^{ap}_{ij}}{\omega - \epsilon_i^{\mathrm{HF}} + \left(\epsilon_a^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}\right) - 2v_M} + \sum_{iab} \frac{\mathcal{K}^{ab}_{ip}}{\omega - \epsilon_a^{\mathrm{HF}} - \left(\epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right) + v_M}. \tag{8.400}$$

For holes, the quasiparticle equation reads (with $S_{kk} = \langle\Phi_k|\hat{S}|\Phi_k\rangle = \epsilon_k^{\mathrm{HF}}[F] - \epsilon_k^{\mathrm{HF}}[F_M] = -\frac{v_M}{2}$)

$$E_k^{(2)} = \epsilon_k^{\mathrm{HF}} + v_M + S_{kk} + \Sigma^{(2)}_{kk}[F_M](\omega = E_k^{(2)}) \tag{8.401}$$

$$\approx \epsilon_k^{\mathrm{HF}} + v_M + Z_k[F_M]\left(-\frac{v_M}{2} + \Sigma^{(2)}_{kk}[F_M](\omega = \epsilon_k^{\mathrm{HF}} + v_M)\right), \tag{8.402}$$

where

$$Z_k[F_M] = \left(1 - \frac{\partial \Sigma^{(2)}_{kk}[F_M](\omega)}{\partial \omega}\Big|_{\omega=\epsilon_k^{\mathrm{HF}}+v_M}\right)^{-1} \tag{8.403}$$

$$= \left(1 + \left[\sum_{ija} \frac{\mathcal{K}^{ap}_{ij}}{\left(\epsilon_k^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}} + \left(\epsilon_a^{\mathrm{HF}} - \epsilon_j^{\mathrm{HF}}\right) - v_M\right)^2} + \sum_{iab} \frac{\mathcal{K}^{ab}_{ip}}{\left(\epsilon_k^{\mathrm{HF}} - \epsilon_a^{\mathrm{HF}} - \left(\epsilon_b^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}\right) + 2v_M\right)^2}\right]\right)^{-1}. \tag{8.404}$$

For particles, the quasiparticle equation reads (with $S_{cc} = \langle\Phi_c|\hat{S}|\Phi_c\rangle = \epsilon_c^{\rm HF}[F] - \epsilon_c^{\rm HF}[F_M] = +\frac{v_M}{2}$)

$$E_c^{(2)} = \epsilon_c^{\rm HF} + S_{cc} + \Sigma_{cc}^{(2)}[F_M](\omega = E_c^{(2)}) \tag{8.405}$$

$$\approx \epsilon_c^{\rm HF} + Z_c[F_M]\left(+\frac{v_M}{2} + \Sigma_{cc}^{(2)}[F_M](\omega = \epsilon_c^{\rm HF})\right), \tag{8.406}$$

where

$$Z_c[F_M] = \left(1 - \frac{\partial\Sigma_{cc}^{(2)}[F_M](\omega)}{\partial\omega}\Big|_{\omega=\epsilon_c^{\rm HF}}\right)^{-1} \tag{8.407}$$

$$= \left(1 + \left[\sum_{ija}\frac{\mathcal{K}_{ij}^{ap}}{\left(\epsilon_c^{\rm HF} - \epsilon_i^{\rm HF} + \left(\epsilon_a^{\rm HF} - \epsilon_j^{HF} - 2v_M\right)\right)^2} + \sum_{iab}\frac{\mathcal{K}_{ip}^{ab}}{\left(\epsilon_c^{\rm HF} - \epsilon_a^{\rm HF} - (\epsilon_b^{\rm HF} - \epsilon_i^{\rm HF}) + v_M\right)^2}\right]\right)^{-1}. \tag{8.408}$$

8.14 Updated treatment of Madelung in plots

All throughout, we take $\eta_\Sigma = 0^+ \approx 1\text{e}-8\text{Hartree}$ and η_G is the actual finite broadening parameter, which is 0.2 eV and 0.8 eV for 14e19o and 114e485o.

8.14.1 What has not changed

We should check that things are equivalent if we instead compute the spectral function as $A_k(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G_k(\omega)$, but redefining the v_q in Σ_k^x to be $v_q = \begin{cases}\frac{4\pi}{q^2}, & q \neq 0\\ -v_M, & q = 0\end{cases}$.

LDA+G0W0

It is known that LDA is a semi-local theory, so Madelung does not play any role in the construction of the correlation self-energy. Therefore, the formula that was used to get the spectral function was

$$A_k(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G_k^R(\omega) \tag{8.409}$$

$$= \frac{1}{\pi}\frac{|\,\mathrm{Im}\,\Sigma_k^c[\hat{F}^{\rm LDA},\eta_\Sigma](\omega)| + \eta_G}{\left(\omega - F_k^{\rm LDA} - \left(\Sigma_k^x + \mathrm{Re}\,\Sigma_k^c[\hat{F}^{\rm LDA},\eta_\Sigma](\omega) - V_k^{xc}\right)\right)^2 + \left(|\,\mathrm{Im}\,\Sigma_k^c[\hat{F}^{\rm LDA},\eta_\Sigma](\omega)| + \eta_G\right)^2}, \tag{8.410}$$

where $F_k^{\rm LDA} = \frac{k^2}{2} + V_k^{xc}$ and V_k^{xc} is the non-dispersive exchange-correlation potential of the LDA. $\Sigma_k^x = -\frac{1}{\Omega}\sum_{\mathbf{q}} v_q n_{\mathbf{k}+\mathbf{q}}$ with $v_q = \begin{cases}\frac{4\pi}{q^2}, & q \neq 0\\ -v_M, & q = 0\end{cases}$ is the non-local exchange self-energy. Finally,

$$\Sigma_k^c[\hat{F}^{\rm LDA},\eta_\Sigma](\omega) = \sum_{j\nu}\frac{W_{kj\nu}W_{kj\nu}}{\omega - \left(F_j^{\rm LDA} - \Omega_\nu\right) + \mathrm{i}\eta_\Sigma} + \sum_{b\nu}\frac{W_{kb\nu}W_{kb\nu}}{\omega - \left(F_b^{\rm LDA} + \Omega_\nu\right) + \mathrm{i}\eta_\Sigma}, \tag{8.411}$$

where $W_{np\nu} = \sqrt{2}\sum_{ia}(np|ia)(\mathbf{X}+\mathbf{Y})_{ia,\nu}$ in the direct approximation, and Ω_ν are solutions of the Casida eigenproblem with

$$A_{ia,jb} = \delta_{ij}\delta_{ab}\left(F_a^{\mathrm{LDA}} - F_i^{\mathrm{LDA}}\right) + (ia|bj) \tag{8.412}$$

$$= \delta_{ij}\delta_{ab}\left(\frac{a^2 - i^2}{2}\right) + (ia|bj) \tag{8.413}$$

$$B_{ia,jb} = (ia|jb). \tag{8.414}$$

LDA+G0W0+C

Here we are solving for the spectral function as

$$A_k(\omega) = -\frac{1}{\pi}\operatorname{Im} G_k^R(\omega) \tag{8.415}$$

$$G_k^R(t) = G_{0,k}^R(t)e^{C_k^R(t)} \tag{8.416}$$

with

$$G_{0,k}^R(t) = -\mathrm{i}\theta(t)e^{-\mathrm{i}\left(F_k^{\mathrm{LDA}} + \Sigma_k^x - V_k^{xc} - i\eta_G\right)t} \tag{8.417}$$

and

$$C_k^R(t) = +\frac{1}{\pi}\int d\omega \frac{|\operatorname{Im}\Sigma_k^c[\hat{F}^{\mathrm{LDA}}, \eta_\Sigma](\omega + F_k^{\mathrm{LDA}})}{\omega^2}\left(e^{-\mathrm{i}\omega t} + \mathrm{i}\omega t - 1\right). \tag{8.418}$$

8.14.2 What I was doing before

I had $F_k^{\mathrm{HF}} = \frac{k^2}{2} + \Sigma_k^x + v_M n_k$ and $\Sigma_k^x = -\frac{1}{\Omega}\sum_{\mathbf{q}} v_q n_{\mathbf{k}+\mathbf{q}}$ with $v_q = \begin{cases} \frac{4\pi}{q^2}, & q \neq 0 \\ 0, & q = 0 \end{cases}$ as the non-local exchange self-energy.

HF+G0W0

$$A_k(\omega) = -\frac{1}{\pi}\operatorname{Im} G_k^R(\omega) \tag{8.419}$$

$$= \frac{1}{\pi}\frac{|\operatorname{Im}\Sigma_k^c[\hat{F}^{\mathrm{HF}}, \eta_\Sigma](\omega)| + \eta_G}{\left(\omega - F_k^{\mathrm{HF}} - \left(\operatorname{Re}\Sigma_k^c[\hat{F}^{\mathrm{HF}}, \eta_\Sigma](\omega)\right)\right)^2 + \left(|\operatorname{Im}\Sigma_k^c[\hat{F}^{\mathrm{HF}}, \eta_\Sigma](\omega)| + \eta_G\right)^2}. \tag{8.420}$$

$$\Sigma_k^c[F^{\mathrm{HF}}, \eta_\Sigma](\omega) = \sum_{j\nu}\frac{W_{kj\nu}W_{kj\nu}}{\omega - \left(F_j^{\mathrm{HF}} - \Omega_\nu\right) + \mathrm{i}\eta_\Sigma} + \sum_{b\nu}\frac{W_{kb\nu}W_{kb\nu}}{\omega - \left(F_b^{\mathrm{HF}} + \Omega_\nu\right) + \mathrm{i}\eta_\Sigma}, \tag{8.421}$$

where $W_{np\nu} = \sqrt{2}\sum_{ia}(np|ia)(\mathbf{X}+\mathbf{Y})_{ia,\nu}$ in the direct approximation, and Ω_ν are solutions of the Casida eigenproblem with

$$A_{ia,jb} = \delta_{ij}\delta_{ab}\left(F_a^{\mathrm{HF}} - F_i^{\mathrm{HF}}\right) + (ia|bj) \tag{8.422}$$

$$B_{ia,jb} = (ia|jb). \tag{8.423}$$

BWs2+C

$$A_p(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,G^{R,(2;\alpha_1)+\mathrm{C}}_{pp}(\omega) \tag{8.424}$$

$$G^{R,(2;\alpha_1)+\mathrm{C}}_{pp}(t) = -i\theta(t)e^{-i\tilde{\varepsilon}_p t}\exp\left\{C^{(2;\alpha_1)}_{pp}(t)\right\} \tag{8.425}$$

$$= -i\theta(t)e^{-iF^{\mathrm{HF}}_p t}\exp\left\{-i\underbrace{W_{pp}}_{\frac{\alpha_1}{2}\Sigma^{(2),2p1h}_{pp}[G^{(0)\prime}](\omega=\tilde{\varepsilon}_p)}t + \sum_{iab}\frac{\mathcal{K}^{ab}_{pi}}{\left(\tilde{\epsilon}^{ab}_{pi}\right)^2}\left(e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\left(1-\alpha_1/2\right)\tilde{\epsilon}^{ab}_{pi}t - 1\right) + \right. \tag{8.426}$$

$$= -i\theta(t)e^{-iF^{\mathrm{HF}}_p t}\exp\left\{\sum_{iab}\frac{\mathcal{K}^{ab}_{pi}}{\left(\tilde{\epsilon}^{ab}_{pi}\right)^2}\left(e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\tilde{\epsilon}^{ab}_{pi}t - 1\right) + \sum_{ija}\frac{\mathcal{K}^{pa}_{ij}}{\left(\tilde{\epsilon}^{ij}_{pa}\right)^2}\left(e^{-i\tilde{\epsilon}^{ij}_{pa}t} + i\tilde{\epsilon}^{ij}_{pa}t - 1\right)\right\} \tag{8.427}$$

with

$$C^{(2;\alpha_1)}_{pp}(t) = \sum_{iab}\frac{\mathcal{K}^{ab}_{pi}}{\left(\tilde{\epsilon}^{ab}_{pi}\right)^2}\left(e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\left(1-\alpha_1/2\right)\tilde{\epsilon}^{ab}_{pi}t - 1\right) + \sum_{ija}\frac{\mathcal{K}^{pa}_{ij}}{\left(\tilde{\epsilon}^{ij}_{pa}\right)^2}\left(e^{-i\tilde{\epsilon}^{ij}_{pa}t} + i\tilde{\epsilon}^{ij}_{pa}t - 1\right) \tag{8.428}$$

where $\tilde{\epsilon}^{ab}_{pi} = \tilde{\varepsilon}_a + \tilde{\varepsilon}_b - \tilde{\varepsilon}_i - \tilde{\varepsilon}_p$ and $\tilde{\epsilon}^{ij}_{pa} = \tilde{\varepsilon}_i + \tilde{\varepsilon}_j - \tilde{\varepsilon}_a - \tilde{\varepsilon}_p$.

8.14.3 What I am doing now

Now, I am taking $F^{\mathrm{HF}}_k = \frac{k^2}{2} + \Sigma^x_k + \frac{v_M}{2} = \frac{k^2}{2} + \Sigma^x_k - \frac{|v_M|}{2}$ and $\Sigma^x_k = -\frac{1}{\Omega}\sum_{\mathbf{q}} v_q n_{\mathbf{k}+\mathbf{q}}$ with $v_q = \begin{cases}\frac{4\pi}{q^2}, & q \neq 0\\ 0, & q = 0\end{cases}$ as the non-local exchange self-energy.

HF+G0W0

I have ($\omega_{\mathrm{p}} = \omega \mp \frac{|v_M|}{2} \implies \omega = \omega_{\mathrm{p}} \pm \frac{|v_M|}{2}$, where ω_p and ω are the frequency variables on the physical spectral function for electron removal/addition and in the simulation cell, respectively. Note that the upper sign corresponds to electron removal, while the lower sign corresponds to electron addition.)

$$A_k(\omega_p) = -\frac{1}{\pi}\,\mathrm{Im}\,G^R_k(\omega) \tag{8.429}$$

$$= \frac{1}{\pi}\frac{|\,\mathrm{Im}\,\Sigma^c_k[\hat{F}^{\mathrm{HF}}](\omega)| + \eta_G}{\left((\omega) - F^{\mathrm{HF}}_k - \left(\mathrm{Re}\,\Sigma^c_k[\hat{F}^{\mathrm{HF}}](\omega)\right)\right)^2 + \left(|\,\mathrm{Im}\,\Sigma^c_k[\hat{F}^{\mathrm{HF}}](\omega)| + \eta_G\right)^2} \tag{8.430}$$

$$= \frac{1}{\pi}\frac{|\,\mathrm{Im}\,\Sigma^c_k[\hat{F}^{\mathrm{HF}}](\omega)| + \eta_G}{\left(\omega - \frac{k^2}{2} - \Sigma^x_k + \frac{|v_M|}{2} - \left(\mathrm{Re}\,\Sigma^c_k[\hat{F}^{\mathrm{HF}}](\omega)\right)\right)^2 + \left(|\,\mathrm{Im}\,\Sigma^c_k[\hat{F}^{\mathrm{HF}}](\omega)| + \eta_G\right)^2} \tag{8.431}$$

with

$$\Sigma_k^c[F^{\mathrm{HF}}](\omega) = \sum_{j\nu} \frac{W_{kj\nu}W_{kj\nu}}{\omega - \left(F_j^{\mathrm{HF}} - \Omega_\nu\right) + \mathrm{i}\eta_G} + \sum_{b\nu} \frac{W_{kb\nu}W_{kb\nu}}{\omega - \left(F_b^{\mathrm{HF}} + \Omega_\nu\right) + \mathrm{i}\eta_G} \tag{8.432}$$

$$= \sum_{j\nu} \frac{W_{kj\nu}W_{kj\nu}}{\omega + \frac{|v_M|}{2} - \left(\frac{j^2}{2} + \Sigma_j^x\right) + \Omega_\nu + \mathrm{i}\eta_G} + \sum_{b\nu} \frac{W_{kb\nu}W_{kb\nu}}{\omega + \frac{|v_M|}{2} - \left(\frac{b^2}{2} + \Sigma_b^x\right) - \Omega_\nu + \mathrm{i}\eta_G}. \tag{8.433}$$

For one-shot electron removal spectral functions, this is equivalent to

$$A_k(\omega_\mathrm{p}) = -\frac{1}{\pi}\,\mathrm{Im}\, G_k^R(\omega_\mathrm{p} + \frac{|v_M|}{2}) \tag{8.434}$$

$$= \frac{1}{\pi} \frac{|\,\mathrm{Im}\,\Sigma_k^c[\hat{F}^{\mathrm{HF}}](\omega_\mathrm{p} + \frac{|v_M|}{2})| + \eta_G}{\left((\omega_\mathrm{p} + \frac{|v_M|}{2}) - F_k^{\mathrm{HF}} - \left(\mathrm{Re}\,\Sigma_k^c[\hat{F}^{\mathrm{HF}}](\omega_\mathrm{p} + \frac{|v_M|}{2})\right)\right)^2 + \left(|\,\mathrm{Im}\,\Sigma_k^c[\hat{F}^{\mathrm{HF}}](\omega_\mathrm{p} + \frac{|v_M|}{2})| + \eta_G\right)^2} \tag{8.435}$$

$$= \frac{1}{\pi} \frac{|\,\mathrm{Im}\,\Sigma_k^c[\hat{F}^{\mathrm{HF}}](\omega_\mathrm{p} + \frac{|v_M|}{2})| + \eta_G}{\left(\omega_p - \frac{k^2}{2} - \Sigma_k^x + |v_M| - \left(\mathrm{Re}\,\Sigma_k^c[\hat{F}^{\mathrm{HF}}](\omega_\mathrm{p} + \frac{|v_M|}{2})\right)\right)^2 + \left(|\,\mathrm{Im}\,\Sigma_k^c[\hat{F}^{\mathrm{HF}}](\omega_\mathrm{p} + \frac{|v_M|}{2})| + \eta_G\right)^2} \tag{8.436}$$

with

$$\Sigma_k^c[F^{\mathrm{HF}}, \eta_G](\omega_\mathrm{p} + \frac{|v_M|}{2}) = \sum_{j\nu} \frac{W_{kj\nu}W_{kj\nu}}{\omega_\mathrm{p} + \frac{|v_M|}{2} - \left(F_j^{\mathrm{HF}} - \Omega_\nu\right) + \mathrm{i}\eta_G} + \sum_{b\nu} \frac{W_{kb\nu}W_{kb\nu}}{\omega_\mathrm{p} + \frac{|v_M|}{2} - \left(F_b^{\mathrm{HF}} + \Omega_\nu\right) + \mathrm{i}\eta_G} \tag{8.437}$$

$$= \sum_{j\nu} \frac{W_{kj\nu}W_{kj\nu}}{\omega_\mathrm{p} + |v_M| - \left(\frac{j^2}{2} + \Sigma_j^x\right) + \Omega_\nu + \mathrm{i}\eta_G} + \sum_{b\nu} \frac{W_{kb\nu}W_{kb\nu}}{\omega_\mathrm{p} + |v_M| - \left(\frac{b^2}{2} + \Sigma_b^x\right) - \Omega_\nu + \mathrm{i}\eta_G} \tag{8.438}$$

$W_{np\nu} = \sqrt{2}\sum_{ia}(np|ia)(\mathbf{X}+\mathbf{Y})_{ia,\nu}$ in the direct approximation, and Ω_ν are solutions of the Casida eigenproblem. To derive the form of these matrices, consider that

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} \left(n_{\mathbf{k}} + :a_{\mathbf{k}}^\dagger a_{\mathbf{k}}:\right) + E_\mathrm{x} + \sum_{\mathbf{k}} \Sigma_{\mathbf{k}}^x :\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}: + \hat{V}_\mathrm{res} \tag{8.439}$$

$$\equiv E_{\mathrm{HF}} + \underbrace{\sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} :\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}:}_{\hat{F}} + \hat{V}_\mathrm{res} \tag{8.440}$$

$$\implies \hat{H}_N = \hat{H} - E_{\mathrm{HF}}. \tag{8.441}$$

with $\epsilon_{\mathbf{k}}^{\mathrm{HF}} = \epsilon_{\mathbf{k}}^{(0)} + \Sigma_{\mathbf{k}}^{x} = \frac{k^2}{2} + \frac{v_M}{2} + \Sigma_{\mathbf{k}}^{x}$, $E_{\mathrm{HF}} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} n_{\mathbf{k}} + E_{\mathrm{x}}$, and $\hat{V}_{\mathrm{res}} = \frac{1}{2\Omega} \sum_{\mathbf{k}_1\mathbf{k}_2\mathbf{q}\neq 0} v_q :\hat{a}^{\dagger}_{\mathbf{k}_1+\mathbf{q}}\hat{a}^{\dagger}_{\mathbf{k}_2-\mathbf{q}}\hat{a}_{\mathbf{k}_2}\hat{a}_{\mathbf{k}_1}:$. Consider the partitioning $\hat{H}_N = \underbrace{\hat{F}}_{\hat{H}_0} + \underbrace{\hat{V}_{\mathrm{res}}}_{\hat{V}}$.

$$A_{\mu\nu} = \frac{\partial^2 E}{\partial \kappa_{\mu}^{*} \partial \kappa_{\nu}} \tag{8.442}$$

$$= \langle \Phi_0 | \left[E_{\mu}, \left[\hat{H}_N, E_{\nu}^{\dagger} \right] \right] | \Phi_0 \rangle \tag{8.443}$$

$$= \langle \Phi_0 | \left[E_{\mu}, \left[\hat{F}^{\mathrm{HF}}, E_{\nu}^{\dagger} \right] \right] | \Phi_0 \rangle + \langle \Phi_0 | \left[E_{\mu}, \left[\hat{V}_{\mathrm{res}}, E_{\nu}^{\dagger} \right] \right] | \Phi_0 \rangle \tag{8.444}$$

$$A_{ia,jb} = \delta_{ij}\delta_{ab} \left(F_a^{\mathrm{HF}} - F_i^{\mathrm{HF}} \right) + (ia|bj) \tag{8.445}$$

$$= \delta_{ij}\delta_{ab} \left(\frac{a^2}{2} + \Sigma_a^x - \frac{i^2}{2} - \Sigma_i^x \right) + (ia|bj) \tag{8.446}$$

$$B_{ia,jb} = (ia|jb). \tag{8.447}$$

HF+G0W0+C

Here I am solving for the spectral function as

$$A_k(\omega - \frac{|v_M|}{2}) = -\frac{1}{\pi} \operatorname{Im} G_k^R(\omega) \tag{8.448}$$

$$G_k^R(t) = G_{0,k}^R(t) e^{C_k^R(t)} \tag{8.449}$$

with

$$G_{0,k}^R(t) = -\mathrm{i}\theta(t) e^{-\mathrm{i}\left(F_k^{\mathrm{HF}} - i\eta_G\right)t} \tag{8.450}$$

and

$$C_k^R(t) = +\frac{1}{\pi} \int d\omega \frac{|\operatorname{Im} \Sigma_k^c[\hat{F}^{\mathrm{HF}}, \eta_{\Sigma}](\omega + F_k^{\mathrm{HF}})}{\omega^2} \left(e^{-\mathrm{i}\omega t} + \mathrm{i}\omega t - 1 \right). \tag{8.451}$$

The idea is to solve for the time-domain cumulant and the time-domain Green's function all in the time frame of the simulation cell, then Fourier transform into the frequency frame of the simulation cell, and only then shift to the frequency frame of the electron removal spectral function.

BWs2

We have solved the non-interacting Hamiltonian as

$$\mathbf{H}_0' = \begin{pmatrix} \mathbf{F}_{oo} + \alpha_1 \mathbf{W}^O & 0 \\ 0 & \mathbf{F}_{vv} + \alpha_2 \mathbf{W}^V \end{pmatrix}, \tag{8.452}$$

where

$$\mathbf{W}_{ii}^O = \frac{1}{2} \mathbf{\Sigma}_{ii}^{(2),2p1h}(\omega = \tilde{\varepsilon}_i) = \frac{1}{2} \sum_{abj} \frac{\mathcal{K}_{ij}^{ab}}{\tilde{\varepsilon}_i + \tilde{\varepsilon}_j - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b}, \tag{8.453}$$

$$\mathbf{W}_{aa}^V = \frac{1}{2} \mathbf{\Sigma}_{aa}^{(2),2h1p}(\omega = \tilde{\varepsilon}_a) = \frac{1}{2} \sum_{ijb} \frac{\mathcal{K}_{ij}^{ab}}{\tilde{\varepsilon}_a + \tilde{\varepsilon}_b - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j}, \tag{8.454}$$

where $\tilde{\varepsilon}$ is the BWs2 ground-state MO energy and we chose the starting point of the self-consistency cycle to be the HF MO energies due to F_k^{HF}. Given that the idea is to define the noninteracting Green's function as

$$G^{(0)'}(\omega) = \left(\omega - (\mathbf{F} + \underbrace{\alpha_1 \mathbf{W}^O \oplus \alpha_2 \mathbf{W}^V}_{\mathbf{W}})\right)^{-1} \tag{8.455}$$

$$\implies G^{-1}(\omega) = \left(G^{(0)'}(\omega)\right)^{-1} - \mathbf{\Sigma}_{\mathrm{eff}}(\omega), \tag{8.456}$$

with $\mathbf{\Sigma}_{\mathrm{eff}}(\omega) = \mathbf{\Sigma}(\omega) - \mathbf{W} \approx \mathbf{\Sigma}^{(2)}[G^{(0)'}(\omega)] - \mathbf{W}$. The correlation self-energy is

$$\Sigma^{(2)}_{pp}[G^{(0)'}(\omega), \eta_\Sigma](\omega) = \sum_{ija} \frac{\mathcal{K}^{ap}_{ij}}{\omega - \tilde{\varepsilon}_i + (\tilde{\varepsilon}_a - \tilde{\varepsilon}_j) + \mathrm{i}\eta_\Sigma} + \sum_{iab} \frac{\mathcal{K}^{ab}_{ip}}{\omega - \tilde{\varepsilon}_a - (\tilde{\varepsilon}_b - \tilde{\varepsilon}_i) + \mathrm{i}\eta_\Sigma}. \tag{8.457}$$

The spectral function is then

$$A_k(\omega - \frac{|v_M|}{2}) = -\frac{1}{\pi} \operatorname{Im} G^R_k(\omega) \tag{8.458}$$

$$= \frac{1}{\pi} \frac{|\operatorname{Im} \Sigma^{(2)}_k[G^{(0)'}, \eta_\Sigma](\omega)| + \eta_G}{\left(\omega - \tilde{\varepsilon}_k - \left(\operatorname{Re} \Sigma^{(2)}_k[G^{(0)'}, \eta_\Sigma](\omega) - \mathbf{W}_{kk}\right)\right)^2 + \left(|\operatorname{Im} \Sigma^{(2)}_k[G^{(0)'}, \eta_\Sigma](\omega)| + \eta_G\right)^2} \tag{8.459}$$

$$= \frac{1}{\pi} \frac{|\operatorname{Im} \Sigma^{(2)}_k[G^{(0)'}, \eta_\Sigma](\omega)| + \eta_G}{\left(\omega - \epsilon^{\mathrm{HF}}_k + \frac{|v_M|}{2} - \operatorname{Re} \Sigma^{(2)}_k[G^{(0)'}, \eta_\Sigma](\omega)\right)^2 + \left(|\operatorname{Im} \Sigma^{(2)}_k[G^{(0)'}, \eta_\Sigma](\omega)| + \eta_G\right)^2}. \tag{8.460}$$

BWs2+C

Here

$$A_p(\omega - \frac{|v_M|}{2}) = -\frac{1}{\pi} \operatorname{Im} G^{R,(2;\alpha_1)+\mathrm{C}}_{pp}(\omega) \tag{8.461}$$

$$G^{R,(2;\alpha_1)+\mathrm{C}}_{pp}(t) = -i\theta(t) e^{-i\tilde{\varepsilon}_p t} \exp\left\{C^{(2;\alpha_1)}_{pp}(t)\right\} \tag{8.462}$$

$$= -i\theta(t) e^{-i(\epsilon^{\mathrm{HF}}_p - \frac{|v_M|}{2})t} \exp\left\{-i \underbrace{W_{pp}}_{\frac{\alpha_1}{2}\mathbf{\Sigma}^{(2),2p1h}_{pp}[G^{(0)'}](\omega=\tilde{\varepsilon}_p)} t + \sum_{iab} \frac{\mathcal{K}^{ab}_{pi}}{\left(\tilde{\epsilon}^{ab}_{pi}\right)^2} \left(e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\left(1 - \alpha_1/2\right)\tilde{\epsilon}^{ab}_{pi}t - \right.\right. \tag{8.463}$$

$$= -i\theta(t) e^{-i(\epsilon^{\mathrm{HF}}_p - \frac{|v_M|}{2})t} \exp\left\{\sum_{iab} \frac{\mathcal{K}^{ab}_{pi}}{\left(\tilde{\epsilon}^{ab}_{pi}\right)^2} \left(e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\tilde{\epsilon}^{ab}_{pi}t - 1\right) + \sum_{ija} \frac{\mathcal{K}^{pa}_{ij}}{\left(\tilde{\epsilon}^{ij}_{pa}\right)^2} \left(e^{-i\tilde{\epsilon}^{ij}_{pa}t} + i\tilde{\epsilon}^{ij}_{pa}t - 1\right)\right. \tag{8.464}$$

with

$$C^{(2;\alpha_1)}_{pp}(t) = \sum_{iab} \frac{\mathcal{K}^{ab}_{pi}}{\left(\tilde{\epsilon}^{ab}_{pi}\right)^2} \left(e^{-i\tilde{\epsilon}^{ab}_{pi}t} + i\left(1 - \alpha_1/2\right)\tilde{\epsilon}^{ab}_{pi}t - 1\right) + \sum_{ija} \frac{\mathcal{K}^{pa}_{ij}}{\left(\tilde{\epsilon}^{ij}_{pa}\right)^2} \left(e^{-i\tilde{\epsilon}^{ij}_{pa}t} + i\tilde{\epsilon}^{ij}_{pa}t - 1\right) \tag{8.465}$$

8.14.4 Plot labels

I use the definition $\Sigma_k^x = -\frac{1}{\Omega}\sum_{\mathbf{q}} v_q n_{\mathbf{k}+\mathbf{q}}$ with $v_q = \begin{cases} \frac{4\pi}{q^2}, & q \neq 0 \\ 0, & q = 0 \end{cases}$. F^{HF} is diagonal.

Before

Before, I had $F_k^{\mathrm{HF}} = \frac{k^2}{2} + \Sigma_k^x - |v_M| n_k$. The spectral function plotted is the spectral function obtained in the simulation cell.

Now

Now, I am taking $F_k^{\mathrm{HF}} = \frac{k^2}{2} + \Sigma_k^x - \frac{|v_M|}{2}$. The spectral function plotted is the spectral function obtained in the simulation cell *shifted* towards higher binding energies by $\frac{|v_M|}{2}$.

Why this new form for HF energies

Partition the normal-ordered Hamiltonian as $\hat{H}_N = F^{\mathrm{HF}} + \hat{V}_{\mathrm{res}}$, where $\hat{H}_N = \hat{H} - E_{\mathrm{HF}}$. I derived

$$\hat{H} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2\Omega} \sum_{\mathbf{q}\neq\mathbf{0}} v_q \sum_{\mathbf{k}\mathbf{k}'} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}'-\mathbf{q}}^\dagger a_{\mathbf{k}'} a_{\mathbf{k}} \tag{8.466}$$

$$\vdots \tag{8.467}$$

$$= \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} \left(n_{\mathbf{k}} + :a_{\mathbf{k}}^\dagger a_{\mathbf{k}}: \right) + E_{\mathrm{x}} + \sum_{\mathbf{k}} \Sigma_{\mathbf{k}}^x :\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}: + \hat{V}_{\mathrm{res}} \tag{8.468}$$

$$= E_{\mathrm{HF}} + \underbrace{\sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\mathrm{HF}} :\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}:}_{\hat{F}^{\mathrm{HF}}} + \hat{V}_{\mathrm{res}} \tag{8.469}$$

where $\epsilon_{\mathbf{k}}^{\mathrm{HF}} = \epsilon_{\mathbf{k}}^{(0)} + \Sigma_{\mathbf{k}}^x = \frac{k^2}{2} + \frac{v_M}{2} + \Sigma_{\mathbf{k}}^x$, $E_{\mathrm{HF}} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{(0)} n_{\mathbf{k}} + E_{\mathrm{x}}$, and $\hat{V}_{\mathrm{res}} = \frac{1}{2\Omega}\sum_{\mathbf{k}_1\mathbf{k}_2\mathbf{q}\neq 0} v_q :\hat{a}_{\mathbf{k}_1+\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}_2-\mathbf{q}}^\dagger \hat{a}_{\mathbf{k}_2} \hat{a}_{\mathbf{k}_1}:$. We defined $\epsilon_{\mathbf{k}}^{(0)} = \frac{k^2}{2} + \frac{v_M}{2}$ as the bare single-particle energy with $v_M = -2.837297 \times \left(\frac{3}{4\pi}\right)^{1/3} N^{-1/3} r_s^{-1}$ the Madelung constant for the cubic simulation cell.

8.14.5 Appealing to the definition of IP and EA

Define $\Sigma_k^x = -\frac{1}{\Omega}\sum_{\mathbf{q}\neq 0} v_q\, n_{\mathbf{k}+\mathbf{q}}$ and $E_x = -\frac{1}{2\Omega}\sum_{\mathbf{k}}\sum_{\mathbf{q}\neq 0} v_q\, n_{\mathbf{k}} n_{\mathbf{k}+\mathbf{q}} = -\frac{1}{2\Omega}\sum_{\mathbf{k}\neq\mathbf{k}'} v_{\mathbf{k}-\mathbf{k}'}\, n_{\mathbf{k}} n_{\mathbf{k}'} = \frac{1}{2}\sum_{\mathbf{k}} \Sigma_k^x\, n_{\mathbf{k}}$. We have

$$E_{\mathrm{HF}}(N) = \sum_{\mathbf{k}} \frac{k^2}{2}\, n_k + E_x - \frac{N|v_M|}{2} \tag{8.470}$$

$$= \sum_{\mathbf{k}} \left(\frac{k^2}{2} - \frac{|v_M|}{2} \right) n_k + E_x \tag{8.471}$$

IP

If we removed an electron from occupied spin-orbital $\mathbf{i}$, then

$$\begin{aligned} \mathrm{IP} &\equiv E_{\mathrm{HF}}(N-1) - E_{\mathrm{HF}}(N) && (8.472)\\ &= -\left(\frac{i^2}{2} + \Sigma_i^x\right) + \frac{|v_M|}{2} && (8.473)\\ &\equiv -F_i^{\mathrm{HF}}, && (8.474) \end{aligned}$$

since $E_x(N-1) = E_x(N) + \frac{1}{\Omega}\sum_{\mathbf{k}'\neq\mathbf{i}} v_{\mathbf{i}-\mathbf{k}'}\, n_{\mathbf{k}'} = E_x(N) - \Sigma_i^x$.

EA

Add an electron to unoccupied spin-orbital $\mathbf{j}$,

$$\begin{aligned} \mathrm{EA} &\equiv E_{\mathrm{HF}}(N+1) - E_{\mathrm{HF}}(N) && (8.475)\\ &= \left(\frac{j^2}{2} + \Sigma_j^x\right) - \frac{|v_M|}{2} && (8.476)\\ &= F_j^{\mathrm{HF}}, && (8.477) \end{aligned}$$

since $E_x(N+1) = E_x(N) - \frac{1}{\Omega}\sum_{\mathbf{k}'} v_{\mathbf{j}-\mathbf{k}'}\, n_{\mathbf{k}'} = E_x(N) + \Sigma_j^x$.

8.14.6 Interpretation of the first part

It is written *The Madelung constant can be either included in the Hamiltonian as written in Eq. (64) or it can be included as an a posteriori correction to the simulation done without it.* I interpret this as saying that it makes no difference whether the Madelung constant gets lumped into the single-particle energies, as in Eq. (8.471), or added afterwards as

$$E_{\mathrm{HF}}(N) + \frac{N|v_M|}{2} = \sum_{\mathbf{k}} \frac{k^2}{2}\, n_k + E_x. \tag{8.478}$$

This is the meaning of *when the latter choice is made, the spectral functions have to be shifted accordingly in order to compare the results obtained from the former approach.*

8.14.7 Interpretation of the second part

Regardless of whether we included the Madelung constant in the Hamiltonian, there is an additional correction of $-\xi/2$ for both IP and EA to remove spurious image interactions coming from the excess charge created. This means

$$\begin{aligned} E_g &\equiv EA + \frac{|v_M|}{2} + IP + \frac{|v_M|}{2} && (8.479)\\ &= \left(\frac{j^2}{2} + \Sigma_j^x - \frac{|v_M|}{2}\right) + \left[-\left(\frac{i^2}{2} + \Sigma_i^x\right) + \frac{|v_M|}{2}\right] + |v_M| && (8.480)\\ &= \frac{j^2}{2} + \Sigma_j^x - \frac{i^2}{2} - \Sigma_i^x + |v_M| && (8.481) \end{aligned}$$

which is consistent with what is done in QCPBC.

8.14.8 What before does

We know **before** gives correct band gaps. Also, it gives the correct HF total energy:

$$E_{\mathrm{HF}}(N) = \sum_{\mathbf{k}} \frac{k^2}{2}\, n_k - \frac{1}{2}\sum_{\mathbf{k}} n_{\mathbf{k}} \left(\frac{1}{\Omega} \sum_{\mathbf{q}} v_q\, n_{\mathbf{k}+\mathbf{q}} \right) \tag{8.482}$$

$$= \sum_{\mathbf{k}} \frac{k^2}{2}\, n_k - \frac{1}{2}\sum_{\mathbf{k}} n_{\mathbf{k}} \left(\frac{1}{\Omega} \left[\sum_{\mathbf{q}\neq 0} v_q\, n_{\mathbf{k}+\mathbf{q}} + \Omega |v_M|\, n_{\mathbf{k}} \right] \right) \tag{8.483}$$

$$= \sum_{\mathbf{k}} \frac{k^2}{2}\, n_k + \frac{1}{2}\sum_{\mathbf{k}} \Sigma^x_k\, n_{\mathbf{k}} - \frac{|v_M|}{2} \sum_{\mathbf{k}} n^2_{\mathbf{k}} \tag{8.484}$$

$$= \sum_{\mathbf{k}} \frac{k^2}{2}\, n_k + E_x - \frac{N|v_M|}{2}, \tag{8.485}$$

where the last step uses $n^2_{\mathbf{k}} = n_{\mathbf{k}}$ for spin-orbital occupations. This is identical to 8.471. The corresponding HF single-particle energies are

$$\varepsilon^{\mathrm{HF}}_k = \frac{k^2}{2} + \Sigma^x_k - |v_M|\, n_{\mathbf{k}}, \tag{8.486}$$

The eigenvalue gap is then $\left(\frac{j^2}{2} + \Sigma^x_j\right) - \left(\frac{i^2}{2} + \Sigma^x_i\right) + |v_M| \equiv E_g$, so **before** reaches the corrected gap directly, without going through the uncorrected IP and EA.

1. The **before** scheme reproduces the total energy because only occupied $n_{\mathbf{k}}$ enter that sum, but a self-energy is evaluated on both occupied and unoccupied states. Is there a way to check this using Galitskii-Migdal?

8.15 Exact answers for the quasiboson Hamiltonian

We can write the quasiboson Hamiltonian as

$$\hat{H} = \hat{H}_{\mathrm{el}} + \hat{H}_{\mathrm{pl}} + \hat{H}_{\mathrm{ep}} \tag{8.487}$$

8.15.1 Just 2h1p

Here

$$\hat{H}_{\mathrm{el}} = \sum_{\boldsymbol{k}\in\mathcal{O}} \epsilon^{\mathrm{HF}}_{\boldsymbol{k}}\, a^\dagger_{\boldsymbol{k}} a_{\boldsymbol{k}}, \tag{8.488}$$

$$\hat{H}_{\mathrm{pl}} = \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}\, b^\dagger_{\nu_{\boldsymbol{q}}} b_{\nu_{\boldsymbol{q}}}, \tag{8.489}$$

and

$$\hat{H}_{\mathrm{ep}} = \sum_{\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}} \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} \left(b^\dagger_{\nu_{-\boldsymbol{q}}} + b_{\nu_{\boldsymbol{q}}} \right) \tag{8.490}$$

ph-RPA

Here, the idea is to get the quasiboson values from dRPA@LDA.

pp-RPA

Here, the idea is to get the quasiboson values from ppRPA@LDA. The Casida equation is

$$\begin{pmatrix} A & B \\ B^\dagger & C \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \Omega \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \tag{8.491}$$

with

$$A_{ab,cd} = \delta_{ac}\delta_{bd}\left(\epsilon_a + \epsilon_b\right) + \frac{1}{\sqrt{\left(1+\delta_{ab}\right)\left(1+\delta_{cd}\right)}}(\langle ab||cd\rangle + \langle ab||dc\rangle) \tag{8.492}$$

$$B_{ab,kl} = \frac{1}{\sqrt{\left(1+\delta_{ab}\right)\left(1+\delta_{kl}\right)}}(\langle ab||kl\rangle + \langle ab||lk\rangle) \tag{8.493}$$

$$C_{ij,kl} = -\delta_{ik}\delta_{jl}\left(\epsilon_i + \epsilon_j\right) + \frac{1}{\sqrt{\left(1+\delta_{ij}\right)\left(1+\delta_{kl}\right)}}(\langle ij||kl\rangle + \langle ij||lk\rangle) \tag{8.494}$$

8.15.2 2h1p+2p1h

Here

$$\hat{H}_{\mathrm{el}} = \sum_{\boldsymbol{k}} \epsilon^{\mathrm{HF}}_{\boldsymbol{k}} a^\dagger_{\boldsymbol{k}} a_{\boldsymbol{k}}, \tag{8.495}$$

$$\hat{H}_{\mathrm{pl}} = \sum_{\nu_{\boldsymbol{q}} \in \mathcal{M}} \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}} b^\dagger_{\nu_{\boldsymbol{q}}} b_{\nu_{\boldsymbol{q}}}, \tag{8.496}$$

and

$$\hat{H}_{\mathrm{ep}} = \sum_{\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q}} \sum_{\nu_{\boldsymbol{q}} \in \mathcal{M}} g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q}) a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} \left(b^\dagger_{\nu_{-\boldsymbol{q}}} + b_{\nu_{\boldsymbol{q}}}\right) \tag{8.497}$$

8.16 Reduction to G0W0+C

Setup

For an occupied orbital p,

$$G^R_p(t) = -i\theta(t)\left\langle \Psi_0 \right| \left\{a_p(t), a^\dagger_p\right\} \left|\Psi_0\right\rangle \tag{8.498}$$

$$= -i\theta(t)\left[\left\langle \Psi_0 \right| a_p(t)\, a^\dagger_p \left|\Psi_0\right\rangle + \left\langle \Psi_0 \right| a^\dagger_p\, a_p(t) \left|\Psi_0\right\rangle\right] \tag{8.499}$$

$$= -i\theta(t)\left[\left\langle \Psi_0 \right| e^{iHt} a_p e^{-iHt} a^\dagger_p \left|\Psi_0\right\rangle + \left\langle \Psi_0 \right| a^\dagger_p e^{iHt} a_p e^{-iHt} \left|\Psi_0\right\rangle\right] \tag{8.500}$$

$$= -i\theta(t)\left[e^{iE_0 t}\left\langle \Psi_0 \right| a_p\, e^{-iHt}\, a^\dagger_p \left|\Psi_0\right\rangle + e^{-iE_0 t}\left\langle \Psi_0 \right| a^\dagger_p\, e^{iHt}\, a_p \left|\Psi_0\right\rangle\right] \tag{8.501}$$

Partition into the non-interacting Hamiltonian $\hat{H}_0 = \hat{H}_{\mathrm{el}} + \hat{H}_{\mathrm{pl}}$ and the perturbation $\hat{V} = \hat{H}_{\mathrm{ep}}$. We can define our reference state to be $|\Phi\rangle \equiv |\mathrm{FS}\rangle \otimes |0\rangle_{\mathrm{b}}$, where $|\mathrm{FS}\rangle$ is the Fermi surface state (the choices for this are HF or LDA) and $|0\rangle_{\mathrm{b}}$ is the bosonic vacuum. Then, $\hat{H}_0|\Phi\rangle = E_\Phi|\Phi\rangle$, with $E_\Phi = \sum_{\boldsymbol{k}\in\mathcal{O}} \epsilon^{\mathrm{HF}}_{\boldsymbol{k}}$.

Definitions

Define $\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q}) \equiv \epsilon_{\boldsymbol{k}+\boldsymbol{q}} - \epsilon_{\boldsymbol{k}} + \Omega_{\nu,-\boldsymbol{q}} > 0$ and $\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q}) \equiv \epsilon_{p+\boldsymbol{q}} - \epsilon_p - \Omega_{\nu,\boldsymbol{q}}$, which use $\boldsymbol{k} \in \mathcal{O}$ with $\boldsymbol{k}+\boldsymbol{q} \in \mathcal{V}$ and $p+\boldsymbol{q} \in \mathcal{O}$, respectively. We have the Hermiticity condition $g_\nu(\boldsymbol{k},\boldsymbol{q})^* = g_\nu(\boldsymbol{k}+\boldsymbol{q},-\boldsymbol{q})$. The positivity of $\Delta^{\mathcal{V}}$ in the finite basis ensures that $|\Phi\rangle$ is a non-degenerate ground state of $\hat{H}_0$, allowing for a well-defined RSPT. The boson parity operator $\hat{P} = (-1)^{\hat{N}_b}$ trivially commutes with the fermionic operators $\hat{H}_{\text{el}}$, a_p, and $a_p^\dagger$, commutes with $\hat{H}_{\text{pl}}$ (even number of bosons), and anticommutes with $\hat{V}$ (odd number of bosons). Any matrix element with an odd number of $\hat{V}$ operators between states of the same parity vanishes.

Ground state properties

With intermediate normalization, $\langle\Phi|\Psi_0\rangle = 1$, and defining the reduced resolvent $\hat{R}_0 \equiv \hat{Q}\,(E_\Phi - \hat{H}_0)^{-1}\hat{Q}$ with $\hat{Q} \equiv 1 - |\Phi\rangle\langle\Phi|$, RSPT gives the ground state as $|\Psi_0\rangle = \mathcal{N}^{-1/2}|\tilde{\Psi}_0\rangle$,

with

$$|\tilde{\Psi}_0\rangle = |\Phi\rangle + |\Psi^{(1)}\rangle + |\Psi^{(2)}\rangle + O(g^3), \tag{8.502}$$

$$|\Psi^{(1)}\rangle \equiv \hat{R}_0\hat{V}|\Phi\rangle \tag{8.503}$$

$$= \sum_{n\neq\Phi} |n\rangle \frac{\langle n|\hat{V}|\Phi\rangle}{E_\Phi - E_n} \tag{8.504}$$

$$= -\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{g_\nu(\boldsymbol{k},\boldsymbol{q})}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu-\boldsymbol{q}}|\Phi\rangle, \tag{8.505}$$

$$|\Psi^{(2)}\rangle \equiv \hat{R}_0\hat{V}|\Psi^{(1)}\rangle - E_0^{(1)}\hat{R}_0|\Psi^{(1)}\rangle = \hat{R}_0\hat{V}\hat{R}_0\hat{V}|\Phi\rangle \qquad \Big(E_0^{(1)} = \langle\Phi|\hat{V}|\Phi\rangle = 0\Big) \tag{8.506}$$

$$= \sum_{n\neq\Phi}\sum_{m\neq\Phi} |n\rangle \frac{\langle n|\hat{V}|m\rangle\langle m|\hat{V}|\Phi\rangle}{(E_\Phi - E_n)(E_\Phi - E_m)} \tag{8.507}$$

$$\mathcal{N} \equiv \langle\tilde{\Psi}_0|\tilde{\Psi}_0\rangle \tag{8.508}$$

$$= \Big(\langle\Phi| + \langle\Psi^{(1)}| + \langle\Psi^{(2)}| + \dots\Big)\Big(|\Phi\rangle + |\Psi^{(1)}\rangle + |\Psi^{(2)}\rangle + \dots\Big) \tag{8.509}$$

$$= \langle\Phi|\Phi\rangle + \underbrace{2\,\mathrm{Re}\,\langle\Phi|\Psi^{(1)}\rangle + 2\,\mathrm{Re}\,\langle\Phi|\Psi^{(2)}\rangle}_{=0\ \text{(intermediate normalization)}} + \langle\Psi^{(1)}|\Psi^{(1)}\rangle + \underbrace{2\,\mathrm{Re}\,\langle\Psi^{(1)}|\Psi^{(2)}\rangle}_{=0\ \text{(boson parity)}} + O(g^4) \tag{8.510}$$

$$= 1 + \langle\Psi^{(1)}|\Psi^{(1)}\rangle + O(g^4) \tag{8.511}$$

$$= 1 + \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2} + \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2} + O(g^4), \tag{8.512}$$

$$E_0 \equiv \langle\Phi|\hat{H}|\tilde{\Psi}_0\rangle \qquad \Big(\text{since } \hat{H}|\tilde{\Psi}_0\rangle = E_0|\tilde{\Psi}_0\rangle \text{ and } \langle\Phi|\tilde{\Psi}_0\rangle = 1\Big) \tag{8.513}$$

$$= \underbrace{\langle\Phi|\hat{H}_0|\tilde{\Psi}_0\rangle}_{E_\Phi} + \langle\Phi|\hat{V}|\tilde{\Psi}_0\rangle \tag{8.514}$$

$$= E_\Phi + \underbrace{\langle\Phi|\hat{V}|\Phi\rangle}_{=0\ \text{(parity)}} + \langle\Phi|\hat{V}|\Psi^{(1)}\rangle + \underbrace{\langle\Phi|\hat{V}|\Psi^{(2)}\rangle}_{=0\ \text{(parity)}} + O(g^4) \tag{8.515}$$

$$= E_\Phi - \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{g_\nu(\boldsymbol{k},\boldsymbol{q})}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} \langle\Phi|\hat{V}\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu-\boldsymbol{q}}|\Phi\rangle + O(g^4) \tag{8.516}$$

$$= E_\Phi - \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} - \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})} + O(g^4). \tag{8.517}$$

8.16.1 Electron removal

We want the removal term to second order in the coupling,

$$R_p(t) = e^{-iE_0 t} \langle \Psi_0 | \, a_p^\dagger \, e^{i\hat{H}t} \, a_p \, | \Psi_0 \rangle \tag{8.518}$$

$$= \frac{e^{-iE_0 t}}{\mathcal{N}} \left\langle \tilde{\Psi}_0 \right| a_p^\dagger \, e^{i\hat{H}t} \, a_p \left| \tilde{\Psi}_0 \right\rangle . \tag{8.519}$$

To remove the need for constructing any $(N-1)$-electron eigenstates, we must expand the propagator $e^{i\hat{H}t}$ acting on the $(N-1)$-electron state to $O(g^2)$.

$$a_p|\tilde{\Psi}_0\rangle = a_p|\Phi\rangle + a_p|\Psi^{(1)}\rangle + a_p|\Psi^{(2)}\rangle + O(g^3) \tag{8.520}$$

$$= \underbrace{|p\rangle}_{O(1)} \underbrace{- \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\} \\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{g_\nu(\boldsymbol{k},\boldsymbol{q})}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} \, |p;\boldsymbol{k}\boldsymbol{q}\nu\rangle}_{a_p|\Psi^{(1)}\rangle = O(g)} + \underbrace{a_p|\Psi^{(2)}\rangle}_{O(g^2)} + O(g^3). \tag{8.521}$$

Expanding on the identity $e^{i\hat{H}t} = e^{i\hat{H}_0 t} + i\int_0^t ds\, e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}s}$ gives

$$e^{i\hat{H}t} = \underbrace{e^{i\hat{H}_0 t}}_{\hat{X}_0} + \underbrace{i\int_0^t ds\, e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}_0 s}}_{\hat{X}_1} \underbrace{- \int_0^t ds \int_0^s ds'\, e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}_0(s-s')}\hat{V}e^{i\hat{H}_0 s'}}_{\hat{X}_2} + O(g^3) \tag{8.522}$$

$$\implies \mathcal{N}\, e^{iE_0 t} R_p(t) = \langle \tilde{\Psi}_0 | a_p^\dagger e^{i\hat{H}t} a_p | \tilde{\Psi}_0 \rangle \tag{8.523}$$

$$= \underbrace{\langle p|\hat{X}_0|p\rangle}_{O(1)} \tag{8.524}$$

$$+ \underbrace{\langle p|\hat{X}_1|p\rangle + \langle p|\hat{X}_0 a_p|\Psi^{(1)}\rangle + \langle \Psi^{(1)}|a_p^\dagger \hat{X}_0|p\rangle}_{=0(\text{odd number of } \hat{V})} \tag{8.525}$$

$$+ \underbrace{\langle p|\hat{X}_2|p\rangle + \langle \Psi^{(1)}|a_p^\dagger \hat{X}_0 a_p|\Psi^{(1)}\rangle + \langle p|\hat{X}_1 a_p|\Psi^{(1)}\rangle + \langle \Psi^{(1)}|a_p^\dagger \hat{X}_1|p\rangle}_{O(g^2)} \tag{8.526}$$

$$+ \underbrace{\langle p|\hat{X}_0 a_p|\Psi^{(2)}\rangle + \langle \Psi^{(2)}|a_p^\dagger \hat{X}_0|p\rangle}_{O(g^2)} + O(g^4) \tag{8.527}$$

$$= \underbrace{\langle p|\hat{X}_0 + \hat{X}_2|p\rangle}_{T_0} + \underbrace{\langle \Psi^{(1)}|a_p^\dagger \hat{X}_0 a_p|\Psi^{(1)}\rangle}_{T_1} + \underbrace{\langle p|\hat{X}_1 a_p|\Psi^{(1)}\rangle + \langle \Psi^{(1)}|a_p^\dagger \hat{X}_1|p\rangle}_{T_2+T_3} + O(g^4). \tag{8.528}$$

T_0

First, using $\hat{H}_0|p\rangle = (E_\Phi - \epsilon_p)|p\rangle$:

$$\langle p|\hat{X}_0|p\rangle = e^{i(E_\Phi - \epsilon_p)t}. \tag{8.529}$$

Then,

$$\langle p|\hat{X}_2|p\rangle = -\int_0^t ds \int_0^s ds'\, \langle p|e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}_0(s-s')}\hat{V}e^{i\hat{H}_0 s'}|p\rangle \tag{8.530}$$

$$= -\int_0^t ds \int_0^s ds'\, e^{i(E_\Phi-\epsilon_p)(t-s)}e^{i(E_\Phi-\epsilon_p)s'}\, \langle p|\hat{V}e^{i\hat{H}_0(s-s')}\hat{V}|p\rangle \tag{8.531}$$

$$= -\int_0^t ds \int_0^s ds'\, e^{i(E_\Phi-\epsilon_p)(t-s+s')} \sum_n \langle p|\hat{V}|n\rangle\, e^{iE_n(s-s')}\, \langle n|\hat{V}|p\rangle \tag{8.532}$$

$$= -\, e^{i(E_\Phi-\epsilon_p)t} \sum_n \left|\langle n|\hat{V}|p\rangle\right|^2 \int_0^t ds \int_0^s ds'\, e^{i[E_n-(E_\Phi-\epsilon_p)](s-s')} \tag{8.533}$$

$$= e^{i(E_\Phi-\epsilon_p)t}\Bigg[\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu |g_\nu(\boldsymbol{k},\boldsymbol{q})|^2\, \frac{e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t} - 1 - i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}$$

$$+ \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}} |g_\nu(p,\boldsymbol{q})|^2\, \frac{e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t} - 1 + i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\Bigg], \tag{8.534}$$

where to get to 8.532 we inserted a complete set of $\hat{H}_0$ eigenstates $|n\rangle$. To get to 8.534, define $\omega_n \equiv E_n - (E_\Phi - \epsilon_p)$ and evaluate the double integral as

$$\int_0^t ds \int_0^s ds'\, e^{i\omega_n(s-s')} = \int_0^t ds\, \frac{e^{i\omega_n s}-1}{i\omega_n} = \frac{1}{i\omega_n}\left[\frac{e^{i\omega_n t}-1}{i\omega_n} - t\right] = -\,\frac{e^{i\omega_n t}-1-i\omega_n t}{\omega_n^2}. \tag{8.535}$$

To understand why only two specific terms survive, write out $\hat{V}|p\rangle$,

$$\hat{V}|p\rangle = \sum_{\boldsymbol{k},\boldsymbol{q}} \sum_\nu g_\nu(\boldsymbol{k},\boldsymbol{q})\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} \left(b^\dagger_{\nu_{-\boldsymbol{q}}} + b_{\nu_{\boldsymbol{q}}}\right) a_p|\Phi\rangle \tag{8.536}$$

$$= \sum_{\boldsymbol{k},\boldsymbol{q}} \sum_\nu g_\nu(\boldsymbol{k},\boldsymbol{q})\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu_{-\boldsymbol{q}}} a_p|\Phi\rangle \tag{8.537}$$

$$= \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu g_\nu(\boldsymbol{k},\boldsymbol{q})\, |p;\boldsymbol{k}\boldsymbol{q}\nu\rangle - \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}} g_\nu(p,\boldsymbol{q})^*\, |m_{\boldsymbol{q}\nu}\rangle, \tag{8.538}$$

where $|p;\boldsymbol{k}\boldsymbol{q}\nu\rangle \equiv a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu_{-\boldsymbol{q}}}|p\rangle$ and $|m_{\boldsymbol{q}\nu}\rangle \equiv a_{p+\boldsymbol{q}} b^\dagger_{\nu_{\boldsymbol{q}}}|\Phi\rangle$. Physically, the first term corresponds to the situation where the hole remains in orbital p, while another electron is excited. The second term corresponds to the situation where the hole hops from orbital p to orbital $p+\boldsymbol{q}$.

T_1

$$T_1 = \langle \Psi^{(1)} | a_p^\dagger \, \hat{X}_0 \, a_p | \Psi^{(1)} \rangle \tag{8.539}$$

$$= \left(a_p | \Psi^{(1)} \rangle \right)^\dagger e^{i\hat{H}_0 t} \left(a_p | \Psi^{(1)} \rangle \right) \tag{8.540}$$

$$= \sum_{\substack{\boldsymbol{k}' \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k}' + \boldsymbol{q}' \in \mathcal{V}}} \sum_{\nu'} \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k} + \boldsymbol{q} \in \mathcal{V}}} \sum_{\nu} \frac{g_{\nu'}(\boldsymbol{k}', \boldsymbol{q}')^* \, g_\nu(\boldsymbol{k}, \boldsymbol{q})}{\Delta^{\mathcal{V}}_{\nu'}(\boldsymbol{k}', \boldsymbol{q}') \, \Delta^{\mathcal{V}}_{\nu}(\boldsymbol{k}, \boldsymbol{q})} \langle p; \boldsymbol{k}' \boldsymbol{q}' \nu' | e^{i\hat{H}_0 t} | p; \boldsymbol{k} \boldsymbol{q} \nu \rangle \tag{8.541}$$

$$= \sum_{\substack{\boldsymbol{k}' \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k}' + \boldsymbol{q}' \in \mathcal{V}}} \sum_{\nu'} \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k} + \boldsymbol{q} \in \mathcal{V}}} \sum_{\nu} \frac{g_{\nu'}(\boldsymbol{k}', \boldsymbol{q}')^* \, g_\nu(\boldsymbol{k}, \boldsymbol{q})}{\Delta^{\mathcal{V}}_{\nu'}(\boldsymbol{k}', \boldsymbol{q}') \, \Delta^{\mathcal{V}}_{\nu}(\boldsymbol{k}, \boldsymbol{q})} e^{i\left(E_\Phi - \epsilon_p + \Delta^{\mathcal{V}}_\nu(\boldsymbol{k}, \boldsymbol{q})\right) t} \, \delta_{\boldsymbol{k}\boldsymbol{k}'} \delta_{\boldsymbol{q}\boldsymbol{q}'} \delta_{\nu\nu'} \tag{8.542}$$

$$= e^{i(E_\Phi - \epsilon_p)t} \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k} + \boldsymbol{q} \in \mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k}, \boldsymbol{q})|^2}{\Delta^{\mathcal{V}}_\nu(\boldsymbol{k}, \boldsymbol{q})^2} \, e^{i\Delta^{\mathcal{V}}_\nu(\boldsymbol{k}, \boldsymbol{q}) t}, \tag{8.543}$$

where to get to 8.541 we used

$$a_p | \Psi^{(1)} \rangle = - \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \\ \boldsymbol{k} + \boldsymbol{q} \in \mathcal{V}}} \sum_{\nu} \frac{g_\nu(\boldsymbol{k}, \boldsymbol{q})}{\Delta^{\mathcal{V}}_\nu(\boldsymbol{k}, \boldsymbol{q})} \, a_p \, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu_{-\boldsymbol{q}}} | \Phi \rangle \tag{8.544}$$

$$= - \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k} + \boldsymbol{q} \in \mathcal{V}}} \sum_{\nu} \frac{g_\nu(\boldsymbol{k}, \boldsymbol{q})}{\Delta^{\mathcal{V}}_\nu(\boldsymbol{k}, \boldsymbol{q})} \, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu_{-\boldsymbol{q}}} \, a_p | \Phi \rangle \tag{8.545}$$

$$\equiv - \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \setminus \{p\} \\ \boldsymbol{k} + \boldsymbol{q} \in \mathcal{V}}} \sum_{\nu} \frac{g_\nu(\boldsymbol{k}, \boldsymbol{q})}{\Delta^{\mathcal{V}}_\nu(\boldsymbol{k}, \boldsymbol{q})} \, | p; \boldsymbol{k} \boldsymbol{q} \nu \rangle. \tag{8.546}$$

T_2 and T_3

$$T_2 = \langle p|\hat{X}_1\, a_p|\Psi^{(1)}\rangle \tag{8.547}$$

$$= i\int_0^t ds\, \langle p|e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}_0 s}a_p|\Psi^{(1)}\rangle \tag{8.548}$$

$$= i\int_0^t ds\, e^{i(E_\Phi-\epsilon_p)(t-s)}\sum_n \langle p|\hat{V}|n\rangle\, e^{iE_n s}\,\langle n|a_p|\Psi^{(1)}\rangle \tag{8.549}$$

$$= i\int_0^t ds\, e^{i(E_\Phi-\epsilon_p)(t-s)}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu g_\nu(\boldsymbol{k},\boldsymbol{q})^*\left(-\frac{g_\nu(\boldsymbol{k},\boldsymbol{q})}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\right)e^{i\left(E_\Phi-\epsilon_p+\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})\right)s} \tag{8.550}$$

$$= -i\, e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\int_0^t ds\, e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})s} \tag{8.551}$$

$$= -i\, e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\,\frac{e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}-1}{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} \tag{8.552}$$

$$= -\, e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}\left(e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}-1\right), \tag{8.553}$$

where to get to 8.550 we used the fact, derived in 8.546, that $\langle n|a_p|\Psi^{(1)}\rangle$ is nonzero only for $|n\rangle = |p;\boldsymbol{k}\boldsymbol{q}\nu\rangle$. Then,

$$T_3 = \langle\Psi^{(1)}|a_p^\dagger\,\hat{X}_1|p\rangle \tag{8.554}$$

$$= i\int_0^t ds\,\langle\Psi^{(1)}|a_p^\dagger\,e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}_0 s}|p\rangle \tag{8.555}$$

$$= i\int_0^t ds\,e^{i(E_\Phi-\epsilon_p)s}\,\langle\Psi^{(1)}|a_p^\dagger\,e^{i\hat{H}_0(t-s)}\hat{V}|p\rangle \tag{8.556}$$

$$= i\int_0^t ds\,e^{i(E_\Phi-\epsilon_p)s}\sum_n\langle\Psi^{(1)}|a_p^\dagger|n\rangle\,e^{iE_n(t-s)}\,\langle n|\hat{V}|p\rangle \tag{8.557}$$

$$= i\int_0^t ds\,e^{i(E_\Phi-\epsilon_p)s}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\left(-\frac{g_\nu(\boldsymbol{k},\boldsymbol{q})^*}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\right)e^{i\left(E_\Phi-\epsilon_p+\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})\right)(t-s)}\,g_\nu(\boldsymbol{k},\boldsymbol{q}) \tag{8.558}$$

$$= -i\,e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\,e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}\int_0^t ds\,e^{-i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})s} \tag{8.559}$$

$$= -i\,e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\,e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}\,\frac{1-e^{-i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}}{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} \tag{8.560}$$

$$= -\,e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}\left(e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}-1\right). \tag{8.561}$$

Therefore,

$$T_2+T_3 = -2\,e^{i(E_\Phi-\epsilon_p)t}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}\left(e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}-1\right). \tag{8.562}$$

Putting it all together gives

$$\mathcal{N}\,e^{iE_0t}R_p(t) = T_0+T_1+T_2+T_3+O(g^4) \tag{8.563}$$

$$= e^{i(E_\Phi-\epsilon_p)t}\Bigg[1+\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}\Big(\underbrace{e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}-1-i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}_{T_0}+\underbrace{e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}}_{T_1}\underbrace{-2e^{i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t}+2}_{T_2+T_3}\Big)$$
$$+\sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}}\frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\left(e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t}+i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t-1\right)\Bigg]+O(g^4) \tag{8.564}$$

$$= e^{i(E_\Phi-\epsilon_p)t}\Bigg[1+\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}}\sum_\nu\frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}\left(1-i\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})t\right)$$
$$+\sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}}\frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\left(e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t}+i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t-1\right)\Bigg]+O(g^4). \tag{8.565}$$

Now using the identities

$$\mathcal{N}^{-1} = 1 - \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2} - \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2} + O(g^4), \tag{8.566}$$

$$e^{-iE_0 t} = e^{-iE_\Phi t}\, e^{-i(E_0-E_\Phi)t} = e^{-iE_\Phi t}\left[1 + it \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} + it \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})}\right] + O(g^4), \tag{8.567}$$

we find

$$R_p(t) = \mathcal{N}^{-1}\, e^{-iE_0 t}\left[\mathcal{N}\, e^{iE_0 t} R_p(t)\right] \tag{8.568}$$

$$\begin{aligned}
&= \underbrace{e^{-iE_\Phi t}\, e^{i(E_\Phi - \epsilon_p)t}}_{=e^{-i\epsilon_p t}} \left[1 - \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2} - \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2}\right]_{\mathcal{N}^{-1}} \\
&\quad \times \left[1 + it \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} + it \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})}\right]_{e^{-i(E_0-E_\Phi)t}} \\
&\quad \times \left[1 + \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2} - it \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\right. \\
&\qquad \left. + \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\left(e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t} + i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t - 1\right)\right]_{T_0+T_1+T_2+T_3} + O(g^4)
\end{aligned} \tag{8.569}$$

$$\begin{aligned}
&= e^{-i\epsilon_p t}\left[1 + \cancel{\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}} - \cancel{\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})^2}}\right. \\
&\quad + \cancel{\left(-it \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})} + it \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\setminus\{p\}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_{\nu} \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(\boldsymbol{k},\boldsymbol{q})}\right)} \\
&\quad - \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2} + it \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})} \\
&\quad \left. + \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\left(e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t} + i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t - 1\right)\right] + O(g^4)
\end{aligned} \tag{8.570}$$

$$\begin{aligned}
&= e^{-i\epsilon_p t}\left[1 + \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\left(e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t} + i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t - 1\right)\right. \\
&\quad \left. + \sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2}\left(i\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})t - 1\right)\right] + O(g^4),
\end{aligned} \tag{8.571}$$

where to get to 8.570, we used the fact that each bracket is $1 + O(g^2)$, so the product of any two $O(g^2)$ pieces is $O(g^4)$.

8.16.2 Electron addition

Since $p \in \mathcal{O}$, $a_p^\dagger|\Phi\rangle = 0$, and only the $\boldsymbol{k} = p$ terms of $|\Psi^{(1)}\rangle$ survive. Using $a_p^\dagger a_{p+\boldsymbol{q}}^\dagger a_p|\Phi\rangle = -a_{p+\boldsymbol{q}}^\dagger|\Phi\rangle$,

$$a_p^\dagger|\Psi^{(1)}\rangle = \sum_{\substack{\boldsymbol{q},\nu \\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{g_\nu(p,\boldsymbol{q})}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})}\, a_{p+\boldsymbol{q}}^\dagger b_{\nu-\boldsymbol{q}}^\dagger|\Phi\rangle, \qquad \hat{H}_0\, a_{p+\boldsymbol{q}}^\dagger b_{\nu-\boldsymbol{q}}^\dagger|\Phi\rangle = \big(E_\Phi + \epsilon_p + \Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})\big)\, a_{p+\boldsymbol{q}}^\dagger b_{\nu-\boldsymbol{q}}^\dagger|\Phi\rangle. \tag{8.572}$$

Because $a_p^\dagger|\tilde{\Psi}_0\rangle = O(g)$, parity implies that every correction to $\langle\Psi^{(1)}|a_p\, e^{-i\hat{H}_0 t}\, a_p^\dagger|\Psi^{(1)}\rangle$ (from $|\Psi^{(2)}\rangle$, from higher terms of $e^{-i\hat{H}t}$, from $\mathcal{N}$, and from $E_0 - E_\Phi$) enters at $O(g^4)$. Hence

$$A_p(t) = e^{iE_\Phi t}\langle\Psi^{(1)}|a_p\, e^{-i\hat{H}_0 t}\, a_p^\dagger|\Psi^{(1)}\rangle + O(g^4) = e^{-i\epsilon_p t} \sum_{\substack{\boldsymbol{q},\nu \\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2}\, e^{-i\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})t} + O(g^4). \tag{8.573}$$

Landau form

Add Eqs. (**??**) and (8.573) and divide by $G_p^{R,0}(t) = -i\theta(t)e^{-i\epsilon_p t}$. The addition sector supplies the missing $e^{-i\Delta t}$ for the virtual branch, so both branches take the same form:

$$\begin{aligned}\frac{G_p^R(t)}{G_p^{R,0}(t)} = 1 &+ \sum_{\substack{\boldsymbol{q},\nu \\ p+\boldsymbol{q}\in\mathcal{O}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})^2}\Big(e^{-i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t} + i\Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})t - 1\Big) \\ &+ \sum_{\substack{\boldsymbol{q},\nu \\ p+\boldsymbol{q}\in\mathcal{V}}} \frac{|g_\nu(p,\boldsymbol{q})|^2}{\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})^2}\Big(e^{-i\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})t} + i\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})t - 1\Big) + O(g^4).\end{aligned} \tag{8.574}$$

Introducing the excitation spectral function

$$\beta_p(\omega) \equiv \sum_{\substack{\boldsymbol{q},\nu \\ p+\boldsymbol{q}\in\mathcal{O}}} |g_\nu(p,\boldsymbol{q})|^2\, \delta\big(\omega - \Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q})\big) + \sum_{\substack{\boldsymbol{q},\nu \\ p+\boldsymbol{q}\in\mathcal{V}}} |g_\nu(p,\boldsymbol{q})|^2\, \delta\big(\omega - \Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q})\big)\,, \tag{8.575}$$

this becomes

$$\frac{G_p^R(t)}{G_p^{R,0}(t)} = 1 + \int_{-\infty}^{\infty} d\omega\, \frac{\beta_p(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right) + O(g^4). \tag{8.576}$$

Defining the cumulant through $G_p^R(t) \equiv G_p^{R,0}(t)\, e^{C_p(t)}$, and using $\ln(1+x) = x + O(x^2)$ with $x = O(g^2)$,

$$C_p(t) = \int_{-\infty}^{\infty} d\omega\, \frac{\beta_p(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right) + O(g^4), \tag{8.577}$$

which is the Landau form. Truncating at second order gives the retarded G0W0+C Green's function,

$$G_p^R(t) \approx -i\theta(t)\, e^{-i\epsilon_p t} \exp\left[\int_{-\infty}^{\infty} d\omega\, \frac{\beta_p(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right)\right]. \tag{8.578}$$

Define $D_\nu(\boldsymbol{k},\boldsymbol{q}) \equiv \epsilon_{\boldsymbol{k}+\boldsymbol{q}} - \epsilon_{\boldsymbol{k}} + \Omega_{\nu,-\boldsymbol{q}} > 0$ for $\boldsymbol{k} \in \mathcal{O}$, $\boldsymbol{k}+\boldsymbol{q} \in \mathcal{V}$. Define

$$|p\rangle \equiv a_p|\Phi\rangle, \qquad E_p^{(0)} \equiv E_\Phi - \epsilon_p, \qquad K(\omega,t) \equiv \frac{e^{-i\omega t} + i\omega t - 1}{\omega^2}, \tag{8.579}$$

$$\Delta_\nu^{\mathcal{V}}(p,\boldsymbol{q}) \equiv \epsilon_{p+\boldsymbol{q}} - \epsilon_p + \Omega_{\nu,-\boldsymbol{q}} \;\; (p+\boldsymbol{q} \in \mathcal{V}), \qquad \Delta_\nu^{\mathcal{O}}(p,\boldsymbol{q}) \equiv \epsilon_{p+\boldsymbol{q}} - \epsilon_p - \Omega_{\nu\boldsymbol{q}} \;\; (p+\boldsymbol{q} \in \mathcal{O}). \tag{8.580}$$

We abbreviate three sums. $\sum_{\mathrm{sp}}$ runs over $\boldsymbol{k} \in \mathcal{O}\setminus\{p\}$, $\boldsymbol{k}+\boldsymbol{q} \in \mathcal{V}$, and ν, with $g \equiv g_\nu(\boldsymbol{k},\boldsymbol{q})$ and $D \equiv D_\nu(\boldsymbol{k},\boldsymbol{q})$. $\sum_{\mathcal{X}}$ runs over $(\boldsymbol{q},\nu)$ with $p+\boldsymbol{q} \in \mathcal{X}$, with $g \equiv g_\nu(p,\boldsymbol{q})$ and $\Delta \equiv \Delta_\nu^{\mathcal{X}}(p,\boldsymbol{q})$. Since $\hat{P} = (-1)^{\hat{N}_b}$ commutes with $\hat{H}_0$ and a_p but anticommutes with $\hat{V}$, every odd-order term below vanishes, and all remainders are $O(g^4)$ at fixed t.

Ground state. In RSPT with intermediate normalization, $|\tilde{\Psi}_0\rangle = |\Phi\rangle + |\Psi^{(1)}\rangle + |\Psi^{(2)}\rangle + O(g^3)$, $\langle\Phi|\Psi^{(n)}\rangle = 0$, and

$$|\Psi^{(1)}\rangle = -\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{g_\nu(\boldsymbol{k},\boldsymbol{q})}{D_\nu(\boldsymbol{k},\boldsymbol{q})}\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu,-\boldsymbol{q}}|\Phi\rangle, \tag{8.581}$$

$$\mathcal{N} \equiv \langle\tilde{\Psi}_0|\tilde{\Psi}_0\rangle = 1 + \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{|D_\nu(\boldsymbol{k},\boldsymbol{q})|^2} + \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{|\Delta_\nu(\boldsymbol{k},\boldsymbol{q})|^2} + O(g^4) \tag{8.582}$$

$$E_0 = E_\Phi - \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{D_\nu(\boldsymbol{k},\boldsymbol{q})} - \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{|g_\nu(\boldsymbol{k},\boldsymbol{q})|^2}{\Delta_\nu(\boldsymbol{k},\boldsymbol{q})} + O(g^4). \tag{8.583}$$

Then, the normalized ground state is $|\Psi_0\rangle = \mathcal{N}^{-1/2}|\tilde{\Psi}_0\rangle$.

Removal. Acting with a_p kills the $\boldsymbol{k} = p$ terms of $|\Psi^{(1)}\rangle$.

$$a_p|\Psi^{(1)}\rangle = -\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu \frac{g_\nu(\boldsymbol{k},\boldsymbol{q})}{D_\nu(\boldsymbol{k},\boldsymbol{q})}\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu,-\boldsymbol{q}} a_p|\Phi\rangle \tag{8.584}$$

$$\hat{V}a_p|\Phi\rangle = -\sum_{\substack{\boldsymbol{q},\nu\\ p+\boldsymbol{q}\in\mathcal{O}}} g_\nu(p,\boldsymbol{q})^*\,|m_{\boldsymbol{q}\nu}\rangle + \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}+\boldsymbol{q}\in\mathcal{V}}} \sum_\nu g_\nu(\boldsymbol{k},\boldsymbol{q})\, a^\dagger_{\boldsymbol{k}+\boldsymbol{q}} a_{\boldsymbol{k}} b^\dagger_{\nu,-\boldsymbol{q}} a_p|\Phi\rangle, \tag{8.585}$$

with excitation energies D and $-\Delta^{\mathcal{O}}$ relative to $E_p^{(0)}$. Iterate the Duhamel identity $e^{i\hat{H}t} = e^{i\hat{H}_0 t} + i\int_0^t ds\, e^{i\hat{H}_0(t-s)}\hat{V}e^{i\hat{H}s}$ to obtain $\hat{X}_0+\hat{X}_1+\hat{X}_2+O(g^3)$. By parity, and using $\langle\Psi^{(2)}|a_p^\dagger|p\rangle =$

$\langle\Psi^{(2)}|\Phi\rangle = 0,$

$$\mathcal{N}\, e^{iE_0 t} R_p(t) = \langle p|\hat{X}_0 + \hat{X}_2|p\rangle + \langle\Psi^{(1)}|a_p^\dagger \hat{X}_0 a_p|\Psi^{(1)}\rangle + \langle\Psi^{(1)}|a_p^\dagger \hat{X}_1|p\rangle + \langle p|\hat{X}_1 a_p|\Psi^{(1)}\rangle + O(g^4) \tag{8.586}$$

$$= e^{iE_p^{(0)}t}\left[1 + \sum_{\text{sp}} |g|^2 K(-D,t) + \sum_{\mathcal{O}} |g|^2 K(\Delta,t) + \sum_{\text{sp}} \frac{|g|^2}{D^2} e^{iDt} - 2\sum_{\text{sp}} \frac{|g|^2}{D^2}\left(e^{iDt} - 1\right)\right] + O(g \tag{8.587}$$

$$= e^{iE_p^{(0)}t}\left[1 + S - it\sum_{\text{sp}} \frac{|g|^2}{D} + \sum_{\mathcal{O}} |g|^2 K(\Delta,t)\right] + O(g^4). \tag{8.588}$$

In the last step the spectator terms collapse to $|g|^2(1 - iDt)/D^2$. Multiplying by $\mathcal{N}^{-1}e^{-iE_0 t}$ cancels S and the spectator secular term, leaving

$$R_p(t) = e^{-i\epsilon_p t}\left[1 + \sum_{\mathcal{O}} |g|^2 K(\Delta,t) + \sum_{\mathcal{V}} \frac{|g|^2}{\Delta^2}\left(i\Delta t - 1\right)\right] + O(g^4). \tag{8.589}$$

Addition. Since $a_p^\dagger|\Phi\rangle = 0$, only the $\boldsymbol{k} = p$ terms of $|\Psi^{(1)}\rangle$ survive, and $a_p^\dagger a_{p+\boldsymbol{q}}^\dagger a_p|\Phi\rangle = -a_{p+\boldsymbol{q}}^\dagger|\Phi\rangle$. Because $a_p^\dagger|\tilde{\Psi}_0\rangle = O(g)$, parity leaves only $\hat{X}_0$ at this order:

$$a_p^\dagger|\Psi^{(1)}\rangle = \sum_{\mathcal{V}} \frac{g}{\Delta}\, a_{p+\boldsymbol{q}}^\dagger b_{\nu_{-\boldsymbol{q}}}^\dagger|\Phi\rangle \quad\Longrightarrow\quad A_p(t) = e^{-i\epsilon_p t}\sum_{\mathcal{V}} \frac{|g|^2}{\Delta^2}\, e^{-i\Delta t} + O(g^4). \tag{8.590}$$

Landau form. Adding the two sectors, the virtual branch completes the kernel K:

$$\frac{G_p^R(t)}{G_p^{R,0}(t)} = 1 + \sum_{\mathcal{X}\in\{\mathcal{O},\mathcal{V}\}}\sum_{\mathcal{X}} |g|^2 K(\Delta,t) + O(g^4) = 1 + \int_{-\infty}^{\infty} d\omega\, \beta_p(\omega)\, \frac{e^{-i\omega t} + i\omega t - 1}{\omega^2} + O(g^4), \tag{8.591}$$

where $G_p^{R,0}(t) = -i\theta(t)e^{-i\epsilon_p t}$ and $\beta_p(\omega) \equiv \sum_{\mathcal{X}}\sum_{\mathcal{X}} |g_\nu(p,\boldsymbol{q})|^2\, \delta\big(\omega - \Delta_\nu^{\mathcal{X}}(p,\boldsymbol{q})\big)$. Defining $C_p(t) \equiv \ln[G_p^R(t)/G_p^{R,0}(t)] = C_p^{(2)}(t) + O(g^4)$ and truncating at second order gives

$$G_p^R(t) \approx -i\theta(t)\, e^{-i\epsilon_p t} \exp\left[\int_{-\infty}^{\infty} d\omega\, \frac{\beta_p(\omega)}{\omega^2}\left(e^{-i\omega t} + i\omega t - 1\right)\right]. \tag{8.592}$$

The effective Hamiltonian can be diagonalized by introducing shifted bosonic operators

$$\widetilde{B}_{pm\nu} = B_{pm\nu} - \beta_{pm\nu}, \qquad \widetilde{B}_{pm\nu}^\dagger = B_{pm\nu}^\dagger - \beta_{pm\nu}^*, \tag{8.593}$$

where, analogous to section 9.1.1, the coherent displacements are determined as

$$\beta_{pm\nu} = -\frac{\mathcal{M}_{pm\nu}}{\Delta_{pm\nu}}. \tag{8.594}$$

This allows us to rewrite the Hamiltonian as

$$H_p^{\text{eff}} = E_p^{\text{QP}} + \sum_{m\nu} \Delta_{pm\nu} \widetilde{B}_{pm\nu}^\dagger \widetilde{B}_{pm\nu}, \tag{8.595}$$

with

$$E_p^{\text{QP}} = \epsilon_p^{\text{HF}} - \sum_{m\nu} \frac{|\mathcal{M}_{pm\nu}|^2}{\Delta_{pm\nu}} \tag{8.596}$$

$$= \epsilon_p^{\text{HF}} + \Sigma_p^{\text{GW}}(\epsilon_p^{\text{HF}}). \tag{8.597}$$

As in section 9.2, we find that the RPA vacuum is a coherent state of the shifted operators,

$$\left(\widetilde{B}_{pm\nu} + \beta_{pm\nu}\right) |0_{\text{RPA}}\rangle = 0, \tag{8.598}$$

$$\widetilde{B}_{pm\nu} |0_{\text{RPA}}\rangle = -\beta_{pm\nu} |0_{\text{RPA}}\rangle. \tag{8.599}$$

The retarded Green's function therefore becomes

$$G_p^{GW,R}(t) = -i\theta(t) \langle 0_{\text{RPA}} | e^{-iH_p^{\text{eff}} t} | 0_{\text{RPA}} \rangle \tag{8.600}$$

$$= -i\theta(t) e^{-iE_p^{\text{QP}} t} \left\langle 0_{\text{RPA}} \left| e^{-i\sum_{m\nu} \Delta_{pm\nu} \widetilde{B}_{pm\nu}^\dagger \widetilde{B}_{pm\nu} t} \right| 0_{\text{RPA}} \right\rangle \tag{8.601}$$

$$= -i\theta(t) e^{-iE_p^{\text{QP}} t} \exp\left[\sum_{m\nu} |\beta_{pm\nu}|^2 \left(e^{-i\Delta_{pm\nu} t} - 1 \right) \right] \tag{8.602}$$

$$= -i\theta(t) Z_p e^{-iE_p^{\text{QP}} t} \prod_{m\nu} \exp\left[|\beta_{pm\nu}|^2 e^{-i\Delta_{pm\nu} t} \right] \tag{8.603}$$

$$= -i\theta(t) Z_p \sum_{\mathbf{n}} \left[\prod_{m\nu} \frac{|\beta_{pm\nu}|^{2n_{pm\nu}}}{n_{pm\nu}!} \right] \exp\left[-i \left(E_p^{\text{QP}} + \sum_{m\nu} n_{pm\nu} \Delta_{pm\nu} \right) t \right], \tag{8.604}$$

where $Z_p = \exp\left[-\sum_{m\nu} |\beta_{pm\nu}|^2 \right]$. This happens to be the G0W0+C Green's function. I am wondering, though, why we even care if we solve 9.36 exactly with G0W0+C. What is the relationship between this independent boson Hamiltonian with the effective excitation energies and the original quasiboson Hamiltonian, where the fermionic portion of the Hamiltonian is a sum over all available states?

Chapter 9

QBA

9.1 Forming the electronic polaron ground state

Start by considering the Hamiltonian of the electronic polaron, due to the sudden creation of a core hole:

$$H_{\rm ch} = E_{\rm ch} + \sum_{\nu} \Omega_{\nu} Q_{\nu}^{\dagger} Q_{\nu} + \sum_{\nu} \left(M_{\nu} Q_{\nu}^{\dagger} + M_{\nu}^{*} Q_{\nu} \right). \tag{9.1}$$

E_{ch} is the mean-field single-particle energy of the fermionic core-hole, Ω_{ν} is the frequency of the boson with index ν, while M_{ν} is the fermion-boson coupling. We postulate that the ground state of this Hamiltonian is the coherent state $|\Theta_{\rm ch}\rangle$, which must satisfy $Q_{\nu}|\Theta_{\rm ch}\rangle = \beta_{\nu}|\Theta_{\rm ch}\rangle$, where β_{ν} is the coherent displacement amplitude for mode ν.

9.1.1 Determining the ground state energy of $H_{\rm ch}$

Using the coherent state property, we can write

$$E_{ch}[\beta] \equiv \langle \Theta_{ch}| H_{\rm ch} |\Theta_{ch}\rangle = E_{\rm ch} + \sum_{\nu} \Omega_{\nu} |\beta_{\nu}|^2 + \sum_{\nu} (M_{\nu}\beta_{\nu}^{*} + M_{\nu}^{*}\beta_{\nu}). \tag{9.2}$$

Minimizing the energy with respect to β_{ν}^{*} gives

$$\frac{\partial E_{ch}[\beta]}{\partial \beta_{\nu}^{*}} = \Omega_{\nu}\beta_{\nu} + M_{\nu} = 0 \tag{9.3}$$

$$\implies \beta_{\nu} = -\frac{M_{\nu}}{\Omega_{\nu}}. \tag{9.4}$$

9.1.2 Coherent state in the quasiboson (QB) approximation

We define the coherent state in the QB approximation as

$$|\Theta_{ch}\rangle = D(\beta)\,|0_{\rm RPA}\rangle\,, \tag{9.5}$$

where $|0_{\rm RPA}\rangle$ is the RPA ground state. The displacement operator is defined as

$$D(\beta) = \exp\left[\sum_{\mu} \left(\beta_{\mu} Q_{\mu}^{\dagger} - \beta_{\mu}^{*} Q_{\mu} \right) \right]. \tag{9.6}$$

Proof of coherence

Let

$$A = \sum_{\mu} \left(\beta_\mu Q_\mu^\dagger - \beta_\mu^* Q_\mu \right) \implies D(\beta) = e^A. \tag{9.7}$$

Then,

$$[Q_\nu, A] = \sum_{\mu} \left(\beta_\mu \left[Q_\nu, Q_\mu^\dagger \right] - \beta_\mu^* \left[Q_\nu, Q_\mu \right] \right). \tag{9.8}$$

Using the QB commutation relations [1], we get

$$[Q_\nu, A] \overset{\text{QBA}}{=} \sum_{\mu} \beta_\mu \delta_{\nu\mu} = \beta_\nu \tag{9.9}$$

$$\implies \left[\left[Q_\nu, A \right], A \right] \overset{\text{QBA}}{=} \left[\beta_\nu, A \right] = 0. \tag{9.10}$$

Therefore, in the BCH expansion

$$e^{-A} Q_\nu e^A = Q_\nu + [Q_\nu, A] \overset{\text{QBA}}{=} Q_\nu + \beta_\nu \tag{9.11}$$

$$\implies Q_\nu e^A \overset{\text{QBA}}{=} e^A \left(Q_\nu + \beta_\nu \right) \tag{9.12}$$

$$\implies Q_\nu |\Theta_{ch}\rangle = Q_\nu e^A |0_{\text{RPA}}\rangle \tag{9.13}$$

$$\overset{\text{QBA}}{=} e^A \left(Q_\nu + \beta_\nu \right) |0_{\text{RPA}}\rangle \tag{9.14}$$

$$\overset{\text{QBA}}{=} e^A \beta_\nu |0_{\text{RPA}}\rangle \tag{9.15}$$

$$= \beta_\nu |\Theta_{ch}\rangle \tag{9.16}$$

where in equation (9.14) we have used the fact that the QB operators Q_ν annihilate the RPA vacuum. This proves that $|\Theta_{ch}\rangle$ is a coherent state for all bosonic modes ν with displacement amplitudes β_ν within the QB approximation.

9.2 Green's function of the electronic polaron

The initial state is the RPA ground state, which was QB vacuum *before* the creation of the polaron. After this event, the time evolution is governed by H_{ch}.

Redefinition of the Hamiltonian

Defining the shifted QB operators

$$\widetilde{Q}_\nu = Q_\nu - \beta_\nu \qquad \widetilde{Q}_\nu^\dagger = Q_\nu^\dagger + \beta_\nu^*, \tag{9.17}$$

we can rewrite the Hamiltonian as

$$H_{\text{ch}} = E_{\text{pol}} + \sum_{\nu} \Omega_\nu \widetilde{Q}_\nu^\dagger \widetilde{Q}_\nu, \tag{9.18}$$

where

$$E_{\text{pol}} = E_{\text{ch}} - \sum_{\nu} \Omega_\nu |\beta_\nu|^2 \tag{9.19}$$

[1] The QB approximation assumes $\left[Q_\mu, Q_\nu^\dagger \right] \overset{\text{QBA}}{=} \delta_{\mu\nu}$, $[Q_\mu, Q_\nu] \overset{\text{QBA}}{=} 0$, and $\left[Q_\mu^\dagger, Q_\nu^\dagger \right] \overset{\text{QBA}}{=} 0$.

RPA vacuum is a coherent state of shifted QB operators

The RPA vacuum satisfies

$$Q_v \left|0_{\mathrm{RPA}}\right\rangle = 0 \tag{9.20}$$

$$\left(\widetilde{Q}_\nu + \beta_\nu\right) \left|0_{\mathrm{RPA}}\right\rangle = 0 \tag{9.21}$$

$$\implies \widetilde{Q}_\nu \left|0_{\mathrm{RPA}}\right\rangle = -\beta_\nu \left|0_{\mathrm{RPA}}\right\rangle . \tag{9.22}$$

So $\left|0_{\mathrm{RPA}}\right\rangle$ is a coherent state of the shifted bosons $\widetilde{Q}_\nu$, with coherent amplitude $-\beta_\nu$.

9.2.1 Deriving the Retarded Green's Function

We can start from the formal definition.

$$G^{QB,R}_{\mathrm{ch}}(t) = -i\theta(t) \left\langle 0_{\mathrm{RPA}}\right| e^{-iH_{\mathrm{ch}}t} \left|0_{\mathrm{RPA}}\right\rangle \tag{9.23}$$

$$= -i\theta(t) e^{-iE_{\mathrm{pol}}t} \left\langle 0_{\mathrm{RPA}}\right| e^{-i\sum_\nu \Omega_\nu \tilde{Q}_\nu \tilde{Q}_\nu t} \left|0_{\mathrm{RPA}}\right\rangle \tag{9.24}$$

$$= -i\theta(t) e^{-iE_{\mathrm{pol}}t} \exp\left[\sum_\nu \left|\beta_\nu\right|^2 \left(e^{-i\Omega_\nu t} - 1\right)\right] \tag{9.25}$$

$$= -i\theta(t) e^{-iE_{\mathrm{pol}}t} \underbrace{\exp\left[-\sum_\nu \left|\beta_\nu\right|^2\right]}_{Z_{\mathrm{pol}}} \prod_\nu \exp\left[\left|\beta_\nu\right|^2 e^{-i\Omega_\nu t}\right] \tag{9.26}$$

$$= -i\theta(t) Z_{\mathrm{pol}} e^{-iE_{\mathrm{pol}}t} \prod_\nu \sum_{n_\nu=0}^{\infty} \frac{\left|\beta_\nu\right|^{2n_\nu}}{n_\nu!} e^{-in_\nu\Omega_\nu t} \tag{9.27}$$

$$= -i\theta(t) Z_{\mathrm{pol}} \sum_{\mathbf{n}} \left[\prod_\nu \frac{\left|\beta_\nu\right|^{2n_\nu}}{n_\nu!}\right] \exp\left[-i\left(E_{\mathrm{pol}} + \sum_\nu n_\nu \Omega_\nu\right) t\right] . \tag{9.28}$$

Note that in equation (9.24), we have used the coherent state identity

$$\left\langle 0_{\mathrm{RPA}}\right| e^{-i\sum_\nu \Omega_\nu \tilde{Q}^t_\nu \tilde{Q}_\nu t} \left|0_{\mathrm{RPA}}\right\rangle = \prod_\nu \exp\left[\left|\beta_\nu\right|^2 \left(e^{-i\Omega_\nu t} - 1\right)\right] . \tag{9.29}$$

n_ν denotes the occupation number of bosonic mode ν, and $\sum_{\mathbf{n}} \equiv \prod_\nu \sum_{n_\nu=0}^{\infty}$ is the sum over all possible combinations of occupation numbers for all bosonic modes. Fourier transforming according to

$$G^{QB,R}_{\mathrm{ch}}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} G^{QB,R}_{\mathrm{ch}}(t), \tag{9.30}$$

gives the spectral function as

$$A_{\mathrm{ch}}(\omega) = -\frac{1}{\pi} \operatorname{Im} G^{QB,R}_{\mathrm{ch}}(\omega) \tag{9.31}$$

$$= e^{-\sum_\nu \left|\beta_\nu\right|^2} \sum_{n_\nu} \left[\prod_\nu \frac{\left|\beta_\nu\right|^{2n_\nu}}{n_\nu!}\right] \frac{1}{\pi} \frac{\eta}{\left(\omega - E_{\mathrm{pol}} - \sum_\nu n_\nu \Omega_\nu\right)^2 + \eta^2}. \tag{9.32}$$

where $\beta_\nu = -M_\nu/\Omega_\nu$ is the coherent displacement amplitude of mode ν, $E_{\rm pol} = E_{\rm ch} - \sum_\nu \Omega_\nu |\beta_\nu|^2$ is the polaron energy And we can take $E_{\rm ch} = \epsilon_{\boldsymbol{k}}^{\rm MF}$ for $\boldsymbol{k} = [0,0,0]$, and η is a small broadening parameter. In our convention, we should take

1. $\eta = 0.2$ eV for 14e19o and 14e57o
2. $\eta = 0.4$ eV for 38e123o
3. $\eta = 0.6$ eV for 54e305o
4. $\eta = 0.8$ eV for 114e485o

Physical intuition

Thus, the spectral function consists of a coherent polaron peak at

$$\omega = E_{\rm pol}, \tag{9.33}$$

with weight

$$Z_{\rm pol} = e^{-\sum_\nu |\beta_\nu|^2}, \tag{9.34}$$

and a series of bosonic shake-up satellites at energies

$$\omega = E_{\rm pol} + \sum_\nu n_\nu \Omega_\nu. \tag{9.35}$$

9.3 UEG

9.3.1 Connection to GW+C

The independent boson Hamiltonian may be generalized to an effective Hamiltonian for a hole created in orbital p,

$$H_p^{\rm eff} = \epsilon_p^{\rm HF} + \sum_{m\nu} \Delta_{pm\nu} B_{pm\nu}^\dagger B_{pm\nu} + \sum_{m\nu} \left(\mathcal{M}_{pm\nu} B_{pm\nu}^\dagger + \mathcal{M}_{pm\nu}^* B_{pm\nu} \right), \tag{9.36}$$

where the index m runs over all occupied (i) and virtual (a) states. The effective excitation energies are $\Delta_{pi\nu} = \epsilon_i^{\rm HF} - \epsilon_p^{\rm HF} - \Omega_\nu - i\eta$ and $\Delta_{pa\nu} = \epsilon_a^{\rm HF} - \epsilon_p^{\rm HF} + \Omega_\nu - i\eta$ with effective couplings $\mathcal{M}_{pm\nu}$ for QB mode ν (these are transition densities in RPA and ERIs in BWs2). The effective Hamiltonian is diagonalized by introducing displaced bosonic operators

$$\widetilde{B}_{pm\nu} = B_{pm\nu} - \beta_{pm\nu}, \qquad \widetilde{B}_{pm\nu}^\dagger = B_{pm\nu}^\dagger - \beta_{pm\nu}^*, \tag{9.37}$$

where, analogous to section 9.1.1, the coherent displacements are determined as

$$\beta_{pm\nu} = -\frac{\mathcal{M}_{pm\nu}}{\Delta_{pm\nu}}. \tag{9.38}$$

This allows us to rewrite the Hamiltonian as

$$H_p^{\text{eff}} = E_p^{\text{QP}} + \sum_{m\nu} \Delta_{pm\nu} \widetilde{B}_{pm\nu}^\dagger \widetilde{B}_{pm\nu}, \tag{9.39}$$

with

$$E_p^{\text{QP}} = \epsilon_p^{\text{HF}} - \sum_{m\nu} \frac{|\mathcal{M}_{pm\nu}|^2}{\Delta_{pm\nu}} \tag{9.40}$$

$$= \epsilon_p^{\text{HF}} + \Sigma_p^{\text{GW}}(\epsilon_p^{\text{HF}}). \tag{9.41}$$

As in section 9.2, we find that the RPA vacuum is a coherent state of the shifted operators,

$$\left(\widetilde{B}_{pm\nu} + \beta_{pm\nu}\right) |0_{\text{RPA}}\rangle = 0, \tag{9.42}$$

$$\widetilde{B}_{pm\nu} |0_{\text{RPA}}\rangle = -\beta_{pm\nu} |0_{\text{RPA}}\rangle. \tag{9.43}$$

The retarded Green's function therefore becomes

$$G_p^{GW,R}(t) = -i\theta(t) \langle 0_{\text{RPA}} | e^{-iH_p^{\text{eff}} t} | 0_{\text{RPA}} \rangle \tag{9.44}$$

$$= -i\theta(t) e^{-iE_p^{\text{QP}} t} \left\langle 0_{\text{RPA}} \left| e^{-i\sum_{m\nu} \Delta_{pm\nu} \widetilde{B}_{pm\nu}^\dagger \widetilde{B}_{pm\nu} t} \right| 0_{\text{RPA}} \right\rangle \tag{9.45}$$

$$= -i\theta(t) e^{-iE_p^{\text{QP}} t} \exp\left[\sum_{m\nu} |\beta_{pm\nu}|^2 \left(e^{-i\Delta_{pm\nu} t} - 1 \right) \right] \tag{9.46}$$

$$= -i\theta(t) Z_p e^{-iE_p^{\text{QP}} t} \prod_{m\nu} \exp\left[|\beta_{pm\nu}|^2 e^{-i\Delta_{pm\nu} t} \right] \tag{9.47}$$

$$= -i\theta(t) Z_p \sum_{\mathbf{n}} \left[\prod_{m\nu} \frac{|\beta_{pm\nu}|^{2n_{pm\nu}}}{n_{pm\nu}!} \right] \exp\left[-i \left(E_p^{\text{QP}} + \sum_{m\nu} n_{pm\nu} \Delta_{pm\nu} \right) t \right], \tag{9.48}$$

where $Z_p = \exp\left[-\sum_{m\nu} |\beta_{pm\nu}|^2 \right]$. So, we establish a connection between the G0W0+C Green's function and that of the electronic polaron in the QB approximation.

9.4 CC for the electronic polaron

We write the quasiboson Hamiltonian in second quantization as

$$\hat{H} = \hat{H}_{\text{el}} + \hat{H}_{\text{pl}} + \hat{H}_{\text{ep}} \tag{9.49}$$

$$= \sum_{\boldsymbol{k} \in \mathcal{O}} \epsilon_{\boldsymbol{k}}^{\text{HF}} a_{\boldsymbol{k}}^\dagger a_{\boldsymbol{k}} + \sum_{\nu_{\boldsymbol{q}} \in \mathcal{M}} \omega_{\nu_{\boldsymbol{q}}}^{\text{pl}} b_{\nu_{\boldsymbol{q}}}^\dagger b_{\nu_{\boldsymbol{q}}} + \sum_{\boldsymbol{k}, \boldsymbol{k}+\boldsymbol{q} \in \mathcal{O}} \sum_{\nu_{\boldsymbol{q}} \in \mathcal{M}} g_\nu^{\text{pl}}(\boldsymbol{k}, \boldsymbol{q}) a_{\boldsymbol{k}+\boldsymbol{q}}^\dagger a_{\boldsymbol{k}} \left(b_{\nu_{-\boldsymbol{q}}}^\dagger + b_{\nu_{\boldsymbol{q}}} \right). \tag{9.50}$$

Here, $\hat{H}_{\text{el}}$, $\hat{H}_{\text{pl}}$, and $\hat{H}_{\text{ep}}$ are the electronic, plasmonic, and electron–plasmon Hamiltonians, respectively. The set $\mathcal{O}$ contains all occupied bands, whose single-particle energies are due to HF. $\mathcal{M}$ denotes the set of modes from the plasmonic bath pl, as obtained through the given quasiboson approximation. Here we investigate dRPA@LDA quasibosons.

9.4.1 Interpretation

Physically, this Hamiltonian describes a single fermion (hole) coupled to a bath of bosons (plasmons) on a lattice (occupied bands of the UEG supercell). Note that the hole creation operator $h_{\boldsymbol{k}}^\dagger$ is equivalent to the electron annihilation operator $a_{\boldsymbol{k}}$ and vice versa.

Electronic Hamiltonian

$\hat{H}_{\text{el}}$ assigns the appropriate HF single-particle energy for the fermion to be at any lattice site.

Plasmonic Hamiltonian

$\hat{H}_{\text{pl}}$ assigns the appropriate RPA excitation energy for the presence of a plasmon. Even though our system now has a hole, these are plasmons computed for the neutral system. This is in accordance with linear response theory, which postulates that the plasmons of the ionized system can be well approximated by those of the neutral system. **Treating plasma oscillations within the harmonic approximation allows us to truncate $\hat{H}_{\text{pl}}$ at the quadratic level.**

Electron-Plasmon Hamiltonian

Finally, $\hat{H}_{\text{ep}}$ describes the process of the fermion hopping to a new lattice site, while simultaneously creating or destroying a plasmon in the bath, conserving momentum in the process. **Expanding the UEG electronic Hamiltonian only up to first order in the plasmonic density fluctuation coordinate is what allows us to truncate $\hat{H}_{\text{ep}}$ at the linear coupling.**

Mean-field reference

Defining $h_{\boldsymbol{k}}^\dagger \equiv a_{\boldsymbol{k}}$, we choose the zeroth-order $(N-1)$-electron reference as

$$|\Phi_0\rangle = h_0^\dagger |0\rangle_F \otimes |0\rangle_{\text{pl}}, \tag{9.51}$$

where $|0\rangle_F$ is the Fermi-sea of the reference and $|0\rangle_{\text{pl}}$ is the bosonic vacuum. The reference is a single hole at $\boldsymbol{k} = 0$ in the Fermi sea, with no bosons excited. Since the electron is removed at $\boldsymbol{k} = 0$, this state belongs to the total-momentum sector $\boldsymbol{K} = 0$. We therefore construct the cluster operators so that every retained term conserves total momentum individually, rather than restoring the symmetry by an explicit momentum projector.

Method labels

To first order, we have

$$\hat{T}^{(1)} = \hat{T}_{\text{ep}}^{(1)} \tag{9.52}$$

$$= \sum_{\substack{\boldsymbol{k}\in\mathcal{O} \\ \boldsymbol{k}\neq 0}} \sum_{\nu} t_{\boldsymbol{k}\nu} h_{\boldsymbol{k}}^\dagger h_0 b_{\nu_{\boldsymbol{k}}}^\dagger \tag{9.53}$$

To second order, we have

$$\hat{T}^{(2)} = \hat{T}^{(1)}_{\mathrm{ep}} + \hat{T}^{(2)}_{\mathrm{pl}} + \hat{T}^{(2)}_{\mathrm{ep}} \tag{9.54}$$

$$= \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}\neq 0}} \sum_{\nu} t_{\boldsymbol{k}\nu} h^{\dagger}_{\boldsymbol{k}} h_0 b^{\dagger}_{\nu_{\boldsymbol{k}}} + \frac{1}{2}\sum_{\boldsymbol{q}}\sum_{\nu\mu} t_{\nu\boldsymbol{q},\mu-\boldsymbol{q}} b^{\dagger}_{\nu_{\boldsymbol{q}}} b^{\dagger}_{\mu_{-\boldsymbol{q}}} + \frac{1}{2}\sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}\neq 0}} \sum_{\nu\boldsymbol{q}_1,\mu\boldsymbol{q}_2} \delta_{\boldsymbol{k},\boldsymbol{q}_1+\boldsymbol{q}_2} t_{\boldsymbol{k};\nu\boldsymbol{q}_1,\mu\boldsymbol{q}_2} h^{\dagger}_{\boldsymbol{k}} h_0 b^{\dagger}_{\nu_{\boldsymbol{q}_1}} b^{\dagger}_{\mu_{\boldsymbol{q}_2}} \tag{9.55}$$

We rank where we are in the CC hierarchy in accordance with Hamlin's paper as CC-X-SY, where X is the highest pure plasmon excitation order retained and Y lists the retained coupled hole–plasmon excitations.

Coupled-cluster state ansatz

We define

$$|\Psi_{\mathrm{CC}}\rangle = e^{\hat{T}}|\Phi_0\rangle. \tag{9.56}$$

At first order,

$$\hat{T}^{(1)} = \hat{T}^{(1)}_{\mathrm{el}} + \hat{T}^{(1)}_{\mathrm{pl}} + \hat{T}^{(1)}_{\mathrm{ep}}. \tag{9.57}$$

A pure hole excitation,

$$h^{\dagger}_{\boldsymbol{k}} h_0, \tag{9.58}$$

changes the total momentum by $-\boldsymbol{k}$. Hence no nontrivial pure hole excitation can remain in the $\boldsymbol{K} = 0$ sector, and therefore

$$\hat{T}^{(1)}_{\mathrm{el}} = 0. \tag{9.59}$$

Similarly, a pure one-boson excitation

$$b^{\dagger}_{\nu_{\boldsymbol{q}}} \tag{9.60}$$

carries momentum $\boldsymbol{q}$. Thus a nonzero one-boson amplitude would require $\boldsymbol{q} = 0$. Since a $\boldsymbol{q} = 0$ density mode is not generated by the QBA in the neutral UEG, we take

$$\hat{T}^{(1)}_{\mathrm{pl}} = 0. \tag{9.61}$$

The only non-trivial first-order cluster operator therefore consists of a hole recoil accompanied by a boson carrying the compensating momentum,

$$\hat{T}^{(1)}_{\mathrm{ep}} = \sum_{\substack{\boldsymbol{k}\in\mathcal{O}\\ \boldsymbol{k}\neq 0}} \sum_{\nu} t_{\boldsymbol{k}\nu} h^{\dagger}_{\boldsymbol{k}} h_0 b^{\dagger}_{\nu_{\boldsymbol{k}}}, \tag{9.62}$$

where $\mathcal{O}$ denotes the set of occupied bands in the N-electron reference.
At second order, there is no pure electronic doubles operator because the effective fermionic Hilbert space contains only one particle. The allowed pure bosonic sector contains pairs of bosons with opposite total momentum,

$$\hat{T}^{(2)}_{\mathrm{pl}} = \frac{1}{2}\sum_{\boldsymbol{q}}\sum_{\nu\mu} t_{\nu\boldsymbol{q},\mu-\boldsymbol{q}} b^{\dagger}_{\nu_{\boldsymbol{q}}} b^{\dagger}_{\mu_{-\boldsymbol{q}}}. \tag{9.63}$$

The coupled hole–two-boson sector is

$$\hat{T}_{\mathrm{ep}}^{(2)} = \frac{1}{2} \sum_{\substack{\boldsymbol{k} \in \mathcal{O} \\ \boldsymbol{k} \neq 0}} \sum_{\nu \boldsymbol{q}_1, \mu \boldsymbol{q}_2} \delta_{\boldsymbol{k}, \boldsymbol{q}_1 + \boldsymbol{q}_2} t_{\boldsymbol{k}; \nu \boldsymbol{q}_1, \mu \boldsymbol{q}_2} h_{\boldsymbol{k}}^{\dagger} h_0 b_{\nu_{\boldsymbol{q}_1}}^{\dagger} b_{\mu_{\boldsymbol{q}_2}}^{\dagger}. \tag{9.64}$$

So, to second order, we may write

$$\hat{T} = \hat{T}_{\mathrm{ep}}^{(1)} + \hat{T}_{\mathrm{pl}}^{(2)} + \hat{T}_{\mathrm{ep}}^{(2)}. \tag{9.65}$$

Since every fermionic excitation contains the factor $h_{\boldsymbol{k}}^{\dagger} h_0$, products of two such fermionic excitations vanish when acting on the one-hole reference. Consequently,

$$e^{\hat{T}} |\Phi_0\rangle = \left(1 + \hat{T}_{\mathrm{ep}}^{(1)} + \hat{T}_{\mathrm{ep}}^{(2)}\right) e^{\hat{T}_{\mathrm{pl}}^{(2)}} |\Phi_0\rangle. \tag{9.66}$$

Variational energy evaluation

The variational coupled-cluster energy is evaluated as

$$E_{\mathrm{CC}}(\boldsymbol{t}) = \frac{\langle \Phi_0 | e^{\hat{T}^{\dagger}} \hat{H} e^{\hat{T}} | \Phi_0 \rangle}{\langle \Phi_0 | e^{\hat{T}^{\dagger}} e^{\hat{T}} | \Phi_0 \rangle}. \tag{9.67}$$

The ground-state amplitudes are obtained variationally from

$$\frac{\partial E_{\mathrm{CC}}}{\partial t_{\mu}^{*}} = 0, \tag{9.68}$$

for all independent cluster amplitudes t_{μ}. Defining

$$|\boldsymbol{k}\nu\rangle = h_{\boldsymbol{k}}^{\dagger} h_0 b_{\nu_{\boldsymbol{k}}}^{\dagger} |\Phi_0\rangle, \tag{9.69}$$

the first-order state, for example, is then

$$|\Psi_{\mathrm{CC}}^{(1)}\rangle = |\Phi_0\rangle + \sum_{\boldsymbol{k}\nu} t_{\boldsymbol{k}\nu} |\boldsymbol{k}\nu\rangle. \tag{9.70}$$

Tangent-space response

Once the ground-state amplitudes have been determined, approximate excited states in the $(N-1)$-electron sector are constructed from tangent vectors to the variational manifold,

$$|\Psi_{\mu}\rangle = \hat{\mathcal{R}}_{\mu} |\Psi_{\mathrm{CC}}(\boldsymbol{t}_{\mathrm{gs}})\rangle. \tag{9.71}$$

The zeroth-order response operator is

$$\hat{\mathcal{R}}_0 = 1, \tag{9.72}$$

so that the optimized coupled-cluster state itself is included in the response manifold. For the $\boldsymbol{K} = 0$ sector, the first-order response manifold is

$$\mathcal{R}_{K=0}^{(1)} = \left\{ 1,\, h_{\boldsymbol{k}}^{\dagger} h_0 b_{\nu_{\boldsymbol{k}}}^{\dagger} \right\}. \tag{9.73}$$

At second order, the response manifold is enlarged to

$$\mathcal{R}^{(2)}_{K=0} = \mathcal{R}^{(1)}_{K=0} \cup \left\{ b^\dagger_{\nu_{\boldsymbol{q}}} b^\dagger_{\mu_{-\boldsymbol{q}}}, \delta_{\boldsymbol{k},\boldsymbol{q}_1+\boldsymbol{q}_2} h^\dagger_{\boldsymbol{k}} h_0 b^\dagger_{\nu_{\boldsymbol{q}_1}} b^\dagger_{\mu_{\boldsymbol{q}_2}} \right\}. \tag{9.74}$$

The Hamiltonian and overlap matrices in this nonorthogonal tangent basis are

$$\widetilde{H}_{\mu\nu} = \langle \Psi_\mu | \hat{H} | \Psi_\nu \rangle, \tag{9.75}$$

and

$$S_{\mu\nu} = \langle \Psi_\mu | \Psi_\nu \rangle. \tag{9.76}$$

The approximate excited-state energies and eigenvectors are obtained from the generalized eigenvalue problem

$$\widetilde{\boldsymbol{H}}\boldsymbol{C} = \boldsymbol{S}\boldsymbol{C}\boldsymbol{E}. \tag{9.77}$$

The generalized eigenvectors may be normalized according to

$$\boldsymbol{C}^\dagger \boldsymbol{S} \boldsymbol{C} = \boldsymbol{1}. \tag{9.78}$$

The approximate $(N-1)$-electron eigenstates are then

$$|\widetilde{\Psi}^{N-1}_m\rangle = \sum_\mu C_{\mu m} |\Psi_\mu\rangle. \tag{9.79}$$

Retarded hole Green's function

The electron-removal part of the retarded Green's function is written in Lehmann form as

$$G^-_{\boldsymbol{k}}(\omega) = \sum_m \frac{\left| \langle \widetilde{\Psi}^{N-1}_m | \Phi_0 \rangle \right|^2}{\omega - (E^N_0 - E^{N-1}_m) + i\eta}. \tag{9.80}$$

Finally, the electron-removal spectral function is

$$A^-_0(\omega) = -\frac{1}{\pi} \operatorname{Im} G^-_0(\omega). \tag{9.81}$$

9.5 SCCE for the electronic polaron

Recall that our quasiboson Hamiltonian is

$$\begin{aligned} \hat{H} &= \hat{H}_{\text{el}} + \hat{H}_{\text{pl}} + \hat{H}_{\text{ep}} \qquad (9.82) \\ &= \sum_{\boldsymbol{k}\in\mathcal{O}} \epsilon^{\text{G0}}_{\boldsymbol{k}} h^\dagger_{\boldsymbol{k}} h_{\boldsymbol{k}} + \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} \omega^{\text{pl}}_{\nu_{\boldsymbol{q}}} b^\dagger_{\nu_{\boldsymbol{q}}} b_{\nu_{\boldsymbol{q}}} \\ &\quad + \sum_{\boldsymbol{k},\boldsymbol{k}+\boldsymbol{q}\in\mathcal{O}} \sum_{\nu_{\boldsymbol{q}}\in\mathcal{M}} g^{\text{pl}}_\nu(\boldsymbol{k},\boldsymbol{q}) h^\dagger_{\boldsymbol{k}+\boldsymbol{q}} h_{\boldsymbol{k}} \left(b^\dagger_{\nu_{-\boldsymbol{q}}} + b_{\nu_{\boldsymbol{q}}} \right). \qquad (9.83) \end{aligned}$$

with $h_{\boldsymbol{k}}$ ($h^\dagger_{\boldsymbol{k}}$) the hole annihilation (creation) operators and $b_{\nu_{\boldsymbol{q}}}$ ($b^\dagger_{\nu_{\boldsymbol{q}}}$) the bosonified particle-hole annihilation (creation) operators. The latter are interpreted as plasmons in the QBA. Therefore, the Hilbert space for this Hamiltonian considers all processes involving the interaction of one hole with n plasmons. In other words, we can accommodate 1h, 2h1p, 3h2p, ... processes, which notably excludes 2p1h processes.

G_0W_0 self-energy for the QB Hamiltonian

In accordance Garnet's data, I can take pl=dRPA@LDA for the determination of my quasi-bosonic quantities and $\epsilon_k^{G0} = \left[\underbrace{\epsilon_k^{\rm LDA} + \Sigma_k^x - V_{xc,k}^{\rm LDA}}_{\epsilon_k^{\rm HF}}\right]$. We can start from

$$\Sigma_{\boldsymbol{k}}^{G_0W_0,R}(t) = -i\theta(t)\sum_{\nu_q\in\mathcal{M}} \left|g_\nu^{p1}(\boldsymbol{k},\boldsymbol{q})\right|^2 e^{-i\epsilon_{\boldsymbol{k}-\boldsymbol{q}}^{\rm LDA}t} e^{+i\omega_{\nu\boldsymbol{q}}^{\rm pl}t} \tag{9.84}$$

$$= i\sum_{\nu_q\in\mathcal{M}} \left|g_{\boldsymbol{k},\boldsymbol{q}}^{\nu}\right|^2 D_{\nu_{\boldsymbol{q}}}^{(-)}(t) G_{\boldsymbol{k}-\boldsymbol{q}}^0(t) \tag{9.85}$$

where the 2p1h contribution vanished because the quasiboson Hamiltonian restricts the fermionic degree of freedom to $\mathcal{O}$. We have identified the bare fermionic and bosonic Green's functions, describing hole and plasmon propagation, respectively, as

$$G_{\boldsymbol{k}}^0(t) = -i\theta(t)e^{-i\epsilon_{\boldsymbol{k}}^{\rm LDA}t}, \quad D_{\nu_{\boldsymbol{q}}}^{(-)}(t) = -i\theta(t)e^{+i\omega_{\nu\boldsymbol{q}}^{\rm pl}t}. \tag{9.86}$$

Now we can pick up from the EOM of the Dyson equation for the causal Green's function. For $0<\tau<t$, the bare propagator satisfies

$$G_{\boldsymbol{k}-\boldsymbol{q}}^0(t-\tau) = -i\left[G_{\boldsymbol{k}-\boldsymbol{q}}^0(\tau)\right]^{-1} G_{\boldsymbol{k}-\boldsymbol{q}}^0(t). \tag{9.87}$$

So,

$$\int_0^t d\tau\, \Sigma_{\boldsymbol{k}}^R(t-\tau)G_{\boldsymbol{k}}(\tau) = i\sum_{\nu q}\left|g_\nu^{\rm pl}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, \mathcal{D}_{\nu_{\boldsymbol{q}}}^{(-)}(t-\tau) G_{\boldsymbol{k}-\boldsymbol{q}}^0(t-\tau) G_{\boldsymbol{k}}(\tau) \tag{9.88}$$

$$= \sum_{\nu q}\left|g_\nu^{\rm pl}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, \mathcal{D}_{\nu_{\boldsymbol{q}}}^{(-)}(t-\tau)\frac{G_{\boldsymbol{k}-\boldsymbol{q}}^0(t)}{G_{\boldsymbol{k}-\boldsymbol{q}}^0(\tau)} G_{\boldsymbol{k}}(\tau) \tag{9.89}$$

$$\implies \frac{dG_{\boldsymbol{k}}(t)}{dt} = -i\epsilon_{\boldsymbol{k}}^{\rm G0}G_{\boldsymbol{k}}(t) - i\sum_{\nu_{\boldsymbol{q}}}\left|g_\nu^{\rm pl}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, \mathcal{D}_{\nu_{\boldsymbol{q}}}^{(-)}(t-\tau)\frac{G_{\boldsymbol{k}}(\tau)G_{\boldsymbol{k}-\boldsymbol{q}}(t)}{G_{\boldsymbol{k}-\boldsymbol{q}}(\tau)} \tag{9.90}$$

where, following PJ's paper, we affected a partial resummation by replacing the bare fermionic propagators with the dressed ones as

$$\frac{G_{\boldsymbol{k}-\boldsymbol{q}}^0(t)}{G_{\boldsymbol{k}-\boldsymbol{q}}^0(\tau)} \longrightarrow \frac{G_{\boldsymbol{k}-\boldsymbol{q}}(t)}{G_{\boldsymbol{k}-\boldsymbol{q}}(\tau)}. \tag{9.91}$$

Then, introducing $y_{\boldsymbol{k}}(t) \equiv e^{C_{\boldsymbol{k}}(t)}$, we also have

$$G_{\boldsymbol{k}}(t) = -i\exp\bigl(-i\epsilon_{\boldsymbol{k}}^{\rm LDA}t\bigr)y_{\boldsymbol{k}}(t) \tag{9.92}$$

$$\implies \frac{dG_{\boldsymbol{k}}(t)}{dt} = -i\epsilon_{\boldsymbol{k}}^{\rm LDA}G_{\boldsymbol{k}}(t) - ie^{-i\epsilon_{\boldsymbol{k}}^{\rm LDA}t}\frac{dy_{\boldsymbol{k}}(t)}{dt}. \tag{9.93}$$

Setting these two expressions for the time derivative of the Green's function equal to each other gives (with $\Delta^x_{\boldsymbol{k}} \equiv \epsilon^{\mathrm{G0}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} = \Sigma^x_{\boldsymbol{k}} - V^{\mathrm{LDA}}_{xc,k}$)

$$-ie^{-i\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} t}\frac{dy_{\boldsymbol{k}}(t)}{dt} = -i\Delta^x_{\boldsymbol{k}} G_{\boldsymbol{k}}(t) - i\sum_{\nu_{\boldsymbol{q}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, \mathcal{D}^{(-)}_{\nu_{\boldsymbol{q}}}(t-\tau)\frac{G_{\boldsymbol{k}}(\tau)G_{\boldsymbol{k}-\boldsymbol{q}}(t)}{G_{\boldsymbol{k}-\boldsymbol{q}}(\tau)}. \quad (9.94)$$

Because $\frac{G_{\boldsymbol{k}}(\tau)G_{\boldsymbol{k}-\boldsymbol{q}}(t)}{G_{\boldsymbol{k}-\boldsymbol{q}}(\tau)} = -ie^{-i\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}}\tau}e^{-i\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}}(t-\tau)}\frac{y_{\boldsymbol{k}}(\tau)y_{\boldsymbol{k}-\boldsymbol{q}}(t)}{y_{\boldsymbol{k}-\boldsymbol{q}}(\tau)}$, we can divide through by $-ie^{-i\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}}t}$ to get

$$\frac{dy_{\boldsymbol{k}}(t)}{dt} = -i\Delta^x_{\boldsymbol{k}} y_{\boldsymbol{k}}(t) - i\sum_{\nu_{\boldsymbol{q}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, \mathcal{D}^{(-)}_{\nu_{\boldsymbol{q}}}(t-\tau)e^{i(\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}}-\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}})(t-\tau)}\frac{y_{\boldsymbol{k}}(\tau)y_{\boldsymbol{k}-\boldsymbol{q}}(t)}{y_{\boldsymbol{k}-\boldsymbol{q}}(\tau)} \quad (9.95)$$

$$\implies \frac{dC_{\boldsymbol{k}}(t)}{dt} = -i\Delta^x_{\boldsymbol{k}} - \sum_{\nu_{\boldsymbol{q}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, e^{i\Delta^{\nu}_{\boldsymbol{k}\boldsymbol{q}}(t-\tau)}\frac{y_{\boldsymbol{k}}(\tau)y_{\boldsymbol{k}-\boldsymbol{q}}(t)}{y_{\boldsymbol{k}}(t)y_{\boldsymbol{k}-\boldsymbol{q}}(\tau)} \quad (9.96)$$

$$= -i\Delta^x_{\boldsymbol{k}} - \sum_{\nu_{\boldsymbol{q}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, e^{i\Delta^{\nu}_{\boldsymbol{k}\boldsymbol{q}}(t-\tau)}e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_{\boldsymbol{k}-\boldsymbol{q}}(t)-C_{\boldsymbol{k}-\boldsymbol{q}}(\tau)}. \quad (9.97)$$

where we used that

$$\Delta^{\nu}_{\boldsymbol{k}\boldsymbol{q}} \equiv \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}} + \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}} \qquad y_{\boldsymbol{k}}(0) = 1. \quad (9.98)$$

9.5.1 Different choices for the quasiboson Hamiltonian

The equation being solved is

$$\frac{dC_{\boldsymbol{k}}(t)}{dt} = -i\Delta^x_{\boldsymbol{k}} - \sum_{\nu_{\boldsymbol{q}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, e^{i\Delta^{\nu}_{\boldsymbol{k}\boldsymbol{q}}(t-\tau)}e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_{\boldsymbol{k}-\boldsymbol{q}}(t)-C_{\boldsymbol{k}-\boldsymbol{q}}(\tau)}. \quad (9.99)$$

It is true that to recover the corresponding G_0W_0+C scheme, all that is needed is to set all $C_{\boldsymbol{k}}(t) = 0$ in the memory kernel on the right-hand side?

LDA band, dRPA@LDA plasmons

Here we are taking $\Delta^{\nu}_{\boldsymbol{k}\boldsymbol{q}} \equiv \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}} + \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}$ and $\Delta^x_{\boldsymbol{k}} \equiv \Sigma^x_{\boldsymbol{k}} - V^{\mathrm{LDA}}_{xc,k}$. As the plasmonic bath is due to dRPA@LDA, our Casida matrix build, for a given momentum transfer $\mathbf{q}$, defines $A_{ia,jb}(\mathbf{q}) = \delta_{ij}\delta_{ab}(\varepsilon^{\mathrm{LDA}}_a - \varepsilon^{\mathrm{LDA}}_i) + 2(ia|bj)$ and $B_{ia,jb}(\mathbf{q}) = 2(ia|jb)$.

HF band, dRPA@HF plasmons

Here we are taking $\Delta^{\nu}_{\boldsymbol{k}\boldsymbol{q}} \equiv \epsilon^{\mathrm{HF}}_{\boldsymbol{k}} - \epsilon^{\mathrm{HF}}_{\boldsymbol{k}-\boldsymbol{q}} + \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}$ and $\Delta^x_{\boldsymbol{k}} \equiv 0$. As the plasmonic bath is due to dRPA@HF, our Casida matrix build, for a given momentum transfer $\mathbf{q}$, defines $A_{ia,jb}(\mathbf{q}) = \delta_{ij}\delta_{ab}(\varepsilon^{\mathrm{HF}}_a - \varepsilon^{\mathrm{HF}}_i) + 2(ia|bj)$ and $B_{ia,jb}(\mathbf{q}) = 2(ia|jb)$. Is it the case that $\varepsilon^{\mathrm{HF}}_k \equiv k^2/2 - \sum_{\boldsymbol{q}\neq\boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}}v(\boldsymbol{q})$, while $\epsilon^{\mathrm{HF}}_k = k^2/2 - n_k v_m - \sum_{\boldsymbol{q}\neq\boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}}v(\boldsymbol{q})$, where $n_{\boldsymbol{k}}$ is the occupation number of the state $\boldsymbol{k}$ at zero temperature, $v(\boldsymbol{q})$ is the Fourier transform of the Coulomb potential, and v_m is the Madelung constant?

HF band, dRPA@LDA plasmons

Here we are taking $\Delta^{\nu}_{\boldsymbol{kq}} \equiv \epsilon^{\mathrm{HF}}_{\boldsymbol{k}} - \epsilon^{\mathrm{HF}}_{\boldsymbol{k}-\boldsymbol{q}} + \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}$ and $\Delta^{x}_{\boldsymbol{k}} \equiv 0$. As the plasmonic bath is due to dRPA@LDA, our Casida matrix build, for a given momentum transfer $\mathbf{q}$, defines $A_{ia,jb}(\mathbf{q}) = \delta_{ij}\delta_{ab}(\varepsilon^{\mathrm{LDA}}_{a} - \varepsilon^{\mathrm{LDA}}_{i}) + 2(ia|bj)$ and $B_{ia,jb}(\mathbf{q}) = 2(ia|jb)$. Is it the case that $\epsilon^{\mathrm{HF}}_{k} \equiv k^2/2 - n_k v_m - \sum_{\boldsymbol{q}\neq\boldsymbol{0}} n_{\boldsymbol{k}+\boldsymbol{q}} v(\boldsymbol{q})$?

G_0W_0 self-energy for electronic structure Hamiltonian

Here we instead use

$$\Sigma^{G_0W_0,R}_{\boldsymbol{k},c}(t) = -i\theta(t)\Bigg[\sum_{\substack{\nu_{\boldsymbol{q}}\in\mathcal{M}\\ \boldsymbol{k}-\boldsymbol{q}\in\mathcal{O}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 e^{-i\left(\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}} - \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}\right)t} + \sum_{\substack{\nu_{\boldsymbol{q}}\in\mathcal{M}\\ \boldsymbol{k}-\boldsymbol{q}\in\mathcal{V}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 e^{-i\left(\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}} + \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}\right)t}\Bigg] \tag{9.100}$$

$$= i\sum_{\substack{\nu_{\boldsymbol{q}}\in\mathcal{M}\\ \boldsymbol{k}-\boldsymbol{q}\in\mathcal{O}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \mathcal{D}^{(-)}_{\nu_{\boldsymbol{q}}}(t) G^0_{\boldsymbol{k}-\boldsymbol{q}}(t) + i\sum_{\substack{\nu_{\boldsymbol{q}}\in\mathcal{M}\\ \boldsymbol{k}-\boldsymbol{q}\in\mathcal{V}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \mathcal{D}^{(+)}_{\nu_{\boldsymbol{q}}}(t) G^0_{\boldsymbol{k}-\boldsymbol{q}}(t). \tag{9.101}$$

Following the same procedure as before leads to

$$\begin{aligned}\frac{dC_{\boldsymbol{k}}(t)}{dt} =& -i\Delta^{x}_{\boldsymbol{k}} \\ &- \sum_{\substack{\nu_{\boldsymbol{q}}\in\mathcal{M}\\ \boldsymbol{k}-\boldsymbol{q}\in\mathcal{O}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, e^{i\Delta^{\nu,-}_{\boldsymbol{kq}}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_{\boldsymbol{k}-\boldsymbol{q}}(t)-C_{\boldsymbol{k}-\boldsymbol{q}}(\tau)} \\ &- \sum_{\substack{\nu_{\boldsymbol{q}}\in\mathcal{M}\\ \boldsymbol{k}-\boldsymbol{q}\in\mathcal{V}}} \left|g^{\mathrm{pl}}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2 \int_0^t d\tau\, e^{i\Delta^{\nu,+}_{\boldsymbol{kq}}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_{\boldsymbol{k}-\boldsymbol{q}}(t)-C_{\boldsymbol{k}-\boldsymbol{q}}(\tau)}.\end{aligned} \tag{9.102}$$

with

$$\Delta^{\nu,-}_{\boldsymbol{kq}} \equiv \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}} + \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}} \qquad \Delta^{\nu,+}_{\boldsymbol{kq}} \equiv \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}-\boldsymbol{q}} - \omega^{\mathrm{pl}}_{\nu_{\boldsymbol{q}}}. \tag{9.103}$$

BWs2 self-energy for the electronic structure Hamiltonian

The BWs2 partition chooses

$$\hat{H} = \hat{H}'_0 + \hat{H}'_1, \tag{9.104}$$

with

$$\mathbf{H}'_0 = \mathbf{F} + \alpha\mathbf{W}, \qquad \mathbf{W} = \mathbf{\Sigma}^{(2),2p1h}(\omega = \tilde{\varepsilon}), \tag{9.105}$$

where $\tilde{\varepsilon}$ is the BWs2 ground-state MO energy. We know that this novel partitioning renormalizes the single-particle energies below the Fermi surface, but leaves the ones above unchanged from their Hartree-Fock value, i.e. $\tilde{\varepsilon}_a = \epsilon^{\mathrm{HF}}_a$. Thus, defining the noninteracting Green's function as $G^{(0)'}(\omega) = \left(\omega - (\mathbf{F} + \alpha\mathbf{W})\right)^{-1}$, we can write

$$G^{-1}(\omega) = \left(G^{(0)'}(\omega)\right)^{-1} - \mathbf{\Sigma}_{\mathrm{eff}}(\omega). \tag{9.106}$$

where

$$\mathbf{\Sigma}^{\text{eff}}(\omega) \approx \mathbf{\Sigma}^{(2)}\left[G^{(0)'}\right](\omega) - \alpha\mathbf{W}, \tag{9.107}$$

$$\begin{aligned}\implies \Sigma^{\text{eff}}_{pp}(t) = -i\theta(t)\left[\sum_{ijb}\mathcal{K}_{pjib}\,e^{-i(\tilde{\varepsilon}_i-\tilde{\Omega}_{jb})t} + \sum_{iab}\mathcal{K}_{pbai}\,e^{-i(\epsilon^{\text{HF}}_a+\tilde{\Omega}_{ib})t}\right] \\ - \delta(t)\alpha\sum_{iab}\frac{\mathcal{K}_{pbai}}{\tilde{\varepsilon}_p - \epsilon^{\text{HF}}_a - \tilde{\Omega}_{ib}}.\end{aligned} \tag{9.108}$$

where

$$\mathcal{K}_{pqrs} = (pr|qs)\left[2(pr|qs) - (pq|rs)\right],$$

and

$$\tilde{\Omega}_{kc} = \epsilon^{\text{HF}}_c - \tilde{\varepsilon}_k.$$

Accordingly, the bare fermionic and bosonic propagators are

$$G^{(0)'}_p(t) = -i\theta(t)e^{-i\tilde{\varepsilon}_p t}. \tag{9.109}$$

and

$$\mathcal{D}^{(-)}_{jb}(t) = -i\theta(t)e^{+i\tilde{\Omega}_{jb}t}, \qquad \mathcal{D}^{(+)}_{ib}(t) = -i\theta(t)e^{-i\tilde{\Omega}_{ib}t}, \tag{9.110}$$

respectively, leading to

$$\begin{aligned}\Sigma^{\text{eff}}_{pp}(t) = i\sum_{ijb}\mathcal{K}_{pjib}\mathcal{D}^{(-)}_{jb}(t)G^{(0)'}_i(t) + i\sum_{iab}\mathcal{K}_{pbai}\mathcal{D}^{(+)}_{ib}(t)G^{(0)'}_a(t) \\ - \delta(t)\alpha\underbrace{\sum_{iab}\frac{\mathcal{K}_{pbai}}{\tilde{\varepsilon}_p - \epsilon^{\text{HF}}_a - \tilde{\Omega}_{ib}}}_{W_{pp}}.\end{aligned} \tag{9.111}$$

The equation of motion therefore becomes

$$\frac{dG_p(t)}{dt} = -i\tilde{\varepsilon}_p G_p(t) + i\alpha W_{pp}G_p(t) - i\int_0^t d\tau\,\Sigma^{(2),R}_{pp}(t-\tau)G_p(\tau) \tag{9.112}$$

$$\begin{aligned}&= -i\tilde{\varepsilon}_p G_p(t) + i\alpha W_{pp}G_p(t) \\ &\quad - i\sum_{ijb}\mathcal{K}_{pjib}\int_0^t d\tau\,\mathcal{D}^{(-)}_{jb}(t-\tau)\frac{G^{(0)'}_i(t)}{G^{(0)'}_i(\tau)}G_p(\tau) \\ &\quad - i\sum_{iab}\mathcal{K}_{pbai}\int_0^t d\tau\,\mathcal{D}^{(+)}_{ib}(t-\tau)\frac{G^{(0)'}_a(t)}{G^{(0)'}_a(\tau)}G_p(\tau)\end{aligned} \tag{9.113}$$

$$\begin{aligned}&= -i\tilde{\varepsilon}_p G_p(t) + i\alpha W_{pp}G_p(t) \\ &\quad - i\sum_{ijb}\mathcal{K}_{pjib}\int_0^t d\tau\,\mathcal{D}^{(-)}_{jb}(t-\tau)\frac{G_p(\tau)G_i(t)}{G_i(\tau)} \\ &\quad - i\sum_{iab}\mathcal{K}_{pbai}\int_0^t d\tau\,\mathcal{D}^{(+)}_{ib}(t-\tau)\frac{G_p(\tau)G_a(t)}{G_a(\tau)},\end{aligned} \tag{9.114}$$

where the last step has affected the partial resummation. Introducing again $y_p(t) \equiv e^{C_p(t)}$, we have

$$G_p(t) = -ie^{-i\tilde{\varepsilon}_p t} y_p(t) \tag{9.115}$$

$$\implies \frac{dG_p(t)}{dt} = -i\tilde{\varepsilon}_p G_p(t) - ie^{-i\tilde{\varepsilon}_p t}\frac{dy_p(t)}{dt}. \tag{9.116}$$

Setting the two expressions for the time derivative equal gives

$$-ie^{-i\tilde{\varepsilon}_p t}\frac{dy_p(t)}{dt} = i\alpha W_{pp} G_p(t) - i\sum_{ijb} \mathcal{K}_{pjib} \int_0^t d\tau\, \mathcal{D}^{(-)}_{jb}(t-\tau)\frac{G_p(\tau)G_i(t)}{G_i(\tau)} - i\sum_{iab} \mathcal{K}_{pbai} \int_0^t d\tau\, \mathcal{D}^{(+)}_{ib}(t-\tau)\frac{G_p(\tau}{G} \tag{9.117}$$

$$\implies \frac{dy_p(t)}{dt} = i\alpha W_{pp} y_p(t) - \sum_{ijb} \mathcal{K}_{pjib} \int_0^t d\tau\, e^{i\Delta^{(-)}_{pi,jb}(t-\tau)}\frac{y_p(\tau)y_i(t)}{y_i(\tau)} - \sum_{iab} \mathcal{K}_{pbai} \int_0^t d\tau\, e^{i\Delta^{(+)}_{pa,ib}(t-\tau)}\frac{y_p(\tau)y_a(t)}{y_a(\tau)}. \tag{9.118}$$

Here we have defined

$$\Delta^{(-)}_{pi,jb} \equiv \tilde{\varepsilon}_p - \tilde{\varepsilon}_i + \tilde{\Omega}_{jb}, \quad \Delta^{(+)}_{pa,ib} \equiv \tilde{\varepsilon}_p - \tilde{\varepsilon}_a - \tilde{\Omega}_{ib}. \tag{9.119}$$

Finally, using $y_p(t) = e^{C_p(t)}$, we obtain

$$\begin{aligned}\frac{dC_p(t)}{dt} &= i\alpha W_{pp} \\ &\quad - \sum_{ijb} \mathcal{K}_{pjib} \int_0^t d\tau\, e^{i\Delta^{(-)}_{pi,jb}(t-\tau)} e^{-C_p(t)+C_p(\tau)+C_i(t)-C_i(\tau)} \\ &\quad - \sum_{iab} \mathcal{K}_{pbai} \int_0^t d\tau\, e^{i\Delta^{(+)}_{pa,ib}(t-\tau)} e^{-C_p(t)+C_p(\tau)+C_a(t)-C_a(\tau)}.\end{aligned} \tag{9.120}$$

9.5.2 Bosonic cumulants

$i, j, \boldsymbol{k}, \ldots$ denote occupied indices, whereas $a, b, \ldots$ denote virtual indices. We can write the SCCE ODE as

$$\begin{aligned}\frac{dC_{\boldsymbol{k}}(t)}{dt} &= -i\mathcal{C}_{\boldsymbol{k}} \\ &\quad - \sum_{i\nu} \mathcal{L}_{\boldsymbol{k}i\nu} \int_0^t d\tau\, e^{i\Delta^{(-)}_{\boldsymbol{k}i,\nu}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_i(t)-C_i(\tau)}\end{aligned} \tag{9.121}$$

where we then determine the Green's function as $G_{\boldsymbol{k}}(t) = -ie^{-i\epsilon^{\mathrm{ref}}_{\boldsymbol{k}} t + C_{\boldsymbol{k}}(t) - \eta t}$.

LDA reference

If we are starting from the LDA reference, then $\mathcal{C}_{\boldsymbol{k}} = 0$ and $\epsilon^{\mathrm{ref}}_{\boldsymbol{k}} = |\boldsymbol{k}|^2/2 + \Sigma^x_{\boldsymbol{k}}$. Also, we have

$$\Delta^{(-)}_{\boldsymbol{k}i,\nu} = \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_i + \omega^{\mathrm{pl}}_\nu \tag{9.122}$$

and

$$\mathcal{L}_{\boldsymbol{k}i\nu} = \left|g^{\rm pl}_{\nu}(\boldsymbol{k},\boldsymbol{q})\right|^2, \quad \boldsymbol{k}-\boldsymbol{q}\in\mathcal{O} \tag{9.123}$$

where we use the LDA single-particle energies $\epsilon^{\rm LDA}_{\boldsymbol{k}}$ along with the dRPA@LDA excitation energies $\omega^{\rm pl}_{\nu}$ and transition densities $g^{\rm pl}_{\nu}(\boldsymbol{k},\boldsymbol{q})$.

BW reference

If we are starting from the BW reference, $\nu \equiv jb$. Then, $\mathcal{C}_{\boldsymbol{k}} = -\alpha W_{\boldsymbol{k}\boldsymbol{k}}$ and $\epsilon^{\rm ref}_{\boldsymbol{k}} = \tilde{\varepsilon}_{\boldsymbol{k}}$. Also, we have

$$\Delta^{(-)}_{\boldsymbol{k}i,\nu} = \tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_i + \tilde{\Omega}_{\nu} \tag{9.124}$$

and

$$\mathcal{L}_{\boldsymbol{k}i\nu} = \mathcal{K}_{\boldsymbol{k}jib} \tag{9.125}$$

where we use the BWs2 single-particle energies $\tilde{\varepsilon}_{\boldsymbol{k}}$ along with the BWs2 particle-hole excitation energies $\tilde{\Omega}_{\nu} \equiv \tilde{\Omega}_{jb} = \epsilon^{\rm HF}_b - \tilde{\varepsilon}_j$ and second-order Coulomb interaction kernels defined by $\mathcal{K}_{pqrs} = (pr|qs)\left[2(pr|qs) - (pq|rs)\right]$.

9.5.3 Fermionic cumulants

In this theory, the particle and hole equations are coupled VIDEs because the derivative at time t depends on the full history. Therefore, they should be solved together by propagating

$$\frac{d\mathbf{C}(t)}{dt} = \mathbf{F}[\mathbf{C}](t) \tag{9.126}$$

with

$$\mathbf{C}(t) = \left(\{C^+_{\boldsymbol{a}}(t)\}_{\boldsymbol{a}\in\mathcal{V}}, \{C^-_{\boldsymbol{i}}(t)\}_{\boldsymbol{i}\in\mathcal{O}}\right). \tag{9.127}$$

We determine the Green's functions for particles and holes as $G_{\boldsymbol{a}}(t) = -ie^{-i\epsilon^{\rm ref}_{\boldsymbol{a}}t + C^+_{\boldsymbol{a}}(t) - \eta t}$ and $G_{\boldsymbol{i}}(t) = -ie^{-i\epsilon^{\rm ref}_{\boldsymbol{i}}t + C^-_{\boldsymbol{i}}(t) - \eta t}$, respectively. For particles, we have

$$\frac{dC^+_{\boldsymbol{a}}(t)}{dt} = -i\mathcal{C}_{\boldsymbol{a}} - \sum_{c\nu}\mathcal{K}_{\boldsymbol{a}c\nu}\int_0^t d\tau\, e^{i\Delta^{(+)}_{\boldsymbol{a}c\nu}(t-\tau)}e^{\Phi_{\boldsymbol{a}c\nu}(t,\tau)}. \tag{9.128}$$

For holes, we have

$$\frac{dC^-_{\boldsymbol{i}}(t)}{dt} = -\mathcal{C}_{\boldsymbol{i}} - \sum_{k\nu}\mathcal{K}_{\boldsymbol{i}k\nu}\int_0^t d\tau\, e^{i\Delta^{(-)}_{\boldsymbol{i}k\nu}(t-\tau)}e^{\Psi_{\boldsymbol{i}k\nu}(t,\tau)}. \tag{9.129}$$

LDA reference

Here, $\mathcal{C}_{\boldsymbol{p}} = 0$ and $\epsilon_{\boldsymbol{p}}^{\text{ref}} = |\boldsymbol{p}|^2/2 + \Sigma_{\boldsymbol{p}}^x$. Also, we have

$$\Delta_{\boldsymbol{p}k,\nu}^{(-)} = \epsilon_{\boldsymbol{p}}^{\text{LDA}} - \epsilon_k^{\text{LDA}} + \omega_\nu^{\text{pl}}, \quad \Delta_{\boldsymbol{p}c,\nu}^{(+)} = \epsilon_{\boldsymbol{p}}^{\text{LDA}} - \epsilon_c^{\text{LDA}} - \omega_\nu^{\text{pl}} \tag{9.130}$$

and (where $\boldsymbol{q}$ is the momentum imparted by plasmon mode ν)

$$\mathcal{K}_{\boldsymbol{p}k\nu} = \left|g_\nu^{\text{pl}}(\boldsymbol{p},\boldsymbol{q})\right|^2, \quad \boldsymbol{p}-\boldsymbol{q} \in \mathcal{O}, \quad \mathcal{K}_{\boldsymbol{p}c\nu} = \left|g_\nu^{\text{pl}}(\boldsymbol{p},\boldsymbol{q})\right|^2, \quad \boldsymbol{p}-\boldsymbol{q} \in \mathcal{V} \tag{9.131}$$

We used the LDA single-particle energies $\epsilon_{\boldsymbol{k}}^{\text{LDA}}$ along with the dRPA@LDA excitation energies ω_ν^{pl} and transition densities $g_\nu^{\text{pl}}(\boldsymbol{k},\boldsymbol{q})$. Here we take

$$\Psi_{\boldsymbol{p}q\nu}(t,\tau) = +C_p^-(t) - C_p^-(\tau) + C_q^-(t) - C_q^-(\tau), \tag{9.132}$$

given that we freeze the plasmon modes at the dRPA@LDA level. This means that if we choose the dRPA@LDA reference, we are forced to do bosonic cumulants. And so we only have one VIDE to solve, if we want the electron-removal spectral function of the state at the bottom of the band.

BW reference

Here, $\tilde{\varepsilon}_k$ denote the BWs2 single-particle energies, which are dressed for occupied states and unaffected for virtual ones. We can try $\nu \equiv jb$ if we want to do fully fermionic cumulants. Then, $\mathcal{C}_{\boldsymbol{p}} = -\alpha W_{\boldsymbol{pp}}$, where $W_{\boldsymbol{pp}} = \Theta(|\boldsymbol{k}_F| - |\boldsymbol{p}|)\Sigma_{\boldsymbol{pp}}^{2p1h}(\tilde{\varepsilon}_{\boldsymbol{p}})$ and $\epsilon_{\boldsymbol{p}}^{\text{ref}} = \tilde{\varepsilon}_{\boldsymbol{p}}$. Also, we have

$$\Delta_{\boldsymbol{p}k\nu}^{(-)} = \tilde{\varepsilon}_{\boldsymbol{p}} - \tilde{\varepsilon}_k + \varepsilon_b^{\text{HF}} - \tilde{\varepsilon}_j, \quad \Delta_{\boldsymbol{p}c\nu}^{(+)} = \tilde{\varepsilon}_{\boldsymbol{p}} - \epsilon_c^{\text{HF}} - \varepsilon_b^{\text{HF}} + \tilde{\varepsilon}_j \tag{9.133}$$

and second-order Coulomb interaction kernels defined by $\mathcal{K}_{pqrs} = (pr|qs)\,[2(pr|qs) - (pq|rs)]$. We can take

$$\Psi_{\boldsymbol{p}k\nu}(t,\tau) = -C_p^-(t) + C_p^-(\tau) + C_k^-(t) - C_k^-(\tau) - C_b^+(t) + C_b^+(\tau) + C_j^-(t) - C_j^-(\tau) \tag{9.134}$$

$$\Psi_{\boldsymbol{p}c\nu}(t,\tau) = -C_p^+(t) + C_p^+(\tau) + C_c^+(t) - C_c^+(\tau) + C_b^+(t) - C_b^+(\tau) - C_j^-(t) + C_j^-(\tau) \tag{9.135}$$

We discretize time according to

$$t_n = n\Delta t, \qquad n = 0, \ldots, N_t, \tag{9.136}$$

with initial conditions

$$C_a^+(0) = 0, \qquad C_i^-(0) = 0. \tag{9.137}$$

At a given time t_n, the histories

$$C^\pm(t_0), C^\pm(t_1), \ldots, C^\pm(t_{n-1}) \tag{9.138}$$

are already known. The particle and hole cumulants at the current time t_n are then determined simultaneously.

A simple predictor for the current-time cumulants is

$$\mathbf{C}_n^{(0)} = \mathbf{C}_{n-1} + \Delta t\, \dot{\mathbf{C}}_{n-1}. \tag{9.139}$$

Using this trial value together with the stored history, we evaluate the memory integrals. For example, using a trapezoidal quadrature,

$$\int_0^{t_n} d\tau\, K(t_n,\tau) \approx \Delta t \left[\frac{1}{2}K(t_n,t_0) + \sum_{m=1}^{n-1} K(t_n,t_m) + \frac{1}{2}K(t_n,t_n)\right]. \tag{9.140}$$

The current-time cumulants are then corrected using, for example, a trapezoidal time step,

$$\mathbf{C}_n^{(m+1)} = \mathbf{C}_{n-1} + \frac{\Delta t}{2}\left[\dot{\mathbf{C}}_{n-1} + \dot{\mathbf{C}}_n^{(m)}\right]. \tag{9.141}$$

Because $\dot{\mathbf{C}}_n$ depends on the unknown particle and hole cumulants at t_n, this update is iterated self-consistently,

$$\mathbf{C}_n^{(0)} \longrightarrow \mathbf{F}[\mathbf{C}_n^{(0)}] \longrightarrow \mathbf{C}_n^{(1)} \longrightarrow \mathbf{F}[\mathbf{C}_n^{(1)}] \longrightarrow \cdots \tag{9.142}$$

until

$$\max_p \left|C_p^{(m+1)}(t_n) - C_p^{(m)}(t_n)\right| < \varepsilon. \tag{9.143}$$

Once convergence is reached, the values at t_n are accepted and the propagation proceeds to t_{n+1}.

Thus, the numerical procedure is

$$\boxed{\text{predict} \longrightarrow \text{evaluate memory kernels} \longrightarrow \text{update particle and hole cumulants} \longrightarrow \text{iterate to convergence}} \tag{9.144}$$

and the particle and hole sectors are propagated as a single coupled object rather than sequentially.

The coupling between the two sectors follows from the fermionic intermediate-state structure: a $2p1h$ configuration contains an internal hole line, whereas a $2h1p$ configuration contains an internal particle line. Hence,

$$\{C_a^+(t)\} \longleftrightarrow \{C_i^-(t)\}, \tag{9.145}$$

and both sets of cumulants must be updated simultaneously at every time step. BW lends itself to fermionic cumulants. As before, $i, j, \ldots$ denote occupied indices, $a, b, \ldots$ denote virtual indices, and $\boldsymbol{k}$ is the external state, which may now lie in either $\mathcal{O}$ or $\mathcal{V}$. For *every* orbital $p \in \mathcal{O} \cup \mathcal{V}$ we keep the same forward-time representation used in the bosonic case,

$$G_p(t) = -i\, e^{-i\epsilon_p^{\mathrm{ref}} t + C_p(t) - \eta t}, \qquad t > 0, \tag{9.146}$$

so that a single family of cumulants $\{C_p(t)\}$ is propagated. *What is the reason we have chosen to stay in the forward time representation? Why are we not computing the lesser concentration for hole states and the greater concentration for particle states? It seems that we are computing the retarded Green's function for all states, regardless of where they are relative to the Fermi level.* The only change relative to the bosonic construction is the intermediate state: instead of one hole i plus one *undressed* quasiboson ν, the fermionic theory propagates the explicit $2h1p$ configurations $(ij;b)$ and $2p1h$ configurations $(ab;j)$, and every line in these configurations is dressed self-consistently.

Forward-time Dyson equation and bare kernel

Following Robinson *et al.*, we start from the forward-time ($t > 0$) Dyson equation for the diagonal element,

$$i\frac{dG_{\boldsymbol{k}}(t)}{dt} = \big(\epsilon^{\rm ref}_{\boldsymbol{k}} + \mathcal{C}_{\boldsymbol{k}}\big)G_{\boldsymbol{k}}(t) + \int_0^t d\tau\, \Sigma_{\boldsymbol{k}}(t-\tau)\, G_{\boldsymbol{k}}(\tau), \tag{9.147}$$

where $\mathcal{C}_{\boldsymbol{k}}$ is the static remainder of the self-energy and $\Sigma_{\boldsymbol{k}}(s)$ is its dynamical second-order part. With both self-energy branches retained (both poles on the same side, i.e. the forward/retarded convention), the time-domain kernel is

$$\Sigma_{\boldsymbol{k}}(s) = -i\,\chi^{(-)}_{\boldsymbol{k}} \sum_{ijb} \mathcal{L}^{(-)}_{\boldsymbol{k}ijb}\, e^{-i(\epsilon_i+\epsilon_j-\epsilon_b)s} - i\,\chi^{(+)}_{\boldsymbol{k}} \sum_{abj} \mathcal{L}^{(+)}_{\boldsymbol{k}abj}\, e^{-i(\epsilon_a+\epsilon_b-\epsilon_j)s}, \qquad s>0, \tag{9.148}$$

with kernels

$$\mathcal{L}^{(-)}_{\boldsymbol{k}ijb} = \mathcal{K}_{\boldsymbol{k}jib}, \qquad \mathcal{L}^{(+)}_{\boldsymbol{k}abj} = \mathcal{K}_{\boldsymbol{k}baj}, \qquad \mathcal{K}_{pqrs} = (pr|qs)\big[2(pr|qs) - (pq|rs)\big], \tag{9.149}$$

which are precisely the second-order Coulomb kernels already used for the BW reference. The switches

$$\chi^{(-)}_{\boldsymbol{k}} = \begin{cases} 1, & \boldsymbol{k}\in\mathcal{O} \\ \chi, & \boldsymbol{k}\in\mathcal{V} \end{cases}, \qquad \chi^{(+)}_{\boldsymbol{k}} = \begin{cases} \chi, & \boldsymbol{k}\in\mathcal{O} \\ 1, & \boldsymbol{k}\in\mathcal{V} \end{cases}, \tag{9.150}$$

play the same role as χ in the bosonic one-shot equations: $\chi = 1$ dresses the full second-order self-energy of every orbital, whereas $\chi = 0$ keeps only the branch natural to the sector of $\boldsymbol{k}$ ($2h1p$ for holes, $2p1h$ for particles).

Inserting Eq. (9.146) into Eq. (9.147) and dividing by $-ie^{-i\epsilon^{\rm ref}_{\boldsymbol{k}}t}$ gives the exact-in-form equation for $y_{\boldsymbol{k}}(t) \equiv e^{C_{\boldsymbol{k}}(t)}$,

$$\begin{aligned}\frac{dy_{\boldsymbol{k}}(t)}{dt} = &-i\mathcal{C}_{\boldsymbol{k}}\, y_{\boldsymbol{k}}(t) - \chi^{(-)}_{\boldsymbol{k}} \sum_{ijb} \mathcal{L}^{(-)}_{\boldsymbol{k}ijb} \int_0^t d\tau\, e^{i\Delta^{(-)}_{\boldsymbol{k}ij,b}(t-\tau)}\, y_{\boldsymbol{k}}(\tau) \\ &- \chi^{(+)}_{\boldsymbol{k}} \sum_{abj} \mathcal{L}^{(+)}_{\boldsymbol{k}abj} \int_0^t d\tau\, e^{i\Delta^{(+)}_{\boldsymbol{k}ab,j}(t-\tau)}\, y_{\boldsymbol{k}}(\tau),\end{aligned} \tag{9.151}$$

with the intermediate-state denominators

$$\Delta^{(-)}_{\boldsymbol{k}ij,b} = \epsilon^{\rm ref}_{\boldsymbol{k}} - \epsilon_i - \epsilon_j + \epsilon_b, \qquad \Delta^{(+)}_{\boldsymbol{k}ab,j} = \epsilon^{\rm ref}_{\boldsymbol{k}} - \epsilon_a - \epsilon_b + \epsilon_j. \tag{9.152}$$

Note that $\Delta^{(-)}_{\boldsymbol{k}ij,b} = \epsilon^{\rm ref}_{\boldsymbol{k}} - \epsilon_i + (\epsilon_b - \epsilon_j)$ coincides with the bosonic $\Delta^{(-)}_{\boldsymbol{k}i,\nu}$ for $\nu \equiv jb$ and $\Omega_\nu = \epsilon_b - \epsilon_j$; likewise $\Delta^{(+)}_{\boldsymbol{k}ab,j} = \Delta^{(+)}_{\boldsymbol{k}a,\nu}$.

Self-consistent dressing of the intermediate lines

The Dunn–Robinson step is to factorize the bare phase of each intermediate line and then replace bare propagators by dressed ones. Write the $2h1p$ phase as a product of single-line phases,

$$e^{-i(\epsilon_i+\epsilon_j-\epsilon_b)(t-\tau)} = \underbrace{e^{-i\epsilon_i(t-\tau)}}_{\text{forward}}\,\underbrace{e^{-i\epsilon_j(t-\tau)}}_{\text{forward}}\,\underbrace{e^{+i\epsilon_b(t-\tau)}}_{\text{backward}}, \tag{9.153}$$

and similarly for $2p1h$, where a, b are forward and j is backward. Introduce $\mathcal{G}_p(s) \equiv e^{-i\epsilon_p s + C_p(s)}$ with bare limit $\mathcal{G}_p^{(0)}(s) = e^{-i\epsilon_p s}$.

Forward lines. Exactly as in Robinson's Eq. (13), $\mathcal{G}_p^{(0)}(t-\tau) = \mathcal{G}_p^{(0)}(t)/\mathcal{G}_p^{(0)}(\tau)$, and the replacement $\mathcal{G}^{(0)} \to \mathcal{G}$ gives

$$e^{-i\epsilon_p(t-\tau)} \;\longrightarrow\; \frac{\mathcal{G}_p(t)}{\mathcal{G}_p(\tau)} = e^{-i\epsilon_p(t-\tau)}\, e^{+\Delta C_p(t,\tau)}, \qquad \Delta C_p(t,\tau) \equiv C_p(t) - C_p(\tau). \tag{9.154}$$

This is the rule that produced $+C_i(t) - C_i(\tau)$ in the bosonic equation.

Backward lines. A backward line is the same propagator evaluated at negative time argument, $e^{+i\epsilon_p(t-\tau)} = \mathcal{G}_p^{(0)}(\tau - t)$. Since the exact single-particle propagator obeys $G_p(-s) = -G_p(s)^*$, the cumulant satisfies $C_p(-s) = C_p(s)^*$, so the dressed backward line is the complex conjugate of the dressed forward line,

$$e^{+i\epsilon_p(t-\tau)} \;\longrightarrow\; \left[\frac{\mathcal{G}_p(t)}{\mathcal{G}_p(\tau)}\right]^* = e^{+i\epsilon_p(t-\tau)}\, e^{+\Delta C_p(t,\tau)^*}. \tag{9.155}$$

The conjugate is essential: it makes the backward line decay whenever $\mathrm{Re}\, C_p < 0$, i.e. the $2h1p$ and $2p1h$ intermediate states are damped by the lifetimes of all three of their constituents. (Using $e^{-\Delta C_p}$ instead would make the backward line grow.)

External line. As in the bosonic case, the factor $y_{\boldsymbol{k}}(\tau)/y_{\boldsymbol{k}}(t)$ in Eq. (9.151) contributes $e^{-\Delta C_{\boldsymbol{k}}(t,\tau)}$ once we pass from $y_{\boldsymbol{k}}$ to $C_{\boldsymbol{k}}$.

Fermionic SCCE ODE

Collecting the three rules, the fermionic SCCE ODE for an arbitrary orbital $\boldsymbol{k}$ reads

$$\begin{aligned}\frac{dC_{\boldsymbol{k}}(t)}{dt} = {} & -i\,\mathcal{C}_{\boldsymbol{k}} \\ & - \chi_{\boldsymbol{k}}^{(-)} \sum_{ijb} \mathcal{L}_{\boldsymbol{k}ijb}^{(-)} \int_0^t d\tau\, e^{i\Delta_{\boldsymbol{k}ij,b}^{(-)}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_i(t)-C_i(\tau)+C_j(t)-C_j(\tau)+C_b^*(t)-C_b^*(\tau)} \\ & - \chi_{\boldsymbol{k}}^{(+)} \sum_{abj} \mathcal{L}_{\boldsymbol{k}abj}^{(+)} \int_0^t d\tau\, e^{i\Delta_{\boldsymbol{k}ab,j}^{(+)}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_a(t)-C_a(\tau)+C_b(t)-C_b(\tau)+C_j^*(t)-C_j^*(\tau)},\end{aligned} \tag{9.156}$$

with $C_p(0) = 0$ for all p, and the Green's function is then $G_{\boldsymbol{k}}(t) = -ie^{-i\epsilon^{\rm ref}_{\boldsymbol{k}}t + C_{\boldsymbol{k}}(t) - \eta t}$ exactly as before. For numerical work the y-form analogous to Robinson's Eq. (C1) is

$$\frac{dy_{\boldsymbol{k}}(t)}{dt} = -i\mathcal{C}_{\boldsymbol{k}}\, y_{\boldsymbol{k}}(t) - \chi^{(-)}_{\boldsymbol{k}} \sum_{ijb} \mathcal{L}^{(-)}_{\boldsymbol{k}ijb} \int_0^t d\tau\, e^{i\Delta^{(-)}_{\boldsymbol{k}ij,b}(t-\tau)}\, y_{\boldsymbol{k}}(\tau)\, \frac{y_i(t)\, y_j(t)\, y^*_b(t)}{y_i(\tau)\, y_j(\tau)\, y^*_b(\tau)} - \chi^{(+)}_{\boldsymbol{k}} \sum_{abj} \mathcal{L}^{(+)}_{\boldsymbol{k}abj} \int_0^t d\tau\, e^{i\Delta^{(+)}_{\boldsymbol{k}ab,j}(t-\tau)}\, y_{\boldsymbol{k}}(\tau)\, \frac{y_a(t)\, y_b(t)\, y^*_j(t)}{y_a(\tau)\, y_b(\tau)\, y^*_j(\tau)}, \tag{9.157}$$

which is a first-order Volterra integro-differential system and can be integrated with the same Adams–Bashforth/Adams–Moulton predictor–corrector scheme used by Robinson *et al.*

Remarks.

1. *Relation to the bosonic SCCE.* Setting $\Delta C_j = \Delta C^*_b = 0$ in the $2h1p$ term (i.e. leaving the quasiboson $\nu = jb$ undressed) and $\chi^{(+)}_{\boldsymbol{k}} \to \chi$ reproduces the bosonic ODE of the previous subsection term by term. The fermionic theory therefore changes only the exponent; the kernels $\mathcal{L}$, the denominators Δ, and the static data $\mathcal{C}_{\boldsymbol{k}}, \epsilon^{\rm ref}_{\boldsymbol{k}}$ are unchanged.

2. *One-shot limit.* Dropping all ΔC factors in Eq. (9.156) gives the commented one-shot equation for $C_{\boldsymbol{k}}(t)$ with $\nu \equiv jb$, so the first self-consistency iteration started from $C \equiv 0$ reproduces the ordinary second-order cumulant, in analogy with Robinson's Appendix B.

3. *Coupled propagation.* The $2h1p$ exponent contains a virtual cumulant C_b and the $2p1h$ exponent contains an occupied cumulant C_j, independently of χ. Hence occupied and virtual cumulants cannot be propagated one after the other; $\{C_i\}_{i\in\mathcal{O}}$ and $\{C_a\}_{a\in\mathcal{V}}$ form a single state vector $\dot{C} = F[t; C(t), C(\tau \le t)]$ and are advanced together. The system remains causal: at time t only $C_p(\tau)$ with $0 \le \tau \le t$ is required.

4. *Static piece.* As in the bosonic case, the choice $\epsilon^{\rm ref}_{\boldsymbol{k}}$ and $\mathcal{C}_{\boldsymbol{k}}$ is fixed by the partition of the Hamiltonian, not by the cumulant formalism. One should still require $\mathrm{Im}\,\mathcal{C}_{\boldsymbol{k}} = 0$.

LDA reference

If we start from the LDA reference, then $\mathcal{C}_{\boldsymbol{k}} = 0$ and $\epsilon^{\rm ref}_{\boldsymbol{k}} = |\boldsymbol{k}|^2/2 + \Sigma^x_{\boldsymbol{k}}$ as before. The quasiboson ν (a dRPA@LDA excitation) is now resolved into bare particle-hole pairs (j, b) interacting through the bare Coulomb interaction, and the LDA single-particle energies are used for all internal lines *But then, is it true that we are not incorporating any screening effects such that the scheme wouldn't really work for metallic systems? Can we keep the sum between EJ and EB as an RPA excitation energy in this theory so that we are working with the screened interaction instead? Does this just mean we are using a dRPA@LDA reference instead of an LDA one?*:,

$$\Delta^{(-)}_{\boldsymbol{k}ij,b} = \epsilon^{\rm LDA}_{\boldsymbol{k}} - \epsilon^{\rm LDA}_i - \epsilon^{\rm LDA}_j + \epsilon^{\rm LDA}_b, \qquad \Delta^{(+)}_{\boldsymbol{k}ab,j} = \epsilon^{\rm LDA}_{\boldsymbol{k}} - \epsilon^{\rm LDA}_a - \epsilon^{\rm LDA}_b + \epsilon^{\rm LDA}_j. \tag{9.158}$$

For plane waves the two-electron integrals are momentum-conserving,

$$(\boldsymbol{pr}|\boldsymbol{qs}) = \frac{1}{\Omega}\, v(\boldsymbol{p}-\boldsymbol{r})\, \delta_{\boldsymbol{p}+\boldsymbol{q},\boldsymbol{r}+\boldsymbol{s}}, \qquad v(\boldsymbol{q}) = \frac{4\pi}{|\boldsymbol{q}|^2}, \tag{9.159}$$

so that the kernels of Eq. (9.149) become

$$\mathcal{L}^{(-)}_{\boldsymbol{k}ijb} = \mathcal{K}_{\boldsymbol{k}jib} = \frac{1}{\Omega^2}\, v(\boldsymbol{k}-\boldsymbol{i})\big[2v(\boldsymbol{k}-\boldsymbol{i}) - v(\boldsymbol{k}-\boldsymbol{j})\big]\, \delta_{\boldsymbol{b},\,\boldsymbol{k}+\boldsymbol{j}-\boldsymbol{i}}, \qquad \boldsymbol{b} \in \mathcal{V}, \tag{9.160}$$

$$\mathcal{L}^{(+)}_{\boldsymbol{k}abj} = \mathcal{K}_{\boldsymbol{k}baj} = \frac{1}{\Omega^2}\, v(\boldsymbol{k}-\boldsymbol{a})\big[2v(\boldsymbol{k}-\boldsymbol{a}) - v(\boldsymbol{k}-\boldsymbol{b})\big]\, \delta_{\boldsymbol{j},\,\boldsymbol{a}+\boldsymbol{b}-\boldsymbol{k}}, \qquad \boldsymbol{j} \in \mathcal{O}. \tag{9.161}$$

The momentum-conservation δ removes one of the three internal sums, e.g. the $2h1p$ term is a double sum over $\boldsymbol{i},\boldsymbol{j} \in \mathcal{O}$ with $\boldsymbol{b} = \boldsymbol{k}+\boldsymbol{j}-\boldsymbol{i}$ restricted to lie outside the Fermi sphere. Compared with the bosonic LDA construction, $\omega^{\rm pl}_\nu \to \epsilon^{\rm LDA}_b - \epsilon^{\rm LDA}_j$ and $|g^{\rm pl}_\nu(\boldsymbol{k},\boldsymbol{q})|^2 \to \mathcal{K}$ with $\boldsymbol{q} = \boldsymbol{k}-\boldsymbol{i}$; the screening that was built into the plasmon pole must now be generated dynamically by the self-consistent dressing of the j and b lines.

BW reference

If we start from the BW reference, then $\mathcal{C}_{\boldsymbol{k}} = -\alpha W_{\boldsymbol{kk}}$ and $\epsilon^{\rm ref}_{\boldsymbol{k}} = \tilde{\varepsilon}_{\boldsymbol{k}}$, exactly as in the bosonic case. *But we know that in BWS2 theory, $W_{\boldsymbol{kk}} = \Theta(\boldsymbol{k}_F - \boldsymbol{k})\Sigma^{(2),2p1h}(\omega = \tilde{\epsilon}_k)$ where $\boldsymbol{k}_F$ is the Fermi wavevector and $\tilde{\epsilon}_k = \Theta(\boldsymbol{k}-\boldsymbol{k}_F)\tilde{\epsilon}_k + \Theta(\boldsymbol{k}_F-\boldsymbol{k})\epsilon^{\rm HF}_k$ is the dressed single-particle energy. So does this carry over to this theory?* Since the BWs2 reference dresses occupied orbitals only, internal hole lines carry $\tilde{\varepsilon}$ and internal particle lines carry $\epsilon^{\rm HF}$,

$$\Delta^{(-)}_{\boldsymbol{k}ij,b} = \tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_i - \tilde{\varepsilon}_j + \epsilon^{\rm HF}_b = \tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_i + \tilde{\Omega}_{jb}, \qquad \Delta^{(+)}_{\boldsymbol{k}ab,j} = \epsilon^{\rm ref}_{\boldsymbol{k}} - \epsilon^{\rm HF}_a - \epsilon^{\rm HF}_b + \tilde{\varepsilon}_j = \epsilon^{\rm ref}_{\boldsymbol{k}} - \epsilon^{\rm HF}_a - \tilde{\Omega}_{jb}, \tag{9.162}$$

with $\tilde{\Omega}_{jb} = \epsilon^{\rm HF}_b - \tilde{\varepsilon}_j$ as before, and

$$\mathcal{L}^{(-)}_{\boldsymbol{k}ijb} = \mathcal{K}_{\boldsymbol{k}jib}, \qquad \mathcal{L}^{(+)}_{\boldsymbol{k}abj} = \mathcal{K}_{\boldsymbol{k}baj}, \tag{9.163}$$

which are identical to the bosonic $\mathcal{L}_{\boldsymbol{k}i\nu}$ and $\mathcal{L}_{\boldsymbol{k}a\nu}$ with $\nu \equiv jb$. Thus the BW bosonic SCCE was already the $2h1p$ diagram with the hole i dressed and the pair (j,b) left bare; the fermionic BW SCCE keeps the same kernels and denominators and only adds $+\Delta C_j + \Delta C^*_b$ to the exponent (and, for $\chi^{(+)}_{\boldsymbol{k}} \neq 0$, the $2p1h$ term with $+\Delta C_a + \Delta C_b + \Delta C^*_j$).

For a virtual external orbital $\boldsymbol{k} \in \mathcal{V}$ the BWs2 construction does not supply $\tilde{\varepsilon}_{\boldsymbol{k}}$ or $W_{\boldsymbol{kk}}$. The consistent choices are either (i) $\epsilon^{\rm ref}_{\boldsymbol{k}} = \epsilon^{\rm HF}_{\boldsymbol{k}}$, $\mathcal{C}_{\boldsymbol{k}} = 0$, matching the treatment of internal particle lines above, or (ii) a particle-sector BW analogue built from the $2p1h$ self-energy, $W^{+}_{\boldsymbol{kk}} \sim \Sigma^{(2),2p1h}_{\boldsymbol{kk}}(\omega = \tilde{\varepsilon}^{+}_{\boldsymbol{k}})$, in which case $\epsilon^{\rm ref}_{\boldsymbol{k}} = \tilde{\varepsilon}^{+}_{\boldsymbol{k}}$ and $\mathcal{C}_{\boldsymbol{k}} = -\alpha W^{+}_{\boldsymbol{kk}}$ should also be used for the internal particle lines. Whichever option is chosen, the same reference energies must be used in Eq. (9.152) and in Eq. (9.146). *I think we should take choice (i) then.*

9.5.4 A unified VIDE

$i, j, \ldots$ denote occupied indices, whereas $a, b, \ldots$ denote virtual indices. We can write the SCCE ODE as

$$\begin{aligned}\frac{dC_{\boldsymbol{k}}(t)}{dt} &= -i\mathcal{C}_{\boldsymbol{k}} \\ &\quad - \sum_{i\nu} \mathcal{L}_{\boldsymbol{k}i\nu} \int_0^t d\tau\, e^{i\Delta^{(-)}_{\boldsymbol{k}i,\nu}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_i(t)-C_i(\tau)} \\ &\quad - \chi \sum_{a\nu} \mathcal{L}_{\boldsymbol{k}a\nu} \int_0^t d\tau\, e^{i\Delta^{(+)}_{\boldsymbol{k}a,\nu}(t-\tau)} e^{-C_{\boldsymbol{k}}(t)+C_{\boldsymbol{k}}(\tau)+C_a(t)-C_a(\tau)},\end{aligned} \tag{9.164}$$

where we then determine the Green's function as $G_{\boldsymbol{k}}(t) = -ie^{-i\epsilon^{\mathrm{ref}}_{\boldsymbol{k}}t + C_{\boldsymbol{k}}(t) - \eta t}$. We take $\chi = 0$ if we are solving the quasiboson Hamiltonian (because we can only allow for $n\mathrm{h}n + 1\mathrm{p}$ configurations, which should not include any 2h1p contributions) and $\chi = 1$ if we are solving the electronic structure one.

How to recover the one-shot variants

To do so, we instead solve the ODE

$$\begin{aligned}\frac{dC_{\boldsymbol{k}}(t)}{dt} &= -i\mathcal{C}_{\boldsymbol{k}} - \sum_{i\nu} \mathcal{L}_{\boldsymbol{k}i\nu} \int_0^t d\tau\, e^{i\Delta^{(-)}_{\boldsymbol{k}i,\nu}(t-\tau)} - \chi \sum_{a\nu} \mathcal{L}_{\boldsymbol{k}a\nu} \int_0^t d\tau\, e^{i\Delta^{(+)}_{\boldsymbol{k}a,\nu}(t-\tau)} \\ &= -i\mathcal{C}_{\boldsymbol{k}} - \sum_{i\nu} \mathcal{L}_{\boldsymbol{k}i\nu} \frac{e^{i\Delta^{(-)}_{\boldsymbol{k}i,\nu}t} - 1}{i\Delta^{(-)}_{\boldsymbol{k}i,\nu}} - \chi \sum_{a\nu} \mathcal{L}_{\boldsymbol{k}a\nu} \frac{e^{i\Delta^{(+)}_{\boldsymbol{k}a,\nu}t} - 1}{i\Delta^{(+)}_{\boldsymbol{k}a,\nu}} \\ \implies \quad C_{\boldsymbol{k}}(t) &= -i\mathcal{C}_{\boldsymbol{k}}t + \sum_{i\nu} \frac{\mathcal{L}_{\boldsymbol{k}i\nu}}{\left(\Delta^{(-)}_{\boldsymbol{k}i,\nu}\right)^2} \left(e^{i\Delta^{(-)}_{\boldsymbol{k}i,\nu}t} - i\Delta^{(-)}_{\boldsymbol{k}i,\nu}t - 1\right) \\ &\quad + \chi \sum_{a\nu} \frac{\mathcal{L}_{\boldsymbol{k}a\nu}}{\left(\Delta^{(+)}_{\boldsymbol{k}a,\nu}\right)^2} \left(e^{i\Delta^{(+)}_{\boldsymbol{k}a,\nu}t} - i\Delta^{(+)}_{\boldsymbol{k}a,\nu}t - 1\right).\end{aligned} \tag{9.165}$$

LDA reference

If we are starting from the LDA reference, then $\mathcal{C}_{\boldsymbol{k}} = 0$ and $\epsilon^{\mathrm{ref}}_{\boldsymbol{k}} = |\boldsymbol{k}|^2/2 + \Sigma^x_{\boldsymbol{k}}$. Also, we have

$$\Delta^{(-)}_{\boldsymbol{k}i,\nu} = \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_i + \omega^{\mathrm{pl}}_\nu, \quad \Delta^{(+)}_{\boldsymbol{k}a,\nu} = \epsilon^{\mathrm{LDA}}_{\boldsymbol{k}} - \epsilon^{\mathrm{LDA}}_a - \omega^{\mathrm{pl}}_\nu \tag{9.166}$$

and

$$\mathcal{L}_{\boldsymbol{k}i\nu} = \left|g^{\mathrm{pl}}_\nu(\boldsymbol{k}, \boldsymbol{q})\right|^2, \quad \boldsymbol{k} - \boldsymbol{q} \in \mathcal{O}, \tag{9.167}$$

$$\mathcal{L}_{\boldsymbol{k}a\nu} = \left|g^{\mathrm{pl}}_\nu(\boldsymbol{k}, \boldsymbol{q})\right|^2, \quad \boldsymbol{k} - \boldsymbol{q} \in \mathcal{V} \tag{9.168}$$

where we use the LDA single-particle energies $\epsilon^{\mathrm{LDA}}_{\boldsymbol{k}}$ along with the dRPA@LDA excitation energies ω^{pl}_ν and transition densities $g^{\mathrm{pl}}_\nu(\boldsymbol{k}, \boldsymbol{q})$.

BW reference

If we are starting from the BW reference, $\nu \equiv jb$. Then, $\mathcal{C}_{\boldsymbol{k}} = -\alpha W_{\boldsymbol{kk}}$ and $\epsilon_{\boldsymbol{k}}^{\mathrm{ref}} = \tilde{\varepsilon}_{\boldsymbol{k}}$. Also, we have

$$\Delta_{\boldsymbol{k}i,\nu}^{(-)} = \tilde{\varepsilon}_{\boldsymbol{k}} - \tilde{\varepsilon}_i + \tilde{\Omega}_\nu, \quad \Delta_{\boldsymbol{k}a,\nu}^{(+)} = \tilde{\varepsilon}_{\boldsymbol{k}} - \epsilon_a^{\mathrm{HF}} - \tilde{\Omega}_\nu \tag{9.169}$$

and

$$\mathcal{L}_{\boldsymbol{k}i\nu} = \mathcal{K}_{\boldsymbol{k}jib}, \quad \mathcal{L}_{\boldsymbol{k}a\nu} = \mathcal{K}_{\boldsymbol{k}baj}, \tag{9.170}$$

where we use the BWs2 single-particle energies $\tilde{\varepsilon}_{\boldsymbol{k}}$ along with the BWs2 particle-hole excitation energies $\tilde{\Omega}_\nu \equiv \tilde{\Omega}_{jb} = \epsilon_b^{\mathrm{HF}} - \tilde{\varepsilon}_j$ and second-order Coulomb interaction kernels defined by $\mathcal{K}_{pqrs} = (pr|qs)\,[2(pr|qs) - (pq|rs)]$.

9.6

9.6.1 Holm–Aryasetiawan self-consistent cumulant derivation

We begin from the cumulant representation of the Green's function,

$$G_{\boldsymbol{k}}(t) = G_{\boldsymbol{k}}^{(0)}(t) e^{C_{\boldsymbol{k}}(t)}. \tag{9.171}$$

Equating the cumulant expansion to the Dyson one to second order in the interaction yields

$$C_{\boldsymbol{k}}(t) = \frac{1}{G_{\boldsymbol{k}}^{(0)}(t)} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi}\, e^{-i\omega t} \Sigma_{\boldsymbol{k}}(\omega) \left[G_{\boldsymbol{k}}^{(0)}(\omega) \right]^2. \tag{9.172}$$

Employing inverse Fourier transforms, we obtain

$$\begin{aligned} C_{\boldsymbol{k}}(t) = \frac{1}{G_{\boldsymbol{k}}^{(0)}(t)} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \int_{-\infty}^{\infty} dt_3 \\ \times\, e^{i\omega(t_1+t_2+t_3-t)} \Sigma_{\boldsymbol{k}}(t_1) G_{\boldsymbol{k}}^{(0)}(t_2) G_{\boldsymbol{k}}^{(0)}(t_3). \end{aligned} \tag{9.173}$$

The frequency integral produces a Dirac delta function,

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{i\omega(t_1+t_2+t_3-t)} = \delta(t_1+t_2+t_3-t) \tag{9.174}$$

$$\implies C_{\boldsymbol{k}}(t) = \frac{1}{G^{(0)}_{\boldsymbol{k}}(t)} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2\, \Sigma_{\boldsymbol{k}}(t_1) G^{(0)}_{\boldsymbol{k}}(t_2) G^{(0)}_{\boldsymbol{k}}(t-t_1-t_2) \tag{9.175}$$

$$= -i \int_0^t dt_1 \int_0^{t-t_1} dt_2\, \Sigma^R_{\boldsymbol{k}}(t_1) e^{i\varepsilon_{\boldsymbol{k}} t_1} \tag{9.176}$$

$$= -i \int_0^t dt_2 \int_0^{t_2} dt_1\, \Sigma^R_{\boldsymbol{k}}(t_1) e^{i\varepsilon_{\boldsymbol{k}} t_1} \tag{9.177}$$

$$= -\int_0^t dt_2 \int_0^{t_2} dt_1\, \frac{\Sigma^R_{\boldsymbol{k}}(t_1)}{G^{(0),R}_{\boldsymbol{k}}(t_1)} \tag{9.178}$$

$$\implies \dot{C}_{\boldsymbol{k}}(t) = -\int_0^t d\tau\, \frac{\Sigma^R_{\boldsymbol{k}}(\tau)}{G^{(0),R}_{\boldsymbol{k}}(\tau)} \tag{9.179}$$

$$\implies \frac{dG^R_{\boldsymbol{k}}(t)}{dt} = \left[-i\varepsilon_{\boldsymbol{k}} + \dot{C}_{\boldsymbol{k}}(t)\right] G^R_{\boldsymbol{k}}(t), \quad t > 0 \tag{9.180}$$

$$= -i\varepsilon_{\boldsymbol{k}} G^R_{\boldsymbol{k}}(t) - \int_0^t d\tau\, \Sigma^R_{\boldsymbol{k}}(\tau) \frac{G^R_{\boldsymbol{k}}(t)}{G^{(0),R}_{\boldsymbol{k}}(\tau)} \tag{9.181}$$

$$\implies 0 = i\frac{\partial G^R_{\boldsymbol{k}}(t)}{\partial t} - \varepsilon_{\boldsymbol{k}} G^R_{\boldsymbol{k}}(t) + \sum_{\boldsymbol{q}} \int_0^t d\tau\, W^{(0),R}_{\boldsymbol{q}}(\tau) G^R_{\boldsymbol{k}+\boldsymbol{q}}(\tau) \frac{G^R_{\boldsymbol{k}}(t)}{G^{(0),R}_{\boldsymbol{k}}(\tau)}, \tag{9.182}$$

where we made the insertion

$$\Sigma^R_{\boldsymbol{k}}(\tau) = i \sum_{\boldsymbol{q}} G^R_{\boldsymbol{k}+\boldsymbol{q}}(\tau) W^{(0),R}_{\boldsymbol{q}}(\tau). \tag{9.183}$$

9.6.2 G_0W_0 is CC-0-S0-(1,1) on the quasiboson Hamiltonian

For CC-0-S0, the cluster operator vanishes, such that the reference is

$$|\Phi_{\boldsymbol{p}}\rangle = h^\dagger_{\boldsymbol{p}} |0\rangle_F \otimes |0\rangle_{\rm pl}. \tag{9.184}$$

The first-order response manifold is

$$\mathcal{R}^{(1)}_{\boldsymbol{p}} = \left\{1,\, h^\dagger_{\boldsymbol{p}-\boldsymbol{q}} h_{\boldsymbol{p}} b^\dagger_{\nu_{\boldsymbol{q}}}\right\}. \tag{9.185}$$

The quasiboson Hamiltonian projected onto this space has the block form

$$\boldsymbol{H}^{(1)} = \begin{pmatrix} \epsilon^{\rm HF}_{\boldsymbol{p}} & \boldsymbol{g}^\dagger \\ \boldsymbol{g} & \boldsymbol{H}_{\rm 1b} \end{pmatrix}, \tag{9.186}$$

where

$$[\boldsymbol{H}_{\rm 1b}]_{\boldsymbol{q}\nu,\boldsymbol{q}'\nu'} = \delta_{\boldsymbol{q}\boldsymbol{q}'}\delta_{\nu\nu'} \left(\epsilon^{\rm HF}_{\boldsymbol{p}-\boldsymbol{q}} - \omega^{\rm pl}_{\nu_{\boldsymbol{q}}}\right), \tag{9.187}$$

and

$$g_{\boldsymbol{q}\nu} = \langle \boldsymbol{p}-\boldsymbol{q};\nu_{\boldsymbol{q}}|\hat{H}_{\rm ep}|\Phi_{\boldsymbol{p}}\rangle, \tag{9.188}$$

where we have defined $|\boldsymbol{p}-\boldsymbol{q};\nu_{\boldsymbol{q}}\rangle = h^\dagger_{\boldsymbol{p}-\boldsymbol{q}}h_{\boldsymbol{p}}b^\dagger_{\nu_{\boldsymbol{q}}}|\Phi_{\boldsymbol{p}}\rangle$. If we define the broadenings as η_G and η_Σ, respectively, then the retarded Green's function for the hole at momentum $\boldsymbol{p}$ is

$$\begin{aligned}
G^R_{\boldsymbol{p}}(\omega) &= \langle\Phi_{\boldsymbol{p}}|\left(\omega - \boldsymbol{H}^{(1)} + i\eta_G\right)^{-1}|\Phi_{\boldsymbol{p}}\rangle && (9.189)\\
&= \frac{1}{\omega - \epsilon^{\rm HF}_{\boldsymbol{p}} - \boldsymbol{g}^\dagger\left(\omega - \boldsymbol{H}_{\rm 1b} + i\eta_\Sigma\right)^{-1}\boldsymbol{g} + i\eta_G} && (9.190)\\
&= \frac{1}{\omega - \epsilon^{\rm HF}_{\boldsymbol{p}} - \sum_{\boldsymbol{q}\nu}\dfrac{|g_{\boldsymbol{q}\nu}|^2}{\omega - \left(\epsilon^{\rm HF}_{\boldsymbol{p}-\boldsymbol{q}} - \omega^{\rm pl}_{\nu_{\boldsymbol{q}}}\right) + i\eta_\Sigma} + i\eta_G} && (9.191)\\
&= \frac{1}{\omega - \epsilon^{\rm HF}_{\boldsymbol{p}} - \Sigma^{2h1p,c}_{\boldsymbol{p}}(\omega) + i\eta_G} && (9.192)\\
&= \frac{1}{\omega - \epsilon^{0}_{\boldsymbol{p}} - \Sigma^{x}_{\boldsymbol{p}} - \Sigma^{2h1p,c}_{\boldsymbol{p}}(\omega) + i\eta_G}. && (9.193)
\end{aligned}$$

To go from the first to the second step we used the Schur complement relation. Then we identified $\Sigma^{2h1p,c}_{\boldsymbol{p}}(\omega) = \sum_{\boldsymbol{q}\nu}\frac{|g_{\boldsymbol{q}\nu}|^2}{\omega-\left(\epsilon^{\rm HF}_{\boldsymbol{p}-\boldsymbol{q}}-\omega^{\rm pl}_{\nu_{\boldsymbol{q}}}\right)+i\eta_\Sigma}$, and $\epsilon^{\rm HF}_{\boldsymbol{p}} = \epsilon^0_{\boldsymbol{p}} + \Sigma^x_{\boldsymbol{p}}$,. *This may look unconventional, but recall that our quasiboson Hamiltonian only allows for a single hole with (bosonized) particle-hole excitations. Thus, we can maximally get a 2h1p configuration, so the 2p1h contribution must vanish from the self-energy.*

9.6.3 Getting to the CCGF

Once again, we start with the same Hamiltonian:

$$H_{\rm ch} = E_{\rm ch} + \sum_\nu \Omega_\nu Q^\dagger_\nu Q_\nu + \sum_\nu\left(M_\nu Q^\dagger_\nu + M^*_\nu Q_\nu\right), \tag{9.194}$$

But now, if we swap out the QB operators by actual particle-hole excitations, we get

$$H_{\rm ch} = E^{\rm ch}_{\rm ref} + \sum_{pq} f^{\rm ch}_{pq}\left\{a^\dagger_p a_q\right\} + \frac{1}{4}\sum_{pqrs} v^{\rm ch}_{pqrs}\left\{a^\dagger_p a^\dagger_q a_s a_r\right\} \tag{9.195}$$

Then the retarded Green's function can be written as

$$G^R_{c,h}(t) = -i\theta(t)\langle\Phi_0|\left(1+\Lambda\right)\overline{a^\dagger_c e^{-iH_N t}\bar{a}_c}\,|\Phi_0\rangle \tag{9.196}$$

9.7 Getting the correlation energy from a two-particle density matrix

Define

$$\hat{\rho}_{pq}(t) = a^\dagger_p(t)a_q(t) \implies \langle\hat{\rho}_{pq}\rangle = \gamma_{qp} \tag{9.197}$$

and

$$\delta\hat{\rho}_{pq}(t) = \hat{\rho}_{pq}(t) - \langle\hat{\rho}_{pq}\rangle\,. \tag{9.198}$$

Then,

$$\begin{aligned}
L^T_{pq,rs}(t) &= -i\,\langle T\left[\delta\hat{\rho}_{pq}(t)\delta\hat{\rho}_{rs}(0)\right]\rangle && (9.199)\\
&= -i\theta(t)\,\langle\delta\hat{\rho}_{pq}(t)\delta\hat{\rho}_{rs}(0)\rangle - i\theta(-t)\,\langle\delta\hat{\rho}_{rs}(0)\delta\hat{\rho}_{pq}(t)\rangle && (9.200)\\
\implies L^T_{pq,rs}\left(0^+\right) &= -i\,\langle\delta\hat{\rho}_{pq}\delta\hat{\rho}_{rs}\rangle && (9.201)\\
\implies iL^T_{pq,rs}\left(0^+\right) &= \langle\hat{\rho}_{pq}\hat{\rho}_{rs}\rangle - \langle\hat{\rho}_{pq}\rangle\,\langle\hat{\rho}_{rs}\rangle && (9.202)\\
&= \left\langle a^\dagger_p a_q a^\dagger_r a_s\right\rangle - \gamma_{qp}\gamma_{sr} && (9.203)\\
&= \delta_{qr}\gamma_{sp} - \Gamma_{pr,sq} - \gamma_{qp}\gamma_{sr} && (9.204)\\
\implies \Gamma_{pr,sq} &= \delta_{qr}\gamma_{sp} - \gamma_{qp}\gamma_{sr} - iL^T_{pq,rs}\left(0^+\right) && (9.205)
\end{aligned}$$

where we simplified using the following manipulations. [2] The idea is to get the 1-RDMs via the linearized GW density matrix. Meanwhile,

$$\begin{aligned}
L^T(\omega) &= \sum_I \left[\frac{R^I R^{I\dagger}}{\omega - \Omega_I + i\eta} - \frac{R^{I\dagger}R^I}{\omega + \Omega_I - i\eta}\right] && (9.206)\\
\implies L^T(t\to 0^+) &= \sum_I R^I R^{I\dagger}. && (9.207)
\end{aligned}$$

But we already know that

$$\Gamma = A(\gamma\otimes\gamma) + \lambda \tag{9.208}$$

so $\lambda = \sum_I R^I R^{I\dagger}$.

[2] Anticommuting a_q through $a^\dagger_r$ gives $a^\dagger_p a_q a^\dagger_r a_s = \delta_{qr}a^\dagger_p a_s - a^\dagger_p a^\dagger_r a_q a_s$. If

$$\Gamma_{pr,sq} \equiv \left\langle a^\dagger_p a^\dagger_r a_q a_s\right\rangle,$$

then

$$\left\langle a^\dagger_p a_q a^\dagger_r a_s\right\rangle = \delta_{qr}\gamma_{sp} - \Gamma_{pr,sq}.$$

9.8 Screening through BW or RPA

The way that I think about these different choices is through the Casida eigenproblem

$$\begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ -\boldsymbol{B} & -\boldsymbol{A} \end{pmatrix} \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix} = \begin{pmatrix} \boldsymbol{X} & \boldsymbol{Y} \\ \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix} \begin{pmatrix} \boldsymbol{\Omega} & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{\Omega} \end{pmatrix}. \tag{9.209}$$

The $\boldsymbol{A}$ and $\boldsymbol{B}$ matrices are defined as

$$A_{ia,jb} = \underbrace{(F_{aa} - F_{ii} - \alpha W_{ii})}_{\epsilon_a^{\mathrm{HF}} - \tilde{\epsilon}_i} \delta_{ij}\delta_{ab} + \beta \mathcal{K}_{ia,bj} \tag{9.210}$$

$$B_{ia,jb} = \beta \mathcal{K}_{ia,jb} \tag{9.211}$$

where $\boldsymbol{F}$ and $\boldsymbol{W}$ are the canonical HF Fock and W matrices from BWs2 ground state theory, respectively. In particular, we have

$$W_{ii} = \Sigma_{ii}^{(2),2p1h}(\tilde{\epsilon}_i) = \frac{1}{2} \sum_{jab} \frac{\langle ij||ab\rangle \langle ab||ij\rangle}{\tilde{\epsilon}_i + \tilde{\epsilon}_j - \epsilon_a^{\mathrm{HF}} - \epsilon_b^{\mathrm{HF}}} \tag{9.212}$$

where

$$\tilde{\epsilon}_i = \epsilon_i^{\mathrm{HF}} + \alpha \Sigma_{ii}^{(2),2p1h}(\tilde{\epsilon}_i) \tag{9.213}$$

is the nonlinear equation for the dressed BW orbital energies that we solve for self-consistently. This is consistent with the intuition that in BW, the occupied orbitals get dressed, while the virtual orbitals are unchanged. The matrix elements $\mathcal{K}_{pq,rs} = (pq||rs)$ or $(pq|rs)$, where we make the latter choice under the direct approximation. The excitation energies are defined by Ω_I and the transition densities are $M_{pq\nu} = \sum_{ia} (X_{ia}^{\nu} + Y_{ia}^{\nu}) \mathcal{K}_{pq,ia}$. It is instructive to look at the self-energies that result from the choice of screening parameters α and β.

9.8.1 GW

To get the GW correlation self-energy, we make the direct approximation (to avoid double counting exchange contributions) and choose the parameter set $\alpha = 0$ and $\beta = 1$. This yields the GW self-energy as

$$\Sigma_{pq}^{GW}(\omega) = \sum_{i\nu} \frac{M_{pi\nu} M_{qi\nu}^*}{\omega - \epsilon_i^{\mathrm{HF}} + \Omega_\nu} + \sum_{a\nu} \frac{M_{pa\nu} M_{qa\nu}^*}{\omega - \epsilon_a^{\mathrm{HF}} - \Omega_\nu}. \tag{9.214}$$

9.8.2 BWs2

To get the BWs2 self-energy, $\alpha \in \mathbb{Z}_+$ and $\beta = 0$. The BWs2 self-energy, which is the second-order expression on a BW reference, is similarly given by

$$\Sigma_{pq}^{BWs2}(\omega) = \frac{1}{2} \sum_{i\nu} \frac{M_{pi\nu} M_{qi\nu}^*}{\omega - \tilde{\epsilon}_i + \Omega_\nu} + \frac{1}{2} \sum_{a\nu} \frac{M_{pa\nu} M_{qa\nu}^*}{\omega - \epsilon_a^{\mathrm{HF}} - \Omega_\nu}. \tag{9.215}$$

This expression may seem foreign, but consider that for $\alpha \in \mathbb{Z}_+$ and $\beta = 0$, the Casida equation is already diagonal, so the transition densities are simply $M_{pq\nu} = (pq||ia)$.

9.8.3 Connection between RPAx and BW through Lowdin downfolding

Since our purpose is to make contact with Lowdin downfolding, we partition into target and rest spaces, denoted P and Q, respectively. We can write the RPAx eigenvalue equation as

$$\mathcal{H}\boldsymbol{Z}_\nu = \Omega_\nu \boldsymbol{Z}_\nu, \tag{9.216}$$

where

$$\mathcal{H} \equiv \begin{pmatrix} \mathcal{H}_{PP} & \mathcal{H}_{PQ} \\ \mathcal{H}_{QP} & \mathcal{H}_{QQ} \end{pmatrix} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ -\boldsymbol{B} & -\boldsymbol{A} \end{pmatrix} = \begin{pmatrix} \boldsymbol{D}+\boldsymbol{K}^A & \boldsymbol{K}^B \\ -\boldsymbol{K}^B & -\boldsymbol{D}-\boldsymbol{K}^A \end{pmatrix} \tag{9.217}$$

$$\boldsymbol{Z}_\nu \equiv \begin{pmatrix} \boldsymbol{Z}_{P,\nu} \\ \boldsymbol{Z}_{Q,\nu} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} \boldsymbol{X}^P_\nu \\ \boldsymbol{Y}^P_\nu \end{pmatrix} \\ \begin{pmatrix} \boldsymbol{X}^Q_\nu \\ \boldsymbol{Y}^Q_\nu \end{pmatrix} \end{pmatrix}. \tag{9.218}$$

where $D_{ia,jb} = (\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}})\delta_{ij}\delta_{ab}$, $K^A_{ia,jb} = \mathcal{K}_{ia,bj}$, and $K^B_{ia,jb} = \mathcal{K}_{ia,jb}$.

Downfolding

Equation (9.216) therefore gives the coupled equations

$$\mathcal{H}_{PP}\boldsymbol{Z}_{P,\nu} + \mathcal{H}_{PQ}\boldsymbol{Z}_{Q,\nu} = \Omega_\nu \boldsymbol{Z}_{P,\nu}, \tag{9.219}$$

$$\mathcal{H}_{QP}\boldsymbol{Z}_{P,\nu} + \mathcal{H}_{QQ}\boldsymbol{Z}_{Q,\nu} = \Omega_\nu \boldsymbol{Z}_{Q,\nu}. \tag{9.220}$$

Solving Eq. (9.220) for the component in the rest space Q gives

$$\boldsymbol{Z}_{Q,\nu} = (\Omega_\nu \boldsymbol{I} - \mathcal{H}_{QQ})^{-1}\mathcal{H}_{QP}\boldsymbol{Z}_{P,\nu}. \tag{9.221}$$

Substituting this expression into Eq. (9.219) yields an eigenvalue problem defined entirely in the target space P,

$$\left[\mathcal{H}_{PP} + \mathcal{H}_{PQ}\left(\Omega_\nu \boldsymbol{I} - \mathcal{H}_{QQ}\right)^{-1}\mathcal{H}_{QP}\right]\boldsymbol{Z}_{P,\nu} = \Omega_\nu \boldsymbol{Z}_{P,\nu}. \tag{9.222}$$

Defining an energy-dependent effective self-energy

$$\boldsymbol{\Sigma}^{\mathrm{eff}}_P(\omega) = \mathcal{H}_{PQ}\left(\omega \boldsymbol{I} - \mathcal{H}_{QQ}\right)^{-1}\mathcal{H}_{QP} \tag{9.223}$$

$$= -\sum_{iajb} \frac{(ia||jb)^2}{\omega - [(\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}) + (ia||bj)]} \tag{9.224}$$

The downfolded RPAx problem can then be written as

$$\left[\mathcal{H}_{PP} + \boldsymbol{\Sigma}^{\mathrm{eff}}_P(\Omega_\nu)\right]\boldsymbol{Z}_{P,\nu} = \Omega_\nu \boldsymbol{Z}_{P,\nu} \tag{9.225}$$

$$\left[(\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}})\delta_{ij}\delta_{ab} + (ia||bj) - \sum_{iajb} \frac{(ia||jb)^2}{\Omega_\nu - [(\epsilon_a^{\mathrm{HF}} - \epsilon_i^{\mathrm{HF}}) + (ia||bj)]}\right]\boldsymbol{Z}_{P,\nu} = \Omega_\nu \boldsymbol{Z}_{P,\nu} \tag{9.226}$$

which is emblematic of the BW theory. This is an eigenvalue problem like $(x + \frac{c}{w-x})v = wv$, where the correction term depends on the eigenvalue itself. If we think of the effective self-energy as a correction to a non-interacting Hamiltonian $\mathcal{H}_{PP}$, then RPAx (or equivalently dRPA) can be seen as a BW theory, given that the frequencies Ω_ν that are being solved for in the eigenproblem appear in the correction to $\mathcal{H}_{PP}$ itself.

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